ME F211 Mechanics of Solids

Exam Study Guide — Chapters 1 to 6 (Crandall, Dahl & Lardner)

Built from the lecture slides, Crandall, and previous-year papers 2014–2024

Contents

How to use this guide **2**

Chapter 1 — Fundamental Principles of Mechanics **3**

Concept 1.1 — What mechanics of solids is, and how to attack any problem 3
Concept 1.2 — Forces, moments, couples and equivalent force systems 4
Concept 1.3 — Supports, free-body diagrams and equilibrium 6
Concept 1.4 — Dry (Coulomb) friction, including belt friction 20
Concept 1.5 — Plane trusses: method of joints, method of sections, zero-force members 28
Chapter 1 — Formula and convention sheet 42

Chapter 2 — Introduction to Mechanics of Deformable Bodies **44**

Concept 2.1 — The three-step method for deformable bodies; springs in parallel and series 44
Concept 2.2 — Uniaxial loading: $\delta = PL/AE$ and bars with several segments 50
Concept 2.3 — Statically indeterminate axial problems 58
Concept 2.4 — Displacement (Williot) diagrams for pin-jointed structures 67
Concept 2.5 — Strain energy and Castigliano's theorem 72
Concept 2.6 — Flexible members: thin rings and belts 81
Chapter 2 — Formula and convention sheet 83

Chapter 3 — Forces and Moments Transmitted by Slender Members **85**

⚠️ Read first: the sign convention for shear force in this course 85
Concept 3.1 — Internal forces at a section: the method of sections 86
Concept 3.2 — Distributed loads, their resultants and the limits of using them 98
Concept 3.3 — Differential relations between q, V and M 110
Concept 3.4 — Singularity (Macaulay) functions 123
Concept 3.5 — Cantilevers, frames, curved and 3-D members 127
Chapter 3 — Formula and convention sheet 133

Chapter 4 — Stress and Strain **135**

⚠️ Read first: sign conventions for this chapter 135
Concept 4.1 — Stress at a point and the stress tensor 135
Concept 4.2 — Stress transformation in plane stress 138
Concept 4.3 — Mohr's circle, principal stresses and maximum shear stress 144
Concept 4.4 — Thin-walled pressure vessels (Crandall Probs 4.10–4.14) 160
Concept 4.5 — Strain and the strain–displacement relations (plane strain) 165
Concept 4.6 — Strain transformation and Mohr's circle of strain 167
Concept 4.7 — Strain rosettes 174
Chapter 4 — Formula and convention sheet 182

Chapter 5 — Stress–Strain–Temperature Relations **184**

Concept 5.1 — The tensile test and material behaviour 184
Concept 5.2 — Poisson's ratio and the generalized Hooke's law 186
Concept 5.3 — Relations among E, G, ν and the bulk modulus K 195
Concept 5.4 — Thermal strain and thermal stresses 197
Concept 5.5 — Complete equations of elasticity and strain energy 206
Concept 5.6 — Yield criteria: Tresca (maximum shear stress) and von Mises (distortion energy) 207
Chapter 5 — Formula and convention sheet 212

Chapter 6 — Torsion **214**

Concept 6.1 — Geometry of deformation of a twisted circular shaft....214
Concept 6.2 — The torsion formula and the angle of twist....215
Concept 6.3 — Power transmission....231
Concept 6.4 — Gears and coupled shafts....232
Concept 6.5 — Statically indeterminate shafts and torsional strain energy....237
Concept 6.6 — Combined stresses: torsion + axial + bending; yielding of shafts....240
Chapter 6 — Formula and convention sheet....249
Course summary....250
Part 1 — One-page formula and convention sheet (Ch 1–6)....250
Part 2 — PYQ question types ranked by frequency....251
Part 3 — Errors found in the slides (double-check with your instructor)....252
Part 4 — PYQ questions not covered here (topics after Ch 6)....253

How to use this guide

Built from the uploaded lecture slides (Ch 1–6), Crandall, Dahl & Lardner *An Introduction to the Mechanics of Solids* (2nd ed., SI), and six previous-year papers (2014-15 Test-1/Test-2/Compre, 2019-20, 2021-22, 2023-24 and 2024-25 Compre).

Each chapter follows the same layout for every concept: **Part A**, eight numbered items (intuition → why it matters → formal treatment and conventions → derivations → picture → special cases → traps → summary box), then **Part B**, the solved questions (slides → Crandall → PYQs, easy → hard). Every solution uses one template: *Problem · Given/Find (with units) · Approach · FBD in words · numbered steps with every substitution · boxed answer · check (often by a second method) · exam takeaway with a likely variation.*

The textbook problems added in this edition (about 45 more, from Crandall's end-of-chapter sets for Ch 1–5) are labelled *Crandall Prob x.y*. The uploaded textbook PDF stops before Chapter 6's problem set, so Ch 6 uses the slide and PYQ problems.

> ⚠️ **Scope warning.** The compre papers cover the **whole course**. About **40% of PYQ marks** come from chapters **after** Ch 6 (bending stress σ = My/I, beam deflection and indeterminate beams, curved beams, Euler buckling). Those slides weren't uploaded, so those questions are listed below but not solved here.

What's in this Word edition: every question whose source has a figure shows that **original figure** directly under its heading (cropped from the lecture slides, from Crandall's textbook, or from the question paper), and most solutions end with a **worked diagram** drawn from the solution itself: free-body diagrams, truss member-force maps (blue = tension, red = compression), shear-force and bending-moment diagrams (Crandall sign convention), axial-force and torque diagrams, load–deflection and design curves, Mohr's circles of stress and strain (+τ plotted down at X) and rotated stress elements.

Chapter 1 — Fundamental Principles of Mechanics

Force systems, free-body diagrams, equilibrium, dry friction, plane trusses

Sources used: Slides "chapter 1 slides" (Dr. Sharad Shrivastava); Crandall, Dahl & Lardner Ch. 1 (§1.1–1.8, Probs 1.7, 1.10, 1.13, 1.21, 1.31, 1.37, 1.39, 1.42, 1.45); PYQ 2014-15 Test-1 (Q1, Q3).

How to read this chapter. Every concept has a **Part A** with eight numbered items (intuition → why it matters → formal treatment → derivation → picture → special cases → traps → summary box). It is followed by **Part B**, the solved questions: slides first, then Crandall, then PYQs, each group running easy → hard. Every solved question uses the same template:

> **Problem** (restated) · **Given / Find** (with units) · **Approach** (which idea and why) · **FBD in words** · **Solution** (numbered steps, every substitution shown) · **Answer** (boxed) · **Check** (does it make sense? second method where possible) · **Exam takeaway** (what is really tested, plus a likely variation).

Concept 1.1 — What mechanics of solids is, and how to attack any problem

A. Concept explanation

1. Intuition. Put four books on a bookshelf. *Statics* tells you how hard each bracket must push up so that the shelf doesn't fall. *Mechanics of solids* asks the next questions. How much does the plank sag? How hard are the wood fibres being stretched at the bottom of the plank? Will it crack, and where first? The slides make the same point with a beam carrying four loads of $w/4$. Equilibrium gives the support reactions. But as those loads travel through the material to the supports, the material "develops resisting forces, generates stresses and undergoes deformation". That internal story is the subject of this course.

2. Why it matters. Every later chapter starts from correct external forces. Ch. 2 turns them into elongations, Ch. 3 into internal shear forces and bending moments, Ch. 4 into stresses, Ch. 5 into strains and failure checks, Ch. 6 into twist. If the free-body diagram (FBD) is wrong, every number after it is wrong. In the compre papers roughly a quarter of the marks on most long questions go to "draw the FBD and find the reactions".

3. Formal treatment.

(a) The slides' five-step procedure (applied mechanics in general):

1. Select the system of interest.
2. Idealize it (simplify: rigid bars, point loads, frictionless pins, weightless members …).
3. Apply the principles of mechanics to predict behaviour.
4. Compare the predictions with the real system (tests and experiments).
5. If they don't agree, construct a different idealized model and repeat.

The slides note that in research, steps 1 and 2 are the hardest.

(b) Crandall's working version for a single problem (§1.8):

1. Select the system and draw it (to scale where possible).
2. Isolate it: show every force the surroundings exert on it. Involve as many of the desired unknowns and given data as possible, and as few other quantities as possible.
3. Postulate the characteristics of the system, i.e. idealize it.
4. Apply the principles of mechanics. **Count unknowns against independent equations**; if there are too many unknowns, pick additional subsystems.
5. Reduce the result to **symbolic form first**. Check limiting cases and dimensional consistency.
6. Substitute numbers and **compare with intuition and order of magnitude**. Ask: "would a bar this size hold a mosquito, a mouse, a man, a car, a truck?"
7. Compare with the real system (test).
8. If anything is unsatisfactory, go back and reconsider the system choice or the idealization.

(c) Two kinds of motion. A body can (i) change position overall or (ii) change shape locally (deformation). This course deals with **static** situations and **small deformations**.

(d) Rigid body. The distance between any two points never changes. No real structure is rigid. But a lever carrying weights bends so little that the lever arm stays $L/2$, so for its **equilibrium** we may treat it as rigid. For its **strength** we must not, because the strength depends on the small deformation.

(e) Principle of transmissibility. A force may be slid along its own line of action without changing its effect on a **rigid** body. The slide's example: pushing a truck at the front bumper has the same effect as pulling it with the same force at the rear bumper.

4. Derivation — why transmissibility works (and when it fails). Slide a force F from point A to point B on the same line of action, so $r_B = r_A + \lambda e$ with $e \parallel F$.

- The force sum is unchanged, because it is the same vector.
- The moment about any point O is $r_B \times F = r_A \times F + \lambda e \times F = r_A \times F$, since $e \times F = 0$.

So the **external** resultant force and moment, which are all a rigid body "feels", are unchanged. ∎

Inside a deformable body the story differs. Pulling a rubber band at its far end stretches the whole band. Pushing at its near end with the same force stretches nothing. The resultants are equal but the deformations differ.

5. Picture. Draw the truck twice: once with an arrow pushing on the rear bumper, once with the same arrow pulling on the front bumper, both on one horizontal line. The truck's motion is identical. Now draw a spring between the bumpers. In one case it is compressed and in the other stretched.

6. Special / limiting cases.

- Stiffness $\to \infty$ is the rigid-body limit. It always gives the correct reactions for **statically determinate** systems.
- It cannot give the forces in **statically indeterminate** systems (more unknown reactions than equations). There, the way the members deform decides how the load is shared (Ch. 2).

7. Common traps.

- Using transmissibility to move a load along a beam before computing bending moment. **Never.** The moment diagram changes.
- Replacing a distributed load by its resultant *before* cutting the beam. This is fine for reactions, wrong for internal forces (Ch. 3).
- Plugging numbers in too early. You lose the chance to check limiting cases and dimensions (Crandall step 5).

8. Summary box

Idea	Valid for	Not valid for
Rigid-body idealization	reactions of determinate structures	deformations; indeterminate structures
Transmissibility	external equilibrium of rigid bodies	internal forces, stresses, deformation
Deformable-body analysis	internal forces, stresses, deflections, indeterminate problems	—

Concept 1.2 — Forces, moments, couples and equivalent force systems

A. Concept explanation

1. Intuition. Loosen a nut with a spanner. Pushing near the nut does almost nothing, and pushing at the end of the handle turns it easily. The *same* force has a different turning effect depending on how far its line of action is from the nut. That turning effect is the **moment**. If you turn a steering wheel with two hands, pushing up with one and down with the other, the wheel turns but is not pushed sideways. That pure turning effect is a **couple**.

2. Why it matters. Every equilibrium equation contains moments. Replacing a load by "a force plus a couple at another point" is exactly what we do at a cut in Ch. 3, where the force-couple system at the cut *is* the set of internal forces (F, V, M_b, M_t). Offset loads (brackets, cranks, arms welded to beams) appear in almost every PYQ and are handled this way.

3. Formal treatment.

- **Force:** a vector with magnitude, direction and line of action. In 2-D, $F = F_x i + F_y j$.
- **Moment of a force about a point O:** $M_O = r \times F$, where r runs from O to *any* point on the line of action. Its magnitude is $|M_O| = F\,d$, with d the **perpendicular** distance from O to the line of action. In 2-D, use $M_O = x\,F_y - y\,F_x$, counter-clockwise positive.
- **Moment about an axis** through O with unit vector n: $M_n = n \cdot (r \times F)$. This is what twists a shaft.
- **Couple:** two equal, opposite, non-collinear forces F a distance d apart. Its moment is $C = F\,d$, the **same about every point**, so a couple can be moved anywhere on a rigid body (it is a "free vector"). The resultant force of a couple is zero.
- **Varignon's theorem:** the moment of a resultant equals the sum of the moments of its components, $r \times (F_1 + F_2) = r \times F_1 + r \times F_2$ (Crandall Prob 1.3).
- **Equivalent force systems:** two systems have the same effect on a rigid body if and only if they have the same resultant force **and** the same resultant moment about one (hence every) point.
- **Force-couple system:** a force F acting at A is equivalent to the same F at O plus a couple $M_O = r_{OA} \times F$.

4. Derivations.

(a) A couple's moment is the same about every point. Forces F at r_1 and $-F$ at r_2:

$$M_O = r_1 \times F + r_2 \times (-F) = (r_1 - r_2) \times F$$

This depends only on the separation $r_1 - r_2$, not on O. ∎

(b) Moving a force sideways. Take F at A. Add $+F$ and $-F$ at O; adding zero changes nothing. The original F at A and the $-F$ at O form a couple $r_{OA} \times F$. What remains is F at O plus that couple. ∎

(c) The reverse (Crandall Prob 1.10). A coplanar force F and couple M at the same point can be replaced by the single force F moved sideways by

$$a = \frac{|M|}{|F|}$$

It is moved to the side that makes F's new moment about the original point have the same sense as M. This is how an eccentric load is created: a column carrying a load P off its axis feels P on the axis plus a couple $P\,e$ (PYQ 2023-24 A-Q2, post pulled by a sling).

5. Picture / graphical method. To get a 2-D moment quickly, resolve the force into F_x and F_y at a convenient point on its line of action. Then take each component times its own perpendicular arm, using a "which way would it spin the page?" sign. This is the same as $M_O = x\,F_y - y\,F_x$ and avoids hunting for the perpendicular distance.

6. Special cases.

- A force whose line of action passes through O has zero moment about O. Choose moment points so that unknown forces pass through them.
- A couple has no force sum, so it never appears in $\sum F_x$ or $\sum F_y$, only in $\sum M$.
- Parallel forces reduce to a single resultant at a definite point (Ch. 3: the resultant of a distributed load acts at its centroid).

7. Common traps.

- Using the slant distance instead of the **perpendicular** distance.
- Dropping an applied couple from $\sum M$, or adding it to $\sum F$.
- Sign slips: decide counter-clockwise positive once and stick to it.
- Unit slips: kN × mm = N·m, not kN·m.

8. Summary box

$$M_O = F\,d = x\,F_y - y\,F_x,\; C = F\,d\,(\text{couple; same about every point}),\; F\,@\,A \equiv F\,@\,O + r_{OA} \times F,\; a = |M|/|F|$$

B. Solved questions

Q1 (Crandall Prob 1.10) — Replace a force and a couple by a single displaced force (easy, derivation)

Problem. A coplanar force F and a moment M act at the same point O. Show that they may be replaced by an equal force displaced sideways a distance $a = |M|/|F|$, and decide which side it goes.

Given: F (magnitude F) at O and a couple M (counter-clockwise, say). **Find:** the single equivalent force and its offset.

Approach. Equivalent systems must have the same resultant force and the same moment about one point (Concept 1.2, item 3).

Solution.

1. *Resultant force.* The couple contributes nothing to the force sum, so the replacement force must be F itself.
2. *Moment about O.* The original system has moment M about O. A force F whose line of action is a perpendicular distance a from O has moment $\pm F\,a$ about O.
3. *Match the moments.* Set $F\,a = |M|$, so $a = |M|/F$.
4. *Side.* With M counter-clockwise, put the line of action on the side where F spins the page counter-clockwise about O. For example, if F points in $+x$, shift it **down** by a, since a $+x$ force below O turns counter-clockwise about O.

$$F \text{ displaced by } a = \frac{|M|}{|F|}, \text{ on the side that reproduces the sense of } M$$

Check. As $M \to 0$, $a \to 0$, so we recover the original force ✓. As $F \to 0$ with M fixed, $a \to \infty$: a pure couple cannot be replaced by a single finite force ✓.

Exam takeaway. This is the eccentric-load idea. A load P applied at offset e equals P on the axis plus $M = P\,e$. **Variation:** "A 26 kN force acts on a bracket 75 mm off the beam axis. Replace it by a force and couple at the section centroid" (PYQ 2019-20 B-Q2 begins exactly like this).

Concept 1.3 — Supports, free-body diagrams and equilibrium

A. Concept explanation

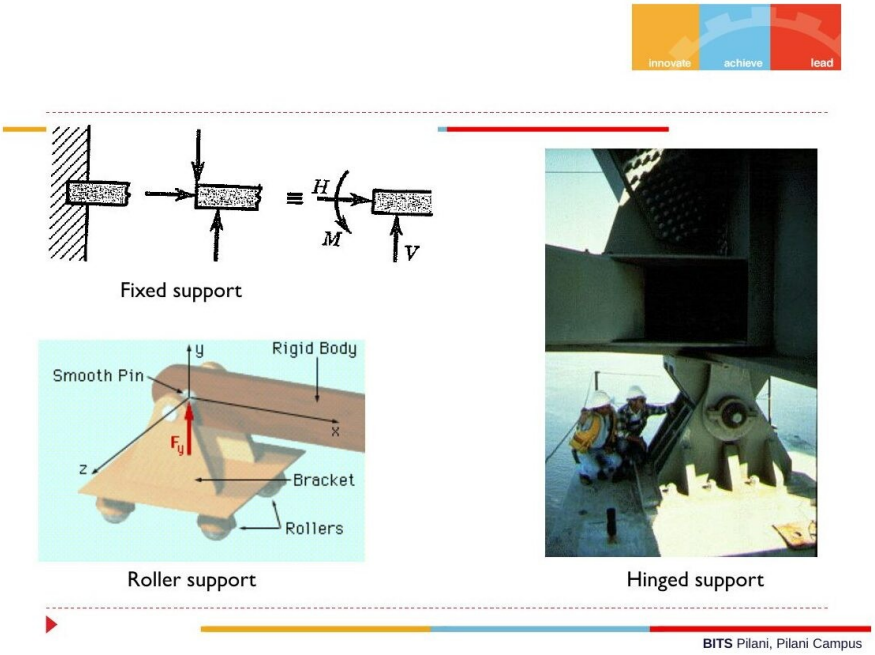
1. Intuition. An FBD is like cutting the object out of a photograph with scissors. Everything that was touching it (the floor, a pin, a rope, a hand, gravity) is removed and replaced by an arrow showing the force it applied. If the object is not accelerating, those arrows must cancel exactly, both as pushes (force sums) and as twists (moment sums).

2. Why it matters. FBDs are step 1 of every problem in the course: trusses here, bars in Ch. 2, beams in Ch. 3, shafts in Ch. 6, columns later. A wrong or incomplete FBD is the single largest source of lost marks.

3. Formal treatment.

Constraint: any restriction on the free motion of a body. **Each prevented motion produces one reaction component.**

Support (slides)	Prevents	Reactions in 2-D	Reactions in 3-D
Roller / rocker / smooth surface	motion ⊥ to the surface	1 force ⊥ surface	1 force
Hinge / pin	both translations	2 forces R_x, R_y	—
Ball-and-socket	all translations	—	3 forces
Journal bearing (short)	translations ⊥ to the shaft	—	2 forces (often treated as no moment)
Built-in / fixed / clamped	translations and rotation	R_x, R_y and a moment M	3 forces and 3 moments
Cable / rope / link	stretching along itself	1 tensile force along it	1 force



Original: Slides Ch 1, slide 8

Equations of equilibrium:

$$\text{2-D: } \sum F_x = 0, \sum F_y = 0, \sum M_O = 0 \quad \text{3-D: } \sum F = 0, \sum M_O = 0 \ (6 \text{ scalar equations})$$

$\sum M_O$ includes the moments of all forces **plus** all applied couples.

Equivalent sets of three equations in 2-D (Crandall Prob 1.7): any of these is a complete set.

- (a) Two force sums in non-parallel directions plus one moment equation.
- (b) Moments about two points P and Q plus the force sum in any direction **not perpendicular** to PQ.
- (c) Moments about three **non-collinear** points.

Whichever set you choose, a 2-D body gives only **three independent** equations.

Statical determinacy. Count the unknown reaction components. More than 3 means the body is **statically indeterminate**. Fewer than 3, or 3 that are all parallel or all concurrent, means it is **improperly constrained**: it can move.

Two-force member: a body loaded at only two points, with no couples. Equilibrium requires the two forces to be **equal, opposite and collinear**, along the line joining the two points.

Three-force member: if three non-parallel forces keep a body in equilibrium, their lines of action **meet at a single point**.

FBD procedure (use every time):

1. Isolate the body and sketch it with its dimensions.
2. Draw every external force: applied loads, **self-weight at the centroid**, and every support reaction with the components from the table above.
3. Give each unknown an assumed direction and a name.
4. Choose axes, and moment points through which as many unknowns as possible pass.
5. Write the equations. Check that the number of unknowns ≤ 3 (2-D).
6. Solve. A negative answer means "opposite to the arrow drawn". State it in words.
7. Check with an equation you did not use.

4. Derivations.

(a) Two-force member. Forces F_A at A and F_B at B.

- $\sum F = 0$ gives $F_B = -F_A$: equal and opposite.
- Moments about A: $r_{AB} \times F_B = 0$. A cross product vanishes only if the vectors are parallel, so $F_B \parallel r_{AB}$: the forces act **along AB**. ∎

(b) Three-force concurrency. Let the lines of action of two of the forces meet at a point K. Take moments about K: those two forces contribute nothing. So the third force's moment about K must be zero too, which means its line of action passes through K. ∎ (If two of the forces are parallel, the third must be parallel to them: the "point at infinity" version.)

(c) Why a 2-D body has only three independent equations. A rigid body in a plane has three independent ways to move: two translations and one rotation. Each equation of equilibrium forbids one of them. Any further equation, such as moments about a fourth point, is a combination of the first three and adds no information.

5. Graphical interpretation. For a three-force body, draw the two known lines of action and mark their intersection K. The third force must pass through K. This gives its direction immediately (lawn roller Q8, rod on a roller Q11). A **force triangle** drawn head-to-tail with the three forces must close. Its geometry gives magnitudes via the sine rule.

6. Special cases.

- **All forces parallel:** one force equation is trivially satisfied, so only 2 useful equations remain.
- **All forces concurrent:** the moment equation is trivially satisfied, again leaving 2.
- **A body hinged at one point with one other unknown:** take moments about the hinge first; it kills two unknowns at once.

7. Common traps.

- Forgetting self-weight (mg, with $g = 9.81$ m/s²) or putting it at the load point instead of the centroid.
- Giving a fixed support only forces and forgetting its **moment**.
- "Correcting" a reaction's assumed direction halfway through. Keep the arrow and let the sign speak.
- Drawing an internal force (such as the force between two parts of one body) on the FBD of the whole body. Internal forces cancel in pairs.
- kg is not N: multiply by 9.81.

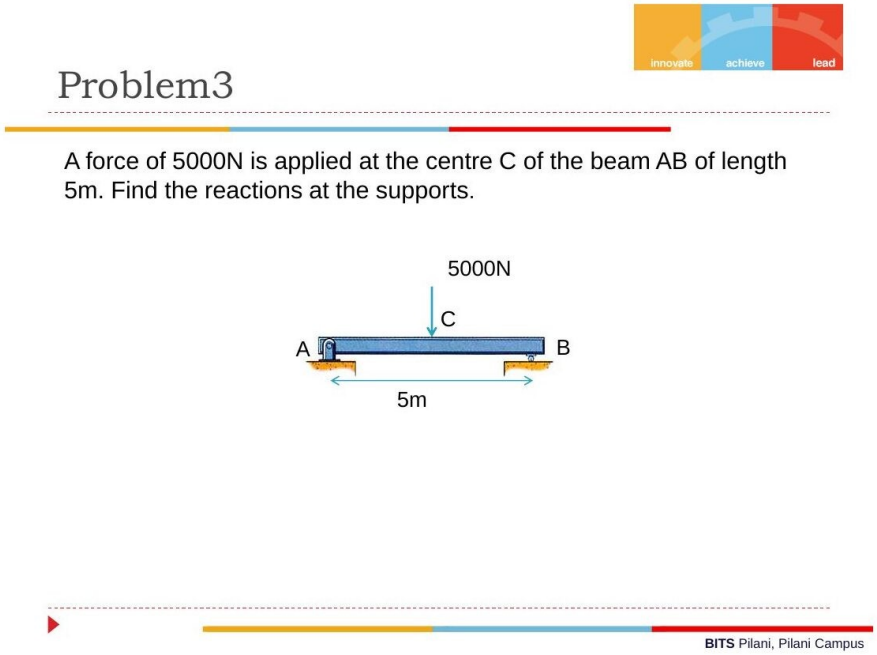
8. Summary box

$$\sum F_x = 0,\ \sum F_y = 0,\ \sum M_{\text{any pt}} = 0 \quad \text{pin} = 2,\ \text{roller} = 1,\ \text{fixed} = 3 \ \text{unknowns}$$

Two-force member → force along the line joining its pins. Three-force member → lines of action concurrent. More than 3 unknowns → indeterminate (Ch. 2).

B. Solved questions

Q2 (Slides Ch1, "Problem 3") — Simply supported beam with a central load (easy)



Original: Slides Ch 1, slide 12

Problem. A 5000 N force acts at the centre C of a beam AB of length 5 m. A is a pin and B is a roller. Find the support reactions.

Given: $L = 5$ m, $P = 5000$ N at $x = 2.5$ m. **Find:** A_x, A_y, B_y.

Approach. Three unknowns and three equations (Concept 1.3), so the beam is statically determinate. Take moments about A first: two unknowns pass through it.

FBD in words. At A draw A_x ($-$) and A_y ($\uparrow$). At B draw B_y ($\uparrow$); a roller gives no horizontal force. At C draw 5000 N $\downarrow$.

Solution.

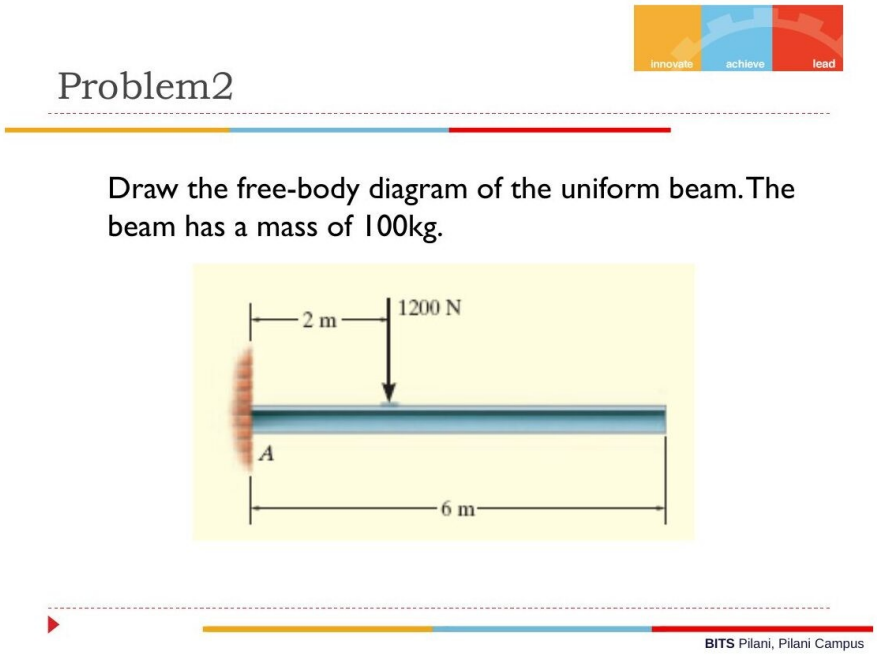
1. $\sum F_x = 0$: $A_x = 0$, since no other horizontal force acts.
2. $\sum M_A = 0$ (counter-clockwise +): $B_y(5) - 5000(2.5) = 0$, so $B_y = 12\,500/5 = 2500$ N.
3. $\sum F_y = 0$: $A_y + B_y - 5000 = 0$, so $A_y = 2500$ N.

$$A_x = 0,\ A_y = B_y = 2500\ \text{N}\,(\uparrow)$$

Check. The load and supports are symmetric, so the reactions must be equal ✓. Moments about B as an independent check: $-A_y(5) + 5000(2.5) = 0$ gives 2500 ✓.

Exam takeaway. With P at distance a from A and b from B, $A_y = Pb/L$ and $B_y = Pa/L$: the nearer support carries more. You will reuse this in every SFD/BMD question. **Variation:** two loads, or a load at $L/3$.

Q3 (Slides Ch1, "Problem 2") — Cantilever with self-weight (easy)



Original: Slides Ch 1, slide 10

Problem. A uniform beam of mass 100 kg and length 6 m is built in at A. A 1200 N load acts 2 m from A. Draw the FBD and find the reactions.

Given: $m = 100$ kg, so $W = 100(9.81) = 981$ N acting at the centroid, 3 m from A; $P = 1200$ N at 2 m. **Find:** A_x, A_y, M_A.

Approach. A fixed support gives three unknowns, including a moment. Three equations solve it.

FBD in words. At A: A_x (→), A_y (↑) and a counter-clockwise moment M_A. Along the beam: 1200 N ↓ at 2 m and 981 N ↓ at 3 m.

Solution.

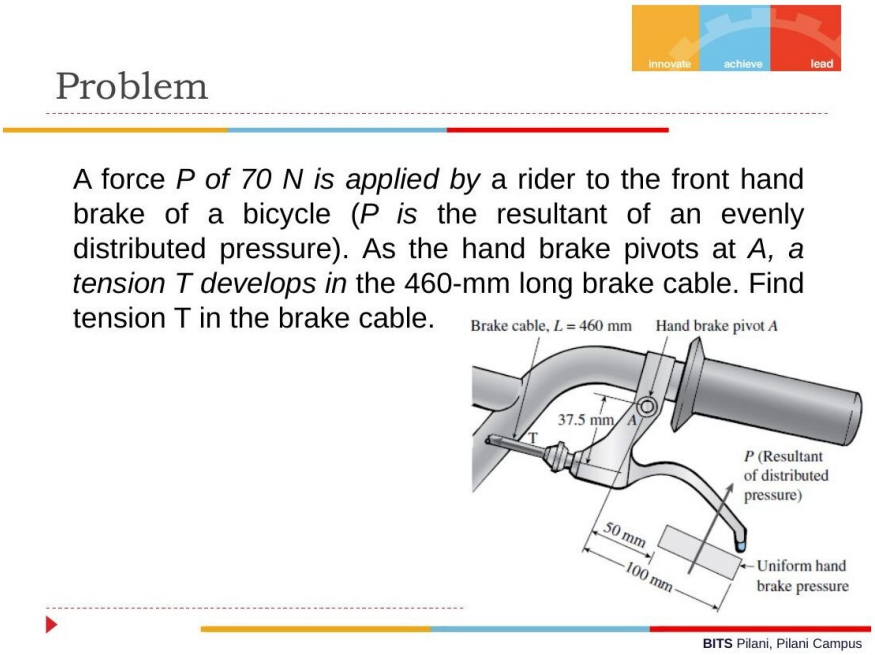
1. $\sum F_x = 0$: $A_x = 0$.
2. $\sum F_y = 0$: $A_y - 1200 - 981 = 0$, so $A_y = 2181$ N.
3. $\sum M_A = 0$: $M_A - 1200(2) - 981(3) = 0$, so $M_A = 2400 + 2943 = 5343$ N·m.

$$A_x = 0,\ A_y = 2181\ \text{N}(\uparrow),\ M_A = 5.34\ \text{kN·m}(\text{counter-clockwise on the beam})$$

Check. The wall must hold the beam up (↑) and stop it drooping clockwise. A counter-clockwise wall moment does exactly that ✓. Magnitude: total load ≈ 2.2 kN at an average arm ≈ 2.45 m gives ≈ 5.3 kN·m ✓.

Exam takeaway. Two traps are planted: the self-weight (981 N) and its position (at $L/2$, not at the load). **Variation:** "the beam has a UDL of its own weight per metre" (same thing, treated in Ch. 3).

Q4 (Slides Ch1) — Bicycle hand brake (easy)



Original: Slides Ch 1, slide 17

Problem. A rider applies $P = 70$ N to the brake lever. P is the resultant of a uniform hand pressure acting between 50 mm and 100 mm from pivot A. The lever pivots at A and the brake cable (460 mm long) is attached 37.5 mm from A, measured perpendicular to the cable. Find the cable tension T.

Given: $P = 70$ N, uniform pressure from 50 to 100 mm from A, so the resultant acts at the middle, 75 mm from A and perpendicular to the lever. Cable arm 37.5 mm. **Find:** T.

Approach. The lever is a body hinged at A. Moments about A eliminate the two unknown pin reactions.

FBD in words. Lever with pin reactions A_x, A_y at A, P at 75 mm from A, and T along the cable at 37.5 mm perpendicular distance from A.

Solution.

1. Resultant of a uniform pressure acts at its centroid: $(50+100)/2 = 75$ mm from A.
2. $\sum M_A = 0$: $T(37.5) - 70(75) = 0$.
3. $T = 5250/37.5 = 140$ N.

$$T = 140\ \text{N}$$

Check. The cable arm is half the hand arm, so the cable force is twice the hand force ✓.

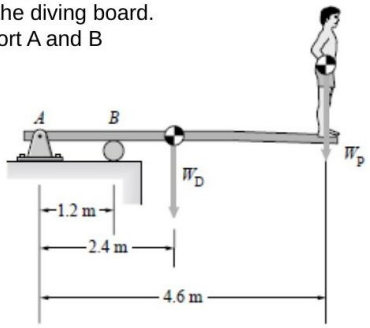
Exam takeaway. The 460 mm cable length is a **distractor**. It would matter only if you were asked how much the cable stretches: $\delta = T\,L/A\,E$ (Ch. 2). **Variation:** find the pin reaction at A, $\sqrt{A_x^2 + A_y^2}$, from the two force equations.

Q5 (Slides Ch1) — Diving board (easy–medium)

The masses of the person and the diving board are 54kg and 36kg respectively. Assume that they are in equilibrium.

a) Draw the Free body diagram of the diving board.

b) Determine the reactions at support A and B



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Original: Slides Ch 1, slide 15

Problem. A person of mass 54 kg stands at the tip of a diving board of mass 36 kg. The board is pinned at A and rests on a roller at B, 1.2 m from A. The board's centroid is 2.4 m from A and the person stands 4.6 m from A. Draw the FBD and find the reactions at A and B.

Given: $W_D = 36(9.81) = 353.2$ N at 2.4 m; $W_P = 54(9.81) = 529.7$ N at 4.6 m; B at 1.2 m. **Find:** A and B.

Approach. Two vertical unknowns, plus $A_x = 0$ since there are no horizontal loads. Moments about A give B directly.

FBD in words. Board with A (↑ assumed) at $x = 0$, B (↑) at 1.2 m, 353.2 N ↓ at 2.4 m and 529.7 N ↓ at 4.6 m.

Solution.

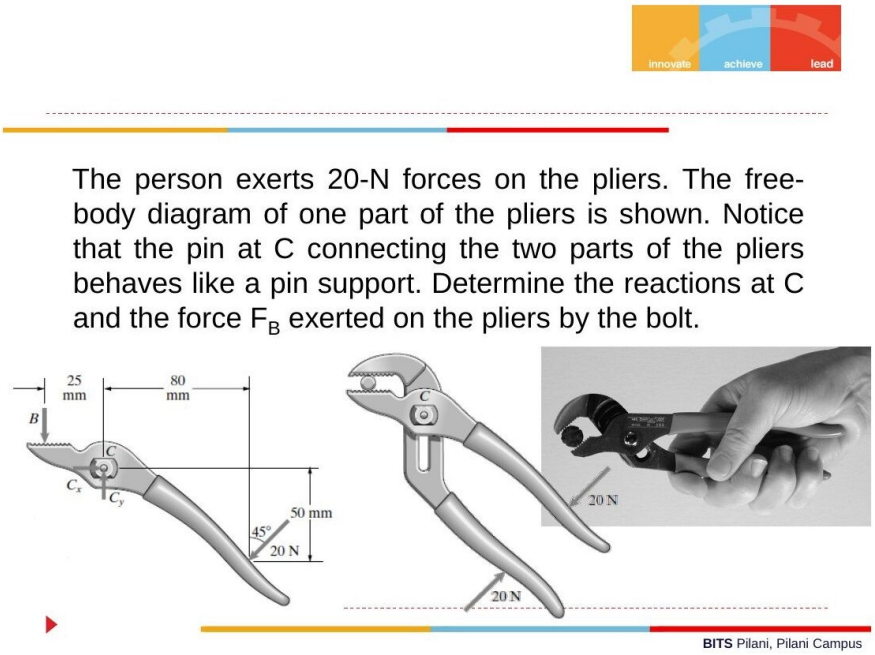
1. $\sum M_A = 0$: $B(1.2) - 353.2(2.4) - 529.7(4.6) = 0$.
2. $B = \dfrac{847.7 + 2436.6}{1.2} = \dfrac{3284.3}{1.2} = 2737$ N.
3. $\sum F_y = 0$: $A + 2737 - 353.2 - 529.7 = 0$, so $A = -1854$ N.

$$B = 2737\ \text{N}(\uparrow),\ A = 1854\ \text{N}(\downarrow)$$

Check. The board is a lever pivoting on B. The loads to the right of B try to lift end A, so the pin at A must pull **down** ✓. B carries more than the total weight (883 N), because A also pulls down ✓.

Exam takeaway. Supports that "pull the wrong way" are popular. Never flip a sign silently; write "the negative sign means A acts downward". **Variation:** where can the diver stand so that the pin force at A is zero? The board's own weight (right of B) turns it clockwise about B, so the diver must stand *between A and B* to balance it. Moments about B: $529.7(1.2 - x) = 353.2(2.4 - 1.2)$, so $1.2 - x = 0.800$ and $x = 0.40$ m.

Q6 (Slides Ch1) — Pliers (medium)



Original: Slides Ch 1, slide 16

Problem. A person squeezes pliers with 20 N forces. In the FBD of one handle, the pin C joins the two halves. The bolt being gripped exerts a force F_B perpendicular to the jaw, 25 mm to the left of C. The 20 N hand force acts at 45° (down and to the left) at a point 80 mm right of and 50 mm below C. Find F_B and the pin reactions at C.

Given: C at the origin; F_B acts vertically at (−25, 0) mm; hand force 20 N in direction $(-\cos 45^\circ, -\sin 45^\circ)$ at (80, −50) mm. **Find:** F_B, C_x, C_y.

Approach. One handle is a rigid body with three unknowns (F_B, C_x, C_y). Moments about C isolate F_B.

FBD in words. Handle with C_x (→) and C_y (↑) at C, F_B ↓ at the jaw and the 20 N hand force.

Solution.

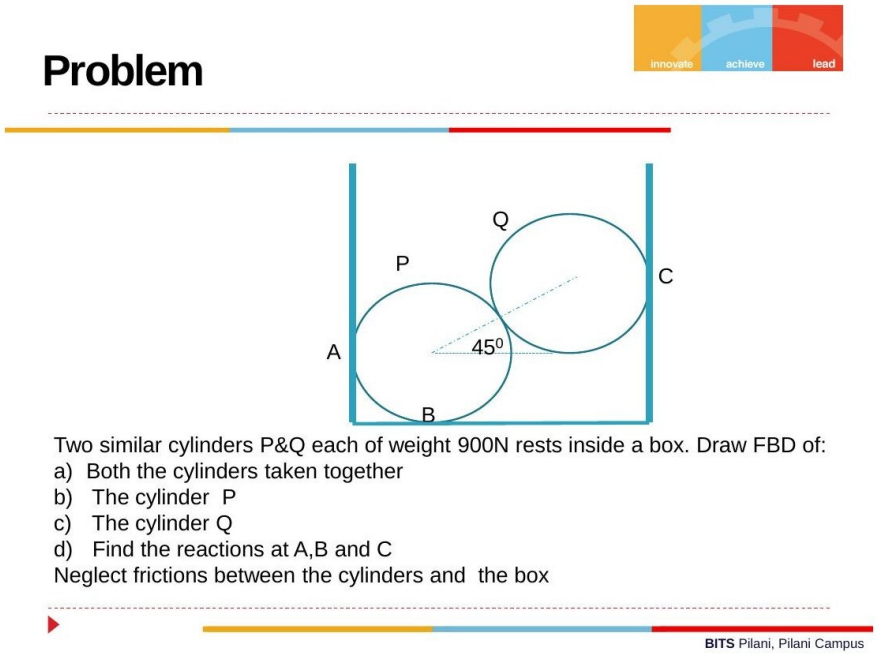
1. Components of the hand force: $F_x = -20\cos 45^\circ = -14.14$ N and $F_y = -14.14$ N.
2. Moment of the hand force about C:
   $M = x F_y - y F_x = 80(-14.14) - (-50)(-14.14) = -1131.4 - 707.1 = -1838.5$ N·mm (clockwise).
3. Moment of F_B (↓ at $x = -25$): $x F_y = (-25)(-F_B) = +25 F_B$ (counter-clockwise).
4. $\sum M_C = 0$: $25 F_B - 1838.5 = 0$, so $F_B = 73.5$ N.
5. $\sum F_x = 0$: $C_x - 14.14 = 0$, so $C_x = 14.1$ N.
6. $\sum F_y = 0$: $C_y - 73.5 - 14.14 = 0$, so $C_y = 87.7$ N.

$$F_B = 73.5\,\text{N},\ C_x = 14.1\,\text{N},\ C_y = 87.7\,\text{N}\ (|C| = 88.8\,\text{N})$$

Check. Mechanical advantage $73.5/20 \approx 3.7$, typical for pliers (handle arm ≈ 94 mm, jaw arm 25 mm, ratio ≈ 3.8) ✓.

Exam takeaway. The component method $M = x F_y - y F_x$ saves you from finding the perpendicular distance of a slanted force. **Variation:** "the jaw can bear at most 100 N; find the maximum hand force", giving $20 \times 100/73.5 = 27.2$ N.

Q7 (Slides Ch1) — Two cylinders in a box (medium)



Original: Slides Ch 1, slide 13

Problem. Two identical cylinders P and Q, each 900 N, rest in a box. P touches the floor (B) and the left wall (A). Q rests on P and touches the right wall (C). The line of centres makes 45° with the horizontal. All surfaces are smooth. Draw FBDs of (a) both together, (b) P, (c) Q, and (d) find the reactions at A, B and C.

Given: $W = 900$ N each; angle of the line of centres 45°; smooth contacts (normal forces only). **Find:** R_A, R_B, R_C and the contact force N between the cylinders.

Approach. Each cylinder is a concurrent force system (all forces pass through its centre, because contact forces on a smooth circle are radial). So each gives two useful equations. Start with Q, which has only two unknowns.

FBD in words.

- *Q:* weight 900 N ↓; R_C horizontal ←; contact force N from P, acting along the line of centres, up and to the right at 45°.
- *P:* weight 900 N ↓; R_A → from the left wall; R_B ↑ from the floor; N from Q, down and to the left at 45° (Newton's third law).
- *Both together:* 1800 N ↓, R_A →, R_B ↑, R_C ←. N is internal and does not appear.

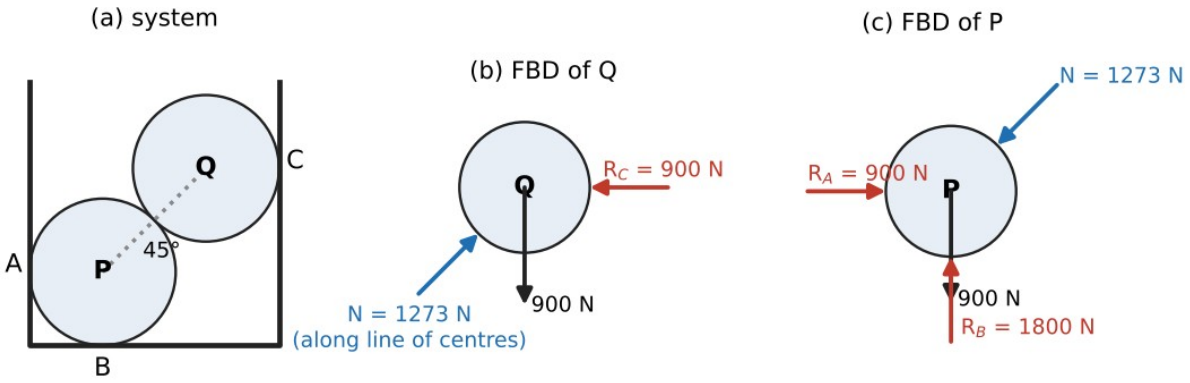
Solution.

1. *Q, vertical:* $N \sin 45^\circ = 900$, so $N = 1272.8$ N.
2. *Q, horizontal:* $R_C = N \cos 45^\circ = 900$ N.
3. *P, horizontal:* $R_A = N \cos 45^\circ = 900$ N.
4. *P, vertical:* $R_B = 900 + N \sin 45^\circ = 1800$ N.

$$R_A = 900\ \text{N},\ R_B = 1800\ \text{N},\ R_C = 900\ \text{N},\ N = 1273\ \text{N}$$

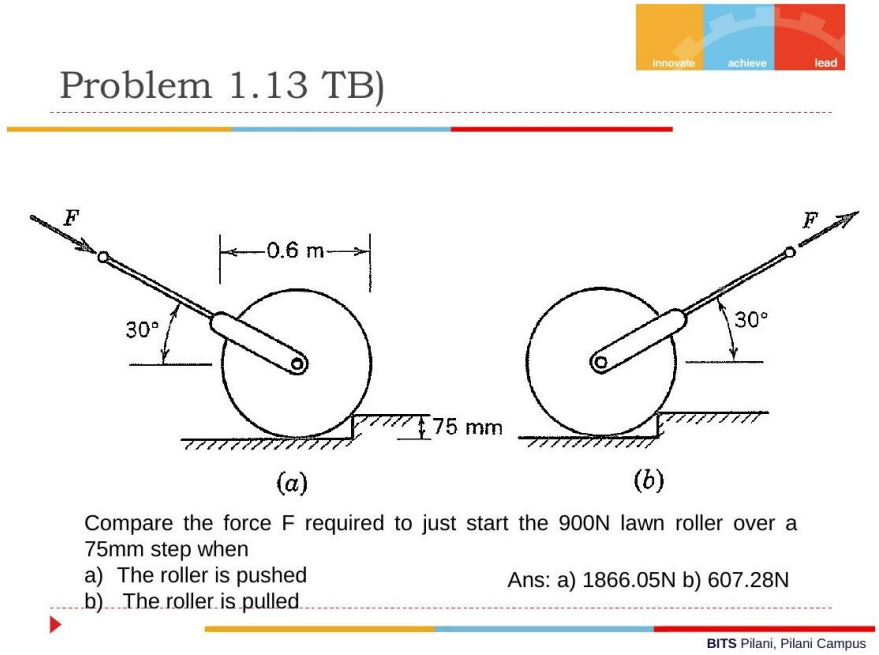
Check (combined FBD). $\sum F_y$: $R_B = 1800$ ✓ (total weight). $\sum F_x$: $R_A = R_C$ ✓. Moment balance of the combined body: the two wall forces form a couple $900 \times$ (vertical distance between contact points $= 2r \sin 45^\circ$), which balances the weight couple $900 \times$ (horizontal offset of the centres $= 2r \cos 45^\circ$). At 45° these are equal ✓.

Exam takeaway. For smooth spheres and cylinders, every contact force passes through the centre, so moment equations give nothing new. Variation: line of centres at angle θ gives $R_A = R_C = W \cot\theta$ and $N = W / \sin\theta$. As θ → 90° (Q directly above P), the wall forces vanish.



Worked FBDs: system, cylinder Q, cylinder P

Q8 (Slides Ch1; Crandall Prob 1.13) — Lawn roller over a step: push vs pull (medium)



Original: Slides Ch 1, slide 14

Problem. Compare the force F needed to just start a 900 N lawn roller (diameter 600 mm) over a 75 mm step when (a) the roller is **pushed** and (b) it is **pulled**. The handle is at 30° to the horizontal and its force acts through the axle.

Given: $W = 900$ N, $r = 300$ mm, $h = 75$ mm, handle at 30°. **Find:** F_{push}, F_{pull}.

Approach. "Just starting to roll over" means the roller is about to leave the ground, so the **ground reaction is zero**. Three forces then act: W, F and the corner reaction R. Take moments about the corner so that R drops out.

FBD in words. Roller centre O. W ↓ at O. F through O along the handle (pushed: down and toward the step at 30° below horizontal; pulled: up and toward the step at 30° above horizontal). R at the step corner K, directed toward O (smooth corner on a circle).

Solution.

1. *Geometry of the corner.* K is $r - h = 225$ mm below O. Its horizontal offset is $a = \sqrt{r^2 - (r-h)^2} = \sqrt{300^2 - 225^2} = \sqrt{39\,375} = 198.4$ mm.
2. *Moment arms about K.* W (vertical) has arm $a = 198.4$ mm. The horizontal component of F has arm 225 mm; the vertical component has arm 198.4 mm.

3. *(a) Pushed* ($F\cos 30^\circ$ toward the step, $F\sin 30^\circ$ downward). The horizontal part tips the roller over; W and the downward part resist.

$$F\cos 30^\circ(225) = W(198.4) + F\sin 30^\circ(198.4)$$

$$F(194.86 - 99.20) = 900(198.4) \Rightarrow F = \frac{178\,560}{95.66} = 1867\ \text{N}$$

4. *(b) Pulled* (vertical part upward, now helping).

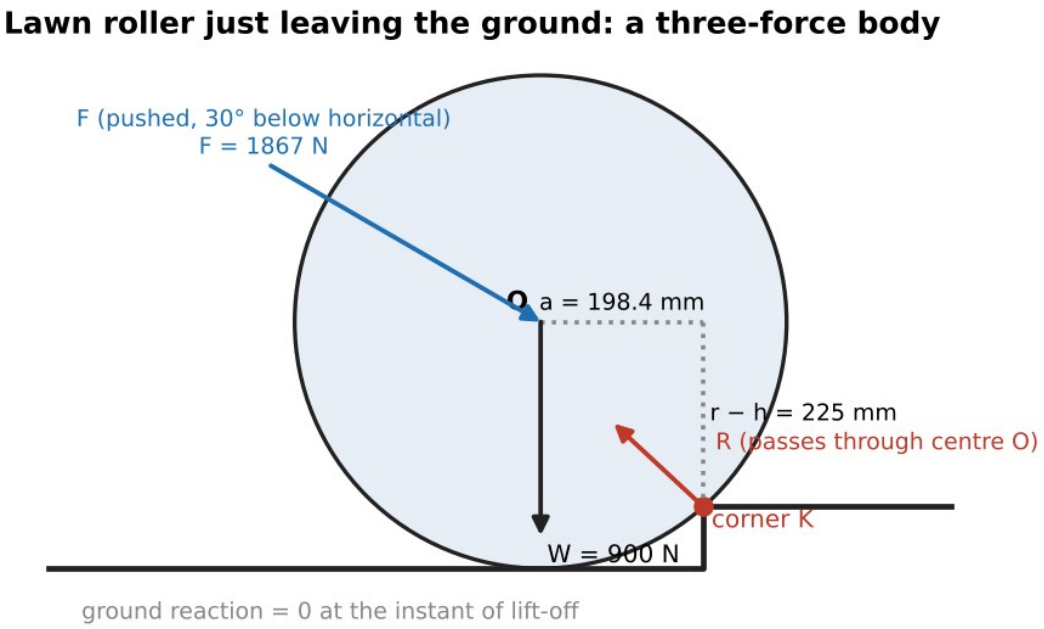
$$F\cos 30^\circ(225) + F\sin 30^\circ(198.4) = W(198.4) \Rightarrow F = \frac{178\,560}{294.06} = 607.2\ \text{N}$$

$$F_{push} \approx 1867\ \text{N},\ F_{pull} \approx 607\ \text{N}$$

(Slides: 1866.05 N and 607.28 N. The difference is rounding.)

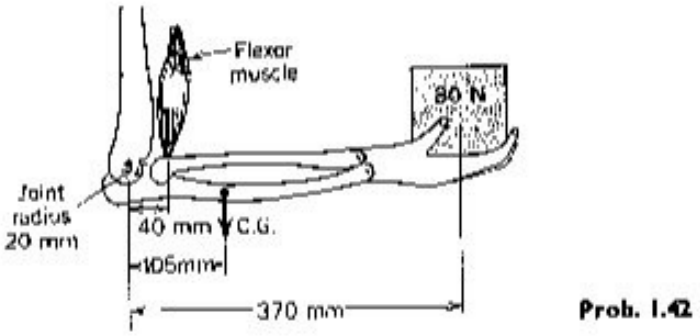
Check. Three-force concurrency: W and F both pass through O, so R must too, which is consistent with a smooth corner pushing radially ✓. Pulling is about 3× easier because it partly lifts the roller, while pushing presses it into the ground ✓.

Exam takeaway. "Just about to" means a contact force becomes zero. **Variation:** find the minimum force and its direction. It is perpendicular to OK, so $F_{min} = W\,a/r = 900(198.4/300) = 595$ N, at angle $\cos^{-1}(225/300) = 41.4^\circ$ above the horizontal.



Geometry and three-force FBD at lift-off (pushed case)

Q9 (Crandall Prob 1.42) — Forearm holding a weight; then elbow friction (medium)



Prob. 1.42

Original: Crandall Prob 1.42 (textbook figure)

Problem. A man holds an 80 N weight in his hand with the elbow at a right angle. The forearm and hand weigh 16 N with C.G. 105 mm from the elbow joint. The weight is 370 mm from the joint. The flexor muscle acts vertically, 40 mm from the joint. (i) Find the muscle force and the joint force from the humerus, neglecting friction. (ii) The elbow joint has radius 20 mm and friction coefficient 0.015. How does the muscle force change (a) to raise the load and (b) to just support it?

Given: loads 80 N at 370 mm and 16 N at 105 mm; muscle line at 40 mm; joint radius 20 mm; $\mu = 0.015$. **Find:** F_m, joint force J, and the friction corrections.

Approach. The forearm is a lever hinged at the elbow. Moments about the joint give the muscle force, and the vertical force sum gives the joint force. Friction adds a resisting moment $\mu J r$ that opposes the motion.

FBD in words. Forearm horizontal. Muscle force F_m ↑ at 40 mm. Joint force J ↓ at the joint (humerus pushing down on the ulna). Weights 16 N ↓ at 105 mm and 80 N ↓ at 370 mm.

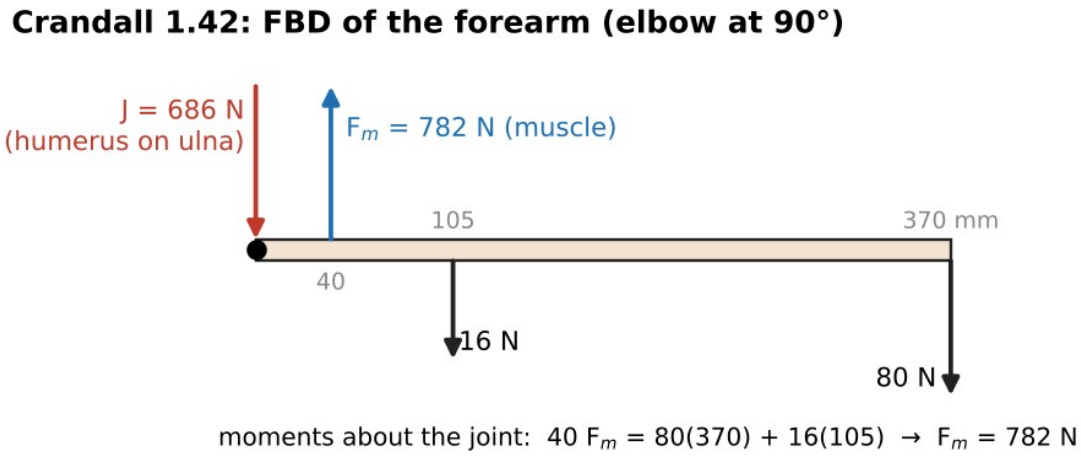
Solution.

1. *(i) Moments about the joint:* $40\,F_m = 80(370) + 16(105) = 29\,600 + 1\,680 = 31\,280$ N·mm, so $F_m = 782$ N.
2. *(i) Vertical sum:* $F_m - J - 80 - 16 = 0$, so $J = 782 - 96 = 686$ N (↓ on the ulna).
3. *(ii) Friction moment.* At the joint surface the friction force is μJ with arm $r = 20$ mm, giving $M_f = \mu J r = 0.015(20)J = 0.3\,J$ N·mm. It always opposes the motion, and $J = F_m - 96$ still holds.
4. *(a) Raising* (friction resists the muscle): $40\,F_m = 31\,280 + 0.3(F_m - 96)$, so $39.7\,F_m = 31\,251.2$ and $F_m = 787.2$ N.
5. *(b) Just supporting* (the load tends to drop, friction helps): $40\,F_m = 31\,280 - 0.3(F_m - 96)$, so $40.3\,F_m = 31\,308.8$ and $F_m = 776.9$ N.

$$F_m = 782\text{ N},\ J = 686\text{ N};\ \text{raise: } 787\text{ N}\,(+5.2\text{ N}),\ \text{hold: } 777\text{ N}\,(-5.1\text{ N})$$

Check. The muscle has a tiny lever arm (40 mm against 370 mm), so it must pull about 10× the load ✓. Joint friction changes the result by only 0.7%, as expected for a lubricated joint ✓.

Exam takeaway. Short-lever "biomechanics" FBDs test moment equations. The friction twist shows that friction always opposes the **impending** motion, so the answer depends on the direction. **Variation:** muscle inclined at 75° to the forearm. Use only its perpendicular component in the moment equation; the joint force then also has a horizontal component.



Worked FBD of the forearm

Q10 (Crandall Prob 1.7) — Alternative sets of equilibrium equations (easy, conceptual)

Problem. Show that, for coplanar forces, each of the following is a complete set of equilibrium conditions: (a) zero force sums in two non-parallel directions plus zero moment about any point; (b) zero moments about two points P and Q plus a zero force sum parallel to PQ; (c) zero moments about three non-collinear points.

Approach. Reduce any coplanar system to a single resultant force R through a chosen point plus a couple. Show that each set forces both to vanish.

Solution.

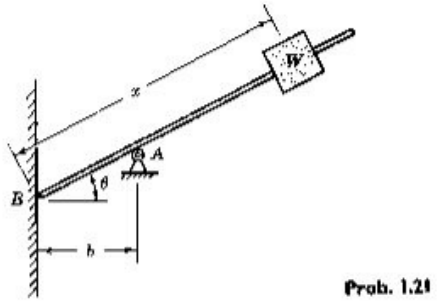
1. *(b)* $\sum M_P = 0$ means the resultant has no couple about P, so it is a single force R through P (or zero). $\sum M_Q = 0$ means that force also has zero moment about Q, so its line of action passes through Q too: **R lies along PQ**. Finally, $\sum F_{/\!/\,PQ} = 0$ says its component along PQ is zero, so $R = 0$. ∎ (The force condition must not be perpendicular to PQ. A perpendicular component would be automatically zero and prove nothing.)
2. *(c)* Moments about P and Q being zero put R along PQ. A zero moment about a third point S, off line PQ, then requires $R \cdot (\text{distance from S to PQ}) = 0$, so $R = 0$. ∎ If S were on PQ the third equation would repeat the first two.
3. *(a)* Two independent force components zero gives $R = 0$, and a zero moment then kills the couple. ∎

Answer. All three sets are equivalent. Each contains exactly three independent conditions, so a 2-D rigid body yields **at most three independent equations** however you choose them.

Exam takeaway. Use set (b) or (c) to write equations with **one unknown each**: take moments about points where other unknowns cross. That is the whole trick of the method of sections (Concept 1.5).

Q11 (Crandall Prob 1.21) — Rod on a roller against a smooth wall: where can the load sit? (medium–hard, three-force member)

1.21. A rigid rod with negligible weight and small transverse dimensions carries a load W whose position is adjustable. The rod rests on a small roller at A and bears against the vertical wall at B. Determine the distance x for any given value of θ such that the rod will be in equilibrium. Assume that friction is negligible.



Prob. 1.21

Original: Crandall Prob 1.21 (textbook figure)

Problem. A weightless rigid rod rests on a small roller A and bears against a smooth vertical wall at its end B. A load W can be placed at distance x from B along the rod. The roller is at horizontal distance b from the wall and the rod makes angle θ with the horizontal. Find x for equilibrium. Neglect friction.

Given: b, θ (rod angle from the horizontal, as drawn), smooth contacts. **Find:** x.

Approach. Exactly three forces act: W (vertical), the wall reaction at B (horizontal, since the wall is smooth) and the roller reaction at A (perpendicular to the rod). A three-force body needs **concurrent** lines of action (Concept 1.3).

FBD in words. B at the origin, rod going up and to the right at θ. N_B → horizontal at B. N_A at A, perpendicular to the rod. W ↓ at distance x along the rod.

Solution.

1. *Coordinates.* A is where the rod reaches $X=b$: $A=(b, b\tan\theta)$. The load is at $(x\cos\theta, x\sin\theta)$.
2. *Meeting point K.* The horizontal line through B ($Y=0$) and the vertical line through W ($X=x\cos\theta$) meet at $K=(x\cos\theta, 0)$.
3. *Roller line through K.* N_A acts along $(-\sin\theta, \cos\theta)$ from A. Points on that line: $(b-s\sin\theta, b\tan\theta+s\cos\theta)$. Setting $Y=0$ gives $s=-b\tan\theta/\cos\theta$, so

$$X=b+\frac{b\tan\theta\sin\theta}{\cos\theta}=b\left(1+\tan^2\theta\right)=\frac{b}{\cos^2\theta}$$

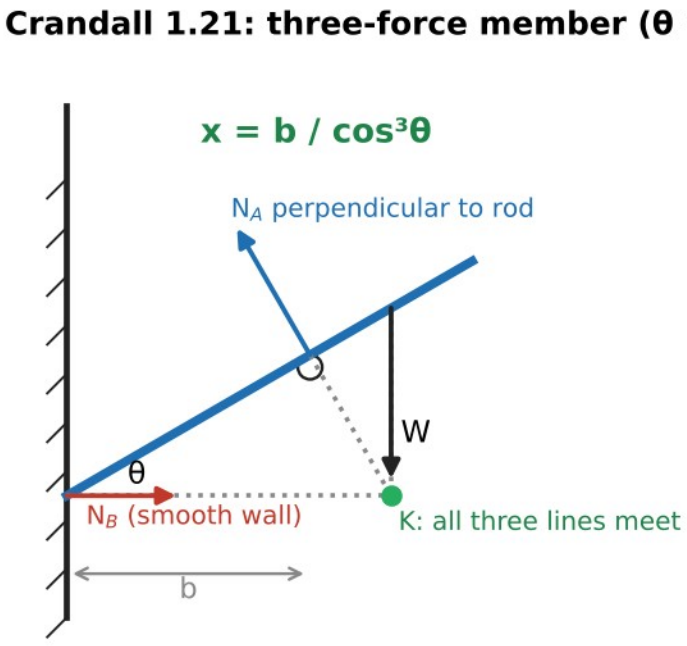
4. *Concurrency:* $x\cos\theta=\dfrac{b}{\cos^2\theta}$.

$$x=\frac{b}{\cos^3\theta}$$

Check. θ → 0: x → b, so the load sits directly over the roller and the wall carries nothing ✓. As θ grows, x grows, because a steeper rod pushes on the wall and the load must move outward to balance ✓.

Second method (equations). Moments about B: $N_A(b/\cos\theta)=W\,x\cos\theta$. Vertical sum: $N_A\cos\theta=W$. Combining gives $x=b/\cos^3\theta$ ✓.

Exam takeaway. When you see "three forces only", look for the concurrency point first; it often makes the problem one line long. **Variation:** if the load is at a fixed x and θ is unknown, $\cos\theta=(b/x)^{1/3}$.



Concurrency point K of the three forces

Concept 1.4 — Dry (Coulomb) friction, including belt friction

A. Concept explanation

1. Intuition. Push a heavy box lightly and it doesn't move. The floor pushes back sideways *exactly as hard as you push*, up to a limit, and beyond that the box slides. The limit is larger for a heavier box (bigger normal force) but does **not** depend on how big its base is. A rope wrapped round a post is the same idea repeated around the curve. Each little bit of contact adds a little friction, and the effect multiplies, which is why a sailor can hold a ship with a few turns of rope.

2. Why it matters. Wedges, brakes, belts, bolts, ladders, vehicles on slopes. In the PYQs: the rope over a drum (2014 T1-Q3) combines belt friction with rope stretch (Ch. 2). In machine design, friction decides whether a joint slips.

3. Formal treatment (slides).

- No relative motion: **static** friction f_s. It takes *whatever value equilibrium needs*, from 0 up to $f_{s,max}$.
- Impending motion (the limiting case): $f_{s,max} = \mu_s N$.
- Sliding: **kinetic** friction $f_k = \mu_k N$. It is roughly constant and slightly less than $\mu_s N$.
- Friction is independent of the contact area (slides). Doubling the area at the same N halves the force per unit area.
- Friction always acts **opposite to the (impending) relative motion** of the contact surfaces.
- *Slide example:* if $f_{s,max} = 10$ N, an 8 N push gives $f = 8$ N and no motion; a 12 N push makes the block move. With no push, $f = 0$, **not** μN.

Angle of friction: $\tan\phi = \mu$. The total contact reaction (N and f combined) can lean from the normal by at most φ, the "friction cone".

4. Derivations.

(a) Block on an incline of angle α, pulled up the slope by a force T parallel to it. Resolve normal to the slope: $N = W\cos\alpha$.

- Impending motion **up** (friction acts down the slope): $T = W\sin\alpha + \mu W\cos\alpha$.

- Impending motion **down** (friction acts up the slope): $T = W\sin\alpha - \mu W\cos\alpha$.

Equilibrium holds for any T between these two values.

(b) Angle of repose. With no T, the block is about to slide when $W\sin\alpha = \mu W\cos\alpha$, i.e. $\tan\alpha = \mu$. Measuring the tilt angle at which a block just slides is a direct measurement of μ.

(c) Belt (capstan) friction. Take a rope element subtending $d\theta$ on a fixed drum, with tension T on one side and $T + dT$ on the other.

- Radial balance: the normal force is $dN = (T + dT)\sin\frac{d\theta}{2} + T\sin\frac{d\theta}{2} \approx T\,d\theta$, dropping second-order terms.
- Tangential balance at impending slip: $(T + dT) - T = \mu\,dN$, so $dT = \mu T\,d\theta$.
- Separate and integrate over the contact angle β:

$$\int_{T_1}^{T_2}\frac{dT}{T} = \mu\int_0^{\beta} d\theta \Rightarrow \ln\frac{T_2}{T_1} = \mu\beta \Rightarrow \frac{T_2}{T_1} = e^{\mu\beta}$$

Here T_2 is the tight side, T_1 the slack side, and β is in **radians** (each full turn is 2π). Along the contact the tension grows exponentially: $T(\theta) = T_1 e^{\mu\theta}$.

5. Graphical picture. Draw the friction cone (half-angle φ) about the contact normal. The resultant of all the *other* forces on the body must fall inside the cone for no slip, and on its surface for impending slip. For a 3-D problem (block pushed sideways on a slope), friction opposes the **resultant** of the in-plane forces, not just the down-slope one (Q14).

6. Special cases.

- $\mu = 0$: a smooth surface; the reaction is purely normal.
- β = π (half wrap): $T_2/T_1 = e^{\mu\pi}$. Four full turns: $e^{8\pi\mu}$, which is huge even for μ = 0.3.
- Tipping versus sliding: a tall box may tip over before it slides. Check both.

7. Common traps.

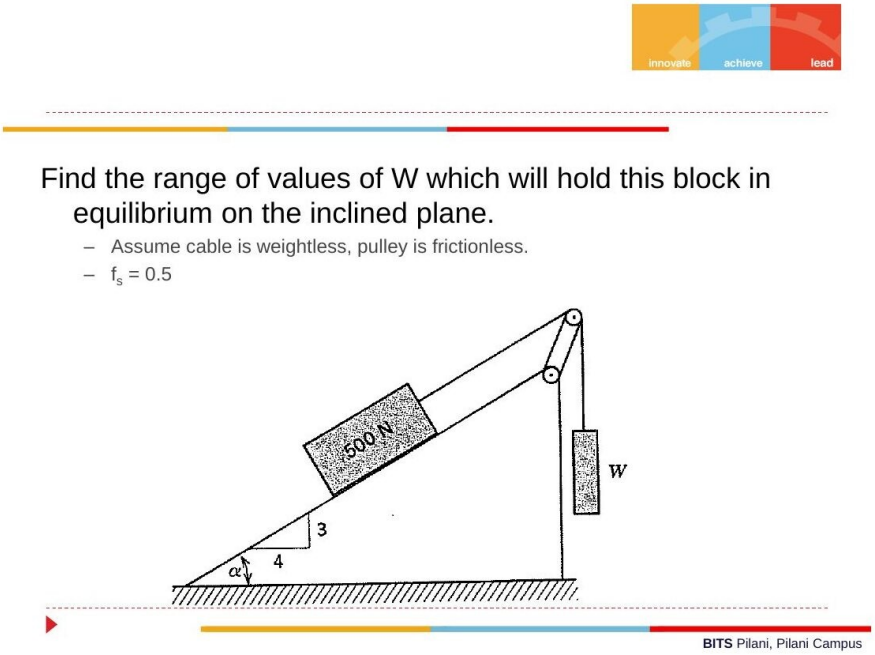
- Writing $f = \mu N$ when the problem does **not** say motion is impending.
- Getting the friction direction wrong. Decide which way the body *would* move without friction.
- Forgetting that N changes when an applied force has a component normal to the surface.
- Degrees instead of radians in $e^{\mu\beta}$.
- In 3-D, balancing friction only against the down-slope weight component and ignoring a sideways push.

8. Summary box

Situation	Equation
No slip	$f \le \mu_s N$; find f from equilibrium
Impending slip	$f = \mu_s N$, opposing the impending motion
Incline, impending up / down	$T = W(\sin\alpha \pm \mu\cos\alpha)$
Angle of repose	$\tan\alpha = \mu$
Rope on drum	$T_2/T_1 = e^{\mu\beta}$ (β in rad), $T(\theta) = T_1 e^{\mu\theta}$

B. Solved questions

Q12 (Slides Ch1) — Range of the hanging weight W for a block on an incline (easy)



Original: Slides Ch 1, slide 24

Problem. A 500 N block rests on a 3-4-5 incline ($\sin\alpha = 0.6, \cos\alpha = 0.8$). A cable parallel to the incline runs over a frictionless pulley to a hanging weight W. $\mu_s = 0.5$. Find the range of W for equilibrium.

Given: block 500 N, α from a 3-4-5 triangle, μ = 0.5, weightless cable, frictionless pulley (so the tension is W).
Find: W_{min}, W_{max}.

Approach. Two impending-motion cases: the block about to slide down (W smallest) and about to slide up (W largest). Friction flips direction between them.

FBD in words. Block with weight 500 N ↓, N normal to the incline, T = W up the slope, and friction f along the slope (direction depends on the case).

Solution.

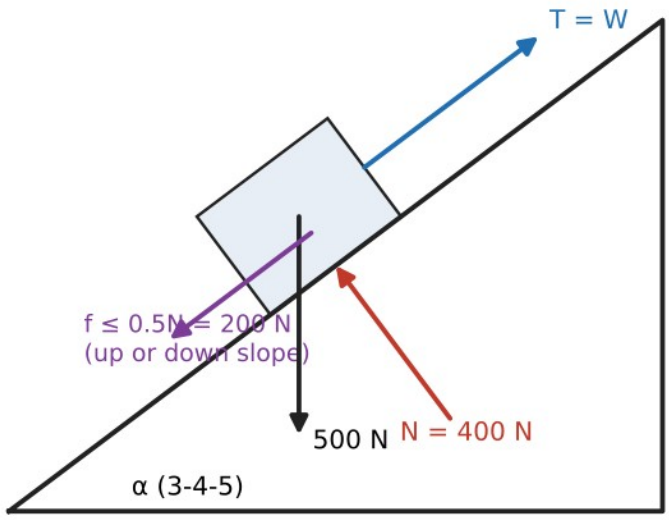
1. Normal force: $N = 500(0.8) = 400$ N, so $f_{max} = 0.5(400) = 200$ N.
2. Weight component down the slope: $500(0.6) = 300$ N.
3. *Impending down* (friction up the slope): $W_{min} + 200 = 300$, so $W_{min} = 100$ N.
4. *Impending up* (friction down the slope): $W_{max} = 300 + 200 = 500$ N.

$$100\,\text{N} \le W \le 500\,\text{N}$$

Check. $\tan\alpha = 0.75 > \mu = 0.5$, so the block would slide down without the cable. That is why $W_{min} > 0$ ✓. When $W = 300$ N, friction is zero (the midpoint of the range) ✓.

Exam takeaway. "Range of values" always means two impending-motion cases with friction reversed.
Variation: cable at angle β above the slope. Then $N = W_b \cos\alpha - T\sin\beta$ changes, and both limits shift.

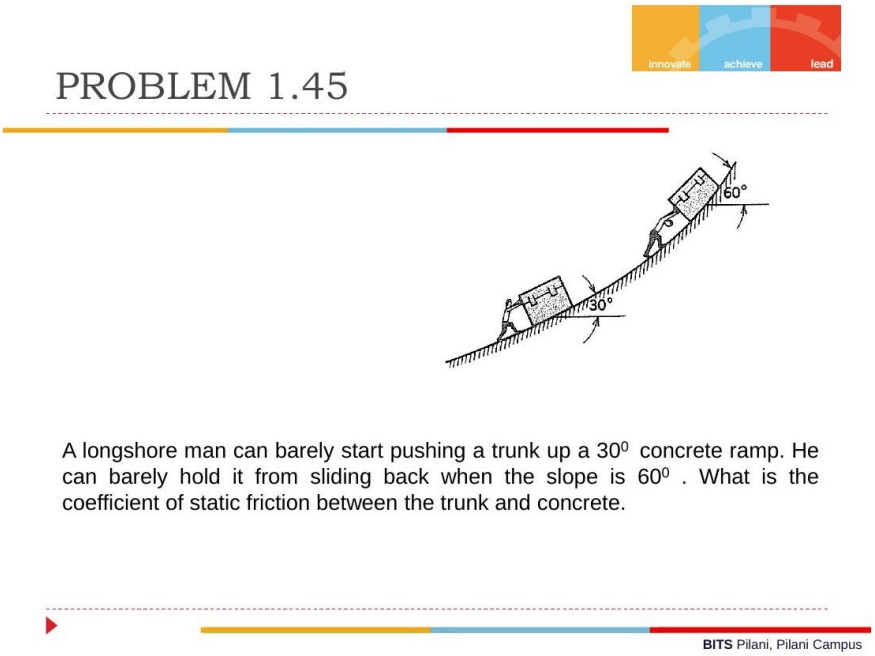
FBD of the block on the incline



$W_{min} = 500(0.6 - 0.5 \cdot 0.8) = 100$ N; $W_{max} = 500(0.6 + 0.5 \cdot 0.8) = 500$ N

Worked FBD of the block

Q13 (Slides Ch1; Crandall Prob 1.45) — Trunk on a ramp: find μ (medium)



Original: Slides Ch 1, slide 25

Problem. A longshoreman can barely start pushing a trunk **up** a 30° ramp. With the same push (parallel to the ramp) he can barely hold it from sliding **back** when the slope is 60°. Find μ_s.

Given: the same push P in both cases, parallel to the ramp. **Find:** μ.

Approach. Write impending-motion equations for both ramps and eliminate the unknown P and W ratio.

FBD in words. Trunk with W ¦, N normal to the ramp, P up the ramp, and friction μN down the ramp (30°, impending up) or up the ramp (60°, impending down).

Solution.

1. 30°, impending up: $P = W\left(\sin 30^\circ + \mu \cos 30^\circ\right) = W\left(0.5 + 0.866\,\mu\right)$.

2. 60°, impending down: $P = W(\sin 60^\circ - \mu \cos 60^\circ) = W(0.866 - 0.5\,\mu)$.
3. Equate: $0.5 + 0.866\,\mu = 0.866 - 0.5\,\mu$, so $1.366\,\mu = 0.366$.

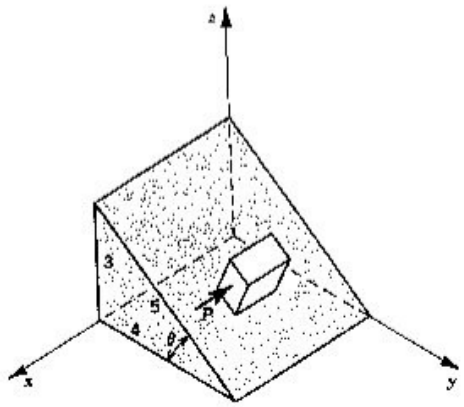
$$\mu_s = 0.268 \left(¿ \tan 15^\circ = 2 - \sqrt{3}\right)$$

Check. $\mu < \tan 30^\circ = 0.577$, so on the 30° ramp the trunk would slide down unaided. That is consistent with him needing to push ✓. The push is $P = W(0.5 + 0.232) = 0.732\,W$ in both cases ✓.

Exam takeaway. "Barely start pushing" (impending up) and "barely hold" (impending down) are the two directions. **Variation:** a single slope with a range of pushes, as in Q12.

Q14 (Crandall Prob 1.39) — Block pushed sideways on an incline (3-D friction) (medium)

1.39. A block of weight W rests on an inclined plane which makes an angle $\theta = \tan^{-1} \frac{3}{4}$ as shown. A force P, parallel to the x axis, is applied to the block and gradually increased from zero; when P reaches the value $0.4W$ the block begins to slide. What is the coefficient of friction between the block and the inclined plane?



Prob. 1.39

Original: Crandall Prob 1.39 (textbook figure)

Problem. A block of weight W rests on a plane inclined at $\theta = \tan^{-1}(3/4)$. A horizontal force P, parallel to the x-axis (the horizontal edge of the incline, so in the plane of the incline and perpendicular to its slope), is applied and slowly increased from zero. The block starts to slide when $P = 0.4\,W$. Find μ.

Given: $\sin\theta = 0.6$, $\cos\theta = 0.8$; $P = 0.4\,W$ along x (in the incline plane, horizontal). **Find:** μ.

Approach. In the plane of the incline the block is pushed by two forces at right angles: the down-slope weight component and P. Friction must balance their **resultant**, and slip starts when that resultant reaches μN.

FBD in words. Normal $N = W\cos\theta$ out of the incline. In the incline plane: $W\sin\theta$ down the slope and P along x. Friction f, opposite to their resultant.

Solution.

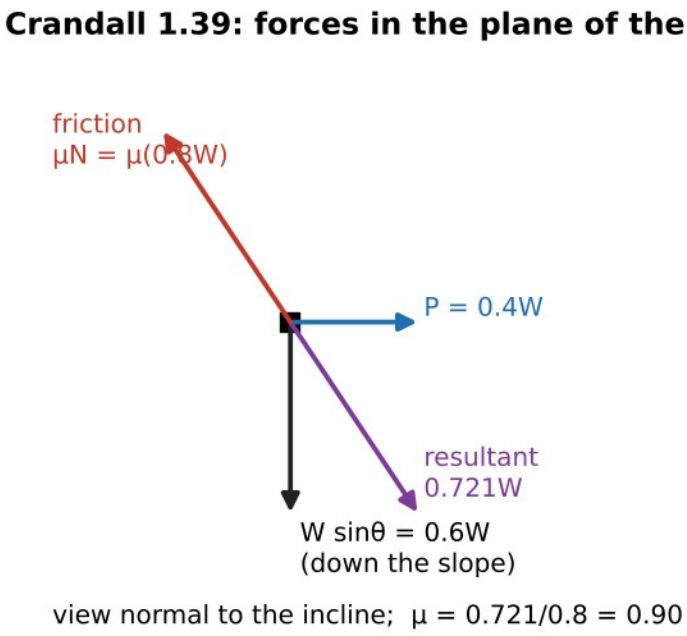
1. Normal force: $N = W\cos\theta = 0.8\,W$.
2. In-plane driving forces: $W\sin\theta = 0.6\,W$ down the slope and $P = 0.4\,W$ along x, at right angles.
3. Resultant: $\sqrt{0.6^2 + 0.4^2}\,W = \sqrt{0.52}\,W = 0.7211\,W$.
4. Impending slip: $\mu N = 0.7211\,W$, so $\mu = 0.7211/0.8 = 0.901$.

$$\mu = 0.90$$

The block starts to move at $\tan^{-1}(0.4/0.6) = 33.7^\circ$ from the line of steepest descent, toward the direction of P.

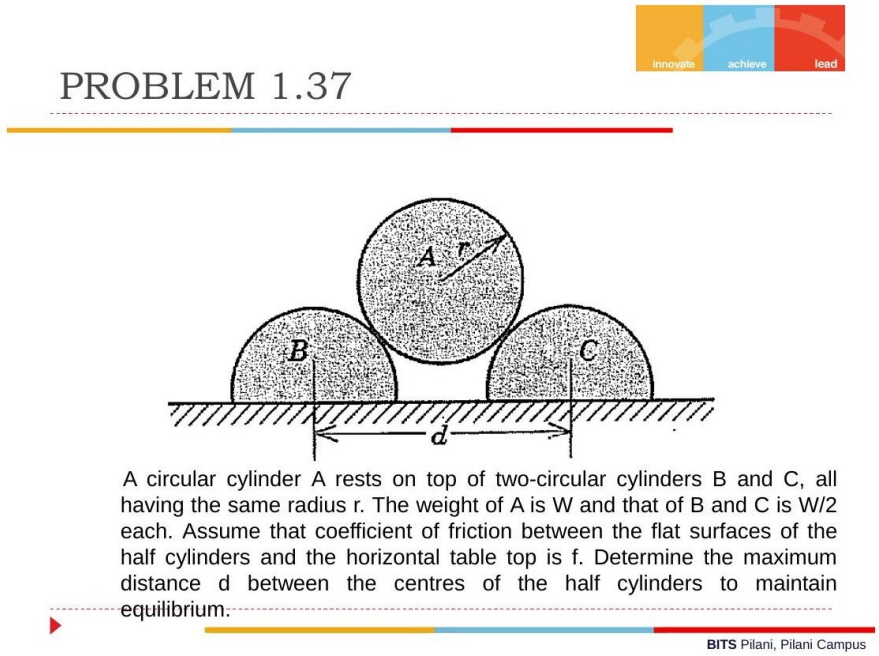
Check. Even with P = 0, $\tan\theta = 0.75 < 0.90$, so the block rests unaided ✓. The answer exceeds 0.75 and P adds the extra drive ✓.

Exam takeaway. Friction resists the vector sum of the in-plane forces. Treating P and $W\sin\theta$ separately (for example $\mu=(0.6+0.4)/0.8=1.25$) is the classic wrong answer. **Variation:** given μ, find the largest P: $P=W\sqrt{\mu^2\cos^2\theta-\sin^2\theta}$.



Friction balances the resultant of the in-plane forces

Q15 (Slides Ch1; Crandall Prob 1.37) — Cylinder on two half-cylinders: maximum spacing (hard)



Original: Slides Ch 1, slide 26

Problem. A cylinder A (weight W, radius r) rests on two half-cylinders B and C (each W/2, radius r). Their flat faces sit on a table with friction coefficient f. The cylinder-to-cylinder contacts are smooth. Find the maximum distance d between the centres of B and C for equilibrium.

Given: W, r, f; smooth curved contacts. **Find:** d_{max}.

Approach. Smooth contacts on circles act along the lines of centres. By symmetry, A pushes B and C equally. B tends to slide outward; table friction must stop it, and the limit is $F=f\,N$.

FBD in words.

- *A:* W ↓ and two contact forces N, one from each half-cylinder, along the lines of centres (each tilted θ from the vertical).
- *B:* W/2 ↓, N from A (down and outward along the line of centres), table normal N_t ↑, and friction F ← (inward).

Solution.

1. *Geometry.* The centres of A and B are $2r$ apart. The horizontal offset is $d/2$, so $\sin\theta = \frac{d/2}{2r} = \frac{d}{4r}$, where θ is measured from the vertical.
2. *A, vertical:* $2N\cos\theta = W$, so $N = \frac{W}{2\cos\theta}$.
3. *B, vertical:* $N_t = \frac{W}{2} + N\cos\theta = \frac{W}{2} + \frac{W}{2} = W$.
4. *B, horizontal:* $F = N\sin\theta = \frac{W}{2}\tan\theta$.
5. *No slip:* $F \le f\,N_t$, so $\frac{W}{2}\tan\theta \le f\,W$, i.e. $\tan\theta \le 2f$.
6. With $\tan\theta = 2f$: $\sin\theta = \frac{2f}{\sqrt{1+4f^2}}$, so $d_{max} = 4r\sin\theta$.

$$d_{max} = \frac{8rf}{\sqrt{1+4f^2}}$$

Check. f → 0 gives d → 0: with no friction the half-cylinders must touch ✓. f → ∞ gives d → 4r, the geometric limit at which A would fall between them ✓.

Exam takeaway. The table normal on B is **W**, not W/2: it carries half of A's weight as well as its own. **Variation:** with f = 0.3, $d_{max} = 8r(0.3)/\sqrt{1.36} = 2.06r$.

Q16 (PYQ 2014-15 Test-1 Q3) — Rope over a fixed drum: μ and the stretch of the wrapped part (medium; combines Ch 1 and Ch 2)

Q.3 A steel rope of diameter 10 mm and modulus of elasticity 200 GPa is hung over a stationary drum with weights on each end (*Fig. 3*). The diameter of the drum is 1.5 m. The rope is just on the point of slipping.
Determine: (*a*) coefficient of friction between rope and drum and (*b*) elongation of rope in segment *AB*. **[15]**

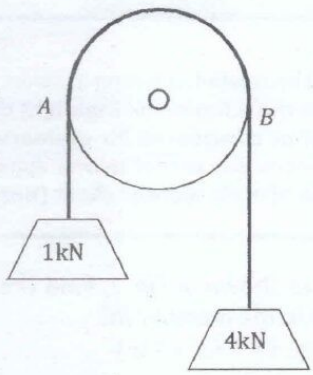


Fig. 3

Original question: PYQ 2014-15 Test-1 Q3

Problem. A steel rope (d = 10 mm, E = 200 GPa) hangs over a stationary drum of diameter 1.5 m, with 1 kN at one end (A) and 4 kN at the other (B), and 180° of contact. The rope is just about to slip. Find (a) the coefficient of friction and (b) the elongation of the rope segment AB in contact with the drum.

Given: $T_1 = 1$ kN, $T_2 = 4$ kN, β = π, r = 750 mm, $A = \frac{\pi}{4}(10)^2 = 78.54$ mm², E = 200 GPa. **Find:** μ and δ_{AB}.

Approach. (a) Belt friction at impending slip. (b) The tension varies along the contact, so the elongation is $\int T\,ds/AE$ (Ch. 2, non-uniform bar).

FBD in words. Rope arc from θ = 0 (slack side, 1 kN) to θ = π (tight side, 4 kN). Normal pressure from the drum points radially outward, and friction acts along the rope toward the slack side.

Solution.

1. *(a)* $T_2/T_1 = e^{\mu\pi}$: $4 = e^{\mu\pi}$, so $\mu = \frac{\ln 4}{\pi} = \frac{1.3863}{3.1416} = 0.441.$
2. *(b) Tension along the arc:* $T(\theta) = T_1 e^{\mu\theta}$, with $ds = r\,d\theta$.
3. *Elongation:*

$$\delta = \int_0^{\pi} \frac{T(\theta)\,r\,d\theta}{AE} = \frac{rT_1}{AE}\int_0^{\pi} e^{\mu\theta}\,d\theta = \frac{rT_1}{AE\mu}\left(e^{\mu\pi} - 1\right) = \frac{r(T_2 - T_1)}{AE\mu}$$

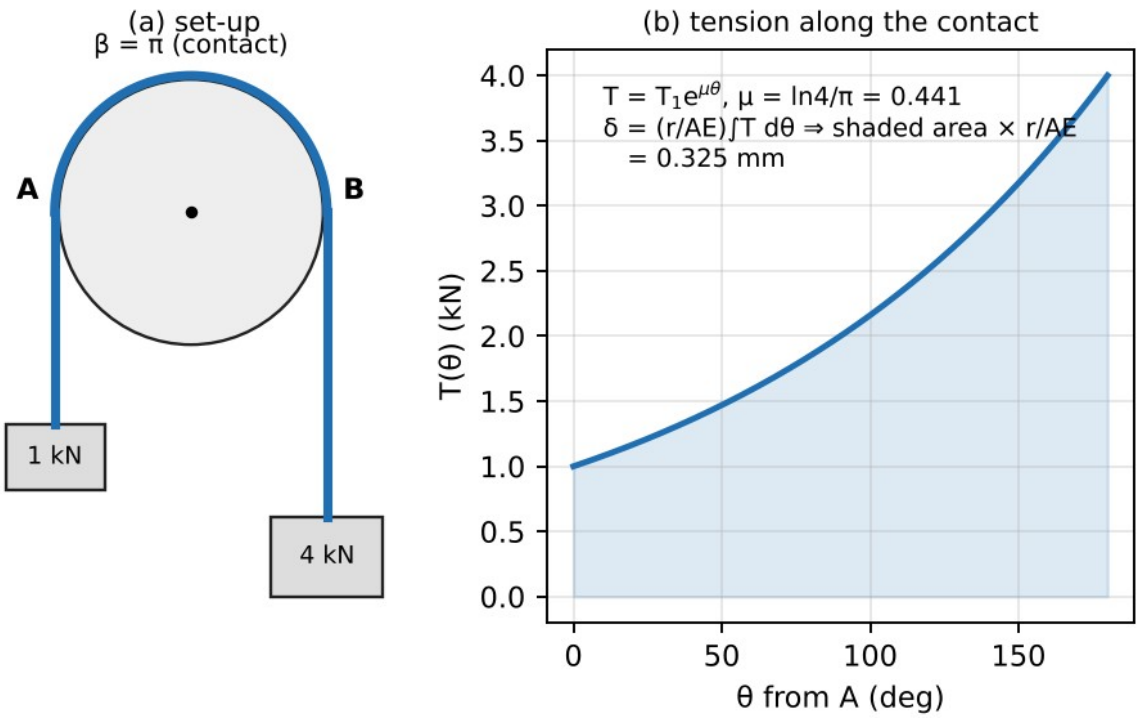
4. *Numbers:* $AE = 78.54(200\,000) = 1.571 \times 10^7$ N, so

$$\delta = \frac{750(4000 - 1000)}{1.571 \times 10^7 (0.441)} = \frac{2.25 \times 10^6}{6.928 \times 10^6} = 0.325\ \text{mm}$$

$$\mu = 0.441,\ \delta_{AB} \approx 0.325\ \text{mm}$$

Check. If the whole arc carried the average tension of 2.5 kN: $\delta = 2500(\pi \times 750)/1.571 \times 10^7 = 0.375$ mm. The exact answer is a little smaller because the exponential curve lies below the straight line joining 1 and 4 kN ✓.

Exam takeaway. This is how Test-1 mixes chapters: friction gives the force distribution, and Ch. 2 gives the stretch. **Variation:** "find the elongation of the free length below the drum" uses ordinary PL/AE with the constant hanging load.



Rope tension varies exponentially along the drum

Concept 1.5 — Plane trusses: method of joints, method of sections, zero-force members

A. Concept explanation

1. Intuition. A truss is a skeleton of straight sticks pinned at their ends, like a Meccano bridge. If loads are applied only at the joints, each stick can only be pulled (tension) or squashed (compression) along its length. A pinned stick with no load along it cannot carry bending. Solving a truss means finding that single number for every stick.

2. Why it matters. Bridges, roof trusses, cranes, towers and space frames. Every member is a two-force member (Concept 1.3). In Ch. 2 the member forces give joint deflections through $\delta = F\,L / A\,E$ and Castigliano's theorem. PYQ 2014 Test-1 Q1 (15 marks) and Q2 (20 marks) are both truss questions.

3. Formal treatment.

Ideal truss assumptions (slides):

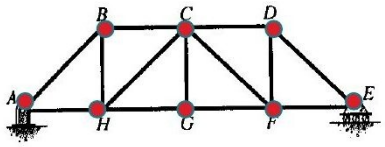
1. Members are connected only at their ends.
2. The connections are **frictionless pins**.
3. Loads are applied only at the joints.
4. Member weights are neglected (or lumped at the joints).

Under these assumptions every member is a two-force member, in pure tension or compression.

Stability / determinacy (slides): with m members, j joints and 3 support reactions:

- $m = 2\,j - 3$: rigid and statically determinate ("perfect").
- $m < 2\,j - 3$: unstable (a mechanism), collapsible.
- $m > 2\,j - 3$: over-rigid, statically indeterminate.

Example



m (Number of members) =13

j (Number of joints) = 8 → m=2j-3

Number of supports reactions=3

Original: Slides Ch 1, slide 33

With r reactions in general, determinacy needs $m+r=2\,j$, and the members must also be arranged sensibly (no collapsible panel).

Zero-force members (slides):

1. Two non-collinear members meet at a joint with no external load or support → **both** carry zero force.
2. Three members meet at an unloaded joint and two of them are collinear → the **third** carries zero force.

Method of joints (slides):

1. Find the support reactions from the whole truss (skip for a cantilever truss if you start at the free end).
2. Pick a joint with at most two unknown member forces. **Assume every unknown force is tensile** (arrow pointing away from the joint).
3. Apply $\sum F_x=0$ and $\sum F_y=0$ and move on joint by joint.
4. Tabulate: member, force, T or C. A negative result means compression.

Method of sections (slides): use it when only a few members are wanted.

1. Find the reactions if needed.
2. Pass a section that cuts **at most three** members, including the ones wanted, and splits the truss into two separate parts. It must pass through members, not joints.
3. Keep the part with fewer forces. Draw the cut members as tensile.
4. Apply the three equilibrium equations. Take moments about the point where two of the unknown forces intersect, so each equation has one unknown.

4. Derivations.

(a) The counting rule. Each pin joint gives 2 equations ($\sum F_x$, $\sum F_y$), so 2j equations in all. The unknowns are m member forces plus r reaction components. Unique solvability needs $m+r=2\,j$, i.e. $m=2\,j-3$ when r = 3. For the slide example, $m=13$ and $j=8$: $13=2(8)-3$ ✓.

(b) Zero-force rule 2. At an unloaded joint with members 1 and 2 collinear, sum forces **perpendicular** to that line. Only member 3 has a component in that direction, so $F_3\sin\alpha=0$ and $F_3=0$. Rule 1 follows the same way: sum perpendicular to member 1, and only member 2 contributes.

(c) Chord force = bending moment / depth. Cut a parallel-chord truss through the top chord (force F_1), a diagonal (F_3) and the bottom chord (F_2). Take moments about the joint where F_2 and F_3 meet: only F_1 and the external

loads on that side remain, so $F_1 h = M_{ext}$ and $F_1 = M/h$. The diagonal's vertical component equals the shear in that panel: $F_3 \sin\theta = V$. A truss works like a beam whose flanges are the chords and whose web is the diagonals. This links directly to Ch. 3.

5. Picture / geometric check.

- A **cantilever** truss loaded downward hogs: top chord in tension, bottom chord in compression.
- A **simply supported** truss sags: top chord in compression, bottom chord in tension.
- Diagonals sloping down toward mid-span carry tension in a simply supported truss.

Sketch the deflected shape before calculating; it predicts every sign.

6. Special cases.

- A joint with only two collinear members and no load: equal forces.
- A load applied directly over a support goes straight into the reaction and causes no member forces (Ch. 2 Q20 bridge).
- Symmetric truss with symmetric load: solve half and mirror.

7. Common mistakes.

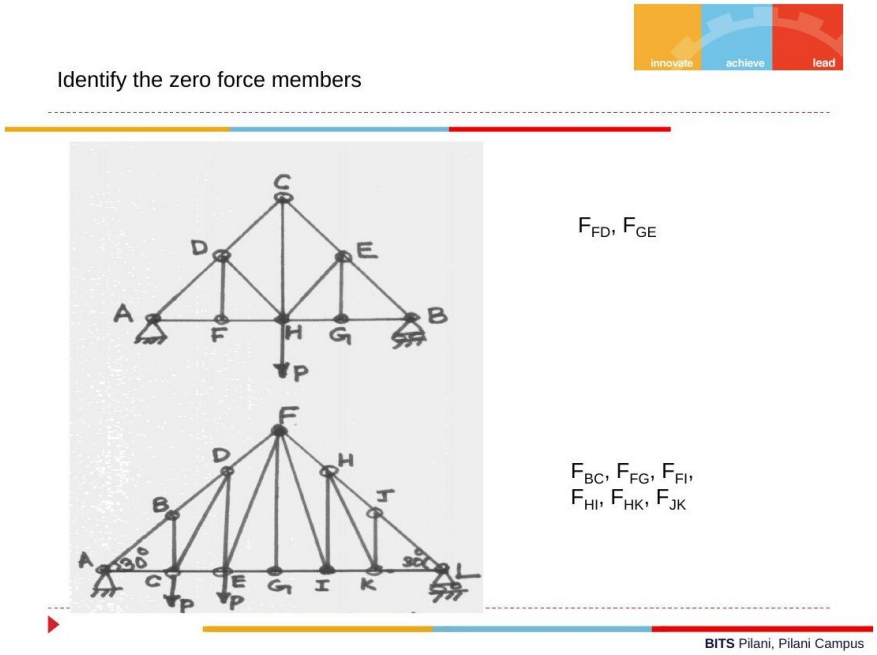
- Mixing up "assumed tension" and "answered compression". Keep the convention fixed throughout.
- Passing a section through a joint, or cutting four unknown members and expecting three equations to solve them.
- Wrong angles. Always compute $\tan\theta$ from the drawn dimensions; never guess 45° or 60°.
- Forgetting that a horizontal load at a pin support changes the horizontal reaction.

8. Summary box

Tool	Use when
$m = 2j - 3$	Check determinacy first
Zero-force rules	Spot them before starting: they save time and catch errors
Joints	All member forces wanted
Sections	One to three specific members wanted; take moments about the intersection of the other two
Chord = M/h, diagonal = V/sinθ	Quick checks of section results

B. Solved questions

Q17 (Slides Ch1) — Identify the zero-force members (easy)



Original: Slides Ch 1, slide 38

Problem. Two trusses are shown. Identify all zero-force members.

Approach. Apply zero-force rule 2 at every unloaded, unsupported joint where two members are collinear. Then remove those members and re-check the neighbouring joints, because the rules cascade.

Solution.

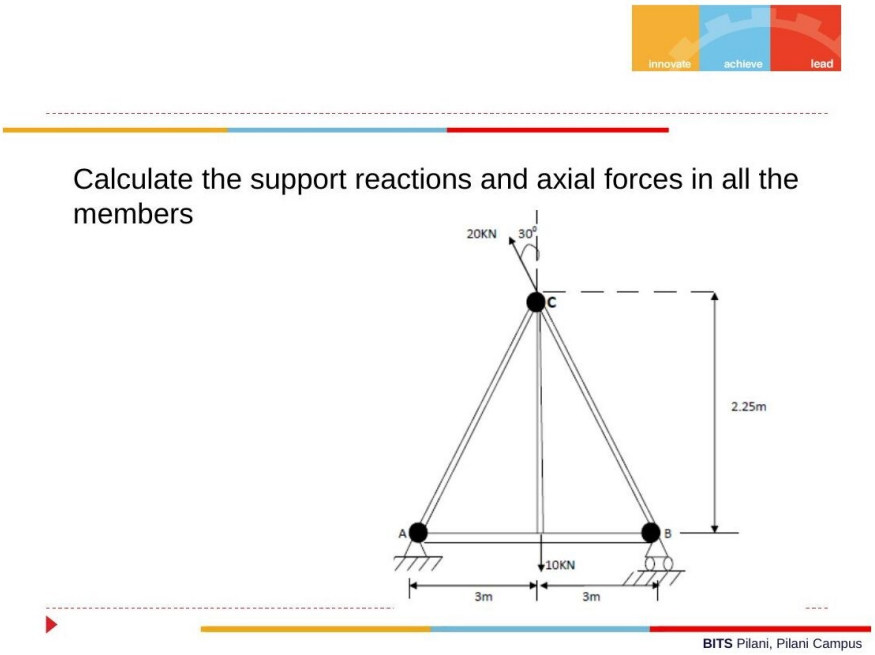
1. *Upper truss:* the unloaded bottom joints F and G each join a straight bottom chord and one vertical, so the verticals FD and GE are zero (rule 2).
2. *Lower truss:* start at the unloaded joints on the free edges and work inward. Each removal of a zero member turns the next joint into a rule 1 or rule 2 case.

$$\text{Upper: } F_{FD} = F_{GE} = 0 \,.\, \text{Lower: } F_{BC}, F_{FG}, F_{FI}, F_{HI}, F_{HK}, F_{JK} = 0$$

Check. Each zero member is “unloaded” in the sense that, if it were removed, the truss would still carry the given loads (though it would become unstable for other loads).

Exam takeaway. Zero-force members often appear as a 2-mark part, and identifying them first shortens every later calculation. **Variation:** add a load at one of those joints; the member is then no longer zero.

Q18 (Slides Ch1) — A-frame truss by the method of joints (easy–medium)



Original: Slides Ch 1, slide 41

Problem. A (pin) and B (roller) are 6 m apart. Apex C is 2.25 m above midpoint D. The members are AC, BC, CD, AD and BD. Loads: 10 kN down at D, and 20 kN at C acting down and to the right at 30° from the vertical. Find the reactions and all member forces.

Given: coordinates A(0,0), D(3,0), B(6,0), C(3,2.25) m. Load at C: $\left(+20\sin 30^\circ, -20\cos 30^\circ\right) = (10, -17.32)$ kN.
Find: reactions and member forces.

Approach. Reactions from the whole truss, then joints D, A and B (each with at most two unknowns).

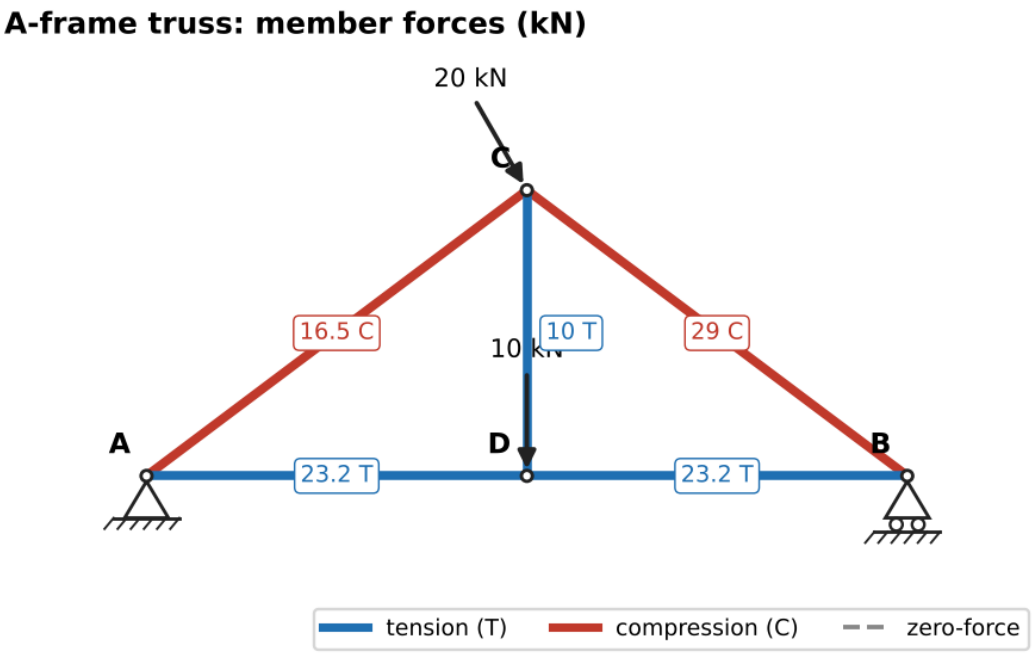
Solution.

1. *Geometry:* $AC = BC = \sqrt{3^2 + 2.25^2} = 3.75$ m, so for AC and BC $\sin\theta = 2.25/3.75 = 0.6$ and $\cos\theta = 0.8$.
2. *Reactions.* $\sum F_x$: $A_x + 10 = 0$, so $A_x = -10$ kN (10 kN $-$). $\sum M_A$: $B_y(6) - 10(3) - \left[17.32(3) + 10(2.25)\right] = 0$. Here the second term is the 10 kN at D, and the 20 kN at C contributes $17.32(3)$ from its vertical part plus $10(2.25)$ from its horizontal part. So $B_y = (30 + 51.96 + 22.5)/6 = 17.41$ kN. $\sum F_y$: $A_y = 10 + 17.32 - 17.41 = 9.91$ kN.
3. *Joint D:* CD is the only member with a vertical component, so $F_{CD} - 10 = 0$ and $F_{CD} = 10$ kN (T). Horizontally, $F_{AD} = F_{BD}$.
4. *Joint A:* vertical: $0.6\,F_{AC} + 9.91 = 0$, so $F_{AC} = -16.52$ kN (C). Horizontal: $0.8(-16.52) + F_{AD} - 10 = 0$, so $F_{AD} = 23.21$ kN (T).
5. *Joint B:* vertical: $0.6\,F_{BC} + 17.41 = 0$, so $F_{BC} = -29.02$ kN (C). Horizontal: $-0.8\,F_{BC} - F_{BD} = 0$, so $F_{BD} = 23.21$ kN (T) ✓ (matches F_{AD}).

Member	Force (kN)	Nature
AC	16.52	C
BC	29.02	C
CD	10.00	T
AD	23.21	T
BD	23.21	T

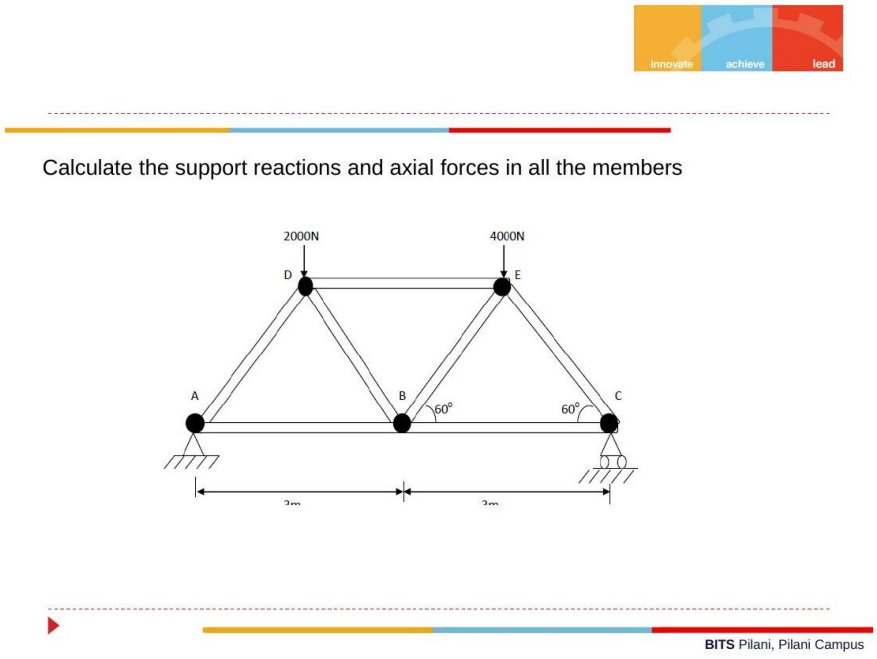
Check. Joint C was not used. Horizontal at C: $0.8(16.52)-0.8(29.02)+10=-10.0+10\approx 0$ ✓. A full computer solve gives the same table ✓.

Exam takeaway. The slide table labels the first row "AB". It means AC, since there is no member AB. **Variation:** make the 20 kN vertical. Then $A_x=0$ and the truss becomes symmetric apart from the load at D.



Solution: member forces (blue = tension, red = compression)

Q19 (Slides Ch1) — Warren truss with 2000 N and 4000 N (medium)



Original: Slides Ch 1, slide 39

Problem. A Warren truss has equilateral panels of 3 m: bottom joints A (pin), B, C (roller) and top joints D (above x = 1.5 m) and E (above x = 4.5 m). 2000 N acts down at D and 4000 N down at E. Find all member forces.

Given: $h=1.5\tan 60^\circ=2.598$ m; all members at 60°. **Find:** forces in AB, BC, AD, BD, BE, CE, DE.

Approach. Reactions, then joints A, C, D and E in turn.

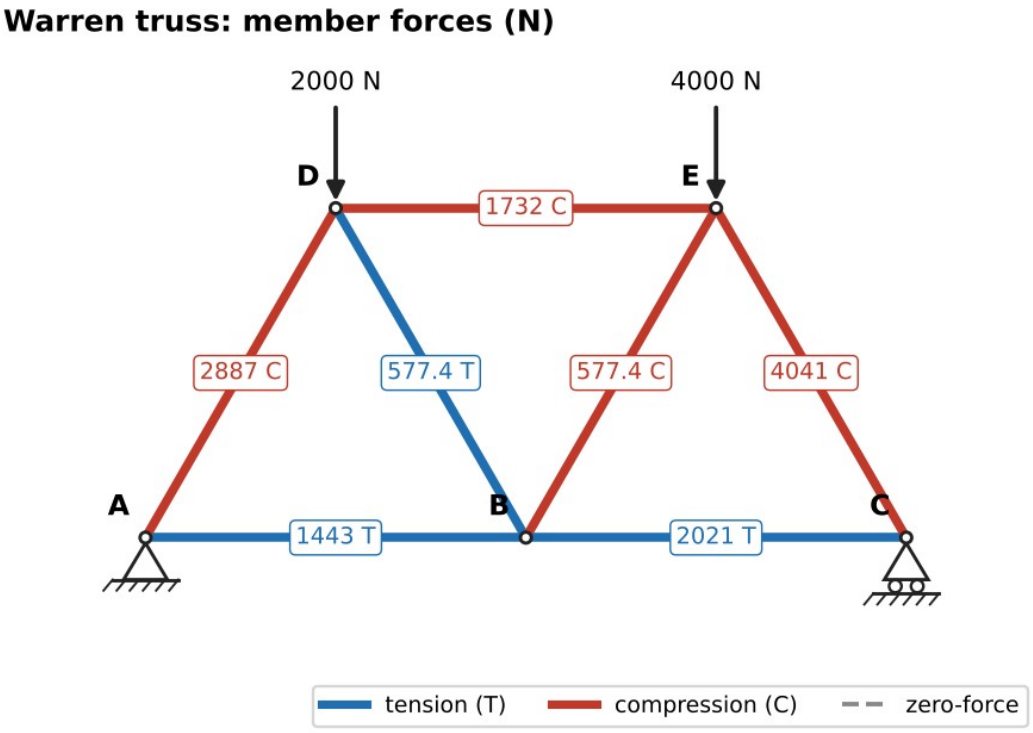
Solution.

1. *Reactions:* $C_y = (2000 \times 1.5 + 4000 \times 4.5)/6 = 3500$ N and $A_y = 6000 - 3500 = 2500$ N.
2. *Joint A:* $F_{AD}\sin 60^\circ + 2500 = 0$, so $F_{AD} = -2886.8$ N (C). $F_{AB} = -F_{AD}\cos 60^\circ = 1443.4$ N (T).
3. *Joint C:* $F_{CE}\sin 60^\circ + 3500 = 0$, so $F_{CE} = -4041.5$ N (C). $F_{BC} = 2020.7$ N (T).
4. *Joint D (vertical):* $-0.866\,F_{DA} - 0.866\,F_{DB} - 2000 = 0$, so $F_{DB} = \dfrac{-2000 + 0.866(2886.8)}{0.866} = 577.4$ N (T). *Horizontal:* $-0.5\,F_{DA} + 0.5\,F_{DB} + F_{DE} = 0$, so $F_{DE} = -1443.4 - 288.7 = -1732.1$ N (C).
5. *Joint E (vertical):* $-0.866\,F_{EB} - 0.866\,F_{EC} - 4000 = 0$, so $F_{EB} = \dfrac{-4000 + 3500}{0.866} = -577.4$ N (C).

Member	AB	BC	AD	BD	BE	CE	DE
Force (N)	1443.4 T	2020.7 T	2886.8 C	577.4 T	**577.4 C**	4041.5 C	1732.1 C

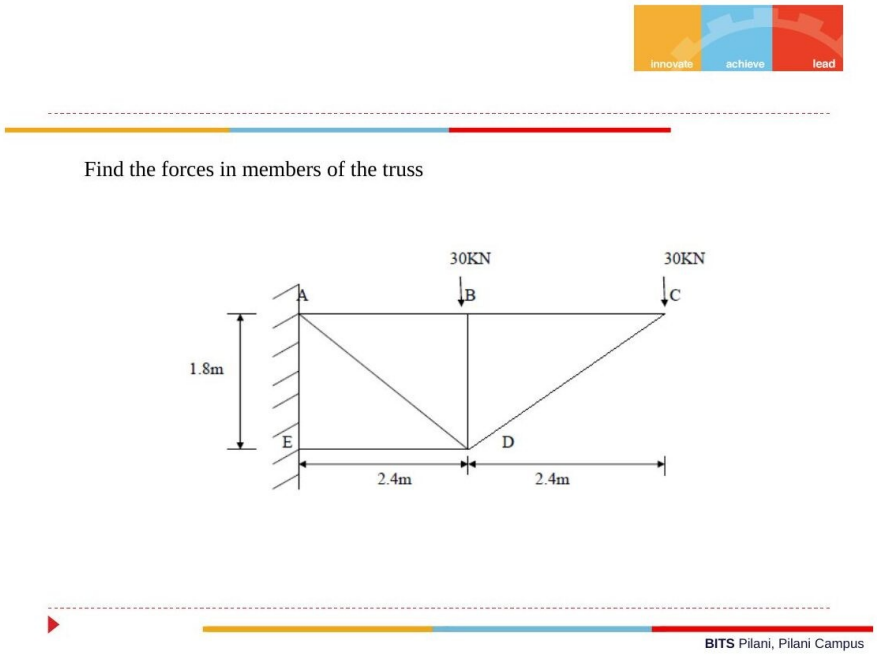
Check. The middle panel carries shear 2500 − 2000 = 500 N, and both diagonals have vertical component 500 N (577.4 × 0.866) ✓. The top chord is in compression and the bottom chord in tension, as for any simply supported truss ✓.

Exam takeaway. ⚠️ The slide table lists BE = 500 N (C). That is a slip. 500 N is the diagonal's *vertical component*; the member force is $500/\sin 60^\circ = 577.4$ N. **Variation:** equal loads at D and E make BD and BE zero (no panel shear).



Solution: member forces; note BE = 577 N (C), not 500 N

Q20 (Slides Ch1) — Cantilever truss by joints (medium)



Original: Slides Ch 1, slide 43

Problem. Wall joints A (top, y = 1.8 m) and E (bottom, y = 0). Top chord A–B–C with 2.4 m panels; bottom joint D below B. Members AB, BC, AD, BD, CD and ED. 30 kN acts down at B and at C. Find all member forces.

Approach. Start at the free joint C, which has only two unknowns. No reactions are needed until the end.

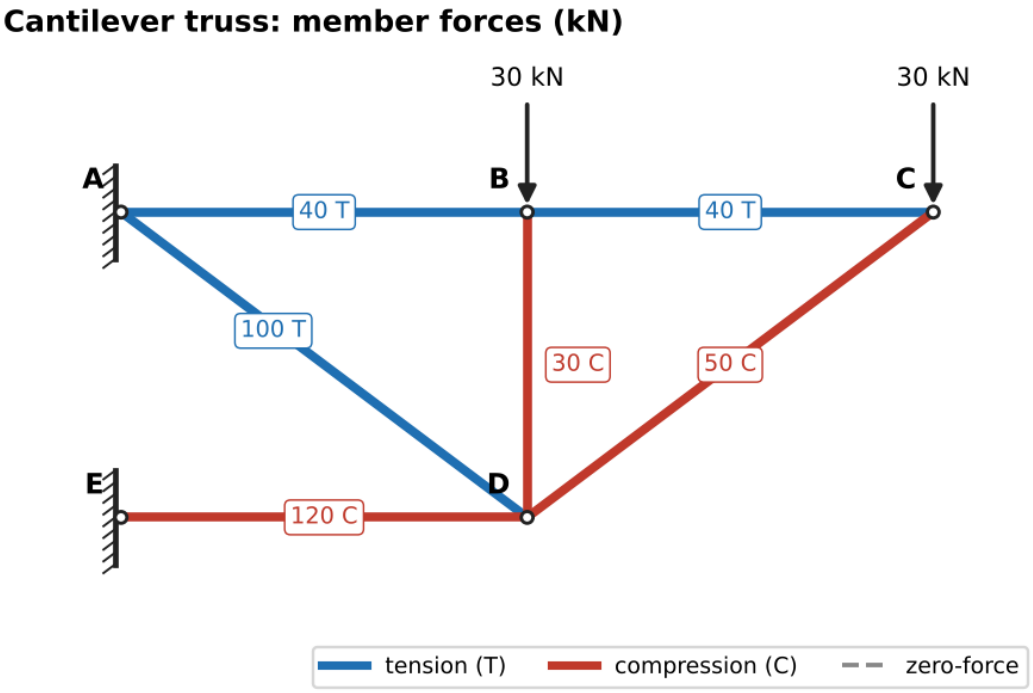
Solution.

1. *Geometry:* $\tan\theta = 1.8/2.4$, so θ = 36.87° ($\sin\theta = 0.6$, $\cos\theta = 0.8$) for AD and CD.
2. *Joint C:* vertical: $30 + F_{CD}(0.6) = 0$ with CD going down-left toward D, so $F_{CD} = -50$ kN (C). Horizontal: $F_{BC} = -F_{CD}(0.8) = 40$ kN (T).
3. *Joint B:* horizontal: $F_{BA} = F_{BC} = 40$ kN (T). Vertical: $F_{BD} + 30 = 0$ with BD pulling down on B when tensile, so $F_{BD} = -30$ kN (C).
4. *Joint D:* vertical: $0.6\,F_{DA} = 30 + 0.6(50)$, so $F_{DA} = 100$ kN (T). Horizontal: $F_{DE} = -120$ kN (C).

$$BC = 40\,T,\ CD = 50\,C,\ BD = 30\,C,\ AB = 40\,T,\ AD = 100\,T,\ DE = 120\,C\ (\text{kN})$$

Check. Moments about E for the whole truss: the wall reaction at A is A_x along the top chord at height 1.8 m. $1.8\,A_x = 30(2.4) + 30(4.8) = 216$, so $A_x = 120$ kN of pull. That equals $F_{AB} + F_{AD}\cos\theta = 40 + 80 = 120$ ✓.

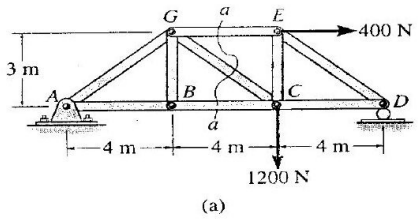
Exam takeaway. The slides show 40.01 and 50.01 because θ was rounded to 36.86°. Exact geometry gives round numbers. **Variation:** move both loads to C only (60 kN), then repeat.



Solution: member forces

Q21 (Slides Ch1) — Method of sections: GE, GC and BC (medium)

Determine the force in members GE, GC and BC of the truss shown in the Figure. Indicate whether the members are in tension or compression.



ANS:
RAx=400N(←)
RAy=300N
RD=900N
FGE=800N(C)
FBC=800N(T)
FGC=500.11N(T)

BITS Pilani, Pilani Campus

Original: Slides Ch 1, slide 52

Problem. A pin at A and a roller at D, with 4 m panels and 3 m depth. 400 N acts horizontally at E and 1200 N downward at C. Find the forces in GE, GC and BC and state T or C.

Given: A(0,0), B(4,0), C(8,0), D(12,0), G(4,3), E(8,3). **Find:** F_{GE}, F_{GC}, F_{BC}.

Approach. Section a–a cuts GE, GC and BC. Keep the left part (A, B, G). Take moments about G and C so that each equation contains one unknown.

Solution.

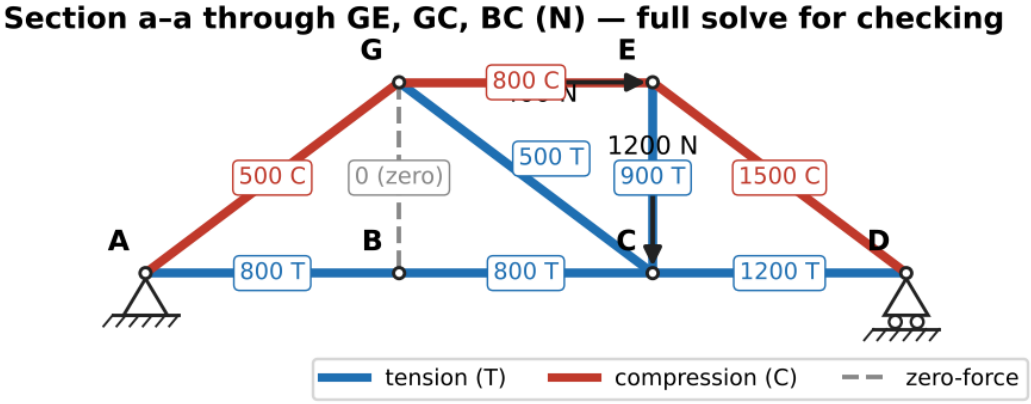
1. *Reactions:* $A_x = 400\ \text{N} \leftarrow$. $\sum M_A$: $D_y(12) - 1200(8) - 400(3) = 0$, so $D_y = 900\ \text{N}$ and $A_y = 300\ \text{N}$.

2. *Moments about G (4, 3)* on the left part: A_y (↑ at 4 m to the left) gives −1200 N·m. A_x (← at 3 m below) gives −1200 N·m. F_{BC} (assumed tensile, pointing →) at 3 m below G gives $+3F_{BC}$. So $-2400+3F_{BC}=0$ and $F_{BC}=800$ N (T).
3. *Moments about C (8, 0):* A_y gives $-300(8)=-2400$. A_x lies on the line through C, giving 0. F_{GE} (→) at 3 m above C gives $-3F_{GE}$. So $F_{GE}=-800$ N (C).
4. *Vertical sum on the left part:* $300-F_{GC}(3/5)=0$, so $F_{GC}=500$ N (T).

$$F_{GE}=800\,\text{N}(\text{C}),\ F_{GC}=500\,\text{N}(\text{T}),\ F_{BC}=800\,\text{N}(\text{T})$$

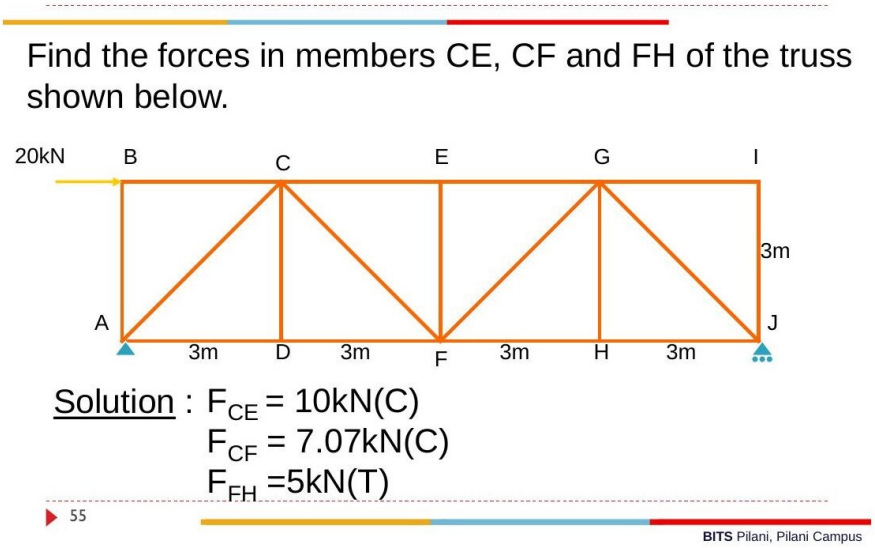
Check. These match the slide's answers. Horizontal sum on the left part: $-400-800+500(0.8)+800=0$ ✓.

Exam takeaway. Moments about the intersection of two cut members give the third directly. That is the reason for choosing the section that way. **Variation:** ask for F_{GB} instead, using a section through GE, GB and AB, or the joint B (where GB is then a zero-force check).



Full joint-by-joint solution, used to check the section answers

Q22 (Slides Ch1) — Sections with a horizontal load: CE, CF and FH (medium)



Original: Slides Ch 1, slide 55

Problem. The top chord is B–C–E–G–I and the bottom chord A–D–F–H–J, with 3 m panels and 3 m depth. The diagonals are AC, CF, FG and GJ. A is a pin and J a roller. 20 kN acts horizontally (→) at B. Find F_{CE}, F_{CF} and F_{FH}.

Approach. Reactions first, then two different sections.

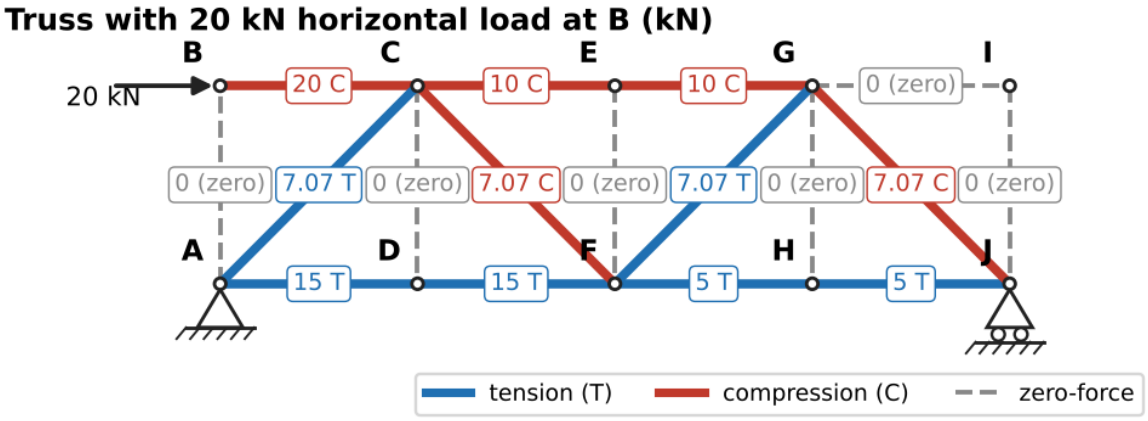
Solution.

1. *Reactions:* $A_x = 20$ kN $\leftarrow$. $\sum M_A$: $J_y(12) = 20(3)$, so $J_y = 5$ kN $\uparrow$ and $A_y = 5$ kN $\downarrow$.
2. *Section through CE, CF and DF, left part.* Moments about F (6, 0): the 20 kN at B (3 m up) gives −60; A_y = 5 $\downarrow$ at 6 m to the left gives +30; F_{CE} ($-$) at 3 m above F gives $-3F_{CE}$. So $F_{CE} = -10$ kN (C).
3. *Same section, vertical sum:* $-5 - F_{CF}/\sqrt{2} = 0$, so $F_{CF} = -7.07$ kN (C).
4. *Section through EG, FG and FH, right part.* Moments about G (9, 3): $J_y = 5$ $\uparrow$ at 3 m to the right gives +15; F_{FH} ($-$, tensile on the right part) at 3 m below G gives $-3F_{FH}$. So $F_{FH} = 5$ kN (T).

$$F_{CE} = 10\,\text{kN}(\text{C}),\ F_{CF} = 7.07\,\text{kN}(\text{C}),\ F_{FH} = 5\,\text{kN}(\text{T})$$

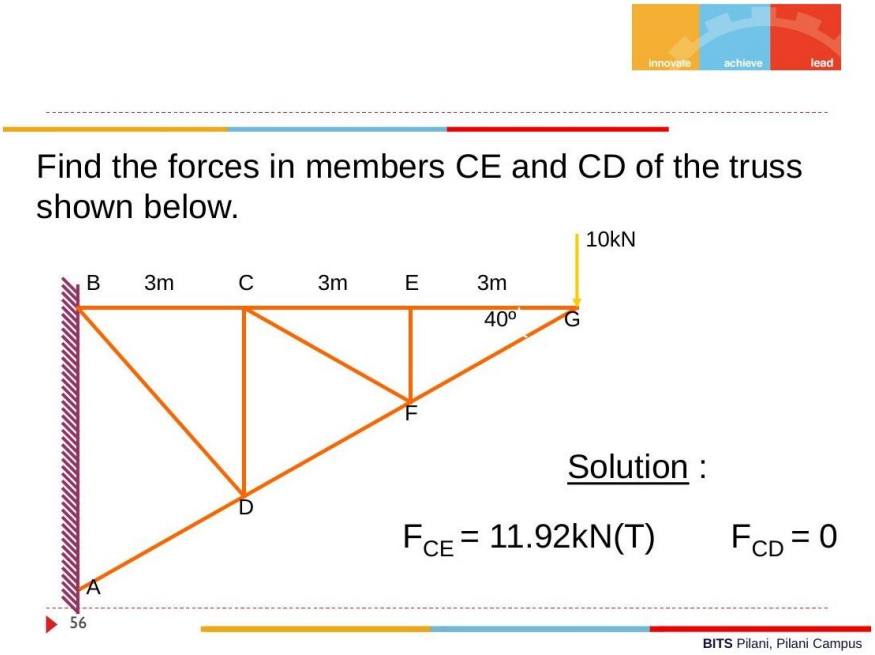
Check. These match the slide's answers ✓. A full solve also shows many zero-force verticals (see the diagram).

Exam takeaway. Horizontal loads create a couple of vertical reactions (A_y down, J_y up). Students who assume both reactions act upward get every sign wrong.



Full solution (zero-force members dashed)

Q23 (Slides Ch1) — Zero-force members plus one joint: CE and CD (medium)



Original: Slides Ch 1, slide 56

Problem. A cantilever truss has top chord B–C–E–G (3 m spacing) and a bottom chord A–D–F–G sloping at 40° near G, with verticals CD and EF and diagonal CF. 10 kN acts at G. Find F_{CE} and F_{CD}.

Approach. Joint G first, then chain the zero-force rules.

Solution.

1. *Joint G:* vertical: $F_{GF}\sin 40^\circ + 10 = 0$, so $F_{GF} = -15.56$ kN (C). Horizontal: $F_{GE} = -F_{GF}\cos 40^\circ = 11.92$ kN (T).
2. *Joint E:* CE and EG are collinear and EF is the third member, with no load → $F_{EF} = 0$ (rule 2). So $F_{CE} = F_{EG} = 11.92$ kN (T).
3. *Joint F:* DF and FG are collinear and EF = 0, so CF is the odd one out → $F_{CF} = 0$.
4. *Joint C:* BC and CE are collinear and CF = 0, so the vertical CD is the odd one out → $F_{CD} = 0$.

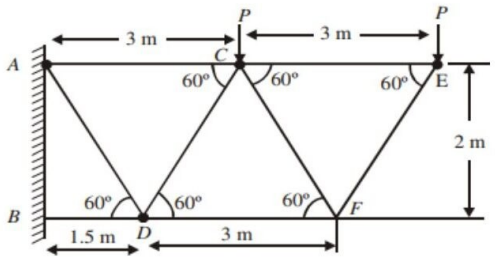
$$F_{CE} = 11.92\,\text{kN}\,(\text{T}),\ F_{CD} = 0$$

Check. These match the slide's answers ✓.

Exam takeaway. Cascading zero-force members is fast and examiners love it. **Variation:** add a load at E; then EF is no longer zero and the cascade stops.

Q24 (Slides Ch1, "Question 2") — Find the load P that gives 3 kN in member AC (medium, sections)

A cantilever truss is loaded and supported as shown in fig. find the values of load P which would produces an axial forces of magnitude 3KN in the member AC



Original: Slides Ch 1, slide 53

Problem. A cantilever truss (60° triangles, height 2 m) carries loads P at C and E. Find P such that the force in AC is 3 kN.

Approach. Cut through AC, CD and the bottom member at D. Take moments about D, where CD and the bottom chord meet, so that only F_{AC} and the loads appear.

Solution.

1. On the outer part, moments about D (horizontal arms 1.5 m to C and 4.5 m to E): $F_{AC}(2) - P(1.5) - P(4.5) = 0$.
2. So $F_{AC} = 6P/2 = 3P$.
3. Set $3P = 3$ kN.

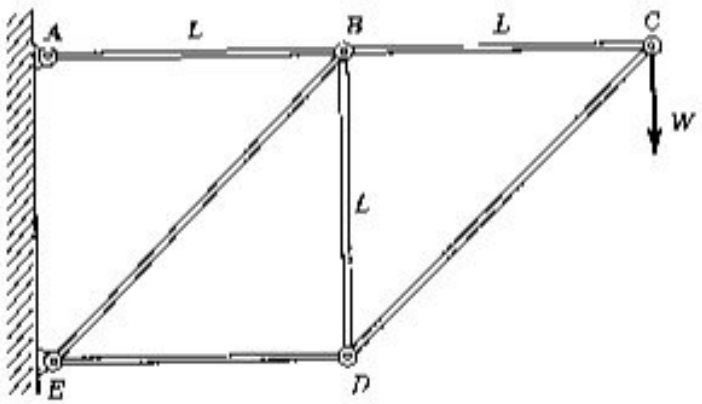
$$P = 1\,\text{kN}\,(AC\ \text{in tension})$$

Check. The top chord of a cantilever truss is in tension ✓.

Exam takeaway. "Find the load that gives force X in member Y" is a section question in disguise. **Variation:** the member has a maximum allowable force. Find the largest P for each member and take the smallest.

Q25 (Crandall Prob 1.31) — Six-member cantilever truss (medium)

1.31. Determine the forces in the six members of the truss shown.



Prob. 1.31

Original: Crandall Prob 1.31 (textbook figure)

Problem. Determine the forces in the six members of the truss. A (top) and E (bottom) are wall pins. B and C lie on the top line with AB = BC = L. D lies below B with BD = L. The members are AB, BC, ED, BD, EB and DC. A load W hangs at C.

Given: A(0,L), B(L,L), C(2L,L), E(0,0), D(L,0); load W ↓ at C. **Find:** all six member forces.

Approach. Joints, starting at the free end C (two unknowns), then D, then B.

Solution.

1. *Joint C* (members CB horizontal and CD diagonal at 45°): vertical: $F_{CD}\sin 45^\circ + W = 0$ with CD pointing down-left toward D, so $F_{CD} = -\sqrt{2}\,W$ (C). Horizontal: $F_{CB} = -F_{CD}\cos 45^\circ = W$ (T).
2. *Joint D* (members DE horizontal, DB vertical, DC diagonal): vertical: $F_{DB} + F_{DC}\sin 45^\circ = 0$, so $F_{DB} = W$ (T). Horizontal: $-F_{DE} + F_{DC}\cos 45^\circ = 0$, so $F_{DE} = -W$ (C).
3. *Joint B* (members BA, BC, BD, BE): vertical: $-F_{BD} - F_{BE}\sin 45^\circ = 0$, so $F_{BE} = -\sqrt{2}\,W$ (C). Horizontal: $-F_{BA} + F_{BC} - F_{BE}\cos 45^\circ = 0$, so $F_{BA} = W + W = 2W$ (T).

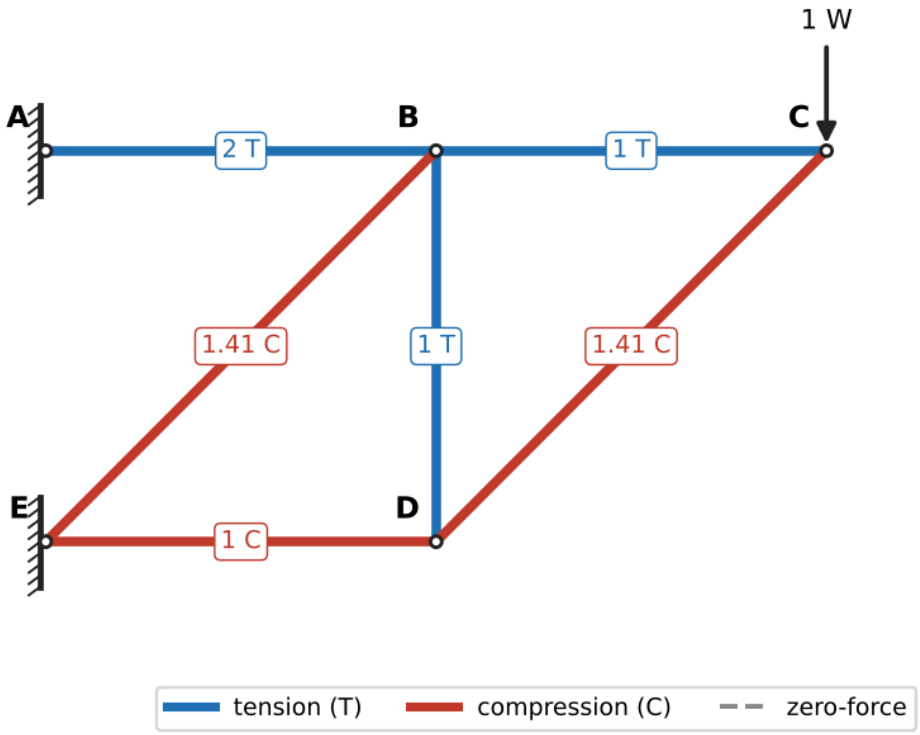
$$AB = 2W\,(T),\ BC = W\,(T),\ BD = W\,(T),\ ED = W\,(C),\ EB = \sqrt{2}\,W\,(C),\ DC = \sqrt{2}\,W\,(C)$$

Wall reactions: A pulls left with 2W. E pushes right with 2W and up with W.

Check. Moments about E for the whole truss: $W(2L) = 2W(L)$ ✓, where 2W is the horizontal force at A acting at height L.

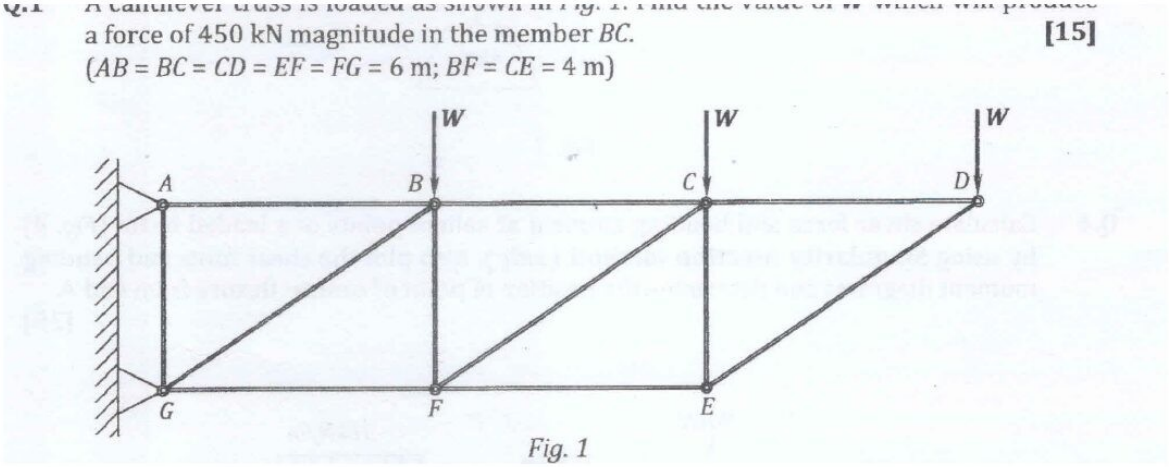
Exam takeaway. This truss reappears in Ch. 2 (Crandall Prob 2.10), where its tip deflection is $12.66\,WL/AE$, found from these forces with Castigliano's theorem. **Variation:** load at B instead of C, which makes BC and CD zero.

Crandall 1.31: member forces in units of W



Solution: member forces in units of W

Q26 (PYQ 2014-15 Test-1 Q1) — Cantilever truss: find W from F_{BC} = 450 kN (medium, 15 marks)



Original question: PYQ 2014-15 Test-1 Q1

Problem. The top chord is A–B–C–D and the bottom chord G–F–E, with AB = BC = CD = EF = FG = 6 m and BF = CE = 4 m. The diagonals are GB, FC and ED. Wall supports at A and G. Loads W at B, C and D. If the force in BC has magnitude 450 kN, find W.

Approach. Method of sections through BC, FC and FE. Keep the right part (C, D, E). Take moments about F, where FC and FE meet, so F_{BC} is the only unknown.

Solution.

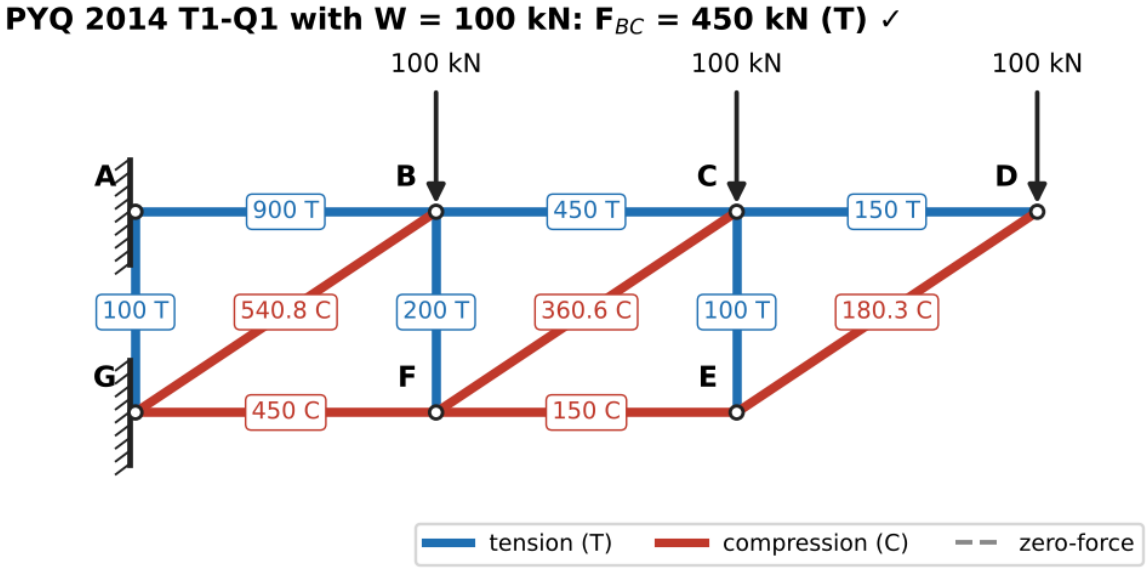
1. *Section:* cut BC (top), FC (diagonal) and FE (bottom). On the right part, the loads are W at C (6 m right of F) and W at D (12 m right of F).
2. *Moments about F:* F_{BC} (assumed tensile, pointing → toward B) acts 4 m above F. Equilibrium: $F_{BC}(4) - W(6) - W(12) = 0$.

3. So $F_{BC} = 18\,W/4 = 4.5\,W$.
4. $4.5\,W = 450$ kN gives W = 100 kN.

$$W = 100\,\text{kN}\,(BC\ \text{in tension})$$

Check. A full joint-by-joint solve with W = 100 kN gives BC = 450 kN T, AB = 900 kN T and GF = 450 kN C. The top chord is in tension, as expected for a cantilever ✓.

Exam takeaway. A 15-mark question that takes three lines once you pick the right moment point. **Variation:** "find the force in GB". Cut AB, GB and GF, take moments about G for AB, then use the vertical sum for GB.



Check: full solve with W = 100 kN

Chapter 1 — Formula and convention sheet

Topic	Key relation
Moment of a force	$M_O = F\,d = x\,F_y - y\,F_x$ (counter-clockwise +)
Couple	$C = F\,d$, same about every point; no force resultant
Force-couple equivalence	F at A ≡ F at O + $r_{OA} \times F$; F and M ≡ F displaced by $a = M/F$
Equilibrium (2-D)	$\sum F_x = \sum F_y = \sum M_O = 0$ (or 2 moments + 1 force, or 3 moments about non-collinear points)
Supports	roller 1, pin 2, fixed 3 (2-D); ball-and-socket 3, fixed 6 (3-D)
Two-force member	Equal, opposite, along the line joining the pins
Three-force member	Lines of action concurrent (or all parallel)
Static friction	$f \le \mu_s N$; at impending motion $f = \mu_s N$, opposing motion
Incline, impending up/down	$T = W(\sin\alpha \pm \mu\cos\alpha)$; angle of repose $\tan\alpha = \mu$
3-D friction on an incline	μN balances the **resultant** of the in-plane forces
Belt friction	$T_2/T_1 = e^{\mu\beta}$ (β in rad); $T(\theta) = T_1 e^{\mu\theta}$
Truss determinacy	$m = 2j - 3$ (with 3 reactions)
Zero-force rules	2 non-collinear unloaded → both 0; 3 with 2 collinear unloaded → third 0

Topic	Key relation
Sections	Cut ≤ 3 members; moments about the intersection of two unknowns
Chord/diagonal checks	chord = M/h; diagonal = $V/\sin\theta$
Sign convention	Assume tension; a negative answer means compression

Other Crandall Ch. 1 problems worth practising (same patterns): 1.8, 1.9 (FBDs of beams and cylinders), 1.12–1.16 (brackets, pedals, welds), 1.22 (frames), 1.24–1.27 (friction, wedges), 1.36 (logs in a tilted truck), 1.43–1.44 (camp stool, rod angle).

Chapter 2 — Introduction to Mechanics of Deformable Bodies

Axial deformation, springs, statically indeterminate bars, displacement diagrams, strain energy and Castigliano's theorem, rings and belts

Sources used: Slides "Chapter 2" (Dr. Sharad Shrivastava); Crandall Ch. 2 (§2.1–2.6; Examples 2.1, 2.3, 2.5; Probs 2.1, 2.6, 2.7, 2.10, 2.11, 2.12, 2.14, 2.17, 2.18, 2.19, 2.20, 2.24, 2.29, 2.31, 2.32, 2.45); PYQs 2014-15 (Test-1 Q2; Compre A Q8, Q11) and 2019-20 (Compre A Q3(b)).

Layout. Part A (eight items) for each concept, then Part B: slides → Crandall → PYQs, easy → hard. Each solution follows *Problem · Given/Find · Approach · FBD · Steps · Answer · Check · Takeaway*.

Concept 2.1 — The three-step method for deformable bodies; springs in parallel and series

A. Concept explanation

1. Intuition. Press down on a mattress resting on two different springs side by side. You can't say how much each spring carries just by balancing forces, because one equation can't fix two unknowns. What settles it is that both springs squash **by the same amount**, and the stiffer one pushes back harder. Every deformable-body problem combines three facts in this way: forces balance, pieces still fit together after deforming, and each piece's force is tied to its stretch by its stiffness.

2. Why it matters. This is the backbone of the whole course. Ch. 4 replaces "force" by stress and "stretch" by strain. Ch. 5 supplies the material law (Hooke's law, thermal strain). Ch. 6 repeats the three steps for twisting. Every indeterminate PYQ (fixed–fixed shafts, beam plus column, bolts in sleeves) is solved this way.

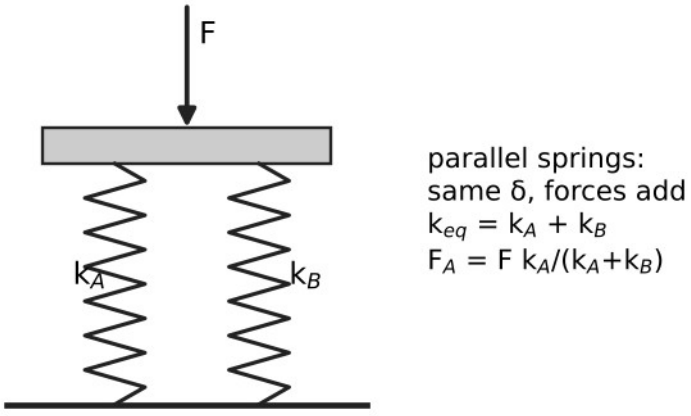
3. Formal treatment (Crandall eq. 2.1; slides). To analyse a deformable body:

1. **Study of forces and equilibrium requirements.**
2. **Study of deformation and conditions of geometric fit (compatibility).**
3. **Force–deformation relation** (the material/component law, e.g. $F = k\,\delta$ or $\delta = F\,L/A\,E$).

Linear spring: a force proportional to deflection, $F = k\,\delta$, where k (N/m or N/mm) is the **spring constant**: the force needed for a unit deflection. Its reciprocal $1/k$ is the **flexibility** (deflection per unit force).

Statically determinate problem: step 1 alone gives every force. Steps 2–3 are needed only for displacements. *Indeterminate problem:* more unknown forces than equilibrium equations, so all three steps are needed **simultaneously**.

4. Derivations.



Parallel springs share the deflection

(a) Two coaxial springs under a piston (Crandall Example 2.1, slides). A piston carrying F compresses springs k_A and k_B side by side.

- Step 1, equilibrium of the piston: $F = F_A + F_B$ ... (1). The slide notes that the spring equilibrium and $\sum M$ are automatically satisfied.
- Step 2, geometric fit: the rigid piston moves both springs equally, $\delta_A = \delta_B = \delta$ ... (2).
- Step 3, force–deflection: $F_A = k_A \delta_A$, $F_B = k_B \delta_B$ ... (3).

Substituting (2) and (3) into (1): $F = (k_A + k_B)\delta$ ... (4). Dividing (4) by $F_A = k_A \delta$:

$$F_A = \frac{k_A}{k_A + k_B} F, \delta = \frac{F}{k_A + k_B}$$

(b) Springs in series (Crandall Prob 2.18). The same force F passes through both springs (equilibrium of each spring and of the connector), and the total deflection is the sum: $\delta = \delta_1 + \delta_2 = F/k_1 + F/k_2$. Hence

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}, k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$$

(c) Springs in parallel: the same δ and forces add, so $k_{eq} = k_1 + k_2$.

5. Picture. Parallel springs look like "‖": same δ, forces add, **stiffnesses add**. Series springs look like a chain "—o—o—": same force, deflections add, **flexibilities add**. A rigid bar supported by springs at different positions is neither. Its deflections vary linearly along the bar (Q4, Q6).

6. Special / limiting cases.

- $k_B \to 0$: $F_A \to F$. $k_A = k_B$: each carries F/2. *The stiffer member attracts the load.* This rule of thumb reappears in composite bars, the RC pier, the bolt and sleeve, and indeterminate shafts.
- In series, the softer spring dominates: if $k_2 \ll k_1$, then $k_{eq} \approx k_2$.

7. Common traps.

- Writing compatibility as "forces equal" instead of "deflections equal" (parallel), or the other way round (series).
- Mixing up the series and parallel formulas.
- For a rigid bar on springs, forgetting that the deflection at each spring is (rotation) × (its distance from the hinge).

8. Summary box

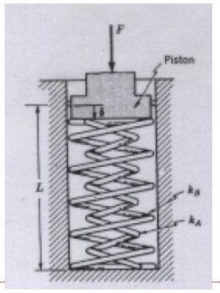
$$\text{equilibrium} + \text{compatibility} + \text{force–deformation}\,;\; k_{par} = \sum k_i, \frac{1}{k_{ser}} = \sum \frac{1}{k_i}, F_i = \frac{k_i}{\sum k} F \,(\text{parallel})$$

B. Solved questions

Q1 (Slides Ch2 / Crandall Example 2.1) — Piston on two coaxial springs (easy)

PROBLEM

A machine part carrying a load F terminates in a piston which fits in to a cavity as shown in fig. Within the cavity are two springs arranged coaxial with each other. Each spring has the characteristic that the force required to deflect it is proportional to the amount of deflection. Such a spring is said to have a linear force-deflection relation, and the amount of force required to produce a unit deflection is called the spring constant of the spring. We use the symbols K_A and K_B to denote the spring constants of the two springs in the cavity. Calculate how much of the load F is carried by the spring with constant K_A.



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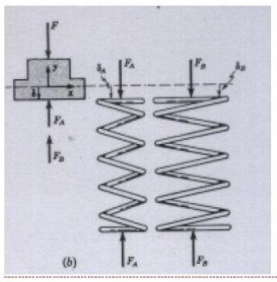
Original: Slides Ch 2, slide 3

Model :
Piston and two springs

Assumption.:
The ends of spring are flat, such that resultant compressive force on the springs are along its axis

STEP 1: STUDY OF FORCES AND EQUILIBRIUM REQUIREMENTS



$$\sum F_Y = 0: \quad F = F_A + F_B \text{ for piston} \quad \text{---------- (1)}$$

$$\sum F_Y = 0: \quad \text{already satisfied for Springs}$$

$$\sum M = 0: \quad \text{already satisfied for both springs and pistons}$$

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Original: Slides Ch 2, slide 4

Problem. A machine part carrying a load F ends in a piston that fits into a cavity containing two coaxial springs of constants k_A and k_B. How much of F is carried by spring A?

Given: F; linear springs k_A, k_B; flat spring ends (slide assumption), so each spring force acts along its axis. **Find:** F_A (and the piston deflection).

Approach. Two unknown spring forces, one useful equilibrium equation, so the problem is indeterminate and needs all three steps.

FBD in words. Piston with F ↓ on top and F_A ↑ and F_B ↑ from the springs underneath.

Solution.

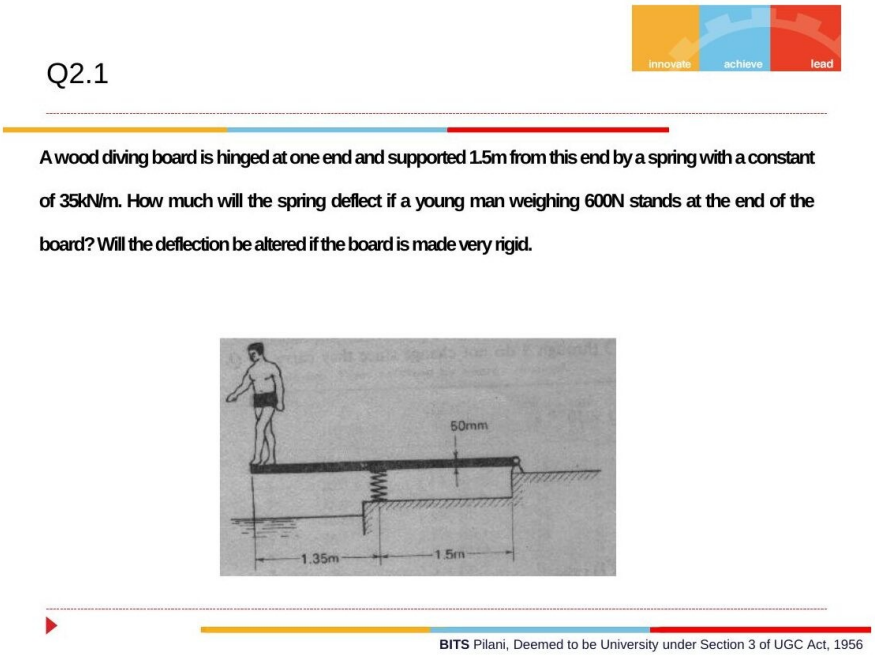
1. Equilibrium: $F_A + F_B = F$.
2. Geometric fit: $\delta_A = \delta_B = \delta$ (the piston is rigid and both springs start in contact).
3. Force–deformation: $F_A = k_A \delta$, $F_B = k_B \delta$.
4. Substitute: $(k_A + k_B)\delta = F$, so $\delta = F/(k_A + k_B)$ and $F_A = k_A \delta$.

$$F_A = \frac{k_A}{k_A + k_B} F, \delta = \frac{F}{k_A + k_B}$$

Check. If $k_A = k_B$, $F_A = F/2$ ✓. Dimensions: k/k is dimensionless ✓.

Exam takeaway. This is the template for every "two members share one load" question. **Variation:** if spring B is initially 1 mm shorter (a gap), A alone carries the load until δ = 1 mm, after which both act. That gives a "kinked" load–deflection curve (Q13).

Q2 (Slides Ch2, Q2.1 = Crandall Prob 2.1) — Diving board on a spring (easy)



Original: Slides Ch 2, slide 29

Problem. A wooden diving board is hinged at one end and supported 1.5 m from the hinge by a spring of constant 35 kN/m. How much does the spring deflect when a 600 N person stands at the end of the board, 1.35 m beyond the spring? Is the spring deflection changed if the board is made very rigid?

Given: k = 35 kN/m at 1.5 m; W = 600 N at 2.85 m from the hinge. **Find:** δ_s, and whether board stiffness matters.

Approach. Hinge plus one spring is statically determinate, so the spring force follows from statics alone.

FBD in words. Board with hinge reactions H_x, H_y at the hinge, spring force F_s ↑ at 1.5 m and 600 N ↓ at 2.85 m.

Solution.

1. $\sum M_{hinge} = 0$: $F_s(1.5) = 600(2.85)$, so $F_s = 1140$ N.
2. $\delta_s = F_s/k = 1140/35\,000 = 0.0326$ m.

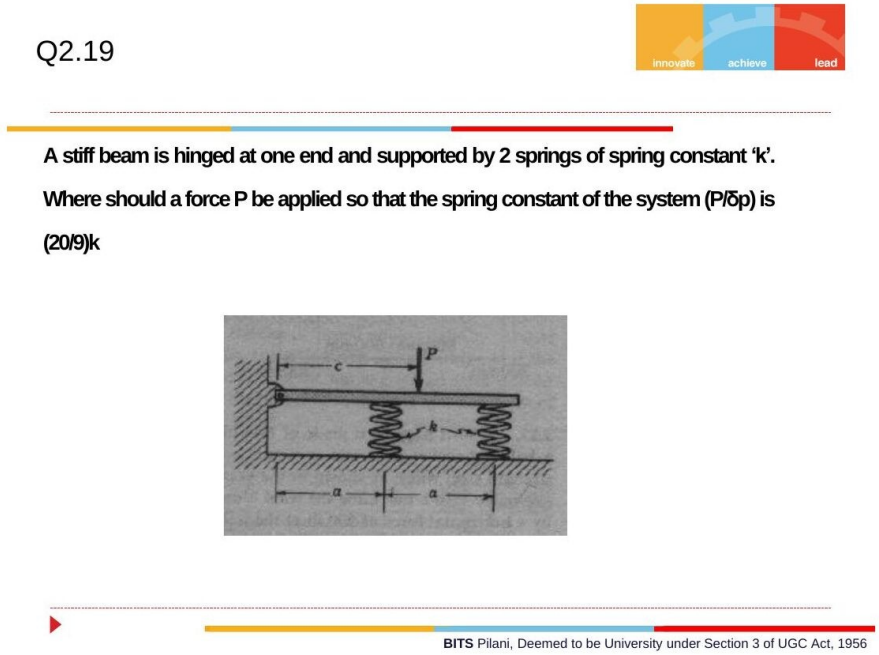
$$\delta_s = 32.6 \text{ mm}$$

No: the spring force (hence its deflection) is fixed by statics, whatever the board's stiffness. The **tip** deflection does change. For a rigid board it is $32.6(2.85/1.5) = 61.9$ mm; a flexible board adds its own bending.

Check. Spring force about 2× the weight because of lever action ✓.

Exam takeaway. Before writing compatibility, check whether the problem is determinate; if it is, deformation never changes the forces. **Variation:** two springs under the board (Q3), which is indeterminate.

Q3 (Slides Ch2, Q2.19 = Crandall Prob 2.19) — Where to put P for system stiffness 20k/9 (medium)



Original: Slides Ch 2, slide 28

Problem. A stiff beam is hinged at its left end and supported by two springs of constant k at distances a and 2a from the hinge. Where should a force P be applied so that the system's spring constant P/δ_P is $20/9\,k$?

Given: springs k at a and 2a; P at unknown distance c from the hinge. **Find:** c.

Approach. The rigid beam rotates by a small angle θ about the hinge. Spring deflections are $a\theta$ and $2a\theta$ (compatibility), forces follow from $F=k\delta$, and moments about the hinge give equilibrium.

FBD in words. Beam with hinge reactions at the hinge, spring forces $k(a\theta)\uparrow$ at a and $k(2a\theta)\uparrow$ at 2a, and P ↓ at c.

Solution.

1. Compatibility: $\delta_1=a\theta,\ \delta_2=2a\theta,\ \delta_P=c\theta$.
2. Force–deflection: $F_1=ka\theta,\ F_2=2ka\theta$.
3. Equilibrium about the hinge: $Pc=F_1a+F_2(2a)=ka^2\theta+4ka^2\theta=5ka^2\theta$.
4. System stiffness: $\dfrac{P}{\delta_P}=\dfrac{5ka^2\theta/c}{c\theta}=\dfrac{5ka^2}{c^2}$.
5. Set $\dfrac{5ka^2}{c^2}=\dfrac{20}{9}k$: $c^2=\dfrac{9a^2}{4}$.

$$c=3/2\,a$$

Check. At $c=2a$ the stiffness is $5k/4$; moving the load inward ($c=1.5a$) makes the system stiffer ($20k/9=2.22k$) ✓. At $c=a$ it is 5k ✓ (very stiff near the hinge).

Exam takeaway. For rigid bars on springs, the stiffness felt at a point depends on $1/c^2$: one factor of c comes from the lever arm for force, the other from the lever arm for displacement. **Variation:** three springs, or springs of different k.

Q4 (Crandall Prob 2.18) — Two springs in series (easy)

Problem. Two linear springs k_1 and k_2 are connected in series and loaded by F. Find the overall spring constant.

Approach. Series: same force, deflections add.

Solution.

1. Equilibrium of each spring: both carry F.
2. Deflections: $\delta_1 = F/k_1$, $\delta_2 = F/k_2$.
3. Total: $\delta = F\left(\frac{1}{k_1}+\frac{1}{k_2}\right)$, so $k = F/\delta$.

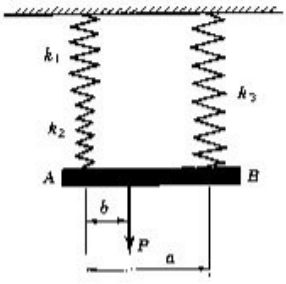
$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$$

Check. $k_1 = k_2 = k$ gives k/2: two equal springs in a chain are twice as soft ✓. $k_2 \to \infty$ (a rigid link) gives k_1 ✓.

Exam takeaway. A bar is a spring with $k = AE/L$. A stepped bar is springs in series, and a composite (concentric) bar is springs in parallel. **Variation:** used inside Q5.

Q5 (Crandall Prob 2.20) — Stiff bar on three springs: keep it level (medium)

arranged as shown. Where should a force P be applied so as to keep the bar AB horizontal? With P in this position how much does the bar move down because of P?



Prob. 2.20

Original: Crandall Prob 2.20 (textbook figure)

Problem. A stiff horizontal bar AB of length a hangs from springs k_1 and k_2 in series at end A and from spring k_3 at end B. Where should a force P be applied (distance b from A) so that the bar stays horizontal? How far does it then move down?

Given: k_1, k_2 in series at A; k_3 at B; length a. **Find:** b and δ.

Approach. "Stays horizontal" is a compatibility condition: both ends move down by the same δ. Combine the series pair into one spring and use moment equilibrium.

FBD in words. Bar with F_A ↑ at A, F_B ↑ at B and P ↓ at b from A.

Solution.

1. Series pair at A: $k_A = \frac{k_1 k_2}{k_1 + k_2}$.
2. Level bar: $\delta_A = \delta_B = \delta$, so $F_A = k_A \delta$ and $F_B = k_3 \delta$.
3. Vertical equilibrium: $F_A + F_B = P$, so $\delta = \frac{P}{k_A + k_3}$.
4. Moments about A: $Pb = F_B a = k_3 \delta a$, so $b = \frac{k_3 a}{k_A + k_3}$.

$$b=\frac{k_3 a}{\frac{k_1 k_2}{k_1+k_2}+k_3},\delta=\frac{P}{\frac{k_1 k_2}{k_1+k_2}+k_3}$$

Check. If all three springs are equal (k): $k_A=k/2$, so $b=\frac{k}{1.5k}a=2/3\,a$. P must sit nearer the stiffer end B ✓.

Exam takeaway. P must go through the "centre of stiffness". This is exactly how a load on a composite column must be placed so that the column shortens without bending. **Variation:** P at the middle; find the tilt of the bar.

Concept 2.2 — Uniaxial loading: δ = PL/AE and bars with several segments

A. Concept explanation

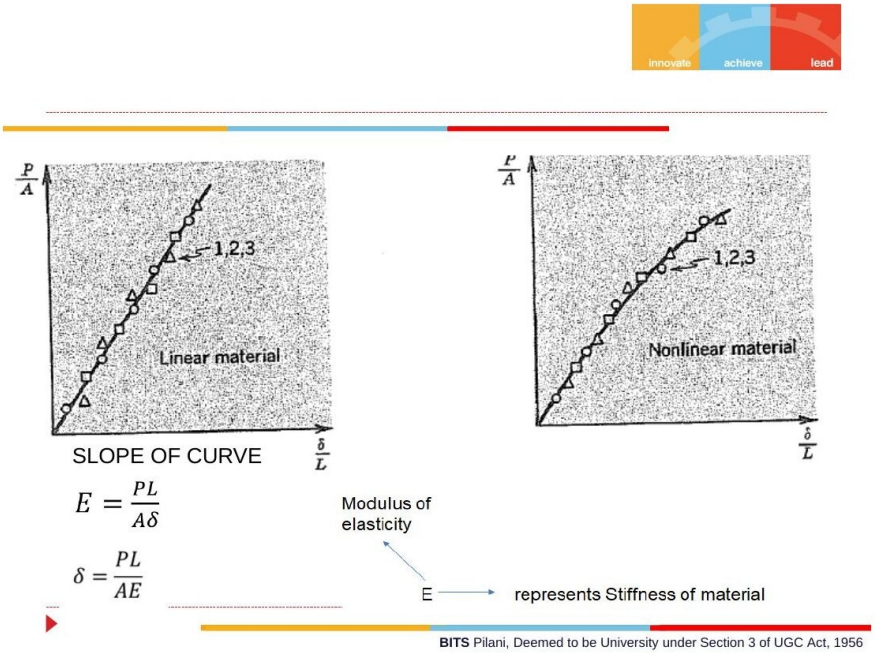
1. Intuition. Pull a rubber band. Doubling the force doubles the stretch. Doubling the length also doubles the stretch, because each half stretches the same amount as before. Doubling the thickness halves the stretch, because twice as much material shares the load. Stretch is proportional to force × length / (area × stiffness of the material).

2. Why it matters. This is the simplest force–deformation law, used in trusses, bolts, cables, columns, links, thermal stress (Ch. 5) and composite members. It also introduces the material constant E, which reappears in every chapter.

3. Formal treatment (slides, Crandall §2.3). Test three bars of different length and area (bars 1, 2, 3 in the slide). Their load–elongation (P–δ) curves differ. But when re-plotted as P/A against δ/L, they **collapse onto a single curve**, which is a property of the material alone (the slide's "slope of curve" figure). For a linear material its slope is the **modulus of elasticity** E:

$$E=\frac{P/A}{\delta/L}\Rightarrow\delta=\frac{PL}{AE}$$

E measures the stiffness of the **material**. $k=AE/L$ measures the stiffness of the **bar**, a spring constant. The slide stresses that E "represents the stiffness of the material" and AE/L that of the member.



Original: Slides Ch 2, slide 10

Several segments or non-uniform conditions:

$$\delta = \sum_i \frac{P_i L_i}{A_i E_i} \text{ or } \delta = \int_0^L \frac{P(x)\,dx}{A(x)E(x)}$$

P_i is the **internal** axial force in segment i, found from an FBD of a cut. Sign: tension +, elongation +.

4. Derivations.

(a) Internal force in a segment. Cut through segment i and keep the left part. Draw the unknown internal force P_i on the cut face as **tension** (pointing away from the cut). Then $\sum F_x = 0$ gives P_i = −(sum of external forces on the left part, + to the right). A positive result means tension.

(b) Bar hanging under its own weight. A bar of specific weight γ hangs from its top. At distance x above the free end the internal force is the weight below: $P(x) = \gamma A x$. Then

$$\delta = \int_0^L \frac{\gamma A x}{A E}\,dx = \frac{\gamma L^2}{2E} = \frac{W L}{2 A E} \quad (W = \gamma A L)$$

This is half what the whole weight would cause if hung at the end.

(c) Tapered bar. With $A(x)$ varying linearly from A_1 to A_2: $\delta = \frac{P L}{E} \frac{\ln(A_2/A_1)}{A_2 - A_1}$. It reduces to $P L / A E$ when $A_1 = A_2$.

5. Picture. Draw an **axial force diagram** (AFD), a step plot of P(x) along the bar. The total elongation is $\frac{1}{E}\sum(\text{AFD area}_i / A_i)$. For a rigid bar hung on rods, draw the bar's displaced position: a straight line joining the rod-end displacements.

6. Special cases.

- Equal and opposite end forces: P is constant and $\delta = P L / A E$.
- A segment with zero internal force does not stretch, even if loads act elsewhere.
- A rigid member on two deformable rods: the end displacements differ, so the rigid member rotates, and points beyond the rods move more (Q8).

7. Common traps.

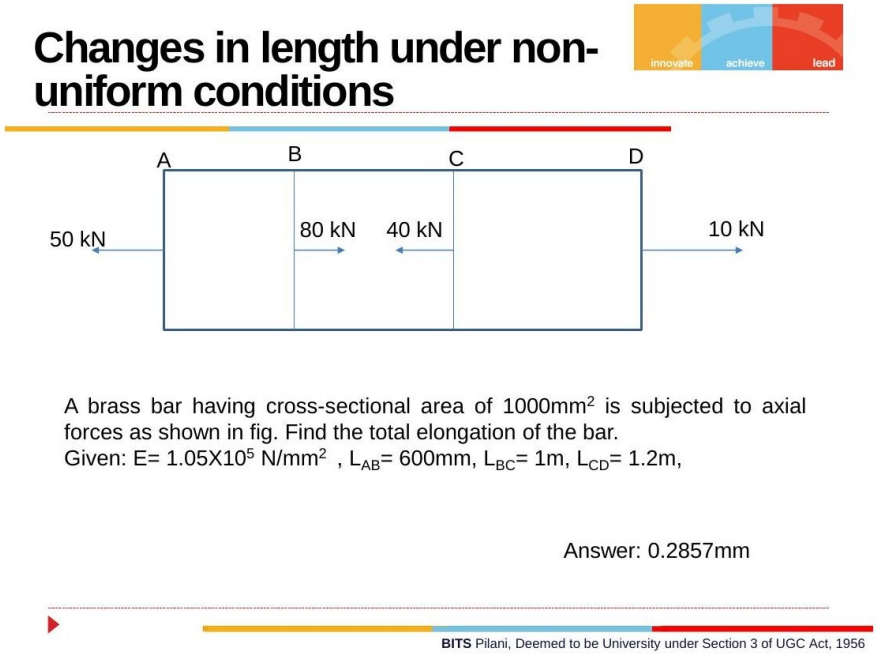
- Using the **external** load at a point instead of the **internal** force in the segment.
- Units: N, mm and MPa (N/mm^2) form a consistent set, as do N, m and Pa. Never mix kN with mm^2 and GPa without converting.
- Forgetting that compression gives a negative δ, which shortens the total.
- Applying δ = PL/AE outside the linear elastic range.

8. Summary box

$$\delta = \sum \frac{P_i L_i}{A_i E_i} ¿$$

B. Solved questions

Q6 (Slides Ch2) — Brass bar with four axial forces (easy)



Original: Slides Ch 2, slide 11

Problem. A brass bar of area 1000 mm² carries axial forces of 50 kN ← at A, 80 kN → at B, 40 kN ← at C and 10 kN → at D. $L_{AB}=600$ mm, $L_{BC}=1000$ mm, $L_{CD}=1200$ mm, $E=1.05\times10^{5}$ N/mm². Find the total elongation.

Given: as above. **Find:** δ_total.

Approach. Segment-by-segment internal forces (AFD), then $\delta=\sum P\,L/A\,E$.

FBD in words. For each segment, cut and keep the part to the left of the cut, drawing the internal force as tension.

Solution.

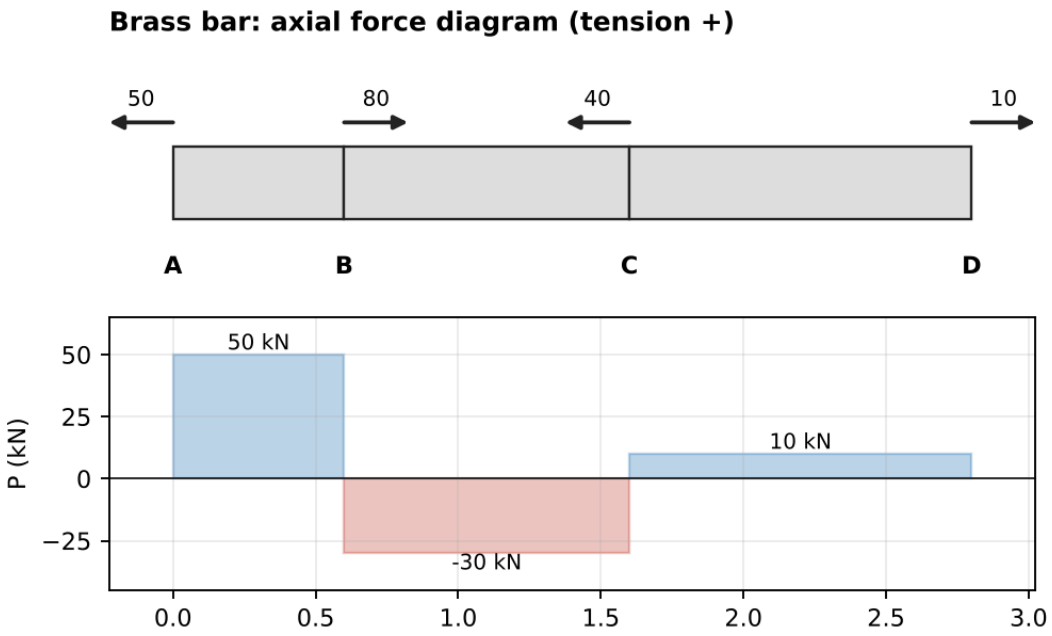
1. *Equilibrium check:* $-50+80-40+10=0$ ✓.
2. *AB:* the left part carries 50 kN ←, so the cut face needs +50 kN → (tension): $P_{AB}=+50$ kN.
3. *BC:* external forces to the left sum to $-50+80=+30$ kN →, so the cut face needs 30 kN ← (compression): $P_{BC}=-30$ kN.
4. *CD:* $-50+80-40=-10$ kN ←, so $P_{CD}=+10$ kN (T). Check: equals the 10 kN pull at D ✓.
5. *Elongation:*

$$\delta=\frac{1}{A\,E}\left[50(600)-30(1000)+10(1200)\right]\times10^{3}=\frac{(30-30+12)\times10^{6}}{1000\times1.05\times10^{5}}=\frac{12\times10^{6}}{1.05\times10^{8}}$$

$$\delta=0.114\,\text{mm}\,(\text{extension})$$

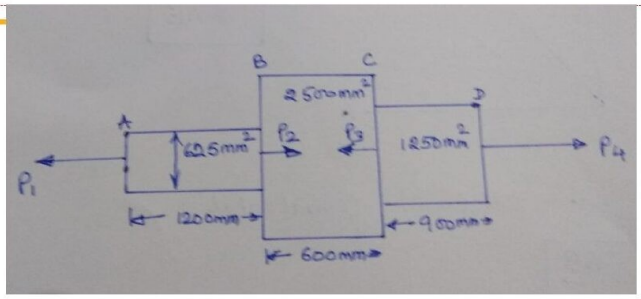
Check. AB stretches 0.286 mm, BC shortens 0.286 mm, CD stretches 0.114 mm. The first two cancel ✓.

Exam takeaway. ⚠️ The slide states 0.2857 mm. That is AB's contribution alone. With the loads as drawn, BC's shortening cancels it, so trust 0.114 mm unless your copy of the figure differs. Most of the marks go for the AFD, so draw it. **Variation:** a different area per segment (Q7).



Axial force diagram (tension +)

Q7 (Slides Ch2) — Stepped member ABCD: find P_2 and the total elongation (easy)



A member ABCD is subjected to point loads P_1, P_2, P_3 and P_4 as shown in fig. Calculate the force P_2 necessary for equilibrium, if P_1=45kN, P_3=450kN and P_4=130kN. Determine the total elongation of the member, assuming the modulus of elasticity to be 2.1X10^5 N/mm^2

$P_{2=}$ 365 kN , 0.4903 m(extension)

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Original: Slides Ch 2, slide 12

Problem. Member ABCD carries $P_1 = 45$ kN ← at A, P_2 → at B, $P_3 = 450$ kN ← at C and $P_4 = 130$ kN → at D. Areas: AB 625 mm² (1200 mm), BC 2500 mm² (600 mm), CD 1250 mm² (900 mm). $E = 2.1 \times 10^5$ MPa. Find P_2 for equilibrium and the total elongation.

Approach. Overall equilibrium for P_2, then the AFD, then $\sum P\,L/A\,E$ with a different A in each segment.

Solution.

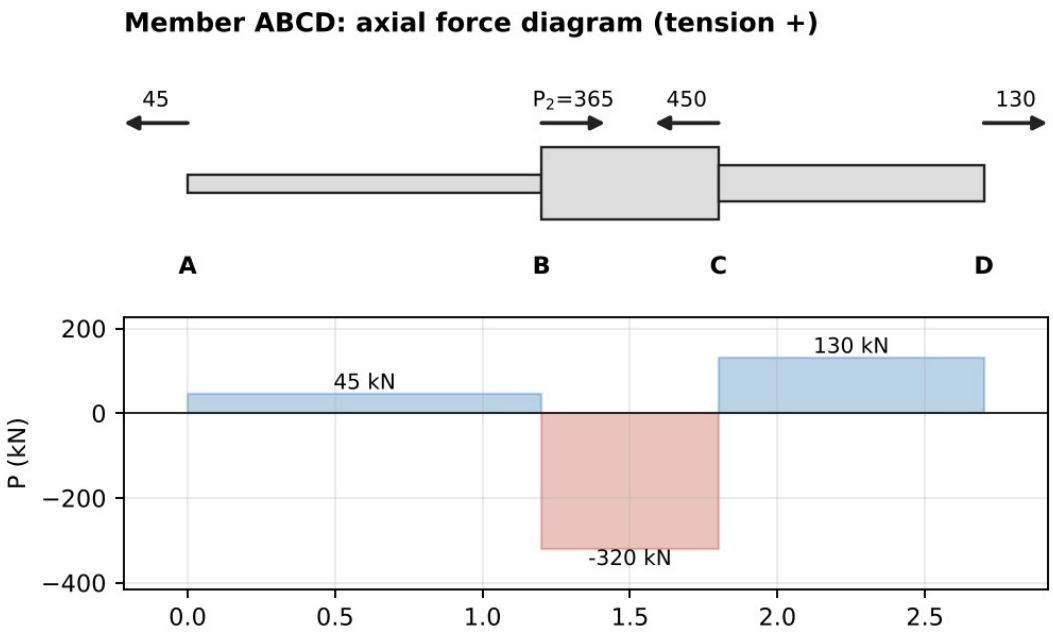
1. *Equilibrium:* $-45 + P_2 - 450 + 130 = 0$, so $P_2 = 365$ kN.
2. *Internal forces:* $P_{AB} = +45$ kN (T); $P_{BC} = 45 - 365 = -320$ kN (C); $P_{CD} = +130$ kN (T).
3. *Elongation:*

$$\delta=\frac{1}{2.1\times10^5}\left[\frac{45\,000(1200)}{625}-\frac{320\,000(600)}{2500}+\frac{130\,000(900)}{1250}\right]=\frac{86\,400-76\,800+93\,600}{2.1\times10^5}$$

$$P_2=365\,\text{kN},\delta=0.491\,\text{mm}\,(\text{extension})$$

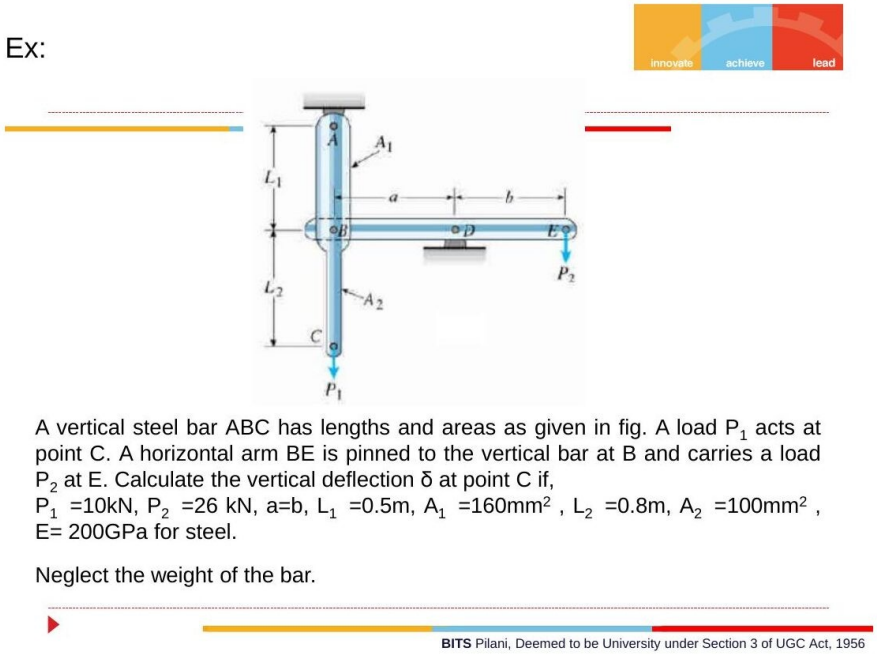
Check. The largest force (BC) is in the fattest segment, so its stress (128 MPa) is no bigger than AB's (72 MPa) or CD's (104 MPa); all are reasonable for steel ✓.

Exam takeaway. The slide writes "0.4903 m". That is a units typo for mm. **Variation:** "find the maximum stress", giving 128 MPa in BC (compressive).



Axial force diagram (bar width shows the area)

Q8 (Slides Ch2, Gere) — Vertical bar with a side arm (medium)



Original: Slides Ch 2, slide 15

Problem. A vertical steel bar ABC hangs from A. Segment AB has length $L_1=0.5$ m and area $A_1=160$ mm²; BC has $L_2=0.8$ m and $A_2=100$ mm². $P_1=10$ kN acts down at C. A horizontal arm BDE is pinned to the bar at B

and rests on a support at D (with a = b), and carries $P_2 = 26$ kN down at E. E = 200 GPa. Find the vertical deflection of C.

Given: as above. **Find:** δ_C.

Approach. First find the force the arm exerts on the bar at B (statics of the arm), then the AFD of the bar, then $\sum P\,L/A\,E$.

FBD in words.

- *Arm BDE:* pin force F_B at B (from the bar), support reaction at D, P_2 ↓ at E.
- *Bar ABC:* reaction at A, F_B from the arm at B (equal and opposite to the arm's), P_1 ↓ at C.

Solution.

1. *Arm, moments about D:* P_2 ↓ at E (distance b) tips the arm clockwise, so end B rises and the bar must hold it down. $F_{B,\text{on arm}}\,a = P_2\,b$ with a = b, so $F_B = 26$ kN. The bar pushes **down** on the arm; by Newton's third law the arm pushes **up** on the bar with 26 kN.
2. *Bar, internal forces:* $P_{BC} = +10$ kN (T). In AB: $10 - 26 = -16$ kN (C).
3. *Deflection of C:*

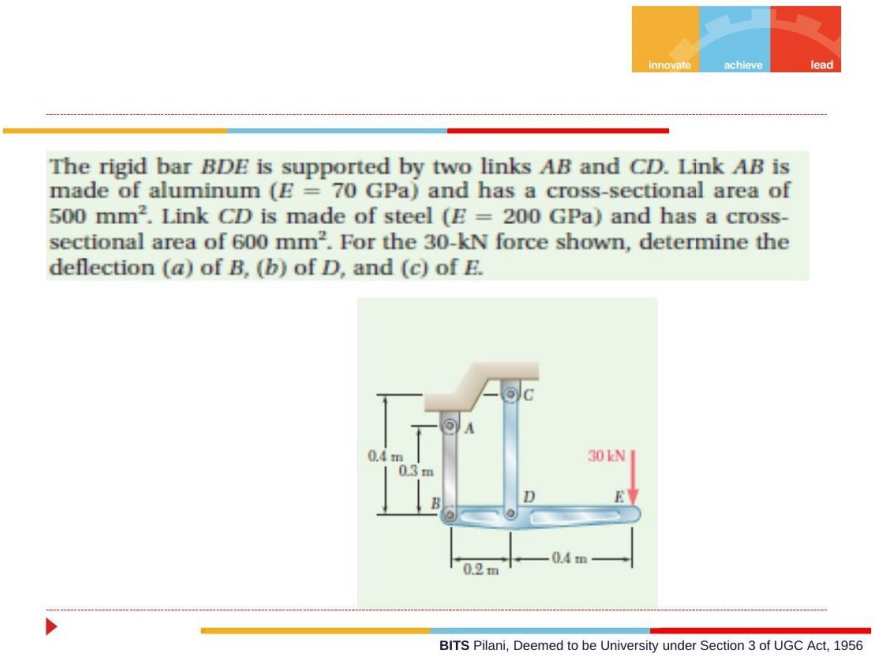
$$\delta_C = \frac{10\,000\,(800)}{100\,(200\,000)} + \frac{(-16\,000)(500)}{160\,(200\,000)} = 0.400 - 0.250$$

$$\delta_C = 0.150\ \text{mm}\ \downarrow$$

Check. BC stretches 0.40 mm while AB shortens 0.25 mm (its upper part is pushed up), so the net drop of C is 0.15 mm ✓.

Exam takeaway. A lever arm can turn a downward load into an upward push on the main bar. Always draw the arm's FBD separately. **Variation:** a = 2b gives $F_B = 13$ kN, so AB carries $10 - 13 = -3$ kN (still slightly compressed) and $\delta_C = 0.400 - 0.047 = 0.353$ mm ↓.

Q9 (Slides Ch2, Beer Sample 2.1) — Rigid bar on two links: deflections of B, D and E (medium)



Original: Slides Ch 2, slide 30

Problem. The rigid bar BDE is supported by two links. Link AB is aluminium (E = 70 GPa, A = 500 mm², L = 0.3 m). Link CD is steel (E = 200 GPa, A = 600 mm², L = 0.4 m). B is at 0, D at 0.2 m and E at 0.6 m along the bar; 30 kN acts down at E. Find the deflections of B, D and E.

Approach. Statics for the two link forces, $\delta = F\,L/A\,E$ for each link, then similar triangles for the rigid bar.

FBD in words. Bar BDE with link force F_{AB} at B, F_{CD} at D and 30 kN ↓ at E. Both links are vertical and attached above the bar.

Solution.

1. *Moments about D:* $F_B(0.2) = 30(0.4)$, so $F_{AB} = 60$ kN. To balance, the force at B must act **down** on the bar, so link AB pushes: **compression**.
2. *Vertical sum:* $F_{CD} = 60 + 30 = 90$ kN ↑ on the bar: CD pulls, **tension**.
3. *Link deformations:*

$$\delta_{AB} = \frac{60\,000(300)}{500(70\,000)} = 0.514\ \text{mm}\ (\text{shortens} \Rightarrow B\ \text{moves up})$$

$$\delta_{CD} = \frac{90\,000(400)}{600(200\,000)} = 0.300\ \text{mm}\ (\text{lengthens} \Rightarrow D\ \text{moves down})$$

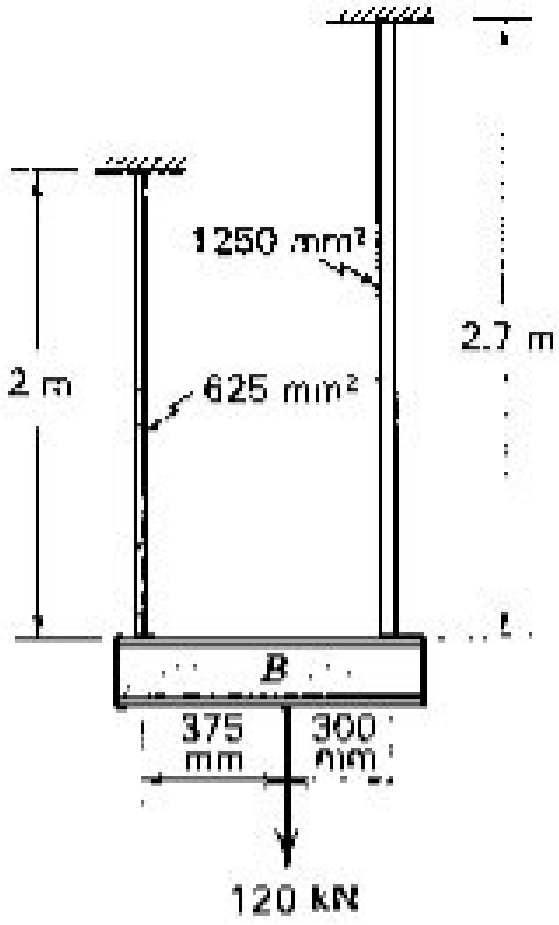
4. *Rigid-bar geometry.* B goes up 0.514 mm and D down 0.300 mm, so the bar pivots about a point H between them: $\frac{x_H}{0.514} = \frac{200 - x_H}{0.300}$, giving $x_H = 126.3$ mm from B.
5. *Point E* (600 mm from B): $\delta_E = 0.300 \cdot \frac{600 - 126.3}{200 - 126.3} = 0.300(6.43) = 1.93$ mm.

$$\delta_B = 0.514\ \text{mm}\uparrow,\ \delta_D = 0.300\ \text{mm}\downarrow,\ \delta_E = 1.93\ \text{mm}\downarrow$$

Check. The slope of the bar is $(0.514 + 0.300)/200 = 4.07 \times 10^{-3}$ rad. From D to E (400 mm): $0.300 + 400\left(4.07 \times 10^{-3}\right) = 1.93$ ✓.

Exam takeaway. Points beyond the supports amplify small link deformations. **Variation:** “the maximum allowable deflection at E is 1 mm; find the maximum load”.

Q10 (Crandall Prob 2.17) — Stiff member on two steel rods of different size (medium)



Prob. 2.17

Original: Crandall Prob 2.17 (textbook figure)

Problem. A very stiff horizontal member hangs from two vertical steel rods. The left rod is 2 m long with area 625 mm^2; the right rod is 2.7 m long with area 1250 mm^2. A vertical load of 120 kN acts at B, 375 mm from the left rod and 300 mm from the right rod. Estimate the vertical deflection of B (E = 200 GPa).

Approach. Determinate: statics gives the rod forces. Then each rod's stretch, and linear interpolation along the rigid member.

FBD in words. Rigid member with F_L ↑ at the left rod, F_R ↑ at the right rod and 120 kN ↓ at B.

Solution.

1. *Rod forces* (moments about each rod): $F_L = 120\,\frac{300}{675} = 53.33$ kN and $F_R = 120\,\frac{375}{675} = 66.67$ kN.
2. *Stretches:*

$$\delta_L = \frac{53\,333\,(2000)}{625\,(200\,000)} = 0.853\text{ mm},\ \delta_R = \frac{66\,667\,(2700)}{1250\,(200\,000)} = 0.720\text{ mm}$$

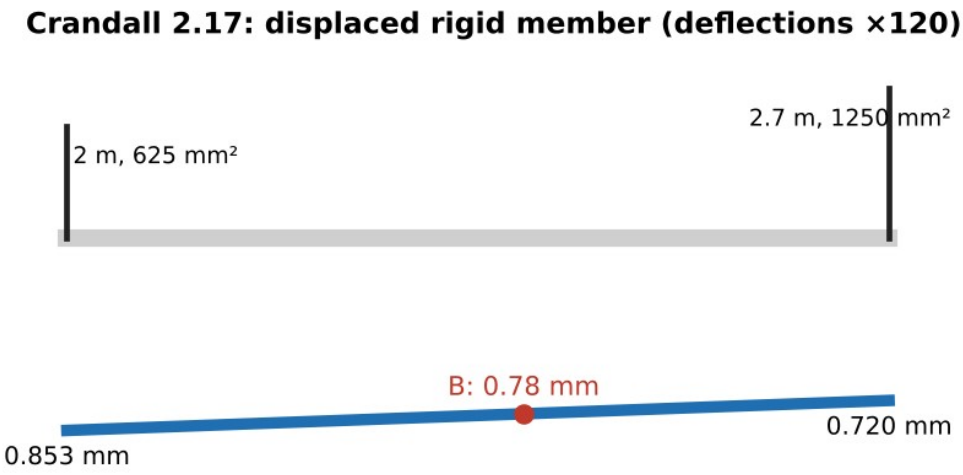
3. *Interpolate to B* (375 mm from the left rod, 675 mm span):

$$\delta_B = 0.853 + (0.720 - 0.853)\,\frac{375}{675} = 0.853 - 0.074.$$

$$\delta_B \approx 0.78\text{ mm}\downarrow$$

Check. The answer lies between the two rod stretches ✓. The member tilts down toward the left.

Exam takeaway. "Very stiff member" means rigid: deflections vary linearly between the rods. **Variation:** "where must the load go so that the member stays level?" The rod stiffnesses are $k_L = AE/L = 62\,500$ N/mm and $k_R = 92\,590$ N/mm. Equal stretches need $F_L/F_R = k_L/k_R = 0.675$, so the load must sit at the "centre of stiffness": $675\,k_R/(k_L + k_R) = 403$ mm from the left rod.



Displaced rigid member (deflections exaggerated)

PYQ on this concept

Q11 (PYQ 2014-15 Compre A Q8) — Stress ratio: solid vs hollow bar (easy)

8 A solid circular bar of diameter D carries an axial load W. If the same load is applied axially on a hollow circular bar of outer diameter D and inner diameter $D/2$, the ratio of stresses in a solid bar to that of hollow bar would be

Original question: PYQ 2014-15 Compre A Q8

Problem. A solid bar of diameter D carries an axial load W. The same load is applied to a hollow bar of outer diameter D and inner diameter D/2. Find the ratio of the stress in the solid bar to that in the hollow bar.

Solution.

1. $A_{solid} = \pi/4\,D^2$ and $A_{hollow} = \pi/4\,(D^2 - D^2/4) = 3/4 \cdot \pi/4\,D^2$.
2. With the same load: $\dfrac{\sigma_s}{\sigma_h} = \dfrac{A_h}{A_s}$.

$$\frac{\sigma_s}{\sigma_h} = 0.75$$

Exam takeaway. A 1-mark "trap": the hollow bar has the higher stress. Ratios invert relative to areas.

Concept 2.3 — Statically indeterminate axial problems

A. Concept explanation

1. Intuition. Wedge a bar tightly between two walls and load it in the middle. Both walls push back, but statics gives one equation for two reactions. The extra fact you need is that **the bar's total length can't change**, because the walls don't move. More generally, whenever members are tied together, their deformations must fit, and that supplies the missing equations.

2. Why it matters. Most real structures are indeterminate: bolted joints, reinforced-concrete columns, composite rods, rings shrunk onto each other, fixed–fixed shafts (Ch. 6), thermal stresses (Ch. 5), beam-plus-column PYQs. "Stiffer attracts more load" lets you sanity-check every answer.

3. Formal treatment.

Indeterminate (slides): the internal forces and reactions cannot be found from statics alone.

Method of superposition (slides, "redundant force method"):

1. Choose a redundant reaction (say R_B) and **release** that support.
2. Find the displacement at B due to the applied loads alone, δ_L.
3. Find the displacement at B due to R_B alone, δ_R (as a function of R_B).
4. Compatibility: $\delta = \delta_L + \delta_R = 0$ (or equal to a known gap or settlement).
5. Solve for R_B, then get the remaining reactions from statics.

This works because the material and geometry are **linear**, so the effects of separate loads add.

Direct (stiffness) method: when several members join at a rigid plate or pin, write each member's force in terms of a common displacement, as in the piston of Q1. This is often quicker for composite members and frameworks.

4. Derivations.

(a) Uniform bar fixed at both ends, load P at distance a from A (b = L − a). Release B. Under P the segment a stretches and b carries nothing, so $\delta_L = P\,a/A\,E$ (B moves down). R_B alone, pushing up, compresses the whole bar: $\delta_R = -R_B\,L/A\,E$. Compatibility: $P\,a - R_B\,L = 0$, giving

$$R_B = \frac{P\,a}{L},\ R_A = \frac{P\,b}{L}$$

This is the same split as a simply supported beam: the support **nearer** the load carries more.

(b) Composite (parallel) members between rigid plates. Members 1 and 2 must shorten equally, $\delta_1 = \delta_2$, so $P_i = \frac{(A\,E)_i}{\sum (A\,E)} P$ and stresses $\sigma_i = E_i \epsilon$, where $\epsilon = P/\sum (A\,E)$. All members have **the same strain**, and the stiffer material has the larger stress.

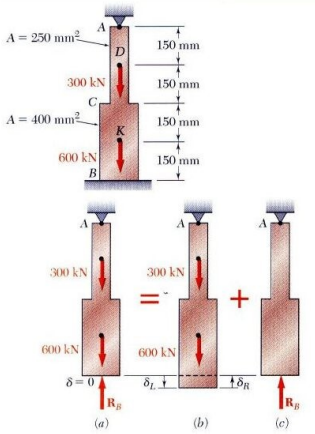
(c) Three-bar framework (Crandall Prob 2.12). An inner vertical bar (area A, length L) and two outer bars (area A_0) at angle θ to the vertical meet at a joint carrying F. The joint moves down by δ. The inner bar stretches δ; each outer bar stretches $\delta\cos\theta$ over length $L/\cos\theta$. Then

$$N_i = \frac{A\,E\,\delta}{L},\ N_o = \frac{A_0\,E\,\delta\cos^2\theta}{L},\ N_i + 2\,N_o\cos\theta = F$$

$$\Rightarrow \delta = \frac{F\,L}{E\left(A + 2\,A_0\cos^3\theta\right)},\ N_i = \frac{F\,A}{A + 2\,A_0\cos^3\theta},\ N_o = \frac{F\,A_0\cos^2\theta}{A + 2\,A_0\cos^3\theta}$$

5. Picture. Use a "release diagram" with three sketches, as on the slide: (a) the actual structure = (b) the structure with the redundant support released, under the loads + (c) the same released structure under the redundant force alone. The displacement at the released point must add up to the real value (usually zero).

Static Indeterminacy



- Structures for which internal forces and reactions cannot be determined from statics alone are said to be *statically indeterminate*.
- Redundant reactions (supports) are replaced with unknown loads which along with the other loads must produce compatible deformations.
- Deformations due to actual loads and redundant reactions are determined separately and then added or *superposed* (method of superposition)

$$\delta = \delta_L + \delta_R = 0$$

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Original: Slides Ch 2, slide 13

6. Special cases.

- Different areas in different segments: use $\delta = \sum P\,L / A\,E$ in each step.
- A **gap** Δ before contact: compatibility becomes $\delta_L + \delta_R = \Delta$ once the gap closes. If the free deflection is less than Δ, the second support carries nothing.
- Pre-tension or misfit (bolt turned by a known amount, Q15): compatibility is "deformation of bolt + deformation of sleeve = misfit".

7. Common traps.

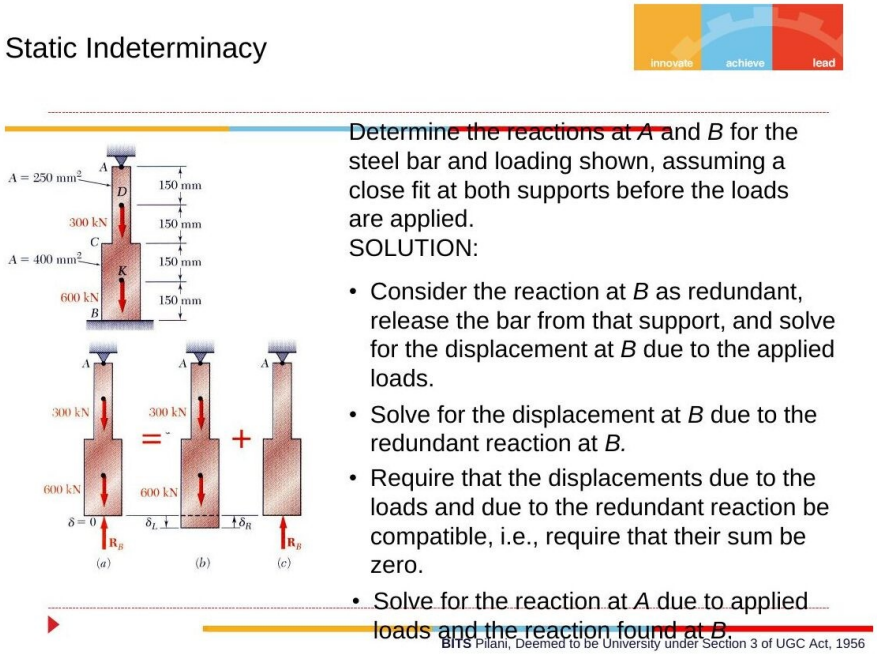
- Sign of δ_R: it acts against δ_L.
- Forgetting that segments beyond the last load carry no force in the released structure.
- In composite members, assuming equal **stress** instead of equal **strain**.
- Using statics alone and "assuming" a share of the load.

8. Summary box

$$\delta_L + \delta_R = 0\,(\text{or gap});\ \text{parallel members: } P_i = \frac{(A\,E)_i}{\sum (A\,E)}\,P,\ \epsilon \text{ common};\ \text{series: same } P,\ \delta\text{'s add}$$

B. Solved questions

Q12 (Slides Ch2, Beer Example 2.04) — Steel bar fixed at both ends (medium)



Original: Slides Ch 2, slide 14

Problem. A vertical steel bar is fixed at A (top) and B (bottom), with a close fit before loading. From the top there are four 150 mm segments: AD and DC with area 250 mm², CK and KB with area 400 mm². Loads: 300 kN down at D and 600 kN down at K. E = 200 GPa. Find the reactions at A and B.

Given: as above. **Find:** R_A, R_B.

Approach. Two unknown reactions, one equilibrium equation: indeterminate. Release B (the redundant) and use superposition.

FBD in words. Bar with R_A ↑ at A, R_B ↑ at B, 300 kN ↓ at D and 600 kN ↓ at K.

Solution.

1. *Released bar, loads only.* Internal forces (tension +): AD = 900, DC = 600, CK = 600, KB = 0 kN.

$$\delta_L = \frac{1}{E}\left[\frac{900\,000(150)}{250} + \frac{600\,000(150)}{250} + \frac{600\,000(150)}{400} + 0\right] = \frac{540\,000 + 360\,000 + 225\,000}{E} = \frac{1.125\times10^{6}}{E}$$ (B moves do

2. *Released bar, R_B alone (↑):* compresses the whole bar.

$$\delta_R = -\frac{R_B}{E}\left[\frac{300}{250} + \frac{300}{400}\right] = -\frac{1.95\,R_B}{E}$$

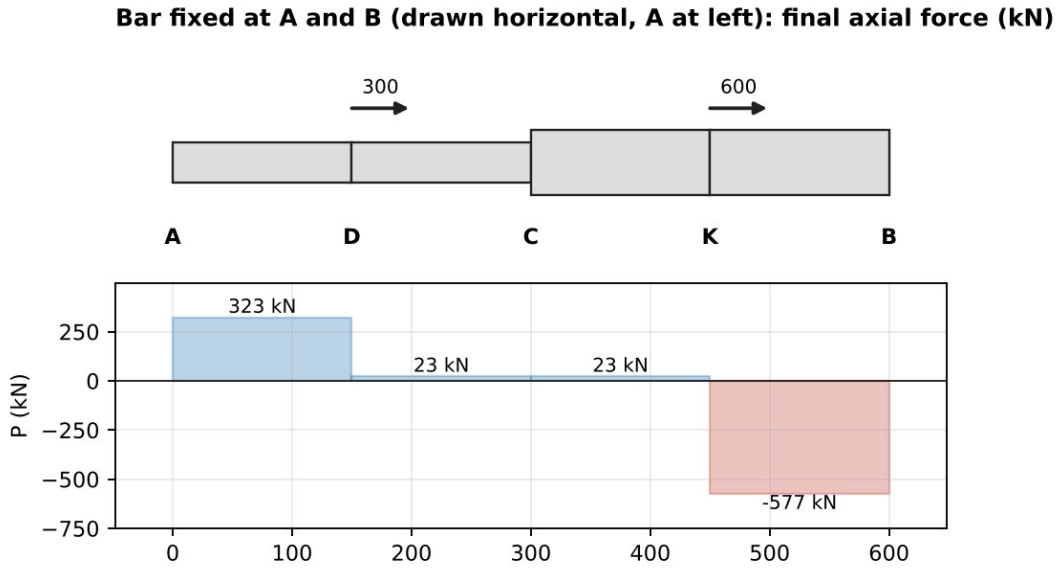
3. *Compatibility:* $\delta_L + \delta_R = 0$, so $1.125\times10^{6} = 1.95\,R_B$ and $R_B = 577$ kN.

4. *Statics:* $R_A = 900 - 577 = 323$ kN.

$$R_B = 577\text{ kN}\uparrow,\ R_A = 323\text{ kN}\uparrow$$

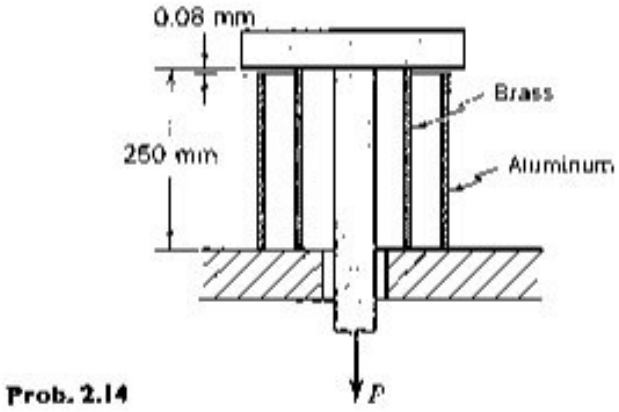
Check. The lower half is stiffer (larger area) and closer to the larger load, so $R_B > R_A$ ✓. These match Beer's answers ✓. Final forces: AD = +323, DC = CK = +23, KB = −577 kN; their $\sum P\,L/A\,E$ is 0 ✓.

Exam takeaway. Write the released displacement as a sum over segments. The slide's three-picture superposition is exactly what examiners want to see. **Variation:** a 4.5 mm gap at B. Then $\delta_L+\delta_R=4.5$ mm before contact matters, and you should check whether the gap closes at all.



Final axial forces: upper part in tension, lower part in compression

Q13 (Crandall Prob 2.14) — A spring with a "kink": two cylinders of different length (medium)



Original: Crandall Prob 2.14 (textbook figure)

Problem. A very stiff spring with a kink in its load–deflection curve is made from a brass cylinder (diameter 150 mm, wall 6.25 mm) and an aluminium cylinder (diameter 250 mm, wall 6.25 mm), both 250 mm long. The aluminium cylinder is 0.08 mm shorter than the brass one. A rigid plate loads both. Sketch the load–deflection relation. Take $E_{brass}=100$ GPa and $E_{Al}=70$ GPa.

Given: thin walls ($A \approx \pi dt$); L = 250 mm; gap 0.08 mm. **Find:** P–δ curve and the kink point.

Approach. Before the gap closes only the brass carries load (a single spring). After that both act in parallel.

FBD in words. Rigid top plate with P ↓; brass reaction ↑ always; aluminium reaction ↑ only after the plate has moved 0.08 mm.

Solution.

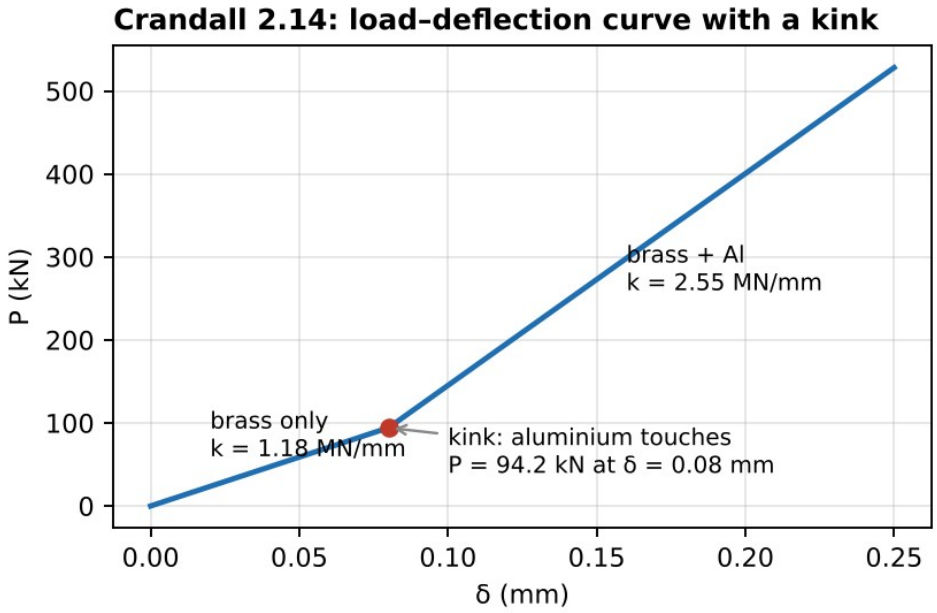
1. *Areas:* $A_b=\pi(150)(6.25)=2945$ mm² and $A_{Al}=\pi(250)(6.25)=4909$ mm².
2. *Stiffnesses:* $k_b=\dfrac{2945(100\,000)}{250}=1.178\times10^6$ N/mm and $k_{Al}=\dfrac{4909(70\,000)}{250}=1.374\times10^6$ N/mm.
3. *Stage 1* ($\delta\le0.08$ mm): $P=k_b\delta$. At the kink, $P_k=1.178\times10^6(0.08)=94.2$ kN.

4. *Stage 2* ($\delta > 0.08$ mm): $P = 94.2\,\text{kN} + (k_b + k_{Al})(\delta - 0.08)$, with slope 2.55×10^6 N/mm.

$$P = 1.18 \times 10^6\,\delta \;(\delta \le 0.08\,\text{mm});\quad P = 94.2\,\text{kN} + 2.55 \times 10^6 (\delta - 0.08)\;(\delta > 0.08)\;[\text{N, mm}]$$

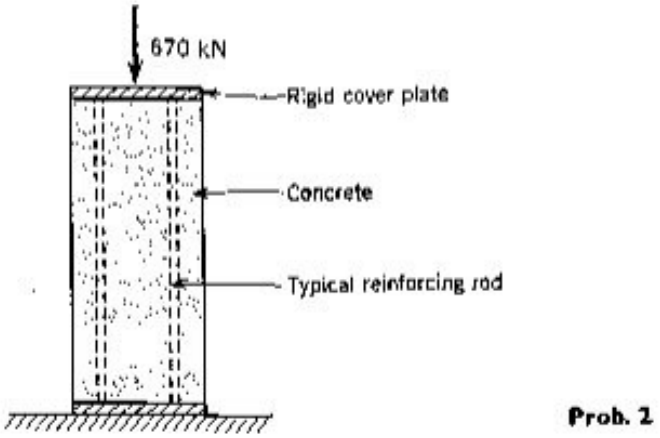
Check. The slope more than doubles after the kink, because aluminium has a larger area even though its E is lower ✓.

Exam takeaway. Gaps make the response piecewise linear, so always ask "does the gap close?" **Variation:** given a load of 300 kN, find each cylinder's force. $\delta = 0.08 + 205.8 \times 10^3 / 2.55 \times 10^6 = 0.161$ mm; brass carries $1.178 \times 10^6 \times 0.161 = 190$ kN and aluminium 110 kN.



Load–deflection curve

Q14 (Crandall Prob 2.11) — Reinforced-concrete pier (medium)



Original: Crandall Prob 2.11 (textbook figure)

Problem. A square reinforced-concrete pier, 0.3 × 0.3 m in cross-section and 1.2 m high, carries 670 kN through a rigid cover plate. It is strengthened by eight vertical 25 × 25 mm square steel bars placed symmetrically. Find the stress in the steel and in the concrete, and the shortening. Take $E_c = 17$ GPa and $E_s = 200$ GPa.

Given: $A_s = 8(625) = 5000$ mm²; $A_c = 90\,000 - 5000 = 85\,000$ mm²; L = 1200 mm; P = 670 kN. **Find:** σ_s, σ_c, δ.

Approach. The rigid cover plate makes steel and concrete shorten equally (compatibility: equal strain). Equilibrium: $P_s + P_c = P$.

FBD in words. Cover plate with 670 kN ↓ on top and P_s ↑ (steel) and P_c ↑ (concrete) from below.

Solution.

1. *Equal strain:*
$$\epsilon = \frac{P}{A_s E_s + A_c E_c} = \frac{670\,000}{5000(200\,000) + 85\,000(17\,000)} = \frac{670\,000}{(1.000 + 1.445) \times 10^9} = 2.740 \times 10^{-4}.$$
2. *Stresses:* $\sigma_s = E_s \epsilon = 54.8$ MPa and $\sigma_c = E_c \epsilon = 4.66$ MPa (both compressive).
3. *Shortening:* $\delta = \epsilon L = 2.74 \times 10^{-4}(1200) = 0.329$ mm.

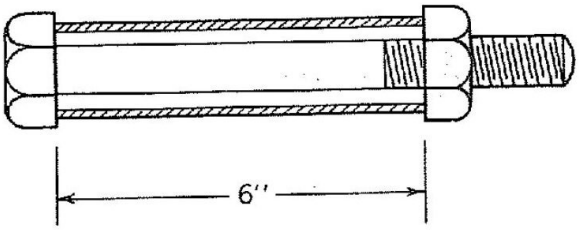
$$\sigma_s = 54.8\,\text{MPa (C)}, \sigma_c = 4.66\,\text{MPa (C)}, \delta = 0.33\,\text{mm}$$

Check. The steel is only 5.6% of the area but carries $54.8(5000)/670\,000 = 41\,\%$ of the load, because it is 11.8× stiffer ✓. $\sigma_s/\sigma_c = E_s/E_c = 11.8$ ✓.

Exam takeaway. Composite parallel members share **strain**, not stress. **Variation:** "the concrete may carry at most 4 MPa", giving $P_{max} = 4/E_c \times \sum A E = 575$ kN.

Q15 (Slides Ch2 = Crandall Prob 2.24) — Bolt and sleeve: a quarter turn of the nut (medium)

Problem 2.24



A bolt is threaded through a tubular sleeve, and the nut is turned up just tight by hand. Using wrenches, the nut is then turned further, the bolt being put in tension and the sleeve in compression. If the bolt has 16 threads per inch, and the nut is given an extra quarter turn (90⁰) by the wrenches, estimate the tensile force in the bolt if both the bolt and sleeve are of steel and the cross-sectional areas are
Bolt area: 1 inch² sleeve area: 0.6 inch², E=30X10⁶ lb/in²

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Original: Slides Ch 2, slide 31

Problem. A bolt passes through a tubular sleeve, and the nut is turned up just tight by hand. The nut is then given an extra quarter turn. The bolt has 16 threads per inch. Both parts are steel ($E = 30 \times 10^6$ psi), the bolt area is 1 in², the sleeve area 0.6 in², and the length 6 in. Estimate the tensile force in the bolt.

Given: misfit Δ = pitch × turns = $\frac{1}{16} \times \frac{1}{4} = \frac{1}{64}$ in; L = 6 in. **Find:** F.

Approach. Bolt (tension) and sleeve (compression) carry the same force: series springs. Compatibility says the misfit is taken up by the bolt stretching plus the sleeve shortening.

FBD in words. Cut the bolt and sleeve between the head and the nut. The bolt carries F (T) and the sleeve F (C). The equilibrium of the head gives the same F in both.

Solution.

1. *Equilibrium:* $F_{bolt} = F_{sleeve} = F$.
2. *Compatibility:* $\delta_b + \delta_s = \Delta = 1/64$ in.

3. *Force–deformation:* $\dfrac{F\,L}{A_b E}+\dfrac{F\,L}{A_s E}=\Delta$.
4. Solve:

$$F=\frac{\Delta E}{L\left(\dfrac{1}{A_b}+\dfrac{1}{A_s}\right)}=\frac{(1/64)\left(30\times10^{6}\right)}{6(1+1.667)}=\frac{468\,750}{16.0}$$

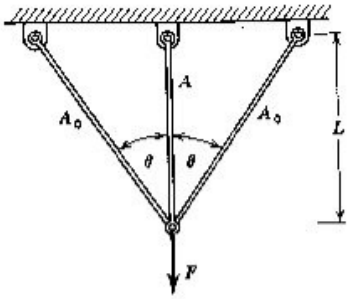
$$F\approx 29\,300\ \text{lb}$$

Check. Bolt stress 29.3 ksi; sleeve stress $29\,300/0.6=48.8$ ksi. Both are high but plausible for steel, and the thinner sleeve is more highly stressed ✓.

Exam takeaway. Misfit problems are series springs: same force, deformations add to the misfit. **Variation:** sleeve of aluminium ($E=10\times10^{6}$ psi), giving $F=\Delta/\left(L/A_b E_b+L/A_s E_s\right)=0.0156/\left(2\times10^{-7}+1\times10^{-6}\right)=13\,000$ lb.

Q16 (Crandall Prob 2.12) — Three-bar pin-connected framework (medium–hard)

2.12. Consider the pin-connected framework loaded as shown in the figure. Find the axial force in each bar. The two outer bars are identical with cross-sectional area A_0; the inner bar has a cross-sectional area A. All bars have the same modulus of elasticity E.



Prob. 2.12

Original: Crandall Prob 2.12 (textbook figure)

Problem. A vertical inner bar of area A and two outer bars of area A_0 (each at angle θ to the vertical) hang from a ceiling and meet at a joint carrying a load F. The inner bar has length L. All bars have the same E. Find the axial force in each bar.

Approach. The joint has two equilibrium equations but three unknown bar forces (two equal by symmetry, so effectively two unknowns with one useful equation). Compatibility: the joint moves straight down by δ, by symmetry.

FBD in words. Joint with F ↓, N_i ↑ along the inner bar and N_o along each outer bar, up and outward at θ from the vertical.

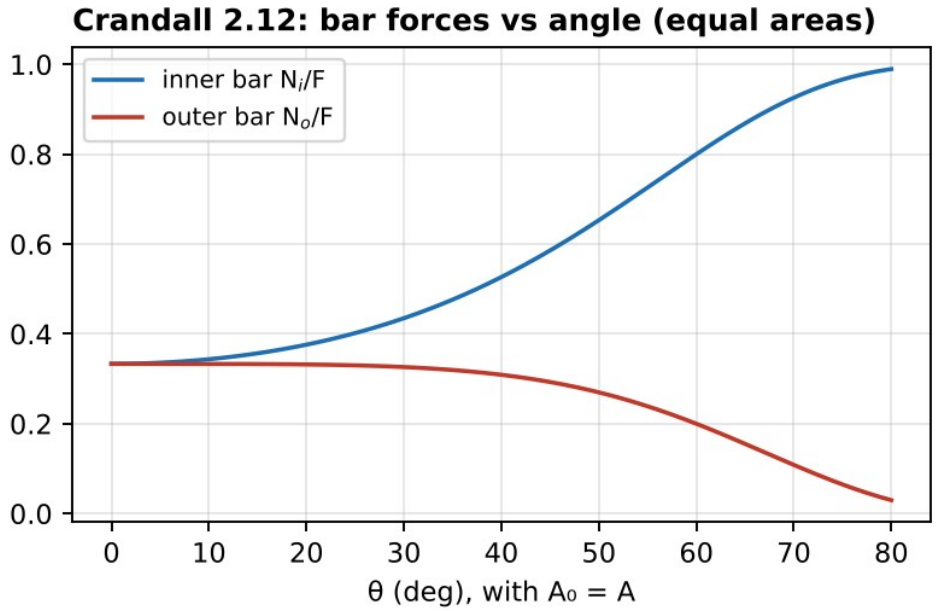
Solution.

1. *Equilibrium (vertical):* $N_i+2N_o\cos\theta=F$. Horizontal is satisfied by symmetry.
2. *Compatibility:* joint displacement δ ↓. Elongation of the inner bar is δ. Elongation of an outer bar is the component of δ along it, $\delta\cos\theta$.
3. *Force–deformation:* $N_i=\dfrac{A\,E\,\delta}{L}$ and $N_o=\dfrac{A_0 E\,\delta\cos\theta}{L/\cos\theta}=\dfrac{A_0 E\,\delta\cos^2\theta}{L}$.
4. *Substitute:* $\dfrac{E\,\delta}{L}\left(A+2A_0\cos^3\theta\right)=F$, so $\delta=\dfrac{F\,L}{E\left(A+2A_0\cos^3\theta\right)}$.

$$N_i = \frac{F\,A}{A + 2\,A_0 \cos^3\theta},\quad N_o = \frac{F\,A_0 \cos^2\theta}{A + 2\,A_0 \cos^3\theta}$$

Check. $\theta \to 0$ (all bars vertical and equal length): $N_i = F\,A/(A + 2\,A_0)$ and $N_o = F\,A_0/(A + 2\,A_0)$, ordinary parallel sharing ✓. $\theta \to 90°$: the outer bars carry no vertical load and $N_i = F$ ✓.

Exam takeaway. The inclined bar's elongation is $\delta \cos\theta$ (the projection of the joint displacement onto the bar). Students who use $\delta/\cos\theta$ get the wrong compatibility. **Variation:** A = A0 and θ = 30°: $\cos^3 30^\circ = 0.6495$, so $N_i = F/2.299 = 0.435\,F$ and $N_o = 0.435\,F \times 0.75 = 0.326\,F$.



How the load split changes with the angle of the outer bars

Q17 (Crandall Probs 2.31 and 2.32) — Composite hoop under internal pressure (medium–hard)

Problem. A brass hoop (internal radius 300 mm, thickness 3 mm) fits snugly inside a steel hoop (internal radius 303 mm, radial thickness 6 mm). Both are 6 mm wide. A radial pressure of 1.4 MPa acts inside the brass hoop. Estimate the tangential (hoop) forces in each hoop, and the radial expansion of the brass hoop. Take $E_b = 100$ GPa and $E_s = 200$ GPa.

Given: $p = 1.4$ MPa on inner radius $r_i = 300$ mm, width w = 6 mm; $A_b = 3 \times 6 = 18$ mm², $A_s = 6 \times 6 = 36$ mm²; mean radii $r_b = 301.5$ mm and $r_s = 306$ mm. **Find:** T_b, T_s, Δr_b.

Approach. Thin rings (Concept 2.6). Equilibrium of a half-ring gives the **total** hoop force. Compatibility: both rings expand by the same radial amount at the interface (they stay in contact). Each ring's radial expansion is $r\,\epsilon_\theta = r\,T/(A\,E)$.

FBD in words. Cut both hoops across a diameter. The pressure on the half-circle pushes up with $2\,p\,r_i\,w$; four hoop forces (T_b and T_s on each side) hold it down.

Solution.

1. *Equilibrium:* $2(T_b + T_s) = 2\,p\,r_i\,w$, so $T_b + T_s = 1.4(300)(6) = 2520$ N.
2. *Compatibility:* $\Delta r_b = \Delta r_s$, i.e. $\dfrac{r_b T_b}{A_b E_b} = \dfrac{r_s T_s}{A_s E_s}$.
3. So $\dfrac{T_s}{T_b} = \dfrac{r_b}{r_s} \cdot \dfrac{A_s E_s}{A_b E_b} = \dfrac{301.5}{306} \cdot \dfrac{36(200)}{18(100)} = 0.985(4) = 3.941.$

4. $T_b = \dfrac{2520}{1+3.941} = 510$ N and $T_s = 2010$ N.

5. *Radial expansion of the brass (2.32):* $\Delta r_b = \dfrac{r_b T_b}{A_b E_b} = \dfrac{301.5(510)}{18(100\,000)} = 0.0854$ mm.

$$T_b \approx 510\,\text{N},\ T_s \approx 2010\,\text{N},\ \Delta r_b \approx 0.085\,\text{mm}$$

Check. The steel hoop is 4× stiffer (twice the area, twice the E), so it carries about 4/5 of the load ✓. Stresses: brass 28.3 MPa, steel 55.8 MPa ✓.

Exam takeaway. Exactly the same "equal deformation, forces add" logic as the RC pier, applied around a circle. **Variation:** with a radial gap of 0.05 mm between the hoops, brass alone resists until it has expanded 0.05 mm.

Concept 2.4 — Displacement (Williot) diagrams for pin-jointed structures

A. Concept explanation

1. Intuition. Take two bars pinned together at B, with their other ends fixed to a wall. Load B and each bar changes length a little. B must end up where both bars, at their new lengths, can still reach it, i.e. at the intersection of two arcs. For **small** deformations each arc can be replaced by a straight line **perpendicular** to its bar. That is the Williot construction.

2. Why it matters. It gives joint deflections of trusses and links without energy methods. It is the independent check for Castigliano answers (Concept 2.5). PYQ 2014 Test-1 Q2 asks for exactly these deflections.

3. Formal treatment (small displacements). If a joint moves by $u = (u, v)$ and the other end of a bar is fixed, the bar's change in length is the component of u along the bar:

$$\delta_{bar} = u \cdot e \quad (e = \text{unit vector from the fixed end toward the joint})$$

If both ends move, use the relative displacement: $\delta = (u_j - u_i) \cdot e_{ij}$. Write one such equation per bar, then solve for the unknown joint displacements.

4. Derivation. The new length is

$$|L+u| = \sqrt{L^2 + 2L \cdot u + u^2} = L\sqrt{1 + \frac{2e \cdot u}{L} + \frac{u^2}{L^2}} \approx L + e \cdot u$$

using $\sqrt{1+x} \approx 1 + x/2$ and dropping u^2/L^2 (small displacements). So $\delta = e \cdot u$. ∎ A displacement **perpendicular** to the bar changes its length only to second order; that is why each arc becomes a perpendicular line.

5. Graphical construction. From the fixed end of each bar, lay off its δ along the bar (toward the joint for stretching, away for shortening). At the tip of each δ vector, draw a line perpendicular to the bar. The intersection of these perpendiculars is the displaced joint. Measure u and v.

Displacement diagram

So far we have discussed the procedure for finding the changes in length of single axially loaded members

When the structure contains 2 or more than 2 axially loaded members having pinned connections at their ends then we can use displacement diagrams for determining displacements.

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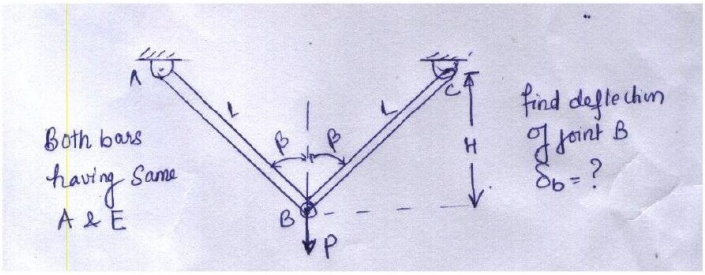

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6. Special cases.

- **Two-bar bracket** (AB horizontal; BC inclined at θ to AB; load P down at B). $F_{AB} = P\cot\theta$ (T) and $F_{BC} = P\csc\theta$ (C). Then $u_B = \delta_{AB}$ (away from the wall) and $v_B = \left(\delta_{BC} + u_B\cos\theta\right)/\sin\theta$ (downward), where δ_{BC} is the shortening of BC.
- **Symmetric V** (two equal bars at ±β from the vertical, load P down at the joint). Each bar force is $P/\left(2\cos\beta\right)$, each stretch is $\delta_b = \dfrac{P\,L}{2\,A\,E\cos\beta}$, and the joint moves $\delta_b/\cos\beta$:

$$\delta_{joint} = \frac{P\,L}{2\,A\,E\cos^2\beta}$$

This is the slides' "find the deflection of joint B" example.

7. Common traps.

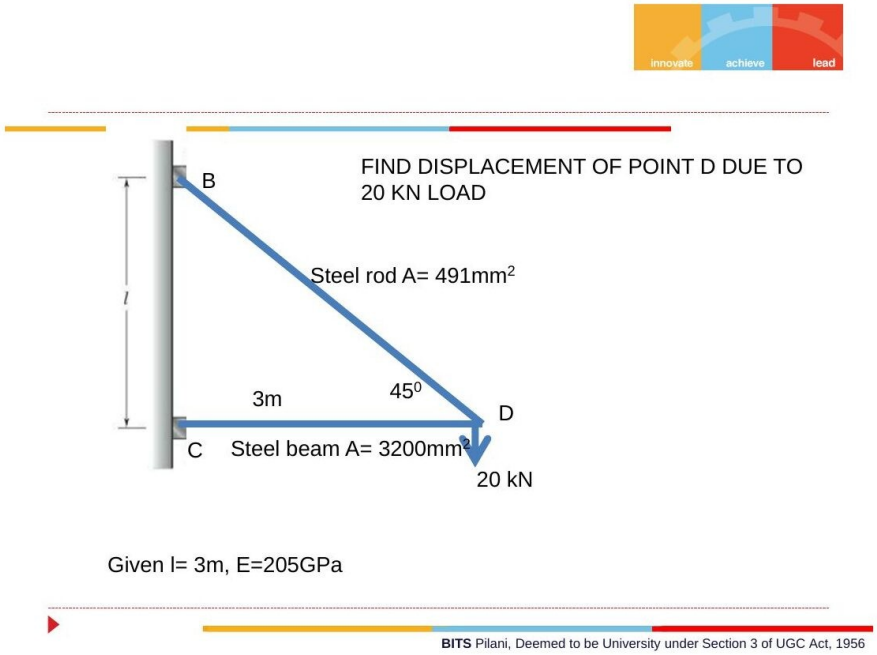
- Forgetting that a **shortening** bar pulls the joint toward its support.
- Adding bar elongations as if they were the joint displacement.
- Using large-angle trigonometry; all angles stay the undeformed ones.

8. Summary box

$$\delta_{bar} = u \cdot e \,(\text{small } u)\,;\ \text{two-bar: } u = \delta_{AB},\ v = \frac{\delta_{BC} + u\cos\theta}{\sin\theta}\,;\ \text{symmetric V: } \delta = \frac{P\,L}{2\,A\,E\cos^2\beta}$$

B. Solved questions

Q18 (Slides Ch2) — Displacement of D in a rod–beam bracket (medium)



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Problem. A horizontal steel beam CD (L = 3 m, A = 3200 mm²) is pinned to a wall at C. A steel rod BD (A = 491 mm²) runs from B on the wall, 3 m above C, to D at 45°. E = 205 GPa. 20 kN hangs at D. Find the displacement of D.

Given: $L_{BD} = 3\sqrt{2} = 4.243$ m. **Find:** u_D, v_D.

Approach. Statics at joint D for the bar forces, then δ = FL/AE, then the Williot conditions for each bar.

FBD in words. Joint D with 20 kN ↓, F_{BD} along DB (up-left at 45°) and F_{CD} along DC (−), both assumed tensile.

Solution.

1. *Joint D:* vertical: $F_{BD}\sin 45^\circ = 20$, so $F_{BD} = 28.28$ kN (T). Horizontal: $F_{CD} + F_{BD}\cos 45^\circ = 0$ (both point toward the wall), so $F_{CD} = -20$ kN (C).
2. *Length changes:*

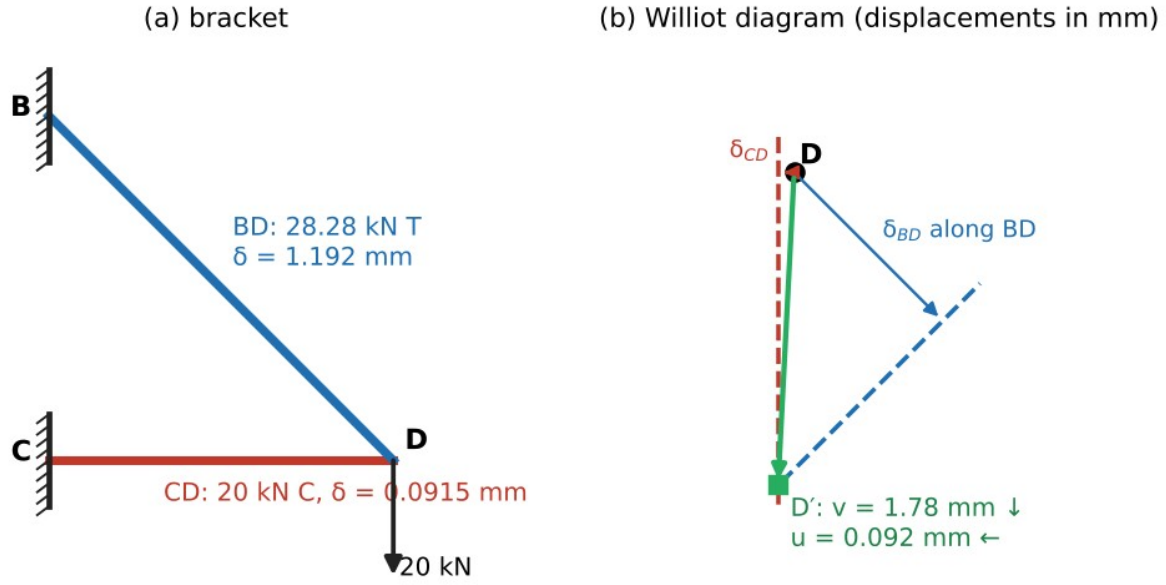
$$\delta_{BD} = \frac{28\,284\,(4243)}{491\,(205\,000)} = 1.192\,\text{mm}\,(\text{lengthens}),\ \delta_{CD} = \frac{20\,000\,(3000)}{3200\,(205\,000)} = 0.0915\,\text{mm}\,(\text{shortens})$$

3. *Compatibility* (u positive away from the wall, v positive downward):
   - CD (horizontal): $u_D = -0.0915$ mm (D moves toward the wall).
   - BD (unit vector from B to D, $(0.7071, +0.7071)$ in (u, v)): $0.7071\,u + 0.7071\,v = 1.192$, so $v = 1.686 + 0.0915 = 1.777$ mm.

$$v_D = 1.78\,\text{mm}\downarrow,\ u_D = 0.092\,\text{mm toward the wall},\ |\Delta_D| = 1.78\,\text{mm}$$

Check (Castigliano, Q22). The vertical deflection comes to 1.777 mm, the same ✓.

Exam takeaway. In a bracket the inclined tie's stretch dominates. **Variation:** "find the angle through which CD rotates", $1.777/3000 = 5.9 \times 10^{-4}$ rad.



Williot (displacement) diagram for joint D

Q19 (Slides Ch2 = Crandall Prob 2.6) — Stiff member on three bars: a zero-force member that still matters (hard)

Problem 2.6

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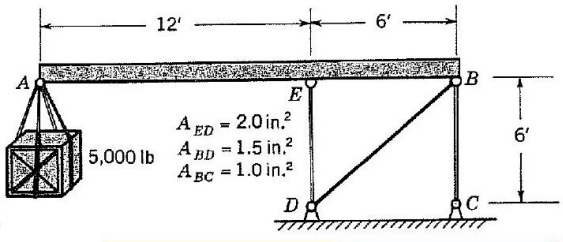
2.6. The stiff member *AB* is horizontal before the load of 5,000 lb is applied at *A*. The three steel bars *ED*, *BD*, and *BC* are fastened with pins at their ends. Find (*a*) the force in the bar *BD*, and (*b*) the horizontal and vertical movement of the point *A*.



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Original: Slides Ch 2, slide 27

Problem. A stiff member AB is horizontal before a 5000 lb load is applied at A. E is 12 ft from A and B is 18 ft from A. Three steel bars ($E = 30 \times 10^6$ psi) are pinned at their ends: ED vertical, 6 ft (2 in²); BC vertical, 6 ft (1 in²); BD diagonal from D up to B at 45° (1.5 in²). Find (a) the force in BD and (b) the horizontal and vertical movement of A.

Approach. Statics first. Then each bar's stretch, then compatibility at B and the rigid-body motion of AB.

FBD in words. Member AB with 5000 lb ↓ at A, F_{ED} at E (vertical), F_{BC} at B (vertical) and F_{BD} at B (along the 45° line).

Solution.

1. *(a) Horizontal equilibrium:* only BD has a horizontal component, so $F_{BD}=0$.
2. *Moments about B:* $5000(18)=F_{ED}(6)$, so $F_{ED}=15\,000$ lb. ED pushes **up** on AB, so it is in **compression**.
3. *Vertical sum:* $15\,000-5000-F_{BC}=0$, so BC pulls B **down** with 10 000 lb (**tension**).
4. *Deformations:* $\delta_{ED}=\dfrac{15\,000(72)}{2(30\times10^{6})}=0.018$ in (shortens, so E moves **down** 0.018 in).

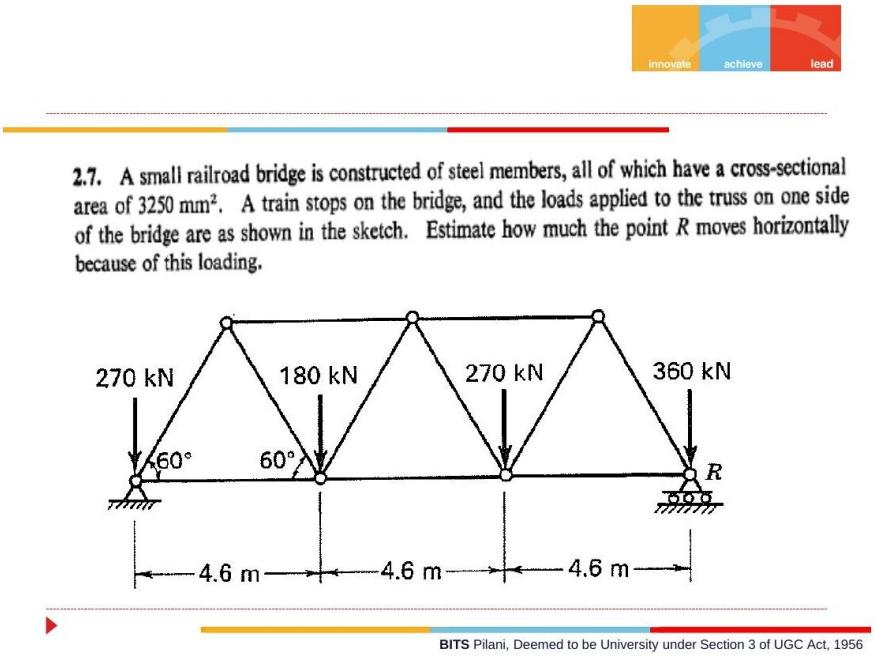
   $\delta_{BC}=\dfrac{10\,000(72)}{1(30\times10^{6})}=0.024$ in (lengthens, so B moves **up** 0.024 in). $\delta_{BD}=0$.
5. *Fit at B:* BD's length is unchanged, with unit vector from D to B $(0.707, 0.707)$: $0.707\,u_B+0.707(0.024)=0$, so $u_B=-0.024$ in (toward A, i.e. left).
6. *Rigid AB:* every point moves the same horizontal amount (small rotation), so $u_A=-0.024$ in. The vertical displacement varies linearly. Slope = $\dfrac{0.024-(-0.018)}{72\text{ in}}=5.83\times10^{-4}$, so $v_A=-0.018-144(5.83\times10^{-4})=-0.102$ in.

$$(a)\,F_{BD}=0;\ (b)\,A\text{ moves }0.024\text{ in left and }0.102\text{ in down}$$

Check. A is 12 ft beyond E on the side away from B, so it must drop more than E ✓.

Exam takeaway. A zero-force member can still **constrain** displacement. That is why Crandall asks for both parts. **Variation:** remove BD; then AB is a mechanism (it can slide horizontally).

Q20 (Slides Ch2 = Crandall Prob 2.7) — Railroad bridge: how far does the roller move? (hard)



Original: Slides Ch 2, slide 25

Problem. A small Warren-truss bridge (60° triangles, bottom panels 4.6 m, span 13.8 m) has every member of area 3250 mm², steel E = 200 GPa. A stopped train loads the bottom joints with 270 kN (at the pin A), 180 kN, 270 kN and 360 kN (at the roller R). Estimate how far R moves horizontally.

Approach. A is pinned and R rolls horizontally, so R's movement equals the total elongation of the bottom chord. Loads directly over supports go straight into the reactions and cause no member forces.

FBD in words. The truss with the two interior loads (180 and 270 kN) and the reactions they cause.

Solution.

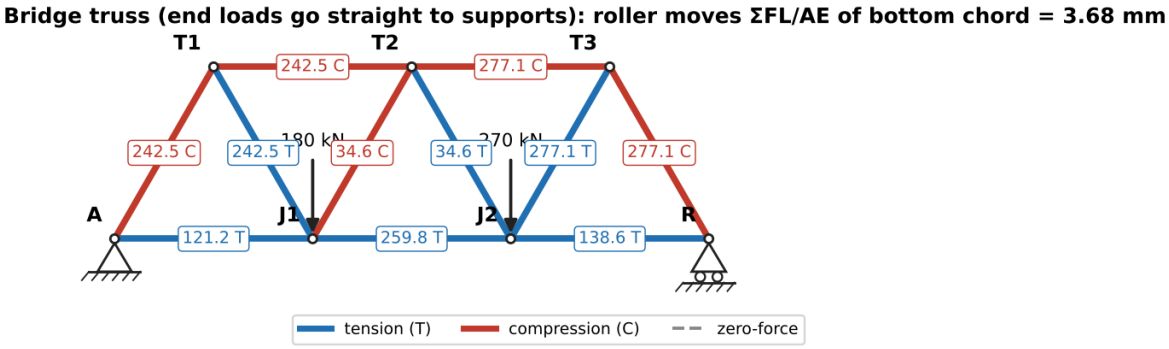
1. *Reactions from the interior loads only:* $R_R' = (180 \times 4.6 + 270 \times 9.2)/13.8 = 240$ kN and $R_A' = 210$ kN.
2. *Height:* $h = 2.3 \tan 60^\circ = 3.984$ m.
3. *Bottom-chord forces* (sections, moments about the top joints at x = 2.3, 6.9 and 11.5 m):
   - $F_1 = 210(2.3)/3.984 = 121.2$ kN (T)
   - $F_2 = [210(6.9) - 180(2.3)]/3.984 = 259.8$ kN (T)
   - $F_3 = 240(2.3)/3.984 = 138.6$ kN (T)
4. *Roller movement:*

$$\delta_R = \frac{(121.2 + 259.8 + 138.6) \times 10^3 (4600)}{3250 (200\,000)} = \frac{2.390 \times 10^9}{6.5 \times 10^8}$$

$$\delta_R \approx 3.68 \text{ mm (to the right)}$$

Check. A full computer solve of the truss gives the same bottom-chord forces ✓.

Exam takeaway. "How far does the roller move" is a sum of member elongations along one line. No energy method is needed. **Variation:** Crandall 2.8 asks for the horizontal movement of a top joint. Add the bottom-chord stretch up to that joint plus the effect of the diagonals (use a dummy load).



Member forces from the 180 kN and 270 kN loads

Concept 2.5 — Strain energy and Castigliano's theorem

A. Concept explanation

1. Intuition. Stretching a spring takes work, and that work is stored like charge in a battery. If you know how the stored energy depends on each load, its rate of change tells you how far that load's point moves. A large rate means a small increase in load stores a lot of extra energy, which means the point moves a long way.

2. Why it matters. It is the fastest route to deflections of trusses, curved members, frames and rings, and it works even where no load acts (using a fictitious load). PYQs: 2014 Test-1 Q2 (truss, 20 marks), 2019-20 A-Q3(b) (curved beam), 2024 B-Q1(c) (torsional energy).

3. Formal treatment (slides, Crandall §2.6).

- *Work of a slowly increasing load:* $dU = P\,dx$, so $U = \int_0^{x_1} P\,dx$, the area under the P–x curve.
- *Linear elastic:* $P = k\,x$ gives $U = 1/2\,k\,x_1^2 = 1/2\,P_1\,x_1$.
- *Axial bar:* $x_1 = P_1 L / A E$, so

$$U = \frac{P^2 L}{2AE}\left(\text{structure: } U = \sum \frac{F_i^2 L_i}{2 A_i E_i};\ \text{varying force: } U = \int \frac{N^2 dx}{2AE}\right)$$

- *Castigliano's theorem (slides):* "When forces act on elastic systems subject to small displacements, the displacement corresponding to any force, in the direction of the force, is equal to the partial derivative of the total strain energy with respect to that force."

$$\delta_i = \frac{\partial U}{\partial F_i}$$

- *Fictitious load (slides):* to find a deflection where no load acts, apply a fictitious force Q there in the desired direction, include it in U, take $\partial U / \partial Q$, and **only then** set Q = 0.

Castigliano's Theorem

When forces act on elastic systems subject to small displacements, the displacement corresponding to any force, in the direction of the force, is equal to the partial derivative of the total strain energy with respect to that force.

$$\delta_i = \frac{\partial U}{\partial F_i}$$

Where:
F_i = Force at i-th application point
δ_i = Displacement at i-th point in the direction of F_i
U = Total strain energy

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Original: Slides Ch 2, slide 22

4. Derivations.

(a) Why the theorem holds. In a linear system each displacement is a linear combination of the loads, $\delta_j = \sum_k c_{jk} P_k$, with $c_{jk} = c_{kj}$ (Maxwell's reciprocal theorem; Crandall Prob 2.13 asks you to verify this). The stored energy is

$$U = 1/2 \sum_j P_j \delta_j = 1/2 \sum_j \sum_k c_{jk} P_j P_k$$

Differentiating with respect to P_i and using symmetry gives $\frac{\partial U}{\partial P_i} = \sum_k c_{ik} P_k = \delta_i$. ∎

(b) Working formula for trusses. Differentiate before squaring, which saves algebra:

$$\delta_i = \sum_{\text{members}} F_j \frac{\partial F_j}{\partial P_i} \frac{L_j}{A_j E_j}$$

(c) Beams and curved members (bending energy; the bending formula comes later in the course):

$$U = \int \frac{M^2}{2EI} ds + \int \frac{N^2}{2AE} ds \Rightarrow \delta_i = \int \frac{M}{EI} \frac{\partial M}{\partial P_i} ds + \int \frac{N}{AE} \frac{\partial N}{\partial P_i} ds$$

(d) Energy of a bar including its own weight (Crandall Prob 2.45). With an end load F_0 and weight w per unit length, measuring x up from the free end, $N(x) = F_0 + w\,x$ and

$$U = \frac{1}{2AE}\int_0^L (F_0 + w\,x)^2\,dx = \frac{1}{2AE}\left(F_0^2 L + F_0\,w\,L^2 + \frac{w^2 L^3}{3}\right)$$

5. Picture. U is the triangle under the straight P–δ line, area ½Pδ. For a linear material the "complementary" triangle above the line has the same area. Castigliano's theorem is really about that complementary area, which is why it needs linearity.

6. Special cases.

- A member whose force doesn't depend on Q ($\partial F/\partial Q = 0$) contributes nothing to that deflection. You only need the members that feel Q.
- A rotation is found with a dummy couple: $\theta = \partial U/\partial C$.
- Indeterminate structures: choose a redundant R and impose $\partial U/\partial R = 0$ (zero displacement at a rigid support). This is Castigliano's second theorem, used for Crandall 2.42.

7. Common traps.

- Using $\partial U/\partial P$ when the same symbol P is applied at **two** points; that gives the *sum* of the two displacements. Give each load its own symbol.
- A negative result means the displacement is opposite to the direction assumed for the load.
- Setting Q = 0 **before** differentiating.
- Mixing units inside $\sum F\,f\,L/A\,E$ (keep N and mm throughout).

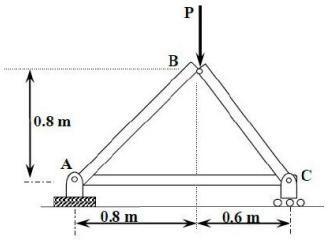
8. Summary box

Item	Formula
Bar energy	$U = P^2 L/2\,A\,E$
Castigliano	$\delta_i = \partial U/\partial F_i$
Truss form	$\delta = \sum F(\partial F/\partial P)L/A\,E$
Beam/curved form	$\delta = \int M(\partial M/\partial P)\,ds/E\,I + \int N(\partial N/\partial P)\,ds/A\,E$
Dummy load	include Q, differentiate, then set Q = 0

B. Solved questions

Q21 (Slides Ch2) — Truss: P needed to move roller C by 0.2 mm (medium)

The truss shown in Fig. is made of three steel members, each having a cross-sectional area of 400 mm^2. Applying the energy method (i.e.Castigliano method), determine the magnitude of P required to displace the roller (i.e. Point C) to the right by 0.2 mm. Take E = 200 GPa.



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Original: Slides Ch 2, slide 26

Problem. A truss of three steel members, each 400 mm^2, has a pin at A and a roller at C, with AC = 1.4 m. Apex B is 0.8 m above the base, 0.8 m horizontally from A (so AB = 1.131 m and BC = 1.0 m). P acts down at B. Using Castigliano's method, find P such that roller C moves 0.2 mm to the right. E = 200 GPa.

Approach. No load acts horizontally at C, so add a fictitious Q (→) there. Express every member force in terms of P and Q, then $\delta_C = \partial U / \partial Q \,¿_{Q=0}$.

FBD in words. Joint C with roller reaction C_y ↑, Q →, F_{CB} toward B and F_{CA} toward A (both assumed tensile).

Solution.

1. *Reactions:* $\sum M_A$: $C_y(1.4) = P(0.8)$, so $C_y = 0.5714\,P$ (Q passes through A's line and adds no moment).
2. *Joint C, vertical:* $0.8\,F_{CB} + 0.5714\,P = 0$, so $F_{CB} = -0.714\,P$ (C). Independent of Q.
3. *Joint C, horizontal:* $-0.6\,F_{CB} - F_{CA} + Q = 0$, so $F_{CA} = Q + 0.4286\,P$ (T).
4. *Joint A:* F_{AB} depends only on P ($F_{AB} = -0.606\,P$). Independent of Q.
5. *Castigliano:* only AC depends on Q, with $\partial F_{AC}/\partial Q = 1$:

$$\delta_C = \frac{F_{AC}(1)\,L_{AC}}{A\,E}\,¿_{Q=0} = \frac{0.4286\,P(1400)}{400(200\,000)} = 0.2\ \text{mm}$$

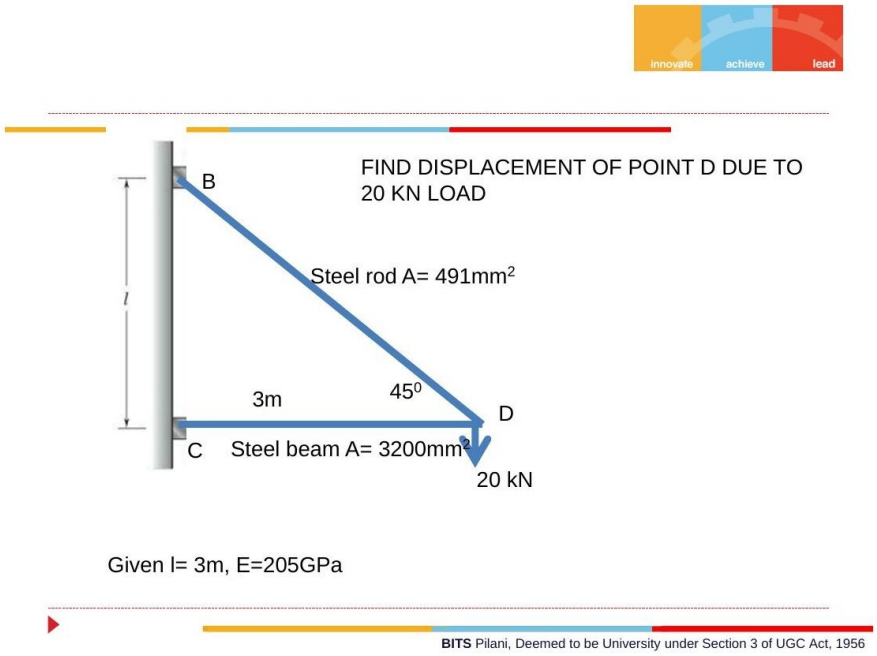
6. $$P = \frac{0.2(8\times10^{7})}{600} = 26\,670\ \text{N}.$$

$$P \approx 26.7\ \text{kN}$$

Check. A is pinned and C slides along AC, so its movement is just the elongation of AC:
$F_{AC}L/AE = 11\,430(1400)/(8\times10^{7}) = 0.2$ mm ✓.

Exam takeaway. Only the members whose force depends on the dummy load contribute. **Variation:** the vertical deflection of B, $\sum F(\partial F/\partial P)L/AE$ over all three members.

Q22 (Slides Ch2) — The rod–beam bracket again, by Castigliano (easy)



Original: Slides Ch 2, slide 24

Problem. Use Castigliano's theorem to find the vertical deflection of D in the bracket of Q18.

Solution.

1. Member forces in terms of P: $F_{BD} = \sqrt{2}\,P$ (T), $F_{CD} = -P$ (C).
2. $\delta_v = \sum F \dfrac{\partial F}{\partial P}\dfrac{L}{AE} = \dfrac{\sqrt{2}\,P(\sqrt{2})(4243)}{491(205\,000)} + \dfrac{(-P)(-1)(3000)}{3200(205\,000)}.$
3. With P = 20 000 N: $1.686 + 0.0915.$

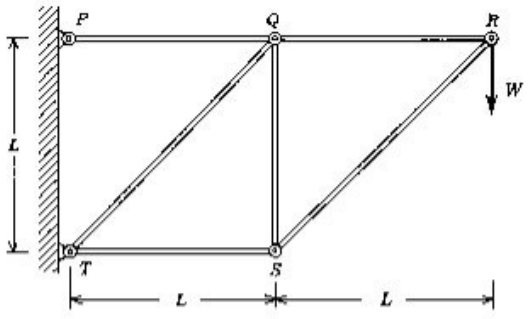
$$\delta_v = 1.78\ \text{mm} \downarrow$$

Check. Same as the Williot diagram in Q18 ✓.

Exam takeaway. Two independent methods, the same number. Use one as the answer and the other as a private check.

Q23 (Crandall Prob 2.10) — Tip deflection of the six-member cantilever truss (medium)

2.10. In the pin-jointed cantilever truss shown, all the members have a cross-sectional area A and elastic modulus E. Find:
(*a*) the forces in the rods due to the load W, distinguishing between tension and compression.
(*b*) the vertical deflection of the loaded joint.



Prob. 2.10

Original: Crandall Prob 2.10 (textbook figure)

Problem. For the pin-jointed cantilever truss of Ch. 1 Q25 (A, E on the wall; AB = BC = BD = L; load W at C), with every member of area A and modulus E, find (a) the member forces and (b) the vertical deflection of the loaded joint C.

Approach. (a) is Ch. 1 Q25. (b) Castigliano: every member force is proportional to W, so $\partial F/\partial W = F/W$.

Solution.

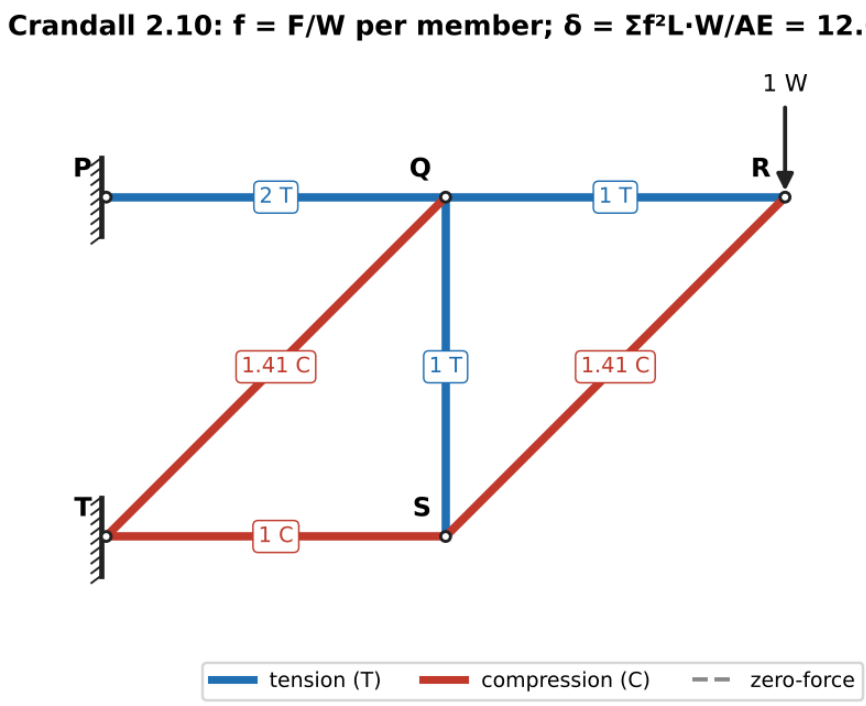
1. *(a) Forces:* AB = 2W (T), BC = W (T), BD = W (T), ED = W (C), EB = √2W (C), DC = √2W (C).
2. *Lengths:* AB, BC, BD and ED are L; EB and DC are √2L.
3. *(b)* $\delta_C = \sum F \dfrac{\partial F}{\partial W}\dfrac{L_i}{AE} = \dfrac{W}{AE}\sum f_i^2 L_i$, with $f_i = F_i/W$:

$$\sum f_i^2 L_i = (4+1+1+1)L + (2+2)\sqrt{2}L = (7+5.657)L = 12.66\,L$$

$$\delta_C = 12.66\,\frac{W\,L}{A\,E}\downarrow$$

Check. All terms are positive (squares), so the deflection is in the direction of W, as it must be ✓. The diagonals (compression) contribute 45% of the deflection.

Exam takeaway. For a single load, $\delta = \sum f^2 L/AE \times W$. Tabulate member, f, L, f²L. **Variation:** the horizontal deflection of C needs a dummy horizontal load. Only members whose force changes with it contribute.



Member forces per unit W used in the Castigliano sum

Q24 (Crandall Prob 2.45) — Strain energy of a bar with and without its own weight (medium)

Problem. A uniform bar of length L hangs vertically, carries an end load F_0 at its lower end and has weight w per unit length. Calculate the stored energy with and without the bar's weight.

Approach. The internal force varies along the bar, so integrate $N^2/2AE$.

Solution.

1. *Internal force* at distance x above the free end: $N(x) = F_0 + w\,x$.
2. *Energy with weight:*

$$U=\int_0^L \frac{(F_0+wx)^2}{2AE}dx=\frac{1}{2AE}\left[F_0^2L+F_0wL^2+\frac{w^2L^3}{3}\right]$$

3. *Without weight* (w = 0): $U_0=\frac{F_0^2L}{2AE}$.

$$U=\frac{F_0^2L}{2AE}+\frac{F_0wL^2}{2AE}+\frac{w^2L^3}{6AE}$$

Check. Castigliano: $\partial U/\partial F_0=\frac{F_0L}{AE}+\frac{wL^2}{2AE}$. That is the end deflection, which matches the end-load stretch plus the self-weight stretch $WL/2AE$ (Concept 2.2) ✓.

Exam takeaway. Energy is **not** additive in loads (the cross term $F_0wL^2/2AE$ appears), because U is quadratic. Only displacements superpose. **Variation:** a mine-hoist cable (Crandall 2.15) with wL comparable to F_0.

Q25 (PYQ 2014-15 Test-1 Q2) — Two-bar bracket: horizontal and vertical deflection by Castigliano only (medium, 20 marks)

Q.2 Using **Castigliano's theorem (only)**, determine vertical and horizontal defection of a loaded joint (*Fig. 2*). All members are made up of steel with modulus of elasticity equal to 200 GPa and each member has a cross sectional area equal to 800 mm². **[20]**
(*AB* = 6 m; *BC* = 7.5 m)

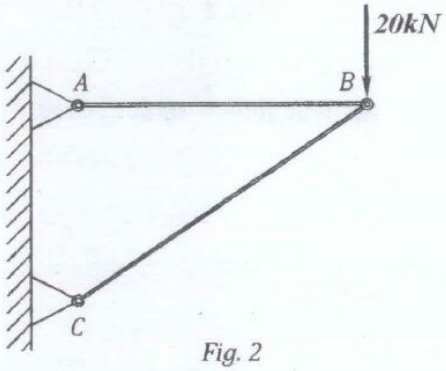


Fig. 2

Original question: PYQ 2014-15 Test-1 Q2

Problem. AB is horizontal, 6 m long, from the wall at A to B. BC runs from C on the wall (4.5 m below A) to B, 7.5 m long. 20 kN acts down at B. E = 200 GPa and A = 800 mm² for each member. Using **Castigliano's theorem only**, find the vertical and horizontal deflections of B.

Given: $\sin\theta=4.5/7.5=0.6$ and $\cos\theta=0.8$ for BC. **Find:** δ_v, δ_h.

Approach. Vertical deflection: differentiate with respect to the real load P. Horizontal deflection: add a fictitious Q → at B and differentiate with respect to Q, then set Q = 0.

FBD in words. Joint B with P ↓, Q →, F_{AB} toward A (←, tensile) and F_{BC} toward C (down-left, tensile).

Solution.

1. *Joint B, vertical:* $-F_{BC}(0.6)-P=0$, so $F_{BC}=-1.667\,P$ (C), independent of Q.
2. *Joint B, horizontal:* $-F_{AB}-F_{BC}(0.8)+Q=0$, so $F_{AB}=Q+1.333\,P$ (T).
3. *Vertical deflection* (Q = 0, P = 20 kN):

$$\delta_v = \frac{1}{AE}\left[F_{AB}\frac{\partial F_{AB}}{\partial P}L_{AB} + F_{BC}\frac{\partial F_{BC}}{\partial P}L_{BC}\right] = \frac{26\,667(1.333)(6000) + 33\,333(1.667)(7500)}{800(200\,000)}$$

$$\delta_v = \frac{2.133\times10^{8} + 4.167\times10^{8}}{1.6\times10^{8}} = 3.94\ \text{mm}$$

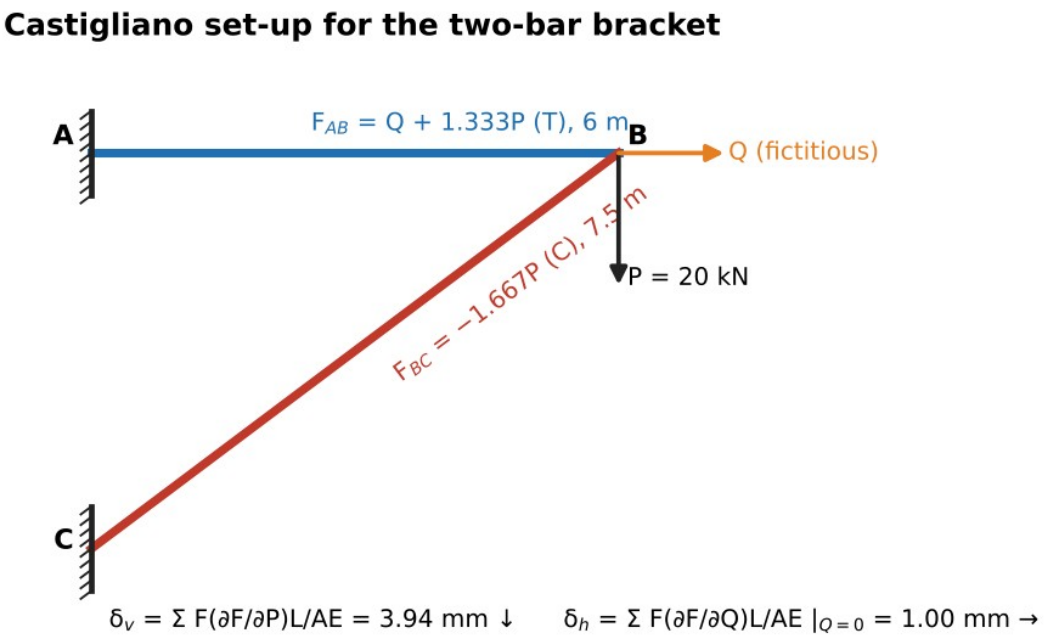
4. *Horizontal deflection:* only AB depends on Q ($\partial F_{AB}/\partial Q = 1$):

$$\delta_h = \frac{F_{AB}(1)L_{AB}}{AE} = \frac{26\,667(6000)}{1.6\times10^{8}} = 1.00\ \text{mm}$$

$$\delta_v = 3.94\ \text{mm}\downarrow,\ \delta_h = 1.00\ \text{mm}\rightarrow$$

Check (Williot, private). $u = \delta_{AB} = 1.00$ mm. BC shortens by $33\,333(7500)/1.6\times10^{8} = 1.5625$ mm. $v = (1.5625 + 1.00(0.8))/0.6 = 3.94$ mm ✓.

Exam takeaway. The paper said "Castigliano only", so a displacement diagram earns no marks there even though it gives the right number. **Variation:** load inclined at 30° to the vertical. Resolve it into P and Q components and use both derivatives.



Castigliano set-up with fictitious load Q

Q26 (PYQ 2019-20 Compre A Q3(b)) — Curved cantilever: vertical tip deflection by Castigliano (hard)

Q3. The curved cantilever beam in FigQ3 has a radius of curvature R and a circular cross section of diameter d. It is subjected to a point load P at point B.

(a) Determine the reactions (i.e. axial force, shear force and bending moment) acting at any section θ in the curved cantilever beam AB.

(b) Using Castigliano's method (the strain energy method), determine the vertical deflection of point B in terms of P, modulus of elasticity E, radius of curvature R, cross-section area A, and moment of inertia of the cross section I. **Neglect the effect of strain energy due to shear** **[8]**

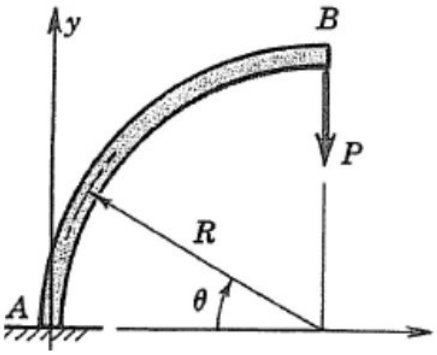


Fig Q3

Original question: PYQ 2019-20 Compre A Q3

Problem. A quarter-circle cantilever of radius R (circular cross-section) is fixed at A (θ = 0, at the base) and free at B (θ = 90°, at the top), with a vertical load P down at B. Using Castigliano's method, find the vertical deflection of B in terms of P, E, R, A and I. Neglect shear energy.

Approach. The internal forces at angle θ come from Ch. 3 Q29: $M = P R \cos\theta$, $N = P \cos\theta$ (compression) and $V = P \sin\theta$. Shear energy is neglected. Use $ds = R\,d\theta$.

Solution.

1. *Derivatives:* $\partial M/\partial P = R\cos\theta$ and $\partial N/\partial P = \cos\theta$.
2. *Castigliano:*

$$\delta_B = \int_0^{\pi/2} \frac{M}{EI}\frac{\partial M}{\partial P} R\,d\theta + \int_0^{\pi/2} \frac{N}{AE}\frac{\partial N}{\partial P} R\,d\theta = \frac{PR^3}{EI}\int_0^{\pi/2} \cos^2\theta\,d\theta + \frac{PR}{AE}\int_0^{\pi/2} \cos^2\theta\,d\theta$$

3. $\int_0^{\pi/2} \cos^2\theta\,d\theta = ¿$.

$$\delta_{B,v} = \frac{\pi P R^3}{4EI} + \frac{\pi P R}{4AE}$$

Check. Ratio of the axial to the bending term: $I/(AR^2) = (d/4R)^2$, tiny for a slender member. That is why it is often dropped, but the question asked for it in terms of A and I, so keep both ✓. Dimensions: PR^3/EI = N·m^3/(N/m^2·m^4) = m ✓.

Exam takeaway. Write M(θ), N(θ), their derivatives and $ds = R\,d\theta$ before integrating. Marks are given for each. **Variation:** horizontal deflection of B, using a dummy horizontal load Q: $M = PR\cos\theta + QR(1-\sin\theta)$, which gives $\delta_h = PR^3/(2EI)$ (bending only).

Concept 2.6 — Flexible members: thin rings and belts

A. Concept explanation

1. Intuition. Blow into a balloon or pump a bicycle tyre. The wall is thin, so it can't bend much; it carries the pressure by being stretched **around** the circumference, like a hoop of rope. A rope wrapped round a post is the opposite case: the post pushes radially on the rope, and friction makes the tension grow around the wrap.

2. Why it matters. Pipes, pressure vessels (Ch. 4 extends this to cylinders and spheres), shrink-fitted hoops (Q17), drum brakes, capstans, belts. PYQ 2014 Compre A Q11 asks for the hoop stress of a thin ring.

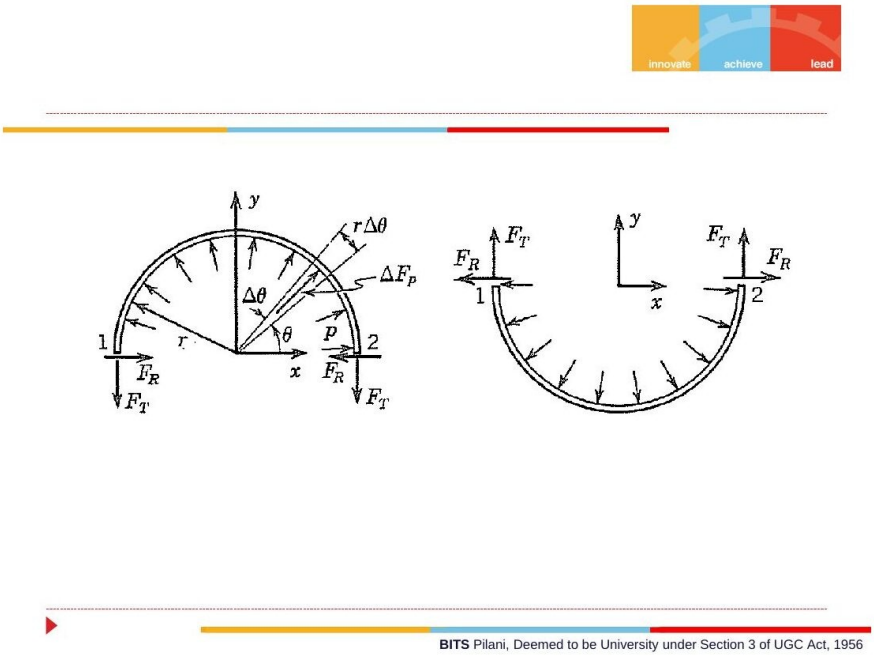
3. Formal treatment.

- *Thin ring* of mean radius r, width b and thickness t $\ll$ r, under internal pressure p (force per unit area). Hoop force $F_T = p\,b\,r$ and hoop stress $\sigma_\theta = \frac{p\,r}{t}$.
- *Radial expansion:* the hoop strain $\epsilon_\theta = \sigma_\theta / E$ stretches the circumference $2\pi r$ by $2\pi r\,\epsilon_\theta$, so the radius grows by $\delta_R = r\,\epsilon_\theta = \frac{p\,r^2}{E\,t}$.
- *Belt/rope on a drum:* $T_2 / T_1 = e^{\mu\beta}$ (derived in Ch. 1 Concept 1.4). The normal pressure per unit length of rope is T / r.

4. Derivation — hoop force (slides, Crandall example). Cut the ring along a diameter and take the upper half as a free body. The pressure on a small arc $r\,d\theta$ (width b) is $p\,b\,r\,d\theta$, acting radially. Its vertical component is $p\,b\,r \sin\theta\,d\theta$. Summing over the half:

$$2F_T = \int_0^{\pi} p\,b\,r \sin\theta\,d\theta = p\,b\,r$$

The shortcut: the net upward pressure force on a half-ring equals p times its **projected** area, $p(2r)b$.



Original: Slides Ch 2, slide 33

5. Picture. The deformed ring (slide figure) is a slightly bigger concentric circle. Each point moves radially out by δ_R, and there is no bending because the load is uniform.

6. Special cases.

- The answer depends only on r/t, not on the width b.
- Use the inner radius for a thin ring unless told otherwise; the mean radius differs by t/2r (about 1% here).
- For a rope, "four turns" means β = 8π, which amplifies tension enormously (Q29).

7. Common traps.

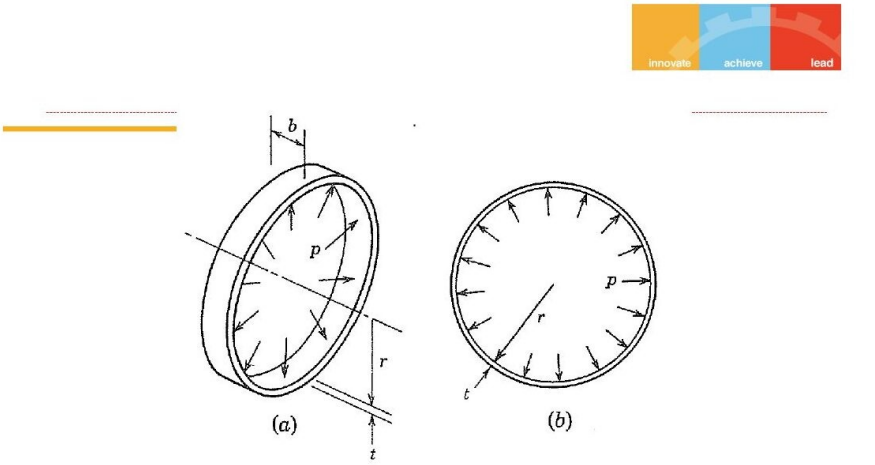
- Using the diameter instead of the radius in pr/t.
- Forgetting that the pressure acts over the **projected** area (2rb), not the curved area (πrb).
- Degrees in $e^{\mu\beta}$.

8. Summary box

$$\sigma_\theta = \frac{pr}{t}, F_T = pbr, \delta_R = \frac{pr^2}{Et}, \frac{T_2}{T_1} = e^{\mu\beta}$$

B. Solved questions

Q27 (Slides Ch2, Crandall example) — Thin ring under internal pressure (easy)



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Problem. Determine the forces in a thin ring (mean radius r, width b, thickness t) under internal pressure p, and its deformation.

Solution. From the derivation above:

$$F_T = pbr, \sigma_\theta = \frac{pr}{t}, \delta_R = \frac{pr^2}{Et}$$

Check. Doubling the pressure doubles everything ✓. A thinner wall gives larger stress and larger expansion ✓.

Exam takeaway. Remember the "projected area" shortcut: $2F_T = p(2r)b$.

Q28 (PYQ 2014-15 Compre A Q11) — Hoop stress in a thin ring (easy)

11 The hoop or circumferential stress in a thin ring of inner diameter 500 mm, width 10 mm and thickness 5 mm, when subjected to an internal pressure 1.5 MPa is equal to _____ MPa

Original question: PYQ 2014-15 Compre A Q11

Problem. A thin ring has inner diameter 500 mm, width 10 mm and thickness 5 mm, and carries an internal pressure of 1.5 MPa. Find the hoop stress.

Solution.

1. $r = 250$ mm (inner radius).
2. $\sigma_\theta = \dfrac{p\,r}{t} = \dfrac{1.5(250)}{5}$.

$$\sigma_\theta = 75\ \text{MPa}$$

Using the mean radius 252.5 mm gives 75.75 MPa; either is accepted at this level.

Exam takeaway. The width (10 mm) is a distractor.

Q29 (Crandall Prob 2.29) — Hawser wrapped four times round a capstan (easy)

Problem. A hawser from a ship is wrapped four times around a rotating capstan. A deckworker pulls on the free end with 200 N. With μ = 0.3 between capstan and hawser, what is the largest force the man can exert on the boat?

Given: $T_1 = 200$ N; β = 4 turns = 8π rad; μ = 0.3. **Find:** T_2.

Approach. Belt friction: the capstan rotates and the hawser slips on it, so friction is fully developed. Use the kinetic value, taken here as μ = 0.3.

Solution.

1. $\mu\beta = 0.3(8\pi) = 7.540$.
2. $T_2 = T_1 e^{\mu\beta} = 200\,e^{7.540} = 200(1881)$.

$$T_2 \approx 3.8 \times 10^5\ \text{N} = 376\ \text{kN}$$

Check. One turn alone gives $e^{0.6\pi} = 6.6$; four turns give $6.6^4 = 1881$ ✓. A person can hold a large ship ✓.

Exam takeaway. The exponential grows with every turn. **Variation:** "how many turns are needed to hold 50 kN with a 100 N pull?" $\beta = \ln(500)/0.3 = 20.7$ rad = 3.3 turns, so 4 turns.

Chapter 2 — Formula and convention sheet

Topic	Formula / rule
Three steps	Equilibrium → geometric fit → force–deformation
Springs	parallel: $k = \sum k_i$, same δ; series: $1/k = \sum 1/k_i$, same F
Rigid bar on springs	δ at a spring = θ × distance; stiffness at a point ∝ $1/c^2$
Axial deformation	$\delta = \sum P_i L_i/(A_i E_i)$, P_i internal, T +
Bar stiffness	$k = A\,E/L$; self-weight $\delta = W\,L/2\,A\,E$
Composite (parallel) member	equal strain: $\epsilon = P/\sum(A\,E)$, $\sigma_i = E_i\,\epsilon$
Misfit (bolt + sleeve)	same F; $\delta_b + \delta_s = \Delta$
Indeterminate (superposition)	release the redundant: $\delta_L + \delta_R = 0$ (or gap)
Fixed–fixed bar, load at a	$R_A = P\,b/L$, $R_B = P\,a/L$
Three-bar framework	$N_i = F\,A/(A + 2\,A_0\cos^3\theta)$, $N_o = F\,A_0\cos^2\theta/(\dots)$

Topic	Formula / rule
Displacement diagram	$\delta_{bar} = u \cdot e$; symmetric V: $\delta = P L / (2 A E \cos^2 \beta)$
Strain energy	$U = P^2 L / 2 A E$; varying: $\int N^2 d x / 2 A E$; beams $\int M^2 d s / 2 E I$
Castigliano	$\delta_i = \partial U / \partial F_i$; truss $\sum F (\partial F / \partial P) L / A E$; dummy load → set 0 after differentiating
Thin ring	$\sigma = p r / t, \delta_R = p r^2 / (E t)$
Belt	$T_2 / T_1 = e^{\mu \beta}$
Units	N, mm, MPa is a consistent set; GPa = 10^3 MPa

Other Crandall Ch. 2 problems worth practising: 2.2–2.5 (determinate links), 2.8–2.9 (truss joint movements), 2.15–2.16 (ropes and rigging), 2.21 (rigid board on three springs), 2.25 (rigid bars on rods), 2.26 (wooden-stave pipe), 2.27–2.28 (rope/band brakes), 2.36–2.38 (struts, screw clamps, coils in rings), 2.42–2.43 (energy methods, indeterminate trusses).

Chapter 3 — Forces and Moments Transmitted by Slender Members

Internal forces, shear-force and bending-moment diagrams, distributed loads, q–V–M relations, singularity functions, frames, curved and 3-D members

Sources used: Slides "Chapter 3" (Dr. Kushagra Tiwari); Crandall Ch. 3 (§3.1–3.8; Examples 3.1, 3.11; Probs 3.1, 3.2, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10, 3.12, 3.14, 3.17, 3.23, 3.25, 3.34); PYQs 2014-15 (Test-1 Q4; Compre A Q6, Q7, Q14, Q17, Q18) and 2019-20 (Compre A Q1, Q3(a)).

Layout. Part A (eight items) for each concept, then Part B: slides → Crandall → PYQs, easy → hard. Each solution follows *Problem · Given/Find · Approach · FBD · Steps · Answer · Check · Takeaway*.

⚠️ Read first: the sign convention for shear force in this course

The slides follow **Crandall's** convention, which is **opposite to Gere/Hibbeler/Beer for the shear force V** (but the same for the bending moment).

Quantity	Crandall (slides, this guide)	Gere / Hibbeler
Axial F	+ tension	+ tension
Bending M_b	+ sagging (concave up, "smile")	+ sagging (same)
Shear V	**+ when it acts on the positive (+x) face in the +y direction**, i.e. the right face of the left segment is pushed **up**	+ when the left face of the right segment is pushed up (element rotates clockwise)
dV/dx	¿ $-q$ (q upward +)	¿ $-w$ (w downward +)
dM/dx	¿ $-V$	¿ $+V$

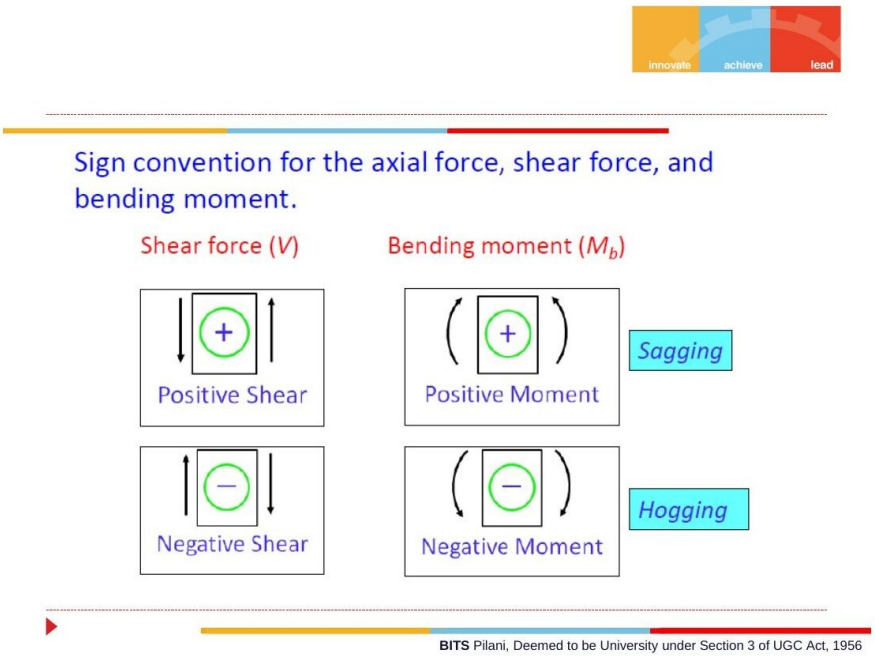
So $V_{Crandall} = -V_{Gere}$ everywhere, while M is identical. In the slide's picture of "positive shear" the arrow is ↓ on the left face and ↑ on the right face of an element.

Working rules used throughout this chapter (cut at x, keep the left part):

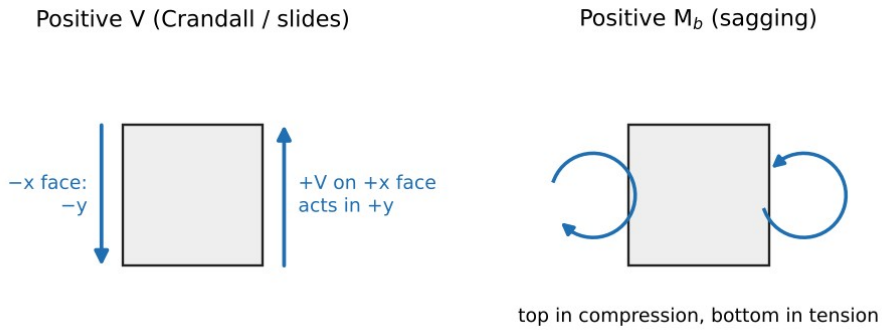
$V(x) = -(\sum$ upward forces to the left of the cut$)$, $M(x) = \sum$ moments of the left-hand forces about the cut, clockwis

An upward force to the left of a cut produces sagging (positive) moment. A clockwise applied couple to the left of the cut adds positive moment.

If you learned shear from Gere, flip the sign of every V in this chapter; nothing else changes.



Original: Slides Ch 3, slide 9



Positive shear and positive bending moment (Crandall)

Concept 3.1 — Internal forces at a section: the method of sections

A. Concept explanation

1. Intuition. Cut a loaded beam with an imaginary saw. Each half would fly off unless something at the cut held it in place. Whatever the other half was supplying (a push or pull along the axis, a sideways shear, a bend, a twist) we draw at the cut. Equilibrium of either half then tells us their values. A slender member transmits load from where it is applied to where it is supported *through* these internal forces.

2. Why it matters. These internal forces feed everything that follows: $\sigma = F/A$ (Ch. 2, 4), $\tau = T\,r/J$ (Ch. 6), later $\sigma = M\,y/I$ for bending and $v'' = M/EI$ for deflection. Every combined-loading PYQ begins by finding F, V, M_b and M_t at a section. The maximum stress occurs where these are largest, which is why we draw diagrams of them along the member.

3. Formal treatment (slides, Crandall §3.2).

- *Slender member* (slides): length at least about five times the lateral dimensions. Examples: beams, columns, shafts, rods, stringers, struts, links.
- *General method* (slides): isolate the system and draw all external forces; find the support reactions from $\sum F = 0, \sum M = 0$; cut at the point of interest, isolate one segment and repeat equilibrium on it.
- *x-section:* a cut whose normal is the member axis x. In 3-D the resultant on it has six components:

Component	Name	Usual symbol	Effect
F_{xx}	axial force	F	elongates/shortens the member
F_{xy}, F_{xz}	shear forces	V_y, V_z	slides one part past the other
M_{xx}	twisting moment	M_t (or T)	twists about the axis
M_{xy}, M_{xz}	bending moments	M_{by}, M_{bz}	bends the member

(First subscript = face normal, second = direction of the component.)

- *Positive face:* outward normal along +x. *Negative face:* outward normal along −x.
- *Sign rule (slides):* a component is **positive** if it acts on a positive face in a positive coordinate direction, **or** on a negative face in a negative coordinate direction. Otherwise it is negative. By this rule tension is positive and compression negative.
- *Planar loading* (x–y plane): only three components survive, $F = F_{xx}$, $V = F_{xy}$ and $M_b = M_{xz}$.

4. Derivation — Crandall Example 3.1 (simply supported beam, load W at distance a from A and b from B).

1. *Reactions* (whole beam; slide assumptions: horizontal reactions negligible, self-weight neglected): $\sum M_B$ gives $R_A = W\,b/L$, and $\sum M_A$ gives $R_B = W\,a/L$.
2. *Cut at $x < a$, keep the left part.* On its right (positive) face draw F →, V ↑ and M_b counter-clockwise, all positive by the sign rule.
   - $\sum F_x$: $F = 0$.
   - $\sum F_y$: $R_A + V = 0$, so $V = -\dfrac{W\,b}{L}$.
   - $\sum M_{cut}$: $M_b - R_A\,x = 0$, so $M_b = \dfrac{W\,b}{L}\,x$.
3. *Cut at $x > a$:* $V = -\dfrac{W\,b}{L} + W = \dfrac{W\,a}{L}$ and $M_b = \dfrac{W\,b}{L}\,x - W\,(x-a) = \dfrac{W\,a}{L}\,(L-x)$.
4. *Peak:* $M_{max} = \dfrac{W\,a\,b}{L}$ under the load. V jumps by W there.

5. Picture. A positive M_b is counter-clockwise on the right face of the left segment and clockwise on the left face of the right segment. The pair "opens upward" like a smile (sagging, top fibres in compression). The **shear-force diagram (SFD)** plots V(x) and the **bending-moment diagram (BMD)** plots $M_b(x)$, each drawn directly below a sketch of the loaded beam with matching x.

6. Special cases.

- $a = b = L/2$: $M_{max} = W\,L/4$.
- A load right at a support produces no internal moment anywhere.
- Point of **contraflexure** (slides): where the BMD **changes sign**. M is zero there, and the curvature of the deflected shape reverses.

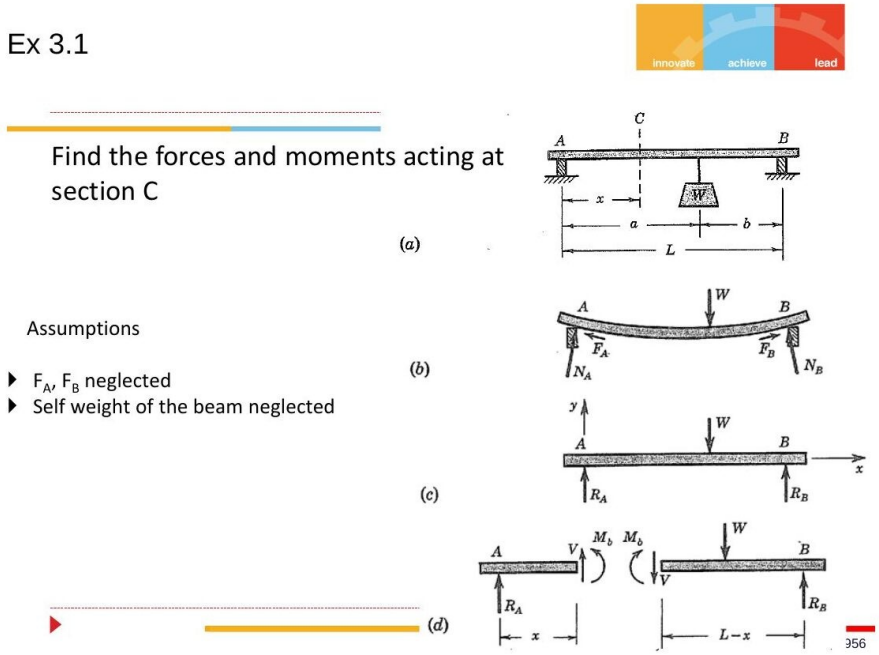
7. Common traps.

- Drawing V and M in *assumed physical* directions instead of the **positive** convention. Always draw them positive and let the algebra give the sign.
- Mixing the Crandall and Gere signs for V (see the box above).
- Using a distributed load's *total* resultant when the cut lies inside it (Concept 3.2).
- Forgetting the axial force F in frames and inclined members.

8. Summary box. Cut → draw F, V, M as positive → equilibrium → read the signs.
$V = -(\sum \text{up-forces on the left})$; $M = \sum ¿$.

B. Solved questions

Q1 (Slides Ch3, Ex 3.1 = Crandall Example 3.1) — Forces and moments at a section C (easy, symbolic)



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Problem. A simply supported beam of span L carries a load W at distance a from A (b from B). Find the forces and moment acting at a section C at distance x from A, and sketch the SFD and BMD.

Given: L, W, a, b = L − a. Slide assumptions: the horizontal reactions F_A, F_B are neglected, and the beam's self-weight is neglected. **Find:** F, V, M_b as functions of x.

Approach. Reactions from the whole beam, then the method of sections on each side of the load.

FBD in words. Beam with R_A ↑ at A, R_B ↑ at B and W ↓ at a. For the cut, the left segment shows R_A and, on its right face, positive F (−), V (↑) and M_b (counter-clockwise).

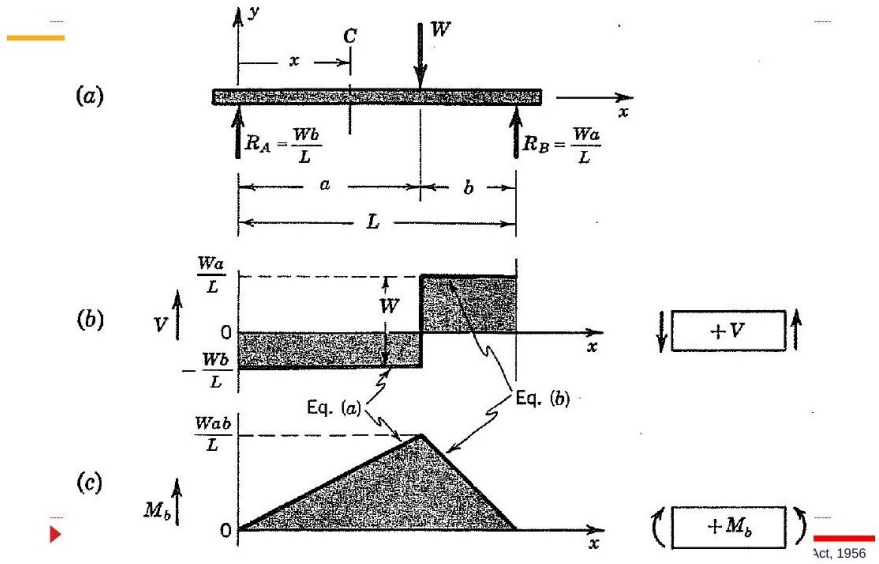
Solution.

1. $R_A = W\,b/L$ and $R_B = W\,a/L$.
2. $0 < x < a$: $F = 0$, $V = -W\,b/L$, $M_b = W\,b\,x/L$.
3. $a < x < L$: $F = 0$, $V = +W\,a/L$, $M_b = W\,a(L-x)/L$.

$$V = \begin{cases} -W\,b/L & x<a \\ +W\,a/L & x>a \end{cases},\quad M_b = \begin{cases} W\,b\,x/L & x<a \\ W\,a(L-x)/L & x>a \end{cases},\quad M_{max} = \frac{W\,a\,b}{L} \text{ at } x=a$$

Check. The jump in V at the load is $W\,a/L + W\,b/L = W$ ✓. $d\,M/d\,x = W\,b/L = -V$ on the left, consistent with Crandall's relation $d\,M/d\,x = -V$ (Concept 3.3) ✓.

Exam takeaway. This one result covers every point-load beam by superposition. **Variation:** W at mid-span gives $M_{max} = W\,L/4$.



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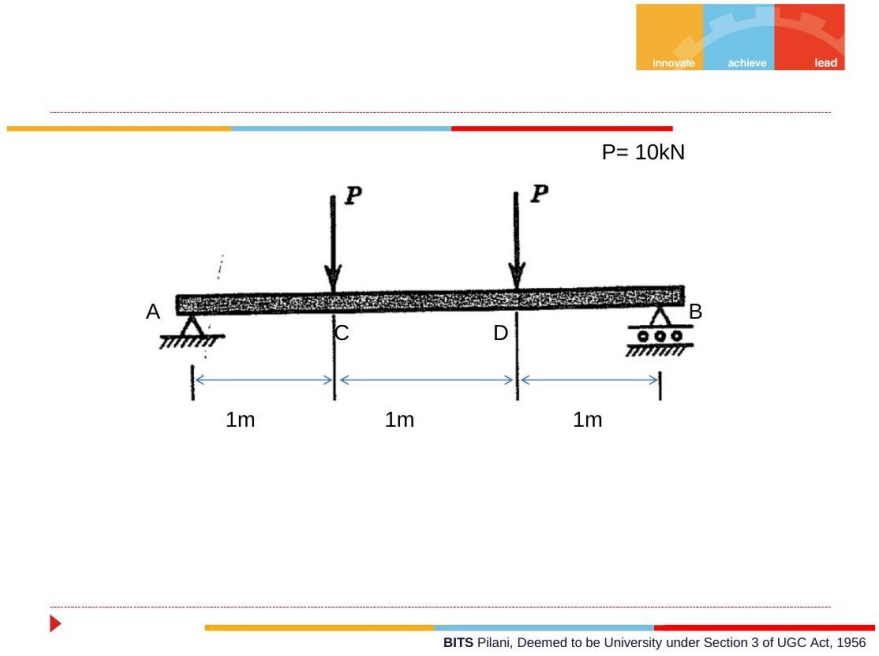
Q2 (Slides Ch3) — Two 10 kN loads on a 3 m span (four-point bending) (easy)



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Problem. A simply supported beam AB of span 3 m carries $P = 10$ kN at C (1 m) and at D (2 m). Draw the SFD and BMD.

Approach. Symmetric loading, so the reactions are equal. Then sections in each of the three regions.

FBD in words. Beam with R_A, R_B ↑ and 10 kN ↓ at 1 m and 2 m.

Solution.

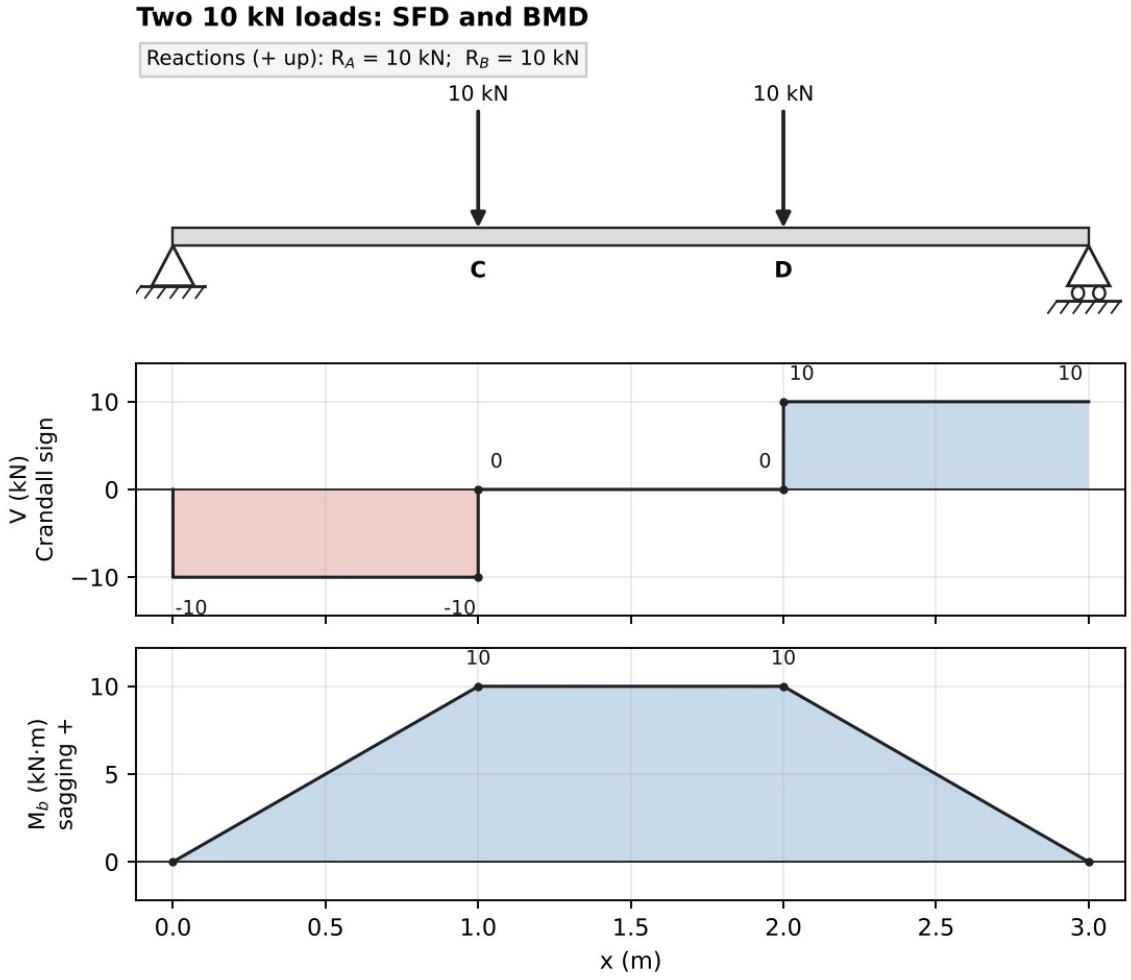
1. $R_A = R_B = 10$ kN.
2. Region by region (Crandall V):

Region	V	M_b
A–C ($0<x<1$)	−10 kN	$10\,x$ (0 → 10 kN·m)
C–D ($1<x<2$)	0	10 kN·m (constant)
D–B ($2<x<3$)	+10 kN	$10(3-x)$ (10 → 0)

$$M_{max} = 10\,\text{kN·m, constant between C and D, where } V = 0$$

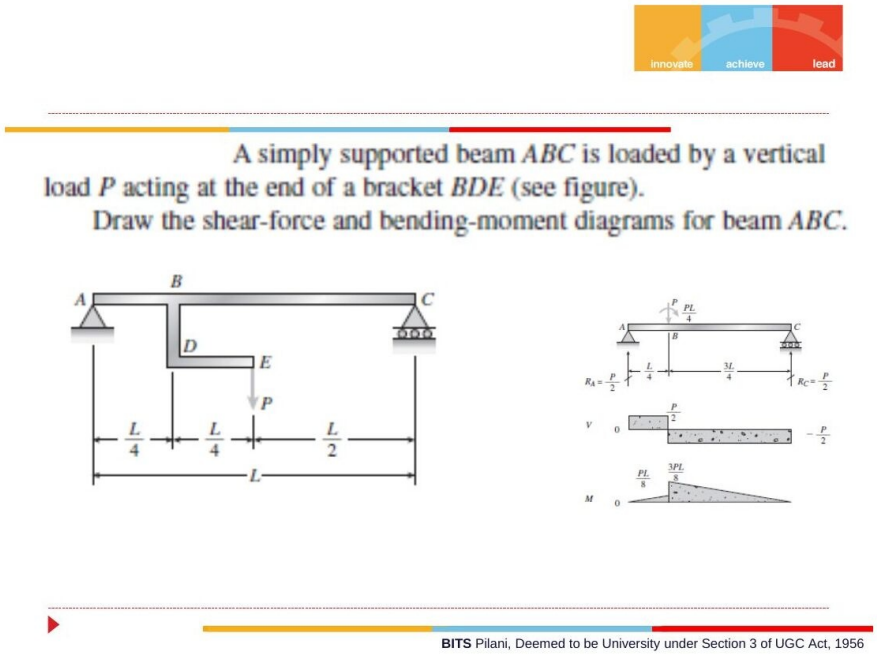
Check. The BMD is a trapezoid. The flat top between the loads is **pure bending** (V = 0) ✓.

Exam takeaway. The four-point-bending test uses exactly this layout. Constant M with zero shear lets you measure bending stiffness without shear effects. **Variation:** Crandall Prob 3.5, loads P at distance a from each end: $M = P\,a$ between the loads.



Solution: loading, SFD, BMD

Q3 (Slides Ch3, Gere) — Beam loaded through a bracket (medium)



Original: Slides Ch 3, slide 17

Problem. A simply supported beam ABC of span L carries a bracket BDE welded at B (at L/4 from A). A vertical load P acts at the end E of the bracket, whose line of action is at mid-span (x = L/2). Draw the SFD and BMD of ABC. (For the plotted diagram, P = 10 kN and L = 4 m.)

Approach. Replace the bracket load by an equivalent force plus a couple at B (Ch. 1, Concept 1.2), then analyse the beam.

FBD in words. Beam with R_A, R_C ↑, P ↓ at B and a clockwise couple $P \cdot \frac{L}{4}$ at B (P is L/4 to the right of B).

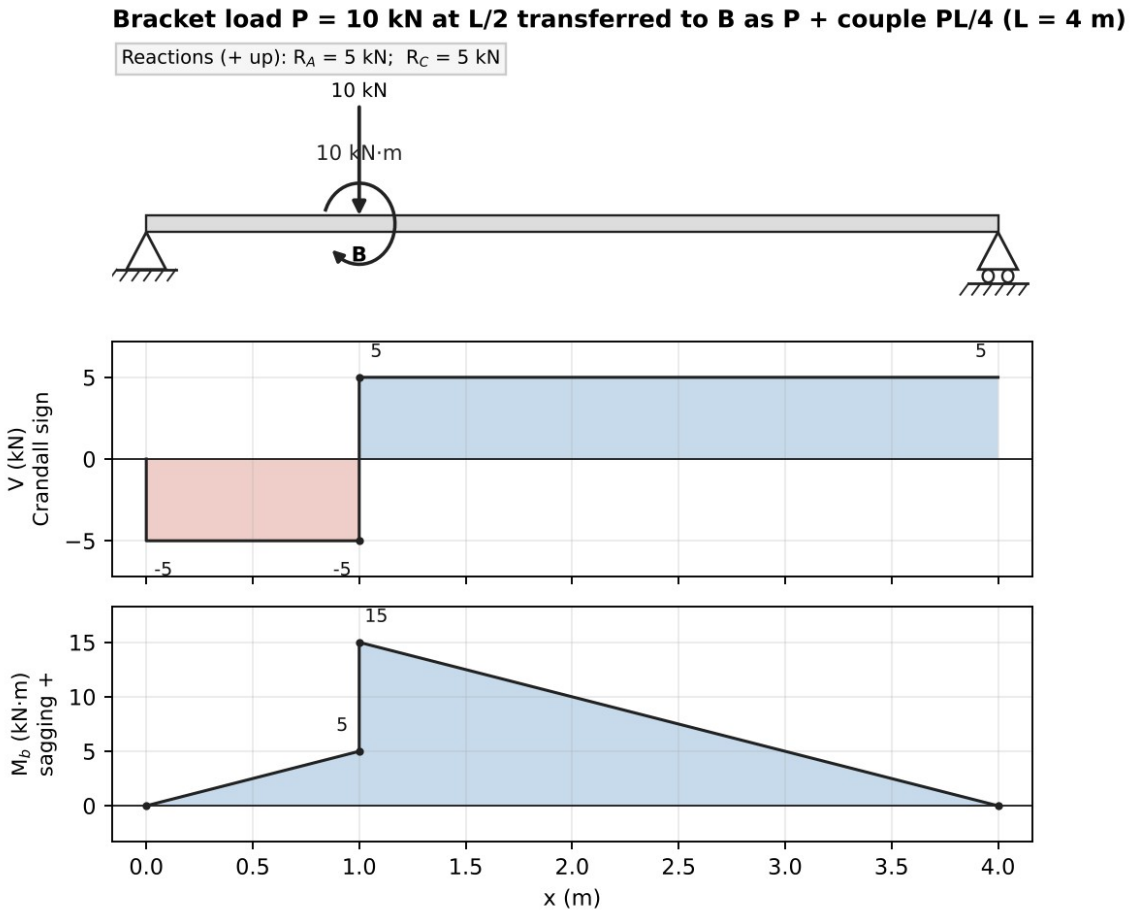
Solution.

1. *Equivalent load at B:* force P ↓ plus a clockwise couple $C = P\left(L/2 - L/4\right) = P\,L/4$.
2. *Reactions:* the line of action of P is at mid-span, so $R_A = R_C = P/2$. (Moments about C: $R_A\,L - P\left(3\,L/4\right) + P\,L/4 = 0$, so $R_A = P/2$ ✓.)
3. *A to B* ⁻: $V = -P/2$; M rises linearly from 0 to $P\,L/8$.
4. *At B:* V jumps by +P (to +P/2), and M **jumps up** by $P\,L/4$ (clockwise couple) to $3\,P\,L/8$.
5. *B* ⁺ *to C:* $V = +P/2$; M falls linearly from $3\,P\,L/8$ to 0.

$$M_{max} = 3\,P\,L/8 \text{ just to the right of B} \left(15\ \text{kN·m for } P = 10\ \text{kN}, L = 4\ \text{m}\right)$$

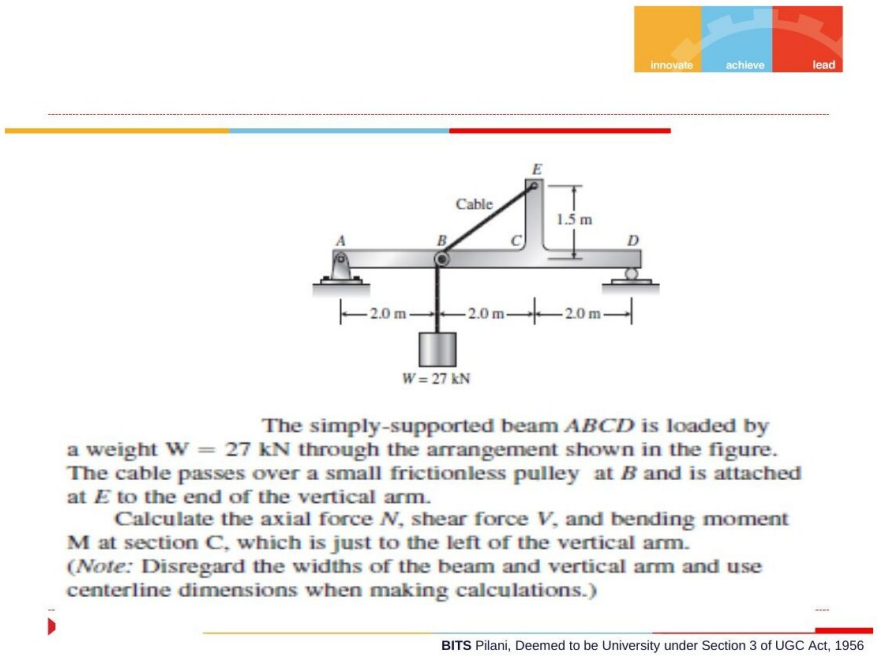
Check. From the right end: M ¿ ✓.

Exam takeaway. An applied **clockwise couple** makes the BMD **jump up** by the couple's magnitude (moving left to right); a counter-clockwise couple makes it jump down. Couples never change V. **Variation:** a bracket pointing the other way (P left of B), which gives a downward jump.



Solution with P = 10 kN, L = 4 m (M jumps up at B because of the clockwise couple)

Q4 (Slides Ch3, Gere) — Beam with a pulley and cable: N, V and M at section C (medium–hard)



Original: Slides Ch 3, slide 28

Problem. A simply supported beam ABCD has a pin at A (x = 0), a small frictionless pulley at B (x = 2 m), section C at x = 4 m and a roller at D (x = 6 m). A vertical arm at C rises 1.5 m to E. A cable carrying a weight W = 27 kN passes over the pulley at B and is tied to E. Find the axial force N, shear V and moment M at a section just to the left of the arm (at C).

Given: W = 27 kN; geometry as above. **Find:** N, V, M at C⁻.

Approach. Reactions from the whole system (beam + pulley + cable + arm), then the force that the pulley exerts on the beam at B, then cut just left of C.

FBD in words.

- *Whole system:* the only external load is W, hanging below B. Reactions A_x, A_y at A and D_y at D.
- *Pulley at B:* two cable segments pull on it, the hanging segment (27 kN ↓) and the segment from B to E.

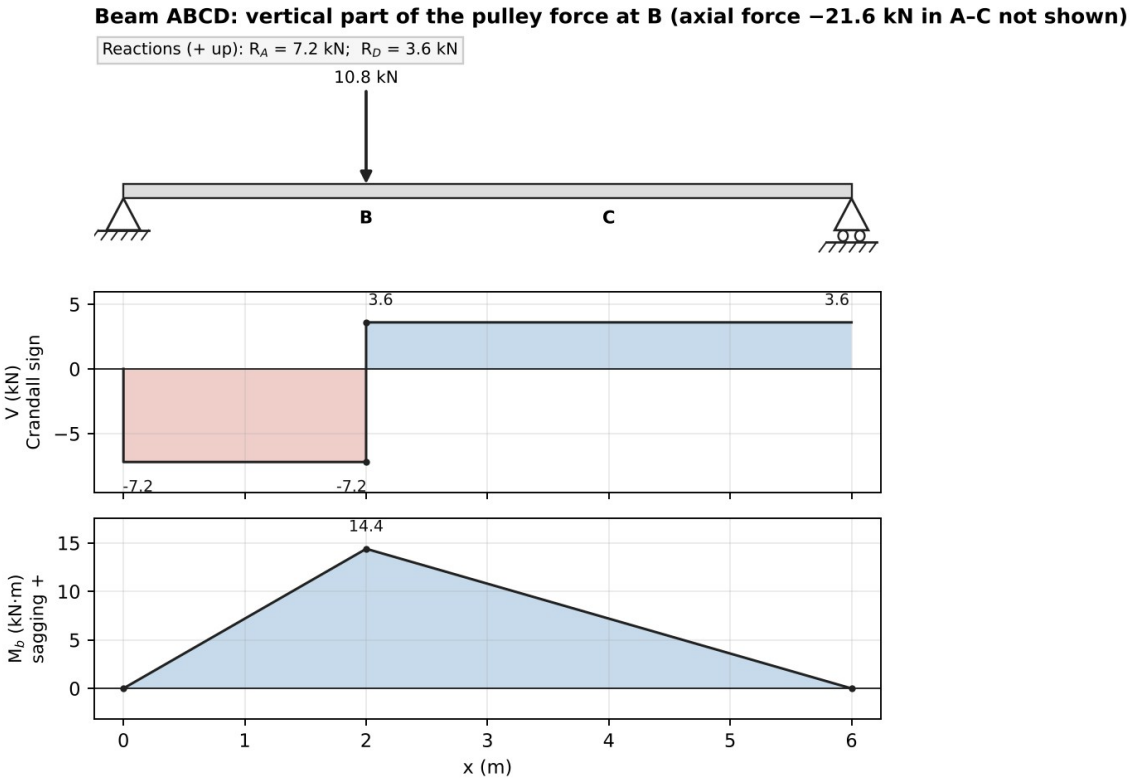
Solution.

1. *Reactions:* $\sum M_A$: $D_y(6)=27(2)$, so $D_y=9$ kN and $A_y=18$ kN. $A_x=0$ (the cable forces are internal to the system).
2. *Cable direction B → E:* from (2, 0) to (4, 1.5): $(2,1.5)/2.5=(0.8,0.6)$. With T = 27 kN throughout (frictionless pulley), the pulley's force on the beam is $(0,-27)+27(0.8,0.6)=(21.6,-10.8)$ kN.
3. *Cut just left of C, keep the left part* (A to C⁻), which carries $A_y=18$ ↑ at 0 and the pulley force $(21.6,-10.8)$ at x = 2.
   - $\sum F_x$: $21.6+N=0$, so N = −21.6 kN (compression).
   - $\sum F_y$: $18-10.8+V=0$, so V = −7.2 kN (Crandall); Gere would write +7.2 kN.
   - $\sum M_C$: $M=18(4)-10.8(2)=72-21.6=50.4$ kN·m.

$$N=-21.6\,\text{kN}\,(\text{C}),\ V=-7.2\,\text{kN}\,(\text{Crandall}),\ M=50.4\,\text{kN·m}\,(\text{sagging})$$

Check. Use the right part instead (C⁻ to D, including the arm, with the cable pulling E toward B with $27(-0.8,-0.6)=(-21.6,-16.2)$ kN). Horizontal: $-21.6-N=0$ gives N = −21.6 ✓. Moments about C: D_y gives $9(2)=18$ and the cable force at E gives $1.5(21.6)=32.4$, total 50.4 kN·m ✓.

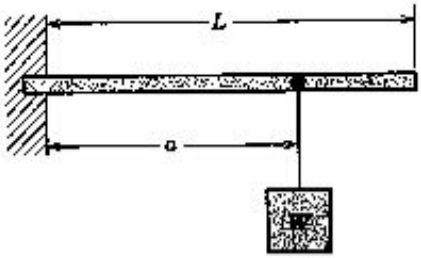
Exam takeaway. Pulleys transfer **both** cable forces to their axle. Forgetting the horizontal component loses the axial force. The slide adds: "disregard the widths of the beam and vertical arm". **Variation:** the section just to the right of the arm, where the cable's moment about C ($27\times0.8\times1.5=32.4$ kN·m) causes a jump in M.



Vertical-load SFD/BMD for ABCD (M at C = 50.4 kNm)

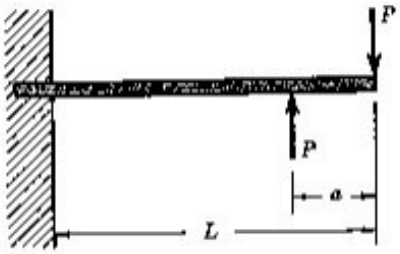
Q5 (Crandall Probs 3.1, 3.2, 3.4, 3.5) — Four quick SFD/BMD drills (easy)

Prob. 3.1

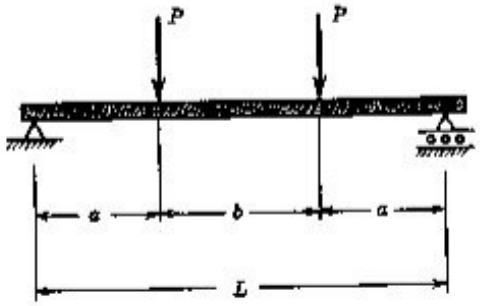


Prob. 3.2

Original: Crandall Prob 3.1–3.2 (textbook figure)



Prob. 3.4



Prob. 3.5

Original: Crandall Prob 3.4–3.5 (textbook figure)

Problem. Sketch V and M for: (3.1) a cantilever of length L, built in at the left, with a weight W hung at the free end; (3.2) the same with W hung at distance a from the wall; (3.4) a cantilever with P down at the tip and P up at distance a from the tip; (3.5) a simply supported beam with P at distance a from each end (span L, b between the loads).

Approach. For each, find the wall reaction or support reactions, then use the working rules ($V = -\sum$ up-forces on the left, M = sum of left-hand moments). x is measured from the left end.

Solution.

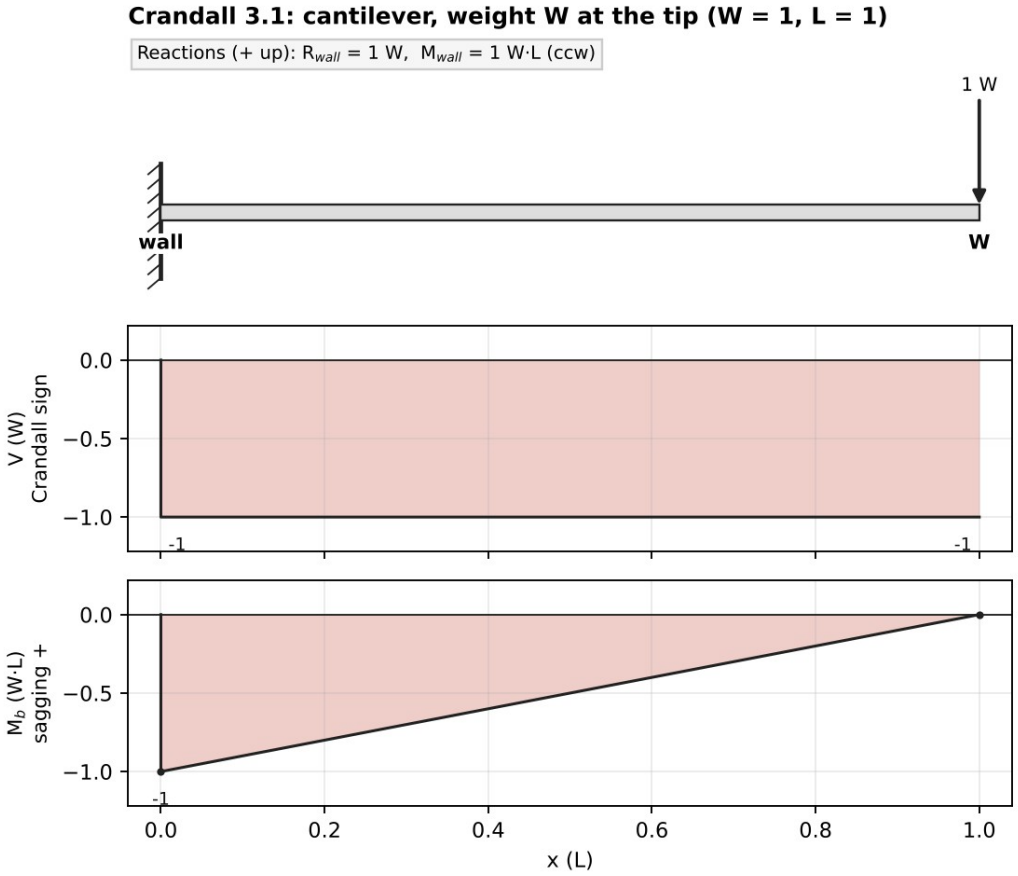
1. *3.1:* wall gives R = W ↑ and a counter-clockwise couple WL. $V = -W$ everywhere; $M = W\,x - W\,L = -W\,(L - x)$, hogging, largest at the wall: $|M|_{max} = W\,L$.
2. *3.2:* $V = -W$ for $x < a$ and 0 beyond; $M = -W\,(a - x)$ for $x < a$ and 0 beyond. The part beyond the load carries nothing.

3. *3.4:* the two forces form a couple, so the wall force is 0 and the wall moment is Pa (counter-clockwise). For $x<L-a$: V = 0 and $M=-Pa$ (constant hogging). For $L-a<x<L$: $V=-P$ (the upward P is now to the left of the cut) and $M=-Pa+P\left[x-(L-a)\right]=-P(L-x)$, reaching 0 at the tip.
4. *3.5:* $R=P$ at each support. $V=-P$, 0, +P in the three regions; M rises to Pa, stays at Pa between the loads, then falls to 0.

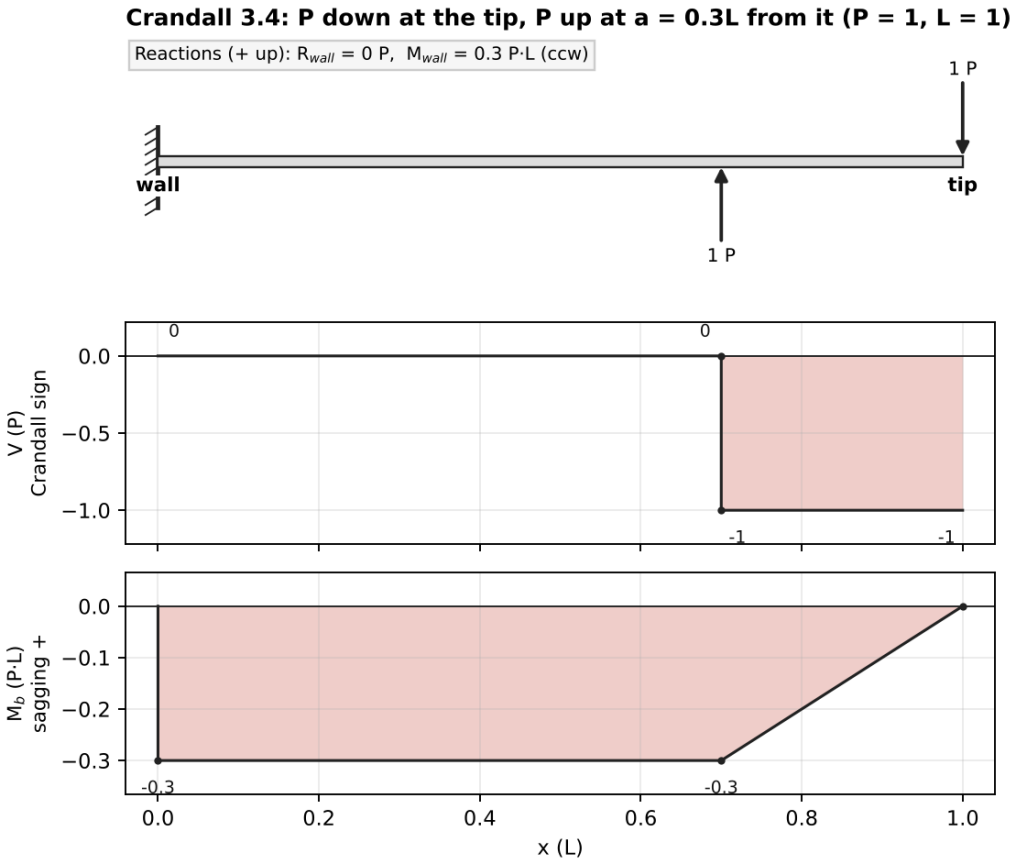
$$3.1:\left|M\right|_{max}=WL;\ 3.2:\left|M\right|_{max}=Wa;\ 3.4:M=-Pa\ (\text{const.});\ 3.5:M_{max}=Pa\ (\text{const.})$$

Check. In 3.4 the section between the wall and the upward load feels only the couple, so V = 0 and M is constant ✓. In 3.1 the magnitude grows linearly toward the wall ✓.

Exam takeaway. Cantilevers: the maximum |M| is at the wall, and its value equals the moment of all loads about the wall. **Variation:** Crandall 3.3, an inclined cantilever (resolve the load into axial and transverse parts).

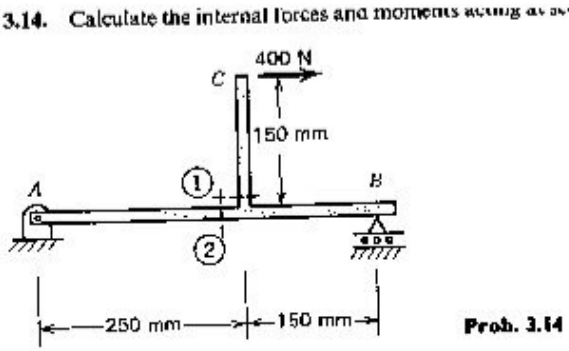


Drill 3.1 (units W and L)



Drill 3.4 (units P and L)

Q6 (Crandall Prob 3.14) — Frame with a vertical arm: forces at two sections (medium)



Original: Crandall Prob 3.14 (textbook figure)

Problem. A horizontal beam AB (pin at A, roller at B) is 400 mm long. A vertical arm 150 mm tall is welded at 250 mm from A, and a 400 N horizontal force (→) acts at its top C. Find the internal forces and moments at section 1 (in the arm just above the beam) and section 2 (in the beam just to the left of the arm).

Given: AB = 400 mm; arm at x = 250 mm; force 400 N → at height 150 mm. **Find:** F, V, M at sections 1 and 2.

Approach. Reactions from the whole frame, then cut each section and keep the simpler side.

FBD in words. Frame with A_x, A_y at A, B_y at B and 400 N → at C.

Solution.

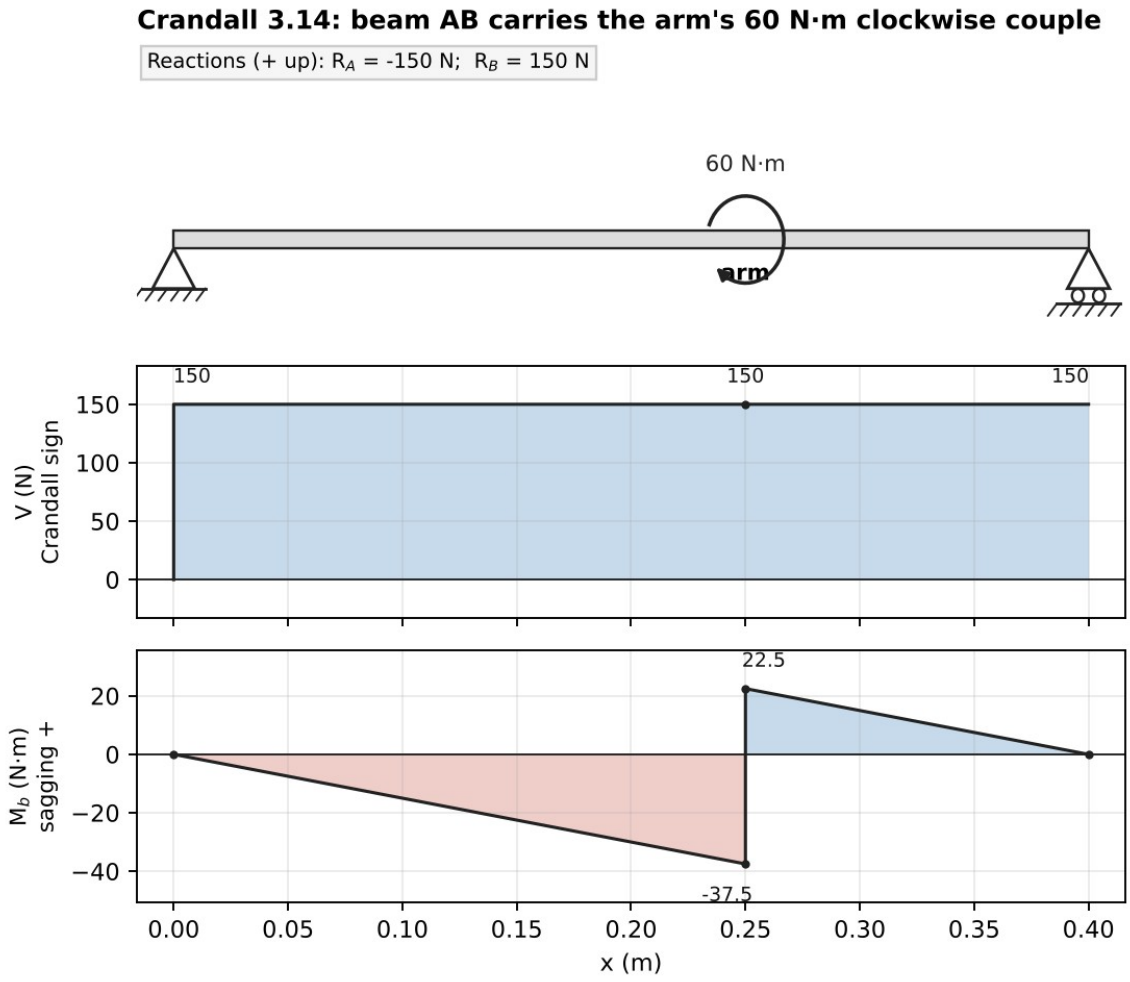
1. *Reactions:* $\sum M_A$: $B_y(400) - 400(150) = 0$, so $B_y = 150$ N ↑. $\sum F_y$: $A_y = -150$ N (↓). $\sum F_x$: $A_x = -400$ N (←).
2. *Section 1 (arm, just above the joint), keep the top part:* the only load is 400 N → at C, 150 mm above the cut. Axial force (along the vertical arm) = 0; shear = 400 N (horizontal); moment = 400(0.150) = 60 N·m.

3. *Section 2 (beam at x = 250⁻ mm), keep the left part:* forces $A_x = 400 \leftarrow$, $A_y = 150 \downarrow$.
   - Axial: the cut face must pull right to balance A_x: F = +400 N (tension).
   - Shear (Crandall): $V = -(\text{up-forces on the left}) = -(-150) = +150$ N.
   - Moment: $M = A_y(0.25) = -150(0.25) = -37.5$ N·m (hogging).
4. *Just to the right of the arm,* the couple from the arm (60 N·m clockwise on the beam) makes M jump to $-37.5 + 60 = +22.5$ N·m. Check from the right: $B_y(0.15) = 22.5$ ✓.

$$\text{Sec. 1: } F = 0,\ V = 400\ \text{N},\ M = 60\ \text{N·m};\ \text{Sec. 2: } F = 400\ \text{N (T)},\ V = 150\ \text{N},\ M = -37.5\ \text{N·m}$$

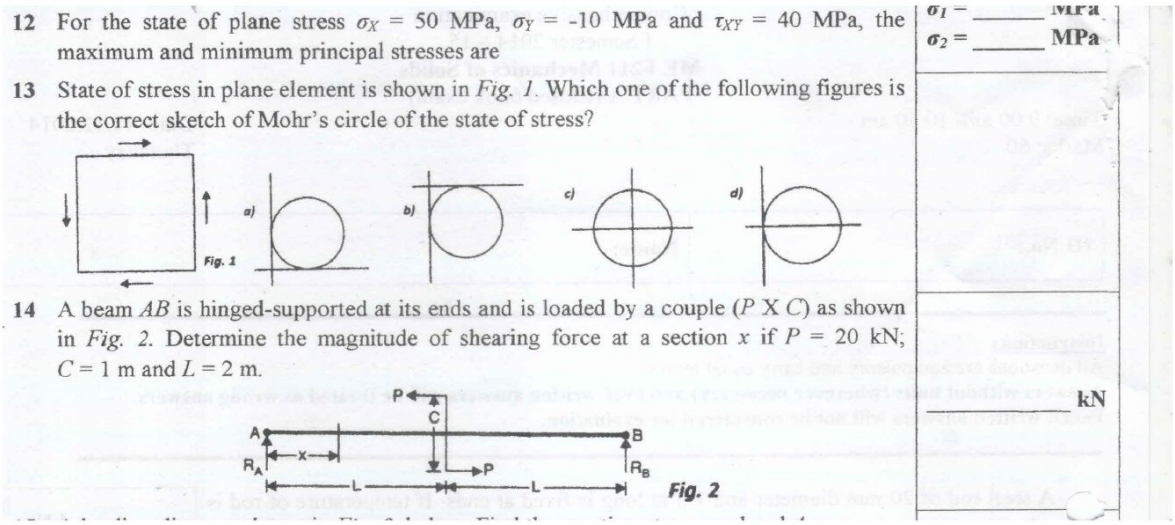
Check. The 400 N horizontal force is carried along the beam as tension between A and the arm, and not at all between the arm and the roller ✓.

Exam takeaway. An offset horizontal load creates an axial force on one side and a moment jump at the joint. This is the same structure as Q18 (5 kN/m segments + couple). **Variation:** reverse the force direction; every sign flips.



Beam AB: the arm's couple makes M jump at x = 0.25 m

Q7 (PYQ 2014-15 Compre A Q14) — Couple on a hinged beam (easy)



12 For the state of plane stress σ_X = 50 MPa, σ_Y = -10 MPa and τ_{XY} = 40 MPa, the maximum and minimum principal stresses are

σ_1 = _____ MPa

σ_2 = _____ MPa

13 State of stress in plane element is shown in *Fig. 1*. Which one of the following figures is the correct sketch of Mohr's circle of the state of stress?

Fig. 1 a) b) c) d)

14 A beam *AB* is hinged-supported at its ends and is loaded by a couple (*P* X *C*) as shown in *Fig. 2*. Determine the magnitude of shearing force at a section *x* if *P* = 20 kN; *C* = 1 m and *L* = 2 m.

P C A B x R_A P R_B L L

Fig. 2

_____ kN

Original question: PYQ 2014-15 Compre A Q12–Q14

Problem. A beam AB, hinged at both ends, spans 2L with L = 2 m. At mid-span a couple is formed by two forces P = 20 kN, C = 1 m apart (20 kN·m). Find the magnitude of the shear force at any section.

Solution.

1. A pure couple needs an equal and opposite couple from the supports: $R_A = -R_B = \frac{20}{2L} = \frac{20}{4} = 5$ kN.
2. No other vertical loads, so V is constant along the beam.

$$|V| = 5 \text{ kN everywhere}$$

Exam takeaway. A couple causes no jump in the SFD, only in the BMD (a jump of 20 kN·m at mid-span, from −10 to +10 kN·m).

Concept 3.2 — Distributed loads, their resultants and the limits of using them

A. Concept explanation

1. Intuition. Spread sand on a plank. Each little strip Δx carries $q\,\Delta x$. For finding the **support reactions** you can pretend all the sand is piled at its centre of gravity. For the **bending inside the plank** you cannot, because sand piled in the middle bends the plank differently from sand spread evenly.

2. Why it matters. Self-weight, snow, water pressure on dams (Q14), sandbags (Q13), stored goods, the triangular and trapezoidal loads of the slides. Most beam PYQs include at least one distributed load.

3. Formal treatment (slides).

$$q = \frac{\lim_{\Delta x \to 0} \Delta F}{\Delta x} \left(\text{force per unit length; positive when acting in} + y\text{, i.e. upward}\right)$$

Common distributions: **uniform**, q constant; **linearly varying**, $q(x) = A\,x + B$.

Resultant and its location (slides):

$$R = \int_0^L q\,dx\,,\; \acute{x} = \frac{\int_0^L x\,q\,dx}{R}$$

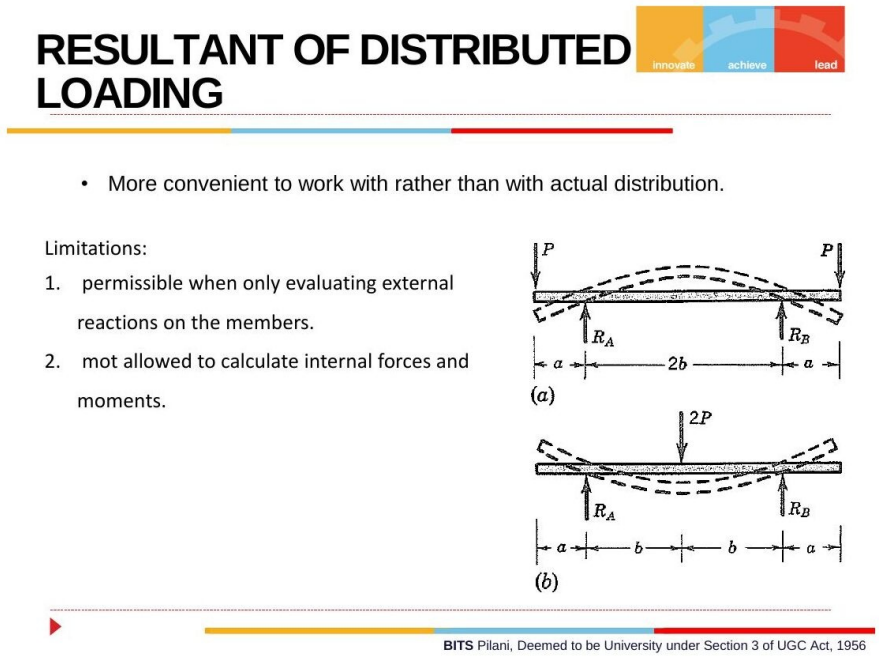
The resultant equals the **area** of the load diagram and acts through its **centroid**.

Load shape	R	$\acute{x}$
Uniform w_0 over L	$w_0 L$	L/2
Triangle from 0 to w_0	$w_0 L/2$	2L/3 from the zero end
Trapezoid $w_1 \to w_2$	$(w_1+w_2)L/2$	$\dfrac{L(w_1+2w_2)}{3(w_1+w_2)}$ from the w_1 end

Limitations of the resultant (slides):

1. Permissible when only evaluating external reactions on the member.
2. **Not** allowed when calculating internal forces and moments.

Slide illustration: beam (a) has P at each overhang tip, a distance a outside supports 2b apart. Beam (b) has 2P at the centre. Same resultant, same reactions ($R_A = R_B = P$). But (a) has **hogging** $M = -Pa$ between the supports, while (b) has **sagging** $M_{max} = +Pb$. The beams even bend in opposite directions.



Original: Slides Ch 3, slide 22

4. Derivation — cantilever with a triangular load (slides). The load $w = w_0 x/L$ acts downward, with the free end A at x = 0 and the wall B at x = L. Cut at x and keep the left part. The load on it is $\int_0^x w_0 \xi/L \, d\xi = w_0 x^2/2L$, acting at $2x/3$ from A, i.e. $x/3$ from the cut.

- $\sum F_y: -\dfrac{w_0 x^2}{2L} + V = 0$, so $V = \dfrac{w_0 x^2}{2L}$.
- $\sum M: M_b + \dfrac{w_0 x^2}{2L} \cdot \dfrac{x}{3} = 0$, so $M_b = -\dfrac{w_0 x^3}{6L}$.

At the wall: $V = w_0 L/2$ (the total load) and $M_b = -w_0 L^2/6$ (hogging).

5. Shape rules. UDL gives a linear V and a parabolic M. A linearly varying load gives a parabolic V and a cubic M. Always use the portion of the load *on your free body*, placed at *its own* centroid.

6. Special cases.

- Simply supported beam with a full UDL: $V = -w_0 L/2 + w_0 x$ and $M = w_0 x(L - x)/2$, so $M_{max} = w_0 L^2/8$ at mid-span, where V = 0.
- Cantilever with a full UDL: $|M|_{max} = w_0 L^2/2$ at the wall.
- Simply supported beam with a triangular load of total W: $M_{max} = \frac{2WL}{9\sqrt{3}} = 0.128\,WL$, at $x = L/\sqrt{3}$ from the zero end (Q13).

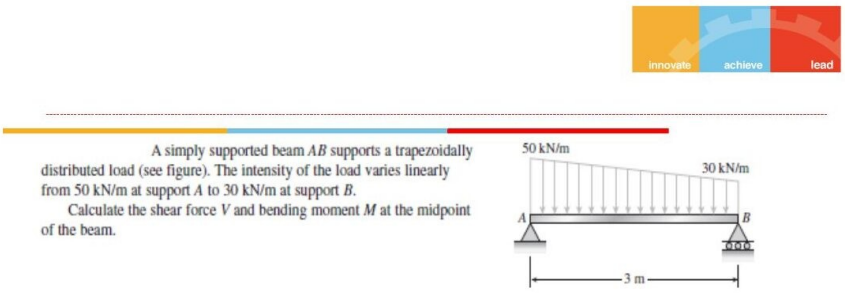
7. Common traps.

- Using the resultant of the *whole* distributed load when cutting inside it.
- Putting a triangle's resultant at L/2 instead of 2L/3 from the zero end.
- Forgetting that the load intensity at the cut changes with x for a varying load.

8. Summary. For reactions, replace the load by its resultant (area at the centroid). For V and M, use only the part of the load on your free body, at its own centroid.

B. Solved questions

Q8 (Slides Ch3, Gere) — Trapezoidal load: V and M at mid-span (medium)



Original: Slides Ch 3, slide 29

Problem. A simply supported beam of span 3 m carries a load decreasing linearly from 50 kN/m at A to 30 kN/m at B. Find the shear force and bending moment at mid-span.

Given: L = 3 m; $w_A = 50$, $w_B = 30$ kN/m. **Find:** V and M at x = 1.5 m.

Approach. Reactions from the resultant of the whole trapezoid. Then cut at mid-span and use only the left part of the load, which is itself a trapezoid from 50 to 40 kN/m.

FBD in words. Left segment: R_A ↑ at 0, a trapezoidal load from 50 to 40 kN/m over 1.5 m, and positive V and M at the cut.

Solution.

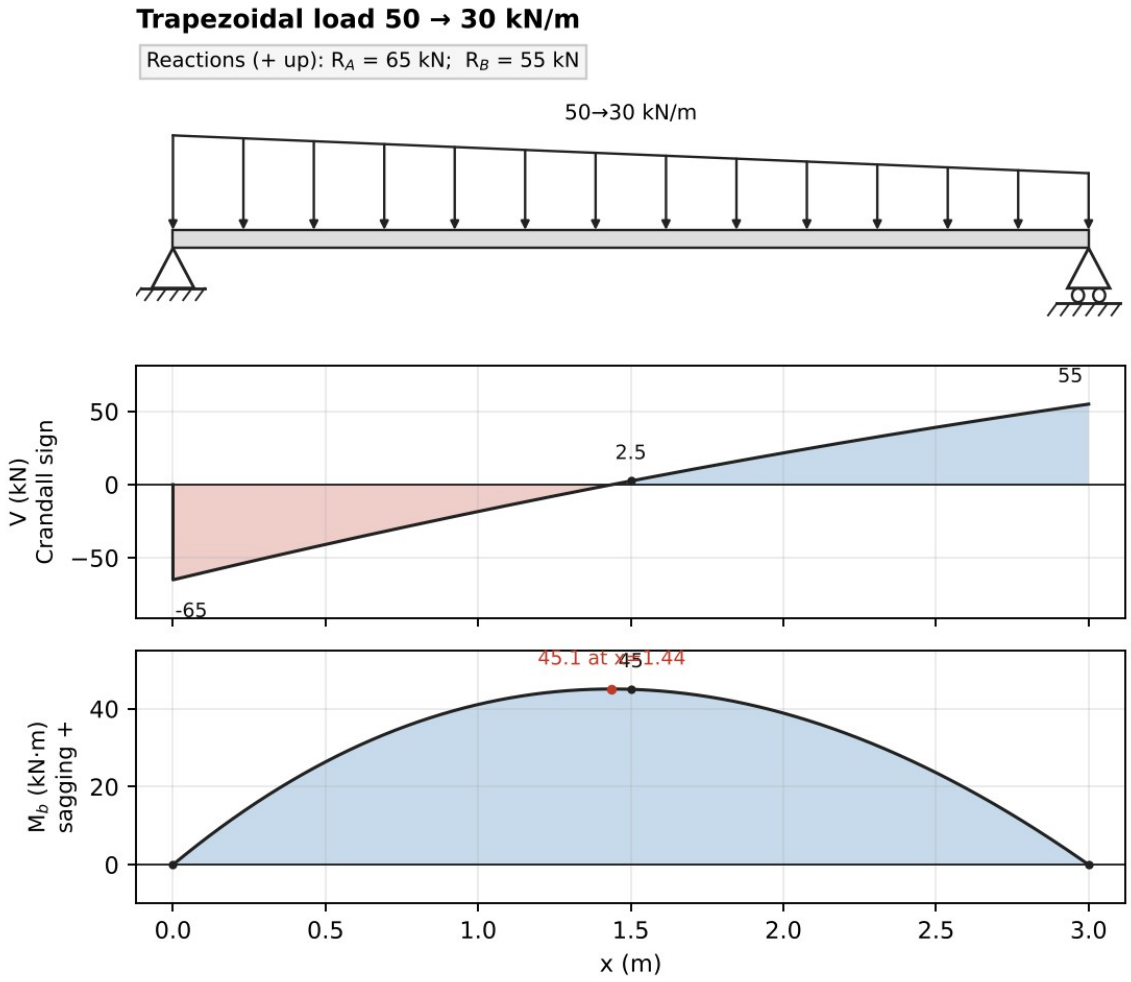
1. *Total load:* $\frac{50+30}{2}(3) = 120$ kN, at $\acute{x} = \frac{3(50+60)}{3(80)} = 1.375$ m from A.
2. *Reactions:* $R_B = 120(1.375)/3 = 55$ kN and $R_A = 65$ kN.

3. *Load on the left part:* intensity at 1.5 m is 40 kN/m. Resultant ¿ $\frac{50+40}{2}(1.5)=67.5$ kN at $\acute{x}=\frac{1.5(50+80)}{3(90)}=0.722$ m from A, i.e. 0.778 m from the cut.
4. *Shear (Crandall):* $V=-(65-67.5)=+2.5$ kN (Gere: −2.5 kN).
5. *Moment:* $M=65(1.5)-67.5(0.778)=97.5-52.5=45.0$ kN·m.

$$V=+2.5\,\text{kN}\,(\text{Crandall}),\ M=45.0\,\text{kN}\cdot\text{m}$$

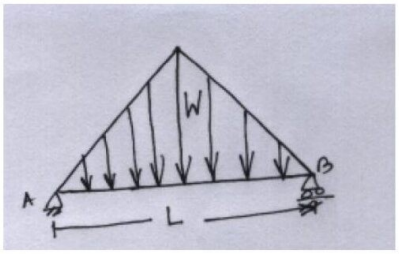
Check. The true maximum (45.08 kN·m) is at x = 1.44 m, where V = 0, just left of mid-span. That makes sense because the load is heavier on the left ✓.

Exam takeaway. Split a trapezoid into a rectangle plus a triangle if the centroid formula slips your mind.
Variation: find the position of M_{max}, by solving V(x) = 0 (a quadratic).



Solution

Q9 (Slides Ch3) — Symmetric triangular load of total W (easy)



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Original: Slides Ch 3, slide 26

Problem. A simply supported beam of span L carries a triangular load of total W, zero at both supports and peaking at mid-span. Find M_{max}. (Plotted for W = 12 kN, L = 6 m.)

Approach. Symmetry: $R_A = R_B = W/2$ and M_{max} is at mid-span.

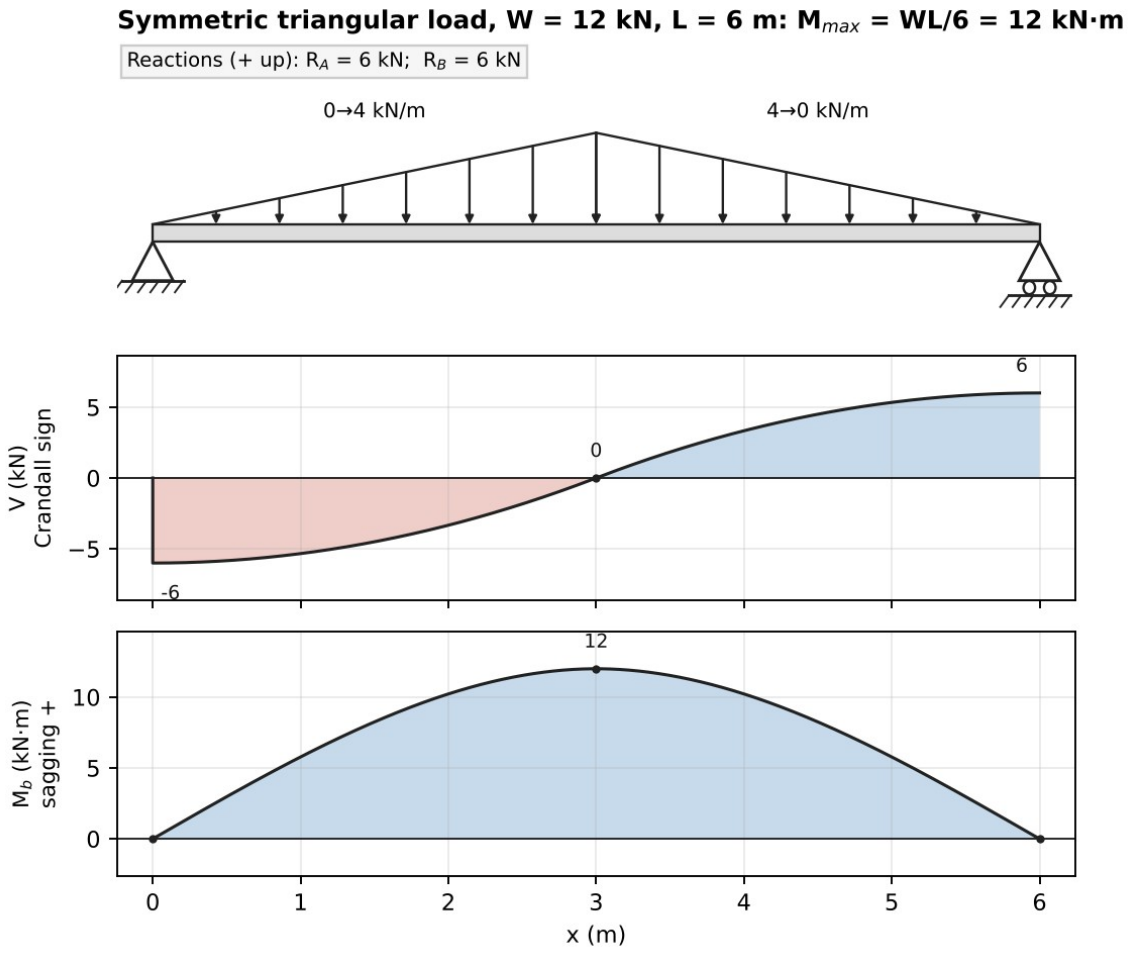
Solution.

1. Peak intensity q_0: $1/2\, q_0 L = W$, so $q_0 = 2W/L$. For $x \le L/2$: $q = 4Wx/L^2$.
2. Load to the left of x: $\int_0^x 4W\xi/L^2\, d\xi = 2Wx^2/L^2$, acting $x/3$ from the cut.
3. $M(x) = \frac{W}{2}x - \frac{2Wx^2}{L^2}\cdot\frac{x}{3} = \frac{W}{2}x - \frac{2W}{3L^2}x^3$.
4. At x = L/2: $M = \frac{WL}{4} - \frac{WL}{12}$.

$$M_{max} = \frac{WL}{6}\,(12\ \text{kN·m for } W = 12\ \text{kN}, L = 6\ \text{m})$$

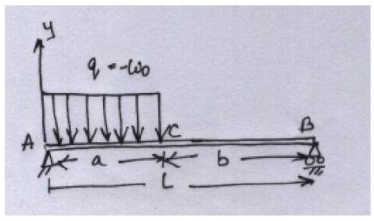
Check. A concentrated W at the centre gives WL/4, and a UDL of total W gives WL/8. A triangle concentrated toward the centre must lie between them: WL/6 ✓.

Exam takeaway. Compare against WL/4 and WL/8 as a quick sanity check on any distributed load. **Variation:** an unsymmetric triangle (Q11).



Solution with W = 12 kN, L = 6 m

Q10 (Slides Ch3) — Partial UDL w_0 over length a from A (simply supported span L) (medium)



Original: Slides Ch 3, slide 27

Problem. A simply supported beam of span L carries a UDL w_0 from A over a length a. Find the maximum bending moment and its position. (Plotted for $w_0 = 10$ kN/m, a = 3 m, L = 5 m.)

Approach. Reactions from the resultant $w_0 a$ at $a/2$. The maximum is where V = 0, which must lie inside the loaded part.

Solution.

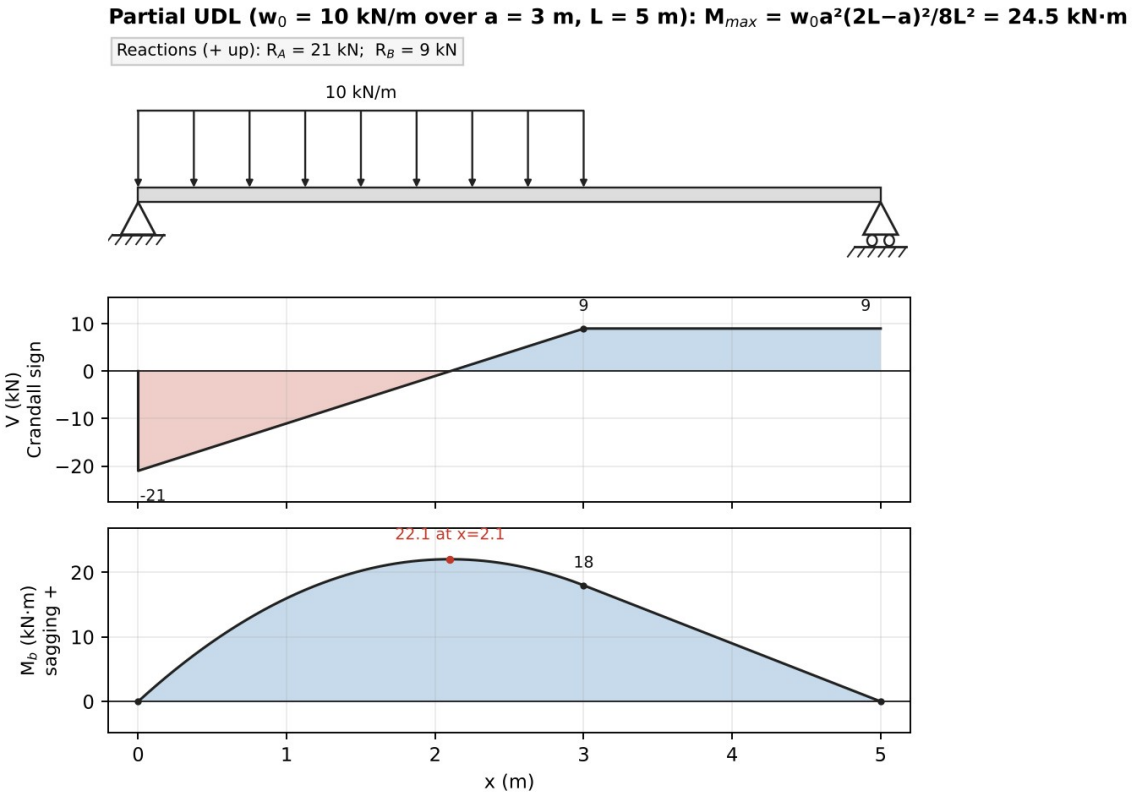
1. $R_B = \dfrac{w_0 a (a/2)}{L} = \dfrac{w_0 a^2}{2L}$ and $R_A = w_0 a - R_B = \dfrac{w_0 a (2L - a)}{2L}$.
2. V = 0 where the load to the left equals R_A: $w_0 x_0 = R_A$, so $x_0 = \dfrac{a(2L - a)}{2L}$. This is always ≤ a ✓.
3. $M_{max} = R_A x_0 - \dfrac{w_0 x_0^2}{2} = \dfrac{R_A^2}{2 w_0}$.

$$x_0 = \frac{a(2L - a)}{2L}, \; M_{max} = \frac{w_0 a^2 (2L - a)^2}{8 L^2}$$

For $w_0 = 10$, a = 3, L = 5: $R_A = 21$ kN, $x_0 = 2.1$ m, $M_{max} = 22.05$ kN·m.

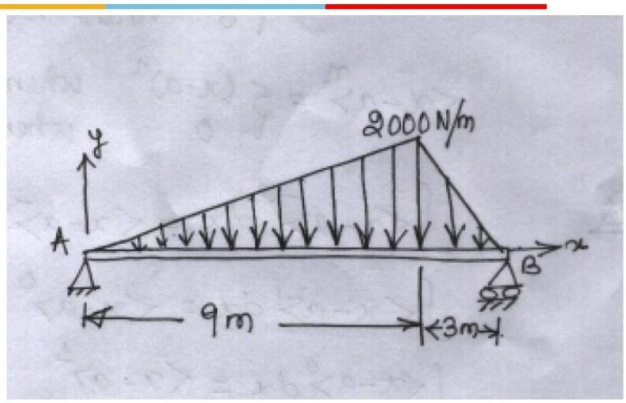
Check. a = L gives $w_0 L^2/8$ ✓.

Exam takeaway. $M_{max} = R_A^2/2 w_0$ is a neat shortcut whenever V reaches zero inside a UDL that starts at a support. **Variation:** UDL over the middle third.



Solution with w_O = 10 kN/m, a = 3 m, L = 5 m

Q11 (Slides Ch3) — Triangular load rising to 2000 N/m over 9 m, falling to zero over 3 m (medium)



BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Original: Slides Ch 3, slide 42

Problem. A simply supported 12 m beam carries a load rising linearly from 0 at A to 2000 N/m at x = 9 m, then falling linearly to 0 at B. Find the reactions and the maximum bending moment.

Approach. The whole load is one triangle with total $1/2(2000)(12)$ at its centroid. For the cut, use the part of the rising triangle to the left.

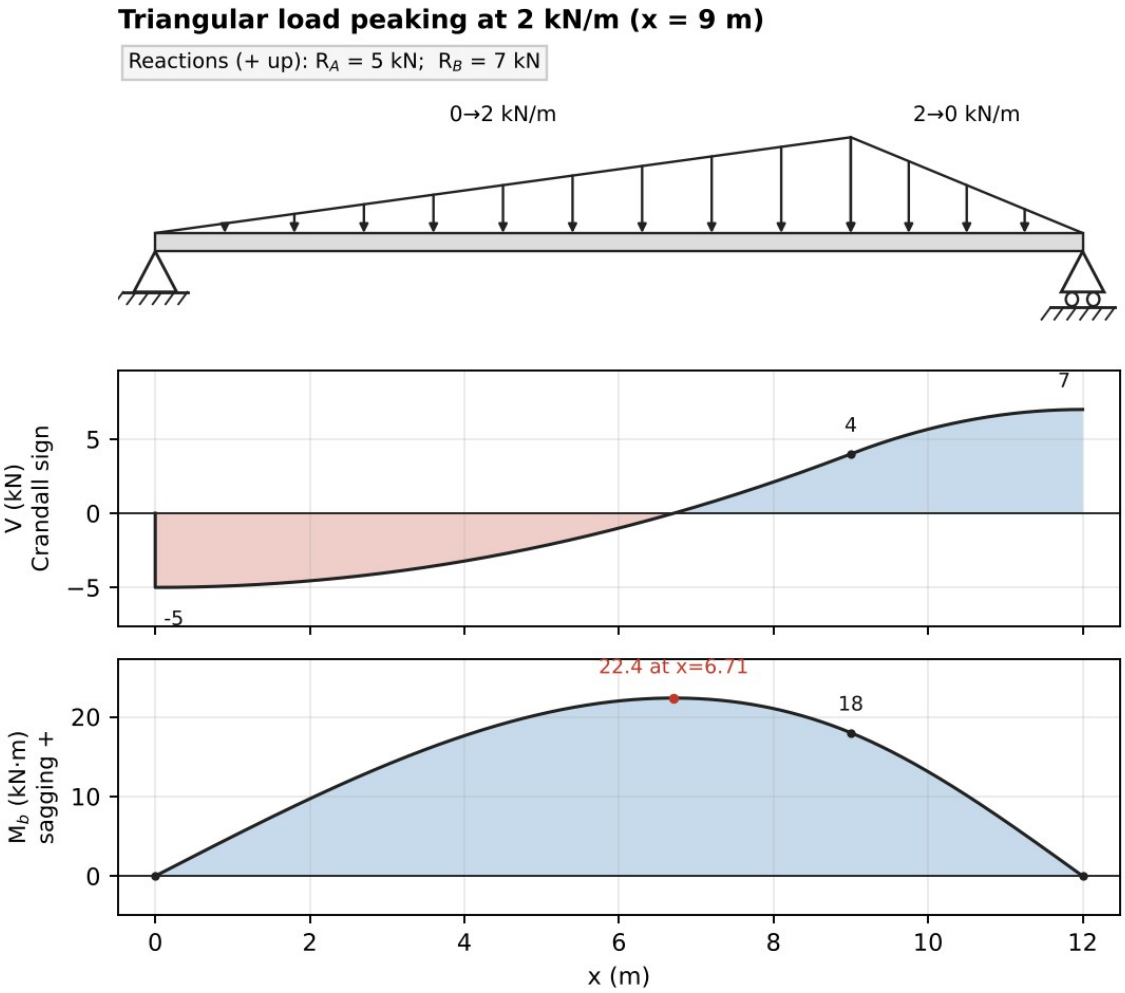
Solution.

1. $R_{total} = 1/2(2000)(12) = 12\,000$ N, at the triangle's centroid $\acute{x} = (0+9+12)/3 = 7$ m.
2. $R_B = 12\,000(7)/12 = 7000$ N and $R_A = 5000$ N.
3. For $x \le 9$: intensity $2000\,x/9$, load to the left $1000\,x^2/9$ at $x/3$ from the cut.
4. V = 0: $5000 = 1000\,x^2/9$, so $x = \sqrt{45} = 6.708$ m.
5. $M = 5000\,x - \dfrac{1000\,x^2}{9} \cdot \dfrac{x}{3}$ at $x = 6.708$: $33\,541 - 11\,180 = 22\,360$ N·m.

$$R_A = 5\,\text{kN},\ R_B = 7\,\text{kN},\ M_{max} = 22.36\,\text{kN·m at } x = 6.71\,\text{m}$$

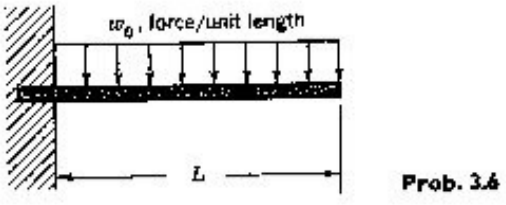
Check. The maximum lies inside the loaded part, slightly left of the load's centroid ✓.

Exam takeaway. The centroid of any triangle is at the average of its three vertices' x-coordinates. **Variation:** peak at mid-span gives Q9.



Solution

Q12 (Crandall Prob 3.6) — Cantilever with a full UDL (easy)



Prob. 3.6

Original: Crandall Prob 3.6 (textbook figure)

Problem. A cantilever of length L, built in at the left end, carries a UDL w_0 (force per unit length, downward) over its full length. Sketch V and M.

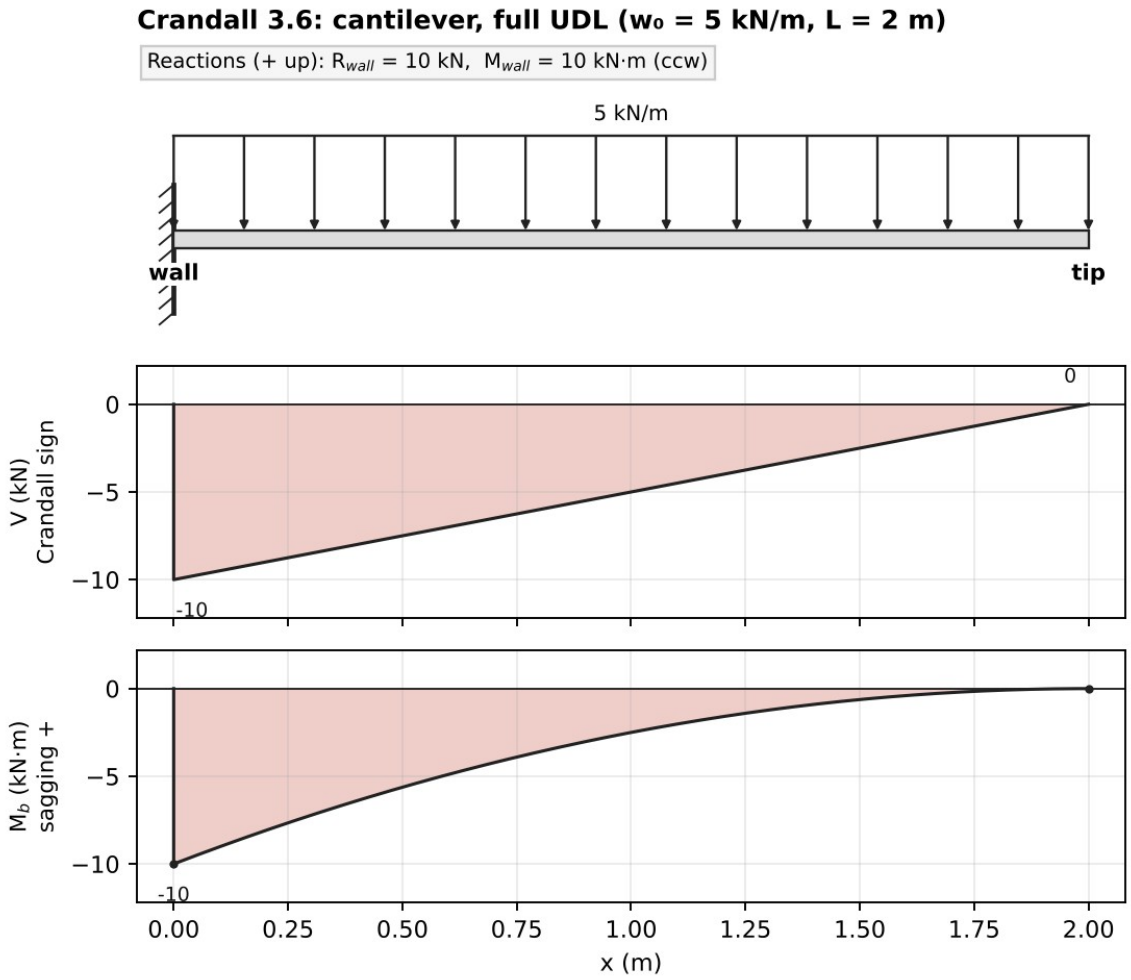
Solution.

1. *Wall reactions:* $R = w_0 L$ ↑ and a counter-clockwise moment $w_0 L^2/2$.
2. *Cut at x (keep the right part, length L − x, for simplicity):* the load on it is $w_0(L-x)$ at $(L-x)/2$ from the cut. Then V (Crandall, from the left part) ¿ $-(w_0 L - w_0 x) = -w_0(L-x)$ and $M = -\dfrac{w_0(L-x)^2}{2}$.

$$V = -w_0(L-x),\ M = -\frac{w_0(L-x)^2}{2},\ |M|_{max} = \frac{w_0 L^2}{2} \text{ at the wall}$$

Check. $dM/dx = w_0(L-x) = -V$ ✓ (Crandall relation).

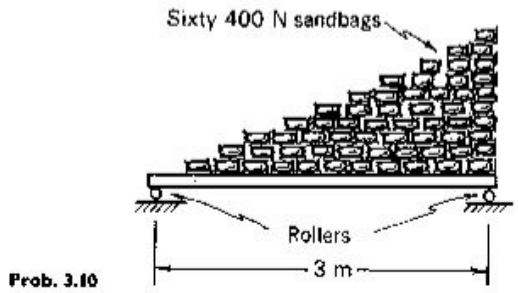
Exam takeaway. Keep whichever segment avoids the reactions; for a cantilever, keep the free end. **Variation:** Crandall 3.12 (Q21) adds a tip load.



Solution with $w_O = 5$ kN/m, $L = 2$ m

Q13 (Crandall Prob 3.10) — Sixty 400 N sandbags piled in a triangle (medium)

3.10. Find the reactions and expressions for the shear force and bending moment as functions of distance along the beam.



Prob. 3.10

Original: Crandall Prob 3.10 (textbook figure)

Problem. Sixty 400 N sandbags are piled on a 3 m beam supported on rollers at both ends, with the pile height increasing linearly from zero at the left to a maximum at the right. Find the reactions and expressions for V and M as functions of x.

Given: total load W = 60(400) = 24 kN, triangular from 0 (left) to q_0 (right); L = 3 m. **Find:** reactions, V(x), M(x), M_{max}.

Approach. Treat the pile as a triangular distributed load: $1/2\,q_0\,L = W$ gives $q_0 = 16$ kN/m. Use the resultant for reactions, and the left portion for V and M.

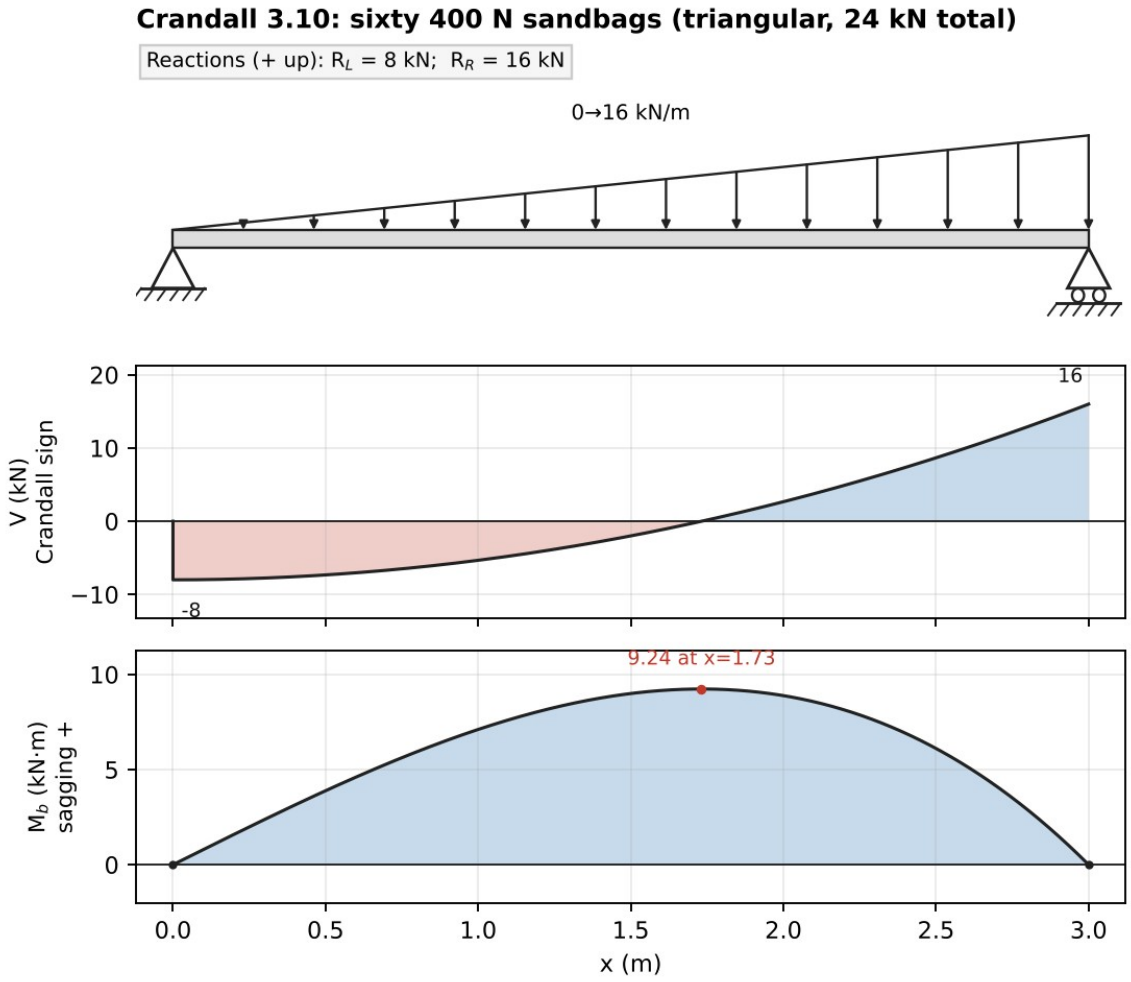
Solution.

1. *Reactions:* the resultant (24 kN) is at 2L/3 = 2 m from the left. $R_R = 24(2)/3 = 16$ kN and $R_L = 8$ kN.
2. *Intensity:* $q(x) = 16x/3$ kN/m. The load to the left of x is $1/2\,x \cdot 16x/3 = 8x^2/3$, at x/3 from the cut.
3. *Shear (Crandall):* $V(x) = -\left(8 - \frac{8x^2}{3}\right)$ kN.
4. *Moment:* $M(x) = 8x - \frac{8x^2}{3}\cdot\frac{x}{3} = 8x - \frac{8x^3}{9}$ kN·m.
5. *Maximum:* V = 0 at $x^2 = 3$, so $x = \sqrt{3} = 1.732$ m, and $M_{max} = 8(1.732) - 8/9(5.196) = 13.86 - 4.62 = 9.24$ kN·m.

$$R_L = 8\,\text{kN},\ R_R = 16\,\text{kN};\ V = -8 + 8/3\,x^2,\ M = 8x - 8/9\,x^3\,(\text{kN, m});\ M_{max} = 9.24\,\text{kN·m at } x = 1.73\,\text{m}$$

Check. The formula $M_{max} = 2WL/(9\sqrt{3}) = 2(24)(3)/15.59 = 9.24$ ✓. M(3) = 24 − 24 = 0 ✓.

Exam takeaway. A "pile of objects" is a distributed load; the slope of the pile gives the shape of q(x). **Variation:** a pile of uniform height gives the UDL result $WL/8 = 9$ kN·m. The triangle gives slightly more, because the maximum moves toward the heavy end where the arm is still large.



Solution

Q14 (Crandall Prob 3.34) — Wooden dam: SFD and BMD of an upright (medium)

Problem. A wooden dam consists of planks fastened to uprights driven into the river bed, 2.4 m apart. The water is 1.8 m deep. Draw the SFD and BMD for an upright.

Given: water (γ = 9.81 kN/m^3), depth h = 1.8 m, spacing s = 2.4 m. Treat each upright as a cantilever fixed at the river bed and carrying the water pressure on a 2.4 m width of planks. **Find:** V and M along the upright.

Approach. Hydrostatic pressure $\gamma(h - y)$ at height y above the bed, times the tributary width s, gives a triangular line load on the upright: zero at the surface, maximum at the bed.

FBD in words. Upright with y measured up from the bed. Load $q(y) = \gamma s(h - y)$ acting horizontally (downstream). Fixed base reactions: shear V_0 and moment M_0.

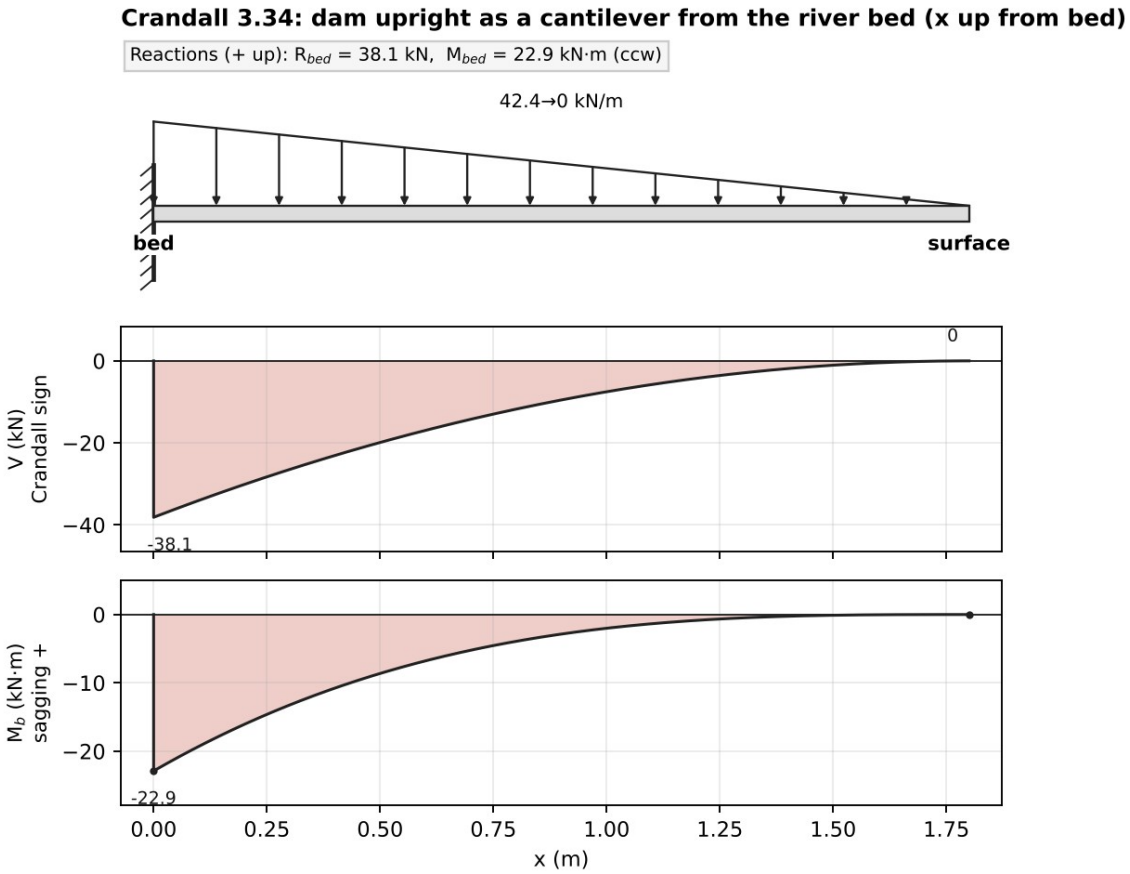
Solution.

1. *Load intensity at the bed:* $q_0 = 9.81(1.8)(2.4) = 42.38$ kN/m. In general $q(y) = 23.54(1.8 - y)$ kN/m.
2. *Shear at height y* (load above y): $V(y) = 1/2\,\gamma s(h - y)^2 = 11.77(1.8 - y)^2$ kN.
3. *Moment at height y:* that load acts at $(h - y)/3$ above y, so $M(y) = 1/6\,\gamma s(h - y)^3 = 3.924(1.8 - y)^3$ kN·m.
4. *At the bed (y = 0):* $V_0 = 11.77(3.24) = 38.1$ kN and $M_0 = 3.924(5.832) = 22.9$ kN·m.

$$V_{max} = 38.1\,\text{kN},\ M_{max} = 22.9\,\text{kN·m, both at the river bed}$$

Check. Total water force on 2.4 m of dam: $1/2\,\gamma h^2 s = 1/2(9.81)(3.24)(2.4) = 38.1$ kN, acting at h/3 = 0.6 m above the bed: $38.1(0.6) = 22.9$ ✓.

Exam takeaway. Hydrostatic loads are triangular, with the resultant at one-third of the depth from the bottom. The V diagram is a parabola and the M diagram a cubic, both maximum at the base. **Variation:** uprights also supported by a strut at the top, which turns the problem into a propped cantilever (indeterminate).



Upright drawn horizontally: x measured up from the river bed

Q15 (PYQ 2014-15 Compre A Q6, Q7) — Simply supported beam with a full UDL (easy)

Problem. A simply supported beam carries w N/m over the whole span L. (Q6) What is the shear force at mid-span? (Q7) What is the bending moment at mid-span?

Solution.

1. $R_A = R_B = wL/2.$
2. At L/2 the load to the left is wL/2, which cancels R_A: V = 0.
3. $M(L/2) = \frac{wL}{2}\cdot\frac{L}{2} - \frac{wL}{2}\cdot\frac{L}{4} = \frac{wL^2}{8}.$

$$V(L/2) = 0,\ M(L/2) = \frac{wL^2}{8}$$

Exam takeaway. A one-line answer; memorise it.

Concept 3.3 — Differential relations between q, V and M

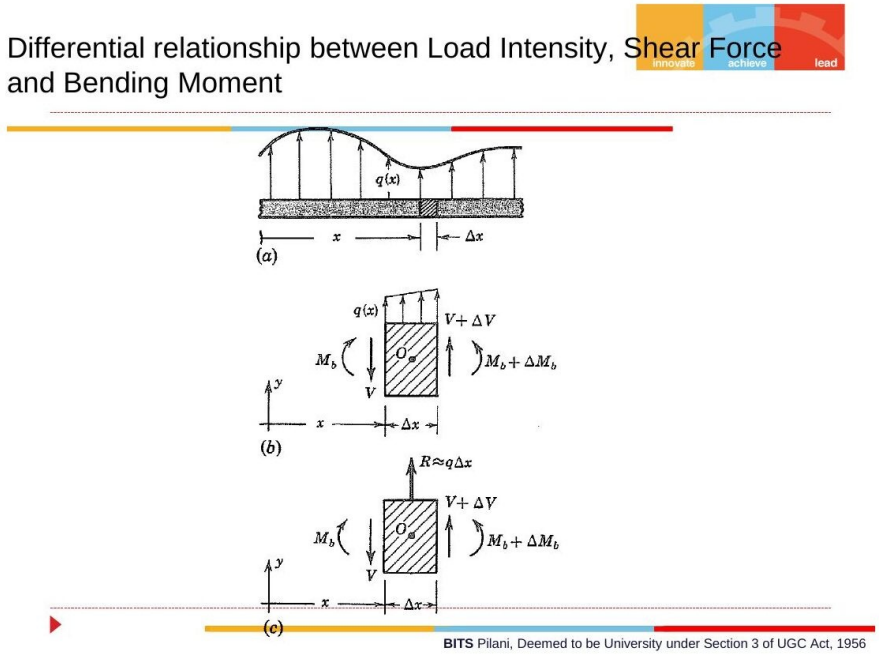
A. Concept explanation

1. Intuition. The load intensity is the "slope-maker" for the shear diagram, and the shear is the "slope-maker" for the moment diagram. Knowing this, you can sketch an SFD and BMD almost without algebra: the load tells you how V changes, V tells you how M changes, and the largest M sits where V crosses zero.

2. Why it matters. Fast, reliable diagrams; finding M_{max} (where V = 0); and the starting point for the beam-deflection chapter ($EIv'''' = q$). The slides derive these relations explicitly.

3–4. Formal treatment and derivation (slides; Crandall §3.5). Take an element Δx with a distributed load q(x) (upward +). On its left (negative) face act V and M_b, drawn ↓ and clockwise (positive on a negative face). On its right (positive) face act $V+\Delta V$ and $M_b+\Delta M_b$, drawn ↑ and counter-clockwise.

- $\sum F_y=0: (V+\Delta V)-V+q\,\Delta x=0$, so $\dfrac{\Delta V}{\Delta x}=-q.$
- $\sum M_O=0$ about the centre of the element: $(M_b+\Delta M_b)-M_b+(V+\Delta V)\dfrac{\Delta x}{2}+V\dfrac{\Delta x}{2}=0.$



Original: Slides Ch 3, slide 31

Let Δx → 0:

$$\frac{dV}{dx}+q=0,\ \frac{dM_b}{dx}+V=0 \Rightarrow \frac{d^2M_b}{dx^2}=q$$

Integral forms: $V_2-V_1=-\int_1^2 q\,dx$ (minus the load area) and $M_2-M_1=-\int_1^2 V\,dx$ (minus the SFD area).

Concentrated load P (↓) at a point: integrate across it, so V jumps by **+P** (Crandall). *Concentrated clockwise couple C:* M jumps by **+C**.

5. Reading diagrams (Crandall signs; a downward load means q < 0).

Load on the segment	SFD	BMD
None	constant	linear
UDL downward	linear, **rising** (slope $-q>0$)	parabola, concave down (∩)
Linearly varying load	parabola	cubic
Concentrated downward P	jumps **up** by P	kink (slope change)
Clockwise couple C	no change	jumps **up** by C

M is extreme where V = 0, or where V jumps through zero at a concentrated load.

6. Special cases.

- At a free end, or a pinned end with no applied couple: M = 0.

- At a free end with no point load: V = 0.
- Where q = 0 and V = 0 over a stretch: M is constant (pure bending).

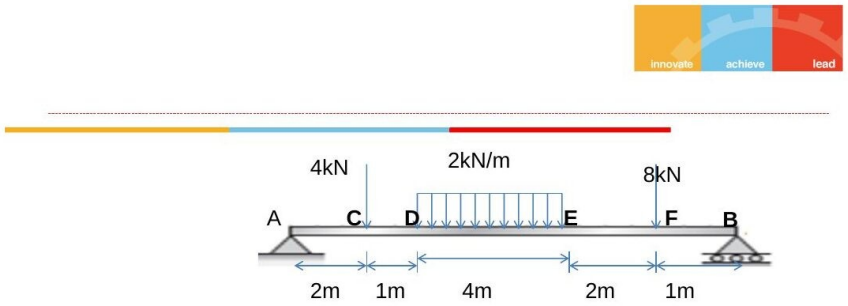
7. Common traps.

- Using Gere's formulas ($dM/dx = +V$) with Crandall's signs for V turns the BMD upside down. Pick one convention and stay with it.
- Missing a maximum inside a distributed load: always solve V = 0 there.
- A **point of contraflexure** is where M **changes sign** (passes through zero), not where V = 0.
- At a support carrying an overhang, V changes sign at the support (Q20), so the hogging maximum is there.

8. Summary. $V' = -q$ and $M' = -V$ (Crandall). The area of the load diagram gives the change in V; minus the area of the SFD gives the change in M.

B. Solved questions

Q16 (Slides Ch3) — 4 kN, 2 kN/m and 8 kN on a 10 m span (medium)



DRAW SFD AND BMD

Original: Slides Ch 3, slide 30

Problem. A simply supported beam (A at 0, B at 10 m) carries 4 kN at C (2 m), a UDL of 2 kN/m from D (3 m) to E (7 m) and 8 kN at F (9 m). Draw the SFD and BMD.

Approach. Reactions, then V at each key point using jumps and slopes, then M from the SFD areas.

Solution.

1. *Reactions:* $\sum M_A$: $R_B(10) = 4(2) + 8(5) + 8(9) = 120$, so $R_B = 12$ kN and $R_A = 20 - 12 = 8$ kN.
2. *Key values:*

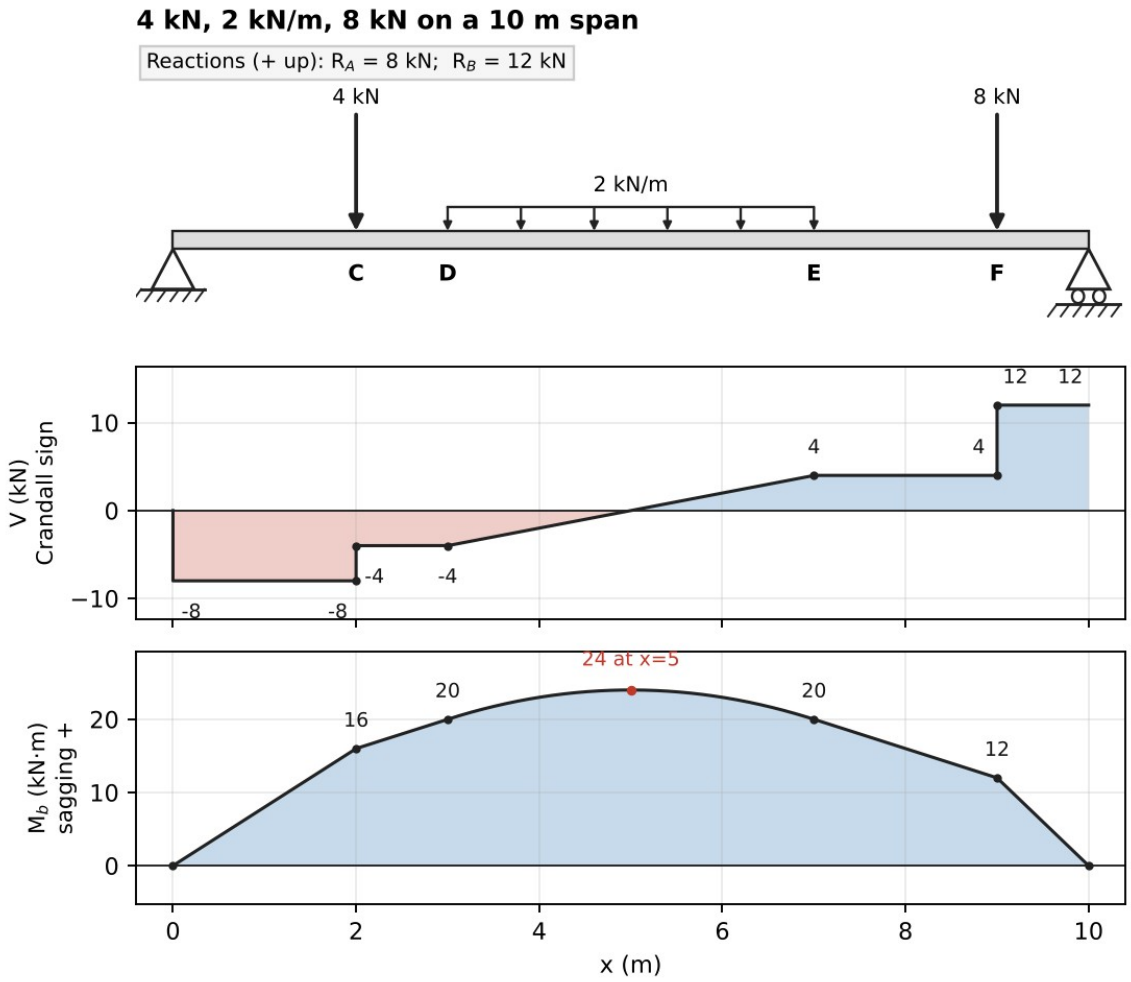
Point	A	C⁻	C⁺	D	x = 5	E	F⁻	F⁺	B
V (kN, Crandall)	−8	−8	−4	−4	0	+4	+4	+12	+12
M (kN·m)	0	16	16	20	**24**	20	12	12	0

3. *Between D and E:* $M = 8x - 4(x-2) - (x-3)^2$, and $dM/dx = 0$ at x = 5, where $M = 40 - 12 - 4 = 24$.

$$M_{max} = 24\ \text{kN·m at mid-span}\ (x = 5\ \text{m});\ \text{no point of contraflexure}$$

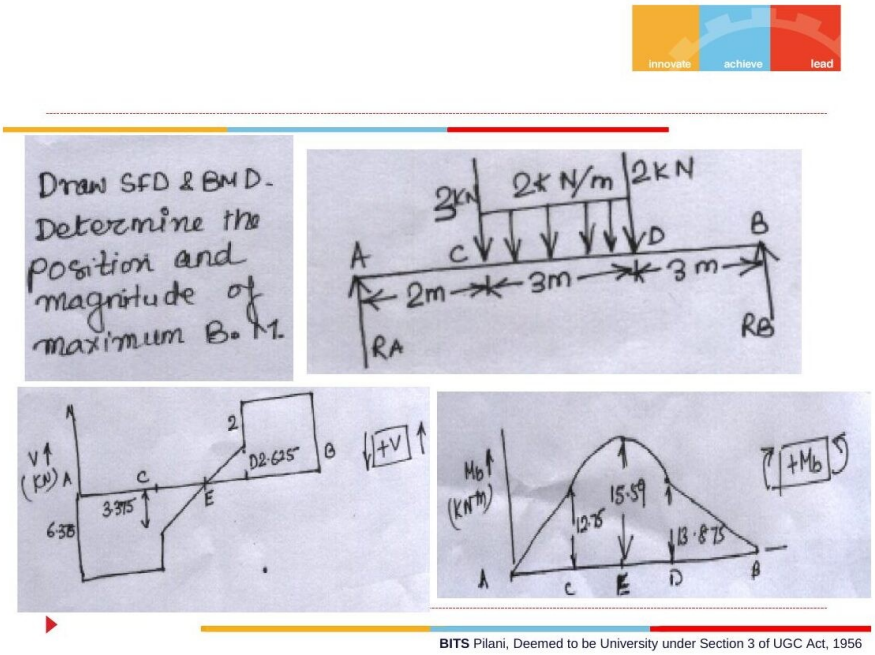
Check. The SFD area from A to the maximum is $8(2)+4(1)+1/2(4)(2)=24$, which equals M_{max} ✓ ($M = -\int V\,dx$ with Crandall's V negative there).

Exam takeaway. Tabulating V and M at every load point (with $^-$ and $^+$ sides) is exactly what the slide's table asks for. **Variation:** move the 8 kN onto an overhang beyond B; a hogging region then appears.



Solution

Q17 (Slides Ch3) — 3 kN + UDL + 2 kN: position and value of the maximum BM (medium)



Original: Slides Ch 3, slide 41

Problem. A simply supported 8 m beam carries 3 kN at C (2 m), a UDL of 2 kN/m from C to D (2–5 m) and 2 kN at D (5 m). Draw the SFD and BMD and find the position and magnitude of the maximum bending moment.

Solution.

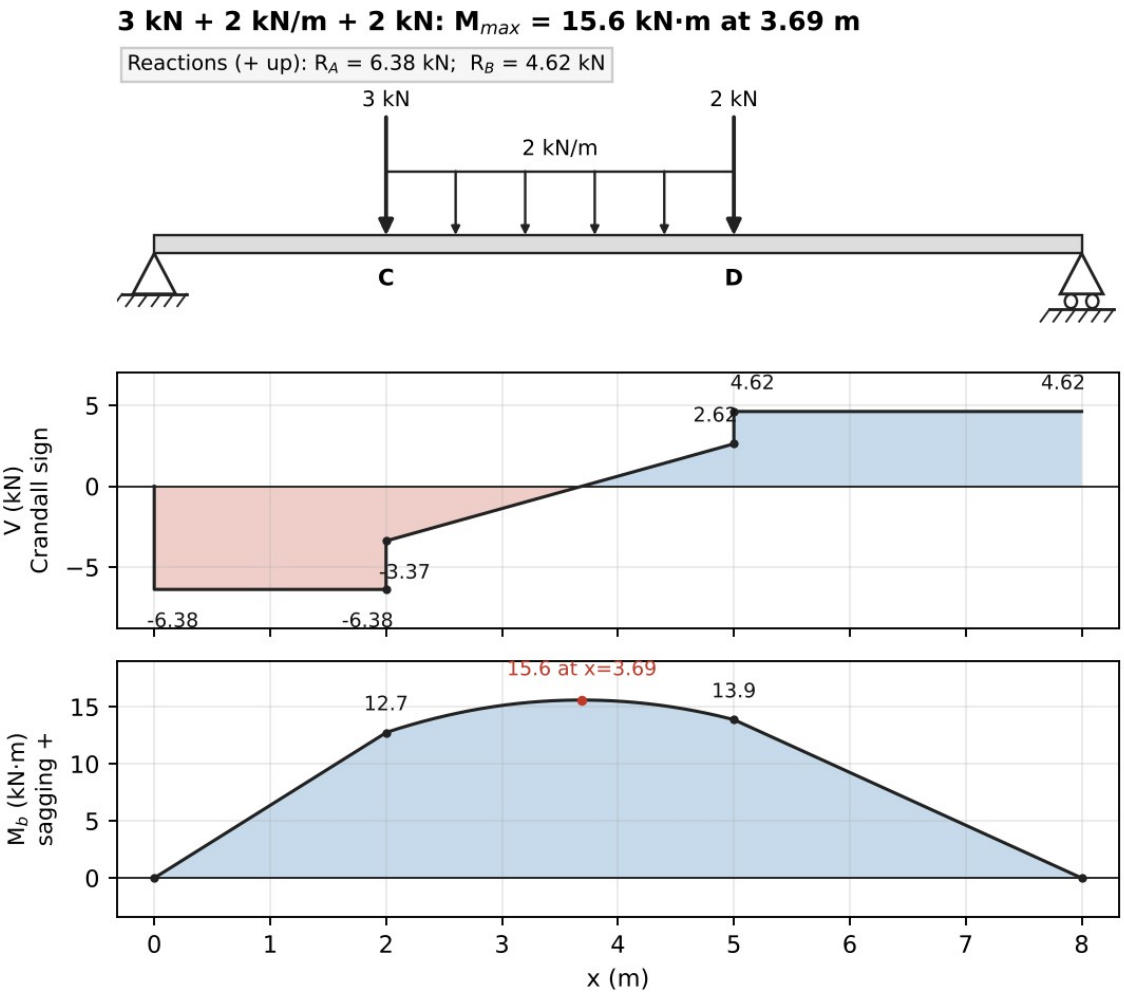
1. *Reactions:* $\sum M_A$: $R_B(8)=3(2)+6(3.5)+2(5)=37$, so $R_B=4.625$ kN and $R_A=11-4.625=6.375$ kN.
2. *V = 0 inside CD:* $6.375-3-2(x-2)=0$, so x = 3.6875 m (point E).
3. *Moments:* $M_C=6.375(2)=12.75$;
   $M_E=6.375(3.6875)-3(1.6875)-2(1.6875)^2/2=23.508-5.063-2.848=15.60$;
   $M_D=4.625(3)=13.875$ kN·m.

Point	A	C⁺	E	D⁻	D⁺	B
V (Crandall)	−6.375	−3.375	0	+2.625	+4.625	+4.625
M (kN·m)	0	12.75	**15.60**	13.875	13.875	0

$$M_{max}=15.6\,\text{kN·m at } x=3.69\,\text{m from A}$$

Check. These match the slide's hand solution (6.38, 3.375, 2.625, 12.75, 15.59, 13.875) ✓.

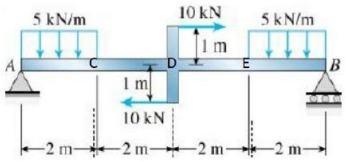
Exam takeaway. "Position and value of maximum BM" means solve V = 0 inside the UDL.



Solution

Q18 (Slides Ch3) — 5 kN/m segments plus an offset couple (medium–hard)

Draw the shear force and bending moment diagram (mention the sign convention) for the beam AB as shown in figure. *Write the values of shear force and bending moment at salient points clearly in a table.* Also write the values of shear force and bending moment at salient points clearly in the shear force and bending moment diagram. ***If any discontinuity point is left in the table please add it.***



	A	C	D		E		B
			D-	D+	E-	E+	
V (kN)							
M_b(kNm)							

Original: Slides Ch 3, slide 25

Problem. A simply supported beam AB, 8 m long, carries 5 kN/m over AC (0–2 m) and over EB (6–8 m). At D (4 m) a vertical arm carries 10 kN → at 1 m above the beam and 10 kN ← at 1 m below it. Fill the V/M table at A, C, D^-, D^+, E and B.

Approach. The two arm forces cancel as forces but form a couple. Replace them by that couple at D (Concept 1.2).

Solution.

1. *Arm loads:* moment about D ¿ $10(1)+10(1)=20$ kN·m **clockwise**, applied at D; no net force.
2. *Reactions:* $\sum M_A$: $R_B(8)-10(1)-10(7)-20=0$, so $R_B=12.5$ kN and $R_A=7.5$ kN.
3. *Moments:* AC: $M=7.5x-2.5x^2$; CD: $M=10-2.5x$, falling to 0 at D^-; at D, +20 (clockwise couple); DE: $M=30-2.5x$; EB: $M=30-2.5x-2.5(x-6)^2$.

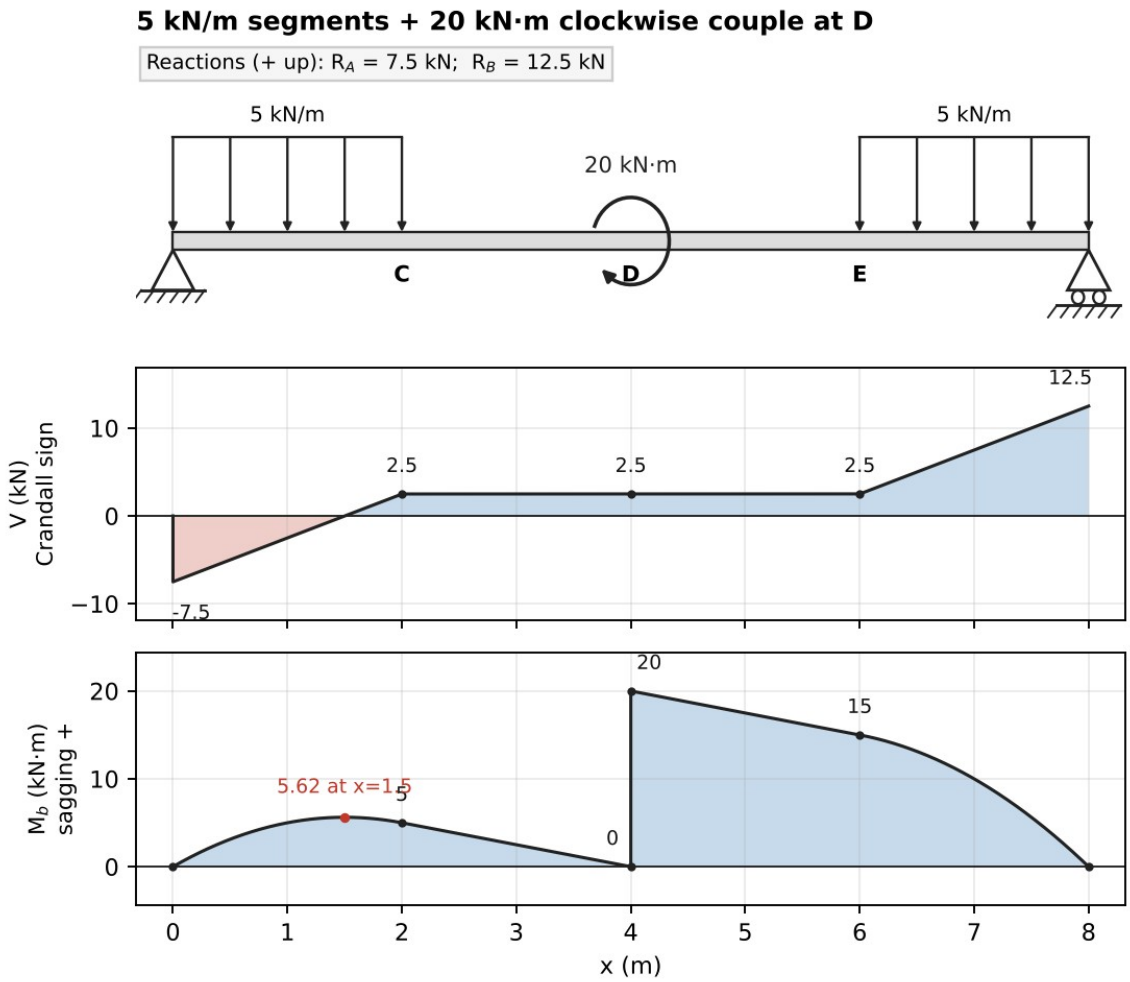
	A	C	D^-	D^+	E	B
V (kN, Crandall)	−7.5	+2.5	+2.5	+2.5	+2.5	+12.5
M_b (kN·m)	0	5	0	**20**	15	0

There is also a local maximum of 5.625 kN·m at x = 1.5 m (where V = 0 in AC).

$$M_{max}=20\text{ kN·m at } D^{+}$$ ¿¿

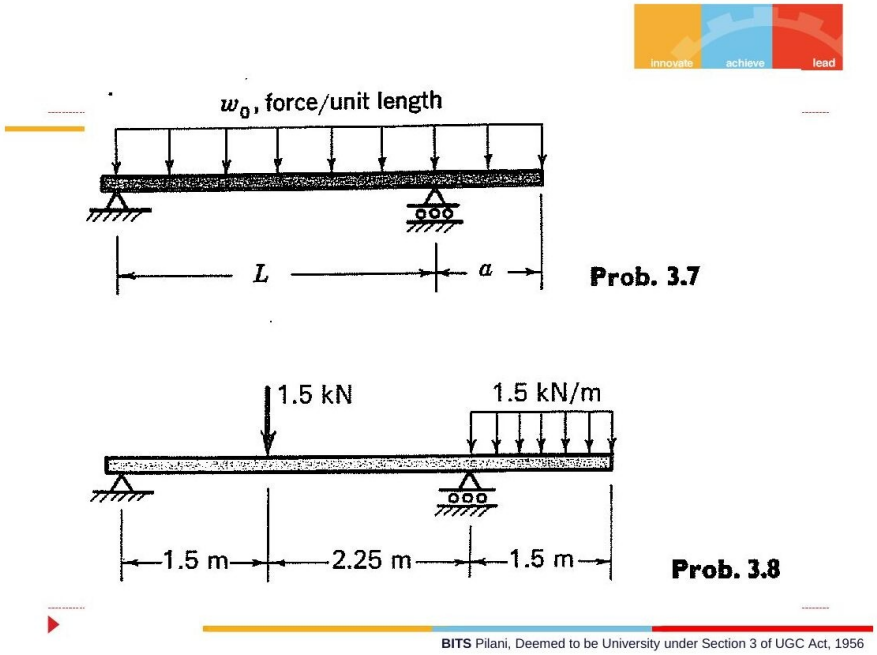
Check. From the right: M ¿ ✓.

Exam takeaway. Horizontal forces on an offset arm add **no** vertical shear but do add a couple. The slide's "if any discontinuity point is left in the table please add it" is a hint about the jump at D.



Solution: the arm forces become a 20 kNm clockwise couple at D

Q19 (Crandall Prob 3.7) — Uniform load over a beam with an overhang (medium)



Original: Slides Ch 3, slide 40

Problem. A beam is supported at x = 0 and x = L and overhangs a distance a beyond the right support. A uniform load w_0 acts over the whole length L + a. Sketch V and M and label the important values.

Solution.

1. *Reactions:* $\sum M_A$: $R_B L = w_0 (L+a)\dfrac{L+a}{2}$, so $R_B = \dfrac{w_0 (L+a)^2}{2L}$ and $R_A = \dfrac{w_0 (L^2 - a^2)}{2L}$.

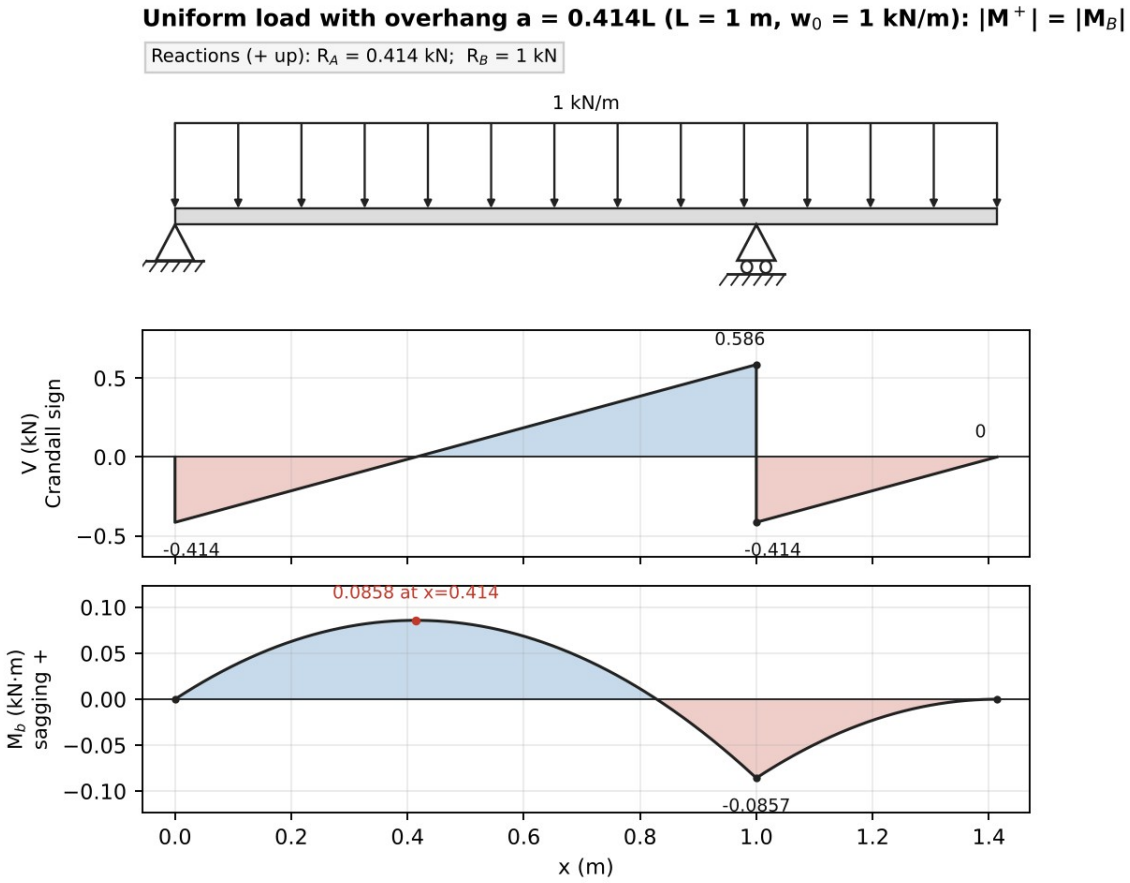
2. *Key values:*

Point	V (Crandall)	M
A	$-R_A$	0
In the span, V = 0 at $x_0 = (L^2 - a^2)/2L$	0	$\dfrac{+w_0 (L^2 - a^2)^2}{8L^2}$
B⁻	$\dfrac{+w_0 (L^2 + a^2)}{2L}$	$-\dfrac{w_0 a^2}{2}$
B⁺	$-w_0 a$	$-\dfrac{w_0 a^2}{2}$
Tip	0	0

$$M_{max} +¿ = \frac{w_0 (L^2 - a^2)^2}{8L^2}, M_B = -\frac{w_0 a^2}{2} ¿$$

Check. a = 0 gives the simple-span result $w_0 L^2/8$ ✓. Equalizing magnitudes gives $a = 0.414\,L$ (plotted).

Exam takeaway. Overhangs create hogging at the support and a contraflexure point in the span. **Variation:** Crandall 3.18, a scaffold plank with overhangs on both sides.



Solution for the optimum overhang a = 0.414L

Q20 (Crandall Prob 3.8) — Overhanging beam with 1.5 kN and 1.5 kN/m (medium)

Problem. A pin at A (x = 0) and a roller at B (x = 3.75 m) support a beam with a 1.5 m overhang. 1.5 kN acts at x = 1.5 m and 1.5 kN/m over the overhang (3.75–5.25 m). Sketch V and M.

Solution.

1. *Reactions:* $R_B(3.75) = 1.5(1.5) + 2.25(4.5)$, so $R_B = 3.30$ kN and $R_A = 3.75 - 3.30 = 0.45$ kN.
2. *Key values:*

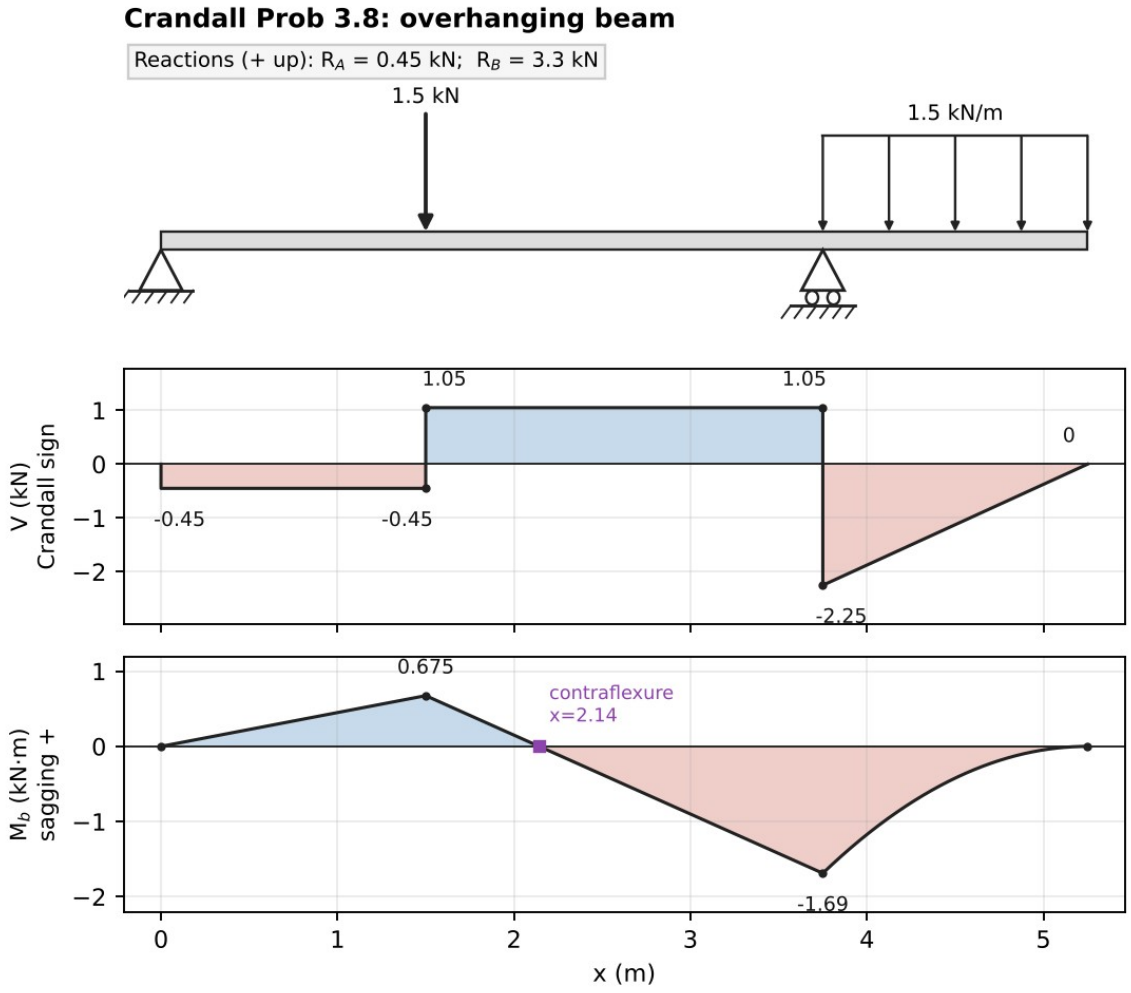
	A	load⁻	load⁺	B⁻	B⁺	tip
V (kN, Crandall)	−0.45	−0.45	+1.05	+1.05	−2.25	0
M (kN·m)	0	0.675	0.675	−1.69	−1.69	0

3. *Contraflexure:* $0.45x - 1.5(x - 1.5) = 0$, so x = 2.14 m.

$$M_{max}^{+} = 0.675 \text{ kN·m at the load; } M_B = -1.69 \text{ kN·m; contraflexure at } x = 2.14 \text{ m}$$

Check. M_B from the overhang: $-1.5(1.5)^2/2 = -1.6875$ ✓.

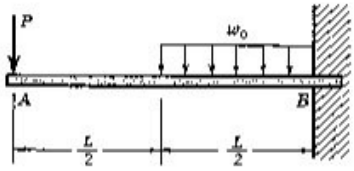
Exam takeaway. A support next to an overhang is usually where the largest moment sits. This same beam is re-solved with singularity functions in Q24.



Solution

Q21 (Crandall Prob 3.12) — Cantilever with a tip load and a UDL on the inner half (medium)

a concentrated force P and a distributed load of intensity w_0 force per unit length.



Prob. 3.12

Original: Crandall Prob 3.12 (textbook figure)

Problem. A cantilever AB is built in at B (right end) and free at A (left end). It carries a concentrated force P at A and a distributed load w_0 over the half nearest the wall (from L/2 to L). Sketch V and M.

Approach. Measure x from the free end A, so no reactions are needed.

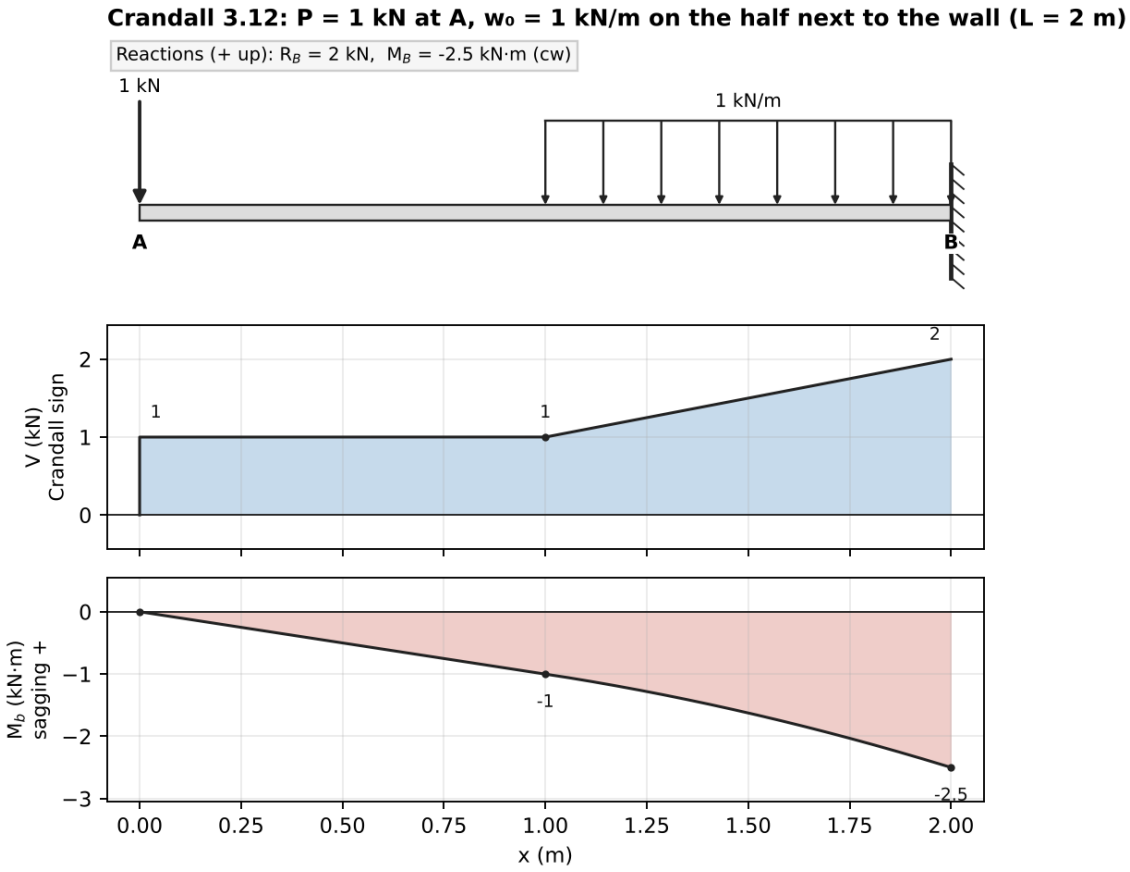
Solution.

1. $0<x<L/2$: only P is to the left. $V=-(-P)=+P$ (Crandall) and $M=-Px$.
2. $L/2<x<L$: $V=P+w_0(x-L/2)$ and $M=-Px-\dfrac{w_0(x-L/2)^2}{2}$.
3. *At the wall B:* $V_B=P+\dfrac{w_0 L}{2}$ and $M_B=-PL-\dfrac{w_0 L^2}{8}$.

$$V_B=P+\frac{w_0 L}{2},\ M_B=-PL-\frac{w_0 L^2}{8}\ (\text{hogging, the maximum})$$

Check. $dM/dx=-P-w_0(x-L/2)=-V$ ✓.

Exam takeaway. For cantilevers, start from the free end. The wall values equal the totals of the loads and of their moments about the wall.



Solution with $P = 1$ kN, $w_0 = 1$ kN/m, $L = 2$ m

Q22 (Crandall Prob 3.17) — The carpenter and the sawhorses (medium, a classic)

Problem. A carpenter has a 6 m plank of uniform weight w_0 per unit length and two sawhorses. He wants to cut off a 1.8 m length at a point where the bending moment is zero, to avoid splitting. If one sawhorse is at one end of the plank, where should he put the other?

Approach. Unknown position s of the second sawhorse. Its reaction from moments about the first, then set M = 0 at the cut (x = 4.2 m from the first sawhorse).

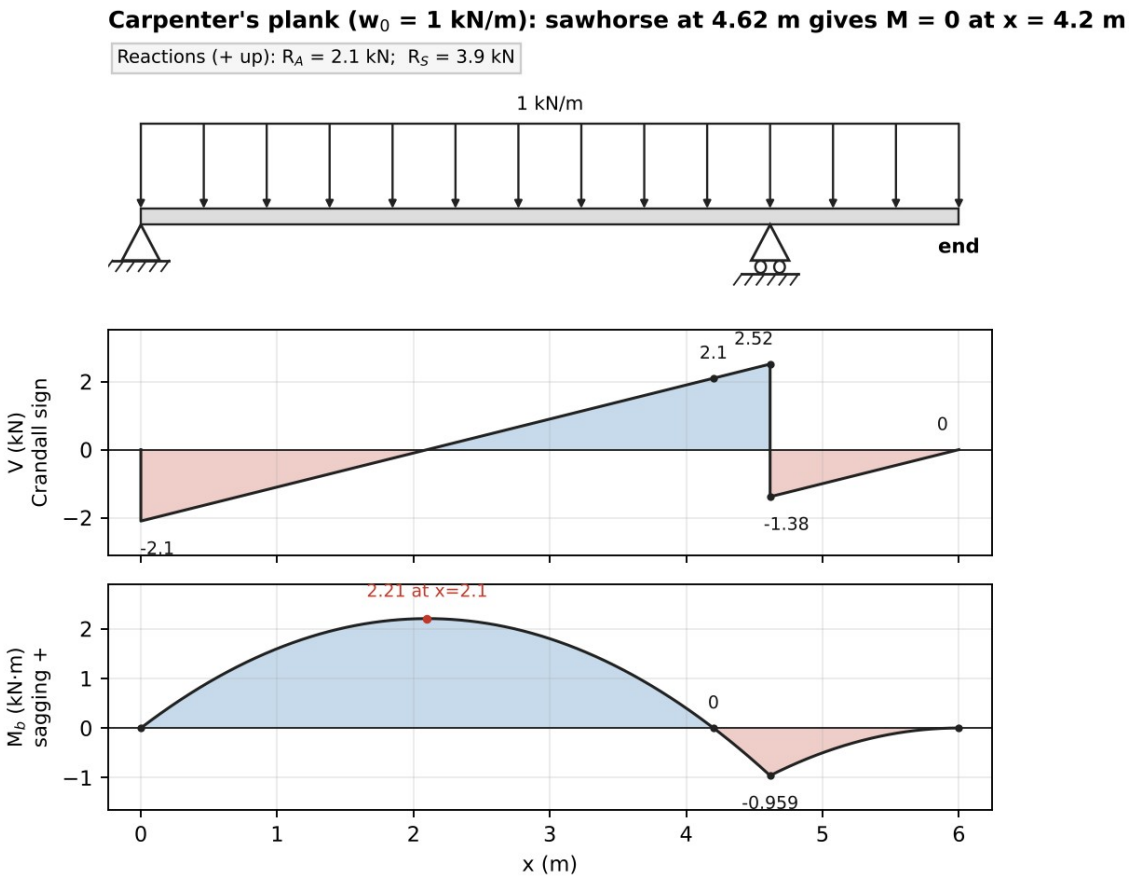
Solution.

1. *Reaction at s:* $R_s = \dfrac{w_0(6)(3)}{s} = \dfrac{18\,w_0}{s}$.
2. *For* $4.2 < s < 6$, take moments of the right part about the cut at 4.2 m: $R_s(s-4.2) - w_0(1.8)(0.9) = 0$.
3. $18 - \dfrac{75.6}{s} = 1.62$, so $s = \dfrac{75.6}{16.38} = 4.615$ m.

$$s = 4.62\text{ m from the first end}\,(1.38\text{ m from the far end})$$

Check. If $s < 4.2$, the cut would lie in the overhang, where $M = -w_0(1.8)^2/2 \neq 0$, so there is no solution there ✓.

Exam takeaway. Zero-moment points (contraflexure) are where a member can be hinged or cut without disturbing equilibrium. **Variation:** Crandall 3.18 (scaffold) and Q23 (bookshelf) optimise the same idea.



Solution: sawhorse at 4.62 m, M = 0 at the cut (x = 4.2 m)

Q23 (Crandall Prob 3.25) — Bookshelf on two bricks: minimise the maximum moment (medium)

Problem. A bookshelf is a plank of length L resting on two bricks placed symmetrically, a distance a from each end. With a uniform load w (books plus plank), where should the bricks go to make the maximum bending moment as small as possible?

Approach. Two candidates for $|M|_{max}$: hogging over a support and sagging at mid-span. The optimum makes them equal in magnitude.

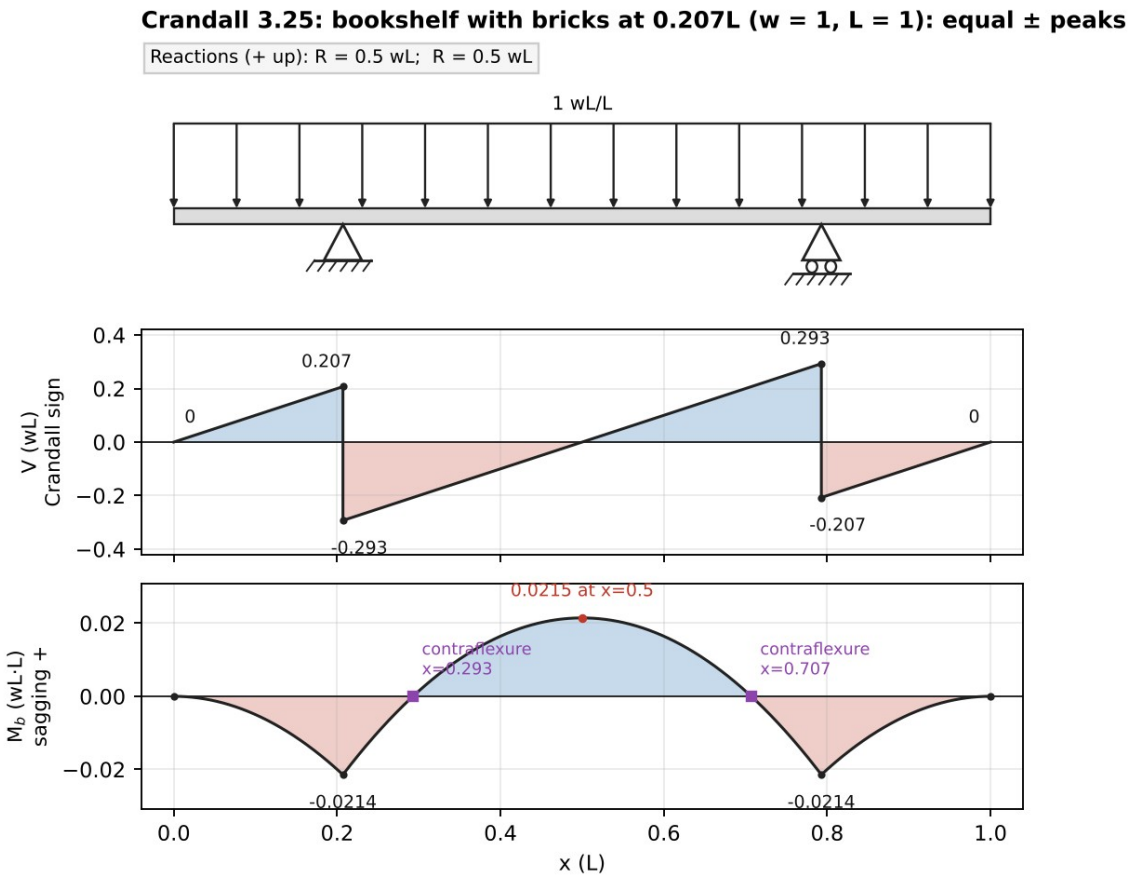
Solution.

1. *Reactions:* each support carries wL/2.
2. *Over a support:* $M_s = -\frac{w a^2}{2}$.
3. *Mid-span:* $M_c = \frac{w L}{2}\left(\frac{L}{2} - a\right) - \frac{w}{2}\left(\frac{L}{2}\right)^2 = \frac{w(L-2a)^2}{8} - \frac{w a^2}{2}$.
4. *Equal magnitudes:* $\frac{(L-2a)^2}{8} - \frac{a^2}{2} = \frac{a^2}{2}$, so $(L-2a)^2 = 8a^2$, $L - 2a = 2\sqrt{2}\,a$ and $a = \frac{L}{2(1+\sqrt{2})}$.

$$a = 0.207\,L, \quad |M|_{max} = \frac{w a^2}{2} = 0.0214\,w L^2$$

Check. Supports at the ends would give $w L^2/8 = 0.125\,w L^2$, nearly 6× larger ✓.

Exam takeaway. Balancing the hogging and sagging peaks is a standard optimisation. **Variation:** supports at the ends but a single load P anywhere on the shelf.



Optimum brick positions: equal hogging and sagging peaks

Concept 3.4 — Singularity (Macaulay) functions

A. Concept explanation

1. Intuition. Instead of writing a new M(x) for every segment, use "switches" $\langle x-a \rangle$ that are off (0) before x = a and on after it. One formula then covers the whole beam. This saves a lot of algebra in the deflection chapter, where you integrate twice.

2. Why it matters. PYQ 2014-15 Test-1 Q4 (25 marks) demanded this method *only*, and 2023-24 Part B Q3 uses it for deflection. Crandall Prob 3.23 asks you to redo Probs 3.1–3.8 with it.

3. Formal treatment (slides).

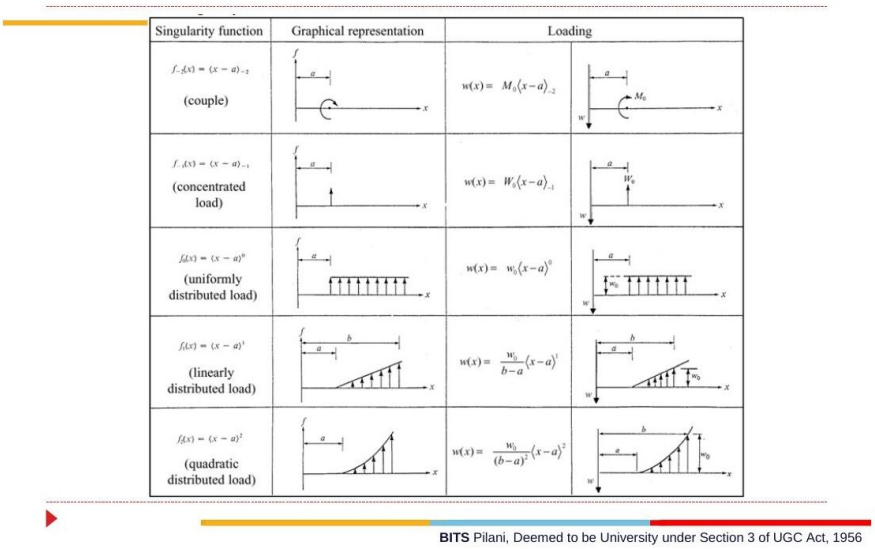
$$\langle x-a \rangle^n = \begin{cases} (x-a)^n & x \geq a \\ 0 & x < a \end{cases} (n \geq 0), \quad \langle x-a \rangle^0 = \text{unit step}$$

$\langle x-a \rangle_{-1}$ is a concentrated (Dirac) load and $\langle x-a \rangle_{-2}$ a concentrated couple (doublet).

Integration rules (slides): for $n \geq 0$, $\int_{-\infty}^{x} \langle x-a \rangle^n dx = \frac{\langle x-a \rangle^{n+1}}{n+1}$. For $n<0$, $\int \langle x-a \rangle_n dx = \langle x-a \rangle_{n+1}$.

Loading representation (slide Rule 2): a distributed load is zero order, $w_0 \langle x-a \rangle^0$; a point load is order –1, $P \langle x-a \rangle_{-1}$; a moment is order –2, $M_0 \langle x-a \rangle_{-2}$. To stop a UDL at b, add $-w_0 \langle x-b \rangle^0$. A ramp is $\frac{w_0}{b-a} \langle x-a \rangle^1$.

Family of Singularity Functions



Original: Slides Ch 3, slide 35

Slide Rule 4: (1) set up w(x) including **all** forces (reactions too) from left to right; (2) integrate w once to get V, with **no** constants; (3) integrate V to get M, with **no** constants.

4. Derivation — why no constants. For x < 0 (just left of the beam) V and M are zero, and every term $\langle x-a\rangle^n$ with a ≥ 0 is zero there too. So the integration constants must be zero. ∎

5. Working form you can use directly (sagging M +, clockwise couples +, upward forces +):

$$M(x)=\sum_{\text{up}} P_i\langle x-a_i\rangle^1-\sum_{\text{down}} P_j\langle x-a_j\rangle^1+\sum_{\text{cw}} C_k\langle x-a_k\rangle^0-\sum_{\text{UDL}\downarrow}\frac{w}{2}\langle x-a\rangle^2+\ldots,\quad V_{Crandall}=-\frac{dM}{dx}$$

6. Special cases. A UDL that runs to the end of the beam needs no "stop" term. A UDL that ends early needs $\frac{+w}{2}\langle x-b\rangle^2$ in M.

7. Traps.

- Evaluating $\langle x-a\rangle$ as $(x-a)$ when $x<a$; it must be 0.
- Forgetting to end a UDL.
- Using $\langle x-a\rangle^1$ for a couple; in M it must be $\langle x-a\rangle^0$.
- Leaving reactions out of w(x).

8. Summary. One equation, switches on at each discontinuity, no constants for V and M.

B. Solved questions

Q24 (Crandall Prob 3.23(h)) — Prob 3.8 again, using singularity functions (medium)

Problem. Obtain the SFD and BMD of Crandall Prob 3.8 (Q20) using singularity-function notation.

Given: pin at 0 ($R_A=0.45$ kN ↑), 1.5 kN ↓ at 1.5 m, roller at 3.75 m ($R_B=3.30$ kN ↑), 1.5 kN/m ↓ from 3.75 m to the end (5.25 m). **Find:** one equation each for M(x) and V(x).

Solution.

1. *Moment:*

$$M(x)=0.45\langle x\rangle^1-1.5\langle x-1.5\rangle^1+3.30\langle x-3.75\rangle^1-0.75\langle x-3.75\rangle^2\,(\text{kN·m})$$

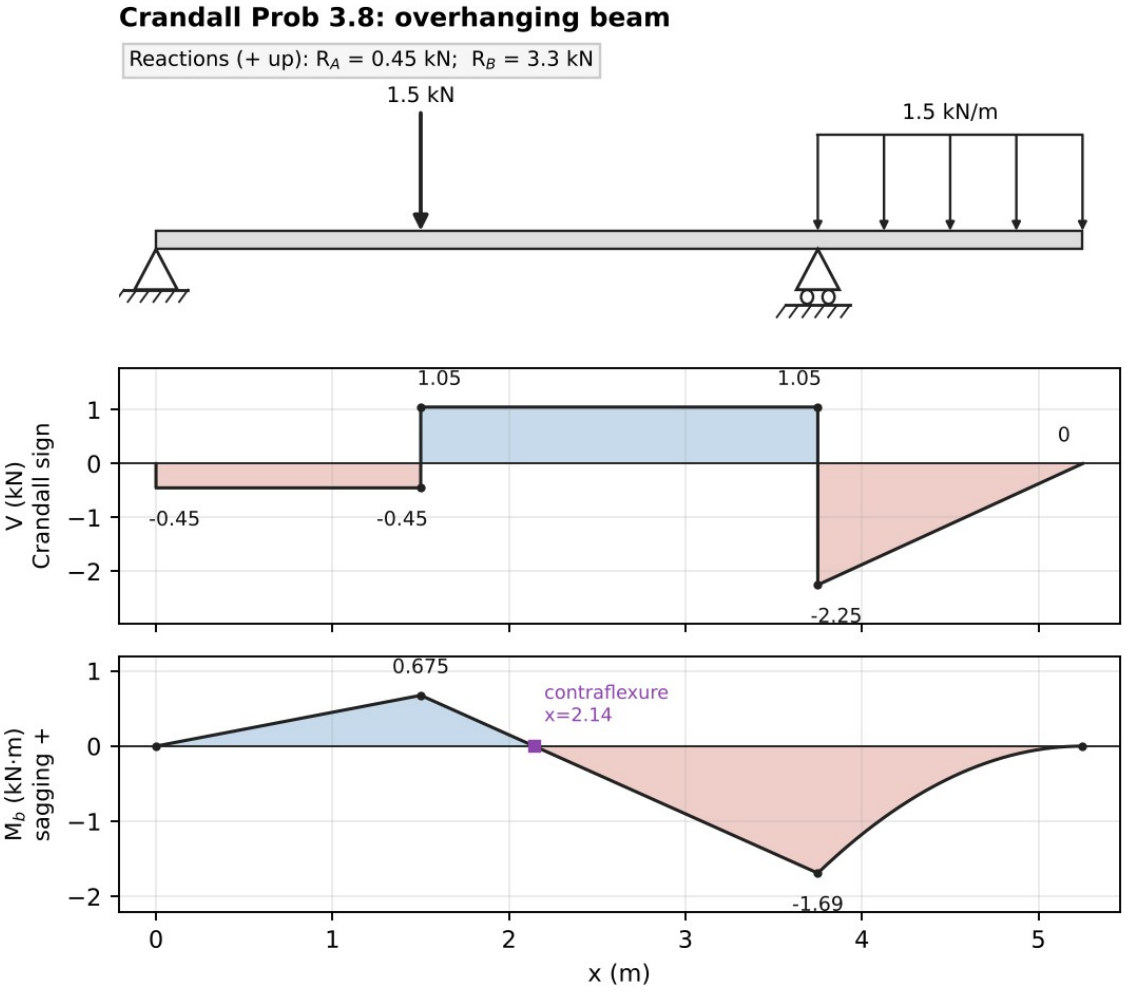
2. *Shear (Crandall):*

$$V(x) = -\frac{dM}{dx} = -0.45\langle x\rangle^0 + 1.5\langle x-1.5\rangle^0 - 3.30\langle x-3.75\rangle^0 + 1.5\langle x-3.75\rangle^1 \,(\text{kN})$$

3. *Checks at key points:* $M(1.5) = 0.675$; $M(3.75) = 1.6875 - 3.375 = -1.6875$; $M(5.25) = 2.3625 - 5.625 + 4.95 - 1.6875 = 0$ ✓.

Same diagrams as Q20: $M_{max}^{+} = 0.675, M_B = -1.69\,\text{kN·m}$

Exam takeaway. The value at the free end must come out zero. That is your built-in check.



The same diagrams as Q20, now from one singularity equation

Q25 (PYQ 2014-15 Test-1 Q4) — Singularity functions: SFD, BMD and contraflexure (hard, 25 marks)

Q.4 Calculate shear force and bending moment at salient points of a loaded beam (*Fig. 4*) by using **Singularity function method (only).** Also plot the shear force and bending moment diagrams and determine the position of point of contra- flexure from end *A*. **[25]**

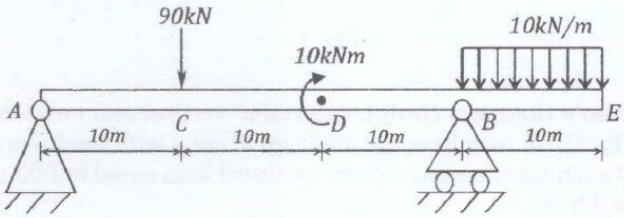


Fig. 4

@@@@@@@@@@

Original question: PYQ 2014-15 Test-1 Q4

Problem. Using the singularity-function method **only**, calculate the shear force and bending moment at the salient points of the beam, draw the SFD and BMD and locate the point of contraflexure. A (pin) at 0; C at 10 m with 90 kN ↓; D at 20 m with a 10 kN·m couple (clockwise as drawn); B (roller) at 30 m; free end E at 40 m; UDL 10 kN/m over BE (30–40 m).

Approach. Reactions first, then one M(x) equation with singularity terms, then V = −dM/dx.

Solution.

1. *Reactions:* $\sum M_A$: $R_B(30)-90(10)-10-100(35)=0$, so $R_B=4410/30=147$ kN and $R_A=190-147=43$ kN.
2. *Moment equation:*

$$M(x)=43\langle x\rangle^1-90\langle x-10\rangle^1+10\langle x-20\rangle^0+147\langle x-30\rangle^1-5\langle x-30\rangle^2\,(\text{kN·m})$$

3. *Shear (Crandall):*

$$V(x)=-43\langle x\rangle^0+90\langle x-10\rangle^0-147\langle x-30\rangle^0+10\langle x-30\rangle^1\,(\text{kN})$$

(The couple term differentiates to $\langle\rangle^{-1}$, which is zero except exactly at x = 20.)

4. *Salient values:*

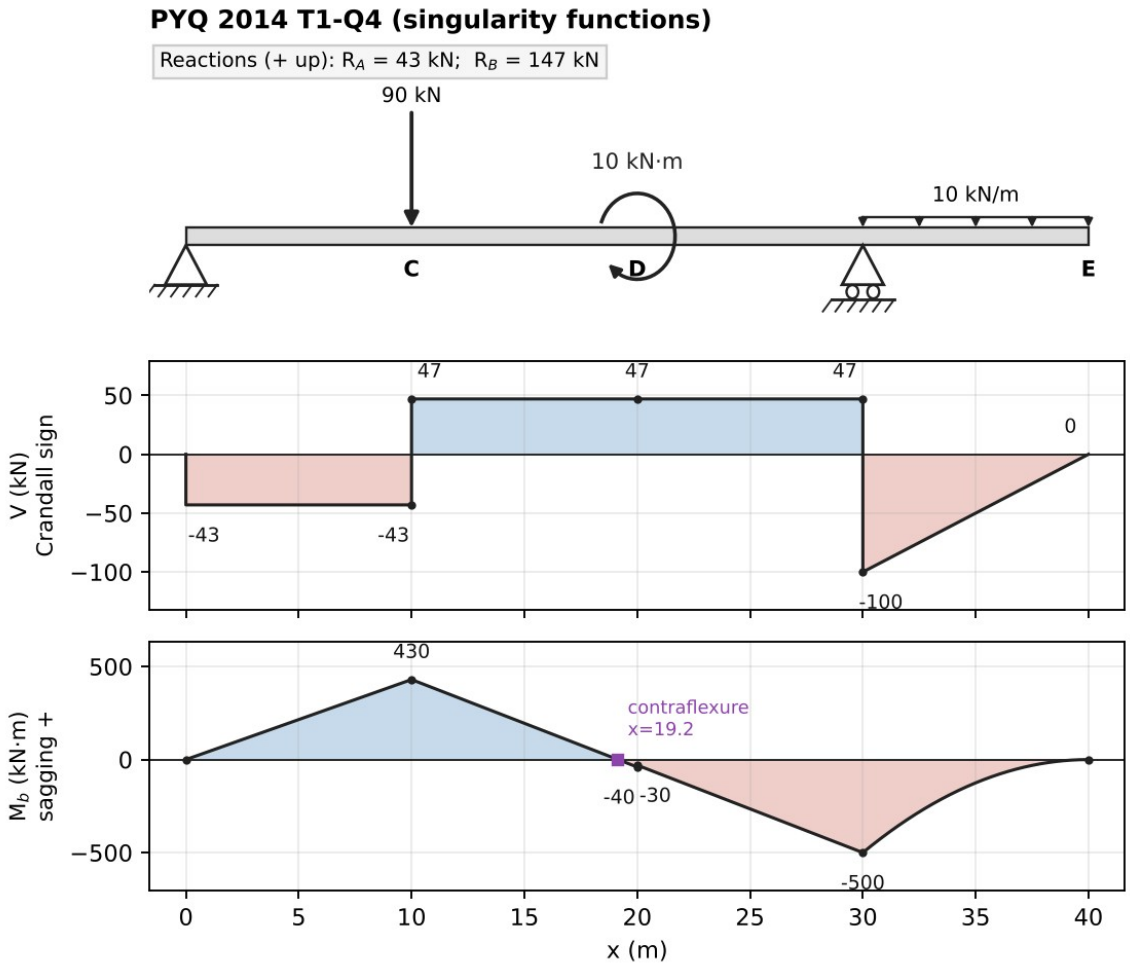
	A	C⁻	C⁺	D⁻	D⁺	B⁻	B⁺	E
V (kN, Crandall)	−43	−43	+47	+47	+47	+47	−100	0
M (kN·m)	0	**430**	430	−40	−30	**−500**	−500	0

5. *Contraflexure:* in CD, $M=43x-90(x-10)=900-47x=0$, so $x=19.15$ m. After D, M stays negative (−30 to −500) and reaches zero only at the free end E, so there is only one point of contraflexure.

$$M_{max}^{+}=430\text{ kN·m at C},\ M_{max}^{-}=-500\text{ kN·m at B},\ \text{contraflexure at } x=19.15\text{ m}$$

Check. $M(40)=43(40)-90(30)+10+147(10)-5(100)=1720-2700+10+1470-500=0$ ✓.

Exam takeaway. The marks are for the singularity equation itself, the table of salient values and the contraflexure location. **Variation:** reverse the couple; M at D then jumps **down** by 10.



Solution: SFD and BMD from the singularity equations

Concept 3.5 — Cantilevers, frames, curved and 3-D members

A. Concept explanation

1. Intuition. The cut method works the same for bent or curved members. Just remember that "axial" and "shear" are measured along and across the **local** direction of the member at the cut. In 3-D, use vectors: F and $M = r \times F$. An out-of-plane load on a curved or bent member makes it **twist** as well as bend.

2. Why it matters. Brackets, cranks, pipes, curved hooks, lamp posts. These are the first part of every combined-stress PYQ (Ch. 6): find F, V, M_b and M_t at a section, then stresses.

3. Formal treatment (Crandall §3.8, Example 3.11). For a cut segment with free-end load P at position r_{CA} relative to the cut C:

$$V_C + P = 0 \, , M_C + r_{CA} \times P = 0$$

Resolve into the component along the local axis (axial force and twist) and the components perpendicular to it (shear and bending).

Crandall's observation: along a straight member under point loads only, the maximum resultant bending moment always occurs at a load point, never between loads (the resultant moment diagram is made of straight lines or curves that are concave outward).

4. Derivations.

(a) Frame moment formula. For a free end D loaded by (F_x, F_y), the in-plane bending moment at any point (x, y) of a planar frame is

$$M(x, y) = (x_D - x) F_y - (y_D - y) F_x$$

One formula covers every straight segment, so you need not redo FBDs for each leg (Q28).

(b) Crandall Example 3.11, quarter-circle pipe with a transverse load (Q26). At angle θ: $V = -P$, $M_b = P a \sin\theta$ and $M_t = -P a (1 - \cos\theta)$.

(c) Curved cantilever with an in-plane tip load (Q29): $N = P\cos\theta$, $V = P\sin\theta$ and $M = P R \cos\theta$.

5. Picture. For a bent bracket, draw the BMD *on* each member, on the tension side. Joints transmit M unchanged around a rigid corner, but the axial and shear forces swap roles when the member turns 90°.

6. Special cases. A bracket loaded so that its free end lies on the line of action of the load from a section: zero moment there. Out-of-plane loads on L-shaped members cause twist in the leg perpendicular to the load's arm.

7. Traps.

- For curved members the "axial" direction rotates with θ. Resolve the load into tangential and radial parts *at the cut*.
- In 3-D, mixing up which moment component is twist (M_{xx}) and which are bending.
- Forgetting the moment of a horizontal load about a vertical leg.

8. Summary. Vectors: $M = r \times F$. Frame: $M = (x_D - x) F_y - (y_D - y) F_x$. Curved cantilever: $N = P\cos\theta$, $V = P\sin\theta$, $M = P R \cos\theta$.

B. Solved questions

Q26 (Crandall Example 3.11) — Quarter-circle pipe with a transverse load (medium, 3-D)

Problem. A piece of refinery piping AB, anchored at B, is bent into a quadrant of a circle of radius a (centre O). A transverse load P acts at the free end A, perpendicular to the plane of the pipe. Find the shear force, bending moment and twisting moment at a general section C at angle θ from A.

Given: radius a; load P along the axis (call it i) perpendicular to the plane. At C use local unit vectors: u_t along the pipe and u_r radial. **Find:** V, M_b, M_t at θ.

Approach. Free body from the cut C to the free end A. Force balance gives V; moment balance about C, with the cross product, gives M_b and M_t.

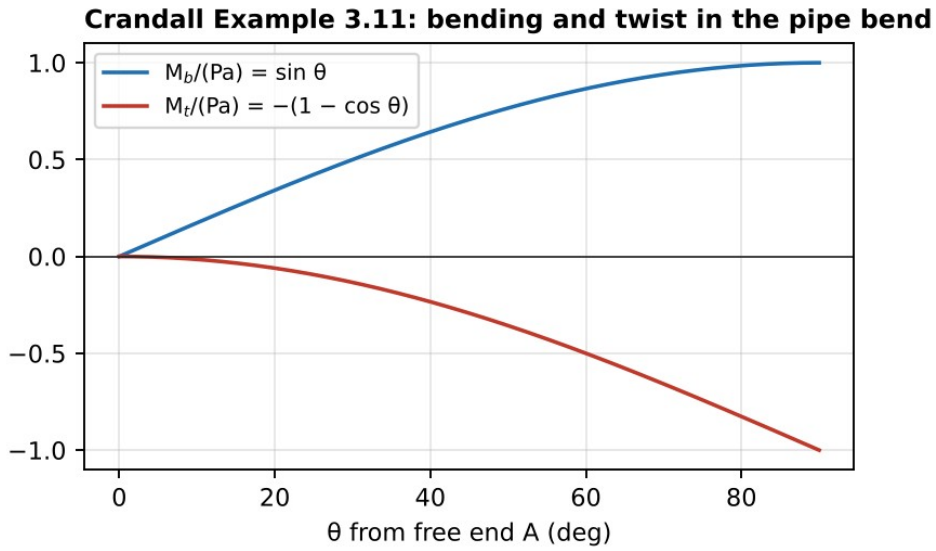
Solution.

1. *Force:* $V + P\,i = 0$, so $V = -P$ (transverse shear, constant).
2. *Position of A relative to C:* $r_{CA} = -a(1 - \cos\theta)\,u_r - a\sin\theta\,u_t$ (Crandall's eq. (e)).
3. *Moment balance:* $M_C + r_{CA} \times (P\,i) = 0$. Working out the cross products and setting each component to zero gives the bending moment about the radial axis and the twist about the tangent.

$$V = -P,\; M_b = P a \sin\theta,\; M_t = -P a (1 - \cos\theta)$$

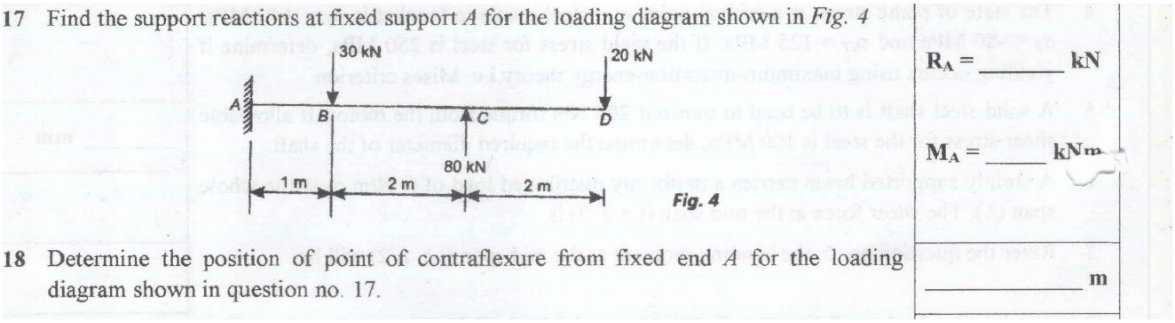
Check. At A (θ = 0) both moments vanish ✓. At the anchor B (θ = 90°): $M_b = P a$ and $|M_t| = P a$; the anchor resists equal bending and twisting moments ✓.

Exam takeaway. An out-of-plane load on a curved member produces **twist**. That is why pipe bends and crankshafts see torsion even without an applied torque. **Variation:** a semicircular pipe (θ up to 180°), where $|M_t|$ reaches 2Pa.



Bending and twisting moments around the bend

Q27 (PYQ 2014-15 Compre A Q17–18) — Cantilever with three loads (easy–medium)



Original question: PYQ 2014-15 Compre A Q17–Q18

Problem. A cantilever is fixed at A, with 30 kN ↓ at B (1 m), 80 kN ↑ at C (3 m) and 20 kN ↓ at D (5 m). Find the reaction and moment at A (Q17) and the position of the point of contraflexure from A (Q18).

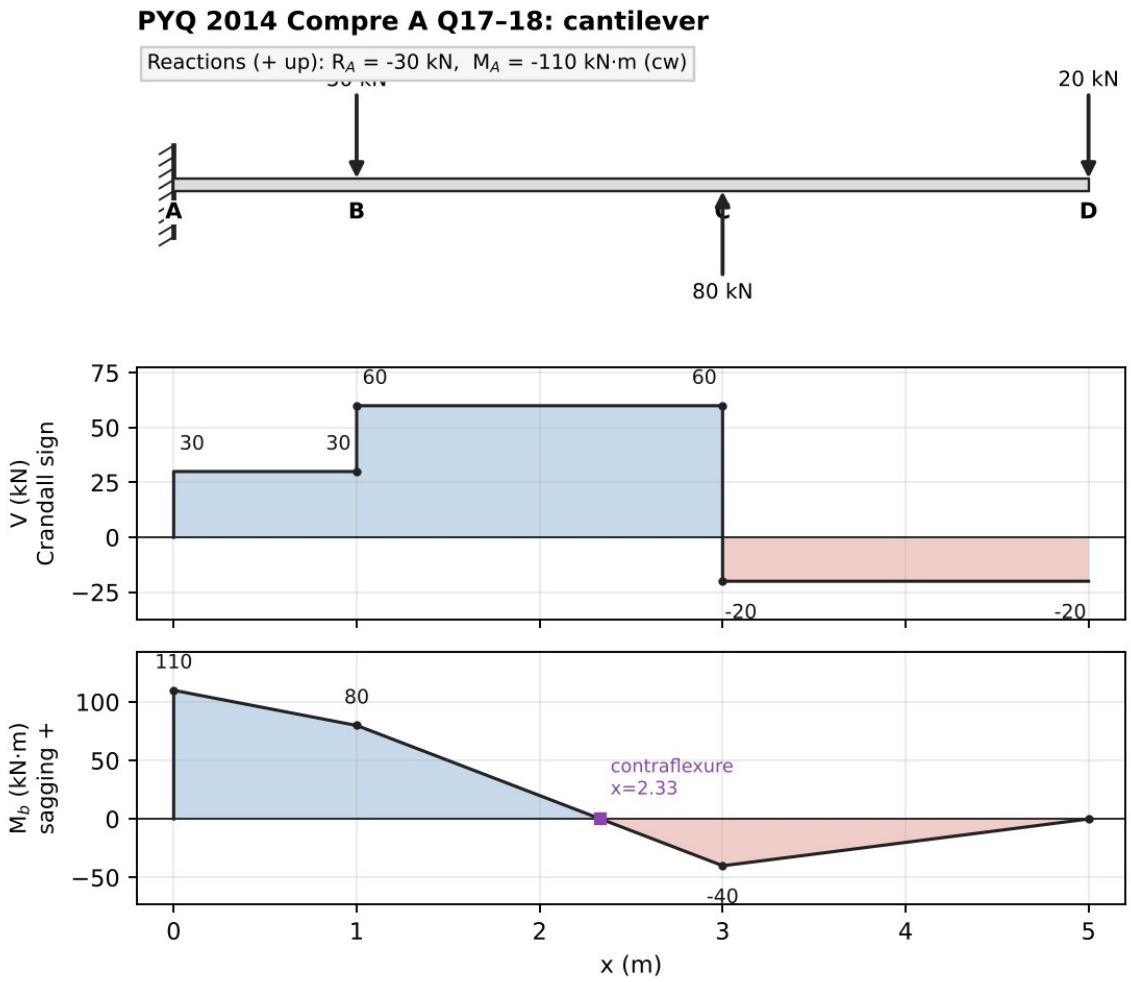
Solution.

1. $\sum F_y$: $R_A - 30 + 80 - 20 = 0$, so $R_A = -30$ kN, i.e. **30 kN ↓**.
2. $\sum M_A$: $M_A - 30(1) + 80(3) - 20(5) = 0$, so $M_A = -110$ kN·m, i.e. **110 kN·m clockwise on the beam**.
3. *M along the beam (from the right):* CD: $M = -20(5 - x)$ (hogging); BC: $M = -20(5 - x) + 80(3 - x) = 140 - 60x$; AB: $M = 140 - 60x - 30(1 - x) = 110 - 30x$.
4. *Zero in BC:* $140 - 60x = 0$, so $x = 2.33$ m.

$$R_A = 30\,\text{kN}\downarrow,\ M_A = 110\,\text{kN·m}\,(\text{clockwise}),\ \text{contraflexure at } 2.33\,\text{m from A}$$

Check. M at A from the formula: 110 kN·m (sagging, since the upward 80 kN dominates near the wall) ✓; at D: 0 ✓.

Exam takeaway. An upward load on a cantilever can make the wall moment sagging. Don't assume "cantilever ⇒ hogging".



Solution

Q28 (PYQ 2019-20 Compre A Q1) — Bracket ABCD: BMD and minimum diameter (medium, 8 marks)

Q1. Bracket *ABCD* of solid circular cross-section has the shape and dimension (b = 37mm) as shown in FigQ1. A vertical and horizontal load P (40 N) acts at the free end D.

(a) Draw the bending moment diagram of bracket *ABCD*. **Hint**: Show the variation of bending moment along segments *AB*, *BC* and *CD* separately.

(b) Determine the minimum permissible diameter d_{min} of the bracket, if the allowable bending stress in the material is 30 MPa. (Neglect the weight of bracket) **[8]**

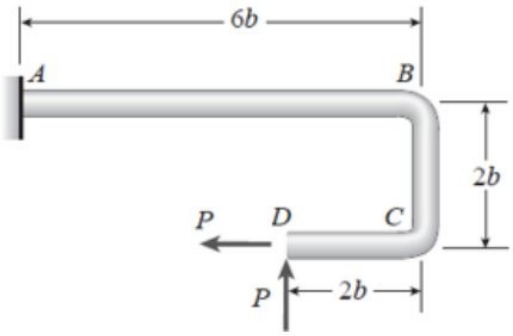


Fig Q1

Original question: PYQ 2019-20 Compre A Q1

Problem. A bracket ABCD of solid circular section is fixed at A. AB is horizontal (6b), BC goes vertically down (2b), and CD runs horizontally back toward A (2b). At the free end D act P = 40 N ← (horizontal) and P = 40 N ↑

(vertical). b = 37 mm and $\sigma_{allow} = 30$ MPa. (a) Draw the BMD of each segment. (b) Find d_{min} (neglect the bracket's weight).

Approach. Use the frame moment formula with D as the loaded free end. Then the bending stress in a solid circle, $\sigma = 32\,M/\pi\,d^3$ (from the flexure formula, covered later in the course).

Solution.

1. *Coordinates:* A(0, 0), B(6b, 0), C(6b, −2b), D(4b, −2b). Loads at D: $(F_x, F_y) = (-P, +P)$.
2. *Frame moment:* $M(x, y) = (x_D - x)F_y - (y_D - y)F_x = P(4b - x) - P(2b + y) = P(2b - x - y)$.

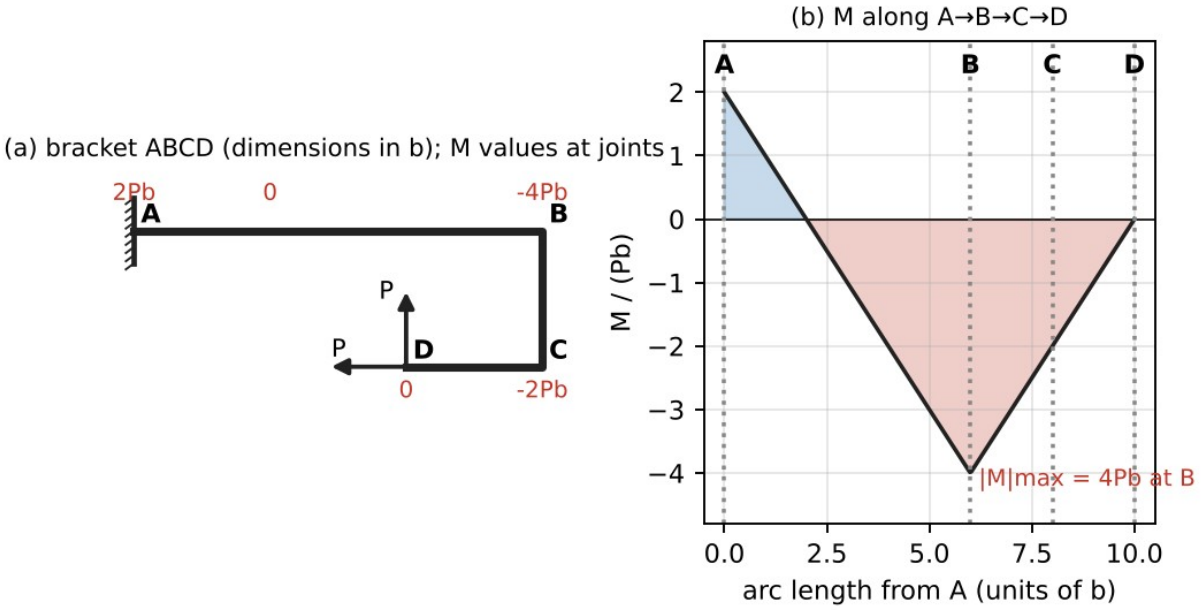
Segment	Expression	End values
CD ($y = -2b$, x from 4b to 6b)	$P(4b - x)$	D: 0 → C: −2Pb (linear)
BC ($x = 6b$, y from −2b to 0)	$P(-4b - y)$	C: −2Pb → B: −4Pb (linear)
AB ($y = 0$, x from 6b to 0)	$P(2b - x)$	B: −4Pb → A: +2Pb; zero at x = 2b

3. *Maximum:* $|M|_{max} = 4\,P\,b = 4(40)(37) = 5920$ N·mm at B. There is a point of contraflexure on AB at 2b from A.
4. *(b) Diameter:* $\sigma = \dfrac{32\,M}{\pi\,d^3} \le 30$, so $d^3 = \dfrac{32(5920)}{\pi(30)} = 2010$ mm³.

$$|M|_{max} = 4\,P\,b = 5.92\ \text{N·m at B};\ d_{min} = 12.6\ \text{mm}$$

Check. The axial force in AB is only P = 40 N, so $P/A \approx 0.3$ MPa, negligible next to 30 MPa ✓.

Exam takeaway. The hint "show the variation along AB, BC, CD separately" is asking for exactly this table. **Variation:** load only the vertical P; then $M = P(4b - x)$ everywhere and the maximum moves to A (4Pb).



Bending moment along the bracket (units of Pb)

Q29 (PYQ 2019-20 Compre A Q3(a) = Crandall Prob 3.9) — Curved cantilever: forces at section θ (medium)

Q3. The curved cantilever beam in FigQ3 has a radius of curvature R and a circular cross section of diameter d. It is subjected to a point load P at point B.

(a) Determine the reactions (i.e. axial force, shear force and bending moment) acting at any section θ in the curved cantilever beam AB.

(b) Using Castigliano's method (the strain energy method), determine the vertical deflection of point B in terms of P, modulus of elasticity E, radius of curvature R, cross-section area A, and moment of inertia of the cross section I. **Neglect the effect of strain energy due to shear** **[8]**

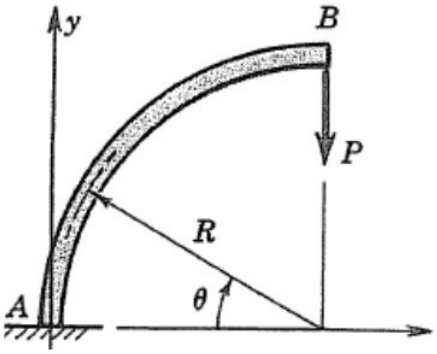
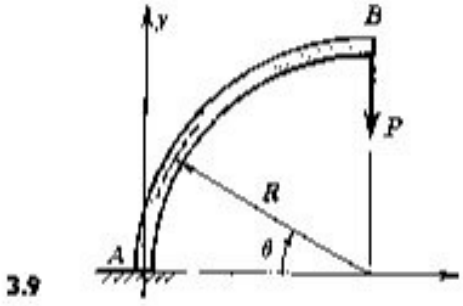


Fig Q3

Original question: PYQ 2019-20 Compre A Q3



Original: Crandall Prob 3.9 (same problem) (textbook figure)

Problem. A quarter-circle cantilever of radius R is fixed at A (θ = 0, at ground level) and free at B (θ = 90°, at the top), with P ↓ at B. θ is measured at the centre O from the horizontal. Find the axial force, shear force and bending moment at any section θ.

Given: R, P. **Find:** N(θ), V(θ), M(θ).

Approach. Free body from the cut to B. Resolve P into tangential and radial components at the cut.

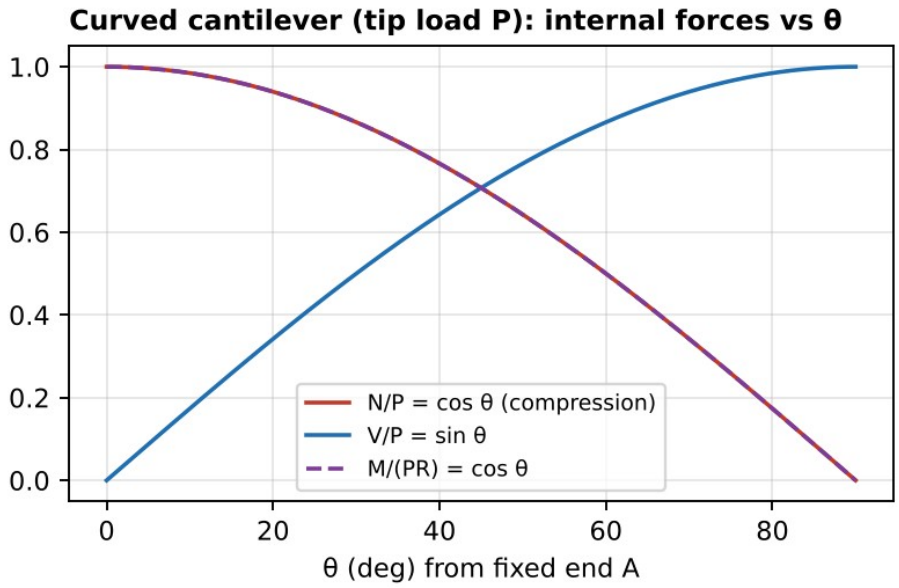
Solution.

1. *Position on the arc* relative to O: $(-R\cos\theta, R\sin\theta)$. Tangent direction (toward B): $t=(\sin\theta,\cos\theta)$. Outward radial direction: $n=(-\cos\theta,\sin\theta)$.
2. *Load* $(0,-P)$ at B = (0, R).
3. *Axial (along t):* $P\cdot t=-P\cos\theta$, so $N=P\cos\theta$ in compression.
4. *Shear (along n):* magnitude $|P\cdot n|=P\sin\theta$.
5. *Moment:* the horizontal lever arm from the cut to B is $R\cos\theta$, so $M=PR\cos\theta$.

$$N=P\cos\theta\,(\text{compression}),\quad V=P\sin\theta,\quad M=PR\cos\theta$$

Check. At B (θ = 90°): N = M = 0 and V = P (the load is purely transverse there) ✓. At A (θ = 0): N = P and M = PR (the maximum) ✓.

Exam takeaway. These expressions feed straight into Castigliano's theorem (Ch. 2 Q26): $\delta_B = \pi P R^3/4EI + \pi P R/4AE$.



Internal forces vs θ

Chapter 3 — Formula and convention sheet

Item	Crandall / slides
Positive face	outward normal along +x
+ component	(+face, +dir) or (−face, −dir)
V	$¿ -\sum$(upward forces to the left of the cut), **opposite sign to Gere**
M_b	sagging +; ¿ $\sum$ moments of the left-hand forces about the cut (clockwise +)
Differential relations	$dV/dx = -q$, $dM_b/dx = -V$, $d^2M_b/dx^2 = q$ (q up +)
Max/min M	where V = 0 or changes sign
Contraflexure	where M changes sign
Concentrated load P¦	V jumps by +P (Crandall); kink in M
Clockwise couple C	M jumps by +C; V unchanged
Resultant of a distributed load	$R = \int q\,dx$ at the centroid, **reactions only**
Standard results	SS + central W: WL/4; SS + UDL: $wL^2/8$; SS + W at a: Wab/L; cantilever + tip P: −PL; cantilever + UDL: $-wL^2/2$; SS + triangle W: 0.128WL
Hydrostatic upright	$V_0 = 1/2\,\gamma s h^2$, $M_0 = 1/6\,\gamma s h^3$
Optimum overhangs	bookshelf a = 0.207L; overhang beam a = 0.414L
Singularity	$\langle x-a \rangle^n$ (0 for x < a); point load $\langle\rangle_{-1}$, couple $\langle\rangle_{-2}$; no constants for V and M
Frame moment	$M = (x_D - x)F_y - (y_D - y)F_x$
Curved cantilever (tip load)	$N = P\cos\theta$, $V = P\sin\theta$, $M = PR\cos\theta$

Item	Crandall / slides
Pipe quadrant, transverse load	$V = -P,\ M_b = P a \sin\theta,\ M_t = -P a\left(1 - \cos\theta\right)$

Other Crandall Ch. 3 problems worth practising: 3.3 (inclined cantilever), 3.11 (semicircular arch), 3.13 and 3.15 (sections in 3-D members and frames), 3.16 (femur muscles), 3.18 (scaffold plank), 3.24 (diving board), 3.26–3.30 (flagpole, bent rods, antenna, trailer bars), 3.31 (cable sag).

Chapter 4 — Stress and Strain

Stress at a point, plane stress, stress transformation, Mohr's circle, principal and maximum shear stresses, thin-walled pressure vessels, plane strain, strain transformation, Mohr's circle of strain, strain rosettes

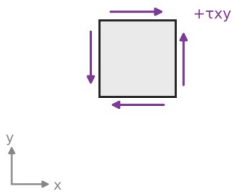
Sources used: Slides "Chapter 4 part 1" (stress) and "Ch 4 Part 2" (strain); Crandall Ch. 4 (§4.2–4.9; the sheet-metal strain example; Probs 4.6–4.15, 4.20–4.23, 4.25, 4.27, 4.28, 4.32, 4.34); PYQs 2019-20 (A-Q2), 2021-22 (A-Q1, B-Q1), 2014-15 (Test-2 Q1, Q2; Compre A-Q12, Q13; Compre B-Q1).

Layout. Part A (eight items) for each concept, then Part B: slides → Crandall → PYQs, easy → hard. Each solution follows *Problem · Given/Find · Approach · Sketch · Steps · Answer · Check · Takeaway*.

⚠️ Read first: sign conventions for this chapter

1. **Stress sign (Crandall, slides):** a stress component is **positive** when it acts on a **positive face in a positive direction**, or on a **negative face in a negative direction**. Tension is +. $\tau_{xy} > 0$ means that on the right (+x) face the shear points **up** (+y).
2. **Angle θ:** measured **counter-clockwise from x to x′**, the outward **normal** of the face of interest.
3. **Mohr's circle (Crandall and slides use the same rule):** σ is plotted to the right (tension positive). **A positive τ_{xy} is plotted *downward* at point x and *upward* at point y.** With this rule, **a counter-clockwise rotation of 2θ on the circle corresponds to a counter-clockwise rotation θ of the element.** Beer/Hibbeler's "positive shear (clockwise on the element) plotted up" gives the same circle.
   - If you plot τ_{xy} *upward* at x (the "maths" convention), the circle is mirrored and the 2θ rotations reverse direction. This is the most common Mohr's-circle mistake in the exam.
4. **Strain Mohr's circle:** the vertical axis is **γ/2**, not γ. Same plotting rule: $+\gamma_{xy}/2$ is plotted downward at x.

Positive τxy (Crandall): ↑ on +x face, → on +y face



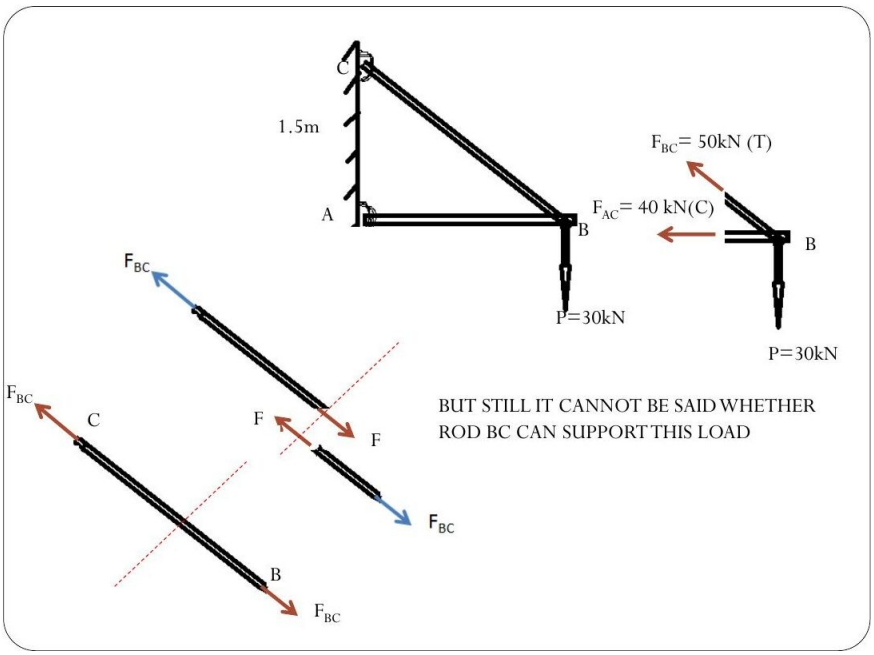
A positive τxy element

Concept 4.1 — Stress at a point and the stress tensor

A. Concept explanation

1. Intuition. Squeeze a sponge in your hand. Every tiny cube inside it is pushed and dragged by its neighbours. "Stress" is the intensity of that push or drag, force per unit area, on a particular imaginary cut. A single number isn't enough, because the answer depends on *which way* you slice. Slice one way and the material is being pulled apart; slice at 45° and it may be sliding.

2. Why it matters (the slides' motivating example). A bracket ABC carries 30 kN at B: AC is 2 m horizontal and BC is a 20 mm diameter rod rising 1.5 m. Statics gives $F_{BC} = 50$ kN (T) and $F_{AC} = 40$ kN (C). "But still it cannot be said whether rod BC can support this load" (slide). That depends on (1) the internal force, (2) the cross-sectional area of the rod and (3) the material. Stress is the bridge from forces (Ch. 1–3) to material failure (Ch. 5).



Original: Slides Ch 4 Part 1, slide 3

3. Formal treatment (Crandall §4.2–4.3; slide figures). Cut the body through point O by a plane with normal **n**. On a small area ΔA the internal force is ΔF. For an x-face ($\Delta A_x = \Delta y\,\Delta z$):

$$\sigma_x = \frac{\lim_{\Delta A_x \to 0} \Delta F_x}{\Delta A_x}, \tau_{xy} = \frac{\lim \Delta F_y}{\Delta A_x}, \tau_{xz} = \frac{\lim \Delta F_z}{\Delta A_x}$$

The first subscript is the **face normal** and the second the **direction**. Doing this on all three faces gives the **stress tensor**:

$$\begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

Plane stress (slides): the only non-zero stress components act in one plane: $\sigma_z = \tau_{zx} = \tau_{zy} = 0$. Two faces of the cubic element are free of stress. This is typical of thin plates loaded in their plane, and of the free surface of any part, which is where every strain gauge sits.

4. Derivation — equilibrium of a differential element (Crandall Fig. 4.10; slides). Take an element Δx × Δy × Δz in plane stress, with stresses varying from face to face (e.g. σ_x on the left, $\sigma_x + \frac{\partial \sigma_x}{\partial x}\Delta x$ on the right).

- *Moments about the z-axis through the centre:* $(\tau_{xy}\Delta y\,\Delta z)\Delta x - (\tau_{yx}\Delta x\,\Delta z)\Delta y + (\text{higher-order terms}) = 0$. Dividing by $\Delta x\,\Delta y\,\Delta z$ and letting the element shrink:

  $$\tau_{xy} = \tau_{yx}$$

  Shear stresses on perpendicular faces are equal, and they point toward or away from the same corner.

- *Force balance in x and y* (no body force):

  $$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0, \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} = 0$$

  In 3-D, only six of the nine stress components are independent (Crandall Prob 4.1 adds the z-equation; Prob 4.2 adds body forces X, Y, Z).

Equilibrium of a differential element in plane stress

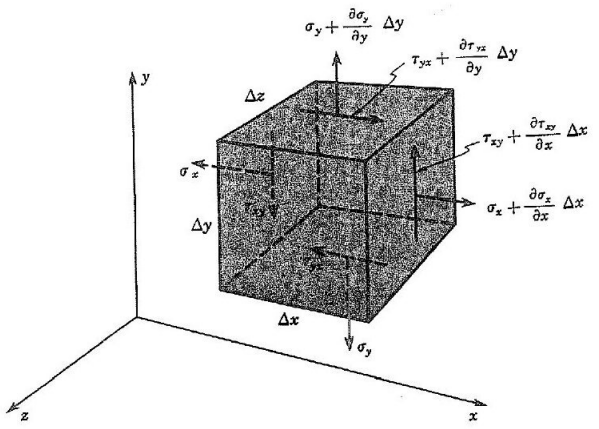


Fig. 4.10 Stress components in plane stress expressed in terms of partial derivatives.

Original: Slides Ch 4 Part 1, slide 11

5. Picture. A positive τ_{xy} element (Crandall Fig. 4.11): arrows ↑ on the right face, ↓ on the left face, → on the top face and ← on the bottom face. The element is pushed into a rhombus that is stretched along the diagonal joining the first-quadrant and third-quadrant corners.

6. Special cases. Uniaxial: only σ_x. Hydrostatic: $\sigma_x = \sigma_y = \sigma_z = -p$, with no shear on any plane. Pure shear: only τ_{xy}.

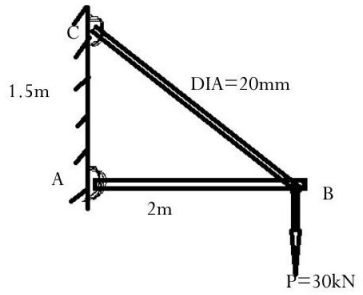
7. Traps.

- Drawing τ_{yx} opposite to τ_{xy} on adjacent faces. They must meet head-to-head or tail-to-tail at the corners.
- Adding stresses that act on differently oriented faces. Stresses only add component by component on the **same** element orientation (PYQ 2014 Compre B Q1).

8. Summary. Stress = force/area on a specified face. Plane stress is described by $(\sigma_x, \sigma_y, \tau_{xy})$ with $\tau_{xy} = \tau_{yx}$.

B. Solved questions

Q1 (Slides Ch4, motivating example) — Can rod BC carry 30 kN? (easy)



Analysis and design of this structure involve the determination of stresses and deformation

To determine whether a load of 30kN may safely be supported at B by this structure

Original: Slides Ch 4 Part 1, slide 2

Problem. A bracket has a horizontal member AC (2 m) and a 20 mm diameter rod BC (B at 1.5 m below C). A load P = 30 kN hangs at B. Find the member forces and the stress in BC; compare with an allowable stress of 165 MPa (structural steel).

Approach. Joint B is a two-member joint; then σ = F/A.

Solution.

1. *Joint B* (BC rising at $\tan^{-1}(1.5/2)$, so $\sin = 0.6$ and $\cos = 0.8$): vertical: $0.6\,F_{BC} = 30$, so $F_{BC} = 50$ kN (T). Horizontal: $F_{AB} = 0.8(50) = 40$ kN (C).
2. *Stress:* $\sigma_{BC} = \dfrac{50\,000}{\pi(10)^2} = \dfrac{50\,000}{314.2} = 159$ MPa.

$$F_{BC} = 50\,\text{kN}\,(\text{T}),\ F_{AB} = 40\,\text{kN}\,(\text{C}),\ \sigma_{BC} = 159\,\text{MPa} < 165\,\text{MPa} \Rightarrow \text{safe}$$

Check. 50 kN on about 314 mm² gives about 160 MPa ✓.

Exam takeaway. "Find the stress" always means find the internal force first. **Variation:** design the rod diameter for 140 MPa: $d = \sqrt{4(50\,000)/(140\,\pi)} = 21.3$ mm.

Concept 4.2 — Stress transformation in plane stress

A. Concept explanation

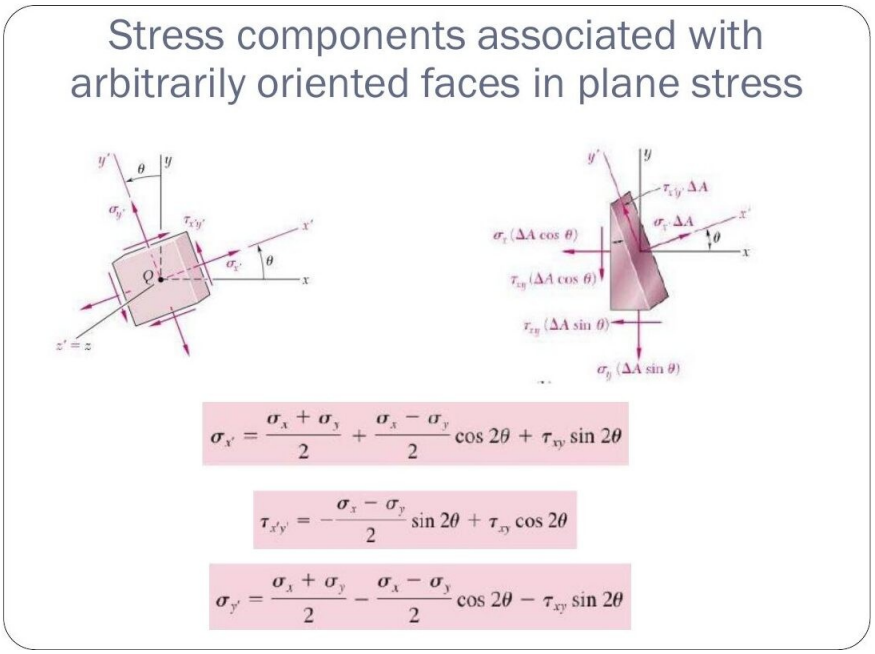
1. Intuition. The material at a point is in one definite physical state. Rotating the imaginary cube is like walking around a statue: you see it differently, but the statue doesn't change. The transformation equations tell you what σ and τ a cut at angle θ "sees".

2. Why it matters. Welded and glued joints at an angle, wood grain, fibre directions, failure planes (ductile metals fail on 45° shear planes, brittle ones on principal planes) and the starting point for principal stresses and Mohr's circle.

3. Formal treatment. Given σ_x, σ_y, τ_{xy}, find $\sigma_{x'}$ and $\tau_{x'y'}$ on the face whose outward normal x′ is at θ counter-clockwise from x.

4. Derivation (wedge equilibrium; slides "stress components associated with arbitrarily oriented faces"). Cut a triangular wedge whose inclined face has area ΔA and normal x′. The vertical (−x) face then has area $\Delta A\cos\theta$ and the horizontal (−y) face $\Delta A\sin\theta$. The forces on the two negative faces are:

- on the −x face: $\sigma_x \Delta A\cos\theta$ (←) and $\tau_{xy}\Delta A\cos\theta$ (↓);
- on the −y face: $\sigma_y \Delta A\sin\theta$ (↓) and $\tau_{xy}\Delta A\sin\theta$ (←).



Original: Slides Ch 4 Part 1, slide 15

Sum forces along x′ (unit vector $(\cos\theta, \sin\theta)$):

$$\sigma_{x'}\Delta A = \sigma_x \Delta A\cos^2\theta + \sigma_y \Delta A\sin^2\theta + 2\tau_{xy}\Delta A\sin\theta\cos\theta$$

Sum forces along y′ (unit vector $(-\sin\theta, \cos\theta)$):

$$\tau_{x'y'} = -(\sigma_x - \sigma_y)\sin\theta\cos\theta + \tau_{xy}(\cos^2\theta - \sin^2\theta)$$

With the double-angle identities $\cos^2\theta = \frac{1+\cos 2\theta}{2}$, $\sin^2\theta = \frac{1-\cos 2\theta}{2}$ and $2\sin\theta\cos\theta = \sin 2\theta$:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2}\cos 2\theta - \tau_{xy}\sin 2\theta$$

Invariant: $\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y$. Use it to check every answer.

Direction-cosine form (Crandall §4.10): $\sigma_{x'} = \sigma_x\cos^2\theta + \sigma_y\sin^2\theta + 2\tau_{xy}\sin\theta\cos\theta$. In index notation $\sigma'_{ij} = l_{ik}l_{jl}\sigma_{kl}$; a quantity that transforms like this is a **tensor**, and Mohr's circle is a graphical picture of a tensor transformation.

5. Picture. All three equations depend on 2θ, so rotating the element by 180° gives the same state, as it must, since a face and its opposite are the same plane.

6. Special cases.

- Uniaxial $\sigma_x = \sigma$: $\sigma_\theta = \sigma\cos^2\theta$ and $\tau = -\frac{\sigma}{2}\sin 2\theta$, so $|\tau|_{max} = \sigma/2$ at 45°. This is why ductile bars yield on 45° planes.
- Pure shear τ: $\sigma_{45^\circ} = +\tau$ and $\sigma_{135^\circ} = -\tau$, so the principal stresses are ±τ at 45°. This is the basis of torsion strain-gauge problems (Ch. 6).

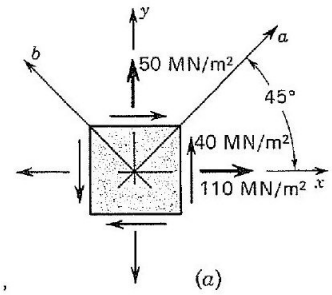
7. Traps.

- Measuring θ from the *plane* instead of from its **normal**. If a weld line is at β to the cross-section, the weld's normal is at β to the axis.
- Using θ instead of 2θ in the trig functions.
- Getting the sign of τ_{xy} wrong. Look at the right face: is the arrow up?

8. Summary. $\sigma_{x'} = \acute{\sigma} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$; $\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$; $\sigma_{x'} + \sigma_{y'}$ = constant.

B. Solved questions

Q2 (Slides Ch4) — Stresses on a plane whose normal is at 45° (easy)



- Find the stress components of a plane whose normal is inclined at 45° to the xy axes.

Original: Slides Ch 4 Part 1, slide 21

Problem. For $\sigma_x = 110$, $\sigma_y = 50$ and $\tau_{xy} = +40$ MN/m², find the stress components on faces a (normal at 45° to x) and b (normal at 135°).

Approach. Direct substitution; θ = 45° makes 2θ = 90°, so cos = 0 and sin = 1.

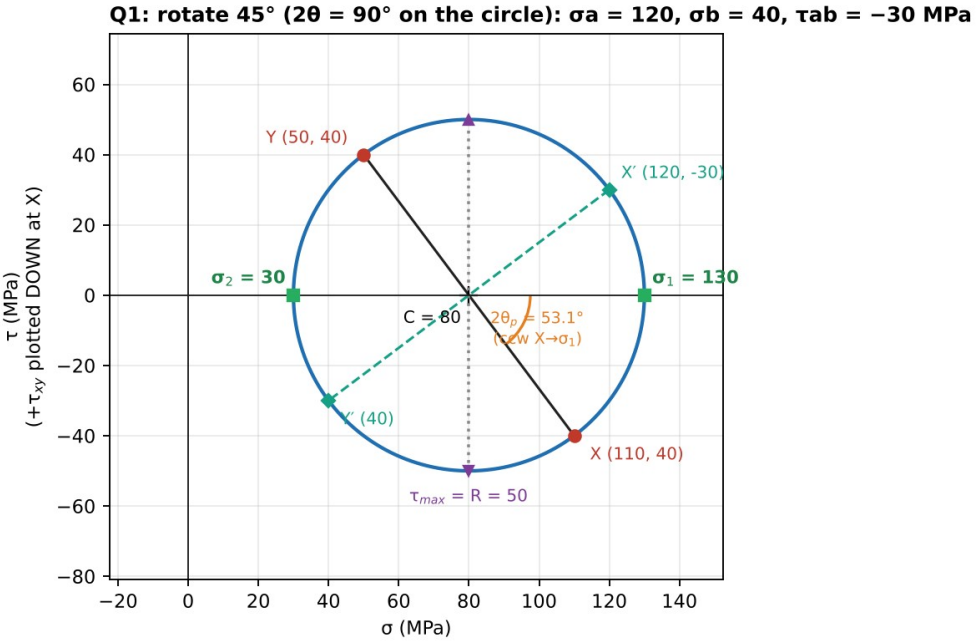
Solution.

1. $\acute{\sigma} = 80$ and $\frac{\sigma_x - \sigma_y}{2} = 30$ MPa.
2. $\sigma_a = 80 + 30(0) + 40(1) = 120$ MPa.
3. $\tau_{ab} = -30(1) + 40(0) = -30$ MPa.
4. $\sigma_b = 80 - 0 - 40 = 40$ MPa.

$$\sigma_a = 120\text{ MPa},\ \sigma_b = 40\text{ MPa},\ \tau_{ab} = -30\text{ MPa}$$

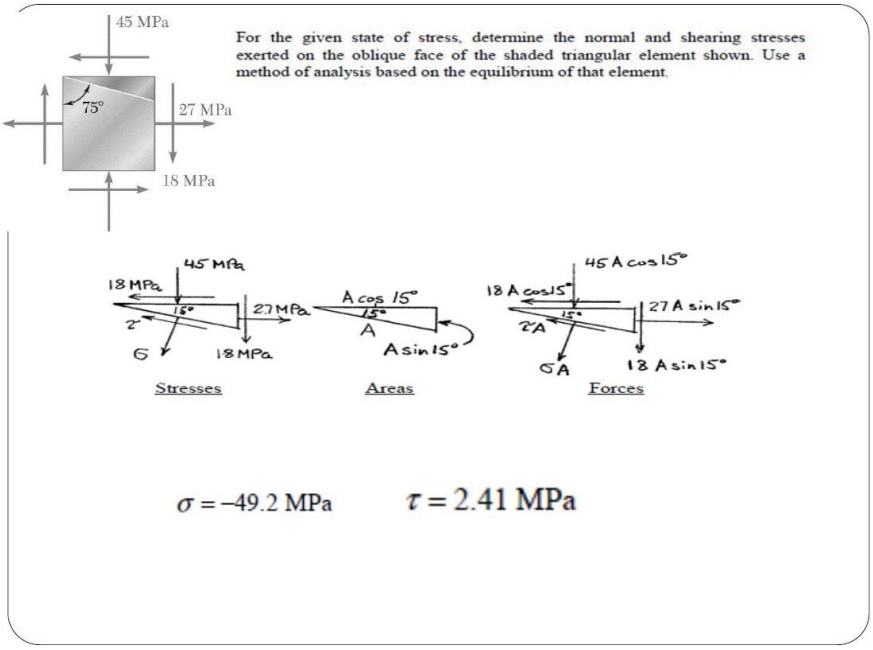
Check. $120+40=160=110+50$ ✓. On Mohr's circle (C = 80, R = 50), X is at (110, 40 below); rotating 90° counter-clockwise gives X′ at (120, 30 above), i.e. $\tau_{x'y'}=-30$ ✓.

Exam takeaway. A negative τ_{ab} means the shear on face a acts in the −b direction (clockwise on the rotated element). **Variation:** θ = 30°.



Mohr's circle: diameter XY rotated 90° to X′Y′

Q3 (Slides Ch4, Beer) — Oblique face by wedge equilibrium (medium)



Original: Slides Ch 4 Part 1, slide 35

Problem. For $\sigma_x=27$ MPa, $\sigma_y=-45$ MPa and $\tau_{xy}=18$ MPa, find the normal and shearing stresses on the oblique face of the shaded triangular element, which is inclined at 15° to the horizontal. Use equilibrium of the element.

Given: the oblique face's outward normal is at θ = 90° + 15° = 105°. **Find:** σ, τ on that face.

Approach. The transformation equations are the wedge equilibrium written in general; using them with θ = 105° is the same as the slide's force balance with areas $A\cos 15^\circ$ and $A\sin 15^\circ$.

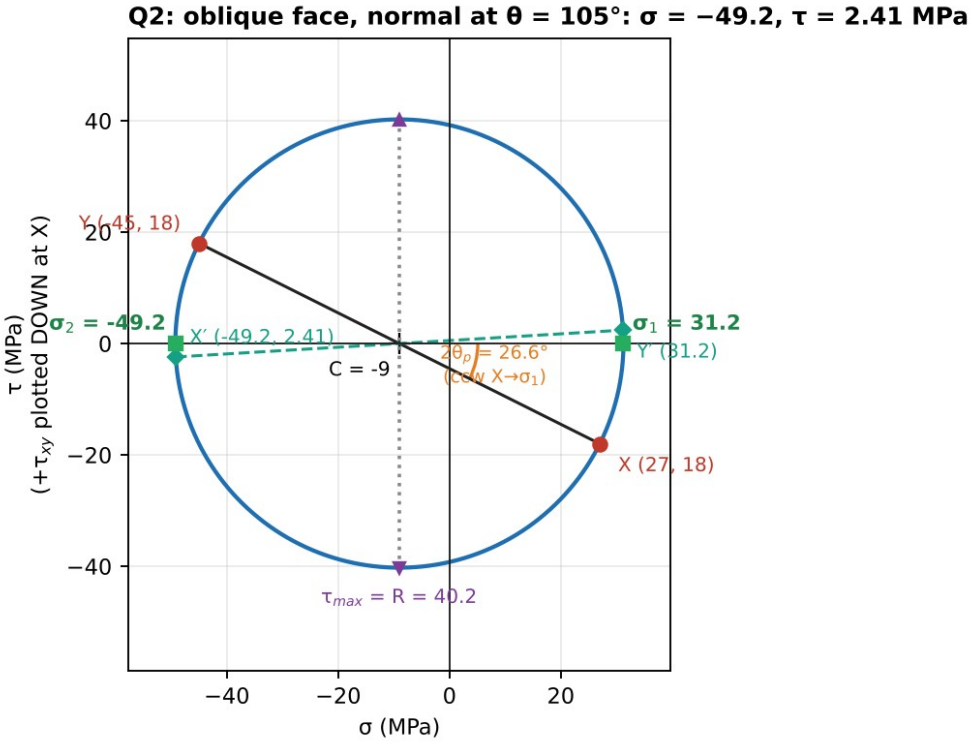
Solution.

1. $\acute{\sigma} = -9, \dfrac{\sigma_x - \sigma_y}{2} = 36, 2\theta = 210^\circ \ (\cos = -0.866, \sin = -0.5).$
2. $\sigma = -9 + 36(-0.866) + 18(-0.5) = -9 - 31.18 - 9 = -49.2 \text{ MPa}.$
3. $\tau = -36(-0.5) + 18(-0.866) = 18 - 15.59 = 2.41 \text{ MPa}.$

$$\sigma = -49.2 \text{ MPa}, \tau = 2.41 \text{ MPa}$$

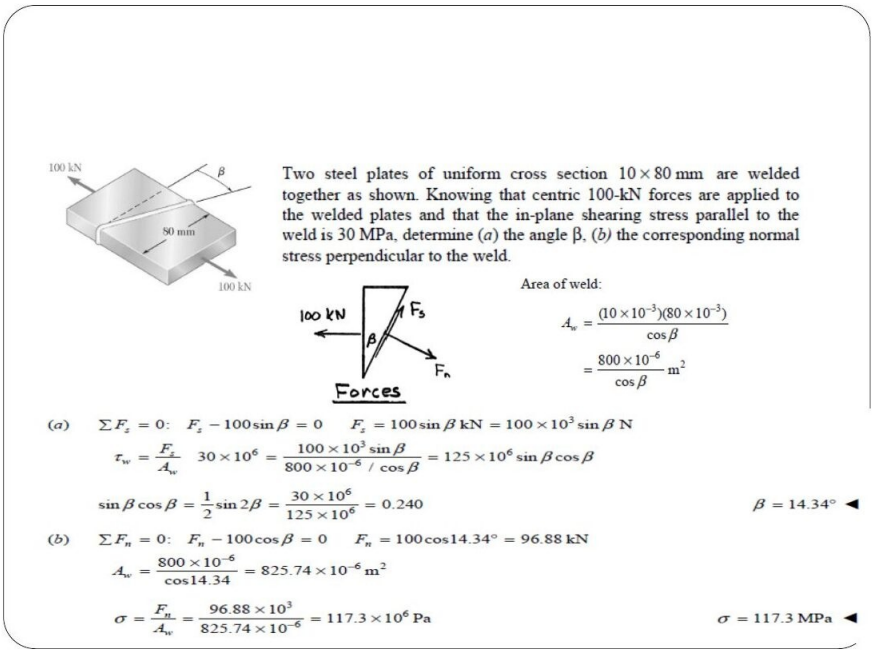
Check. This matches the slide's answer ✓. Mohr check: C = −9, R = 40.2; the face sits close to the leftmost point ($\sigma_2 = -49.2$) ✓.

Exam takeaway. The slide method (stresses → areas → forces → equilibrium) is required when the question says "use equilibrium of the element". The transformation formula is your check.



Mohr's circle check

Q4 (Slides Ch4, Beer) — Welded plates: find the weld angle (medium)



Original: Slides Ch 4 Part 1, slide 33

Problem. Two steel plates of uniform cross-section 10 × 80 mm are welded along a line at angle β to the cross-section. Centric 100 kN forces act on the plates. The in-plane shear stress parallel to the weld is 30 MPa. Find (a) β and (b) the normal stress perpendicular to the weld.

Given: $\sigma_x = 100\,000/(10\times80) = 125$ MPa (uniaxial); the weld's normal is at β to the axis. **Find:** β, σ.

Approach. Uniaxial transformation, or the slide's direct force balance on the weld.

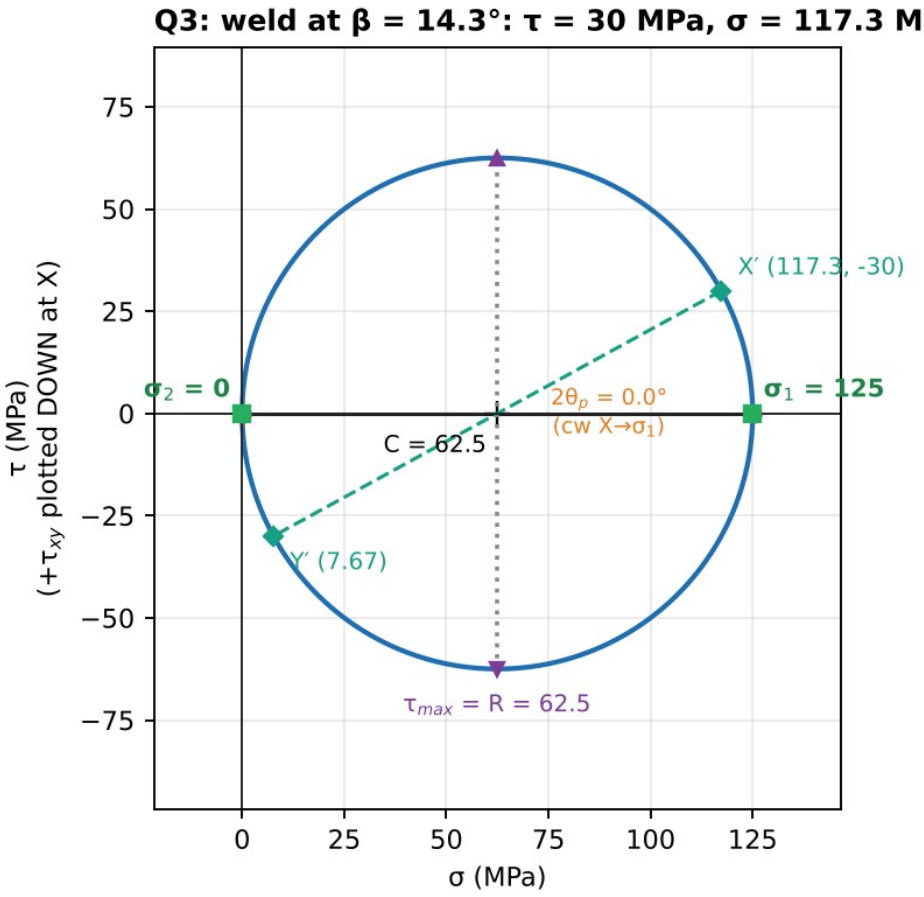
Solution.

1. *(a)* $|\tau| = \frac{\sigma_x}{2}\sin2\beta = 62.5\sin2\beta = 30$, so $\sin2\beta = 0.48$ and β = 14.34°.
2. *(b)* $\sigma = \frac{\sigma_x}{2}(1+\cos2\beta) = 62.5(1+0.8773) = 117.3$ MPa.

$$\beta = 14.3^\circ,\ \sigma = 117.3\ \text{MPa}$$

Check (slide's direct method). $F_s = 100\sin\beta = 24.77$ kN, $A_{weld} = 800/\cos\beta = 825.7$ mm², so $\tau = 24\,770/825.7 = 30$ MPa ✓. $F_n = 100\cos\beta = 96.88$ kN gives $\sigma = 117.3$ MPa ✓.

Exam takeaway. The second solution of sin 2β = 0.48 (β = 75.7°) also gives 30 MPa shear but a normal stress of only 7.7 MPa. Both are valid weld angles; state both if asked "find β".



Uniaxial circle with the weld plane marked

Q5 (Crandall Prob 4.9) — Rotate the state of 4.7(c) by 30° (easy)

Problem. For $\sigma_x = -120$, $\sigma_y = 50$ and $\tau_{xy} = 100$ MN/m² (Prob 4.7(c)), find the stress components on an element inclined at 30° (counter-clockwise) to the x–y axes.

Solution.

1. $\acute{\sigma} = -35$ and $\dfrac{\sigma_x - \sigma_y}{2} = -85$; $2\theta = 60°$.
2. $\sigma_{x'} = -35 + (-85)(0.5) + 100(0.866) = -35 - 42.5 + 86.6 = 9.1$ MPa.
3. $\sigma_{y'} = -70 - 9.1 = -79.1$ MPa.
4. $\tau_{x'y'} = -(-85)(0.866) + 100(0.5) = 73.6 + 50 = 123.6$ MPa.

$$\sigma_{x'} = 9.1,\ \sigma_{y'} = -79.1,\ \tau_{x'y'} = 123.6\ \text{MN/m}^2$$

Check. $9.1 - 79.1 = -70 = -120 + 50$ ✓. $|\tau_{x'y'}| = 123.6 < R = 131.2$ ✓.

Exam takeaway. Always report the result on a sketched rotated element with arrows. That is half the marks.

Concept 4.3 — Mohr's circle, principal stresses and maximum shear stress

A. Concept explanation

1. Intuition. As you spin the cut through 180°, the pair $(\sigma_{x'}, \tau_{x'y'})$ traces out a **circle**. The rightmost point of the circle is the largest normal stress any cut can feel, the leftmost the smallest, and the top and bottom the largest shear. Mohr's circle is a picture of *every possible cut at once*.

2. Why it matters. Failure criteria (Ch. 5) need principal stresses. PYQs ask for "principal stresses and orientation" or "sketch the Mohr's circle" every year.

3–4. Formal treatment and derivation (slides). Move the average to the left side and square both transformation equations:

$$\left(\sigma_{x'}-\frac{\sigma_x+\sigma_y}{2}\right)^2=\left(\frac{\sigma_x-\sigma_y}{2}\cos 2\theta+\tau_{xy}\sin 2\theta\right)^2,\tau_{x'y'}^2=\left(-\frac{\sigma_x-\sigma_y}{2}\sin 2\theta+\tau_{xy}\cos 2\theta\right)^2$$

Adding them, the cross terms cancel and $\sin^2+\cos^2=1$:

$$\left(\sigma_{x'}-\sigma_{avg}\right)^2+\tau_{x'y'}^2=R^2,\sigma_{avg}=\frac{\sigma_x+\sigma_y}{2},R=\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$$

This is a circle with centre $\left(\sigma_{avg},0\right)$ and radius R.

Principal stresses (zero shear): set $\tau_{x'y'}=0$:

$$\tan 2\theta_p=\frac{2\tau_{xy}}{\sigma_x-\sigma_y},\sigma_{1,2}=\sigma_{avg}\pm R$$

This gives two angles 90° apart. **To tell which gives σ₁, substitute back into the $\sigma_{x'}$ equation.** Equivalently, $d\sigma_{x'}/d\theta=0$ gives the same condition (Crandall Prob 4.31), so principal stresses are the extremes of normal stress.

Maximum in-plane shear:

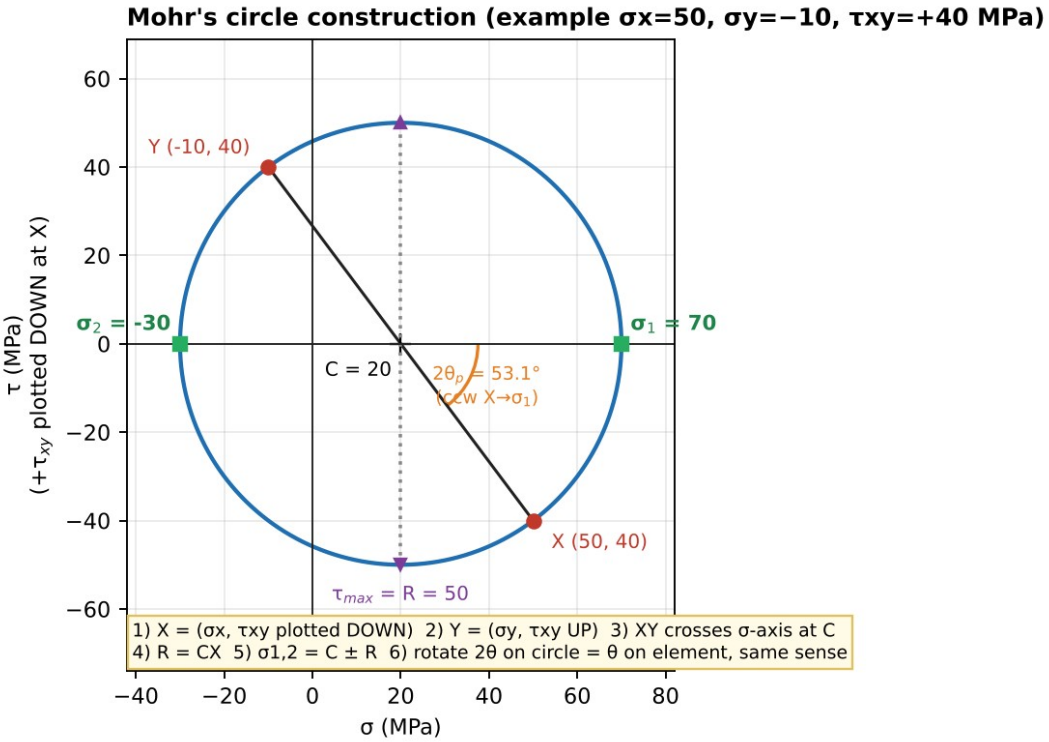
$$\tau_{max}=R=\frac{\sigma_1-\sigma_2}{2},\tan 2\theta_s=-\frac{\sigma_x-\sigma_y}{2\tau_{xy}},\theta_s=\theta_p\pm 45^\circ$$

On the max-shear planes the normal stress is **not** zero; it equals $\sigma'=\sigma_{avg}$.

Absolute maximum shear (3-D view of plane stress). The out-of-plane principal stress is $\sigma_3=0$. Order the three principal stresses as $\sigma_{max}\geq\sigma_{mid}\geq\sigma_{min}$; then $\tau_{abs,max}=\left(\sigma_{max}-\sigma_{min}\right)/2$. If σ₁ and σ₂ have the **same sign**, τ_{abs} is $\left|\sigma_1\right|/2$ or $\left|\sigma_2\right|/2$, which is **larger than the in-plane R**. Examiners love this.

5. Graphical construction (Crandall §4.5; slides). You must be able to draw this.

1. Axes: σ horizontal (+ right) and τ vertical. *Positive τ is plotted downward at X* (Crandall).
2. Plot **X** = (σ_x, τ_{xy} *downward* if positive).
3. Plot **Y** = (σ_y, τ_{xy} *upward* if positive).
4. Join XY. It crosses the σ-axis at the centre C = σ_{avg}. The radius is R = CX.
5. A (rightmost) and B (leftmost) give σ₁ and σ₂. D and E (top and bottom) give ±τ_max, both at σ_{avg}.
6. For the face rotated θ counter-clockwise, rotate the diameter XY by **2θ in the same sense**. The new ends are X′ and Y′.
7. The angle on the circle from CX to CA is $2\theta_p$; the element rotates by θ_p in the **same** sense.



Mohr's circle construction (Crandall convention)

6. Special cases (Crandall Prob 4.6).

State	Circle
Uniaxial tension σ_0	passes through the origin and σ_0; R = $\sigma_0/2$
Uniaxial compression	through the origin and $-\sigma_0$
Equal biaxial σ_0, σ_0	shrinks to a point: every direction is principal, no in-plane shear
Opposite biaxial σ_0, $-\sigma_0$	centred at the origin, R = σ_0: the same circle as pure shear
Pure shear $\pm\tau_0$	centred at the origin, R = τ_0; principal stresses $\pm\tau_0$ at 45°
Hydrostatic	a point, no shear on any plane

7. Common traps.

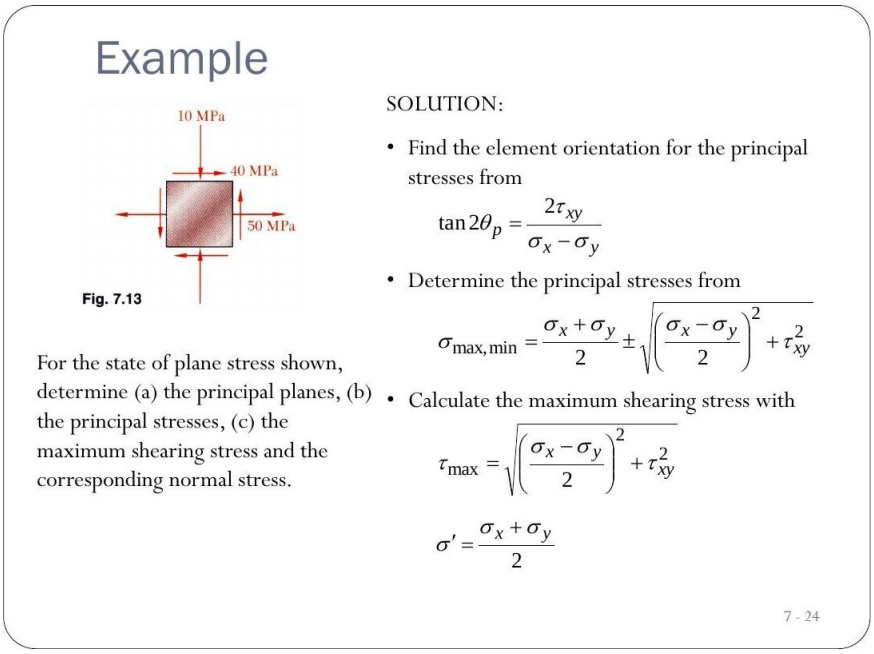
- **θ vs 2θ:** angles on the circle are double the element angles.
- **Direction of rotation:** consistent only if +τ is plotted downward at X (Crandall convention).
- Using arctan blindly: $\tan 2\theta_p$ has two solutions 180° apart. Pick by checking the quadrant on the circle, or substitute back.
- Reporting τ_max = R when both principal stresses have the same sign (the true maximum is out of plane).
- Forgetting σ_{avg} on the max-shear faces when drawing the element.

8. Summary box

$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2},\ \tan 2\theta_p=\frac{2\tau_{xy}}{\sigma_x-\sigma_y},\ \tau_{max}=R,\ \theta_s=\theta_p\pm 45^\circ,\ \sigma'=\sigma_{avg}$$

B. Solved questions

Q6 (Slides Ch4, Beer Example 7.1) — Principal planes, principal stresses and maximum shear (easy, classic)



Original: Slides Ch 4 Part 1, slide 24

Problem. For $\sigma_x = 50$, $\sigma_y = -10$ and $\tau_{xy} = +40$ MPa, find (a) the principal planes, (b) the principal stresses, (c) the maximum shearing stress and the corresponding normal stress. Then construct Mohr's circle.

Solution (equations).

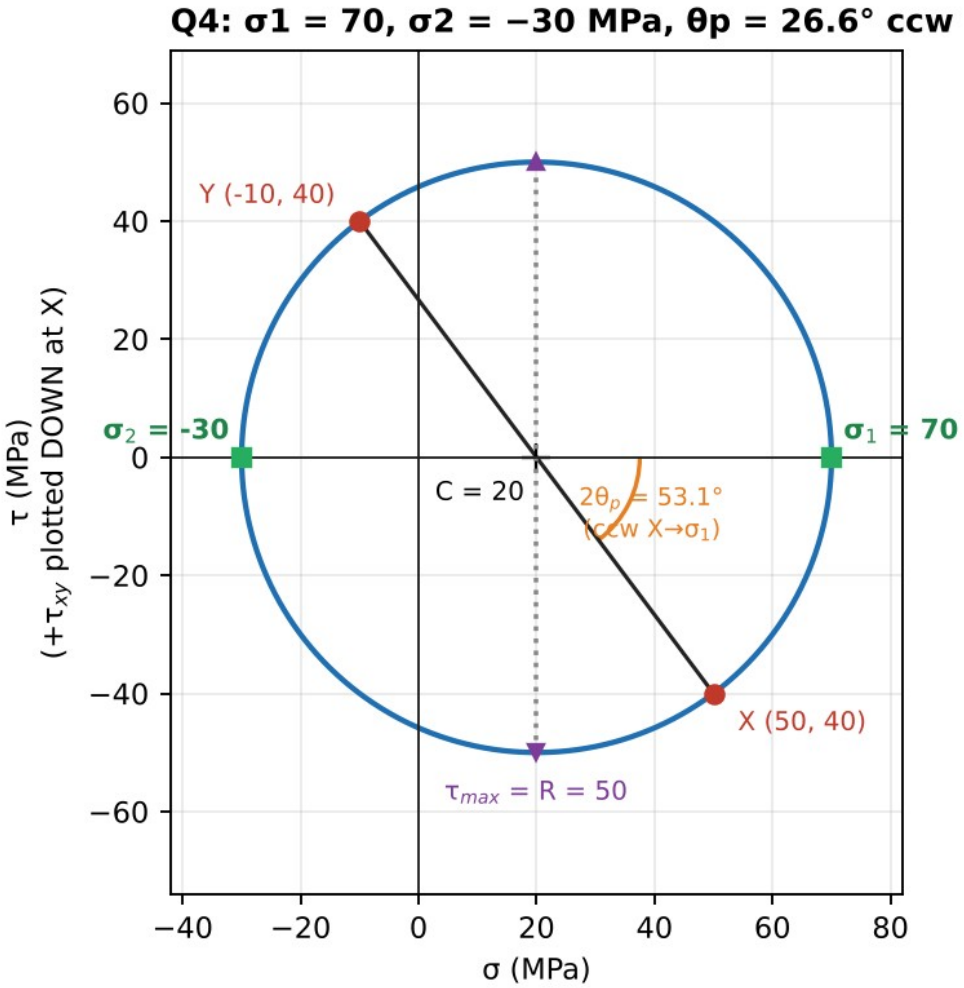
1. $\tan 2\theta_p = \frac{2(40)}{50-(-10)} = 1.333$, so 2θp = 53.1° (or 233.1°) and θp = 26.6° (or 116.6°).
2. Substituting θ = 26.6°: $\sigma_{x'} = 20 + 30(0.6) + 40(0.8) = 70$, so this angle carries σ1.
3. $R = \sqrt{30^2 + 40^2} = 50$, so σ1 = 20 + 50 = 70 and σ2 = 20 − 50 = −30 MPa.
4. τ_max = R = 50 MPa at θs = θp − 45° = −18.4° (and 71.6°), with σ′ = 20 MPa on those faces.

$$\sigma_1 = 70\text{ MPa at } 26.6^\circ, \sigma_2 = -30\text{ MPa at } 116.6^\circ, \tau_{max} = 50\text{ MPa at } -18.4^\circ, \sigma' = 20\text{ MPa}$$

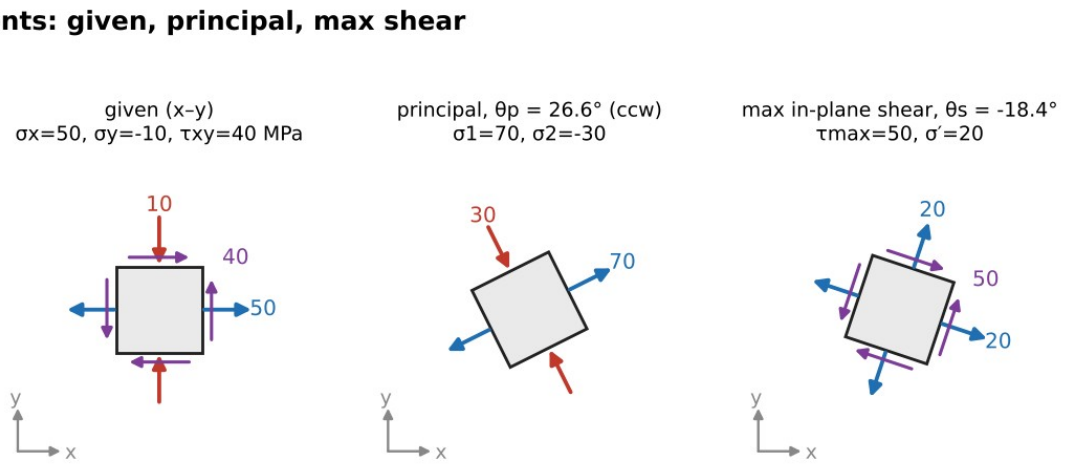
Solution (Mohr's circle). C = 20 and R = CX = 50 (CF = 30, FX = 40). X = (50, 40 *below*) and Y = (−10, 40 *above*). From X to A is 53.1° counter-clockwise, so the element rotates 26.6° counter-clockwise ✓ (slide: "tan 2θp = FX/CF = 40/30").

Check. Absolute max shear: σ1 = 70, σ2 = −30, σ3 = 0, so $\tau_{abs} = (70+30)/2 = 50$, the same as in-plane because the signs are opposite ✓.

Exam takeaway. Show both the equations and the circle; the slides do exactly that.

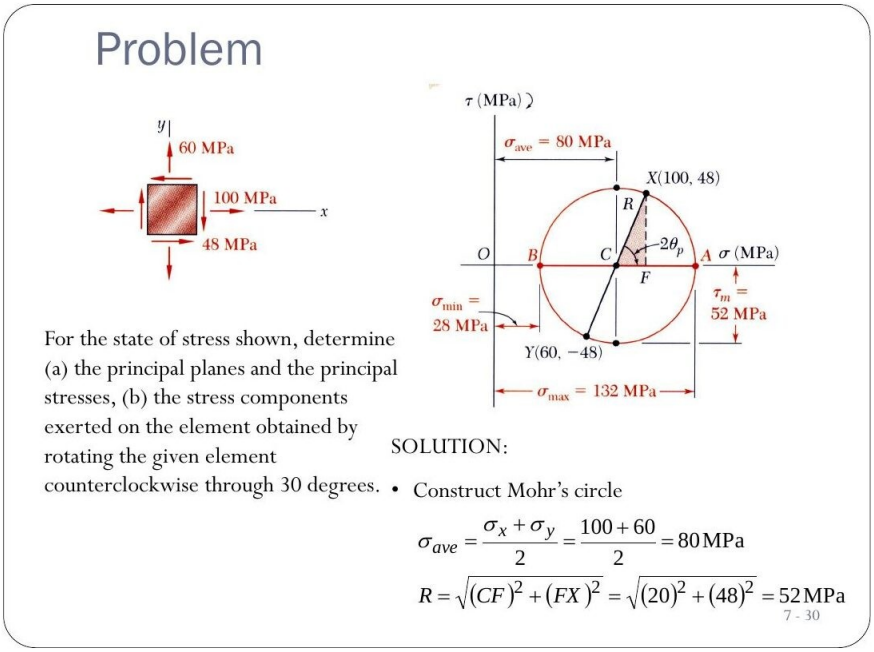


Mohr's circle



Stress elements: given, principal, maximum shear

Q7 (Slides Ch4, Beer Sample 7.2) — Principal stresses, then rotate the element 30° counter-clockwise (medium)



Original: Slides Ch 4 Part 1, slide 30

Problem. For $\sigma_x = 100$, $\sigma_y = 60$ and $\tau_{xy} = -48$ MPa (shear on the right face points down), find (a) the principal planes and stresses and (b) the stresses on the element rotated 30° counter-clockwise.

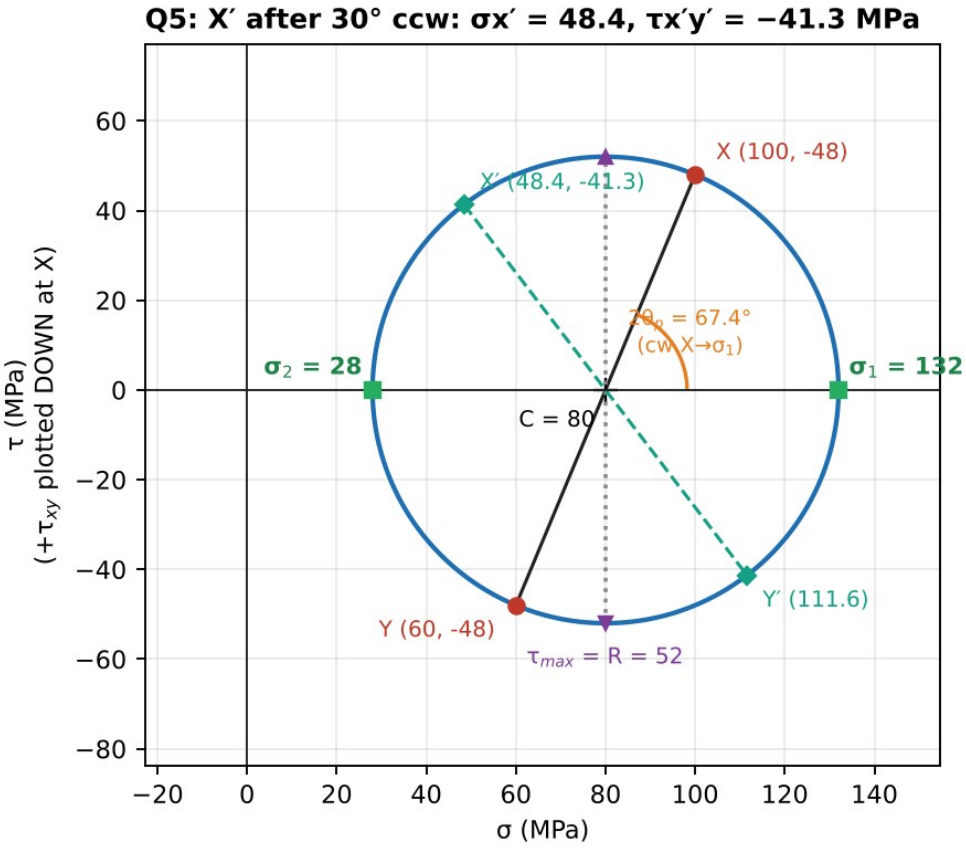
Solution.

1. C = 80; $R = \sqrt{20^2 + 48^2} = 52$.
2. $\sigma_1 = 132$ and $\sigma_2 = 28$ MPa. $\tan 2\theta_p = -96/40$, so $2\theta_p = -67.4°$ and $\theta_p = 33.7°$ **clockwise** for σ_1.
3. *Rotate 30° counter-clockwise* ($2\theta = 60°$):
   - $\sigma_{x'} = 80 + 20\cos 60^\circ - 48\sin 60^\circ = 80 + 10 - 41.6 = 48.4$ MPa
   - $\sigma_{y'} = 160 - 48.4 = 111.6$ MPa
   - $\tau_{x'y'} = -20\sin 60^\circ - 48\cos 60^\circ = -17.3 - 24 = -41.3$ MPa

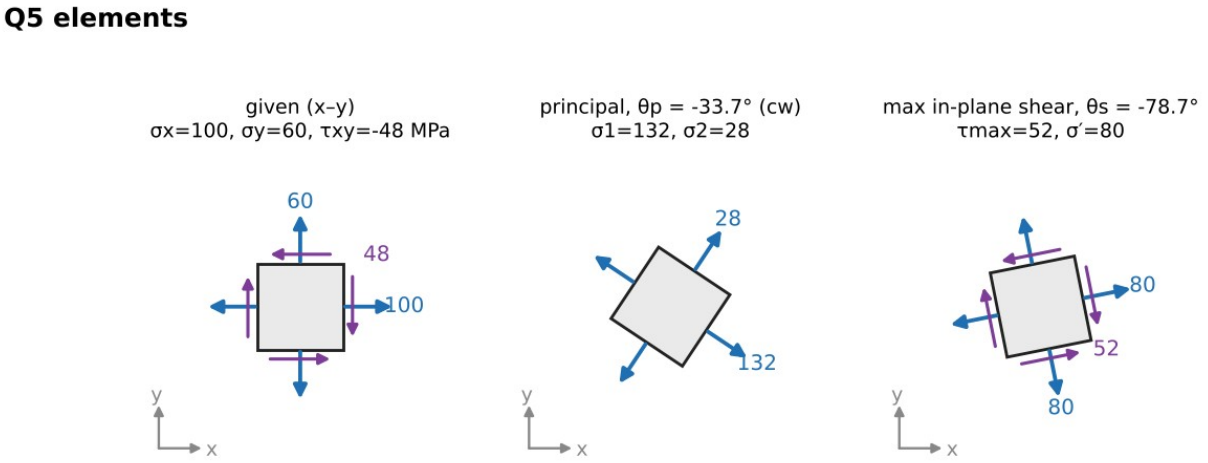
$$\sigma_1 = 132,\ \sigma_2 = 28\ \text{MPa},\ \theta_p = 33.7^\circ\ \text{cw};\ \sigma_{x'} = 48.4,\ \sigma_{y'} = 111.6,\ \tau_{x'y'} = -41.3\ \text{MPa}$$

Check (slide's circle method). X starts 67.4° *clockwise* from A (it sits 48 above the axis, because τ is negative). Rotating 60° counter-clockwise puts X′ at φ = 180° − 60° − 67.4° = 52.6° from the negative σ-axis. So $\sigma_{x'} = 80 - 52\cos 52.6^\circ = 48.4$ ✓ and $|\tau| = 52\sin 52.6^\circ = 41.3$ ✓. Invariant: 48.4 + 111.6 = 160 ✓.

Exam takeaway. "Rotate the element" means "rotate the diameter by twice that angle, in the same sense".



Mohr's circle with the rotated diameter X′Y′



Stress elements

Q8 (Crandall Prob 4.6) — Sketch Mohr's circle for standard states (easy, conceptual)

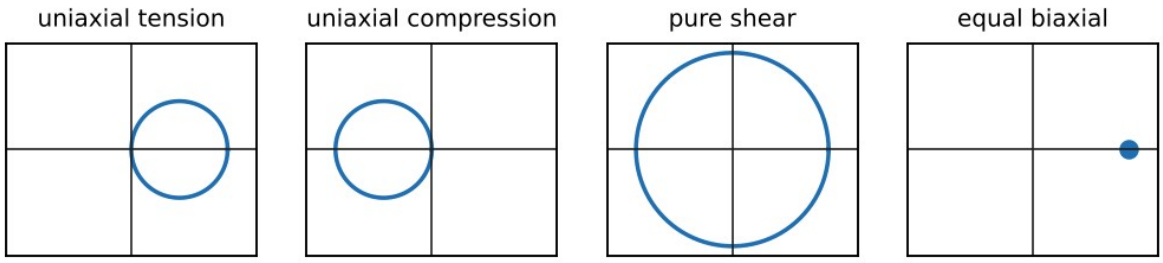
Problem. Sketch Mohr's circle for each of the plane-stress cases drawn. They are built from normal stresses $\pm\sigma_0$ on the x and/or y faces and shear $\pm\tau_0$.

Solution. Use the table in item 6.

1. One normal stress only: the circle passes through the origin; $R = \sigma_0/2$.
2. Equal normal stresses of the same sign on x and y: a single point (no shear on any plane).
3. Equal normal stresses of **opposite** sign: centred at the origin with $R = \sigma_0$, identical to the pure-shear circle.
4. Pure shear $\pm\tau_0$: centred at the origin with $R = \tau_0$. Only the positions of X and Y swap between $+\tau_0$ and $-\tau_0$.

uniaxial: through O; equal biaxial: a point; $(\sigma_0, -\sigma_0) \equiv$ pure shear $\tau_0 = \sigma_0$

Exam takeaway. "σ_0 in x plus $-\sigma_0$ in y" is pure shear of σ_0 on 45° planes. It is a favourite one-mark question (see Q14).



Signature circles: uniaxial tension, uniaxial compression, pure shear, equal biaxial

Q9 (Crandall Prob 4.7) — Principal stresses and orientations for five states (medium)

4.7. Find the principal stresses and the orientation of the principal axes of stress for the following cases of plane stress.

(a) $\sigma_x = 40$ MN/m², $\sigma_y = 0$, $\tau_{xy} = 80$ MN/m²

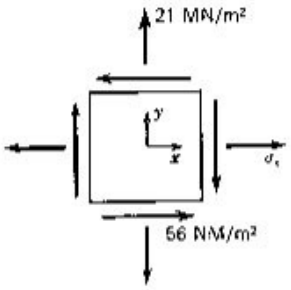
(b) $\sigma_x = 140$ MN/m², $\sigma_y = 20$ MN/m², $\tau_{xy} = -60$ MN/m²

(c) $\sigma_x = -120$ MN/m², $\sigma_y = 50$ MN/m², $\tau_{xy} = 100$ MN/m²

(d) $\sigma_x = 10{,}000$ psi, $\sigma_y = -4{,}000$ psi, $\tau_{xy} = 8{,}000$ psi

(e) $\sigma_x = -10{,}000$ psi, $\sigma_y = 20{,}000$ psi, $\tau_{xy} = -6{,}000$ psi

4.8. If the minimum principal stress is -7 MN/m², find σ_x and the angle which the principal stress axes make with the xy axes for the case of plane stress illustrated.



Prob. 4.8

Original: Crandall Prob 4.7 and 4.8 (textbook figure)

Problem. Find the principal stresses and the orientation of the principal axes for: (a) 40, 0, 80; (b) 140, 20, −60; (c) −120, 50, 100 (all MN/m², listed as $\sigma_x, \sigma_y, \tau_{xy}$); (d) 10 000, −4 000, 8 000 psi; (e) −10 000, 20 000, −6 000 psi.

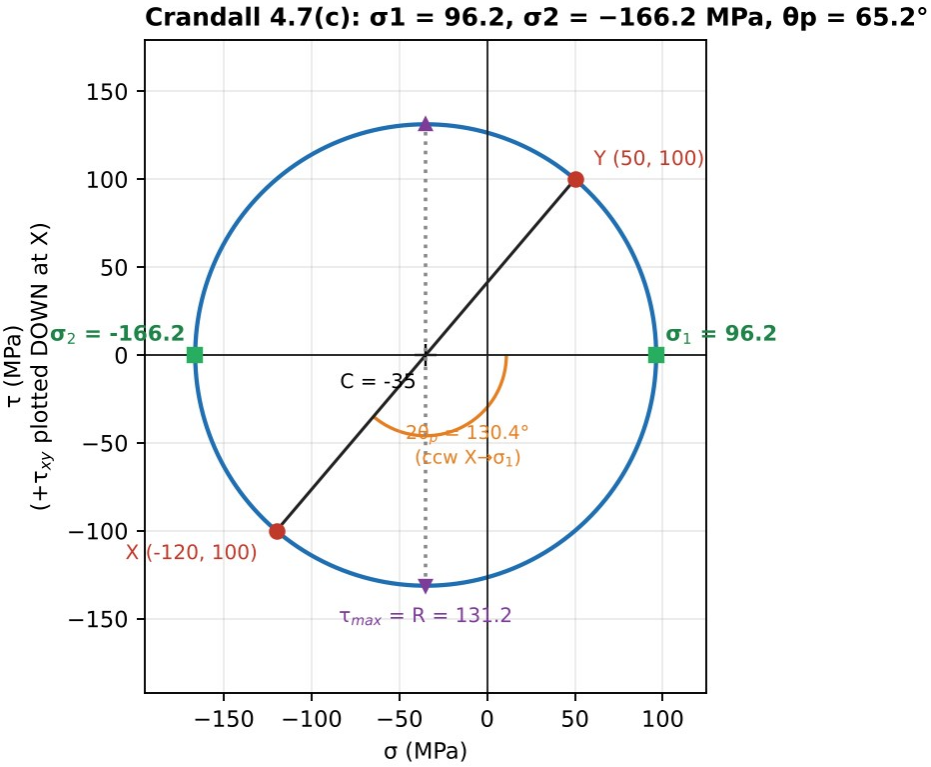
Approach. For each case: $\acute{\sigma}$, R, $\sigma_{1,2} = \acute{\sigma} \pm R$, and $\theta_p = 1/2 \tan^{-1} \dfrac{2\tau_{xy}}{\sigma_x - \sigma_y}$, choosing the branch (atan2) that gives σ_1.

Solution.

Case	$\acute{\sigma}$	R	σ_1	σ_2	θ (σ_1 from x)
(a)	20	82.46	**102.5**	**−62.5**	38.0°
(b)	80	84.85	**164.9**	**−4.9**	−22.5°
(c)	−35	131.2	**96.2**	**−166.2**	65.2°
(d) psi	3000	10 630	**13 630**	**−7 630**	24.4°
(e) psi	5000	16 155	**21 155**	**−11 155**	−79.1° (σ_2 at 10.9°)

Check. Each row: $\sigma_1 + \sigma_2 = \sigma x + \sigma y$ ✓. For (b), the σ_1 direction is clockwise from x because τ is negative ✓.

Exam takeaway. A table is the fastest way to present a multi-case question. The angle always belongs to σ_1 unless you say otherwise.



Mohr's circle for case (c)

Q10 (Crandall Prob 4.8) — Find σx when the minimum principal stress is known (medium)

Problem. In a state of plane stress, $\sigma_y = 21$ MN/m² (tension) and the shear on the x faces is 56 MN/m², acting downward on the right face. If the minimum principal stress is −7 MN/m², find σx and the angle the principal axes make with x–y.

Given: $\sigma_y = 21$, $\tau_{xy} = -56$, $\sigma_2 = -7$ MPa. **Find:** σx, θp.

Approach. Write $\sigma_2 = \acute{\sigma} - R$ with σx unknown, and solve.

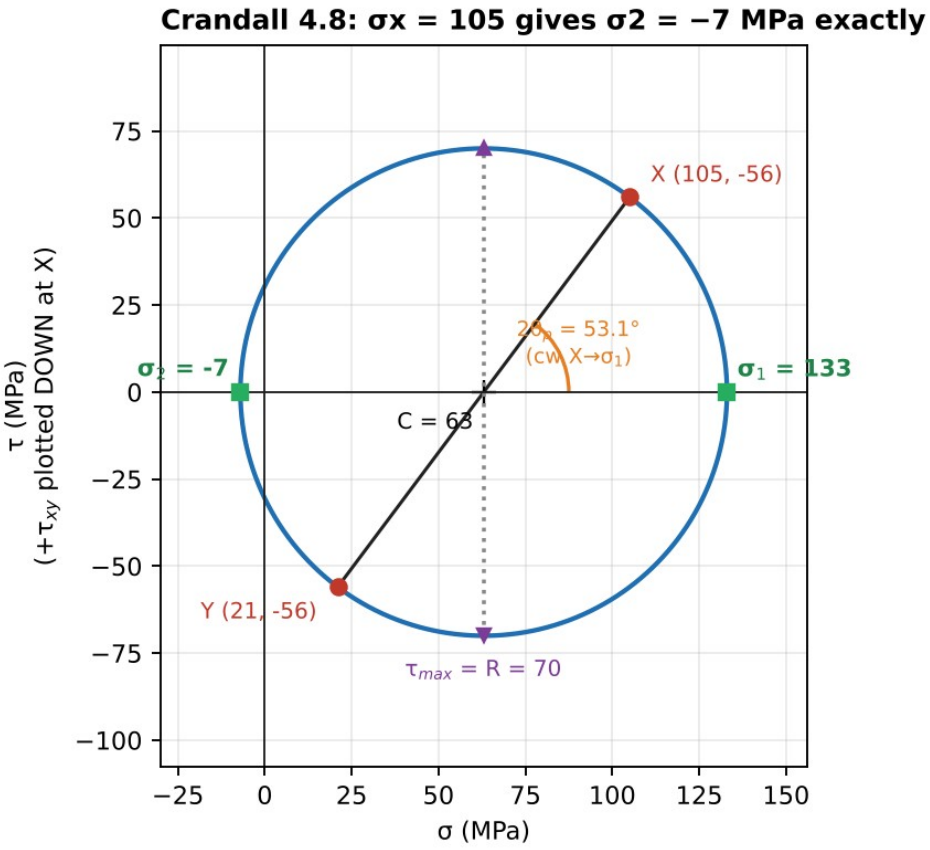
Solution.

1. $\dfrac{\sigma_x + 21}{2} - \sqrt{\left(\dfrac{\sigma_x - 21}{2}\right)^2 + 56^2} = -7.$
2. Rearrange: $\dfrac{\sigma_x + 21}{2} + 7 = \sqrt{\left(\dfrac{\sigma_x - 21}{2}\right)^2 + 3136}.$
3. Square both sides. Using $\left(\dfrac{\sigma_x + 21}{2}\right)^2 - \left(\dfrac{\sigma_x - 21}{2}\right)^2 = 21\sigma_x$: $21\sigma_x + 7(\sigma_x + 21) + 49 = 3136$, so $28\sigma_x = 2940$ and σx = 105 MPa.
4. Check: C = 63 and R = $\sqrt{42^2 + 56^2} = 70$, so σ_2 = −7 ✓ and σ_1 = 133 MPa.
5. $\tan 2\theta_p = \dfrac{2(-56)}{105 - 21} = -1.333$, so 2θp = −53.1° and θp = −26.6°.

$\sigma_x = 105$ MN/m²; $\sigma_1 = 133$ at 26.6° clockwise from x, $\sigma_2 = -7$ at 63.4° counter-clockwise

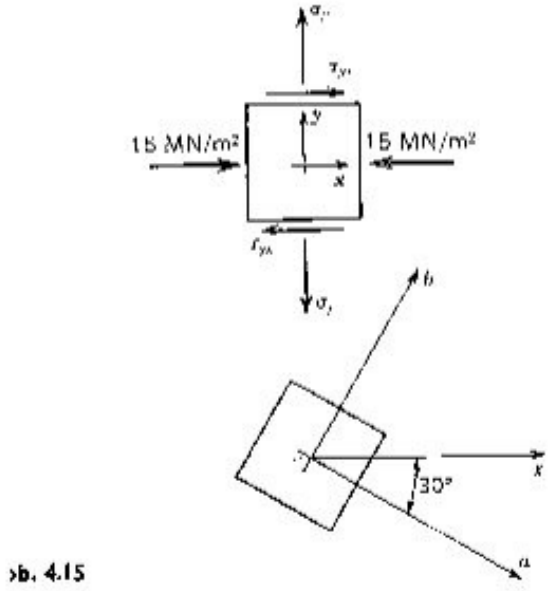
Check. The R is a 3-4-5 triangle scaled by 14 (42-56-70) ✓.

Exam takeaway. "Given one principal stress, find a component" is a squared-root equation; the difference-of-squares trick makes it linear. **Variation:** given σ_1 instead.



Mohr's circle

Q11 (Crandall Prob 4.15) — Deduce the state, then find stresses on rotated faces (medium)



Original: Crandall Prob 4.15 (textbook figure)

Problem. A plate is in uniform plane stress. The maximum tensile stress on any face (normal in the x–y plane) is 75 MN/m². On the face perpendicular to x there is a compressive stress of 15 MN/m² and **no** shear. Find the stresses on faces perpendicular to axes a and b, where a is 30° below x ($\theta = -30°$) and b is 60° above x.

Approach. Zero shear on the x face means x is a principal direction, so σx = −15 is a principal stress. The other principal stress is the 75 MPa maximum, acting along y. Then transform.

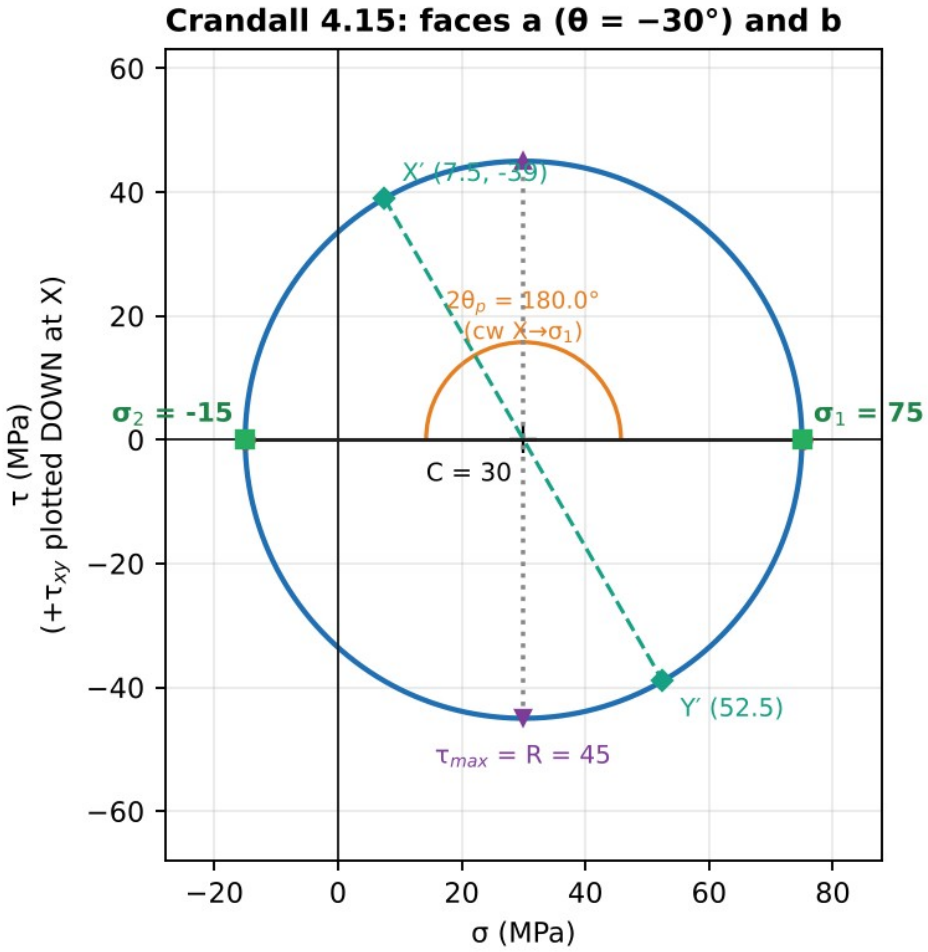
Solution.

1. *State:* $\sigma_x = -15, \sigma_y = 75, \tau_{xy} = 0$ (principal axes x, y).
2. $\acute{\sigma} = 30, \dfrac{\sigma_x - \sigma_y}{2} = -45$; for face a, θ = −30°, so 2θ = −60°.
3. $\sigma_a = 30 + (-45)\cos(-60^\circ) = 30 - 22.5 = 7.5$ MPa.
4. $\sigma_b = 60 - 7.5 = 52.5$ MPa.
5. $\tau_{ab} = -(-45)\sin(-60^\circ) = 45(-0.866) = -39.0$ MPa.

$$\sigma_a = 7.5\,\text{MPa}, \sigma_b = 52.5\,\text{MPa}, \tau_{ab} = -39.0\,\text{MPa}$$

Check. $|\tau_{ab}| \le R = 45$ ✓; 7.5 + 52.5 = 60 = −15 + 75 ✓.

Exam takeaway. "No shear on a face" ⇒ that face is a principal plane. Reading this from the wording is the whole problem.



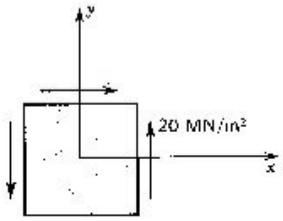
Mohr's circle with faces a and b

Q12 (Crandall Prob 4.28) — Complete the picture from the principal stresses (medium)

4.28. At a point in a body in plane stress the shear stress on the xy faces is as sketched. Also, it is known that the principal stresses at this point are

$$\sigma_1 = 20 \text{ MN/m}^2 \qquad \sigma_2 = -45 \text{ MN/m}^2$$

Complete the stress picture for the xy faces; that is, calculate σ_x and σ_y and show them in proper direction and magnitude in a suitable sketch.



Original: Crandall Prob 4.28 (textbook figure)

Problem. At a point in plane stress, the shear on the x–y faces is 20 MN/m² as sketched (↑ on the right face, so $\tau_{xy} = +20$). The principal stresses are σ1 = 20 and σ2 = −45 MN/m². Find σx and σy.

Approach. The invariant gives σx + σy; the radius gives |σx − σy|. Expect two answers.

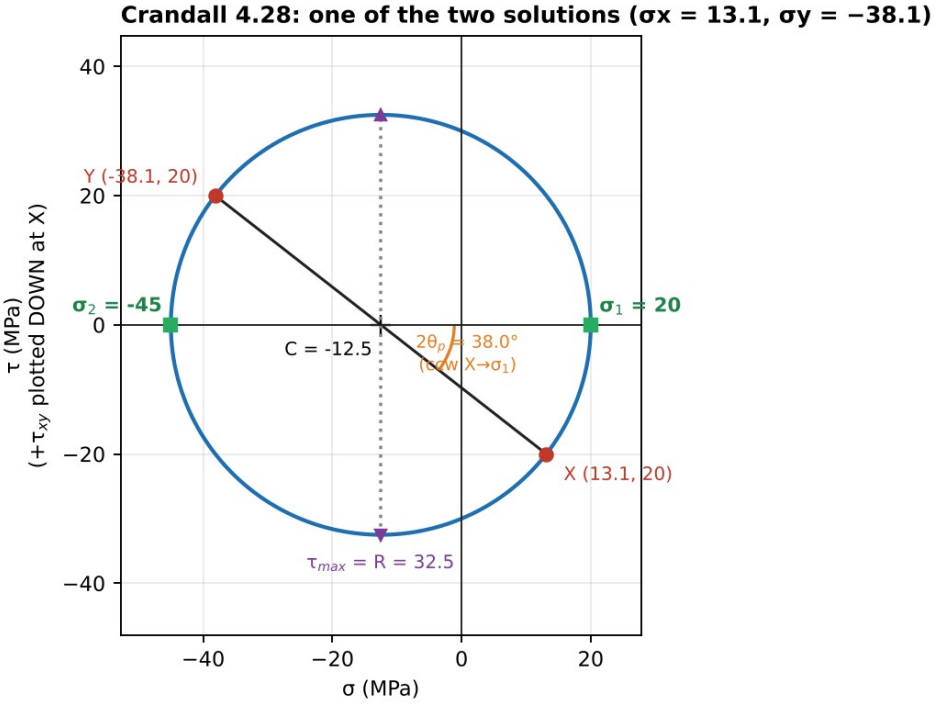
Solution.

1. *Sum:* $\sigma_x + \sigma_y = \sigma_1 + \sigma_2 = -25$, so $\acute{\sigma} = -12.5$.
2. *Radius:* $R = (20+45)/2 = 32.5$, so $\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 = 32.5^2 - 20^2 = 656.25$ and $\frac{\sigma_x - \sigma_y}{2} = \pm 25.62$.
3. Two solutions: σx = −12.5 + 25.62 = 13.1 with σy = −38.1, **or** σx = −38.1 with σy = 13.1.

$$(\sigma_x, \sigma_y) = (13.1, -38.1) \text{ or } (-38.1, 13.1) \text{ MN/m}^2 \quad (\tau_{xy} = +20)$$

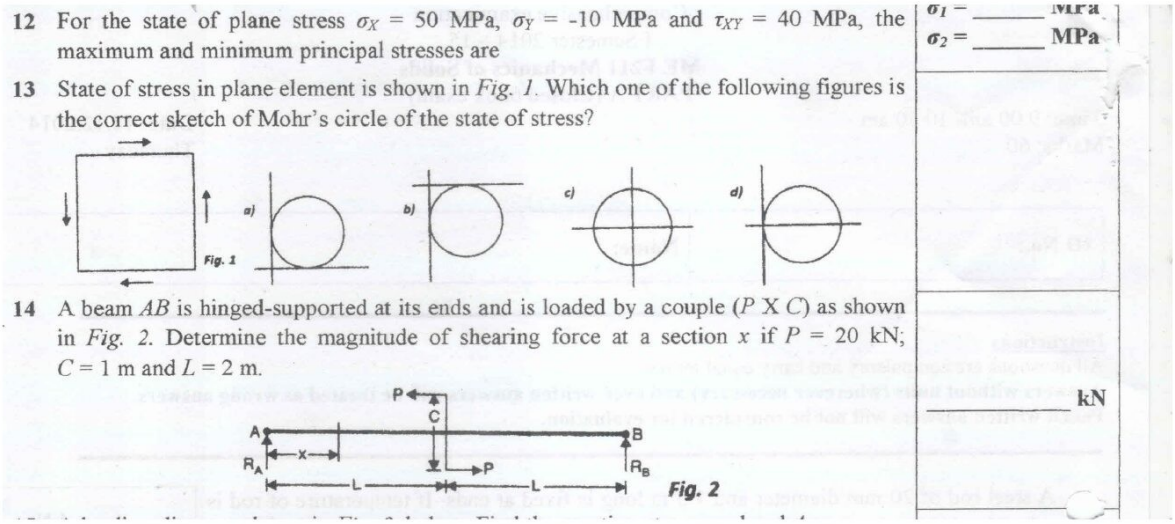
Check. Both give a circle with C = −12.5 and R = 32.5 ✓. They are the same physical state with the principal axes rotated 90°.

Exam takeaway. Without more information (e.g. the principal angle) both answers are correct; showing both earns full marks.



Mohr's circle (first solution)

Q13 (PYQ 2014-15 Compre A Q12) — Principal stresses (easy, one-line answer)


12 For the state of plane stress σ_X = 50 MPa, σ_Y = -10 MPa and τ_{XY} = 40 MPa, the maximum and minimum principal stresses are

σ_1 = _____ MPa
σ_2 = _____ MPa

13 State of stress in plane element is shown in *Fig. 1*. Which one of the following figures is the correct sketch of Mohr's circle of the state of stress?

Fig. 1 a) b) c) d)

14 A beam *AB* is hinged-supported at its ends and is loaded by a couple (*P* X *C*) as shown in *Fig. 2*. Determine the magnitude of shearing force at a section *x* if *P* = 20 kN; *C* = 1 m and *L* = 2 m.

P C A B x R_A P L L R_B Fig. 2

_____ kN

Original question: PYQ 2014-15 Compre A Q12–Q13

Problem. For $\sigma_x = 50$, $\sigma_y = -10$ and $\tau_{xy} = 40$ MPa, find the maximum and minimum principal stresses.

Solution. The same state as Q6: C = 20, R = 50.

$$\sigma_1 = 70\ \text{MPa}, \sigma_2 = -30\ \text{MPa}$$

Q14 (PYQ 2014-15 Compre A Q13) — Which Mohr's circle? (easy, conceptual)

Problem. The element shown carries only shear stresses (no normal arrows). Which of the four sketches is its Mohr's circle?

Solution. Pure shear gives a circle **centred at the origin**, with radius τ.

$$\text{the circle centred on the origin}\ (\text{option c in the paper})$$

Exam takeaway. Know the four signature circles (uniaxial tension, uniaxial compression, pure shear, equal biaxial) from Q8.

Q15 (PYQ 2014-15 Test-2 Q1) — Axial force + torque on a 40 mm shaft: principal stresses and Mohr's circle (medium, 20 marks)

Q.1 An axial force of 90 kN and a torque of 250 Nm are applied to the shaft as shown in *Fig. 1*. If the shaft has a diameter of 40 mm, determine the principal stresses and orientation of principal axes (with respect to *x*-axis) at a point *P* on its surface. Sketch the Mohr's circle for stress, representing all salient points for given combined state of stress. **[20]**

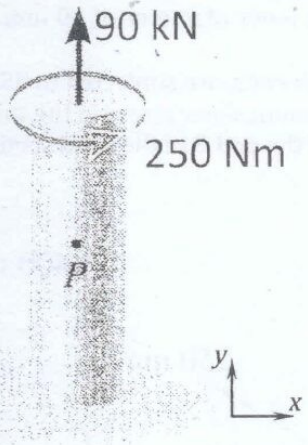


Fig. 1

Original question: PYQ 2014-15 Test-2 Q1

Problem. An axial force of 90 kN and a torque of 250 N·m act on a 40 mm diameter shaft. Find the principal stresses and the orientation of the principal axes (relative to the shaft axis) at a point P on the surface, and sketch Mohr's circle.

Approach. Axial stress P/A and torsional shear $16T/\pi d^3$ (Ch. 6) at the surface point, then the plane-stress formulas.

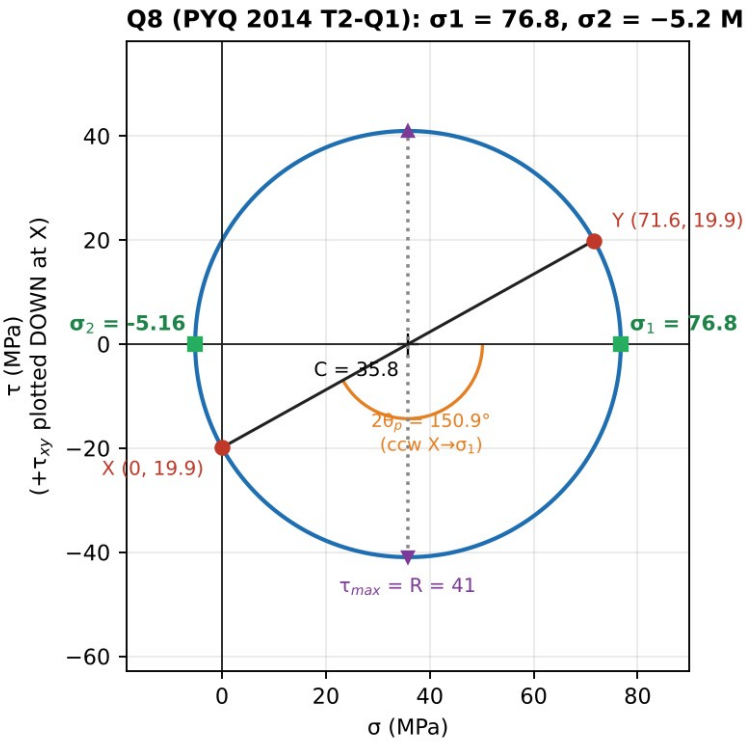
Solution.

1. $\sigma = \dfrac{90\,000}{\pi (20)^2} = 71.6$ MPa along the shaft axis (y in the figure).
2. $\tau = \dfrac{16(250\,000)}{\pi (40)^3} = 19.9$ MPa.
3. With $\sigma_y = 71.6$, $\sigma_x = 0$, $|\tau_{xy}| = 19.9$: C = 35.8 and $R = \sqrt{35.8^2 + 19.9^2} = 41.0$.
4. σ_1 = 76.8 MPa and σ_2 = −5.2 MPa; τ_{max} = 41.0 MPa.
5. $\tan 2\theta_p = \dfrac{2(19.9)}{71.6}$, so θ_p = 14.5°, measured from the **shaft axis** to the σ_1 direction; the sense follows the direction of the torque.

$$\sigma_1 = 76.8\ \text{MPa},\ \sigma_2 = -5.2\ \text{MPa},\ \tau_{max} = 41.0\ \text{MPa},\ \theta_p = 14.5^\circ \text{ from the axis}$$

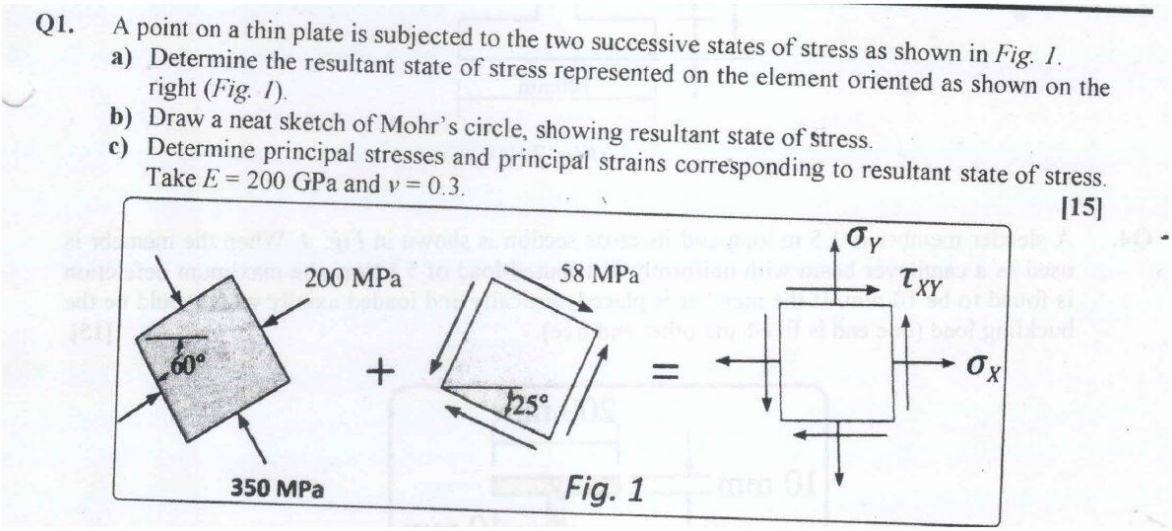
Check. $\sigma_1\sigma_2 = -398 = -\tau^2$ (as required when one normal stress is zero) ✓. The circle crosses the τ-axis because $\sigma_2 < 0$ ✓.

Exam takeaway. The axial stress tilts the principal direction toward the axis; pure torsion alone would give 45°.
Variation: compressive axial load, which gives σ_2 the larger magnitude.



Mohr's circle

Q16 (PYQ 2014-15 Compre B Q1) — Superposing two stress states (hard, 15 marks)



Original question: PYQ 2014-15 Compre B Q1

Problem. A point on a thin plate is subjected to two successive states of stress. State 1 is biaxial compression (−350 MPa and −200 MPa) on an element rotated 60°; state 2 is pure shear of 58 MPa on an element rotated 25°. (a) Find the resultant stress on the x–y element. (b) Sketch Mohr's circle. (c) Find the principal stresses and principal strains. E = 200 GPa, ν = 0.3.

Approach. You **cannot add stresses on differently oriented elements**. Transform each state to the same x–y axes (θ = −60° and θ = −25° respectively), *then* add component by component.

Solution.

1. *State 1* (−350 along x′ at 60°, −200 along y′): $\sigma_x = -275 + (-75)\cos(-120^\circ) = -237.5$, $\sigma_y = -312.5$, $\tau_{xy} = -(-75)\sin(-120^\circ) = -65.0$.

2. *State 2* (pure shear τ′ = −58 on an element at 25°; the sign follows the arrows): $\sigma_x = \tau' \sin(-50^\circ) = 44.4$, $\sigma_y = -44.4$, $\tau_{xy} = \tau' \cos(-50^\circ) = -37.3$.
3. *Sum:* σx ≈ −193 MPa, σy ≈ −357 MPa, |τxy| ≈ 102 MPa. (This matches the textbook answer to the Hibbeler problem this question is taken from. The sign of τ depends on how the arrows are read; the principal values don't.)
4. *Principal stresses:* $\acute{\sigma} = -275$, $R = \sqrt{81.9^2 + 102.2^2} = 131.0$, so σ1 = −144 and σ2 = −406 MPa (σ3 = 0).
5. *Principal strains (Hooke, plane stress; Ch. 5):*

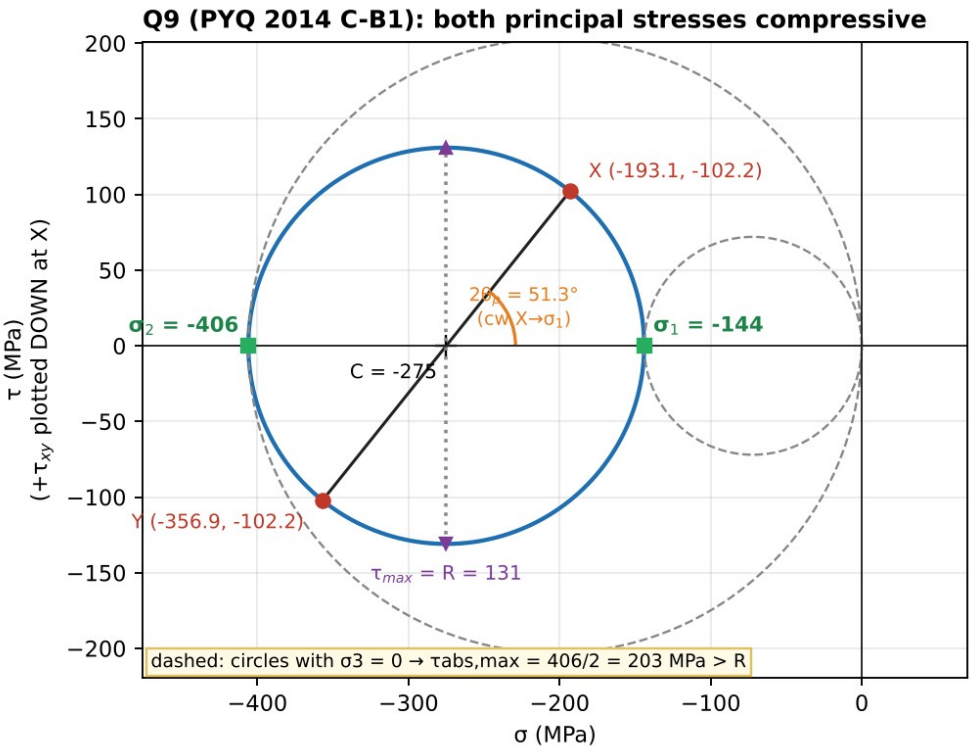
$$\epsilon_1 = \frac{\sigma_1 - \nu\sigma_2}{E} = \frac{-144 + 121.8}{200\,000} = -111 \times 10^{-6}, \epsilon_2 = \frac{-406 + 43.2}{200\,000} = -1814 \times 10^{-6}$$

$$\epsilon_3 = -\frac{\nu}{E}(\sigma_1 + \sigma_2) = +825 \times 10^{-6} \text{ (the plate gets thicker)}$$

$$\sigma_x \approx -193, \sigma_y \approx -357, |\tau_{xy}| \approx 102\,\text{MPa}; \sigma_{1,2} = -144, -406\,\text{MPa}; \epsilon_{1,2,3} = -111, -1814, +825\,\mu$$

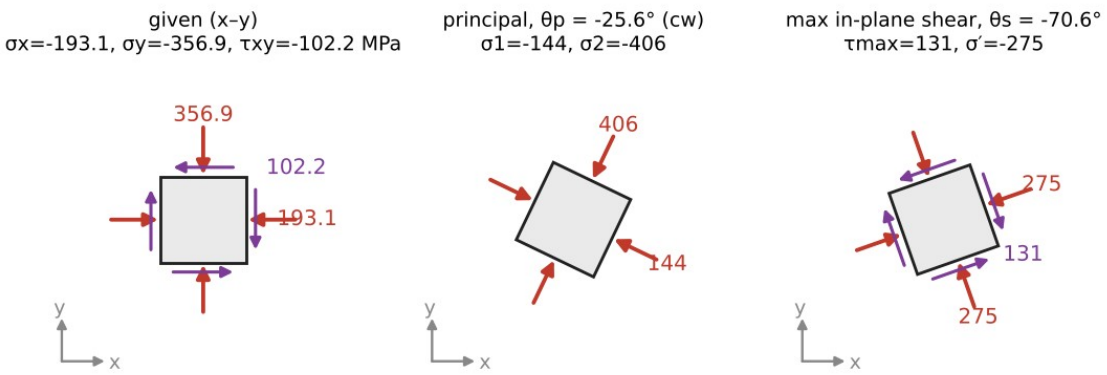
Check. $\sigma_x + \sigma_y = -550 = \sigma_1 + \sigma_2$ ✓. Both principal stresses are compressive, so the circle lies entirely left of the τ-axis and $\tau_{abs,max} = 406/2 = 203$ MPa > R ✓.

Exam takeaway. The rule "transform first, then add" is what is being tested. **Variation:** add a third, uniaxial state.



Mohr's circle, with the σ3 = 0 circles dashed

Q9 resultant element (x-y) and principal element



Resultant x–y element and principal element

Concept 4.4 — Thin-walled pressure vessels (Crandall Probs 4.10–4.14)

A. Concept explanation

1. Intuition. A soft-drink can, a gas cylinder, a boiler, a balloon. The wall is thin compared with the radius, so it cannot resist pressure by bending; it does so by stretching like a membrane. A cylinder is stretched twice as hard around its circumference as along its length. That is why an overheated sausage splits lengthwise.

2. Why it matters. Pressure vessels, pipes, tanks, filament-wound cases, biological cells (Q23). They provide a ready-made biaxial stress state for Mohr's-circle and Hooke's-law problems (Ch. 5 Crandall 5.12, 5.20), and the ring case appeared in PYQ 2014 Compre A Q11.

3. Formal treatment. For a thin wall ($t \ll r$), use cylindrical coordinates (r, θ, z). By symmetry there are no shear stresses on r, θ and z faces, so **r, θ and z are principal directions** (Crandall Prob 4.10). The radial stress varies from −p on the inner surface to 0 on the outer, which is negligible against pr/t, so $\sigma_r \approx 0$.

Vessel	σ_θ (hoop)	σ_z or σ_ϕ (axial/meridional)
Open cylinder with axial force F (4.10)	pr/t	$F/2\pi rt$
Closed cylinder (4.12)	pr/t	$pr/2t$
Sphere (4.14)	$pr/2t$	$pr/2t$

4. Derivations.

(a) Hoop stress. Cut a cylinder of length L along a diameter and keep half. Pressure acts on the projected area 2rL; the two cut walls carry $\sigma_\theta t\,L$ each:

$$2\sigma_\theta t\,L = p(2r)L \Rightarrow \sigma_\theta = \frac{pr}{t}$$

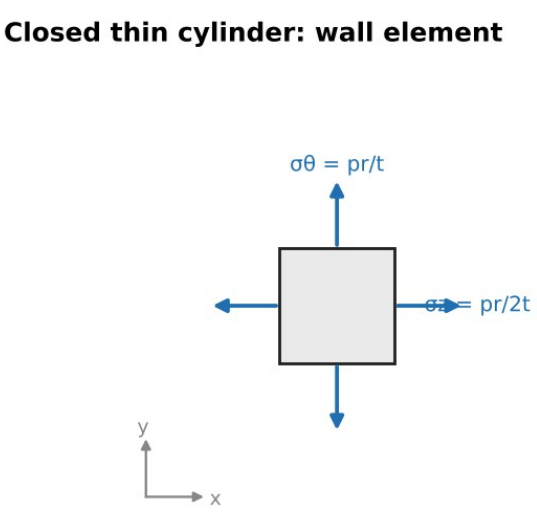
(b) Axial stress, closed ends. Cut across the cylinder. The pressure on the end cap, $p\pi r^2$, is carried by the wall ring of area $2\pi rt$:

$$\sigma_z(2\pi rt) = p\pi r^2 \Rightarrow \sigma_z = \frac{pr}{2t}$$

(c) Sphere. Any great-circle cut gives the same balance as (b): $\sigma = pr/2t$ in every tangential direction.

(d) Maximum in-plane shear, closed cylinder (Crandall Prob 4.13): $\tau_{\theta z}=\frac{\sigma_\theta-\sigma_z}{2}=\frac{pr}{4t}$, one quarter of the hoop stress.

5. Picture. A wall element of a closed cylinder carries σ_θ around and $\sigma_z=\sigma_\theta/2$ along. Its Mohr's circle runs from pr/2t to pr/t, and the 3-D circle including $\sigma_r=0$ is larger (absolute τ_max = pr/2t).



Closed cylinder: σθ = pr/t around, σz = pr/2t along

6. Special cases.

- An open-ended pipe with no axial force has $\sigma_z=0$: uniaxial hoop stress.
- A sphere has equal biaxial stress, so its in-plane Mohr's circle is a point; it is the most efficient shape (half the hoop stress of a cylinder of the same r and t).
- Filament-wound vessels: fibres at ±54.7° to the axis carry closed-end pressure with equal fibre tension (Q22).

7. Traps.

- Using the diameter instead of the radius.
- Writing σz = pr/t (that is the hoop value).
- Forgetting that a strain gauge on the surface reads strain, not stress: use Hooke's law with both stresses (Ch. 5).

8. Summary box

$$\text{cylinder: } \sigma_\theta=\frac{pr}{t},\ \sigma_z=\frac{pr}{2t}\ (\text{closed});\ \text{sphere: } \sigma=\frac{pr}{2t};\ \tau_{\theta z}=\frac{pr}{4t};\ \sigma_r\approx 0$$

B. Solved questions

Q17 (Crandall Probs 4.10, 4.12, 4.14) — Derive the thin-wall stresses (easy, derivation)

Problem. (4.10) For a thin-walled cylinder with internal pressure p and axial force F, show that r, θ and z are principal directions and find the stresses. (4.12) Repeat with closed ends carrying the pressure. (4.14) Repeat for a thin sphere.

Solution.

1. *Principal directions:* the geometry and loading are symmetric about the axis and about any plane through it. A shear stress $\tau_{\theta z}$ on a wall element would have to reverse sign under reflection in such a plane, which leaves the loading unchanged. So it is zero; similarly for the other shears. Therefore r, θ and z are principal.

2. *Hoop:* derivation (a) gives $\sigma_\theta = pr/t$.
3. *Axial:* with F only (open ends), $\sigma_z = F/(2\pi rt)$. With closed ends, the cap force $p\pi r^2$ gives $\sigma_z = pr/2t$.
4. *Sphere:* $\sigma_\theta = \sigma_\phi = pr/2t$.
5. *Radial:* $|\sigma_r| \le p$, which is t/r times smaller than σθ, so $\sigma_r \approx 0$.

$$\sigma_r \approx 0;\ \sigma_\theta = \frac{pr}{t};\ \sigma_z = \frac{F}{2\pi rt}\ (\text{open}),\ \frac{pr}{2t}\ (\text{closed});\ \text{sphere}\ \sigma = \frac{pr}{2t}$$

Exam takeaway. The "projected area" balance is a two-line derivation that examiners like to see.

Q18 (Crandall Prob 4.13) — Closed cylinder: τ_max in the θz plane is σmax/4 (easy)

Solution.

$$\tau_{\theta z,max} = \frac{\sigma_\theta - \sigma_z}{2} = \frac{pr/t - pr/2t}{2} = \frac{pr}{4t} = \frac{1}{4}\sigma_\theta \text{ shown}$$

Exam takeaway. In 3-D (with σr = 0), the absolute maximum shear is $\sigma_\theta/2 = pr/2t$, on planes at 45° between θ and r.

Q19 (Crandall Prob 4.11) — Relation between F and p for two shear conditions (medium)

Problem. For the open cylinder of 4.10 (pressure p and axial force F), what relation between F and p makes (a) the maximum shear stress in the wall equal in magnitude to the maximum normal stress, and (b) half of it?

Approach. In the wall (θz plane): σθ = pr/t and σz = F/2πrt. The in-plane maximum shear is |σθ − σz|/2.

Solution.

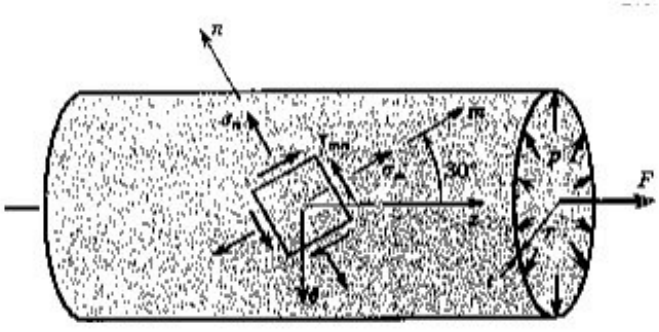
1. *(a)* |σθ − σz|/2 = max(|σθ|, |σz|) requires σz = −σθ: equal and opposite. So $F/2\pi rt = -pr/t$, giving $F = -2\pi r^2 p$ (an axial **compression** of magnitude 2πr²p).
2. *(b)* |σθ − σz|/2 = ½ max(|σθ|, |σz|) requires σz = 0, so F = 0 (an open pipe with no axial force).

$$(a)\ F = -2\pi r^2 p\ (\text{compressive});\ (b)\ F = 0$$

Check. (b) is uniaxial hoop stress, with τ_max = σ/2 ✓. If the 3-D view with σr = 0 is used for (b), any tensile F with 0 ≤ σz satisfies "τ_max = σmax/2". State which interpretation you use.

Exam takeaway. Pure-shear-like states arise when two principal stresses are equal and opposite.

Q20 (Crandall Prob 4.27) — Open cylinder: find p and F from rotated-element stresses (medium)



Original: Crandall Prob 4.27 (textbook figure)

Problem. An open-ended thin cylinder (r = 10 in, t = 0.1 in) carries internal pressure p and axial force F. On an element whose m-axis is at 30° to the cylinder axis: (a) σm = 15 000 psi and σn = 5 000 psi; (b) σm = 15 000 psi and σn = 15 000 psi. Find p, F and τmn in each case.

Approach. z (axial) and θ (hoop) are principal. Transform from (z, θ) to (m, n) with m at 30° from z, and solve two equations for σz and σθ.

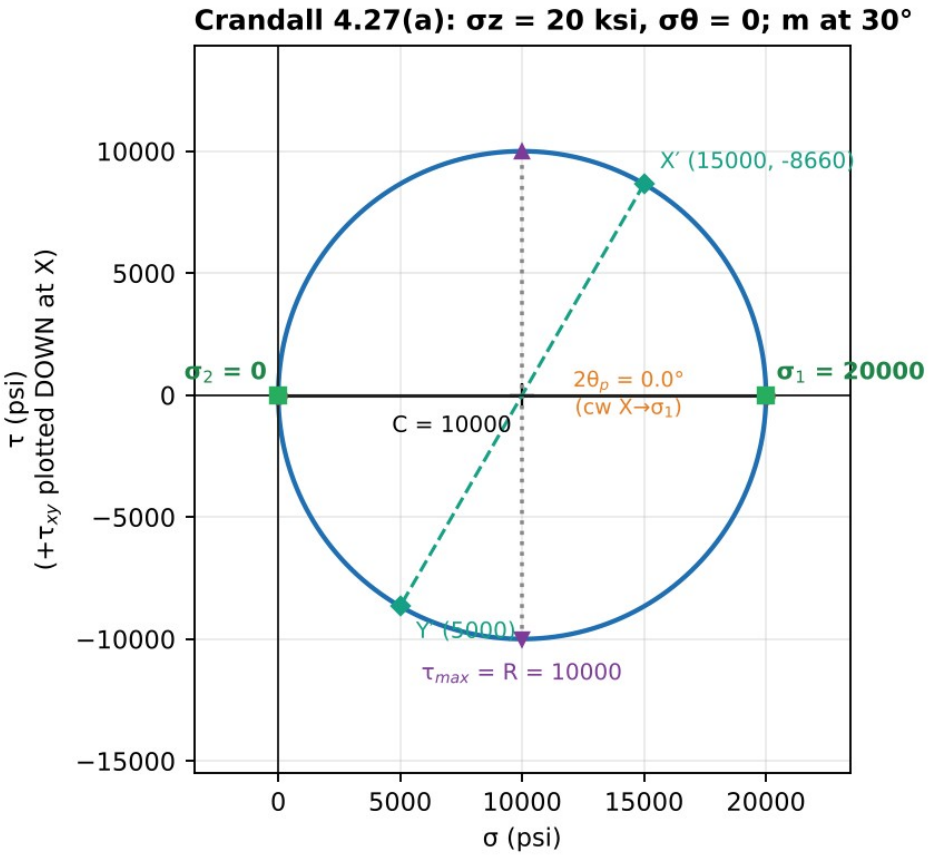
Solution.

1. $\sigma_m = \sigma_z \cos^{230^\circ} + \sigma_\theta \sin^{230^\circ} = 0.75\,\sigma_z + 0.25\,\sigma_\theta$ and $\sigma_n = 0.25\,\sigma_z + 0.75\,\sigma_\theta$.
2. *(a)* Sum: σz + σθ = 20 000. Then $0.75\,\sigma_z + 0.25(20\,000 - \sigma_z) = 15\,000$, so σz = 20 000 and σθ = 0. Hence p = 0 and $F = \sigma_z(2\pi r t) = 20\,000(2\pi)(10)(0.1) = 125\,700$ lb. $|\tau_{mn}| = \dfrac{\sigma_z - \sigma_\theta}{2}\sin 60^\circ = 8\,660$ psi.
3. *(b)* σm = σn gives σz = σθ = 15 000 psi. So $p = \sigma_\theta t/r = 150$ psi and $F = 15\,000(2\pi)(1) = 94\,250$ lb; τmn = 0.

$$(a)\ p = 0,\ F = 125.7\ \text{kip},\ |\tau_{mn}| = 8.66\ \text{ksi};\ (b)\ p = 150\ \text{psi},\ F = 94.2\ \text{kip},\ \tau_{mn} = 0$$

Check. In (b) the in-plane state is equal biaxial, so every direction is principal and τmn must be zero ✓.

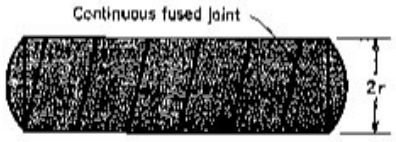
Exam takeaway. Reverse problems ("given rotated stresses, find the loads") are just two linear equations. *(Angle read from the figure: m at 30° to the axis.)*



Mohr's circle for case (a)

Q21 (Crandall Prob 4.25) — Helically wound pressure vessel: maximum strip width (medium–hard)

desired to subject the fused joint to a tensile stress only 80 percent of the maximum in the parent plastic. What is the maximum allowable width w of the strip?



Prob. 4.25

Original: Crandall Prob 4.25 (textbook figure)

Problem. A long closed-end pressure vessel of radius r is made by rolling a plastic strip of thickness t and width w into a helix and fusing the edges continuously. The fused joint may carry a tensile stress of only 80% of the maximum stress in the parent plastic. Find the maximum allowable strip width w.

Approach. Wall stresses σθ = pr/t (the maximum) and σz = pr/2t. The joint is a helix at angle α to the circumferential direction. Find the normal stress on the joint, limit it, and relate α to w.

Solution.

1. *Geometry:* one turn advances the helix axially by $2\pi r\tan\alpha$. The strip width perpendicular to the joint is $w = 2\pi r\tan\alpha\cos\alpha = 2\pi r\sin\alpha$.
2. *Normal stress on the joint:* the joint's normal makes angle α with z, so

$$\sigma_n = \sigma_z\cos^2\alpha + \sigma_\theta\sin^2\alpha = \frac{pr}{2t}\left(1+\sin^2\alpha\right)$$

3. *Limit:* $\frac{pr}{2t}\left(1+\sin^2\alpha\right) \le 0.8\frac{pr}{t}$, so $\sin^2\alpha \le 0.6$ and $\sin\alpha \le 0.775$.
4. $w_{max} = 2\pi r\left(0.775\right)$.

$$w_{max} = 2\pi r\sqrt{0.6} \approx 4.87\,r$$

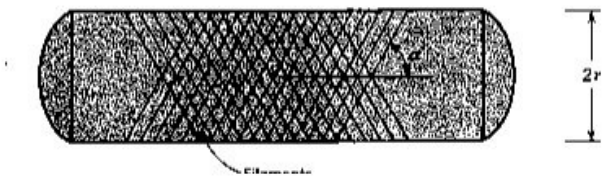
Check. α → 0 (circumferential joints) gives σn = σz = 0.5σθ, comfortably allowed but needing w → 0 ✓. α → 90° (an axial seam) gives σn = σθ, not allowed ✓.

Exam takeaway. A joint carries the normal stress on its own plane, found by transformation. This is the cylinder version of the welded-plate problem (Q4).

Q22 (Crandall Prob 4.32) — Filament-wound vessel: the winding angle for equal fibre tension (medium)

4.31. Show from (4.25) that the angles which make σ_x a maximum or minimum correspond to the principal directions.

4.32. Lightweight pressure vessels often use glass filaments for resisting tensile forces and use epoxy resin as a binder. Find the angle of winding, α, of the filaments when the ends of the vessel are closed such that the tensile forces in the filaments are equal (see Prob. 4.12).



Original: Crandall Prob 4.32 (textbook figure)

Problem. Glass filaments in an epoxy binder resist the tensile forces in a closed-end pressure vessel. Find the winding angle α of the filaments (to the axis) such that the tensile forces in the filaments are equal.

Approach. Netting analysis: the filaments (at ±α to the axis) carry all the load and the binder carries none. A filament stress σf contributes $\sigma_f \cos^2\alpha$ axially and $\sigma_f \sin^2\alpha$ circumferentially.

Solution.

1. Axial: $\sigma_f \cos^2\alpha = \sigma_z = pr/2t$. Hoop: $\sigma_f \sin^2\alpha = \sigma_\theta = pr/t$.
2. Divide: $\tan^2\alpha = \sigma_\theta/\sigma_z = 2$.

$$\alpha = \tan^{-1}\sqrt{2} = 54.7^\circ \text{ to the axis}$$

Check. The same "magic angle" appears in garden hoses and braided pressure tubes ✓.

Exam takeaway. The fibres must follow the ratio of the principal stresses. **Variation:** a sphere gives $\tan^2\alpha = 1$, i.e. 45°.

Q23 (Crandall Prob 4.34) — Membrane stress in a sea-urchin egg (easy)

Problem. A sea-urchin egg, considered a thin-walled sphere, has a volume of 35 × 10^{-5} mm^3 and wall thickness 1 µm. Its internal pressure is about 150 mm Hg. Find the membrane stresses.

Given: $V = 3.5 \times 10^{-4}$ mm^3; t = 10^{-3} mm; p = 150 mm Hg = 150(133.3) Pa = 0.0200 MPa. **Find:** σ.

Solution.

1. $r = \left(\frac{3V}{4\pi}\right)^{1/3} = \left(\frac{3(3.5\times10^{-4})}{4\pi}\right)^{1/3} = 0.0437$ mm (an egg about 87 µm across).
2. $\sigma = \frac{pr}{2t} = \frac{0.0200(0.0437)}{2(0.001)}$.

$$\sigma \approx 0.44\,\text{MPa}\,(\text{equal in all tangential directions})$$

Check. About 440 kPa, small in absolute terms but 22× the internal pressure, because r/2t ≈ 22 ✓.

Exam takeaway. Unit conversion is the real test (mm Hg → Pa; µm → mm).

Concept 4.5 — Strain and the strain–displacement relations (plane strain)

A. Concept explanation

1. Intuition. Draw a small circle on a rubber sheet and stretch the sheet. The circle becomes an ellipse. Some diameters get longer (normal strain) and right angles between lines change (shear strain). Strain is the **geometry** of deformation, just as stress is its statics.

2. Why it matters. Strain is what we actually **measure** (gauges, extensometers, digital image correlation). Hooke's law (Ch. 5) turns measured strains into stresses. The "circle becomes an ellipse" PYQs (2021 B-Q1, 2024 A-Q2) are strain problems.

3. Formal treatment (Crandall §4.6–4.7; slides). Let u(x, y) and v(x, y) be the displacements in x and y. For a small rectangle OCDE (OC along x, length Δx; OE along y, length Δy):

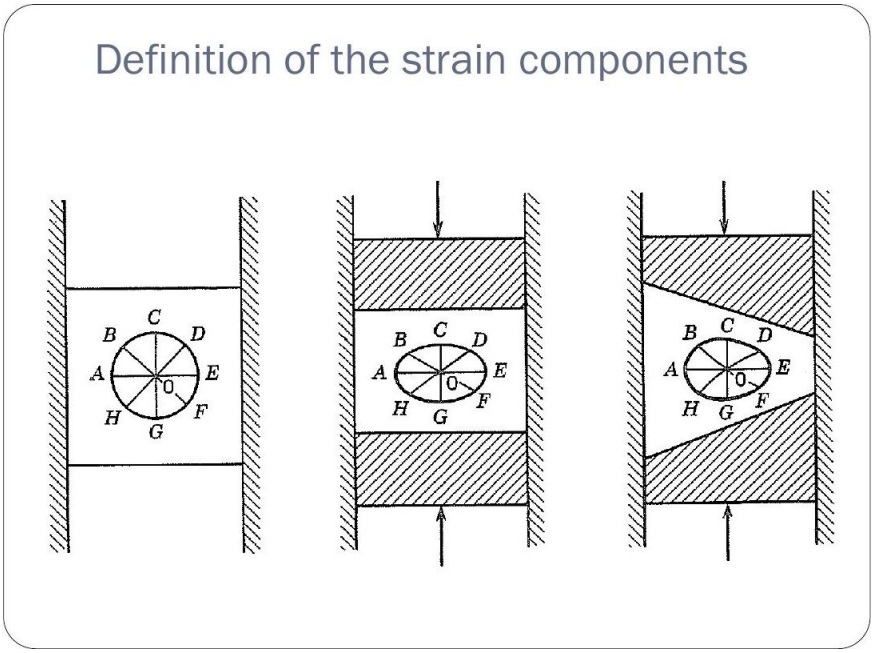
- **Normal strain** = change in length per unit original length:

$$\epsilon_x = \frac{\lim O'C' - OC}{OC} = \frac{\partial u}{\partial x}, \epsilon_y = \frac{\partial v}{\partial y}$$

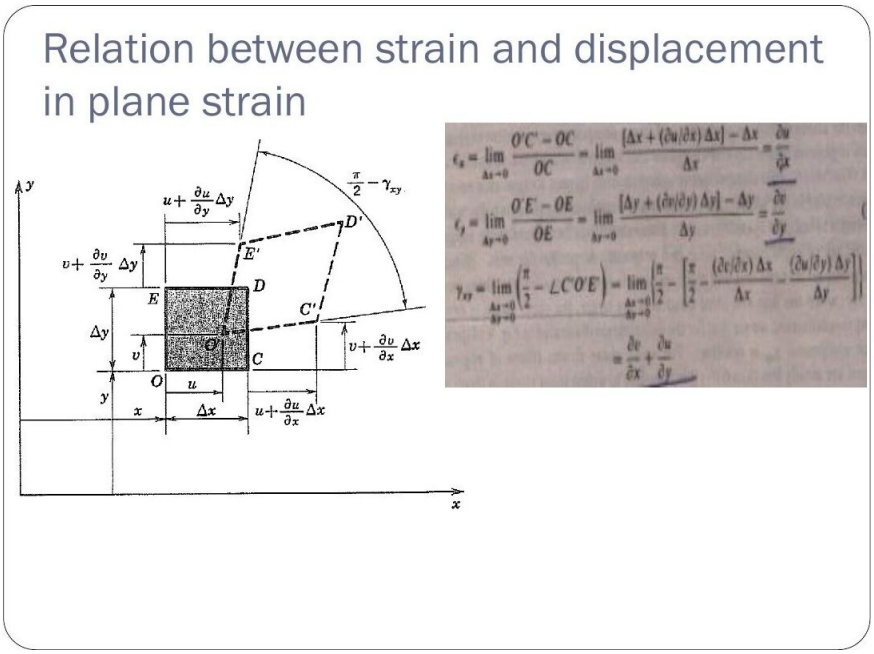
- **Shear strain** = the decrease of the originally right angle COE:

$$\gamma_{xy}=\lim\left(\frac{\pi}{2}-\angle C'O'E'\right)=\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}$$

Plane strain: $\epsilon_z=\gamma_{xz}=\gamma_{yz}=0$. ⚠️ Plane strain ≠ plane stress. A thin plate in plane stress has $\epsilon_z\neq 0$ (it thins). A long dam or a plate between rigid walls is in plane strain, with $\sigma_z\neq 0$.



Original: Slides Ch 4 Part 2, slide 1



Original: Slides Ch 4 Part 2, slide 3

Three-dimensional (Crandall Prob 4.16): $\epsilon_z=\partial w/\partial z$, $\gamma_{yz}=\partial w/\partial y+\partial v/\partial z$, $\gamma_{zx}=\partial u/\partial z+\partial w/\partial x$. *Polar, axisymmetric (Crandall Prob 4.17):* with radial displacement u(r) only, $\epsilon_r=du/dr$, $\epsilon_\theta=u/r$ and $\gamma_{r\theta}=0$. This is why a ring whose radius grows by u has hoop strain u/r (Ch. 2 Concept 2.6).

4. Derivation of εx and γxy. Point C at (x + Δx, y) moves by $u+\frac{\partial u}{\partial x}\Delta x$ in x and $v+\frac{\partial v}{\partial x}\Delta x$ in y. To first order the new length of OC is $\Delta x\left(1+\partial u/\partial x\right)$, which gives εx. Line OC rotates counter-clockwise by $\partial v/\partial x$ and line

OE rotates clockwise by $\partial u/\partial y$; the angle between them closes by the sum, which gives γxy. ▮ Sign: γxy > 0 means the angle between +x and +y **decreases**.

5. Picture. Sketch the rectangle OCDE and its deformed shape O′C′D′E′: shifted (rigid translation), rotated (rigid rotation $\omega = \frac{1}{2}(\partial v/\partial x - \partial u/\partial y)$), stretched (ε) and sheared (γ). Only the last two are strains.

6. Special cases. A rigid-body rotation gives u = −ωy and v = ωx, so ε = γ = 0 (Crandall Prob 4.29). A uniform stretch $u = \epsilon_0 x$ gives εx = ε0 only.

7. Traps. Engineering shear strain γ is *twice* the tensor shear strain. Mohr's circle uses γ/2.

8. Summary. $\epsilon_x = u_{,x}$, $\epsilon_y = v_{,y}$, $\gamma_{xy} = u_{,y} + v_{,x}$; polar axisymmetric: $\epsilon_\theta = u/r$.

Concept 4.6 — Strain transformation and Mohr's circle of strain

A. Concept explanation

1. Intuition. The same idea as for stress. Rotate the measuring axes and the stretch along each direction changes, following *exactly the same algebra* with σ → ε and τ → γ/2.

2. Why it matters. Gauges measure strain in their own direction; to get strains along any other direction (and principal strains) you transform. Combined with Hooke's law, this turns gauge readings into stresses.

3–4. Formal treatment and derivation (slides; Crandall §4.8). With $x = x'\cos\theta - y'\sin\theta$, $y = x'\sin\theta + y'\cos\theta$, $u' = u\cos\theta + v\sin\theta$ and $v' = -u\sin\theta + v\cos\theta$, the chain rule gives

$$\epsilon_{x'} = \frac{\partial u'}{\partial x'} = \left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial v}{\partial x}\sin\theta\right)\cos\theta + \left(\frac{\partial u}{\partial y}\cos\theta + \frac{\partial v}{\partial y}\sin\theta\right)\sin\theta = \epsilon_x\cos^2\theta + \epsilon_y\sin^2\theta + \gamma_{xy}\sin\theta\cos\theta$$

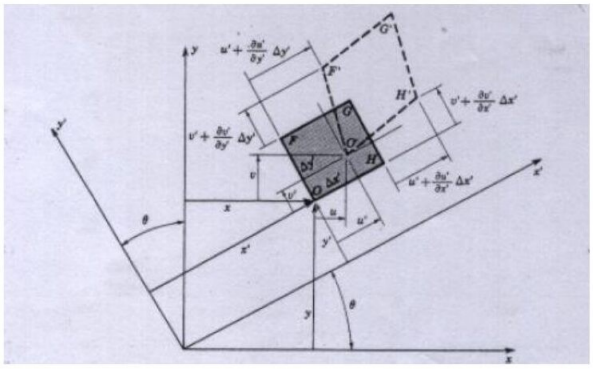
In double-angle form:

$$\epsilon_{x'} = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\frac{\gamma_{x'y'}}{2} = -\frac{\epsilon_x - \epsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}, \quad \tan 2\theta_p = \frac{\gamma_{xy}}{\epsilon_x - \epsilon_y}, \quad \frac{\gamma_{max}}{2} = R$$

Strain components associated with arbitrary set of axes



$$\epsilon_{x'} = \frac{\partial u'}{\partial x'}$$

$$\epsilon_{y'} = \frac{\partial v'}{\partial y'}$$

$$\gamma_{x'y'} = \frac{\partial v'}{\partial x'} + \frac{\partial u'}{\partial y'}$$

Original: Slides Ch 4 Part 2, slide 4

5. Mohr's circle of strain: horizontal axis ε, vertical axis **γ/2** (+γ/2 plotted **downward at x**, as for stress). Centre $\frac{\epsilon_x + \epsilon_y}{2}$, radius $R = \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$.

6. Special cases and the 3-D view.

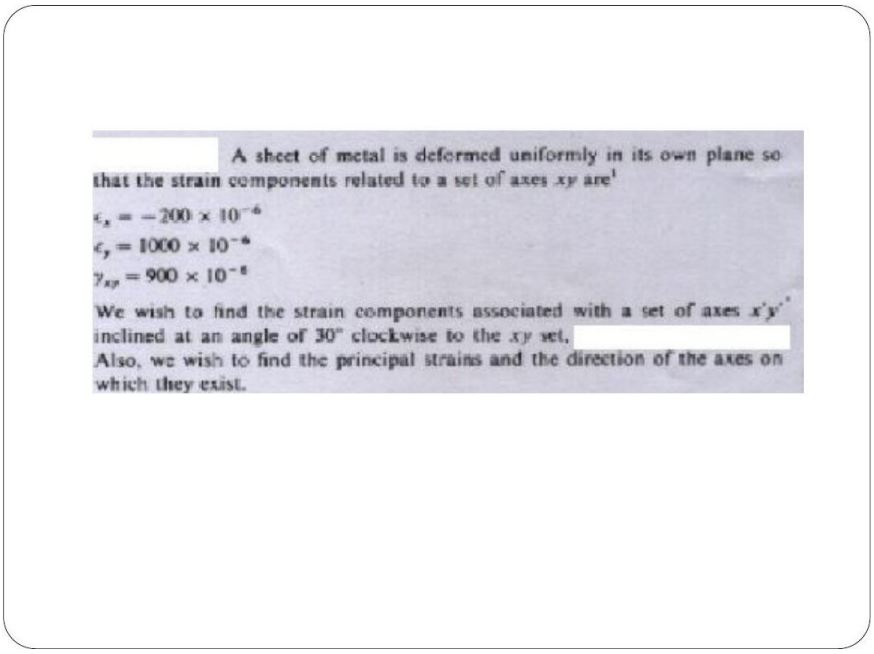
- For an isotropic material, **principal strain axes = principal stress axes** (Ch. 5), so either circle gives the directions.
- *Absolute maximum shear strain:* with all three principal strains, $\gamma_{abs,max} = \epsilon_{max} - \epsilon_{min}$. In plane strain the third principal strain is 0; in plane stress it is $\epsilon_z = -\frac{\nu}{1-\nu}(\epsilon_x + \epsilon_y)$, generally not zero.

7. Traps. Plotting γ instead of γ/2 doubles the radius. Remember $\gamma_{max} = 2R = \epsilon_1 - \epsilon_2$ (in-plane), and check whether the out-of-plane direction gives more.

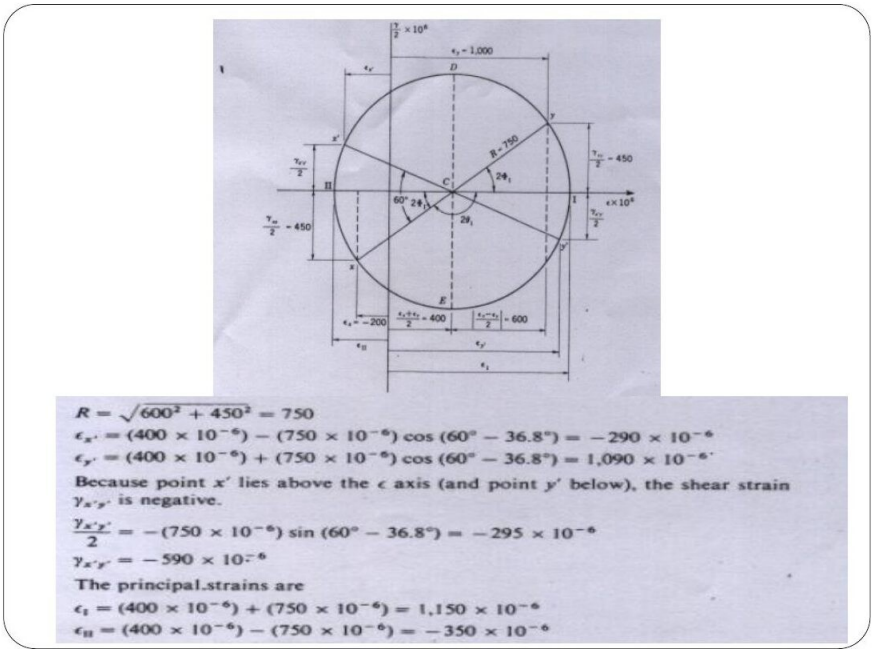
8. Summary. Replace σ → ε and τ → γ/2 in every stress formula.

B. Solved questions

Q24 (Slides Ch4 Part 2 = Crandall worked example) — Sheet metal: strains on axes 30° clockwise; principal strains (medium)



Original: Slides Ch 4 Part 2, slide 6



Original: Slides Ch 4 Part 2, slide 7

Problem. A sheet of metal is deformed uniformly in its own plane with $\epsilon_x = -200\,\mu$, $\epsilon_y = 1000\,\mu$ and $\gamma_{xy} = 900\,\mu$ (μ = 10^{-6}). Find the strain components for axes x′y′ inclined 30° clockwise to xy, and the principal strains and their directions.

Solution.

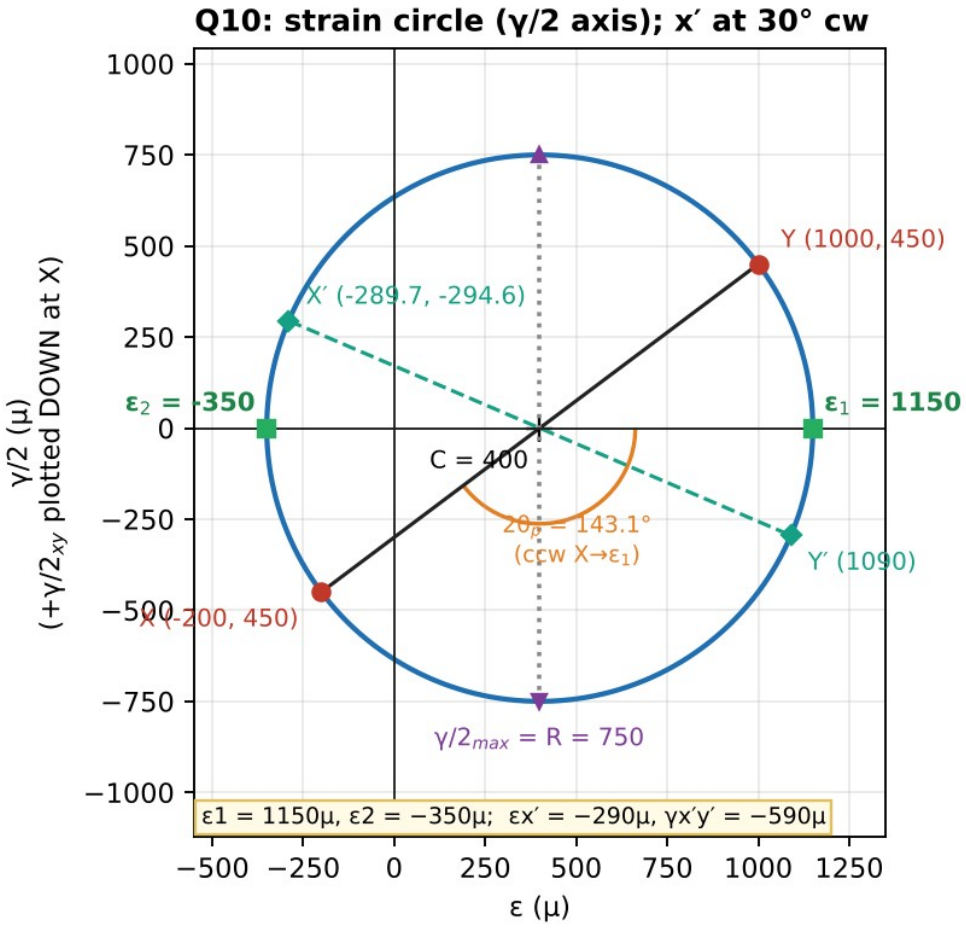
1. Centre = 400μ; $(\epsilon_x - \epsilon_y)/2 = -600\,\mu$; γ/2 = 450μ; $R = \sqrt{600^2 + 450^2} = 750\,\mu$.
2. θ = −30°: cos(−60°) = 0.5 and sin(−60°) = −0.866.
3. $\epsilon_{x'} = 400 + (-600)(0.5) + 450(-0.866) = -290\,\mu$; $\epsilon_{y'} = 800 - (-290) = 1090\,\mu$.

4. $\frac{\gamma_{x'y'}}{2} = -(-600)(-0.866)+450(0.5) = -295\,\mu$, so $\gamma_{x'y'} = -590\,\mu$.
5. *Principal:* $\varepsilon_1 = 1150\mu$ and $\varepsilon_2 = -350\mu$. $\tan 2\theta_p = 900/(-1200)$, so ε_1 lies at $\theta = 71.6°$ from x (ε_2 at −18.4°).

$$\epsilon_{x'} = -290\,\mu,\ \epsilon_{y'} = 1090\,\mu,\ \gamma_{x'y'} = -590\,\mu;\ \epsilon_1 = 1150\,\mu\,(71.6^\circ),\ \epsilon_2 = -350\,\mu$$

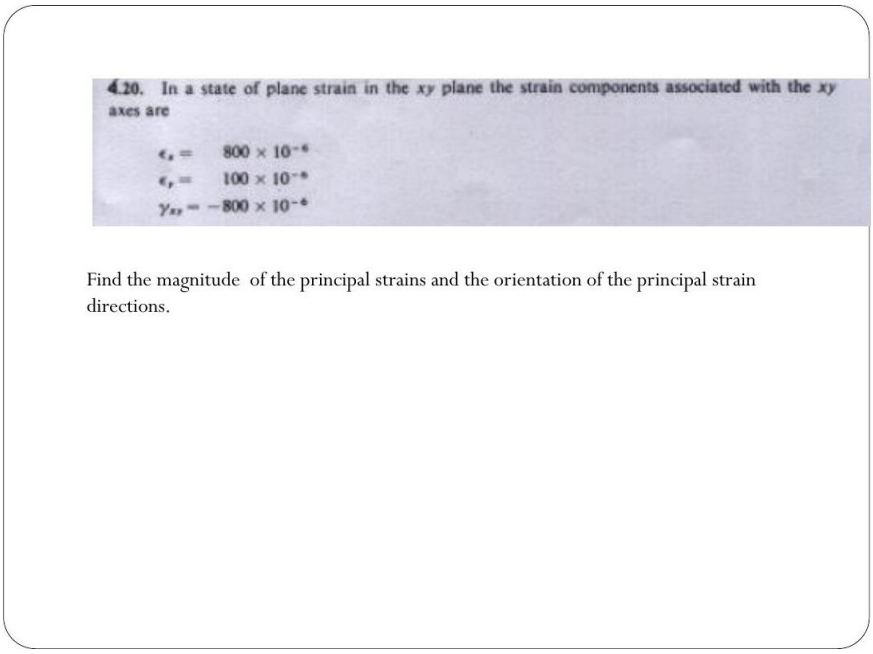
Check (slide's circle). The angle at the centre is 60° − 36.8°, and point x′ lies above the axis, so $\gamma_{x'y'} < 0$ ✓.

Exam takeaway. Mind the direction: "30° clockwise" means $\theta = -30°$.



Strain Mohr's circle (vertical axis $\gamma/2$)

Q25 (Crandall Prob 4.20) — Principal strains (easy)



4.20. In a state of plane strain in the xy plane the strain components associated with the xy axes are

$$\epsilon_x = 800 \times 10^{-6}$$
$$\epsilon_y = 100 \times 10^{-6}$$
$$\gamma_{xy} = -800 \times 10^{-6}$$

Find the magnitude of the principal strains and the orientation of the principal strain directions.

Original: Slides Ch 4 Part 2, slide 8

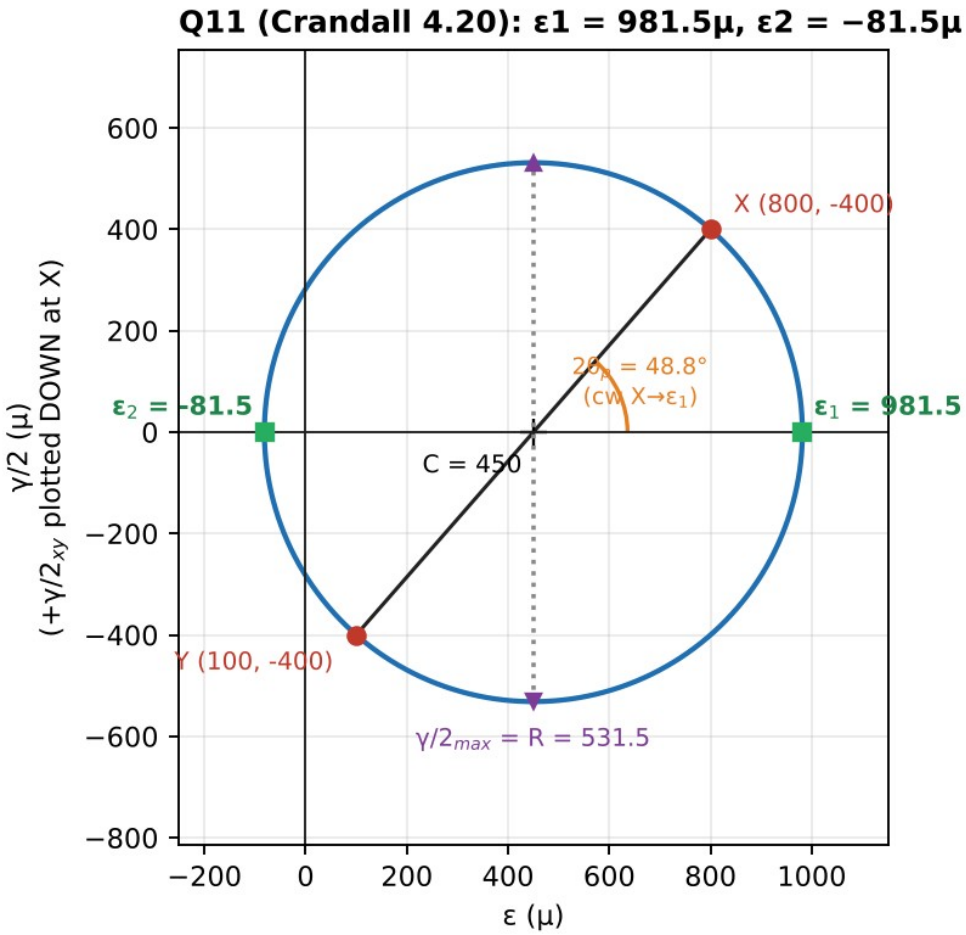
Problem. For $\epsilon_x = 800\,\mu$, $\epsilon_y = 100\,\mu$ and $\gamma_{xy} = -800\,\mu$, find the principal strains and their directions.

Solution.

1. Centre 450μ; $R = \sqrt{350^2 + 400^2} = 531.5\,\mu$.
2. $\varepsilon_1 = 981.5\mu$ and $\varepsilon_2 = -81.5\mu$.
3. $\tan 2\theta_p = \dfrac{-800}{700}$, so $2\theta_p = -48.8°$ and $\theta_{p1} = -24.4°$ (24.4° clockwise from x).

$$\epsilon_1 = 981.5\,\mu \text{ at } 24.4^\circ \text{ clockwise}, \epsilon_2 = -81.5\,\mu, \gamma_{max} = 1063\,\mu$$

Check. Substitute $\theta = -24.4°$: $450 + 350\cos(-48.8^\circ) - 400\sin(-48.8^\circ) = 450 + 230.5 + 301 = 981.5$ ✓.



Strain Mohr's circle

Q26 (Crandall Prob 4.21) — Maximum shear strain from three principal strains (easy)

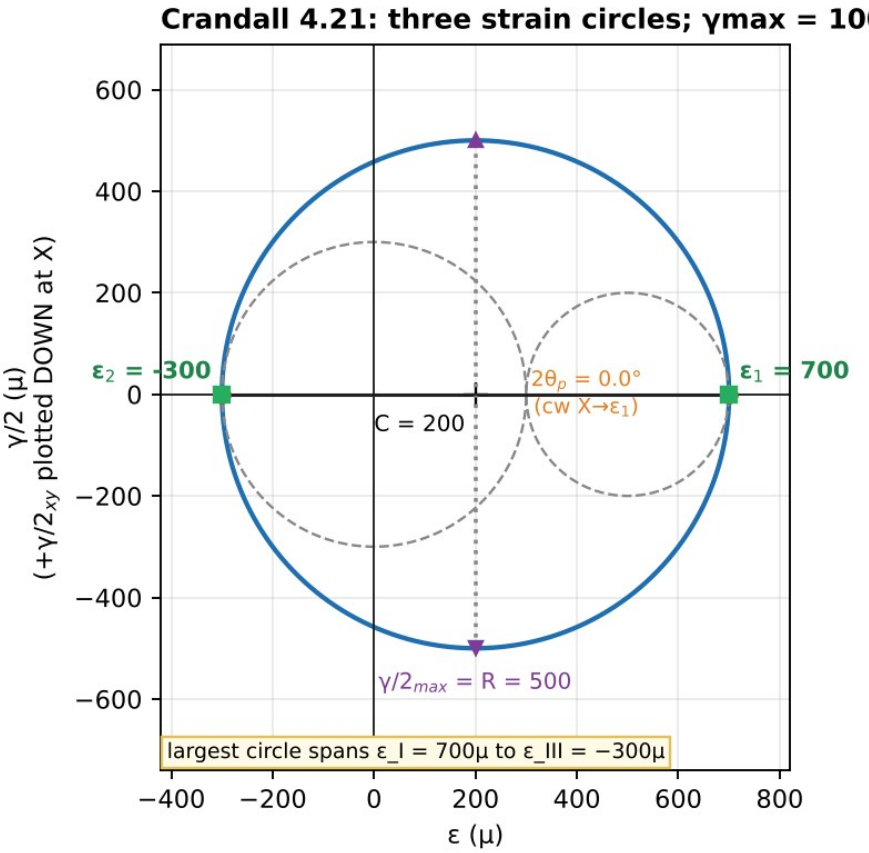
Problem. At a point, the principal strains are $\epsilon_I = 700\,\mu$, $\epsilon_{II} = 300\,\mu$ and $\epsilon_{III} = -300\,\mu$. Find the maximum shear strain and the orientation of the axes that experience it.

Solution.

1. $\gamma_{max} = \epsilon_{max} - \epsilon_{min} = 700 - (-300) = 1000\,\mu$.
2. It occurs in the I–III plane, on the pair of perpendicular axes bisecting the I and III directions (at 45° to each).

$$\gamma_{max} = 1000 \times 10^{-6}\ \text{rad, on axes at } 45^\circ \text{ between the I and III axes}$$

Exam takeaway. The largest Mohr's circle is the one spanning the extreme principal values.



The three Mohr's circles of strain

Q27 (Crandall Prob 4.23) — Plane strain: axes of maximum shear strain and the deformed element (medium)

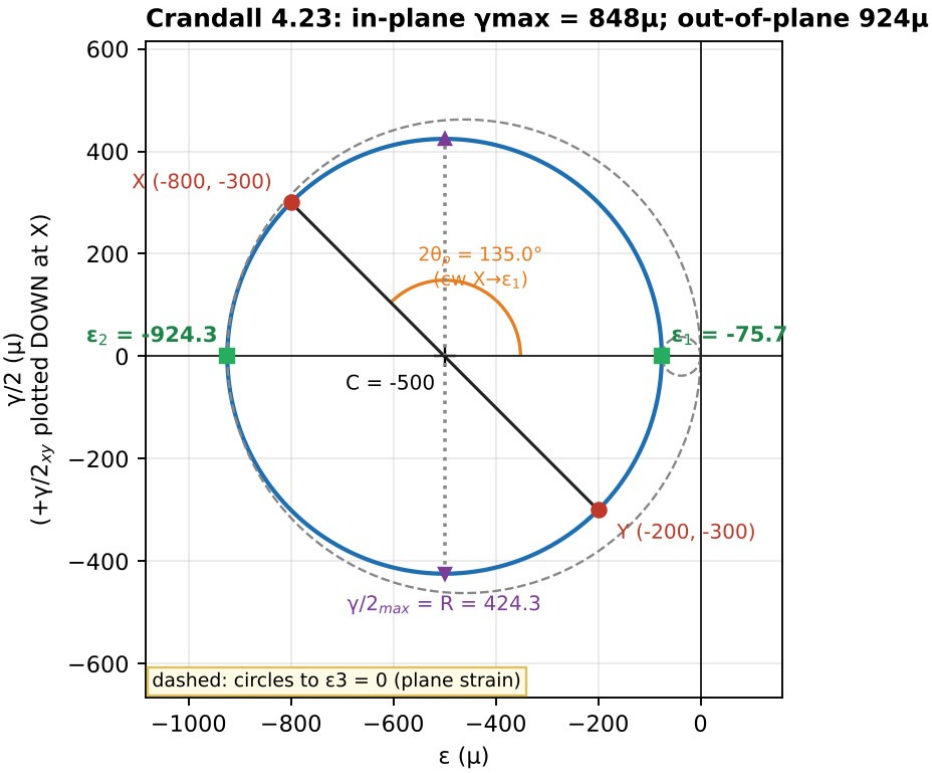
Problem. A body is in plane strain in the x–y plane with $\epsilon_x = -800\,\mu$, $\epsilon_y = -200\,\mu$ and $\gamma_{xy} = -600\,\mu$. Show the axes associated with the maximum shear strain and the deformed shape of an element aligned with them.

Solution.

1. Centre −500μ; $R = \sqrt{300^2 + 300^2} = 424.3\,\mu$, so ε1 = −75.7μ and ε2 = −924.3μ.
2. $\tan 2\theta_p = \dfrac{-600}{-600}$, so 2θp = −135° and ε1 lies at θ = −67.5° (i.e. 112.5°); ε2 lies at 22.5°.
3. *In-plane* $\gamma_{max} = 2R = 848.5\,\mu$, on axes at θp ± 45°, i.e. at −22.5° and 67.5°. An element aligned with these axes shrinks equally on both axes by the centre value (−500μ) and loses (or gains) 848.5 μrad of its right angle.
4. *Absolute (3-D):* in plane strain ε3 = εz = 0, so $\gamma_{abs} = 0 - (-924.3) = 924.3\,\mu$, in the plane containing z and the ε2 direction, on axes at 45° to them.

$$\gamma_{\text{in-plane, max}} = 848.5\,\mu \text{ on axes at } -22.5^\circ \text{ and } 67.5^\circ;\ \gamma_{\text{abs, max}} = 924.3\,\mu\,(\text{out of plane})$$

Exam takeaway. When both in-plane principal strains have the same sign, the out-of-plane direction wins.



In-plane circle (solid) and the circles to ε3 = 0 (dashed)

Concept 4.7 — Strain rosettes

A. Concept explanation

1. Intuition. One strain gauge measures stretch in one direction only. A point in plane strain has *three* unknowns (εx, εy, γxy), so glue **three gauges** at different angles (a rosette) and solve three equations.

2. Why it matters. This is how real structures (bridges, aircraft, the bracket in the slide example) are monitored. PYQs: 2021 A-Q1 (one gauge at 45° on a shaft), 2019 A-Q2 (from principal strains back to x–y).

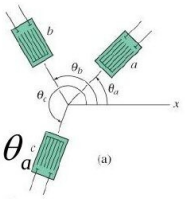
3. Formal treatment (slides). Each gauge at angle θ_i reads

$$\epsilon_i = \epsilon_x \cos^2\theta_i + \epsilon_y \sin^2\theta_i + \gamma_{xy} \sin\theta_i \cos\theta_i$$

45° rectangular rosette (0°, 45°, 90°): $\epsilon_x = \epsilon_a$, $\epsilon_y = \epsilon_c$, $\gamma_{xy} = 2\epsilon_b - (\epsilon_a + \epsilon_c)$. **60° delta rosette** (0°, 60°, 120°): $\epsilon_x = \epsilon_a$, $\epsilon_y = 1/3\,(2\epsilon_b + 2\epsilon_c - \epsilon_a)$, $\gamma_{xy} = 2/\sqrt{3}\,(\epsilon_b - \epsilon_c)$.

STRAIN ROSETTES

- Apply strain transformation to each gauge:



$$\varepsilon_a = \varepsilon_x \cos^2\theta_a + \varepsilon_y \sin^2\theta_a + \gamma_{xy} \sin\theta_a \cos\theta_a$$

$$\varepsilon_b = \varepsilon_x \cos^2\theta_b + \varepsilon_y \sin^2\theta_b + \gamma_{xy} \sin\theta_b \cos\theta_b$$

$$\varepsilon_c = \varepsilon_x \cos^2\theta_c + \varepsilon_y \sin^2\theta_c + \gamma_{xy} \sin\theta_c \cos\theta_c$$

- We determine the values of ε_x, ε_y γ_{xy} by solving the three equations simultaneously.

45

Original: Slides Ch 4 Part 2, slide 10

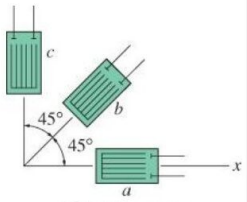
STRAIN ROSETTES

- For rosettes arranged in the 45° pattern, Eqn becomes

$$\varepsilon_x = \varepsilon_a$$

$$\varepsilon_y = \varepsilon_c$$

$$\gamma_{xy} = 2\varepsilon_b - (\varepsilon_a + \varepsilon_c)$$



45° strain rosette

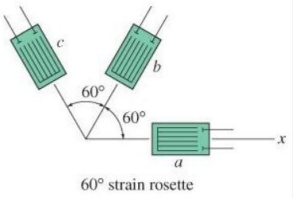
(b)

- For rosettes arranged in the 60° pattern, Eqn becomes

$$\varepsilon_x = \varepsilon_a$$

$$\varepsilon_y = \frac{1}{3}(2\varepsilon_b + 2\varepsilon_c - \varepsilon_a)$$

$$\gamma_{xy} = \frac{2}{\sqrt{3}}(\varepsilon_b - \varepsilon_c)$$



60° strain rosette

(c)

46

Original: Slides Ch 4 Part 2, slide 11

4. Derivations.

- *45°*: at θ = 45°, cos² = sin² = sin·cos = ½, so $\epsilon_b = 1/2(\epsilon_x + \epsilon_y + \gamma_{xy})$, giving $\gamma_{xy} = 2\epsilon_b - \epsilon_a - \epsilon_c$. ∎
- *60°*: at 60°: $\epsilon_b = 0.25\epsilon_x + 0.75\epsilon_y + 0.433\gamma$; at 120°: $\epsilon_c = 0.25\epsilon_x + 0.75\epsilon_y - 0.433\gamma$. Subtracting gives $\gamma = (\epsilon_b - \epsilon_c)/0.866$; adding gives $\epsilon_y = (\epsilon_b + \epsilon_c - 0.5\epsilon_a)/1.5$. ∎

5. Picture. Mohr's circle of strain through the three gauge readings: each gauge's ε is the horizontal coordinate of a point on the circle at 2θ apart.

6. Special cases. A single gauge at 45° on a shaft in pure torsion reads γ/2 (since εx = εy = 0).

7. Traps.

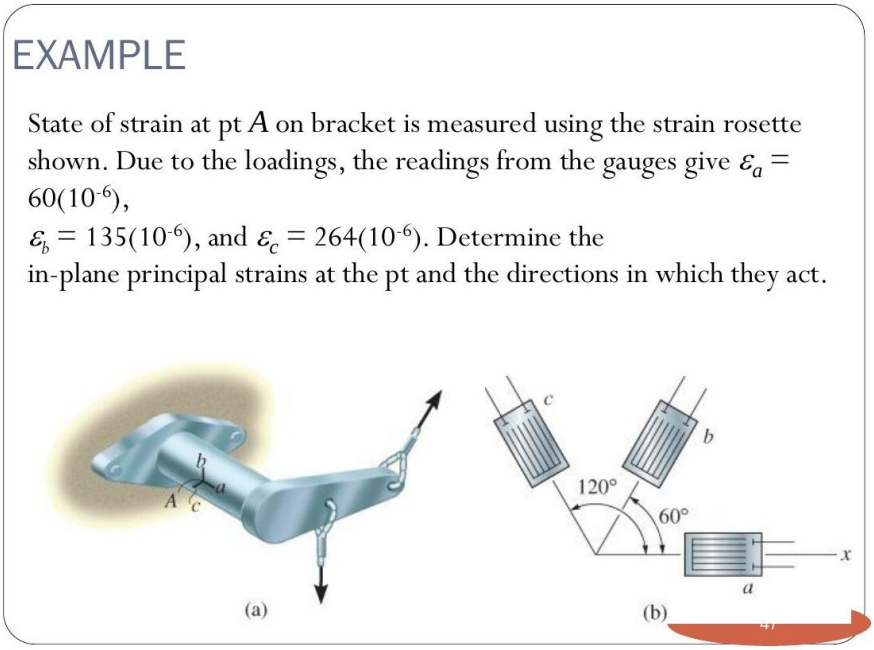
- Angles must be measured counter-clockwise from the gauge chosen as x (gauge a).
- Mixing degrees and radians on calculators.

- Final principal *stresses* need Hooke's law in **plane stress** (the surface is free).

8. Summary. 45°: γ = 2εb − εa − εc. 60°: εy = (2εb + 2εc − εa)/3 and γ = 2(εb − εc)/√3. Gauge at 45° on a shaft: γ/2.

B. Solved questions

Q28 (Slides Ch4 Part 2, Hibbeler Example 10.8) — 60° rosette on a bracket (medium)



Original: Slides Ch 4 Part 2, slide 12

Problem. A 60° rosette at point A on a bracket gives $\epsilon_a = 60\,\mu$ (0°), $\epsilon_b = 135\,\mu$ (60°) and $\epsilon_c = 264\,\mu$ (120°). Find the in-plane principal strains and their directions.

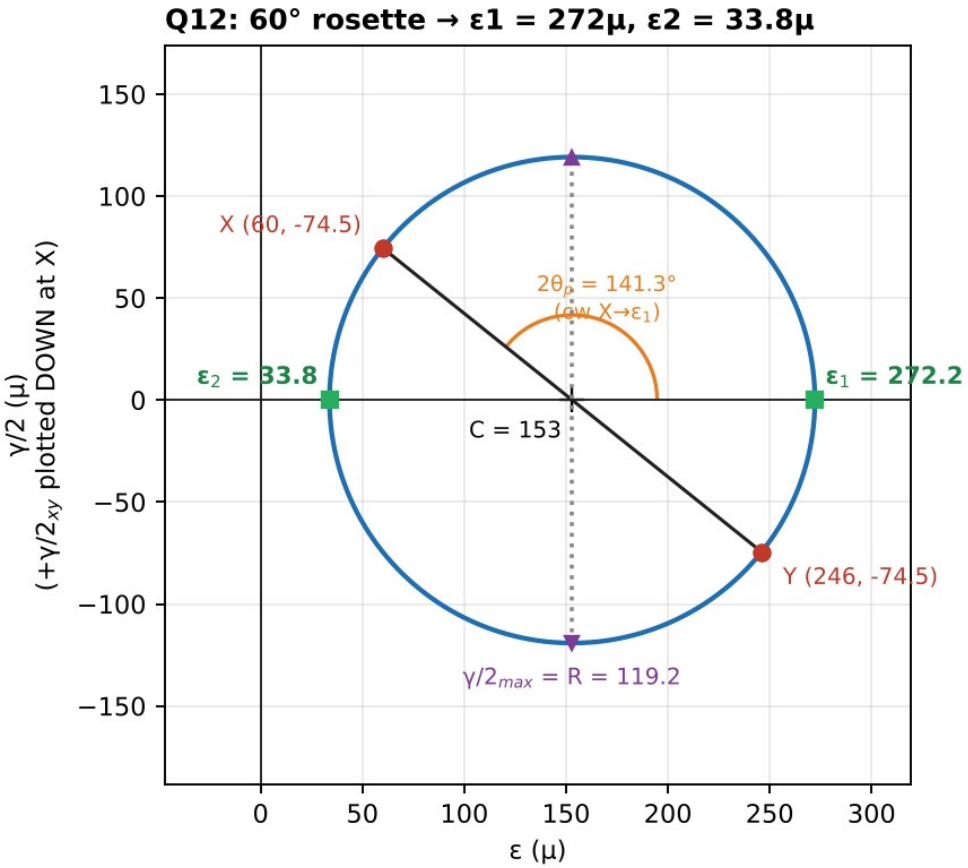
Solution.

1. $\epsilon_x = 60\,\mu$.
2. 60° and 120° gauges: $135 = 0.25\,\epsilon_x + 0.75\,\epsilon_y + 0.433\,\gamma$ and $264 = 0.25\,\epsilon_x + 0.75\,\epsilon_y - 0.433\,\gamma$.
3. Subtract: $0.866\,\gamma = -129$, so γxy = −149μ. Add: $1.5\,\epsilon_y = 399 - 30$, so εy = 246μ.
4. Centre 153μ; $R = \sqrt{93^2 + 74.5^2} = 119.2\,\mu$.

$$\epsilon_1 = 272\,\mu,\ \epsilon_2 = 33.8\,\mu,\ \theta_{p2} = 19.3^\circ \text{ (counter-clockwise from x, direction of } \epsilon_2)$$

Check. Formulas: $\epsilon_y = 1/3\,(270 + 528 - 60) = 246$ ✓ and $\gamma = 2/\sqrt{3}\,(135 - 264) = -149$ ✓.

Exam takeaway. ⚠️ The slide writes "ε2 = 246(10⁻⁶) − 119.2(10⁻⁶) = 33.8". The 246 is a typo for **153** (the centre); the result 33.8 is correct.



Strain Mohr's circle

Q29 (Crandall Prob 4.22) — 45° rosette readings to principal strains (easy–medium)

Problem. A 45° rosette reads (a) εa = 100μ, εb = 200μ, εc = 900μ; (b) εa = 1200μ, εb = 400μ, εc = 60μ. Find the principal strains in the plane of the rosette.

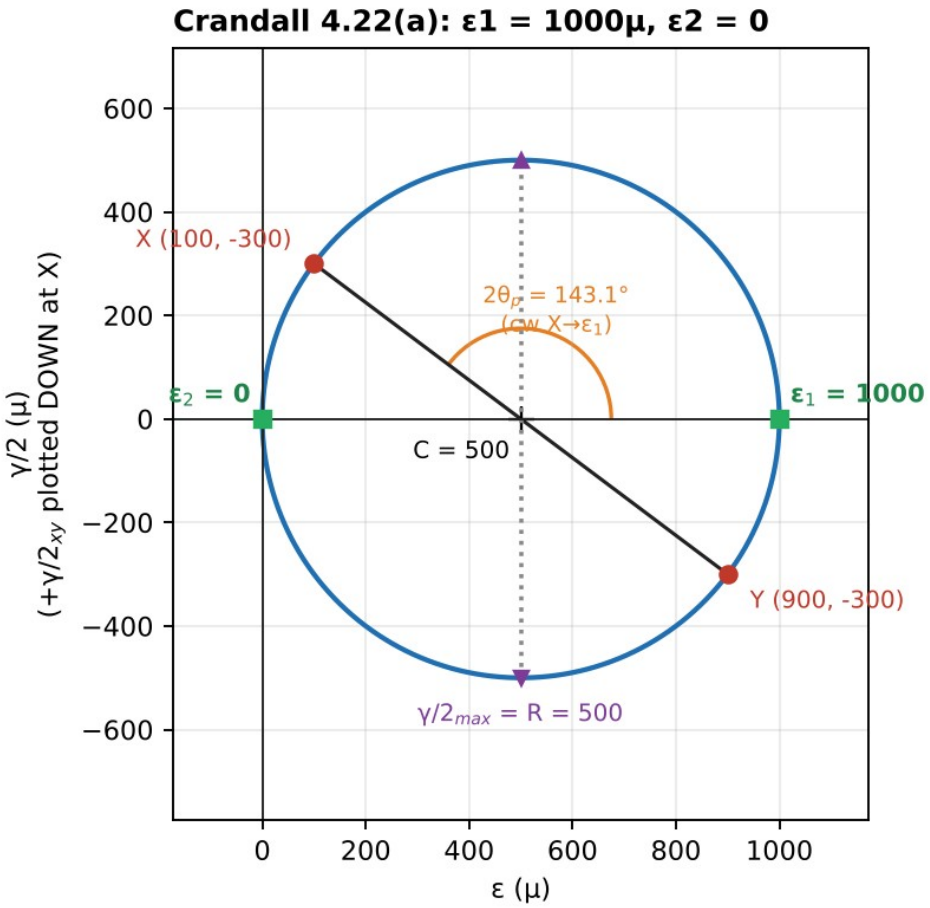
Solution.

1. *(a)* εx = 100, εy = 900, γ = 2(200) − 100 − 900 = −600μ. Centre 500; $R=\sqrt{400^2+300^2}=500$, so ε1 = 1000μ and ε2 = 0.
2. *(b)* εx = 1200, εy = 60, γ = 2(400) − 1260 = −460μ. Centre 630; $R=\sqrt{570^2+230^2}=614.7$, so ε1 = 1244.7μ and ε2 = 15.3μ.

$$(a)\,\epsilon_1=1000\,\mu,\epsilon_2=0\,;(b)\,\epsilon_1=1245\,\mu,\epsilon_2=15\,\mu$$

Check. In (a) the circle is a 3-4-5 triangle scaled by 100 ✓.

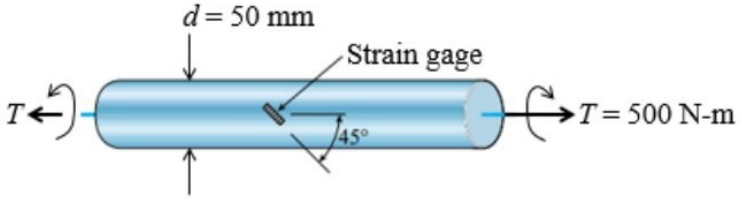
Exam takeaway. The middle gauge supplies the shear; the outer two supply εx and εy directly.



Strain Mohr's circle for case (a)

Q30 (PYQ 2021-22 Compre A Q1) — Strain gauge at 45° on a twisted bar: find G (easy, 10 marks)

Q1. A solid circular bar (see **FigQ1**) of diameter ***d*** = 50 mm is twisted in a testing machine until the applied torque reaches the value ***T*** = 500 N-m. At this value of torque, a strain gage oriented at 45° to the axis of the bar gives a reading ε = 350 x 10^{-6}. What is the shear modulus (***G***) of the material? **[10]**



FigQ1

Original question: PYQ 2021-22 Compre A Q1

Problem. A solid 50 mm bar is twisted until T = 500 N·m. A strain gauge at 45° to the axis reads $\varepsilon = 350 \times 10^{-6}$. Find the shear modulus G.

Approach. Surface shear from torsion (Ch. 6); in pure shear the 45° direction is principal and $\epsilon_{45} = \gamma/2$.

Solution.

1. $\tau = \dfrac{16T}{\pi d^3} = \dfrac{16(500\,000)}{\pi(50)^3} = 20.37$ MPa.
2. $\gamma = 2\epsilon_{45} = 700 \times 10^{-6}$.
3. $G = \tau/\gamma = \dfrac{20.37}{700 \times 10^{-6}}$.

$$G = 29.1\,\text{GPa}$$

Check (via Hooke). Principal stresses ±τ give $\epsilon_{45} = \frac{\tau(1+\nu)}{E} = \frac{\tau}{2G}$ ✓. The slide version of this problem (ε = 339μ) gives G = 30.0 GPa ✓.

Exam takeaway. "Gauge at 45° on a shaft" ⇒ reading = γ/2. **Variation:** given G, find T from the reading.

Q31 (PYQ 2019-20 Compre A Q2) — From principal strains to σx, σy, τxy (medium, 7 marks)

Q2. The principal strains in the plane of a flat aluminum plate loaded in its plane are $\varepsilon_1 = -8.1\times10^{-4}$ $\varepsilon_2 = 5.7\times10^{-4}$. Using Mohr's circle, find σ_x, σ_y, and τ_{xy} where the *x* and *y* axes are as shown in FigQ2. Take *E* = 72GPa and ν = 0.33. **[7]**

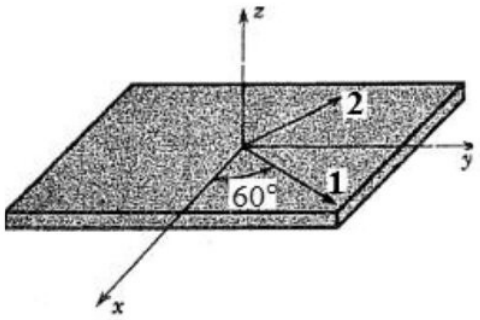


Fig Q2

Original question: PYQ 2019-20 Compre A Q2

Problem. The principal strains in an aluminium plate loaded in its plane are $\varepsilon_1 = -8.1 \times 10^{-4}$ and $\varepsilon_2 = 5.7 \times 10^{-4}$, with principal direction 1 at 60° from x (toward y). Using Mohr's circle, find σx, σy and τxy. E = 72 GPa, ν = 0.33.

Approach. Principal strains → principal stresses (plane-stress Hooke, Ch. 5), then rotate the stress circle back to x–y.

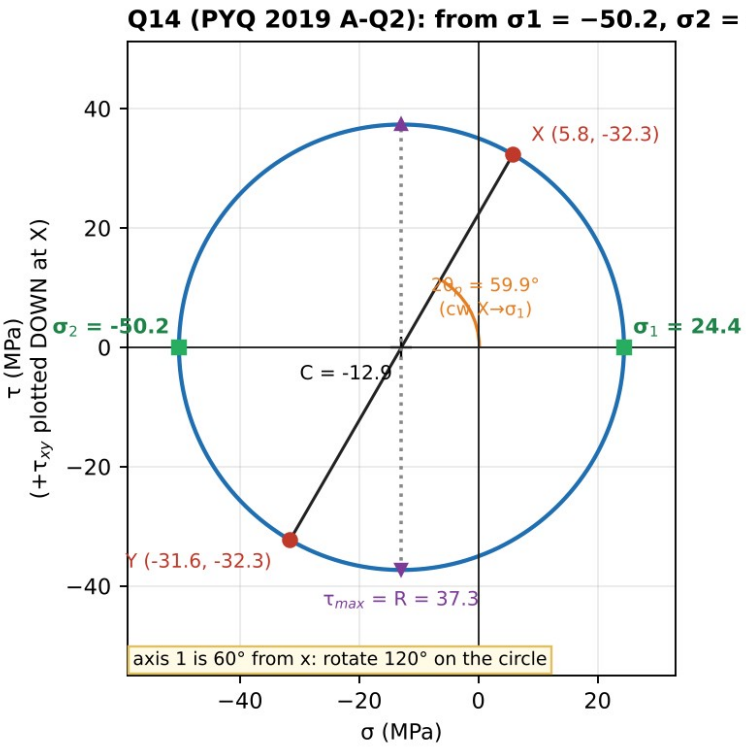
Solution.

1. *Principal stresses:* $\frac{E}{1-\nu^2} = 80\,799$ MPa.
   - $\sigma_1 = 80\,799(\epsilon_1 + \nu\epsilon_2) = 80\,799(-8.1 + 1.881)\times10^{-4} = -50.2$ MPa
   - $\sigma_2 = 80\,799(\epsilon_2 + \nu\epsilon_1) = 80\,799(5.7 - 2.673)\times10^{-4} = 24.5$ MPa
2. *Mohr's circle of stress:* centre −12.9, R = 37.4. The x-axis is at θ = −60° from axis 1, so rotate 120° clockwise from point "1" on the circle.
   - $\sigma_x = -12.9 + (-37.35)\cos(-120^\circ) = 5.8$ MPa
   - $\sigma_y = -12.9 - 18.7 = -31.6$ MPa
   - $\tau_{xy} = -(-37.35)\sin(-120^\circ) = -32.3$ MPa

$$\sigma_x = 5.8\,\text{MPa},\ \sigma_y = -31.6\,\text{MPa},\ \tau_{xy} = -32.3\,\text{MPa}$$

Check. σx + σy = −25.8 = σ1 + σ2 ✓; $\sqrt{18.7^2 + 32.3^2} = 37.4 = R$ ✓.

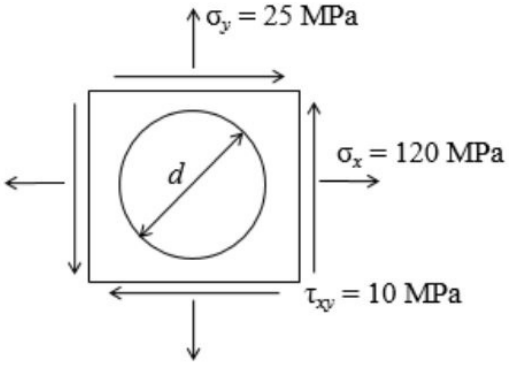
Exam takeaway. Alternatively, transform the *strains* to x–y first (strain circle), then apply Hooke's law; the answer is the same. Crandall Prob 5.10 is the same question with $\varepsilon_1 = 3.2 \times 10^{-4}$ and $\varepsilon_2 = -5.4 \times 10^{-4}$.



Stress Mohr's circle used to return to x–y

Q32 (PYQ 2021-22 Compre B Q1) — Circle → ellipse: lengths and orientation of the axes (medium–hard, 10 marks)

Q1. A circle of diameter (***d***) 250 mm is scribed on a mild steel plate before it is subjected to the stresses as shown in **FigQ1**. In stressing, the circle deforms to an ellipse. If E = 200 GPa and ν = 0.3, calculate the lengths of the major and minor axes of the ellipse. Sketch the ellipse and show the orientation (**θ**) of major and minor axes wrt *xy* axis. **[10]**



FigQ1

Original question: PYQ 2021-22 Compre B Q1

Problem. A 250 mm circle is scribed on a mild-steel plate, which is then stressed with σx = 120, σy = 25 and τxy = 10 MPa. The circle becomes an ellipse. Find the lengths of the major and minor axes and their orientation. E = 200 GPa, ν = 0.3.

Approach. The ellipse's axes lie along the **principal strain** directions, which equal the principal stress directions for an isotropic material.

Solution.

1. *Principal stresses:* centre 72.5, $R=\sqrt{47.5^2+10^2}=48.54$, so σ1 = 121.04 and σ2 = 23.96 MPa.

   $\theta_p=1/2\tan^{-1}\frac{20}{95}=5.94^\circ$ counter-clockwise from x (for σ1).

2. *Principal strains (plane stress):*

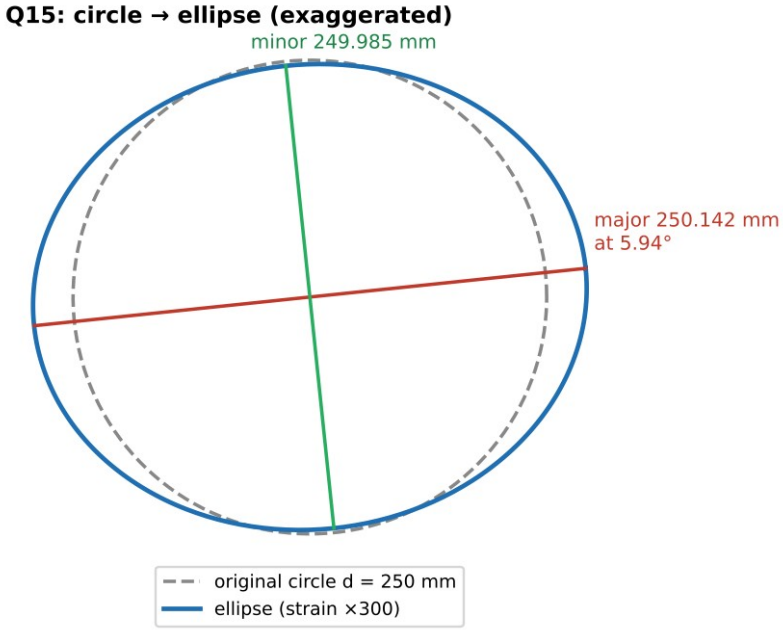
$$\epsilon_1 = \frac{121.04 - 0.3(23.96)}{200\,000} = 569.3\,\mu, \epsilon_2 = \frac{23.96 - 0.3(121.04)}{200\,000} = -61.8\,\mu$$

3. *Axes:* major = 250(1 + ε1) = 250.142 mm; minor = 250(1 + ε2) = 249.985 mm.

$$\text{Major} = 250.142\,\text{mm at } 5.94^\circ \text{ from x ; Minor} = 249.985\,\text{mm at } 95.94^\circ$$

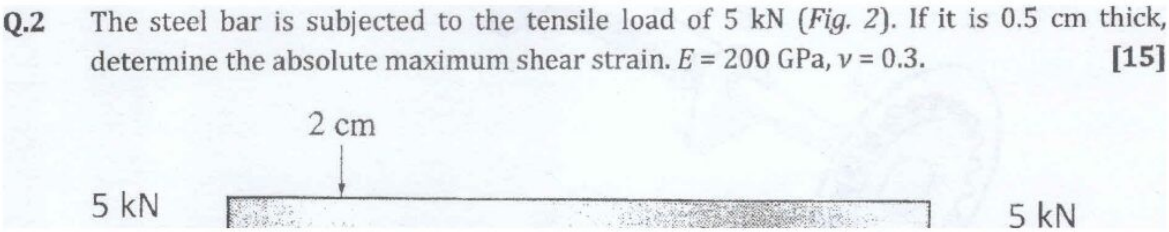
Check. Using σ/E without the Poisson term would give minor = 250.030 mm (longer than d), which is wrong: the contraction from σ1 wins ✓.

Exam takeaway. A sketch of the tilted ellipse is required for full marks.



The deformed circle (strains exaggerated ×300)

Q33 (PYQ 2014-15 Test-2 Q2) — Absolute maximum shear strain in a bar (easy–medium, 15 marks)



Original question: PYQ 2014-15 Test-2 Q2

Problem. A steel bar 2 cm × 0.5 cm carries a tensile load of 5 kN. E = 200 GPa, ν = 0.3. Find the absolute maximum shear strain.

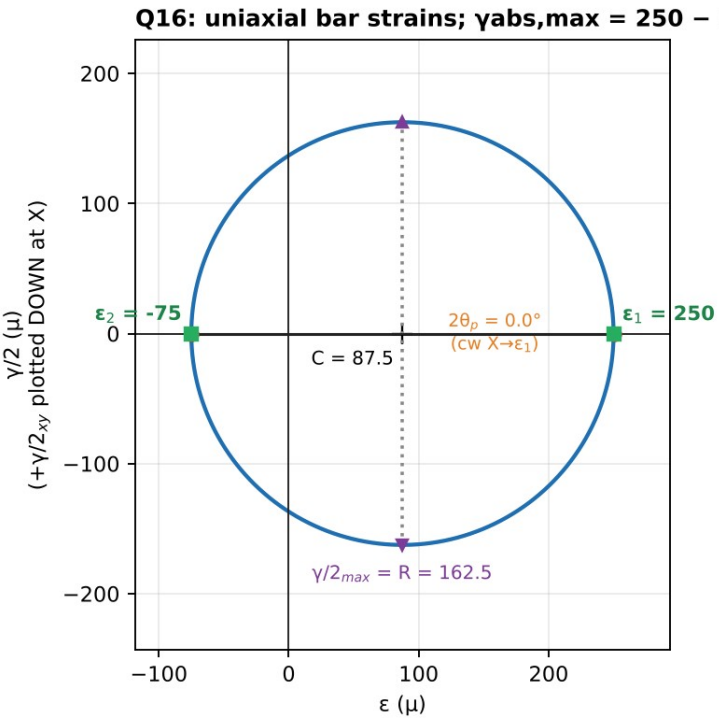
Solution.

1. $\sigma = \frac{5000}{20 \times 5} = 50$ MPa (uniaxial).
2. Principal strains: $\epsilon_1 = \sigma/E = 250\,\mu$ and $\epsilon_2 = \epsilon_3 = -\nu\sigma/E = -75\,\mu$.
3. $\gamma_{abs,max} = \epsilon_{max} - \epsilon_{min} = 250 - (-75)$.

$$\gamma_{abs,max} = 325 \times 10^{-6}\,\text{rad}$$

Check. $\tau_{max} = \sigma/2 = 25$ MPa and $G = E/2(1+\nu) = 76.9$ GPa give γ = 25/76 900 = 325μ ✓.

Exam takeaway. "Absolute" means include the out-of-plane direction; here in-plane and absolute coincide because ε2 = ε3.



Strain circle for the uniaxial bar

Chapter 4 — Formula and convention sheet

Item	Formula / rule
Sign of stress	+face & +direction, or −face & −direction
Symmetry	$\tau_{xy} = \tau_{yx}$
Equilibrium	$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y = 0,\ \partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y = 0$
Transformation	$\sigma_{x'} = \acute{\sigma} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta;$ $\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$
Invariant	$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y$
Principal	$\sigma_{1,2} = \acute{\sigma} \pm R,\ R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2},$ $\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$ (check which θ gives σ1)
Max in-plane shear	$\tau_{max} = R$ at θp ± 45°, with normal stress $\acute{\sigma}$ there
Absolute max shear	$(\sigma_{max} - \sigma_{min})/2$ including σ3 = 0
Mohr's circle	+τ plotted **down at X**, up at Y; 2θ on the circle = θ on the element, same sense
Thin cylinder	$\sigma_\theta = pr/t,\ \sigma_z = pr/2t$ (closed) or $F/2\pi rt$ (open); $\tau_{\theta z} = pr/4t$

Item	Formula / rule
Thin sphere	$\sigma = pr/2t$
Filament angle (closed cylinder)	$\tan^2\alpha = 2$, α = 54.7°
Strains	$\epsilon_x = \partial u/\partial x$, $\epsilon_y = \partial v/\partial y$, $\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$; polar $\epsilon_\theta = u/r$
Strain transformation	replace σ → ε and τ → γ/2 in the stress equations
Strain circle	vertical axis γ/2; $\gamma_{max} = \epsilon_1 - \epsilon_2$ (in-plane); absolute uses all three
45° rosette	εx = εa, εy = εc, γ = 2εb − εa − εc
60° rosette	εy = (2εb + 2εc − εa)/3, γ = 2(εb − εc)/√3
Gauge at 45° on a shaft	$\epsilon_{45} = \gamma/2 = \tau/2G$

Other Crandall Ch. 4 problems worth practising: 4.1–4.4 (equilibrium equations in Cartesian and polar form), 4.5 (sign convention independence), 4.16–4.19 (strain–displacement in 3-D and polar), 4.24 and 4.29 (rigid-body motions), 4.26 (superposed states), 4.30 (stress-free boundary), 4.33 (beam equations from equilibrium).

Chapter 5 — Stress–Strain–Temperature Relations

Tensile test, Hooke's law, Poisson's ratio, generalized Hooke's law, E–G–ν–K, volume change, thermal strain and thermal stresses, complete equations of elasticity, strain energy, yield criteria (Tresca and von Mises)

Sources used: Slides "Chapter 5"; Crandall Ch. 5 (§5.2–5.8, 5.16; Probs 5.1, 5.3–5.5, 5.7–5.9, 5.11–5.13, 5.17, 5.18, 5.20); PYQs 2023-24 (A-Q3), 2024-25 (A-Q2), 2019-20 (A-Q4), 2014-15 (Compre A-Q1–Q4; Test-2 Q3).

Layout. Part A (eight items) for each concept, then Part B: slides → Crandall → PYQs, easy → hard. Each solution follows *Problem · Given/Find · Approach · Sketch · Steps · Answer · Check · Takeaway*.

Material values used when a problem doesn't state them: steel E = 200 GPa, $\nu = 0.3$, $\alpha = 12 \times 10^{-6}/°C$; aluminium alloy 2024-T4 E = 73 GPa (10.6×10^6 psi), $\nu = 0.33$, $\alpha = 23 \times 10^{-6}/°C$ ($12.6 \times 10^{-6}/°F$). Each solution states the values it uses.

Concept 5.1 — The tensile test and material behaviour

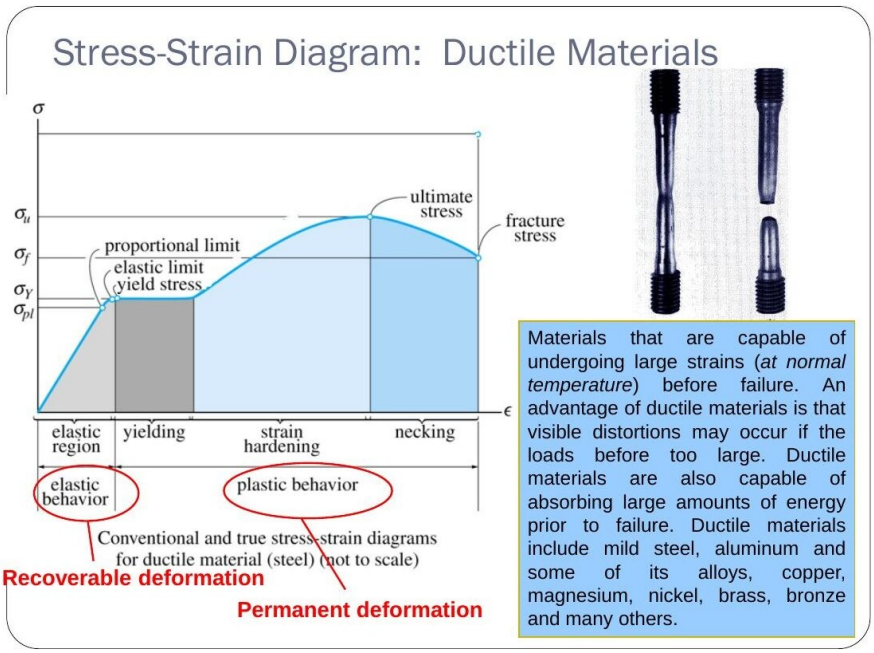
A. Concept explanation

1. Intuition. Pull a steel bar in a testing machine and plot force against stretch. At first it behaves like a stiff spring (a straight line). Then it suddenly "gives" and stretches a lot for little extra force: it **yields**. Then it hardens, necks down and breaks. Dividing force by area and stretch by length removes the size of the specimen, leaving a curve that belongs to the **material**.

2. Why it matters. Every design number (E, σY, σu, ν) comes from this test. Yield strength is the limit in every factor-of-safety PYQ. Recognising ductile vs brittle behaviour tells you which failure theory and which failure plane apply.

3. Formal treatment (slides).

- *Test:* a load is applied along the axis of a circular specimen and stretched at a constant rate; force and elongation (extensometer) are recorded.
- *Engineering stress and strain:* $\sigma = P/A_0$ (Pa) and $\epsilon = \delta/L_0$ (dimensionless), using the **original** dimensions. *True stress and strain* use the current area and length, $\sigma_t = P/A$ and $\epsilon_t = \int dL/L = \ln\left(L/L_0\right)$; they differ only after large deformation.
- *Key points on the curve (slides):*
  - **Proportional limit**: the stress above which stress is no longer proportional to strain.
  - **Elastic limit**: the maximum stress that can be applied without permanent deformation on unloading.
  - **Yield point**: large increases in strain with little or no increase in stress.
  - **Ultimate strength**: the peak of the engineering curve, after which necking begins.
- *Ductile materials* (mild steel, aluminium and its alloys, copper, magnesium, nickel, brass, bronze) undergo large strains before failure, absorb a lot of energy and give visible warning. They usually fail on planes of **maximum shear (45°)**, giving the typical **cup-and-cone** fracture.
- *Brittle materials* (concrete, stone, cast iron, glass, plaster) fail at small strains, by **tensile (normal) stress**, on a surface perpendicular to the load.
- *Elastic vs plastic:* if the strain disappears when the stress is removed the behaviour is elastic; if not, it is plastic.
- *Fatigue:* failure at stresses well below the ultimate strength under many load cycles. *Creep:* time-dependent deformation under constant stress.
- *Hooke's law (uniaxial):* $\sigma = E\epsilon$, where Young's modulus E is the slope of the linear part. Young observed that this slope is a material constant, independent of specimen geometry. Typical values (slides, GPa): mild steel 210, aluminium 70, concrete 18.5, wood 12.5, nylon 2.8, rubber 0.004.
- *Idealized curves (slides, Crandall §5.3):* linear elastic; rigid–perfectly plastic; elastic–perfectly plastic; linear strain hardening. "It is impossible to describe the entire stress–strain curve with a simple mathematical expression." Choose the model that captures what matters for the problem at hand.



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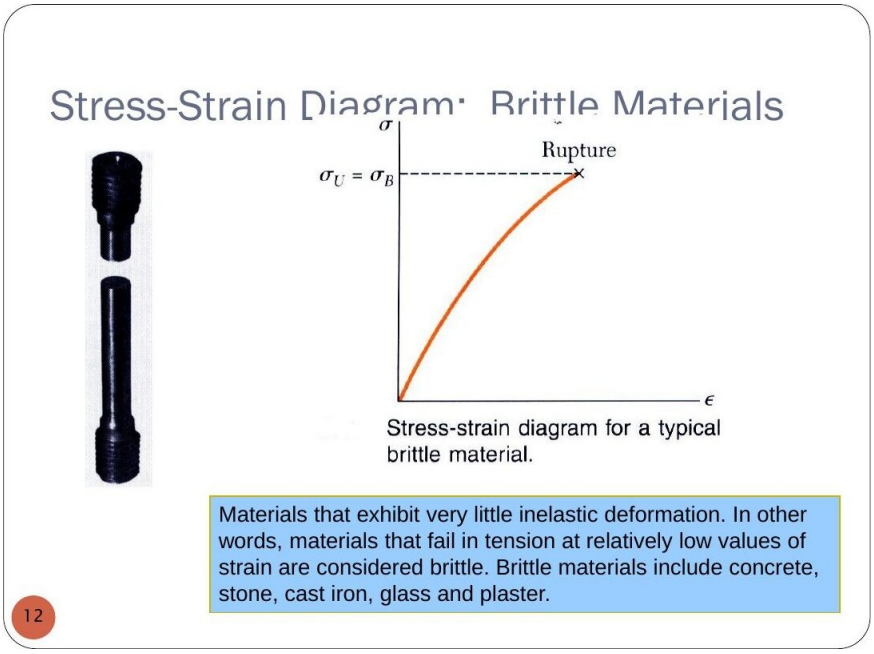

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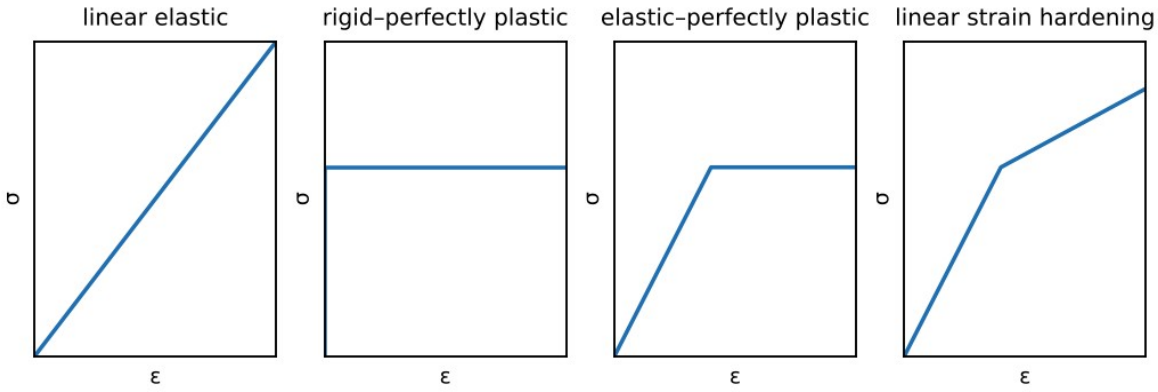
4. Derivation — reduction of area in terms of true strain (Crandall Prob 5.11). When plastic strains dominate, the material deforms at constant volume: $A_0 L_0 = A\,L$. The true strain is $\epsilon_t = \ln\left(L/L_0\right) = \ln\left(A_0/A\right)$, so $A/A_0 = e^{-\epsilon_t}$ and

$$\text{Reduction of area} = \frac{A_0 - A}{A_0} = 1 - e^{-\epsilon_t}$$

5. Picture. Sketch two curves on the same axes: a ductile one (linear, a yield plateau, hardening, necking down to fracture at large strain) and a brittle one (linear almost to fracture at small strain). Beside them, the four idealized models (diagram below).



Idealized stress–strain curves

6. Special cases. Materials without a sharp yield point (aluminium) use the 0.2% offset yield strength. Rubber is nonlinear elastic; E is the initial slope.

7. Traps. Using σu where the question asks for yield; using true stress where engineering stress is meant; quoting E in MPa when the formula needs GPa (or vice versa).

8. Summary. $\sigma = P/A_0$, $\epsilon = \delta/L_0$, $\sigma = E\epsilon$; ductile → shear failure at 45°; brittle → tensile failure on the cross-section.

B. Solved question

Q1 (Crandall Prob 5.11) — Reduction of area and true strain (easy, derivation)

Problem. A tensile specimen has original area A0 and area A at some point in the test. If the elastic strains can be neglected compared with the plastic strains, show that the reduction of area is $1 - e^{-\epsilon_t}$, where εt is the true strain.

Solution. As derived in item 4: plastic flow conserves volume, so $A/A_0 = L_0/L = e^{-\epsilon_t}$.

$$\mathrm{RA} = 1 - e^{-\epsilon_t}$$

Check. A true strain of 1 (length × 2.72) gives RA = 63% ✓, typical of a necked ductile bar.

Exam takeaway. Large-strain plasticity conserves volume ($\nu \to 0.5$); small elastic strains do not (Concept 5.2).

Concept 5.2 — Poisson's ratio and the generalized Hooke's law

A. Concept explanation

1. Intuition. Stretch a rubber band and it gets thinner. The ratio of sideways shrinking to lengthwise stretching is Poisson's ratio. Most metals are about 0.3, rubber about 0.5 (nearly incompressible) and cork about 0. When a plate is pulled in two directions, each pull also thins the plate and shrinks the other direction, and all these effects simply add.

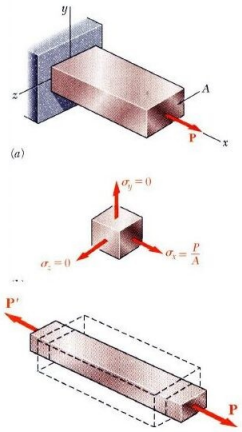
2. Why it matters. It explains why a plate under biaxial stress changes thickness, and it appears in every "change in thickness/volume" PYQ (2023 A-Q3 and 2024 A-Q2 are the same question type two years running). It is also the step that turns gauge readings (strains) into stresses.

3. Formal treatment (slides). For uniaxial σx: $\epsilon_x = \sigma_x/E$ and $\epsilon_y = \epsilon_z = -\nu\epsilon_x$, where

$$\nu = -\frac{\text{lateral strain}}{\text{axial strain}}$$

Assumptions (slides): (1) the material is linearly elastic, (2) strain is linearly related to stress, (3) deformations are small, (4) the material is **isotropic** (no directional dependence).

Poisson's Ratio



- For a slender bar subjected to axial loading: $\varepsilon_x = \frac{\sigma_x}{E} \quad \sigma_y = \sigma_z = 0$
- The elongation in the x-direction is accompanied by a contraction in the other directions. Assuming that the material is isotropic (no directional dependence),

  $\varepsilon_y = \varepsilon_z \neq 0$
- Poisson's ratio is defined as

  $\nu = \left| \frac{\text{lateral strain}}{\text{axial strain}} \right| = -\frac{\varepsilon_y}{\varepsilon_x} = -\frac{\varepsilon_z}{\varepsilon_x}$

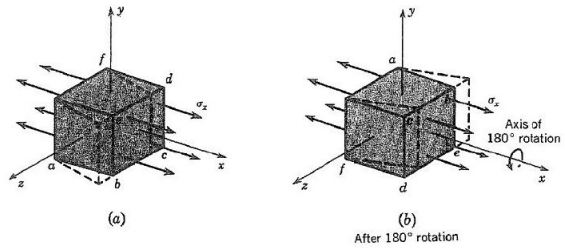
Original: Slides Ch 5, slide 16

Isotropy arguments (slides; Crandall §5.4, Probs 5.5–5.6):

- A normal stress produces **no shear strain**. Rotating the element 180° about the loading axis leaves the load unchanged but would flip the sign of any shear strain, which contradicts isotropy.
- A shear stress produces **no normal strain**. Reversing τ would reverse the normal strain, but by symmetry the result must be the same; by linearity it must therefore be zero.
- (Prob 5.5) τxy cannot produce γyz, and (Prob 5.6) τxy cannot produce a uniform expansion, by the same symmetry reasoning.

Possibility of shear strains from normal stress σ_x

Assumption of isotropy of material simplifies the relation
Suppose shear strain is present (Fig.a)



(a) (b) After 180° rotation

Normal stress produces only normal strain

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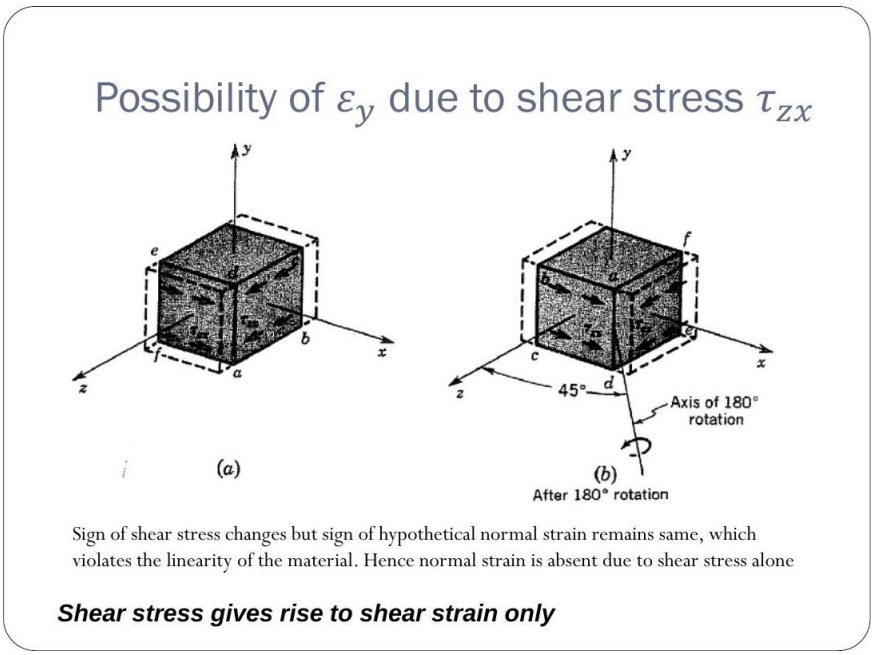


Original: Slides Ch 5, slide 21

So normal stresses cause only normal strains, and each shear stress causes only its own shear strain.

4. Derivation — superposition gives the generalized Hooke's law (slides).

$$\epsilon_x=\frac{1}{E}\left[\sigma_x-\nu\left(\sigma_y+\sigma_z\right)\right]+\alpha\,\Delta T,\epsilon_y=\frac{1}{E}\left[\sigma_y-\nu\left(\sigma_z+\sigma_x\right)\right]+\alpha\,\Delta T,\epsilon_z=\frac{1}{E}\left[\sigma_z-\nu\left(\sigma_x+\sigma_y\right)\right]+\alpha\,\Delta T$$

$$\gamma_{xy}=\frac{\tau_{xy}}{G},\gamma_{yz}=\frac{\tau_{yz}}{G},\gamma_{zx}=\frac{\tau_{zx}}{G}$$

(The $\alpha\Delta T$ term is the thermal strain, Concept 5.4.)

Plane stress ($\sigma_z=0$), inverted:

$$\sigma_x=\frac{E}{1-\nu^2}\left(\epsilon_x+\nu\,\epsilon_y\right),\sigma_y=\frac{E}{1-\nu^2}\left(\epsilon_y+\nu\,\epsilon_x\right),\epsilon_z=-\frac{\nu}{E}\left(\sigma_x+\sigma_y\right)=-\frac{\nu}{1-\nu}\left(\epsilon_x+\epsilon_y\right)$$

Full inversion (Crandall Prob 5.7), with $e=\epsilon_x+\epsilon_y+\epsilon_z$:

$$\sigma_x=\lambda e+2G\epsilon_x-\frac{E\,\alpha\,\Delta T}{1-2\nu},\lambda=\frac{\nu E}{(1+\nu)(1-2\nu)}$$

and similarly for y and z.

Isotropy ⇒ the principal axes of strain coincide with the principal axes of stress (slides). Zero shear stress gives zero shear strain, so either Mohr's circle may be used to locate them.

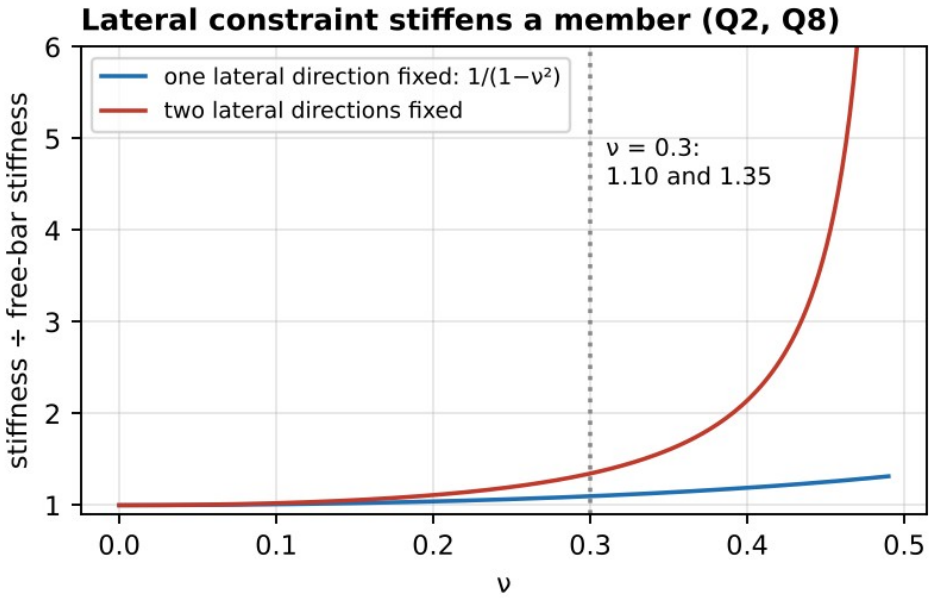
Volume change (Crandall Prob 5.2–5.3): for small strains, $\frac{\Delta V}{V}=\epsilon_x+\epsilon_y+\epsilon_z=\frac{1-2\nu}{E}\left(\sigma_x+\sigma_y+\sigma_z\right)$.

5. Picture. A circle drawn on a plate becomes an ellipse whose axes follow the principal directions (Ch. 4 Q32). A thin plate under in-plane tension gets thinner.

6. Special cases.

- *Uniaxial:* $\epsilon_y=-\nu\,\sigma/E$.
- *Laterally constrained* (plate between rigid walls, Q2; elastic material in a rigid cavity, Q8): one or two strains are zero, so stresses appear in those directions and the member becomes stiffer.

- $\nu = 0.5$: $\Delta V = 0$, constant volume.



How much lateral constraint stiffens a member

7. Traps.

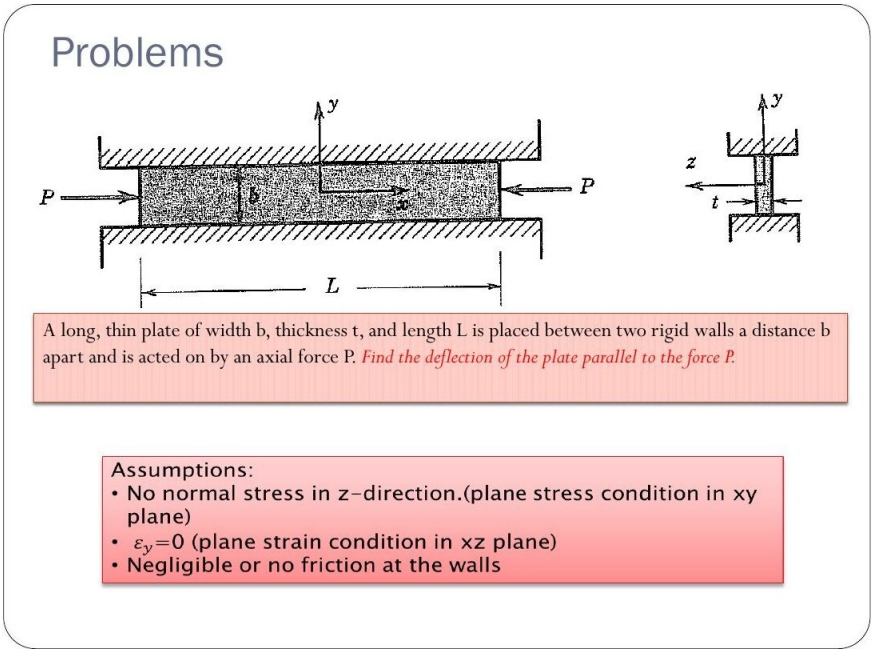
- Using the plane-**strain** formula where plane **stress** applies, or the other way round.
- Forgetting the Poisson terms when finding a dimension change. In the 2024 PYQ the y diameter actually **shrinks** although σy is tensile.
- Mixing E in GPa with σ in MPa. Convert so both are in MPa.
- Using the shear stress τxy to find a normal strain. It has no effect on normal strains (Q4).

8. Summary box

$$\epsilon_i = \frac{1}{E}\left[\sigma_i - \nu\left(\sigma_j + \sigma_k\right)\right] + \alpha\,\Delta T,\ \gamma = \tau/G,\ \frac{\Delta V}{V} = \frac{1-2\nu}{E}\sum\sigma,\ \text{plane stress: } \sigma_x = \frac{E}{1-\nu^2}\left(\epsilon_x + \nu\,\epsilon_y\right)$$

B. Solved questions

Q2 (Slides Ch5) — Plate squeezed between rigid walls (medium)



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Problem. A long thin plate of width b, thickness t and length L is placed between two rigid walls a distance b apart and loaded by an axial (compressive) force P along its length. Find the deflection of the plate parallel to P.

Given: walls prevent lateral strain (εy = 0, y across the width); the faces are free (σz = 0, plane stress through the thickness); no friction. **Find:** δ along x.

Approach. Generalized Hooke's law with one stress unknown (σy) and one strain known (εy = 0).

Solution.

1. *Axial stress:* $\sigma_x = -P/(bt)$.
2. *Wall condition:* $\epsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x) = 0$, so $\sigma_y = \nu\sigma_x$ (the walls push back).
3. *Axial strain:* $\epsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y) = \frac{\sigma_x(1-\nu^2)}{E}$.
4. *Deflection:* $\delta = |\epsilon_x| L$.

$$\delta = \frac{PL(1-\nu^2)}{Ebt}\,(\text{shortening}),\quad \sigma_y = -\frac{\nu P}{bt}$$

Check. ν = 0 gives the free-plate result PL/AE ✓. The walls make the plate stiffer by $1/(1-\nu^2) \approx 1.1$ ✓.

Exam takeaway. "Rigid walls" or "fits exactly between" is the signal for a zero strain. **Variation:** walls in both lateral directions (Q8).

Q3 (Crandall Prob 5.8) — Strains in a steel plate in plane stress (easy)

Problem. The stresses in a flat steel plate in plane stress are σx = 130, σy = −70 and τxy = 80 MN/m². Find the magnitude and orientation of the principal strains in the plane of the plate, and the third principal strain (perpendicular to the plate). E = 200 GPa, ν = 0.3.

Approach. Hooke's law for εx, εy and γ, then the strain circle (or principal stresses first, since the axes coincide).

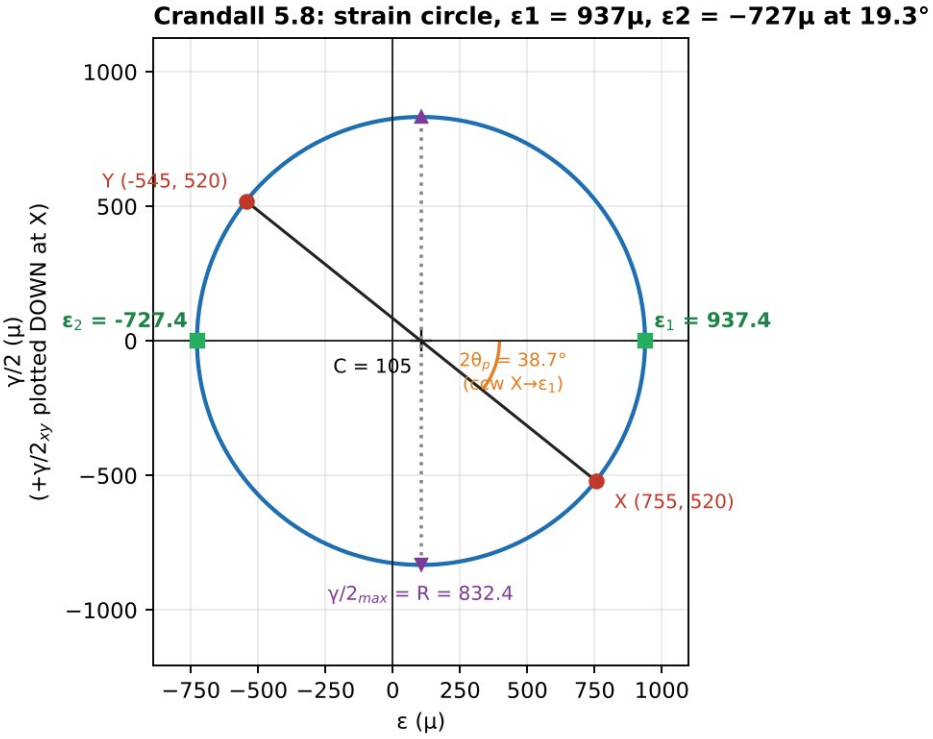
Solution.

1. $G = \dfrac{E}{2(1+\nu)} = 76.9$ GPa.
2. $\epsilon_x = \dfrac{130 - 0.3(-70)}{200\,000} = 755\,\mu;\ \epsilon_y = \dfrac{-70 - 0.3(130)}{200\,000} = -545\,\mu;\ \gamma_{xy} = \dfrac{80}{76\,923} = 1040\,\mu.$
3. *Strain circle:* centre 105μ; $R = \sqrt{650^2 + 520^2} = 832.4\,\mu$; so ε_1 = 937.4μ and ε_2 = −727.4μ.
   $\tan 2\theta_p = \dfrac{1040}{1300}$, so θp = 19.3° (counter-clockwise from x).
4. *Third principal strain:* $\epsilon_z = -\dfrac{\nu}{E}(\sigma_x + \sigma_y) = -\dfrac{0.3(60)}{200\,000} = -90\,\mu.$

$$\epsilon_1 = 937\,\mu \text{ at } 19.3^\circ,\ \epsilon_2 = -727\,\mu,\ \epsilon_3 = \epsilon_z = -90\,\mu$$

Check. Principal stresses first: σ_1 = 158.1, σ_2 = −98.1 MPa at the same 19.3°; $\epsilon_1 = (158.1 + 29.4)/200\,000 = 937\,\mu$ ✓.

Exam takeaway. The principal directions of stress and strain coincide, so you may work with either circle.



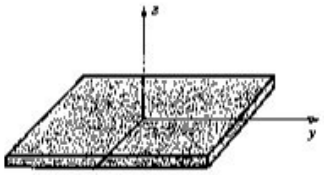
Strain Mohr's circle

Q4 (Crandall Prob 5.9) — Find σy from the thickness strain (easy, with a trap)

5.9. In a flat steel plate which is loaded in the *xy* plane, it is known that

$\sigma_x = 145\ \text{MN/m}^2$
$\tau_{xy} = 42\ \text{MN/m}^2$
$\epsilon_z = -3.6 \times 10^{-4}$

What is the value of the stress σ_y?



Original: Crandall Prob 5.9 (textbook figure)

Problem. In a flat steel plate loaded in the x–y plane, σx = 145 MN/m², τxy = 42 MN/m² and εz = −3.6 × 10⁻⁴. Find σy. E = 200 GPa, ν = 0.3.

Approach. In plane stress the thickness strain depends only on the in-plane **normal** stresses.

Solution.

1. $\epsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y)$, so $\sigma_x + \sigma_y = \frac{-\epsilon_z E}{\nu} = \frac{3.6 \times 10^{-4}(200\,000)}{0.3} = 240$ MPa.
2. $\sigma_y = 240 - 145$.

$$\sigma_y = 95\ \text{MN/m}^2$$

Check. The 42 MPa shear stress plays no part, because shear produces no normal strain in an isotropic material ✓.

Exam takeaway. The shear stress is a **distractor**. Spotting that is the point of the question.

Q5 (PYQ 2014-15 Compre A Q2) — Find ν and E from a tension test (easy)

1	A steel rod of 20 mm diameter and 1.5 m long is fixed at ends. If temperature of rod is raised by 60°C, what will be reaction at supports? Take E = 110 GPa; $\alpha = 1.2\text{x}10^{-5}$ / °C	_____ kN
2	A bar of cross-section 8 mm x 8 mm is subjected to axial pull of 10 kN. The lateral dimensions of the bar are found to be reduced by $2\text{x}10^{-3}$ mm. Find Poisson's ratio and modulus of elasticity. Take G = 80 GPa.	E = _____ GPa ν = _____
3	A machine part is made of zirconium-magnesium alloy (σ_Y = 120 MPa) and a critical point in the material is subjected to in-plane principal stresses σ_1 and $\sigma_2 = -0.5\sigma_1$, determine the magnitude of σ_1 that will cause yielding according to the maximum-shear-stress theory i.e. Tresca criterion.	σ_1 = _____ MPa
4	The state of plane stress at a critical point in a steel machine bracket is σ_X = 140 MPa, σ_Y = -80 MPa and τ_{XY} = 125 MPa. If the yield stress for steel is 250 MPa, determine if yielding occurs using maximum-distortion-energy theory i.e. Mises criterion.	
5	A solid steel shaft is to be used to transmit 205 Nm torque from the motor. If allowable shear stress for the steel is 100 MPa, determine the required diameter of the shaft.	_____ mm

Original question: PYQ 2014-15 Compre A Q1–Q5

Problem. A bar 8 mm × 8 mm carries an axial pull of 10 kN. Its lateral dimensions reduce by 2×10^{-3} mm. Find ν and E, given G = 80 GPa.

Solution.

1. $\sigma = 10\,000/64 = 156.25$ MPa.
2. Lateral strain ¿ $-2 \times 10^{-3}/8 = -2.5 \times 10^{-4}$.

3. $|\epsilon_{lat}| = \nu\sigma/E$ with $E = 2G(1+\nu)$: $2.5\times10^{-4} = \dfrac{156.25\,\nu}{2(80\,000)(1+\nu)}$, so $0.256(1+\nu) = \nu$ and $\nu = 0.344$.
4. $E = 2(80)(1.344) = 215$ GPa.

$$\nu = 0.344,\ E = 215\,\text{GPa}$$

Check. Axial strain ¿ $156.25/215\,000 = 7.27\times10^{-4}$, and $2.5/7.27 = 0.344$ ✓.

Exam takeaway. G given, ν wanted: combine $\epsilon_{lat} = -\nu\sigma/E$ with $E = 2G(1+\nu)$.

Q6 (Crandall Prob 5.12) — Pressure in a tank from an axial strain gauge (medium)

Reduction of area $= \dfrac{A_o - A}{A_o} = 1 - e^{-\epsilon_x}$

where ϵ_x is the true strain [defined by Eq. (5.27)] and e is the base of natural logarithms.

5.12. A long, thin-walled cylindrical tank has a radius r and a wall thickness t. Its ends are closed, and when a pressure p is put in the tank a strain gage mounted on the outside surface in a direction parallel to the axis of the tank measures a strain of ϵ_a. What is the pressure in the tank?

Original: Crandall Prob 5.11–5.12 (textbook figure)

Problem. A long, thin-walled cylindrical tank of radius r and wall thickness t has closed ends. With pressure p inside, a strain gauge on the outside surface, parallel to the axis, reads εa. Find p.

Approach. Wall stresses from Ch. 4 (σθ = pr/t, σa = pr/2t), then Hooke's law in the axial direction.

Solution.

1. $\epsilon_a = \dfrac{1}{E}(\sigma_a - \nu\sigma_\theta) = \dfrac{pr}{Et}\left(\dfrac{1}{2} - \nu\right)$.
2. Solve for p.

$$p = \frac{2Et\epsilon_a}{r(1-2\nu)}$$

Check. ν = 0.5 makes the axial gauge read zero whatever the pressure (the denominator vanishes): a hoop gauge would be needed ✓. For steel (ν = 0.3) the axial gauge reads only $0.2\,pr/Et$, a fifth of the hoop gauge's $0.85\,pr/Et$.

Exam takeaway. A hoop gauge is far more sensitive; mention it.

Q7 (Crandall Prob 5.20) — Tank just fitting between rigid walls: force on the walls (medium)

Problem. A long thin-walled cylindrical tank of length L just fits between two rigid end walls when it is unpressurised. Estimate the force on the walls when the pressure in the tank is p (Hooke's law material).

Approach. The walls prevent axial strain: εa = 0. Equilibrium of an end cap relates the axial wall stress to p and the wall force F.

Solution.

1. *End-cap equilibrium:* the pressure pushes the cap out with $p\pi r^2$; the wall pushes in with F; the tank wall carries the difference: $\sigma_a(2\pi rt) = p\pi r^2 - F$.
2. *Constraint:* $\epsilon_a = \dfrac{1}{E}(\sigma_a - \nu\sigma_\theta) = 0$, so $\sigma_a = \nu\sigma_\theta = \nu\,pr/t$.
3. *Combine:* $p\pi r^2 - F = 2\pi rt\cdot\nu\,pr/t = 2\nu\,p\pi r^2$.

$$F = \pi r^2 p(1-2\nu)$$

Check. For $\nu = 0.3$, $F = 0.4\pi r^2 p$. If ν were 0.5 the tank would not try to lengthen and $F = 0$ ✓.

Exam takeaway. Constraint problems always use "known strain → unknown stress".

Q8 (Crandall Prob 5.1) — Elastic block confined in a rigid cavity (medium)

Problem. An elastic material (E, ν) fills a square cavity of side 2a and height L in a rigid block. A rigid cap on top carries a force F, and the height decreases by δ. Find F.

Approach. Lateral strains are zero in both horizontal directions ($\varepsilon x = \varepsilon y = 0$); $\sigma z = -F/4a^2$.

Solution.

1. By symmetry $\sigma x = \sigma y$. $\epsilon_x = \frac{1}{E}\left[\sigma_x - \nu(\sigma_x + \sigma_z)\right] = 0$, so $\sigma_x(1-\nu) = \nu\sigma_z$ and $\sigma_x = \frac{\nu}{1-\nu}\sigma_z$.
2. $\epsilon_z = \frac{1}{E}(\sigma_z - 2\nu\sigma_x) = \frac{\sigma_z}{E}\left(1 - \frac{2\nu^2}{1-\nu}\right) = \frac{\sigma_z(1+\nu)(1-2\nu)}{E(1-\nu)}$.
3. With $\varepsilon z = -\delta/L$ and $\sigma z = -F/4a^2$:

$$F = \frac{4a^2 E\delta}{L} \cdot \frac{(1-\nu)}{(1+\nu)(1-2\nu)}$$

Check. $\nu = 0$ gives the free result $4a^2 E\delta/L$ ✓. For $\nu = 0.3$ the confined block is 1.35× stiffer. As $\nu \to 0.5$ it becomes infinitely stiff, because an incompressible material cannot shorten if it cannot expand sideways ✓.

Exam takeaway. This is the "constrained modulus" used in soil mechanics. **Variation:** confinement in one direction only gives the factor $1/(1-\nu^2)$ of Q2.

Q9 (PYQ 2023-24 Compre A Q3) — Plate in biaxial stress: γmax, Δt, ΔV (easy–medium, 10 marks)

Q3. A rectangular plate in *biaxial stress* (see Fig.Q3) is subjected to normal stresses σ_x = 67 *MPa* (*Tension*) and $\sigma_y = -23$ *MPa* (*Compression*). The plate has dimension $400mm \times 550mm \times 20mm$ and is made of steel with $E = 200MPa$ and Poisson's ratio $\vartheta = 0.30$.

a) Determine the maximum in-plane shear strain γ_{max} in the plate.
b) Determine the change Δt in the thickness of the plate.
c) Determine the change ΔV in the volume of the plate **[10M]**

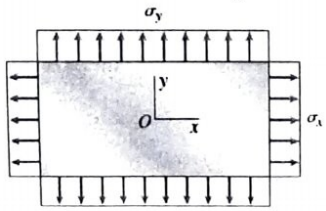


Fig.Q3

Original question: PYQ 2023-24 Compre A Q3

Problem. A rectangular steel plate (400 × 550 × 20 mm) carries $\sigma x = 67$ MPa (tension) and $\sigma y = -23$ MPa (compression). E = 200 GPa (the paper's "200 MPa" is a typo), $\nu = 0.30$. Find (a) the maximum in-plane shear strain, (b) the change in thickness Δt and (c) the change in volume ΔV.

Solution.

1. *(a)* No shear stress, so x and y are principal. $\gamma_{max}=\epsilon_x-\epsilon_y=\frac{(\sigma_x-\sigma_y)(1+\nu)}{E}=\frac{90(1.3)}{200\,000}=585\times10^{-6}$.
   (Or τmax = 45 MPa and G = 76.9 GPa give the same.)
2. *(b)* $\epsilon_z=-\frac{\nu}{E}(\sigma_x+\sigma_y)=-\frac{0.3(44)}{200\,000}=-66\times10^{-6}$, so $\Delta t=20(-66\times10^{-6})=-1.32\times10^{-3}$ mm.
3. *(c)* $\Delta V=V\frac{1-2\nu}{E}(\sigma_x+\sigma_y)=(4.4\times10^{6})\frac{0.4(44)}{200\,000}=387$ mm³.

$$\gamma_{max}=585\times10^{-6},\Delta t=-1.32\times10^{-3}\,\text{mm}\,(\text{thinner}),\Delta V=+387\,\text{mm}^3$$

Check by strains. εx = (67 + 6.9)/2 × 10⁵ = 369.5μ and εy = (−23 − 20.1)/2 × 10⁵ = −215.5μ. Sum with εz: 369.5 − 215.5 − 66 = 88μ, and 88μ × 4.4 × 10⁶ mm³ = 387 mm³ ✓.

Exam takeaway. Three standard parts; the volume change uses the sum of the stresses, not their difference.

Q10 (PYQ 2024-25 Compre A Q2) — Etched circle on a brass plate (easy–medium, 10 marks)

Q2. A circle of diameter d = 200 *mm* is etched on a brass plate (refer **FigQ2**). The plate has dimensions $400\times400\times20$ *mm*. Forces are applied to the plate, producing uniformly distributed normal stresses $\sigma_x=42$ MPa and $\sigma_y=14$ MPa. *Assume* $E=100$ *GPa* and $\mu=0.34$. **[10M]**

Calculate the following:

(a) the change in length Δac of diameter *ac* **(b)** the change in length Δbd of diameter *bd*
(c) the change Δt in the thickness of plate **(d)** the change ΔV in the volume of the plate

Original question: PYQ 2024-25 Compre A Q2

Problem. A 200 mm circle is etched on a 400 × 400 × 20 mm brass plate. σx = 42 MPa and σy = 14 MPa. E = 100 GPa, ν = 0.34. Find the changes in length of diameter ac (along x) and bd (along y), the change in thickness and the change in volume.

Solution.

1. $\epsilon_x=\frac{42-0.34(14)}{100\,000}=372.4\,\mu;\ \epsilon_y=\frac{14-0.34(42)}{100\,000}=-2.8\,\mu;\ \epsilon_z=-\frac{0.34(56)}{100\,000}=-190.4\,\mu.$
2. $\Delta_{ac}=200\,\epsilon_x=0.0745$ mm; $\Delta_{bd}=200\,\epsilon_y=-5.6\times10^{-4}$ mm; $\Delta t=20\,\epsilon_z=-3.81\times10^{-3}$ mm.
3. $\Delta V=(3.2\times10^{6})(372.4-2.8-190.4)\times10^{-6}=573$ mm³.

$$\Delta_{ac}=0.0745\,\text{mm},\Delta_{bd}=-5.6\times10^{-4}\,\text{mm},\Delta t=-3.81\times10^{-3}\,\text{mm},\Delta V=573\,\text{mm}^3$$

Check. $\Delta V/V=179.2\,\mu=(1-2\nu)(56)/E=0.32(56)/100\,000$ ✓.

Exam takeaway. bd **shrinks** even though σy is tensile, because $\nu\sigma_x>\sigma_y$. Examiners set the numbers up for exactly this surprise.

Concept 5.3 — Relations among E, G, ν and the bulk modulus K

A. Concept explanation

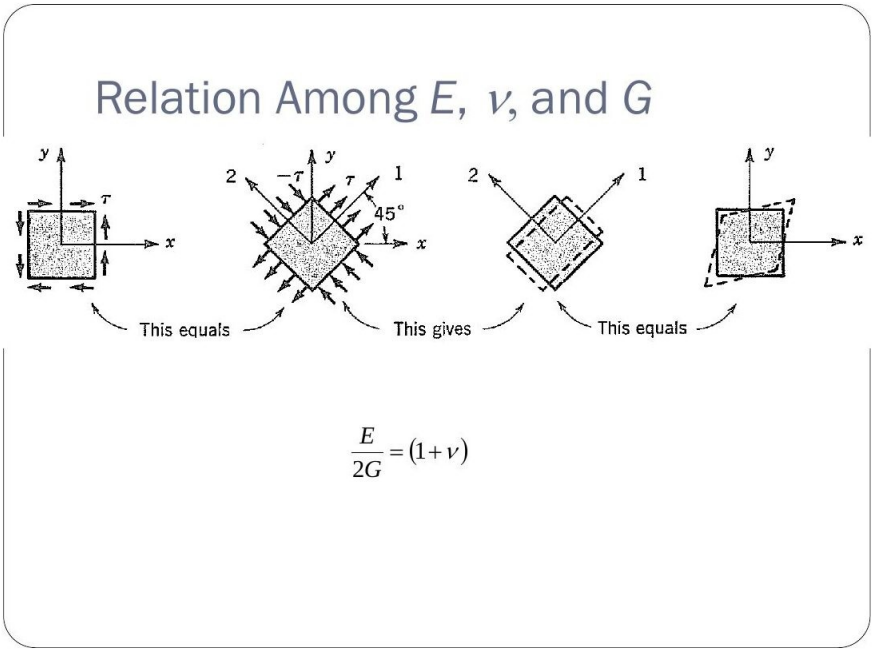
1. Intuition. Shear and stretch are not independent. A square under pure shear is, viewed at 45°, a square stretched along one diagonal and squashed along the other. So the shear stiffness G must be fixed by E and ν. Likewise, squeezing equally from all sides (hydrostatic pressure) shrinks the volume by an amount set by E and ν.

2. Why it matters. Torsion problems give G, while axial problems give E; you often need to convert. The bulk modulus K governs how compressible a material is, and it explains why ν can never exceed ½.

3. Formal treatment. Only **two** of E, G, ν and K are independent for an isotropic material:

$$G=\frac{E}{2(1+\nu)},\ K=\frac{E}{3(1-2\nu)}\left(\text{bulk modulus } K=\frac{p}{-\Delta V/V}\right)$$

4. Derivations.



Original: Slides Ch 5, slide 25

(a) E–G–ν (slides; Crandall §5.4). Pure shear τ on the x–y element ⇔ principal stresses $\sigma_1 = +\tau$ and $\sigma_2 = -\tau$ on the 45° element.

- Strain along direction 1 from Hooke's law: $\epsilon_1=\frac{\tau}{E}-\nu\frac{-\tau}{E}=\frac{\tau(1+\nu)}{E}$.
- The same strain from the shear strain, by transformation (Ch. 4) with εx = εy = 0: $\epsilon_{45}=\gamma/2=\tau/2G$.
- Equate: $\frac{\tau}{2G}=\frac{\tau(1+\nu)}{E}$, so $G=\frac{E}{2(1+\nu)}$. ∎

(b) Bulk modulus (Crandall Prob 5.3). Under hydrostatic pressure, σx = σy = σz = −p. Then

$\Delta V/V=\frac{1-2\nu}{E}(-3p)$, so $K=\frac{p}{-\Delta V/V}=\frac{E}{3(1-2\nu)}$. ∎

5. Picture. A shear-distorted square with its 45° diagonals: one stretched by ε_1, one shortened by the same amount.

6. Special cases. ν = 0: G = E/2 and K = E/3. ν = 0.5: G = E/3 and K → ∞ (incompressible).

7. Traps. Using G = E/2(1 − ν) (wrong sign). Forgetting that the formulas need an isotropic material (not wood or composites).

8. Summary. $E=2G(1+\nu)=3K(1-2\nu)$.

B. Solved question

Q11 (Crandall Probs 5.3, 5.4) — Bulk modulus, and why ν cannot exceed ½ (easy, conceptual)

Problem. (5.3) Show that $\Delta V/V=(\sigma_x+\sigma_y+\sigma_z)(1-2\nu)/E$ and that the bulk modulus is $B=E/3(1-2\nu)$. (5.4) Discuss the stability under hydrostatic pressure of a hypothetical material with ν > ½.

Solution.

1. *(5.3)* Add the three Hooke's-law strains (no temperature change): $\epsilon_x+\epsilon_y+\epsilon_z=\frac{1}{E}(1-2\nu)(\sigma_x+\sigma_y+\sigma_z)$, which is $\Delta V/V$ (Prob 5.2). With all $\sigma = -p$: $B=E/3(1-2\nu)$.
2. *(5.4)* If $\nu > ½$, then $B < 0$: hydrostatic **compression** would make the volume **increase**. Such a body would release energy as it expanded under pressure, so it would be unstable. Equivalently, the strain energy would not be positive for all deformations. Therefore $\nu \le 1/2$ for stable isotropic materials. (Positive G likewise needs $\nu > -1$.)

$$B=\frac{E}{3(1-2\nu)};\ -1<\nu\le 1/2 \text{ for a stable material}$$

Exam takeaway. $\nu > 0.5$ is a red flag in any data you are given.

Concept 5.4 — Thermal strain and thermal stresses

A. Concept explanation

1. Intuition. Heat a rail and it wants to grow. If it is free, it grows and **no stress** develops. If it is clamped between walls, the walls must push it back to its original length, and that push is the thermal stress. A rail can buckle on a hot day for exactly this reason.

2. Why it matters. Rails, pipelines, bimetal strips, bolts in sleeves, shrink fits, aircraft skins, thermostats (the "thermo gate" PYQ). PYQ 2014 Compre A Q1 and Test-2 Q3 are thermal problems.

3. Formal treatment (slides).

$$\epsilon_T=\alpha(T-T_0)=\alpha\,\Delta T,\ \text{total strain}\ \epsilon=\frac{\sigma}{E}+\alpha\,\Delta T$$

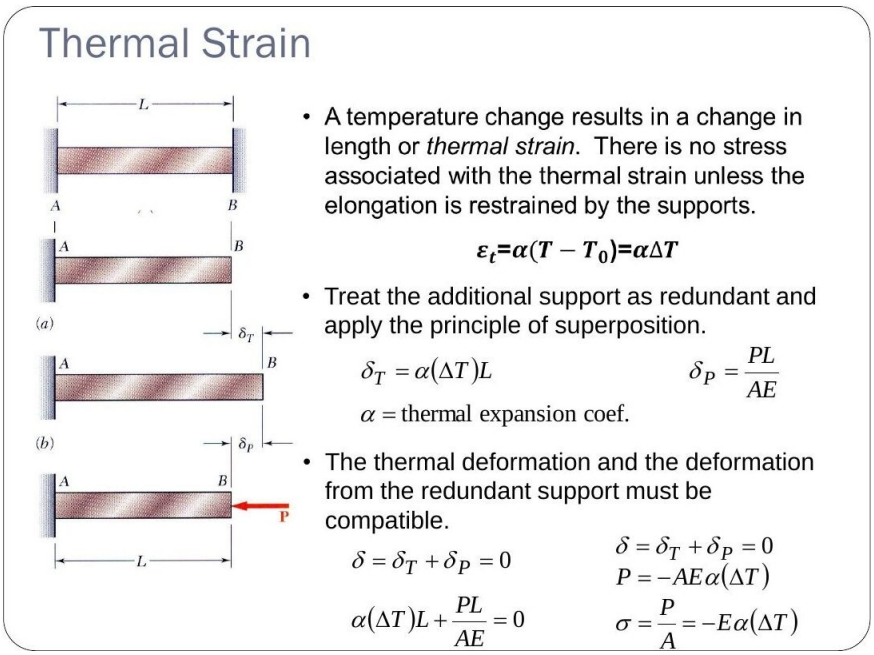
For isotropic materials the thermal strain is the same in all directions and **produces no shear strain**.

4. Derivation — bar fixed at both ends (slides, superposition).

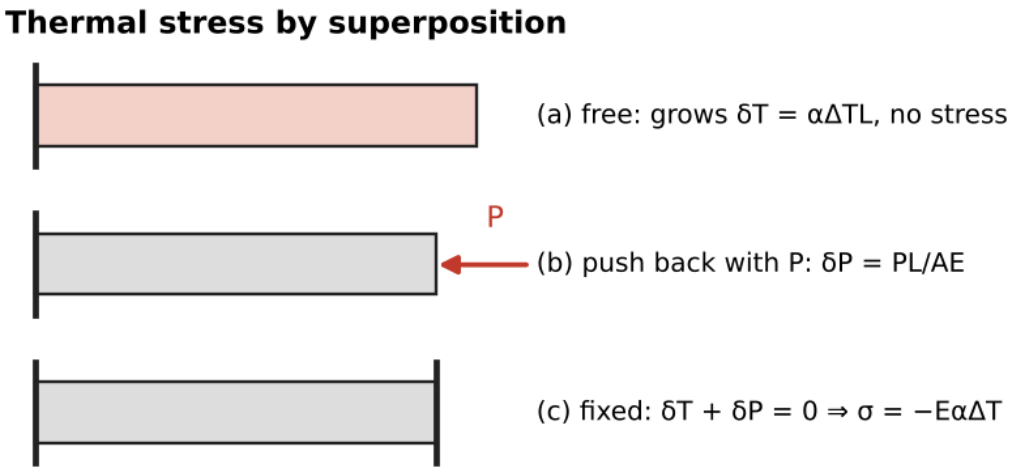
- Release B and let the bar expand freely: $\delta_T=\alpha\,\Delta T\,L$.
- Apply the redundant P to push it back: $\delta_P=P\,L/A\,E$.
- Compatibility: $\delta_T+\delta_P=0$, so $P=-A\,E\,\alpha\,\Delta T$.

$$\sigma=-E\,\alpha\,\Delta T\ (\text{compressive for heating; independent of L and A})$$

With a gap Δ: stress appears only after the gap closes; then $\alpha\,\Delta T\,L-\Delta=|P|L/A\,E$.



Original: Slides Ch 5, slide 26



Thermal stress by superposition

General recipe (the three steps of Ch. 2): (1) equilibrium of the member forces; (2) compatibility of **total** deformations, $\delta_i = \frac{F_i L_i}{A_i E_i} + \alpha_i \Delta T\, L_i$; (3) solve.

5. Picture. Three sketches (diagram below): (a) the free bar grows by αΔTL; (b) the force P pushes it back by PL/AE; (c) together the length is unchanged.

6. Special cases.

- A uniformly heated, unrestrained body of one material has no thermal stress, whatever its shape; even a hole simply grows (Q19).
- In composite systems (bolt + sleeve), the member with the **larger α** ends up in **compression**.
- A temperature **drop** in a fixed or pre-stressed member adds **tension**.

7. Traps.

- Using ΔT in °F with α in /°C.
- Forgetting the gap: close it first, then the stress builds.
- Forgetting the thermal term in **only one** of two members.

- Sign of ΔT for cooling problems.

8. Summary. $\epsilon = \sigma/E + \alpha\,\Delta T$; fixed ends: $\sigma = -E\,\alpha\,\Delta T$; composite: equilibrium + equal total deformation.

B. Solved questions

Q12 (PYQ 2014-15 Compre A Q1) — Rod fixed at both ends, heated (easy)

Problem. A 20 mm diameter steel rod, 1.5 m long, is fixed at both ends. The temperature rises by 60°C. Find the reaction at the supports. E = 110 GPa, α = 1.2 × 10^{-5}/°C.

Solution.

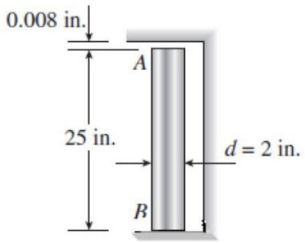
1. $\sigma = E\,\alpha\,\Delta T = 110\,000\,(1.2\times10^{-5})(60) = 79.2$ MPa (compression).
2. $R = \sigma\,A = 79.2\,\pi\,(10)^2 = 24\,880$ N.

$$R = 24.9\ \text{kN}$$

Exam takeaway. L does not matter. That is the classic trap.

Q13 (Slides Ch5, Gere) — Copper bar with a gap (easy)

Problem



- A copper bar AB of length 25 in. is placed in position at room temperature with a gap of 0.008 in. between end A and a rigid restraint (see figure).
- Calculate the axial compressive stress σ_c in the bar if the temperature rises 50°F. (For copper, use $\alpha = 9.6\text{x}10^{-6}$/°F and $E = 16\text{x}10^{6}$ psi.)

Original: Slides Ch 5, slide 48

Problem. A copper bar AB, 25 in long, is placed with a 0.008 in gap between end A and a rigid restraint. Find the axial compressive stress when the temperature rises by 50°F. α = 9.6 × 10^{-6}/°F, E = 16 × 10^{6} psi.

Solution.

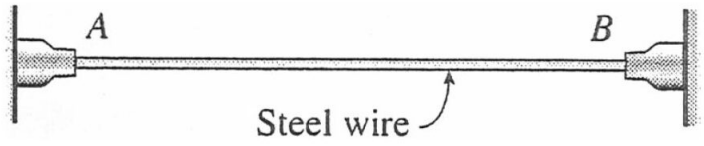
1. Free expansion ¿ $9.6\times10^{-6}(50)(25) = 0.012$ in, which exceeds the 0.008 in gap.
2. Excess suppressed: 0.004 in, so the strain is 0.004/25.
3. $\sigma_c = E\,(0.004/25) = 16\times10^{6}\,(1.6\times10^{-4})$.

$$\sigma_c = 2560\ \text{psi}\ (\text{compression})$$

Exam takeaway. Always compare the free expansion with the gap first.

Q14 (Slides Ch5, Gere) — Pre-stressed steel wire (easy)

Problem



- A steel wire AB is stretched between rigid supports. The initial pre-stress in the wire is 42 MPa when the temperature is 20°C.
- (a) What is the stress in the wire when the temperature drops to 0°C?
- (b) At what temperature T will the stress in the wire become zero?
- (Assume $\alpha = 14 \times 10^{-6}/°C$ and E=200 GPa.)

Original: Slides Ch 5, slide 49

Problem. A steel wire stretched between rigid supports has a pre-stress of 42 MPa at 20°C. (a) What is the stress at 0°C? (b) At what temperature does the stress become zero? $\alpha = 14 \times 10^{-6}/°C$, E = 200 GPa.

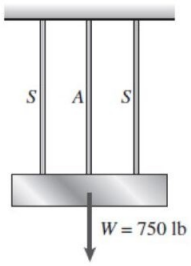
Solution.

1. *(a)* Cooling adds tension: $\Delta\sigma = E\alpha\Delta T = 200\,000\left(14\times10^{-6}\right)(20) = 56$ MPa, so σ = 98 MPa.
2. *(b)* Zero stress when $E\alpha(T-20)=42$: $T-20=42/2.8=15$, so T = 35°C.

$$\sigma(0^\circ\text{C}) = 98\,\text{MPa},\ T_{\sigma=0} = 35^\circ\text{C}$$

Q15 (Slides Ch5, Gere) — Three wires: when does steel carry everything? (medium)

Problem



- A rigid bar of weight W 750 lb hangs from three equally spaced wires, two of steel and one of aluminum. The diameter of the wires is 1/8 in . Before they were loaded, all three wires had the same length.
- What temperature increase T in all three wires will result in the entire load being carried by the steel wires? (Assume E_s= 30x10^6 psi, α_s=6.5x10^{-6}/°F, and α_a=12x10^{-6}/°F.)

Original: Slides Ch 5, slide 47

Problem. A rigid bar weighing 750 lb hangs from three equally spaced wires (two steel, one aluminium in the middle), each 1/8 in diameter and initially of equal length. What temperature rise makes the entire load carried by the steel wires? $E_s = 30 \times 10^6$ psi, $\alpha_s = 6.5 \times 10^{-6}/°\text{F}$, $\alpha_a = 12 \times 10^{-6}/°\text{F}$.

Approach. Aluminium expands more. When its force drops to zero it is just slack; the bar stays level (symmetry), so all three wires have the same length change.

Solution.

1. *Equilibrium:* each steel wire carries W/2 = 375 lb; the aluminium carries 0.
2. *Compatibility:* $\alpha_s \Delta T\, L + \dfrac{(W/2)L}{A_s E_s} = \alpha_a \Delta T\, L.$
3. $\Delta T = \dfrac{W/2}{A_s E_s(\alpha_a - \alpha_s)}$, with $A_s = \dfrac{\pi}{4}(0.125)^2 = 0.01227$ in²:

$$\Delta T = \frac{375}{0.01227\,(30 \times 10^6)\,(5.5 \times 10^{-6})}$$

$$\Delta T \approx 185^\circ\text{F}$$

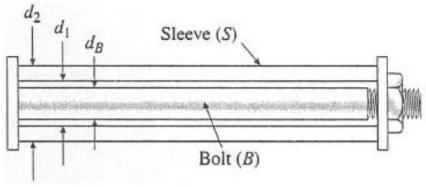
Check. The aluminium modulus is not needed, because its force is zero at this instant ✓.

Exam takeaway. "Entire load carried by X" means "force in the other member = 0". That kills one unknown.

Q16 (Slides Ch5, Gere) — Brass sleeve on a steel bolt (medium)

Problem A brass sleeve S is fitted over a steel bolt B (see figure), and the nut is tightened until it is just snug. The bolt has a diameter $d_B = 25$ mm, and the sleeve has inside and outside diameters $d_1 = 26$ mm and $d_2 = 36$ mm, respectively.

Calculate the temperature rise ΔT that is required to produce a compressive stress of 25 MPa in the sleeve. (Use material properties as follows: for the sleeve, $\alpha_S = 21 \times 10^{-6}/°\text{C}$ and $E_S = 100$ GPa; for the bolt, $\alpha_B = 10 \times 10^{-6}/°\text{C}$ and $E_B = 200$ GPa.)



Original: Slides Ch 5, slide 50

Problem. A brass sleeve S is fitted over a steel bolt B and the nut is tightened until just snug. Bolt: $d_B = 25$ mm, $E_B = 200$ GPa, $\alpha_B = 10 \times 10^{-6}/°\text{C}$. Sleeve: $d_1 = 26$ mm, $d_2 = 36$ mm, $E_S = 100$ GPa, $\alpha_S = 21 \times 10^{-6}/°\text{C}$. Find the temperature rise ΔT that produces a compressive stress of 25 MPa in the sleeve.

Approach. The sleeve (larger α) wants to grow more and is held back by the bolt. Equal forces, equal final lengths.

Solution.

1. *Equilibrium:* $P_S = P_B = P$ (sleeve in compression, bolt in tension).

2. *Compatibility:* $\alpha_S \Delta T - \dfrac{P}{E_S A_S} = \alpha_B \Delta T + \dfrac{P}{E_B A_B}$.

3. So $\sigma_S = \dfrac{P}{A_S} = \dfrac{(\alpha_S - \alpha_B)\Delta T\, E_S}{1 + \dfrac{E_S A_S}{E_B A_B}}$ and $\Delta T = \dfrac{\sigma_S}{E_S(\alpha_S - \alpha_B)}\left(1 + \dfrac{E_S A_S}{E_B A_B}\right)$.

4. $A_S = \dfrac{\pi}{4}(36^2 - 26^2) = 487$ mm² and $A_B = \dfrac{\pi}{4}(25)^2 = 491$ mm², so the bracket is $1 + \dfrac{100(487)}{200(491)} = 1.496$.

5. $\Delta T = \dfrac{25(1.496)}{100\,000(11 \times 10^{-6})}$.

$$\Delta T \approx 34^\circ\text{C}$$

Exam takeaway. The larger-α member always ends in compression. **Variation:** "find the bolt stress", $\sigma_B = 25(487/491) = 24.8$ MPa (T).

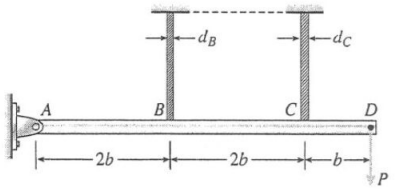
Q17 (Slides Ch5, Gere) — Rigid bar on two cables with a temperature rise: allowable P (hard)

Problem A rigid bar *ABCD* is pinned at end *A* and supported by two cables at points *B* and *C* (see figure). The cable at *B* has nominal diameter d_B = 12 mm and the cable at *C* has nominal diameter d_C = 20 mm. A load *P* acts at end *D* of the bar.

What is the allowable load *P* if the temperature rises by 60°C and each cable is required to have a factor of safety of at least 5 against its ultimate load?

(*Note:* The cables have effective modulus of elasticity *E* = 140 GPa and coefficient of thermal expansion $\alpha = 12 \times 10^{-6}$/°C.)

$(T_B)_{ULT}$ = 102,000 N $(T_C)_{ULT}$ = 231,000



Original: Slides Ch 5, slide 53

Problem. A rigid bar ABCD is pinned at A and supported by cables at B (2b from A, $d_B = 12$ mm) and C (4b, $d_C = 20$ mm). A load P acts at D (5b). The temperature rises by 60°C. Each cable must have a factor of safety of at least 5 against its ultimate load ($(T_B)_{ult} = 102$ kN, $(T_C)_{ult} = 231$ kN). Find the allowable P. E = 140 GPa, α = 12 × 10^{-6}/°C.

Approach. Equilibrium about A; compatibility from the rigid bar's rotation; each cable's deformation includes the thermal term.

Solution.

1. *Equilibrium:* $\sum M_A$: $2T_B + 4T_C = 5P$ … (1).
2. *Compatibility (rigid bar):* $\delta_C = 2\delta_B$ … (2).
3. *Force–displacement–temperature:* $\delta_B = \dfrac{T_B L}{E A_B} + \alpha \Delta T L$ and $\delta_C = \dfrac{T_C L}{E A_C} + \alpha \Delta T L$ … (3), (4).
4. *Substitute into (2):* $\dfrac{T_C}{E A_C} + \alpha \Delta T = \dfrac{2T_B}{E A_B} + 2\alpha \Delta T$, so $2T_B A_C - T_C A_B = -E\alpha \Delta T A_B A_C$ … (5).

5. *Numbers* (effective cable areas from the slide: $A_B = 76.7$ mm², $A_C = 173$ mm²): $T_B(346) - T_C(76.7) = -1\,338\,000$ … (6).
6. *Solve (1) and (6):* $T_B = 0.2494\,P - 3480$ and $T_C = 1.1253\,P + 1740$ (N).
7. *Allowables:* $(T_B)_{allow} = 20.4$ kN gives $P_B = 4.0096(20\,400) + 13\,953 = 95.7$ kN; $(T_C)_{allow} = 46.2$ kN gives $P_C = 0.8887(46\,200) - 1546 = 39.5$ kN.

$$P_{allow} = 39.5\,\text{kN}\,(\text{cable C governs})$$

Exam takeaway. Heating *relieves* cable B (the −3480 term) and *loads* cable C, because compatibility forces C to stretch twice as much as B. Thermal terms appear in both cables, and examiners check that you include them in both.

Q18 (Crandall Prob 5.17) — Shrink fit: heat the pulley or cool the shaft? (medium)

Problem. A steel shaft (25.000 mm) is to be tight-fitted into a steel pulley (hole 24.950 mm). The pulley can be heated, or the shaft cooled, to give a clearance of 0.025 mm for easy assembly. Which is more effective, and what temperature change is needed in each case? ($\alpha = 12 \times 10^{-6}$/°C.)

Solution.

1. *Heat the pulley:* the hole must grow from 24.950 to 25.025 mm ($\Delta d = 0.075$ mm).
$$\Delta T = \frac{0.075}{12 \times 10^{-6}(24.950)} = +250.5^{\circ}\text{C}.$$
2. *Cool the shaft:* the shaft must shrink from 25.000 to 24.925 mm ($\Delta d = -0.075$ mm).
$$\Delta T = \frac{-0.075}{12 \times 10^{-6}(25.000)} = -250^{\circ}\text{C}.$$

$$\text{heat the pulley by} \approx 250^{\circ}\text{C, or cool the shaft by} \approx 250^{\circ}\text{C}$$

The magnitudes are practically the same, but cooling by 250°C below room temperature (to about −230°C) is beyond even liquid nitrogen (−196°C). So **heating the pulley** (or combining a heated pulley with a cooled shaft) is the practical choice.

Check. A hole expands like the material around it, by $\alpha \times$ diameter $\times \Delta T$ ✓ (see Q19).

Exam takeaway. "Does a hole get bigger or smaller when heated?" Bigger, exactly as if it were filled with the same material.

Q19 (Crandall Prob 5.13) — A heated sheet with a hole (easy, conceptual)

Problem. A circular sheet of metal has a small circular hole cut out of it. If the sheet is initially stress-free, unrestrained and heated uniformly by T, what shape do the circular boundaries take?

Solution.

1. A uniform temperature rise in an unrestrained, homogeneous body produces uniform thermal strain αT in every direction and **no stress**.
2. Every length scales by $(1 + \alpha T)$: the outer circle and the hole both remain circles, and the hole **grows**.

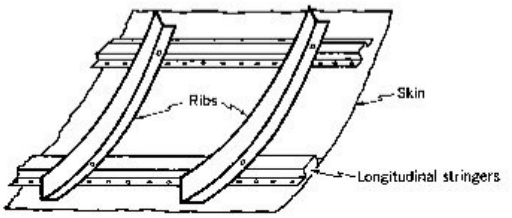
$$\text{Both boundaries stay circular; all radii (hole included) increase by the factor } (1 + \alpha T)$$

Exam takeaway. Uniform heating is a pure "photographic enlargement". Stresses arise only from restraint or from temperature gradients.

Q20 (Crandall Prob 5.18) — Aircraft skin cooled relative to the ribs (medium)

5.17. It is desired to produce a tight fit of a steel shaft in a steel pulley. The internal diameter of the hole in the pulley is 24.950 mm, while the outside diameter of the shaft is 25.000 mm. The pulley will be assembled on the shaft by either heating the pulley or cooling the shaft and then putting the shaft in the pulley hole and allowing the assembly to reach a uniform temperature. Is it more effective to heat the pulley or to cool the shaft? What temperature change would be required in each case to produce a clearance of 0.025 mm for easy assembly?

5.18. An aircraft fuselage is constructed with circular ribs connected by longitudinal stringers and an outer skin riveted to the stringers but not in contact with the ribs. If insulation is placed between the skin and the ribs, the temperature of the skin and stringers may become quite different from that of the ribs. Find the state of stress and strain in the skin due to cooling it and the stringers 50°F below the temperature of the ribs. Note that the skin is free to expand longitudinally, but that lateral expansion is at least partly constrained by the ribs. Neglect the effects of curvature. For simplicity, consider two extreme cases, the first in which the ribs are so heavy as to prevent any lateral strain, and the second in which the ribs are so light that they offer practically no constraint at all. Consider the material to be 2024-T4 aluminum alloy.



Original: Crandall Prob 5.18 (textbook figure)

Problem. An aircraft skin (2024-T4 aluminium) and its stringers are cooled 50°F below the ribs. The skin is free to contract longitudinally (x), but lateral contraction (y) may be constrained by the ribs. Find the stress and strain in the skin for (1) ribs heavy enough to prevent all lateral strain and (2) ribs so light they offer no constraint. Take $E = 10.6 \times 10^6$ psi, $\nu = 0.33$, $\alpha = 12.6 \times 10^{-6}$/°F.

Approach. Plane stress in the skin ($\sigma z = 0$), $\sigma x = 0$ (free longitudinally) and $\Delta T = -50$°F. Case 1 adds $\varepsilon y = 0$.

Solution.

1. *Case 1 (rigid ribs):* $\epsilon_y = \frac{\sigma_y}{E} + \alpha \Delta T = 0$, so $\sigma_y = -E\alpha\Delta T = 10.6\times10^6\left(12.6\times10^{-6}\right)(50) = 6680$ psi (tension).
   - $\epsilon_x = -\frac{\nu\sigma_y}{E} + \alpha\Delta T = -\frac{0.33(6680)}{10.6\times10^6} - 630\times10^{-6} = -208\,\mu - 630\,\mu = -838\,\mu$
   - ϵ_z is the same, −838μ.
2. *Case 2 (no constraint):* no stress; $\epsilon_x = \epsilon_y = \epsilon_z = \alpha\Delta T = -630\,\mu$.

$$\text{Case 1: } \sigma_y = 6.68\ \text{ksi (T)},\ \sigma_x = 0,\ \epsilon_x = -838\,\mu,\ \epsilon_y = 0;\ \text{Case 2: } \sigma = 0,\ \epsilon = -630\,\mu$$

Check. The constrained skin is pulled sideways by the ribs, and its Poisson contraction adds to the thermal contraction along x ✓.

Exam takeaway. Bracketing a real problem between "fully constrained" and "free" limits is good engineering judgment, and Crandall asks for exactly that.

Q21 (PYQ 2014-15 Test-2 Q3) — Thermo-gate: aluminium and magnesium plates closing a gap (medium, 20 marks)

Q.3 A thermo gate consists of a 6061-T6-aluminum plate *AB* and an Am-1004-T61-magnesium plate *CD*, each having a width of 15 mm and fixed supported at their ends (*Fig. 3*). If the gap between them is 1.5 mm when the temperature is T_1 = 25°C, determine the temperature required to just close the gap. Also, what is the axial force in each plate if the temperature becomes T_2 = 100°C? Assume bending or buckling will not occur. E_{al} = 68.9 GPa, E_{am} = 44.7 GPa, α_{al} = 24 x 10^{-6} /°C, α_{am} = 26 x 10^{-6} /°C. **[20]**

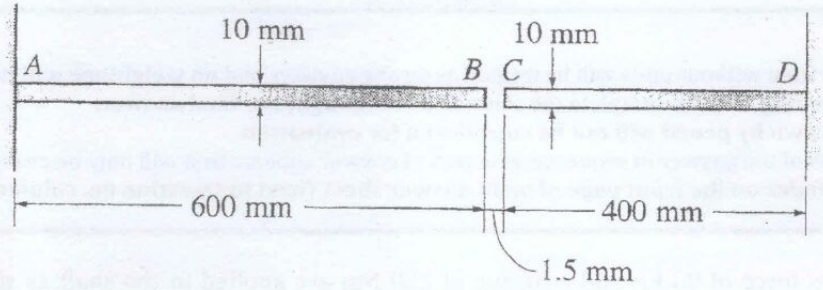


Fig. 3

Original question: PYQ 2014-15 Test-2 Q3

Problem. A thermo-gate consists of an aluminium plate AB (600 mm; $E_{al}=68.9$ GPa, $\alpha_{al}=24\times10^{-6}$/°C) and a magnesium plate CD (400 mm; $E_{am}=44.7$ GPa, $\alpha_{am}=26\times10^{-6}$/°C), each 15 mm wide and 10 mm thick, fixed at their outer ends. The gap between them is 1.5 mm at T_1 = 25°C. Find (a) the temperature that just closes the gap and (b) the axial force in each plate at T_2 = 100°C.

Approach. (a) Free expansions add up to the gap. (b) Beyond that, the excess expansion is suppressed by a common compressive force F (equilibrium: the same F in both plates).

Solution.

1. *(a)* $\left(24\times10^{-6}(600)+26\times10^{-6}(400)\right)\Delta T=1.5$, so $0.0248\,\Delta T=1.5$ and ΔT = 60.5°C. Hence T = 85.5°C.
2. *(b)* At ΔT = 75°C the free expansion is 0.0248(75) = 1.86 mm, an excess of 0.36 mm over the gap. With A = 150 mm²:

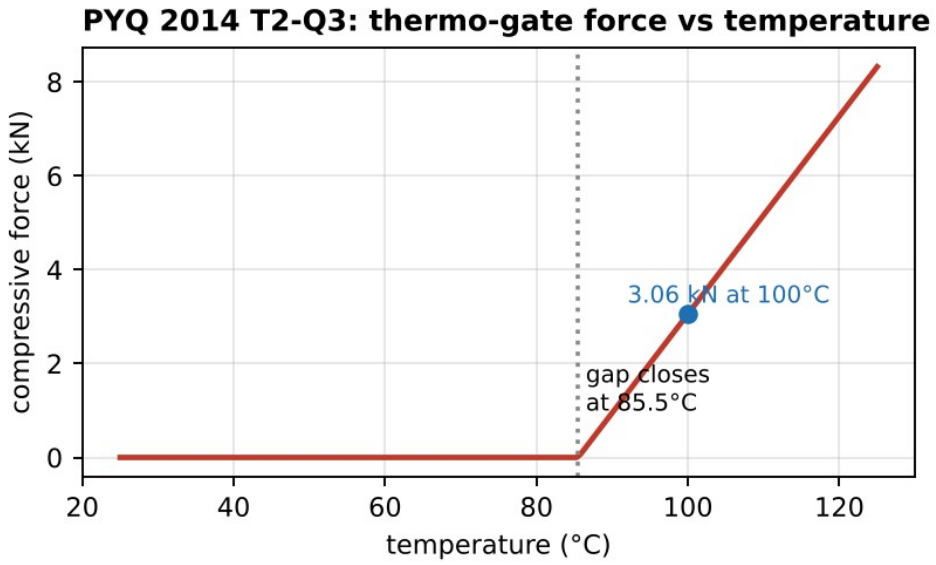
$$0.36=F\left(\frac{600}{150(68\,900)}+\frac{400}{150(44\,700)}\right)=F(5.806+5.966)\times10^{-5}$$

so F = 3058 N.

$$(a)\,T=85.5^{\circ}\text{C};(b)\,F=3.06\,\text{kN}(\text{compression in both plates}),\sigma=20.4\,\text{MPa}$$

Check. The two plates act as springs in series once the gap closes ✓.

Exam takeaway. Gap problems have two stages: free expansion until contact, then series springs. **Variation:** find T at which the stress reaches 50 MPa.



Force in the plates as the temperature rises

Concept 5.5 — Complete equations of elasticity and strain energy

A. Concept explanation

1. Intuition. Every elasticity problem, however complicated, is the three steps of Ch. 2 written for a continuum: forces balance at every point, the deformations fit together, and stress is tied to strain (and temperature) by the material law.

2. Why it matters. It shows that the course's methods are complete. The strain-energy density is the basis of Castigliano's theorem (Ch. 2) and of the von Mises criterion (Concept 5.6).

3. Formal treatment (slides; Crandall §5.6, §5.8).

- **Equilibrium:** $\frac{\partial \sigma_x}{\partial x}+\frac{\partial \tau_{xy}}{\partial y}+\frac{\partial \tau_{zx}}{\partial z}+X=0$, and similarly for y and z (X, Y, Z are body forces per unit volume).
- **Geometric compatibility:** $\epsilon_x=\partial u/\partial x$, $\gamma_{xy}=\partial v/\partial x+\partial u/\partial y$, ... (6 relations).
- **Stress–strain–temperature:** the generalized Hooke's law with $\alpha(T - T_0)$ (6 relations).

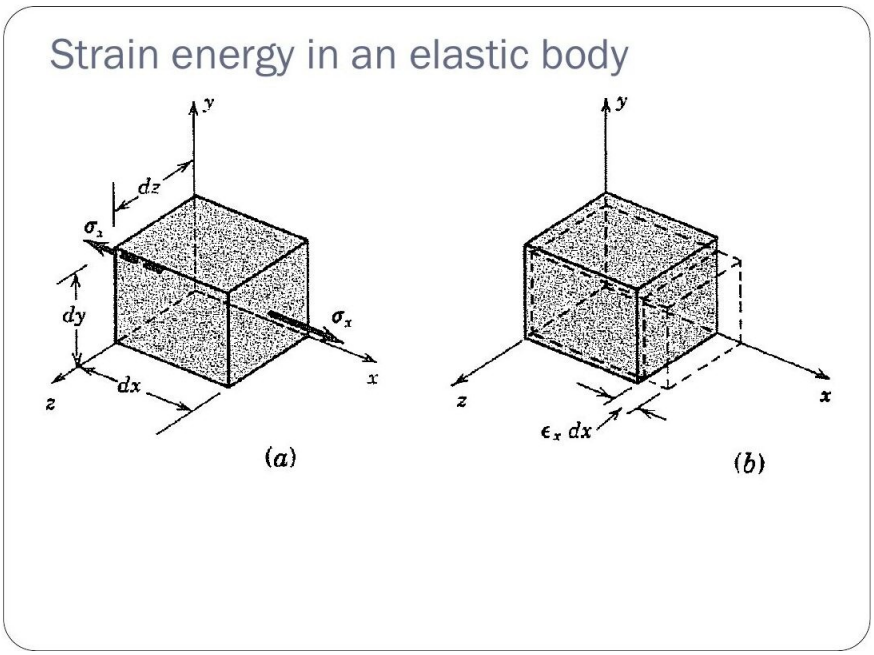
That is 15 equations for 15 unknowns (6 stresses, 6 strains, 3 displacements).

4. Derivation — strain-energy density. Work stored per unit volume, for each component $1/2\,\sigma\epsilon$:

$$u=1/2\left(\sigma_x\epsilon_x+\sigma_y\epsilon_y+\sigma_z\epsilon_z+\tau_{xy}\gamma_{xy}+\tau_{yz}\gamma_{yz}+\tau_{zx}\gamma_{zx}\right)$$

In principal stresses, substituting Hooke's law:

$$u=\frac{1}{2E}\left[\sigma_1^2+\sigma_2^2+\sigma_3^2-2\nu\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1\right)\right]$$



Original: Slides Ch 5, slide 30

5. Picture. u is the area under each stress–strain line, ½σε, added over the components.

6. Special cases. Uniaxial: $u = \sigma^2/2E$. Pure shear: $u = \tau^2/2G$. Integrating over a bar gives Ch. 2's $U = P^2 L/2AE$; over a shaft, Ch. 6's $U = T^2 L/2GJ$.

7. Traps. Forgetting the ½. Adding energies of separate loads (energy is quadratic: cross terms appear).

8. Summary. 15 equations, 15 unknowns; $u = \frac{1}{2E}\left[\sum \sigma_i^2 - 2\nu \sum \sigma_i \sigma_j\right]$.

Concept 5.6 — Yield criteria: Tresca (maximum shear stress) and von Mises (distortion energy)

A. Concept explanation

1. Intuition. A tensile test tells you when a material yields under *one* stress. Real parts feel combinations. A yield criterion is a rule that turns any combination into a single "equivalent" number you can compare with the tensile yield strength. Ductile metals yield by **slip** (shear), so both criteria are built on shear-like quantities, and hydrostatic pressure, which causes no slip, cannot cause yield.

2. Why it matters. Almost every compre has a "factor of safety by Tresca/von Mises" part (2019, 2021, 2024, 2014).

3. Formal treatment (slides). *Fracture vs yield:* fracture is driven by normal stress (bonds break and cannot reform); yield is driven by shear stress (planes slide and bonds re-form).

Tresca (maximum shear stress): yield starts when τmax reaches its value at yield in the tension test, τY = σY/2:

$$\sigma_{max} - \sigma_{min} = \sigma_Y \left(\text{all three principal stresses, including } \sigma_3 = 0 \text{ in plane stress}\right)$$

Plane-stress cases (slides): σ1 and σ2 of **opposite** sign: |σ1 − σ2| = σY. **Same** sign: max(|σ1|, |σ2|) = σY. The yield surface in the σ1–σ2 plane is a **hexagon** (slides: "Tresca yield surface for plane stress problems").

von Mises (distortion energy): yield starts when the distortion part of the strain-energy density reaches its value at yield in simple tension:

$$\sigma_{vm}=\sqrt{\frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}{2}}\geq\sigma_Y$$

Plane stress: $\sigma_{vm}=\sqrt{\sigma_1^2-\sigma_1\sigma_2+\sigma_2^2}$, an **ellipse** circumscribing the Tresca hexagon (slides: "Von Mises yield surface"). In x–y components: $\sigma_{vm}=\sqrt{\sigma_x^2-\sigma_x\sigma_y+\sigma_y^2+3\tau_{xy}^2}$. A shaft or beam point with only σ and τ: $\sigma_{vm}=\sqrt{\sigma^2+3\tau^2}$ and $\sigma_{Tresca}=\sqrt{\sigma^2+4\tau^2}$.

Factor of safety: $n=\sigma_Y/\sigma_{eq}$.

4. Derivation of von Mises (slides). Split each principal stress into a hydrostatic part $\sigma_{av}=(\sigma_1+\sigma_2+\sigma_3)/3$ and a distortional part $(\sigma_i-\sigma_{av})$.

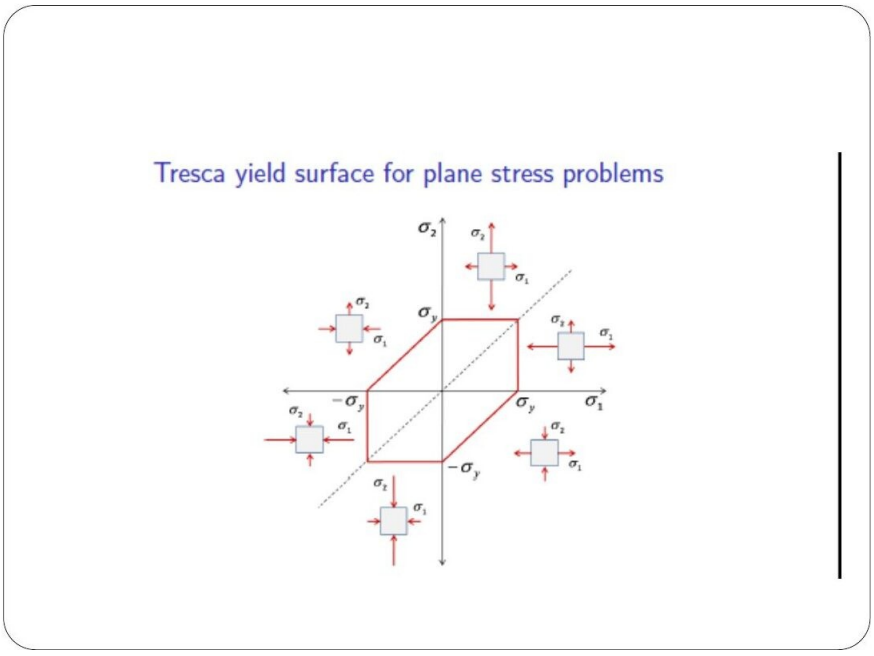
- Total energy: $u=\frac{1}{2E}\left[\sum\sigma_i^2-2\nu\sum\sigma_i\sigma_j\right]$.
- Volume-change energy (replace each σ_i by σ_{av}): $u_v=\frac{3\sigma_{av}^2}{2E}(1-2\nu)=\frac{1-2\nu}{6E}(\sigma_1+\sigma_2+\sigma_3)^2$.
- Distortion energy:

$$u_d=u-u_v=\frac{1+\nu}{3E}\left[\frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}{2}\right](1)$$

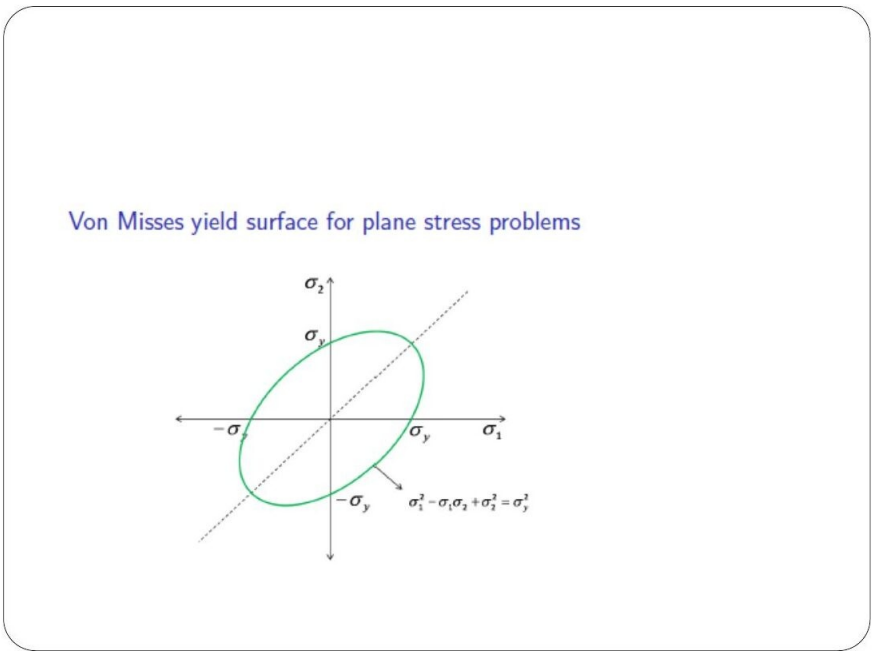
- In simple tension at yield ($\sigma_1=\sigma_Y$, $\sigma_2=\sigma_3=0$): $u_{d,Y}=\frac{1+\nu}{3E}\sigma_Y^2$ … (2).
- Setting (1) = (2) gives the criterion. ∎

Tresca from Mohr's circle: the tension-test circle at yield has diameter σY, so τY = σY/2.

5. Picture. In the σ_1–σ_2 plane the hexagon (Tresca) sits inside the ellipse (von Mises); they touch at the six points of uniaxial and equal-biaxial stress. The largest gap is in pure shear: Tresca predicts τY = 0.5σY and von Mises τY = σY/√3 = 0.577σY. Test data for steel, copper and nickel lie closer to von Mises.



Original: Slides Ch 5, slide 39



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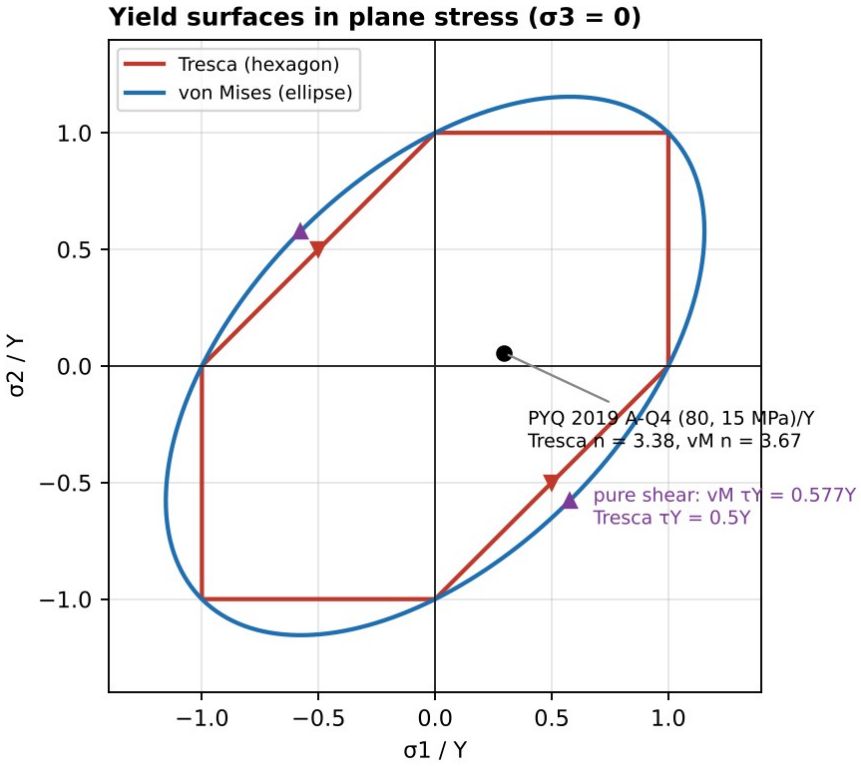


Tresca hexagon inside the von Mises ellipse

6. Special cases. Tresca is **always conservative** (it predicts yield earlier or at the same point), so its FS ≤ the von Mises FS. If the question asks "will it yield?" and Tresca says no, von Mises says no too.

7. Common traps.

- Using the in-plane value $(\sigma_1 - \sigma_2)/2$ for Tresca when both principal stresses have the **same sign**. You must include $\sigma_3 = 0$.
- Plugging x–y stresses into $\sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2}$ without first finding principal stresses (or use the $3\tau^2$ form).
- Using σu instead of σY.

8. Summary box

Criterion	Equivalent stress	Pure-shear yield
Tresca	$\sigma_{max} - \sigma_{min}$ (including σ_3); $\sqrt{\sigma^2+4\tau^2}$ for a σ–τ state	$0.5\sigma_Y$
von Mises	$\sqrt{\sigma_x^2-\sigma_x\sigma_y+\sigma_y^2+3\tau_{xy}^2}$; $\sqrt{\sigma^2+3\tau^2}$ for a σ–τ state	$0.577\sigma_Y$

B. Solved questions

Q22 (Slides Ch5) — Bolt diameter by Tresca and von Mises (easy–medium)

- The load on a bolt subjected to an axial pull of 15 kN together with a shear of 7.5 kN. Determine the diameter of the bolt according to
- Tresca's criterion
- Mises yield criterion
- In uniaxial tension, material yields at the stress of 95 N/mm^2 and poisson's ratio is 0.3.

Original: Slides Ch 5, slide 58

Problem. A bolt carries an axial pull of 15 kN together with a shear of 7.5 kN. The material yields at 95 N/mm² in uniaxial tension ($\nu = 0.3$). Find the bolt diameter by (a) Tresca's criterion and (b) the von Mises criterion.

Approach. $\sigma = P/A$ and $\tau = V/A$ (average shear), then the σ–τ forms of the criteria. ν is not needed for these two theories (it is for the maximum-principal-strain theory).

Solution.

1. *(a) Tresca:* $\dfrac{\sqrt{15^2+4(7.5)^2}\times 10^3}{A} = \dfrac{21\,213}{A} = 95$, so A = 223.3 mm² and d = 16.9 mm.
2. *(b) von Mises:* $\dfrac{\sqrt{15^2+3(7.5)^2}\times 10^3}{A} = \dfrac{19\,843}{A} = 95$, so A = 208.9 mm² and d = 16.3 mm.

$$d_{Tresca} = 16.9\text{ mm}, d_{vM} = 16.3\text{ mm}$$

Check. Tresca is more conservative, so it gives the larger diameter ✓.

Exam takeaway. Always say which criterion is conservative. **Variation:** the maximum-principal-strain theory needs ν.

Q23 (PYQ 2014-15 Compre A Q3) — Tresca with $\sigma_2 = -0.5\sigma_1$ (easy)

Problem. A zirconium–magnesium alloy (σ_Y = 120 MPa) is subjected to in-plane principal stresses σ_1 and $\sigma_2 = -0.5\sigma_1$. Find σ_1 at yield by Tresca.

Solution. The signs are opposite, so $\sigma_{max} - \sigma_{min} = \sigma_1 - \sigma_2 = 1.5\sigma_1 = 120$.

$$\sigma_1 = 80\ \text{MPa}$$

Q24 (PYQ 2014-15 Compre A Q4) — Does it yield by von Mises? (easy)

Problem. At a critical point, σx = 140, σy = −80 and τxy = 125 MPa. Yield stress 250 MPa. Does yielding occur (distortion-energy theory)?

Solution.

$$\sigma_{vm} = \sqrt{140^2 - (140)(-80) + 80^2 + 3(125)^2} = \sqrt{19\,600 + 11\,200 + 6\,400 + 46\,875} = \sqrt{84\,075} = 290\ \text{MPa}$$

$$290 > 250 \Rightarrow \text{yields}\ (\text{FS} = 0.86)$$

Q25 (PYQ 2019-20 Compre A Q4) — Factor of safety by Tresca and von Mises (medium, 7 marks)

Q4. The state of plane stress shown (in **FigQ4**) occurs at a critical point of a steel machine components. As a result of several tensile tests, it has been found that the tensile yield strength is Y = 270 MPa for the grade of steel used. Determine the factor of safety with respect to yield, using
(a) maximum-shearing-stress (MSS) criterion and (b) Distortion energy criterion **[7]**

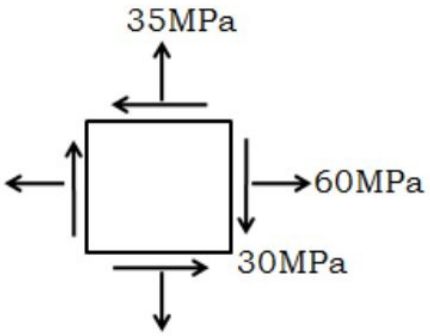


Fig Q4

Original question: PYQ 2019-20 Compre A Q4

Problem. A critical point of a steel machine component carries σx = 60 MPa (T), σy = 35 MPa (T) and τxy = 30 MPa (downward on the right face). σY = 270 MPa. Find the factor of safety by (a) the maximum-shear-stress criterion and (b) the distortion-energy criterion.

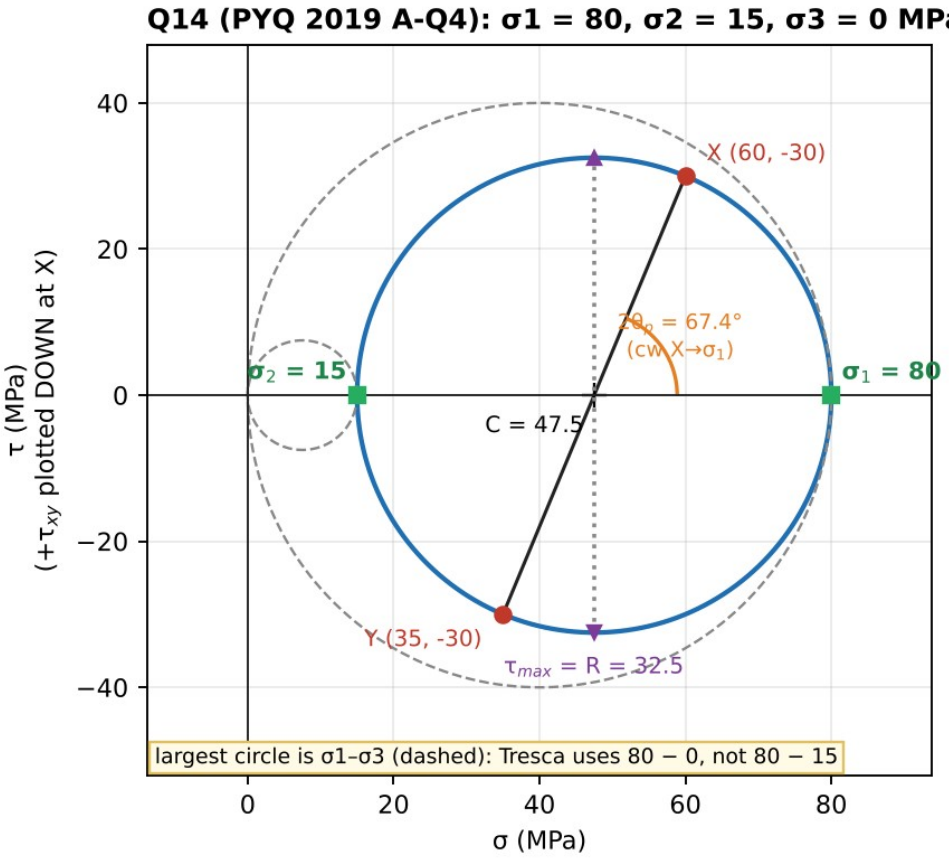
Solution.

1. *Principal stresses:* centre 47.5, $R = \sqrt{12.5^2 + 30^2} = 32.5$, so σ1 = 80, σ2 = 15, σ3 = 0 MPa.
2. *(a) Tresca:* same sign, so σmax − σmin = 80 − 0 = 80 and n = 270/80 = 3.38.
3. *(b) von Mises:* $\sqrt{80^2 - 80(15) + 15^2} = 73.7$ MPa, so n = 270/73.7 = 3.67.

$$n_{Tresca} = 3.38,\ n_{vM} = 3.67$$

Check. Using the in-plane value σ1 − σ2 = 65 would give n = 4.15. That is wrong (unsafe), because both principal stresses are tensile ✓. $n_{Tresca} < n_{vM}$ ✓.

Exam takeaway. The out-of-plane Mohr's circle (σ1 to σ3 = 0) is the largest one here.



Mohr's circles including σ3 = 0

PYQs on this concept solved in Ch. 6

- **2021-22 A-Q2(c):** does the axle yield by von Mises? (σ = 140 MPa, τ = 54.4 MPa, σY = 350 MPa): σvm = 169 MPa, so **no** (FS 2.07). See Ch. 6 Q25.
- **2024-25 B-Q2(d):** factor of safety at point B of the post by von Mises. See Ch. 6 Q27.

Chapter 5 — Formula and convention sheet

Item	Formula
Engineering stress/strain	$\sigma = P/A_0$, $\epsilon = \delta/L_0$; true strain $\ln(L/L_0)$; RA = $1 - e^{-\epsilon_t}$
Hooke + Poisson	$\epsilon_x = \frac{1}{E}\left[\sigma_x - \nu(\sigma_y + \sigma_z)\right] + \alpha \Delta T$ (cyclic), $\gamma = \tau/G$
Plane-stress inverse	$\sigma_x = \frac{E}{1-\nu^2}(\epsilon_x + \nu\epsilon_y)$, $\epsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y)$
Lamé form	$\sigma_x = \lambda e + 2G\epsilon_x$, $\lambda = \frac{\nu E}{(1+\nu)(1-2\nu)}$
Elastic constants	$G = \frac{E}{2(1+\nu)}$, $K = \frac{E}{3(1-2\nu)}$; $-1 < \nu \le ½$
Volume change	$\Delta V/V = \epsilon_x + \epsilon_y + \epsilon_z = \frac{1-2\nu}{E}\sum\sigma$
Lateral constraint	one direction: factor $1/(1-\nu^2)$; two directions: $(1-\nu)/[(1+\nu)(1-2\nu)]$
Thin tank from gauge	axial gauge: $p = 2Et\epsilon_a/[r(1-2\nu)]$; tank between

Item	Formula
	walls: $F = \pi r^2 p(1-2\nu)$
Thermal (fixed ends)	$\sigma = -E\alpha\Delta T$; with a gap, stress only after it closes
Composite thermal	equilibrium + equal total deformation; larger α goes into compression
Shrink fit	$\Delta d = \alpha d \Delta T$ (holes grow like the material)
Energy density	$u = \frac{1}{2E}\left[\sum \sigma_i^2 - 2\nu \sum \sigma_i \sigma_j\right]$; uniaxial $\sigma^2/2E$; shear $\tau^2/2G$
Tresca	σmax − σmin = σY (include σ3 = 0); σ–τ: $\sqrt{\sigma^2 + 4\tau^2}$
von Mises	$\sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2}$; x–y: $\sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau^2}$; σ–τ: $\sqrt{\sigma^2 + 3\tau^2}$
Factor of safety	$n = \sigma_Y/\sigma_{eq}$; Tresca n ≤ von Mises n

Other Crandall Ch. 5 problems worth practising: 5.2 (volume strain), 5.5–5.7 (isotropy proofs, inverting Hooke's law), 5.10 (principal strains → stresses, like PYQ 2019 A-Q2), 5.14 (bolted pressure vessel), 5.15–5.16 (rails, clad plates), 5.19 (bridge sun/shade).

Chapter 6 — Torsion

Circular shafts: geometry of deformation, $\tau = T\,r\,/\,J$, angle of twist, solid vs hollow, power transmission, gears, statically indeterminate shafts, strain energy, combined torsion + axial/bending, yielding

Sources used: Slides "6.pdf" (Chapter 6, Torsion); Crandall Ch. 6 (§6.1–6.5, Figs 6.5–6.14; the uploaded textbook PDF ends before this chapter's problem set, so the solved problems here come from the slides and PYQs); PYQs 2023-24 (A-Q1; B-Q2), 2024-25 (B-Q1; B-Q2), 2019-20 (B-Q4), 2021-22 (A-Q1; A-Q2), 2014-15 (Test-2 Q1, Q4; Compre A-Q5; Compre B-Q2).

Layout. Part A (eight items) for each concept, then Part B: slides → PYQs, easy → hard. Each solution follows *Problem · Given/Find · Approach · Sketch · Steps · Answer · Check · Takeaway*.

Concept 6.1 — Geometry of deformation of a twisted circular shaft

A. Concept explanation

1. Intuition. Draw straight lines along a rubber hose, then twist it. The lines become helices, but each circular cross-section stays a flat circle and simply rotates as a rigid disc, like a stack of coins each turned slightly more than the one below. Points far from the axis move farther, so the shear strain grows **linearly with radius**.

2. Why it matters. Drive shafts, axles, turbine–generator couplings (slide: "turbine exerts torque T on the shaft; the shaft transmits it to the generator; the generator creates an equal and opposite torque T'"), screwdrivers, torsion bars. Torsion appears in **every** compre in your set. The slide's design aim: "we are interested primarily in the twisting moment which can be transmitted by the shaft without damage to the material".

3. Formal treatment (Crandall §6.2; slide figures 6.5–6.7). Use cylindrical coordinates (r, θ, z). Crandall uses M_t for torque; the slides and Beer use T. Crandall proves by **symmetry** (not by assumption) that:

1. Each cross-section originally normal to the axis **remains plane and normal**, and **does not distort in its own plane**. (Fig. 6.5: if the section warped, rotating the shaft 180° about a diameter X–X would give a different deformation under the same loading, which is impossible. Fig. 6.6/6.7: diameters must stay straight for the same reason.)
2. Radial lines stay straight and radial.

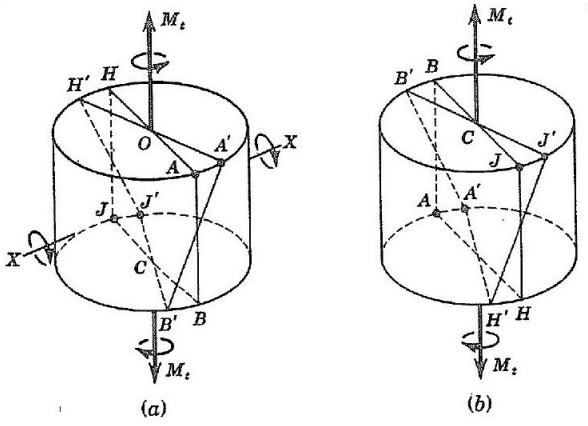


Fig. 6.7 If the diameter *HA* remains straight during deformation, then rotation of (*a*) about *X*-*X* produces (*b*) which is identical in terms of deformation.

- Each plane cross section originally normal to the axis remains plane and normal and does not distort within its own plane.

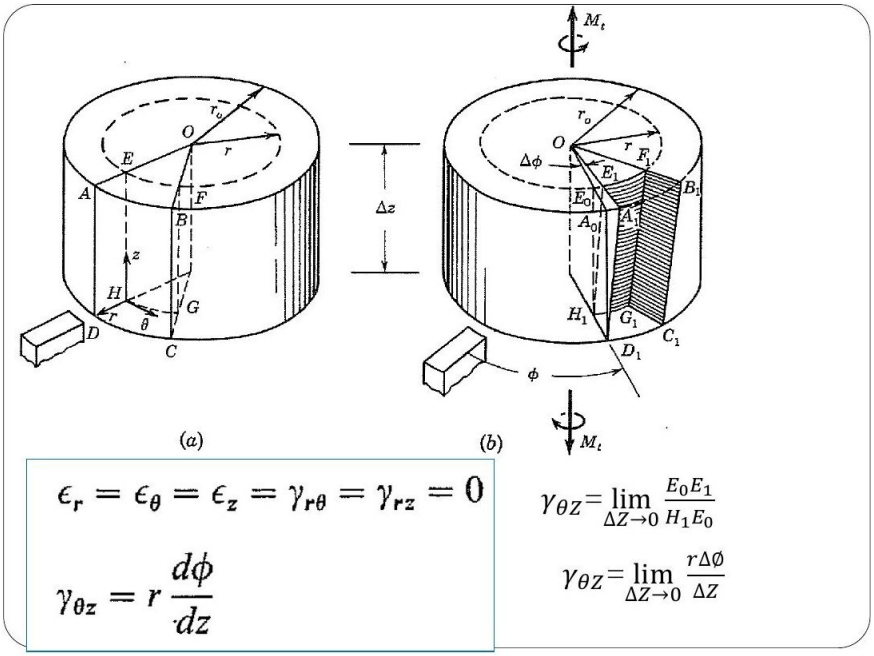
Original: Slides Ch 6, slide 7

No dilation (slides): $\epsilon_r = \epsilon_\theta = \epsilon_z = \gamma_{r\theta} = \gamma_{rz} = 0$. "The only remaining possible mode of deformation is one in which the cross sections of the shaft remain undeformed but rotate relative to each other."

4. Derivation of the shear strain (slides; Crandall Fig. 6.8). Two sections Δz apart rotate relative to each other by Δφ. On the surface element at radius r, point E_0 moves to E_1 through an arc $r\Delta\varphi$:

$$\gamma_{\theta z}=\frac{\lim_{\Delta z\to 0} E_0E_1}{H_1E_0}=\frac{\lim_{\Delta z\to 0} r\,\Delta\phi}{\Delta z}\Rightarrow \gamma_{\theta z}=r\frac{d\phi}{dz}$$

dφ/dz is the **rate of twist** (rad per unit length), the same for all r. So γ varies linearly from 0 at the centre to its maximum at the surface.



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5. Picture. A line drawn along the surface tilts by $\gamma_{max}=r_o\,d\phi/dz$ after twisting: the "helix angle" in the slide figure. Section circles stay circles; radii stay straight.

6. Special cases. These results hold for solid **and** hollow circular sections. They do **not** hold for non-circular sections, which warp (outside this course).

7. Traps. Writing γ = φ (the twist angle). In fact γ = rφ/L. γ is the angle a surface line tilts through; φ is the rotation of one end of the bar relative to the other.

8. Summary. Sections stay plane and rigid; $\gamma_{\theta z}=r\,d\phi/dz$; all other strains are zero.

Concept 6.2 — The torsion formula and the angle of twist

A. Concept explanation

1. Intuition. Material far from the axis is strained more and also has a longer lever arm, so it does most of the work. That is why hollow shafts are efficient: material near the centre carries little and mostly adds weight.

2. Why it matters. The formula τ = Tr/J sizes every shaft, and φ = TL/GJ controls stiffness (gear alignment, vibration). Both appear in every torsion PYQ.

3. Formal treatment / 4. Derivation (slides; Crandall §6.3).

- *Step 1, stress–strain* (Hooke in shear; slide: "stresses obtained from stress–strain relations"): $\sigma_r=\sigma_\theta=\sigma_z=\tau_{r\theta}=\tau_{rz}=0$ and

$$\tau_{\theta z}=G\gamma_{\theta z}=Gr\frac{d\phi}{dz}\quad(A)$$

- *Step 2, equilibrium* (slide: “equilibrium requirements”): the shear stresses on a section must add up to the internal torque:

$$M_t = \int_A r\left(\tau_{\theta z}\, dA\right) = G\frac{d\phi}{dz}\int_A r^2\, dA = G\frac{d\phi}{dz} I_z \quad (1,2)$$

where $I_z = J = I_P$ is the **polar moment of inertia**:

$$J_{solid} = \frac{\pi r^4}{2} = \frac{\pi d^4}{32},\ J_{hollow} = \frac{\pi\left(d_o^4 - d_i^4\right)}{32} \quad (3)$$

- *Step 3, combine:* $\frac{d\phi}{dz} = \frac{M_t}{G I_z}$ (4), and substituting (4) in (A): $\tau_{\theta z} = \frac{M_t r}{I_z}$ (6).

- *Step 4, integrate along the length:* $\phi = \int_0^L \frac{M_t}{G I_z}\, dz = \frac{M_t L}{G I_z}$ (5) for a uniform shaft.

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\phi}{L}$$

Maximum (surface) stress, solid shaft: $\tau_{max} = \frac{T c}{J} = \frac{16 T}{\pi d^3}$. **Hollow shaft:** $\tau_{max} = \frac{T\left(d_o/2\right)}{J}$ at the outer surface and $\tau_{min} = \frac{T\left(d_i/2\right)}{J} = \tau_{max}\frac{d_i}{d_o}$ at the inner surface. **Thin-walled tube** (mean radius r_m, thickness t): $J \approx 2\pi r_m^3 t$ and $\tau \approx \frac{T}{2\pi r_m^2 t}$. **Torsional stiffness:** $k_T = T/\phi = GJ/L$; GJ is the torsional rigidity.

Several segments or varying torque:

$$\phi_{A/B} = \sum_i \frac{T_i L_i}{G_i J_i}\left(T_i = \text{internal torque in segment } i, \text{signed}\right)$$

Sign convention (slides/Beer): right-hand rule. An internal torque is positive if its vector points **away** from the cut face (outward normal), the same “positive face, positive direction” rule as in Ch. 3. *Angle of twist (slides):* for a shaft fixed at one end it equals the rotation of the free end; in general it is the angle through which one end rotates relative to the other.

5. Stress picture. τ is linear in r on the cross-section, zero at the centre. By $\tau_{xy} = \tau_{yx}$, an equal shear acts on **longitudinal** planes too, which is why wooden shafts split lengthwise.

Stress and deformation in a twisted elastic circular shaft

From

$$\tau_{\theta z} = G\gamma_{\theta z} = Gr\frac{d\phi}{dz} \quad \text{------ (A)}$$

and $\int_A r(\tau_{\theta z} dA) = M_t$

$$M_t = G\frac{d\varphi}{dz}\int_A r^2\, dA - - - - - - - - 1$$

$$M_t = G\frac{d\varphi}{dz} I_Z - - - - - - - - - - 2 \qquad I_z = \frac{\pi r^4}{2} = \frac{\pi d^4}{32} - - - - - 3$$

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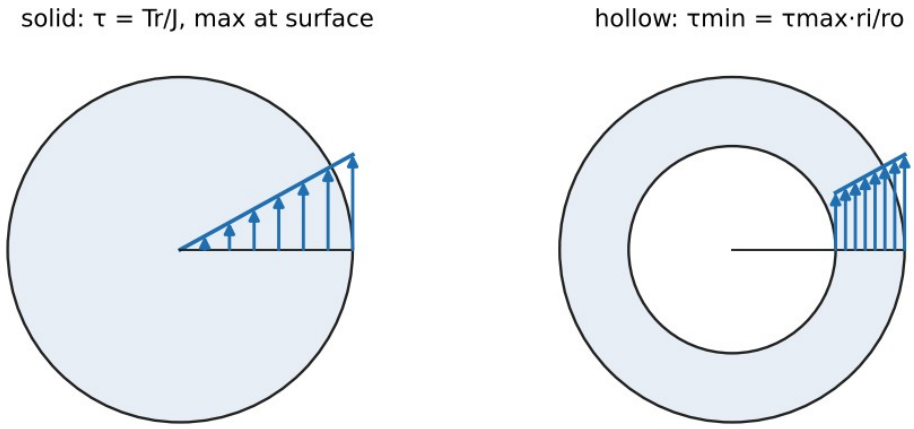
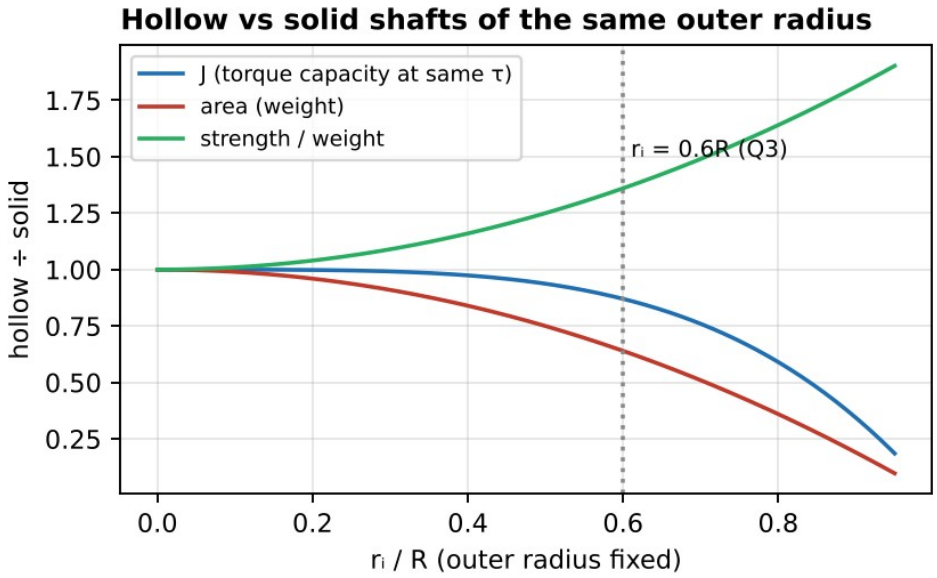


Shear stress varies linearly with radius

6. Special cases.

- Same weight (same area): a hollow shaft carries more torque and is stiffer than a solid one (Q3, Q6).
- Same strength requirement: the hollow shaft is lighter (Q4).



Effect of boring out a shaft (outer radius fixed)

7. Common traps.

- Using r = d where it should be d/2.
- Units: T in N·mm with r in mm and J in mm^4 gives MPa. φ comes out in **radians**; convert to degrees only at the end.
- Using the *external* torque instead of the *internal* torque of a segment. Draw a torque diagram, just like an axial-force diagram.
- $J_{hollow} = \frac{\pi}{32}\left(d_o^4 - d_i^4\right)$, **not** $\frac{\pi}{32}\left(d_o - d_i\right)^4$.

8. Summary box

$$\tau = \frac{T\,r}{J},\ \tau_{max,solid} = \frac{16\,T}{\pi\,d^3},\ \phi = \sum \frac{T\,L}{G\,J},\ J_s = \frac{\pi\,d^4}{32},\ J_h = \frac{\pi\left(d_o^4 - d_i^4\right)}{32},\ k_T = \frac{G\,J}{L}$$

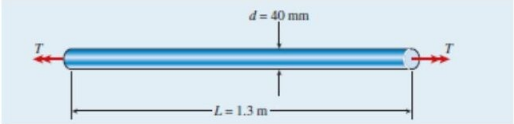
B. Solved questions

Q1 (Slides Ch6, Gere) — Solid steel bar: τmax, φ and the allowable torque (easy)

A solid steel bar of circular cross-section has diameter d=40mm, length L=1.3m, and G=80GPa.

a) if T=340Nm, what is the max shear stress in the bar? What is the angle of twist between the ends.

b) if the allowable shear stress is 42MPa and the allowable angle of twist is 2.5⁰, what is the maximum permissible torque?



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Problem. A solid steel bar has d = 40 mm, L = 1.3 m and G = 80 GPa. (a) With T = 340 N·m, find the maximum shear stress and the angle of twist between the ends. (b) If the allowable shear stress is 42 MPa and the allowable twist is 2.5°, find the maximum permissible torque.

Solution.

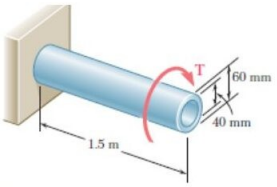
1. *(a) Stress:* $\tau_{max} = \dfrac{16(340\,000)}{\pi(40)^3} = 27.1$ MPa.
2. *(a) Twist:* $J = \pi(40)^4/32 = 251\,327$ mm⁴, so $\phi = \dfrac{340\,000(1300)}{80\,000(251\,327)} = 0.0220$ rad = 1.26°.
3. *(b) Strength:* $T = \tau\,\pi\,d^3/16 = 42\,\pi(40)^3/16 = 528$ N·m.
4. *(b) Stiffness:* $T = \phi\,G\,J/L = 0.04363(80\,000)(251\,327)/1300 = 675$ N·m.

$$(a)\,\tau_{max} = 27.1\text{ MPa},\ \phi = 1.26^\circ;\ (b)\,T_{max} = 528\text{ N·m (stress governs)}$$

Exam takeaway. Always check both strength and stiffness and report the **smaller** torque.

Q2 (Slides Ch6, Beer) — Hollow shaft 40/60 mm (easy)

A hollow cylindrical steel shaft is 1.5 m long and has inner and outer diameters respectively equal to 40 and 60 mm . (*a*) What is the largest torque that can be applied to the shaft if the shearing stress is not to exceed 120 MPa? (*b*) What is the corresponding minimum value of the shearing stress in the shaft?



What torque should be applied to the end of the shaft of to produce a twist of 2°? Use the value $G = 77$ GPa for the modulus of rigidity of steel.

What angle of twist will create a shearing stress of 70 MPa on the inner surface of the hollow steel shaft

Original: Slides Ch 6, slide 18

Problem. A hollow steel shaft (L = 1.5 m, $d_i = 40$ mm, $d_o = 60$ mm, G = 77 GPa). (a) Largest torque if $\tau \le 120$ MPa? (b) The corresponding minimum shear stress? (c) Torque for a twist of 2°? (d) Angle of twist that creates 70 MPa on the inner surface?

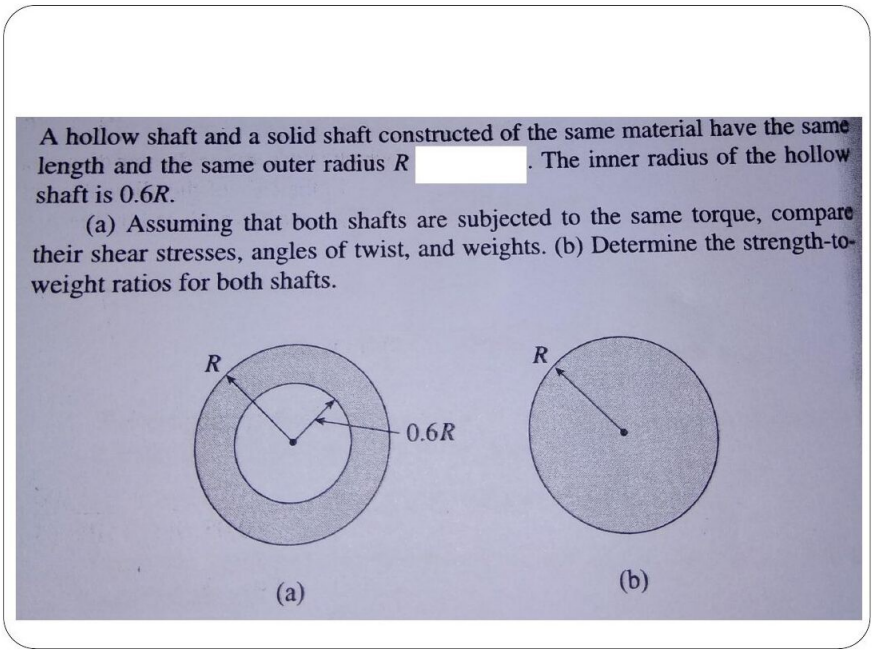
Solution.

1. $J = \dfrac{\pi}{32}\left(60^4 - 40^4\right) = 1.021 \times 10^6$ mm⁴.
2. *(a)* $T = \tau J / c = 120\left(1.021 \times 10^6\right)/30 = 4.08$ kN·m.
3. *(b)* $\tau_{min} = 120\left(20/30\right) = 80$ MPa.
4. *(c)* $T = \phi G J / L = 0.03491\left(77\,000\right)\left(1.021 \times 10^6\right)/1500 = 1.83$ kN·m.
5. *(d)* From $\tau = G r \phi / L$: $\phi = \tau L / \left(G r_i\right) = 70\left(1500\right)/\left(77\,000 \times 20\right) = 0.0682$ rad = 3.91°.

$$(a)\,4.08\,\text{kN·m}\,;(b)\,80\,\text{MPa}\,;(c)\,1.83\,\text{kN·m}\,;(d)\,3.91^\circ$$

Exam takeaway. $\tau \propto r$: the inner surface always has the smaller stress, $\tau_{min} = \tau_{max}\, d_i / d_o$.

Q3 (Slides Ch6, Gere) — Hollow vs solid shaft of the same outer radius (medium, conceptual)



Original: Slides Ch 6, slide 16

Problem. A hollow shaft (outer radius R, inner radius 0.6R) and a solid shaft (radius R) of the same material and length carry the same torque. Compare their shear stresses, angles of twist and weights, and their strength-to-weight ratios.

Solution.

1. $J_h/J_s = 1 - 0.6^4 = 0.8704$ and $A_h/A_s = 1 - 0.36 = 0.64$.
2. Stresses and twists scale as 1/J: $\tau_h/\tau_s = \phi_h/\phi_s = 1/0.8704 = 1.15$.
3. Weight ratio: $W_h/W_s = 0.64$.
4. Strength-to-weight (torque each can carry at the same τallow, per unit weight): $0.8704/0.64 = 1.36$.

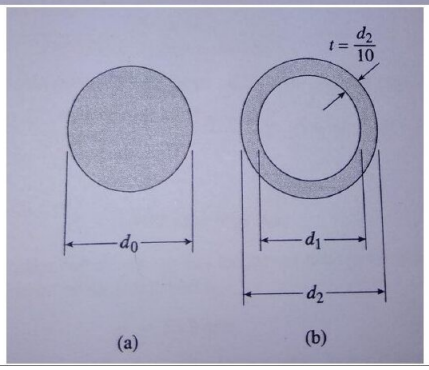
$$\tau_h/\tau_s = \phi_h/\phi_s = 1.15,\ W_h/W_s = 0.64,\ \text{strength/weight}\,(\text{hollow/solid}) = 1.36$$

Exam takeaway. Removing the inner 60% of the radius costs only 13% of J but saves 36% of the weight.

Q4 (Slides Ch6, Gere) — Design a solid vs a hollow shaft for 1200 N·m (medium)

A steel shaft is to be manufactured either as a solid circular bar or as a circular tube . The shaft is required to transmit a torque of 1200 N·m without exceeding an allowable shear stress of 40 MPa nor an allowable rate of twist of 0.75°/m. (The shear modulus of elasticity of the steel is 78 GPa.)

(a) Determine the required diameter d_0 of the solid shaft. (b) Determine the required outer diameter d_2 of the hollow shaft if the thickness t of the shaft is specified as one-tenth of the outer diameter. (c) Determine the ratio of diameters (that is, the ratio d_2/d_0) and the ratio of weights of the hollow and solid shafts.



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Problem. A steel shaft must transmit 1200 N·m without exceeding τallow = 40 MPa or a rate of twist of 0.75°/m (G = 78 GPa). (a) Find the diameter d_0 of a solid shaft. (b) Find the outer diameter d_2 of a hollow shaft with wall thickness $t = d_2/10$. (c) Find d_2/d_0 and the weight ratio.

Solution.

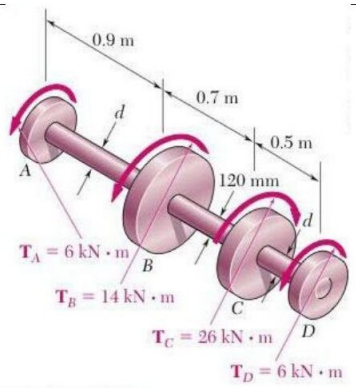
1. Rate limit: $\theta = 0.75°/\text{m} = 1.309 \times 10^{-5}$ rad/mm.
2. *(a) Solid:* strength gives $d^3 = 16T/(\pi\tau)$, so d = 53.5 mm. Stiffness gives $d^4 = 32T/(\pi G\theta)$, so d = 58.8 mm. Take d_0 = 58.8 mm.
3. *(b) Hollow* ($d_1 = 0.8\,d_2$, so $J = \frac{\pi d_2^4}{32}(1 - 0.8^4) = 0.5904\,\frac{\pi d_2^4}{32}$): strength gives 63.7 mm; stiffness gives 67.1 mm. Take d_2 = 67.1 mm.
4. *(c)* $d_2/d_0 = 1.14$. Weight ratio ¿ $\frac{d_2^2(1 - 0.64)}{d_0^2} = 0.47.$

$$d_0 = 58.8\,\text{mm},\ d_2 = 67.1\,\text{mm},\ d_2/d_0 = 1.14,\ W_h/W_s = 0.47$$

Exam takeaway. Here stiffness governs both designs; the hollow shaft is 14% larger in diameter but weighs less than half.

Q5 (Slides Ch6, Beer Sample 3.1) — Four-gear shaft: hollow BC, size AB and CD (medium)

Problem



Shaft *BC* is hollow with inner and outer diameters of 90 mm and 120 mm, respectively. Shafts *AB* and *CD* are solid and of diameter *d*. For the loading shown, determine (*a*) the maximum and minimum shearing stress in shaft *BC*, (*b*) the required diameter *d* of shafts *AB* and *CD* if the allowable shearing stress in these shafts is 65 MPa.

Original: Slides Ch 6, slide 25

Problem. Torques $T_A = 6$, $T_B = 14$, $T_C = 26$ and $T_D = 6$ kN·m act on shaft ABCD (balanced). BC is hollow ($d_i = 90$, $d_o = 120$ mm). AB and CD are solid with diameter d. Find (a) the maximum and minimum shear stress in BC and (b) d for τallow = 65 MPa.

Approach. Internal torques from free bodies (slides: "T_AB" and "T_BC" sketches), then τ = Tr/J.

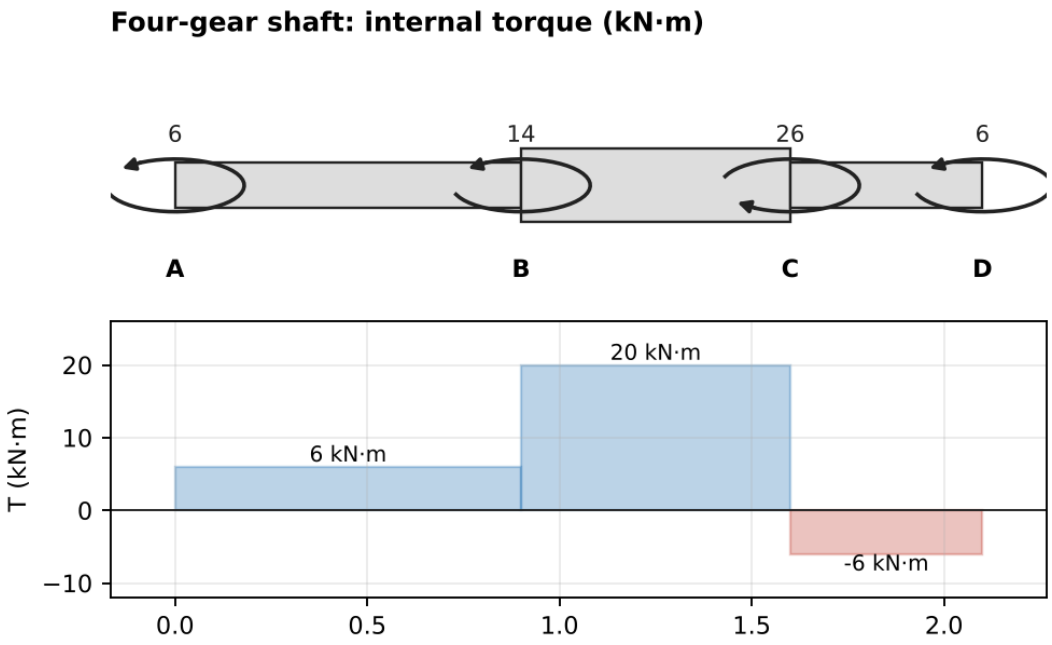
Solution.

1. *Internal torques:* $T_{AB} = 6$ kN·m; $T_{BC} = 6 + 14 = 20$ kN·m; $T_{CD} = 20 - 26 = -6$ kN·m (magnitude 6).
2. $J_{BC} = \frac{\pi}{32}\left(120^4 - 90^4\right) = 13.92 \times 10^6$ mm⁴.
3. *(a)* $\tau_{max} = \frac{20 \times 10^6 (60)}{13.92 \times 10^6} = 86.2$ MPa and $\tau_{min} = 86.2(45/60) = 64.7$ MPa.
4. *(b)* $d = \left(\frac{16(6 \times 10^6)}{\pi(65)}\right)^{1/3} = 77.8$ mm.

$$(a)\ \tau_{max} = 86.2\ \text{MPa},\ \tau_{min} = 64.7\ \text{MPa};\ (b)\ d = 77.8\ \text{mm}$$

Check. Matches the slide (d = 77.8 mm) ✓.

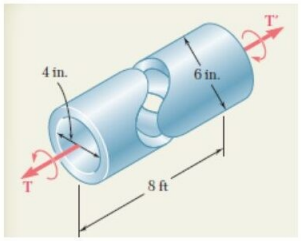
Exam takeaway. The torque diagram decides which segment is critical.



Internal torque diagram

Q6 (Slides Ch6, Beer) — Hollow 4 in/6 in shaft vs equal-weight alternatives (medium)

Problem



The preliminary design of a large shaft connecting a motor to a generator calls for the use of a hollow shaft with inner and outer diameters of 4 in. and 6 in., respectively. Knowing that the allowable shearing stress is 12 ksi, determine the maximum torque that can be transmitted (*a*) by the shaft as designed, (*b*) by a solid shaft of the same weight, (*c*) by a hollow shaft of the same weight and of 8-in. outer diameter.

Original: Slides Ch 6, slide 28

Problem. A motor–generator shaft is hollow, 4 in inner and 6 in outer diameter, with τ_{allow} = 12 ksi. Find the maximum torque (a) for the shaft as designed, (b) for a solid shaft of the same weight and (c) for a hollow shaft of the same weight with an 8 in outer diameter.

Solution.

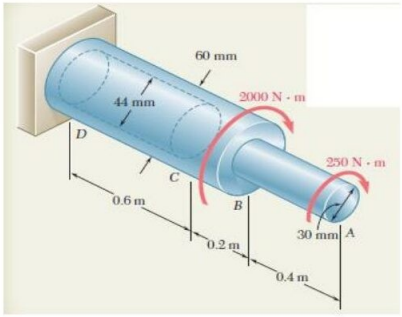
1. *(a)* $J=\frac{\pi}{32}\left(6^4-4^4\right)=102.1$ in⁴, so $T=12\left(102.1\right)/3=408$ kip·in.
2. *(b)* Same area 5π in², so d = 4.47 in and $T=12\,\pi\,d^3/16=211$ kip·in.
3. *(c)* Same area with $d_o=8$: $d_i^2=64-20=44$, so $d_i=6.63$ in and $T=12\cdot\frac{\pi\left(8^4-6.63^4\right)}{32}/4=636$ kip·in.

$$(a)\,408\,;(b)\,211\,;(c)\,636\,\text{kip}\cdot\text{in}$$

Exam takeaway. For the same material, moving it outward raises the torque capacity: 211 → 408 → 636.

Q7 (Slides Ch6, Beer) — Stepped shaft AD with a bored section: twist at A (medium)

The horizontal shaft *AD* is attached to a fixed base at *D* and is subjected to the torques shown. A 44-mm-diameter hole has been drilled into portion *CD* of the shaft. Knowing that the entire shaft is made of steel for which $G = 77$ GPa, determine the angle of twist at end *A*.



Original: Slides Ch 6, slide 19

Problem. Shaft AD is fixed at D. AB is solid, 30 mm, 0.4 m long, with 250 N·m at A. BC is solid, 60 mm, 0.2 m, with 2000 N·m at B. CD is 60 mm with a 44 mm bore, 0.6 m long. G = 77 GPa. Find the angle of twist at A.

Solution.

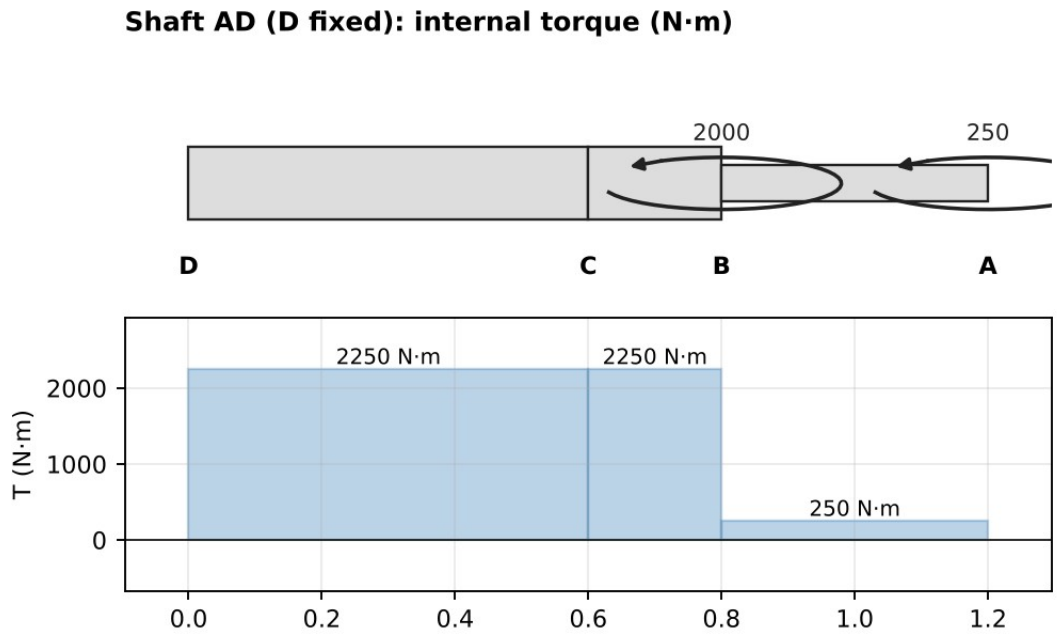
1. *Internal torques:* $T_{AB} = 250$ N·m; $T_{BC} = T_{CD} = 2250$ N·m (same sense).
2. *Twists:*

$$\phi_A = \frac{1}{G}\left[\frac{250\times10^3(400)}{\frac{\pi}{32}30^4} + \frac{2250\times10^3(200)}{\frac{\pi}{32}60^4} + \frac{2250\times10^3(600)}{\frac{\pi}{32}(60^4-44^4)}\right] = 0.0163+0.0046+0.0194$$

$$\phi_A = 0.0403\,\text{rad} = 2.31^\circ$$

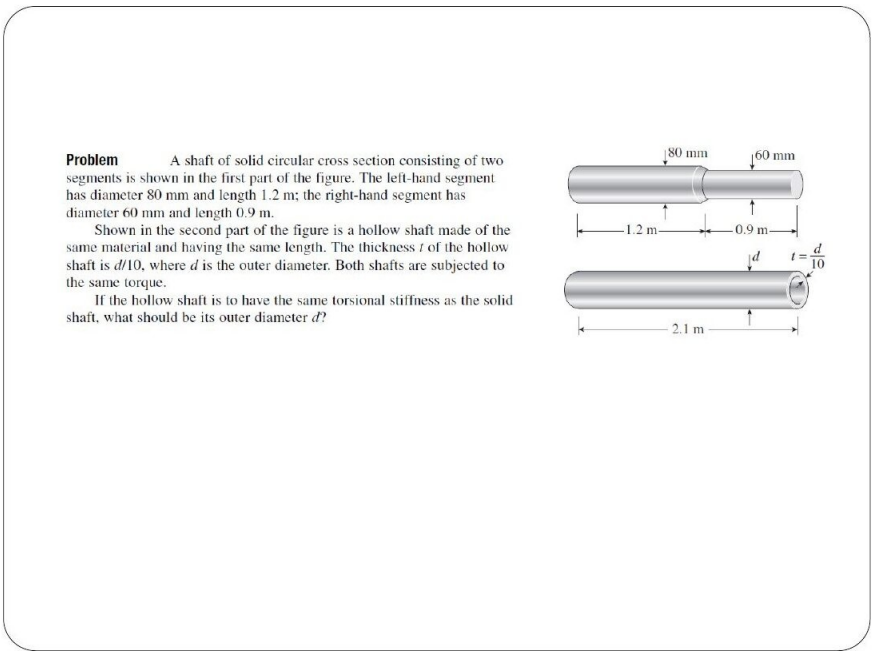
Check. The bored section CD, longest and with the largest torque, contributes most (48%) ✓.

Exam takeaway. Write one term per segment with its own T, L and J.



Internal torque diagram

Q8 (Slides Ch6, Gere) — Two-segment solid shaft vs an equivalent hollow shaft (medium)



Original: Slides Ch 6, slide 45

Problem. A solid shaft has segments of 80 mm × 1.2 m and 60 mm × 0.9 m. A hollow shaft of the same material is 2.1 m long with t = d/10 (so $d_i = 0.8\,d$). For equal torsional stiffness, find the hollow shaft's outer diameter d.

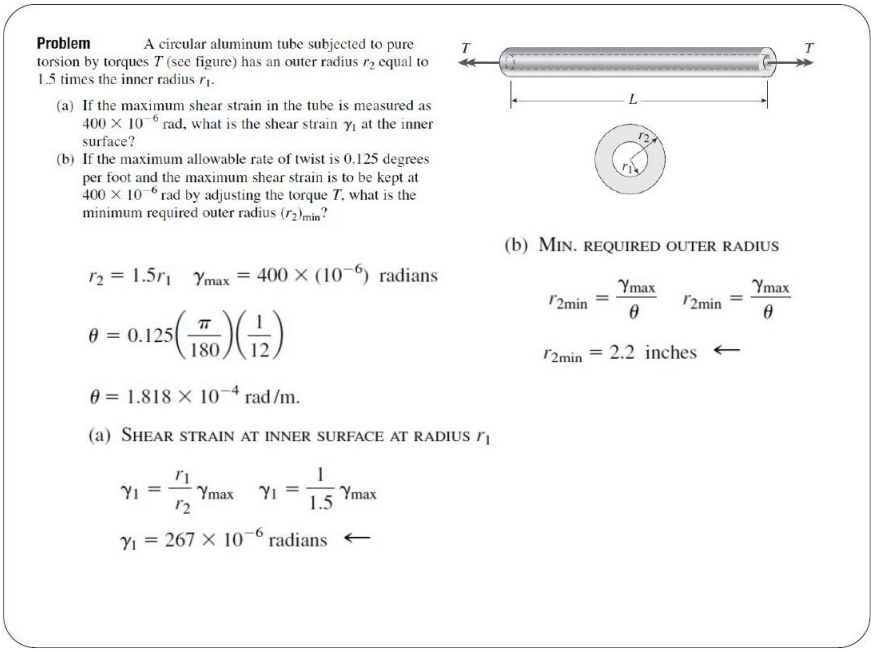
Solution.

1. $$\phi_1 = \frac{32\,T}{\pi\,G}\left[\frac{1.2}{0.08^4} + \frac{0.9}{0.06^4}\right] = \frac{32\,T}{\pi\,G}\left(29\,297 + 69\,444\right) = \frac{32\,T}{\pi\,G}\left(98\,741\ \mathrm{m}^{-3}\right).$$
2. $$\phi_2 = \frac{32\,T}{\pi\,G}\cdot\frac{2.1}{0.5904\,d^4} = \frac{32\,T}{\pi\,G}\cdot\frac{3.5569}{d^4}.$$
3. Equal stiffness means $\phi_1 = \phi_2$, so $d^4 = 3.5569/98\,741 = 36.02\times10^{-6}\ \mathrm{m}^4$.

$$d = 77.5\ \mathrm{mm}$$

Exam takeaway. "Same torsional stiffness" means the same T/φ, i.e. the same φ under the same T.

Q9 (Slides Ch6, Gere) — Aluminium tube: shear strain at the inner surface; minimum outer radius (easy)



Problem A circular aluminum tube subjected to pure torsion by torques T (see figure) has an outer radius r_2 equal to 1.5 times the inner radius r_1.

(a) If the maximum shear strain in the tube is measured as 400×10^{-6} rad, what is the shear strain γ_1 at the inner surface?

(b) If the maximum allowable rate of twist is 0.125 degrees per foot and the maximum shear strain is to be kept at 400×10^{-6} rad by adjusting the torque T, what is the minimum required outer radius $(r_2)_{min}$?

$r_2 = 1.5r_1 \quad \gamma_{max} = 400 \times (10^{-6})$ radians

$$\theta = 0.125\left(\frac{\pi}{180}\right)\left(\frac{1}{12}\right)$$

$\theta = 1.818 \times 10^{-4}$ rad/m.

(a) SHEAR STRAIN AT INNER SURFACE AT RADIUS r_1

$$\gamma_1 = \frac{r_1}{r_2}\gamma_{max} \qquad \gamma_1 = \frac{1}{1.5}\gamma_{max}$$

$\gamma_1 = 267 \times 10^{-6}$ radians ←

(b) MIN. REQUIRED OUTER RADIUS

$$r_{2min} = \frac{\gamma_{max}}{\theta} \qquad r_{2min} = \frac{\gamma_{max}}{\theta}$$

$r_{2min} = 2.2$ inches ←

Original: Slides Ch 6, slide 44

Problem. An aluminium tube in pure torsion has $r_2 = 1.5\,r_1$. (a) If the maximum shear strain is 400×10^{-6} rad, what is the shear strain at the inner surface? (b) If the allowable rate of twist is 0.125°/ft and the maximum shear strain is kept at 400×10^{-6} by adjusting T, find the minimum outer radius.

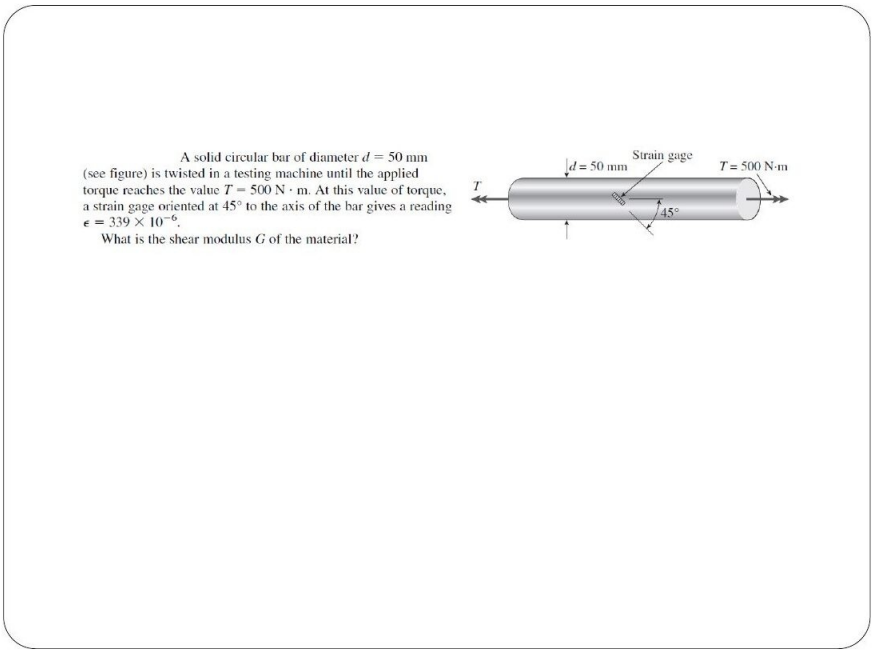
Solution.

1. *(a)* $\gamma \propto r$: $\gamma_1 = \gamma_{max}(r_1/r_2) = 400/1.5 = 267 \times 10^{-6}$ rad.
2. *(b)* $\theta = 0.125(\pi/180)/12 = 1.818 \times 10^{-4}$ rad/in, and $\gamma_{max} = r_2\theta$, so $r_{2,min} = \gamma_{max}/\theta = 400 \times 10^{-6}/1.818 \times 10^{-4}$.

$$(a)\,\gamma_1 = 267 \times 10^{-6}\ \text{rad};\ (b)\,r_{2,min} = 2.2\ \text{in}$$

Exam takeaway. Strain problems in torsion need no G: $\gamma = r\theta$ is pure geometry.

Q10 (Slides Ch6, Gere) — Strain gauge at 45° gives G (easy)



Original: Slides Ch 6, slide 49

Problem. A solid 50 mm bar is twisted to T = 500 N·m; a gauge at 45° to the axis reads $\varepsilon = 339 \times 10^{-6}$. Find G.

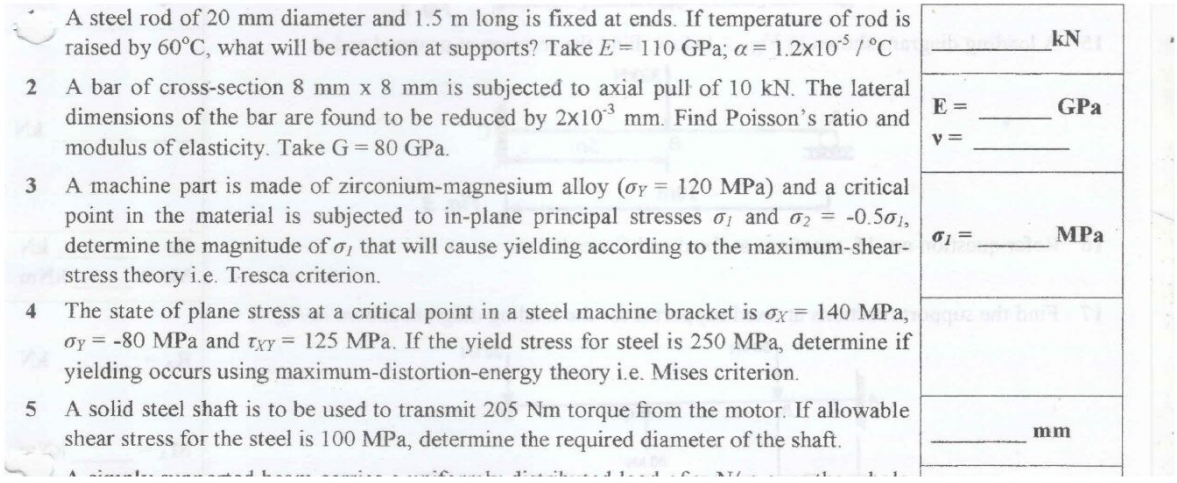
Solution.

1. $\tau_{max} = 16(500)/\pi(0.05)^3 = 20.37$ MPa.
2. Pure shear: $\gamma_{max} = 2\epsilon_{45} = 678 \times 10^{-6}$.
3. $G = \tau/\gamma$.

$$G = 30.0\,\text{GPa}$$

Exam takeaway. PYQ 2021-22 A-Q1 is the same bar with $\varepsilon = 350\mu$, giving G = 29.1 GPa (Ch. 4 Q30).

Q11 (PYQ 2014-15 Compre A Q5) — Size a shaft for 205 N·m (easy)



Original question: PYQ 2014-15 Compre A Q5

Problem. A solid steel shaft transmits 205 N·m; τallow = 100 MPa. Find the required diameter.

Solution. $d = \left(\dfrac{16(205\,000)}{\pi(100)}\right)^{1/3}$.

$$d = 21.9\ \text{mm}$$

Q12 (PYQ 2014-15 Test-2 Q4) — Axle of tubes plus a solid section (easy–medium, 20 marks)

Q.4 The A-36 steel axle is made from tubes *AB* and *CD* and a solid section *BC*. It is supported on smooth bearings that allow it to rotate freely. The tubes have an outer diameter of 30 mm and an inner diameter of 20 mm. The solid section has a diameter of 40 mm. G = 75 GPa. **[20]**

If the gears, fixed to the axle ends, are subjected to 85 Nm torques, determine

a. maximum and minimum shear stress in the tubes *AB* and *CD*

b. the angle of twist of the end *B* of the solid section relative to end *C*.

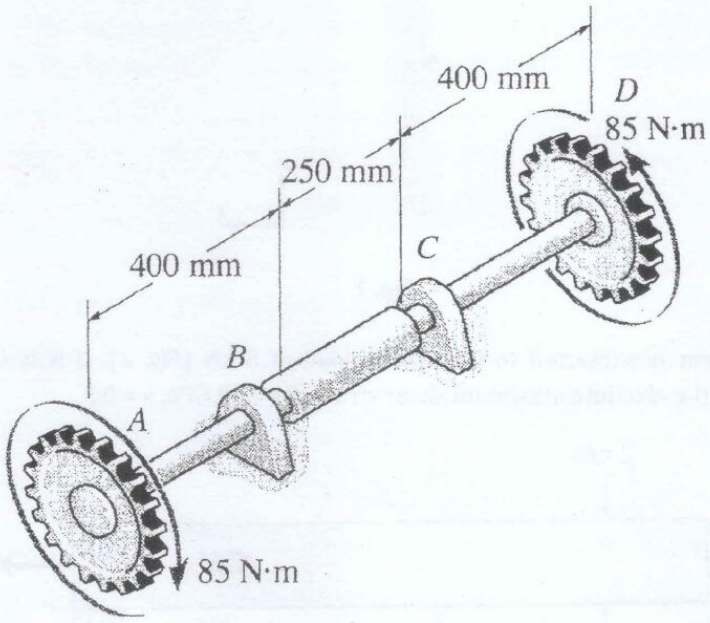


Fig. 4

Original question: PYQ 2014-15 Test-2 Q4

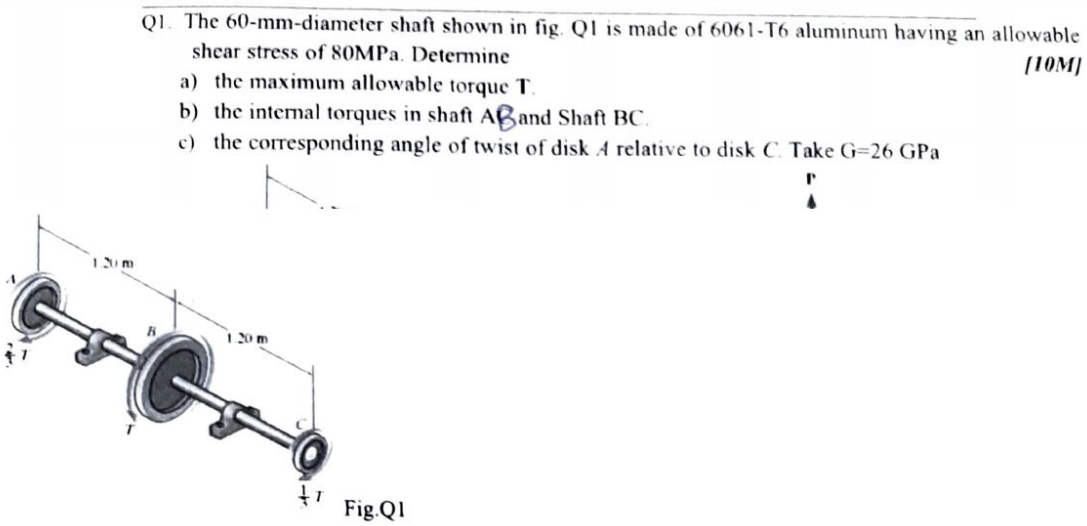
Problem. An A-36 steel axle has tubes AB and CD (outer 30 mm, inner 20 mm) and a solid section BC (40 mm, 250 mm long), on smooth bearings. G = 75 GPa. 85 N·m torques act on the end gears. Find (a) the maximum and minimum shear stress in the tubes and (b) the twist of B relative to C.

Solution.

1. Smooth bearings, so the internal torque is 85 N·m everywhere.
2. $J_{tube} = \frac{\pi}{32}\left(30^4 - 20^4\right) = 63\,814\ \text{mm}^4$; $J_{BC} = \frac{\pi}{32}(40)^4 = 251\,327\ \text{mm}^4$.
3. *(a)* $\tau_{max} = \frac{85\,000(15)}{63\,814} = 20.0\ \text{MPa}$ and $\tau_{min} = 20.0(10/15) = 13.3\ \text{MPa}$.
4. *(b)* $\phi_{B/C} = \frac{85\,000(250)}{75\,000(251\,327)} = 1.13 \times 10^{-3}\ \text{rad} = 0.0646°$.

$$(a)\ \tau_{max} = 20.0\ \text{MPa},\ \tau_{min} = 13.3\ \text{MPa};\ (b)\ \phi_{B/C} = 0.0646^\circ$$

Q13 (PYQ 2023-24 Compre A Q1) — Shaft with three disks: allowable T and twist A relative to C (medium, 10 marks)



Original question: PYQ 2023-24 Compre A Q1

Problem. A 60 mm solid 6061-T6 aluminium shaft (τ_{allow} = 80 MPa, G = 26 GPa) carries disks A (⅔T), B (T, opposite sense) and C (⅓T); AB = BC = 1.20 m. Find (a) the maximum allowable T, (b) the internal torques in AB and BC and (c) the twist of disk A relative to disk C.

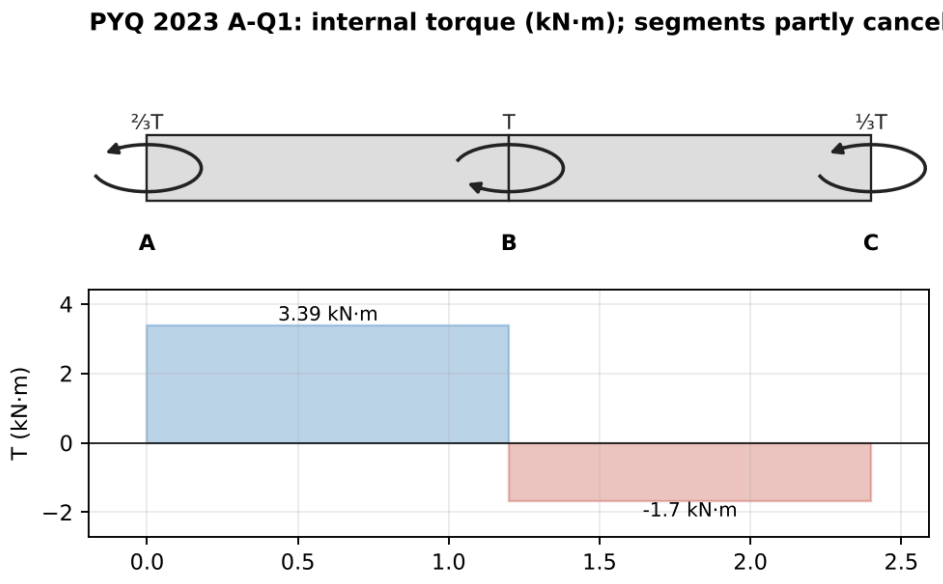
Solution.

1. *Internal torques:* $T_{AB} = 2/3\,T$ and $T_{BC} = 2/3\,T - T = -1/3\,T$. The maximum is in AB.
2. *(a)* $2/3\,T = \tau\,\pi\,d^3/16 = 80\,\pi\,(60)^3/16 = 3.393$ kN·m, so T = 5.09 kN·m.
3. *(b)* $T_{AB} = 3.39$ kN·m; $T_{BC} = 1.70$ kN·m in the opposite sense.
4. *(c)* $J = \pi\,(60)^4/32 = 1.272 \times 10^6$ mm⁴ and $GJ = 3.308 \times 10^{10}$ N·mm²:

$$\phi_{A/C} = \frac{(T_{AB} + T_{BC})\,L}{GJ} = \frac{(3.393 - 1.696) \times 10^6\,(1200)}{3.308 \times 10^{10}} = 0.0615\ \text{rad}$$

$$(a)\ T = 5.09\ \text{kN·m};\ (b)\ 3.39 \text{ and } 1.70\ \text{kN·m (opposite)};\ (c)\ \phi_{A/C} = 3.53^\circ$$

Exam takeaway. Opposite-signed segments partly cancel; students who add magnitudes get 10.6°.



Internal torque diagram

Concept 6.3 — Power transmission

A. Concept explanation

1. Intuition. A shaft spinning fast can carry a lot of power with a small torque, which is why high-speed shafts are thin. Power = torque × angular speed.

2. Why it matters. Drive shafts, motors and generators are specified by power and speed; the designer has to convert to torque first.

3. Formal treatment.

$$P = T\,\omega = \frac{2\pi n T}{60}\,(P \text{ in W}, T \text{ in N·m}, \omega \text{ in rad/s}, n \text{ in rpm}),\quad T = \frac{60\,P}{2\pi n}$$

With frequency f (Hz): $P = 2\pi f T$.

4. Derivation. Work done by a torque turning through dφ is T dφ; power is its rate, $T\,d\phi/dt = T\,\omega$; and one revolution per minute is 2π/60 rad/s.

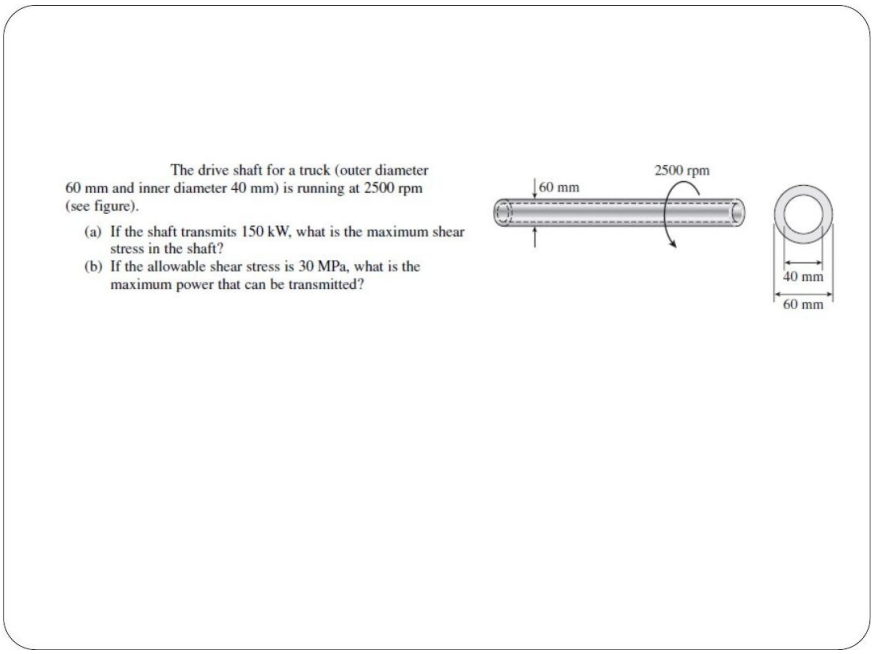
5–6. Picture / special cases. At constant power, T ∝ 1/n: halving the speed doubles the torque and needs a shaft of $2^{1/3} = 1.26$ times the diameter.

7. Traps. Using rpm directly in P = Tω; kW vs W.

8. Summary. $T = 60\,P/(2\pi n)$, then the torsion formula.

B. Solved question

Q14 (Slides Ch6, Gere) — Truck drive shaft (easy)



Original: Slides Ch 6, slide 42

Problem. A truck drive shaft (outer diameter 60 mm, inner 40 mm) runs at 2500 rpm. (a) If it transmits 150 kW, what is the maximum shear stress? (b) If τallow = 30 MPa, what is the maximum power?

Solution.

1. $J = \frac{\pi}{32}\left(60^4 - 40^4\right) = 1.021 \times 10^{-6}$ m⁴.

2. *(a)* $T=\frac{60(150\,000)}{2\pi(2500)}=573$ N·m, so $\tau_{max}=\frac{T\,d_2}{2J}=\frac{573(0.060)}{2(1.021\times10^{-6})}=16.8$ MPa.

3. *(b)* τ ∝ P, so $P_{max}=150(30/16.835)=267$ kW.

$$(a)\,\tau_{max}=16.8\,\text{MPa};(b)\,P_{max}=267\,\text{kW}$$

Exam takeaway. P = Tω needs ω in rad/s, **not** rpm.

Concept 6.4 — Gears and coupled shafts

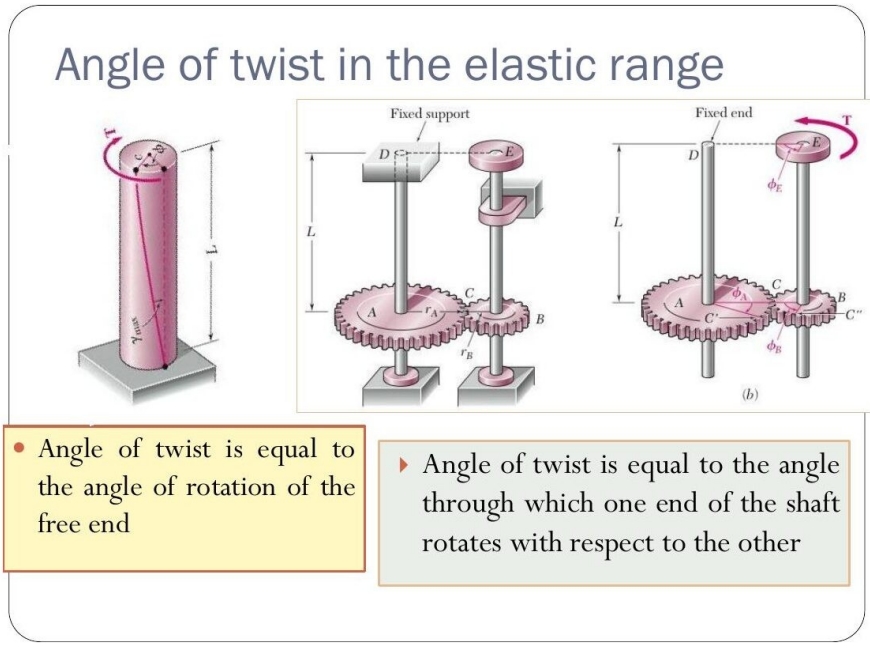
A. Concept explanation

1. Intuition. Two meshing gears share one tooth force F. The bigger gear gets the bigger torque (T = Fr) but turns through a smaller angle. The contact point moves the same arc length on both: $r_B\phi_B=r_C\phi_C$.

2. Why it matters. Gearboxes, coupled shafts, PYQs 2014 Compre B Q2 and 2019 B Q4.

3. Formal treatment (slides, Beer Sample 3.4).

- **Statics** (slide): $F=T_B/r_B=T_C/r_C$, so $T_C=T_B(r_C/r_B)$.
- **Kinematics** (slide: "peripheral motions of the gears are equal"): $r_B\phi_B=r_C\phi_C$.
- **Twist of the free end:** $\phi_A=\phi_B+\phi_{A/B}$, where φB comes from the *other* shaft's twist through the gear ratio.



Original: Slides Ch 6, slide 30

4. Derivation. Tooth contact: equal and opposite tangential forces F on the two gears (Newton's third law). The arc length rolled out at the pitch circles must be the same, otherwise the teeth would separate or interpenetrate.

5. Picture. Slide: two gears with F acting down on A and up on B at the contact point; arcs $r_A\phi_A=r_B\phi_B$.

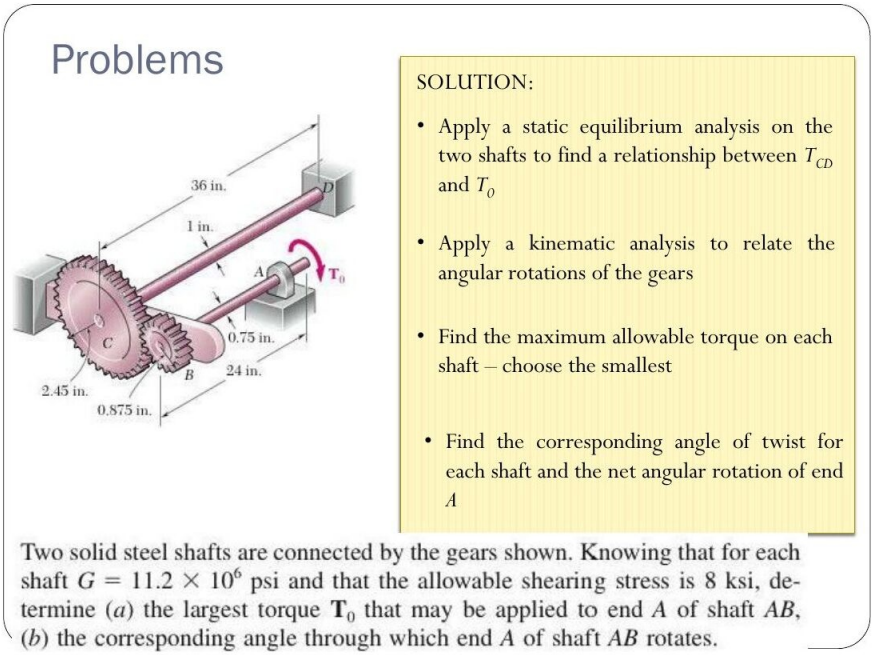
6. Special cases. Gear ratio n = r_large/r_small: torque multiplies by n, rotation divides by n. The stiffness of the far shaft, seen through the gears, scales by n^2.

7. Traps. Using the ratio the wrong way (torque scales with r, rotation with 1/r). Forgetting that the free shaft's end rotation includes the gear's own rotation, not just its twist.

8. Summary. $T_C/T_B=r_C/r_B$; $\phi_B/\phi_C=r_C/r_B$; total rotation = gear rotation + own twist.

B. Solved questions

Q15 (Slides Ch6, Beer Sample 3.4) — Two shafts connected by gears: largest T_0 and rotation of A (medium)



Original: Slides Ch 6, slide 32

Problem. Shaft AB (0.75 in diameter, 24 in long, gear B with $r_B = 0.875$ in) meshes with shaft CD (1 in diameter, 36 in long, fixed at D, gear C with $r_C = 2.45$ in). G = 11.2 × 10⁶ psi and τallow = 8 ksi. Find (a) the largest torque T₀ that may be applied at A and (b) the corresponding rotation of A.

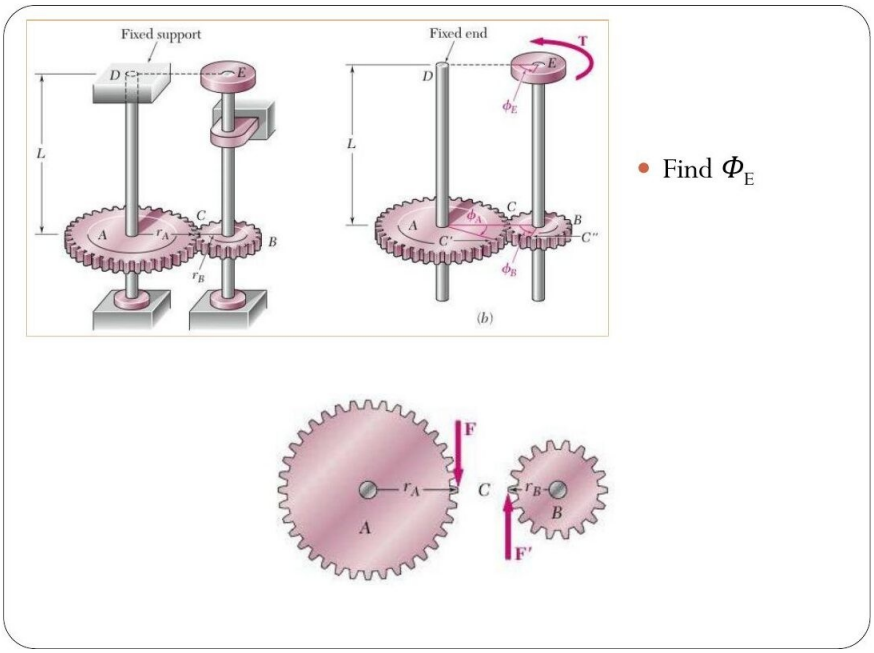
Solution.

1. *Statics:* $F(0.875) = T_0$ and $F(2.45) = T_{CD}$, so $T_{CD} = 2.8\,T_0$ (1).
2. *Kinematics:* $r_B\phi_B = r_C\phi_C$, so $\phi_B = 2.8\,\phi_C$ (2).
3. *(a)* AB: $T_0 = 8000 \cdot \frac{\pi}{2}(0.375)^3 = 663$ lb·in. CD: $2.8\,T_0 = 8000 \cdot \frac{\pi}{2}(0.5)^3$, so $T_0 = 561$ lb·in. The smaller governs: T₀ = 561 lb·in.
4. *(b)* $\phi_{A/B} = \dfrac{561(24)}{\frac{\pi}{2}(0.375)^4(11.2\times10^6)} = 0.0387$ rad = 2.22°. $\phi_C = \dfrac{2.8(561)(36)}{\frac{\pi}{2}(0.5)^4(11.2\times10^6)} = 0.0514$ rad = 2.95°. So $\phi_B = 2.8(2.95^\circ) = 8.26^\circ$.

$$T_0 = 561\ \text{lb}\cdot\text{in},\ \phi_A = 8.26^\circ + 2.22^\circ = 10.48^\circ$$

Exam takeaway. The slide's four steps (statics, kinematics, allowable torque in each shaft, twist) are the marking scheme.

Q16 (Slides Ch6) — Find φE for gear-coupled shafts (symbolic) (medium)



Original: Slides Ch 6, slide 31

Problem. Shaft AD (fixed at D) carries gear A of radius r_A. Shaft BE carries gear B (r_B) and torque T at E. Both have length L and rigidity GJ. Find φE.

Solution.

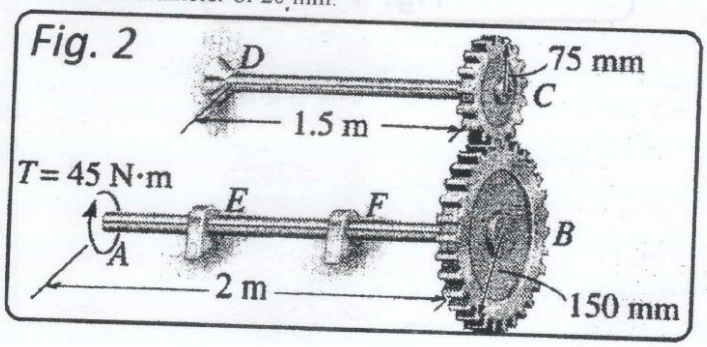
1. *Statics:* tooth force $F = T/r_B$, so $T_{AD} = F\,r_A = T\,r_A/r_B$.
2. *Twist of AD:* $\phi_A = \dfrac{T\,r_A\,L}{r_B\,G\,J}$.
3. *Gear B:* $\phi_B = \dfrac{r_A}{r_B}\phi_A = \dfrac{r_A^2}{r_B^2}\dfrac{T\,L}{G\,J}$.
4. *E:* $\phi_E = \phi_B + \dfrac{T\,L}{G\,J}$.

$$\phi_E = \frac{T\,L}{G\,J}\left(1 + \frac{r_A^2}{r_B^2}\right)$$

Exam takeaway. The far shaft appears $\left(r_A/r_B\right)^2$ times as flexible: torque and rotation each bring one factor of the gear ratio.

Q17 (PYQ 2014-15 Compre B Q2) — Coupled shafts: twist of end A relative to D (medium, 15 marks)

Q2. The two solid shafts shown in *Fig. 2* are coupled together using the meshed gears. Determine the angle of twist of end *A* (wrt end *D*) of shaft *AB* when the torque *T* = 45 Nm is applied. Take *G* = 80 GPa. Shaft *AB* is free to rotate with in the bearings *E* and *F*, whereas shaft *DC* is fixed at *D*. Each shaft has a diameter of 20 mm. **[15]**



Original question: PYQ 2014-15 Compre B Q2

Problem. Solid shafts, each 20 mm diameter, are coupled by meshed gears. T = 45 N·m acts at A. AB (2 m) turns freely in bearings E and F, with gear B ($r_B = 150$ mm). DC (1.5 m) is fixed at D, with gear C ($r_C = 75$ mm). G = 80 GPa. Find the twist of A relative to D.

Solution.

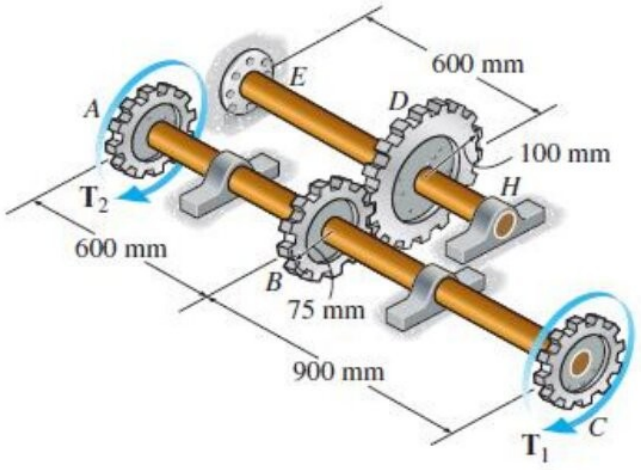
1. $J = \pi(20)^4/32 = 15\,708$ mm⁴, so $GJ = 1.2566 \times 10^9$ N·mm².
2. *Tooth force:* $F = 45\,000/150 = 300$ N, so $T_{DC} = 300(75) = 22.5$ N·m.
3. $\phi_C = \dfrac{22\,500(1500)}{1.2566 \times 10^9} = 0.02686$ rad.
4. *Kinematics:* $\phi_B = \phi_C(r_C/r_B) = 0.01343$ rad.
5. $\phi_{A/B} = \dfrac{45\,000(2000)}{1.2566 \times 10^9} = 0.07162$ rad.

$$\phi_{A/D} = 0.01343 + 0.07162 = 0.0850\,\text{rad} = 4.87^\circ$$

Exam takeaway. Here the larger gear is on the driven shaft, so DC sees *half* the torque but turns gear B only half as far.

Q18 (PYQ 2019-20 Compre B Q4) — Gear-coupled shafts: find T_1 and T_2 for 0.05 rad at A and C (hard, 12 marks)

Q4. The 60-mm diameter shaft *ABC* is supported by two journal bearings, while the 80-mm diameter shaft *EH* is fixed at *E* and supported by a journal bearing at *H* (FigQ4). If the angle of twist at gears A and C is required to be 0.05 rad, determine the magnitudes of the torques T_1 and T_2. The shafts are made of A-36steel (G = 80 GPa). **[12]**



Original question: PYQ 2019-20 Compre B Q4

Problem. Shaft ABC (60 mm, on journal bearings) carries gear B ($r_B = 75$ mm), with AB = 600 mm and BC = 900 mm; torques T_2 at A and T_1 at C act in the same sense. Shaft EH (80 mm) is fixed at E and carries gear D ($r_D = 100$ mm) at 600 mm from E. G = 80 GPa. Find T_1 and T_2 so that the twists at gears A and C are both 0.05 rad.

Solution.

1. $GJ_1 = 80\,000 \cdot \frac{\pi}{32}(60)^4 = 1.018 \times 10^{11}$ and $GJ_2 = 80\,000 \cdot \frac{\pi}{32}(80)^4 = 3.217 \times 10^{11}$ N·mm². Gear ratio $n = r_D/r_B = 4/3$.
2. *Statics:* shaft ABC is in equilibrium, so gear B carries $T_B = T_1 + T_2$, and gear D gets $T_D = n(T_1 + T_2)$.
3. *Shaft EH:* $\phi_D = \dfrac{n(T_1+T_2)(600)}{GJ_2}$. *Gears:* $\phi_B = n\phi_D$.
4. *Shaft ABC:* $\phi_A = \phi_B + \dfrac{T_2(600)}{GJ_1}$ and $\phi_C = \phi_B + \dfrac{T_1(900)}{GJ_1}$.
5. $\phi_A = \phi_C$ gives $600\,T_2 = 900\,T_1$, so $T_2 = 1.5\,T_1$.
6. Then $\dfrac{(16/9)(2.5\,T_1)(600)}{3.217 \times 10^{11}} + \dfrac{1.5\,T_1(600)}{1.018 \times 10^{11}} = 0.05$, i.e. $T_1(8.29 + 8.84) \times 10^{-9} = 0.05$.

$$T_1 = 2.92\ \text{kN·m}, \quad T_2 = 4.38\ \text{kN·m}$$

Check. $\phi_B = 0.0242$ rad; $\phi_A = 0.0242 + 4.38 \times 10^6(600)/1.018 \times 10^{11} = 0.0500$ ✓.

Exam takeaway. Write φA and φC each as "gear rotation + own twist", then the equal-twist condition gives T_2/T_1 at once.

Concept 6.5 — Statically indeterminate shafts and torsional strain energy

A. Concept explanation

1. Intuition. A shaft clamped at both ends with a torque in the middle: both walls resist, but statics gives only $T_A + T_D = T$. The extra fact: **the total twist between the walls is zero**. It is the exact twin of the axially loaded bar fixed at both ends (Ch. 2).

2. Why it matters. Fixed–fixed shafts, composite (tube + core) shafts and gear trains with locked ends. PYQ 2024 B-Q1 (12 marks) is a fixed–fixed shaft with strain energy.

3. Formal treatment (the three steps of Ch. 2):

1. Equilibrium: $T_A + T_D = \sum T_{applied}$.
2. Compatibility: $\phi_{A/D} = \sum \frac{T_i L_i}{G J_i} = 0$.
3. Torque–twist: φ = TL/GJ.

Strain energy (slides): $U = \int \frac{M_t^2}{2 G I_z} dz = \sum \frac{T_i^2 L_i}{2 G_i J_i}$.

4. Derivations. *(a) Single torque T_0 at distance a from A (b from D), uniform GJ.* Release D: φD = T0a/GJ. The redundant T_D alone gives $-T_D L/GJ$. Compatibility gives

$$T_D = T_0 \frac{a}{L}, T_A = T_0 \frac{b}{L}$$

As for axial bars, the nearer wall carries more. *(b) Composite shaft (core + tube, bonded).* Both twist equally, so $T_i = T \frac{(GJ)_i}{\sum (GJ)}$.

5. Picture. The internal-torque diagram changes sign at a loading point; the twist diagram is piecewise linear and peaks there (Q20).

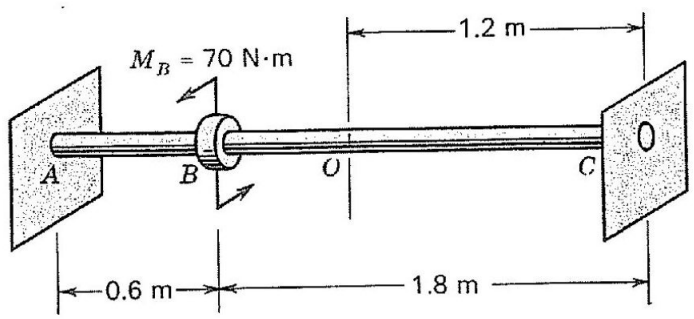
6. Special cases. Several torques: superpose the single-torque results.

7. Traps. Signs of internal torques; forgetting that the twists of all segments must add to zero, not just two.

8. Summary. $T_A = T_0 b/L$; $\sum T L/GJ = 0$; $U = \sum T^2 L/2GJ$.

B. Solved questions

Q19 (Slides Ch6, Crandall example) — Fixed–fixed shaft: rotation of point O (medium)



- Dia = 25 mm; aluminum alloy shaft.
- Find the angle through which O rotates.
- G = 26.5 GN/m^2
- Ends A and C are built in

Original: Slides Ch 6, slide 21

Problem. A 25 mm aluminium-alloy shaft (G = 26.5 GPa) has both ends A and C built in. A 70 N·m torque acts at B, 0.6 m from A; BC = 1.8 m. Find the rotation of point O, 1.2 m from C.

Solution.

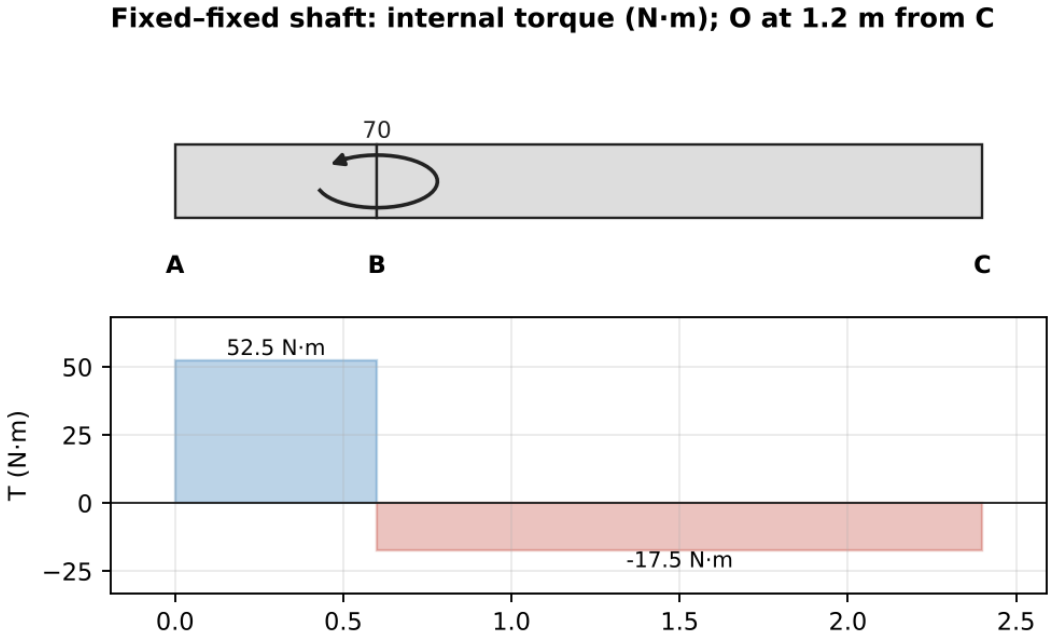
1. *Reactions:* $T_A = 70(1.8/2.4) = 52.5$ N·m and $T_C = 70(0.6/2.4) = 17.5$ N·m.
2. Segment OC carries 17.5 N·m, so

$$\phi_O = \frac{17\,500(1200)}{26\,500 \cdot \frac{\pi}{32}(25)^4} = 0.02066 \text{ rad}$$

$$\phi_O = 1.18^\circ$$

Check (from the A side). $\phi_B = 52\,500(600)/GJ = 0.0310$ rad, then O (0.6 m beyond B) comes back by $17\,500(600)/GJ = 0.0103$, giving 0.0207 ✓.

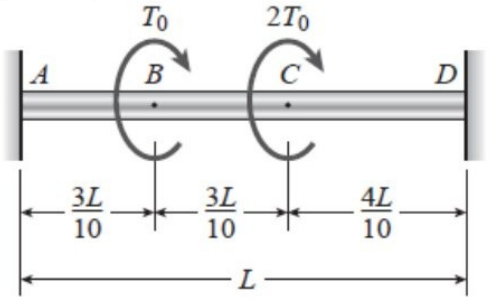
Exam takeaway. Rotations measured from either fixed end must agree; that is the check.



Internal torque diagram

Q20 (PYQ 2024-25 Compre B Q1) — Fixed–fixed bar with two torques: reactions, maximum twist, strain energy (medium, 12 marks)

Q1. A solid circular bar *ABCD* of diameter 40 mm with fixed supports at *A* and *D* is acted upon by torques T_o and $2T_o$ at point *B* and *C* as shown in **FigQ1**.
Take T_o = 800 N-m, L = 1.2 m and G = 80 GPa **[12M]**
Determine
(a) the reactions at end *A* and *D*
(b) the maximum angle of twist of the bar (in **degree**)
(c) the total strain energy (in **Joule**)



FigQ1

Original question: PYQ 2024-25 Compre B Q1

Problem. A solid 40 mm bar ABCD is fixed at A and D. Torques T_0 = 800 N·m at B (x = 0.3L) and $2T_0$ at C (x = 0.6L) act in the same sense. L = 1.2 m, G = 80 GPa. Find (a) the reactions at A and D, (b) the maximum angle of twist (degrees) and (c) the total strain energy (J).

Solution.

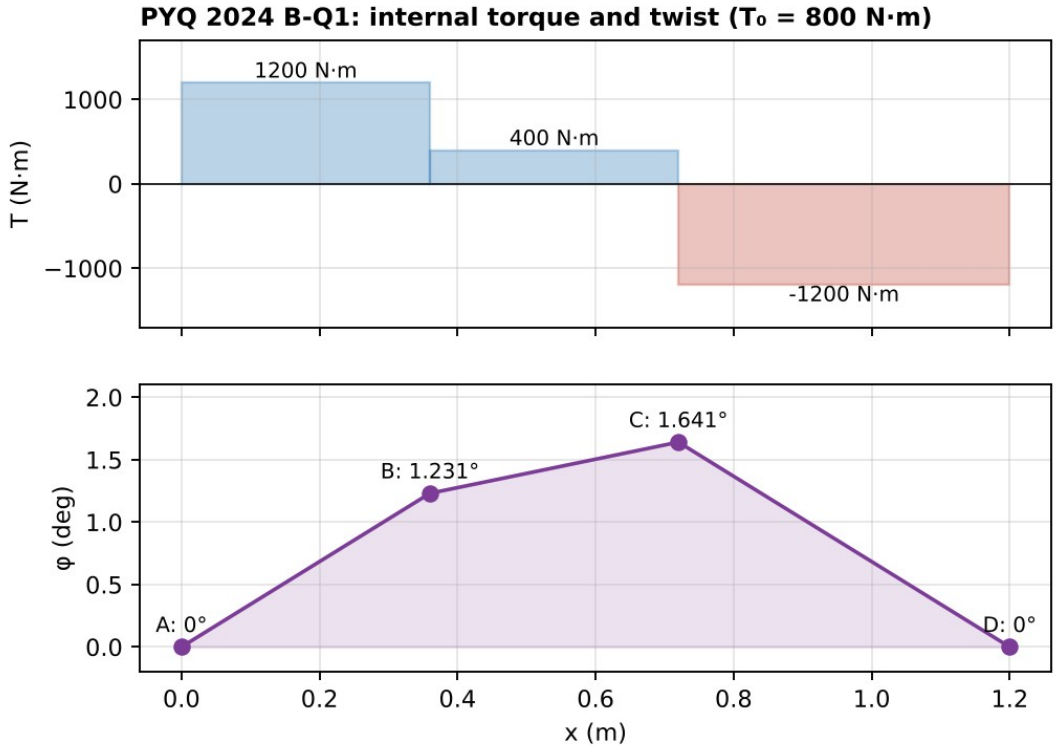
1. $J = 251\,327$ mm⁴ and $GJ = 2.0106 \times 10^{10}$ N·mm².
2. *(a) Superposition of the single-torque result:* $T_A = T_0(0.7) + 2T_0(0.4) = 1.5T_0 = 1200$ N·m and $T_D = 3T_0 - 1.5T_0 = 1200$ N·m, both opposite to the applied torques.
3. *Internal torques:* $T_{AB} = 1200$, $T_{BC} = 1200 - 800 = 400$, $T_{CD} = 400 - 1600 = -1200$ N·m. Compatibility check: $1200(360) + 400(360) - 1200(480) = 0$ ✓.

4. *(b) Twists:* $\phi_B = \dfrac{1200 \times 10^3 (360)}{2.0106 \times 10^{10}} = 0.02149$ rad = 1.23°; $\phi_C = \phi_B + \dfrac{400 \times 10^3 (360)}{2.0106 \times 10^{10}} = 0.02865$ rad = 1.64°.

5. *(c) Energy:* $U = \dfrac{1}{2GJ}\left[1200^2(0.36) + 400^2(0.36) + 1200^2(0.48)\right] \times 10^9$ N·mm.

$$(a)\, T_A = T_D = 1200\,\text{N·m};\ (b)\, \phi_{max} = 1.64^\circ \text{ at C};\ (c)\, U = 31.5\,\text{J}$$

Exam takeaway. The maximum twist occurs at a loading point where the internal torque changes sign, just as a beam's maximum deflection occurs where its slope is zero.



Internal torque and angle of twist along the bar

Concept 6.6 — Combined stresses: torsion + axial + bending; yielding of shafts

A. Concept explanation

1. Intuition. Real shafts are rarely just twisted. A propeller shaft is twisted and pushed; a crank is twisted and bent. At a surface point you get a normal stress σ (from the axial force and bending) **plus** a shear stress τ (from the torque and the transverse shear). Put them on one small square element, then use Mohr's circle (Ch. 4) and a yield criterion (Ch. 5).

2. Why it matters. This is **the most-repeated long question** in the compre papers (2019 B-Q2, 2021 A-Q2, 2023 B-Q2, 2024 B-Q2, 2021 B-Q4, 2014 Test-2 Q1), usually 15–20 marks.

3. Formal treatment (slides; Crandall §6.5, Fig. 6.14).

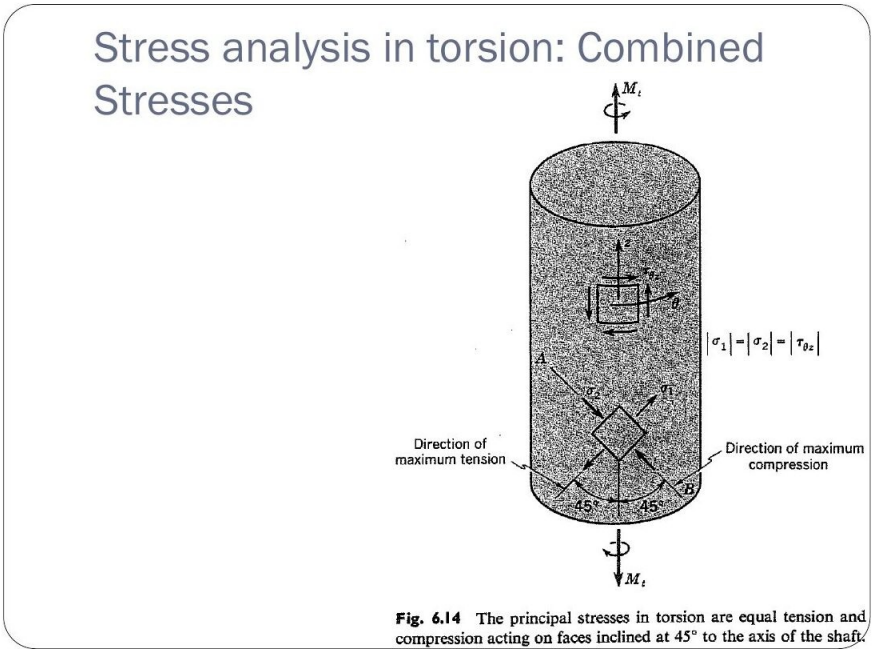
- *Pure torsion:* the surface element is in **pure shear** τ. The principal stresses are ±τ, acting at **45°** to the axis (slide: "the principal stresses in torsion are equal tension and compression acting on faces inclined at 45° to the axis of the shaft"). Brittle shafts (chalk!) break on a 45° helix (maximum tension); ductile shafts shear off on a cross-section (maximum shear).
- *Torsion + axial P* (slide: "combined state of stress"): $\sigma = P/A$ and $\tau = Tc/J$, so

$$\sigma_{1,2}=\frac{\sigma}{2}\pm\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2},\tau_{max}=\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}$$

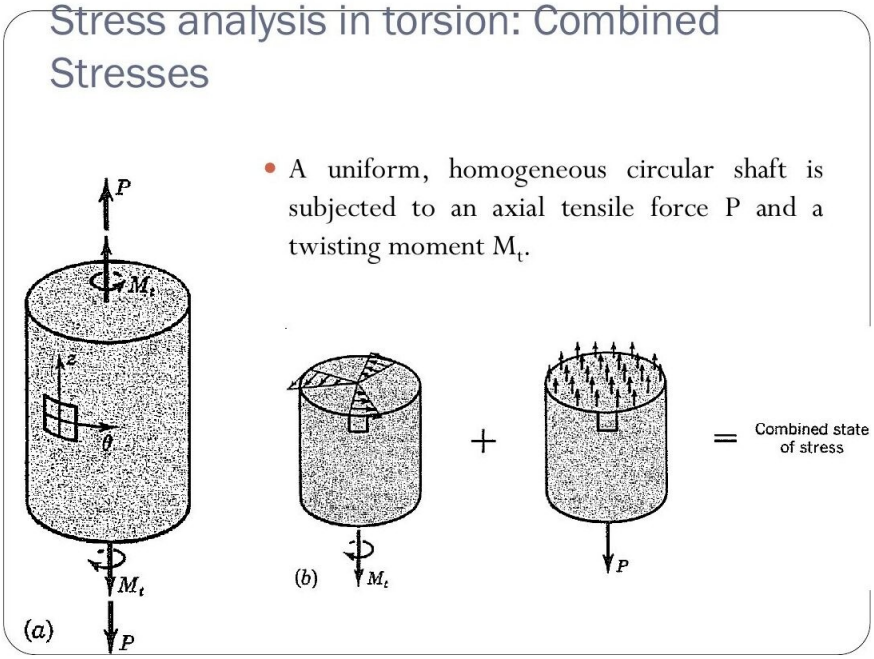
- *Yield (slides):* von Mises $\sqrt{\sigma^2+3\tau^2}=Y$; Tresca $\sqrt{\sigma^2+4\tau^2}=Y$.
- *Torsion + bending (solid shaft):* $\sigma_b=\frac{32M}{\pi d^3}$ and $\tau=\frac{16T}{\pi d^3}$, so

$$\tau_{max}=\frac{16}{\pi d^3}\sqrt{M^2+T^2},\sigma_1=\frac{16}{\pi d^3}\left(M+\sqrt{M^2+T^2}\right)$$

These are the "equivalent torque" and "equivalent moment" formulas.



Original: Slides Ch 6, slide 22



Original: Slides Ch 6, slide 23

4. General procedure for combined-loading PYQs:

1. FBD of the whole member, then reactions (Ch. 1).
2. Cut at the section and find F, V, M_b and T (Ch. 3, 3-D vectors $M = r \times F$).
3. At the requested surface point: $\sigma = F/A \pm M_b c/I$ (bending stress is zero at the neutral axis; I = πd⁴/64 = J/2), and $\tau = T c/J \pm$ transverse shear (for a solid circle $\tau_V = 4V/3A$ at the neutral axis, 0 at the extreme fibre).
4. Draw the 2-D element with the correct directions.
5. Mohr's circle → principal stresses and τmax (Ch. 4).
6. Yield criterion → factor of safety (Ch. 5).

5. Picture. Around the cross-section: bending σ is largest at the top and bottom fibres and zero at the sides (neutral axis); torsional τ is the same all round; transverse-shear τ is largest at the sides and zero at the top and bottom. The critical point is usually where bending σ peaks.

6. Special cases. For a σ–τ state (one normal stress, one shear), the principal stresses always have opposite signs, so in-plane τmax = R is also the absolute maximum.

7. Traps.

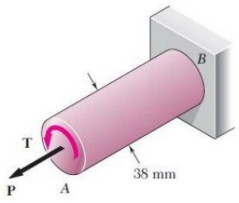
- Putting the maximum bending stress and the maximum transverse shear at the same point. They occur 90° apart.
- Using J where I is needed: for a circle I = J/2.
- Forgetting that transverse shear and torsion shear add on one side and subtract on the other.

8. Summary. σ from F and M, τ from T and V, at the same point; then Mohr; then Tresca/von Mises.

B. Solved questions

Q21 (Slides Ch6, Beer) — Torque for yield by Tresca with an axial load of 240 kN (medium)

The 38 mm diameter shaft *AB* is made of a grade of steel for which the yield strength is *Y* = 250 MPa. Using the maximum shearing stress criterion (MSS), determine the magnitude of torque *T* for which yield occurs when *P* = 240 kN



Original: Slides Ch 6, slide 37

Problem. A 38 mm shaft AB of steel (Y = 250 MPa) carries P = 240 kN. Using the maximum-shear-stress criterion, find the torque T at which yield occurs.

Solution.

1. $A = \frac{\pi}{4}(38)^2 = 1134.1$ mm², so σ = 240 000/1134.1 = 211.6 MPa.
2. Tresca: $2R = \sqrt{\sigma^2 + 4\tau^2} = Y$, so $\tau = 1/2\sqrt{250^2 - 211.6^2} = 66.6$ MPa.

3. $T = \tau J/c = \tau \pi d^3/16 = 66.6\,\pi\,(38)^3/16.$

$$T = 717\ \text{N·m}$$

Exam takeaway. A large axial stress leaves little room for torsion: here σ alone uses 85% of Y.

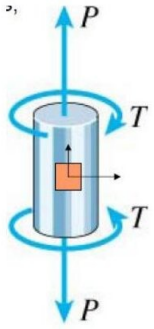
Q22 (Slides Ch6, Hibbeler) — Helicopter rotor shaft (easy)

The rotor shaft of an helicopter drives the rotor blades that provide the lifting force to support the helicopter in the air. As a consequence, the shaft is subjected to a combination of torsion and axial loading.

For a ***50mm*** diameter shaft transmitting a torque T **= 2.4kN.m** and a tensile force **P = 125kN**, determine the maximum tensile stress, maximum compressive stress, and maximum shear stress in the shaft.

The tensile stress $\sigma = \dfrac{P}{A} = \dfrac{125kN}{\pi/4\,(0.05m)^2} = 63.66MPa$

$$\tau_{Torsion} = \frac{Tr}{I_P} = \frac{(2.4kN.m)\left(\frac{0.05}{2}\right)}{\frac{\pi(0.05)^4}{32}} = 97.78MPa$$



Original: Slides Ch 6, slide 40

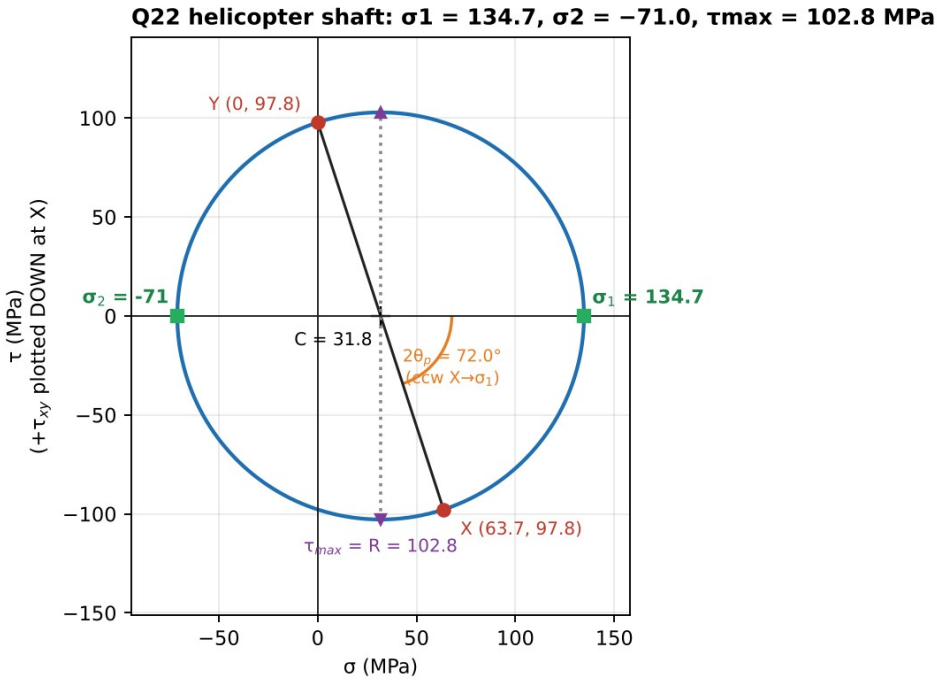
Problem. A 50 mm rotor shaft transmits T = 2.4 kN·m with a tensile force P = 125 kN. Find the maximum tensile stress, maximum compressive stress and maximum shear stress.

Solution.

1. $\sigma = \dfrac{125\,000}{\frac{\pi}{4}(50)^2} = 63.66$ MPa and $\tau = \dfrac{16\left(2.4\times10^6\right)}{\pi(50)^3} = 97.78$ MPa.
2. $R = \sqrt{31.83^2 + 97.78^2} = 102.8$ MPa.

$$\sigma_1 = 134.7\ \text{MPa},\ \sigma_2 = -71.0\ \text{MPa},\ \tau_{max} = 102.8\ \text{MPa}$$

Check. Matches the slide (135 and −71 MPa) ✓.



Mohr's circle

Q23 (PYQ 2014-15 Test-2 Q1) — Axial force + torque: principal stresses

Solved in Ch. 4 Q15: $\sigma_1 = 76.8$ MPa and $\sigma_2 = -5.2$ MPa, with $\theta_p = 14.5°$ from the axis.

Q24 (PYQ 2021-22 Compre A Q1) — Strain gauge at 45°: find G

Solved in Ch. 4 Q30: G = 29.1 GPa.

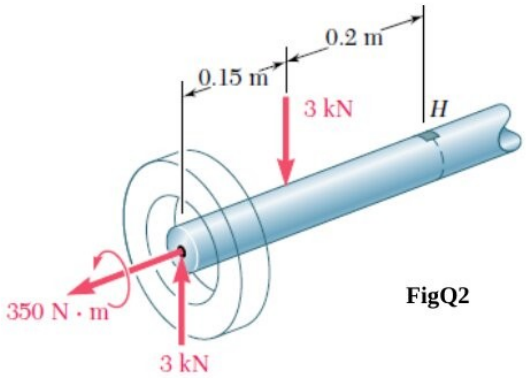
Q25 (PYQ 2021-22 Compre A Q2) — Automobile axle: stress at H, principal stresses, von Mises (medium, 10 marks)

Q2. The axle of an automobile is acted upon by the forces and couple as shown in **Fig Q2**. Knowing that the diameter of the solid axle is 32 mm and yield strength of the material is 350 MPa. Determine: **[10]**

(a) the state of stress at point ***H*** located on top of the axle (on vertical diameter).

(b) the principal stresses at the same point.

(c) will the axle yield as per Von-Mises criteria?



FigQ2

Original question: PYQ 2021-22 Compre A Q2

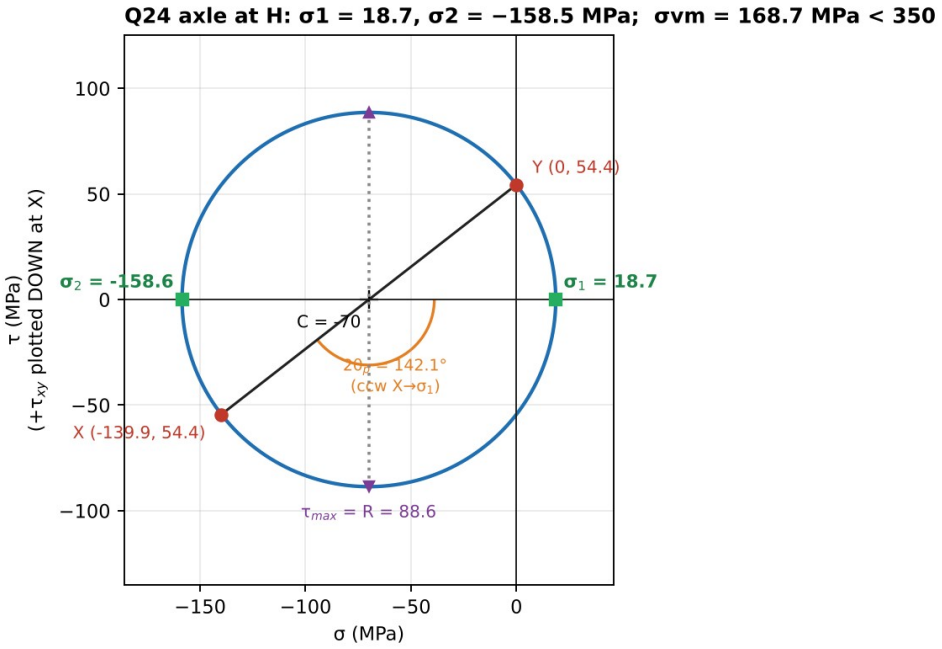
Problem. A solid 32 mm axle carries, at the wheel end, 3 kN ↑ and a 350 N·m torque, and 3 kN ↓ at 0.15 m from the wheel. H is on top of the axle, 0.2 m beyond the 3 kN ↓ load. σY = 350 MPa. Find (a) the state of stress at H, (b) the principal stresses and (c) whether the axle yields by von Mises.

Solution.

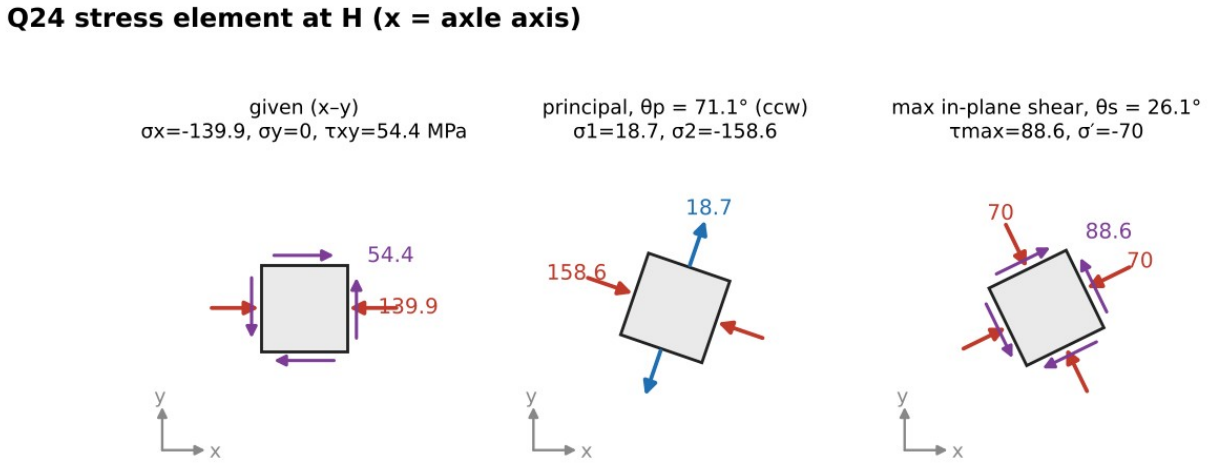
1. *Internal forces at H* (from the wheel side): V = 3 − 3 = 0 (no transverse shear). M = 3(0.35) − 3(0.20) = 0.45 kN·m (the two 3 kN forces form a constant couple beyond the second load). T = 350 N·m.
2. *(a) Stresses at H (top fibre):* $\sigma = \dfrac{32(450\,000)}{\pi(32)^3} = 139.9$ MPa, compression at the top (sagging), and $\tau = \dfrac{16(350\,000)}{\pi(32)^3} = 54.4$ MPa.
3. *(b)* Centre −69.9, R = 88.6, so σ_1 = 18.7 MPa, σ_2 = −158.5 MPa and τ_{max} = 88.6 MPa.
4. *(c)* $\sigma_{vm} = \sqrt{139.9^2 + 3(54.4)^2} = 168.7$ MPa < 350.

$$(a)\ \sigma = -139.9\ \text{MPa},\ \tau = 54.4\ \text{MPa};\ (b)\ 18.7,\ -158.5\ \text{MPa};\ (c)\ \text{no yield}\ (\text{FS} = 2.07)$$

Exam takeaway. Two equal and opposite forces make a constant bending moment between them, so V = 0 at H.



Mohr's circle at H



Stress elements at H

Q26 (PYQ 2023-24 Compre B Q2) — Crank on a 20 mm shaft: reactions and elements E_1 and E_2 (hard, 20 marks)

Q2. ***Fig.Q2*** shows a crank loaded by a force ***F= 2 k N*** that causes twisting and bending of a ***20mm*** diameter shaft fixed to a support at the origin of a reference system. In actuality, the support may be an inertia that we wish to rotate, but for the purpose of stress analysis consider this as a static problem. **[20M]**

a) Draw ***Free body diagram of the shaft AB and compute support Reactions at A***.

b) Consider two **elements** ***E_1 and E_2***, both of which are on the outer surface of the shaft at end A. Element ***E_1 is on y-axis*** *and* ***Element E_2 is on the Z-axis***. Calculate all the ***stress components that act upon this elements E_1 and E_2.***

c) ***Show the stresses on*** elements ***E_1 and E_2*** on properly oriented 2D element.

d) Determine the ***maximum normal stress, minimum normal stress and shear stress at*** element ***E_1.***

pinned conditions at the ends of the spreader.) **[15M]**

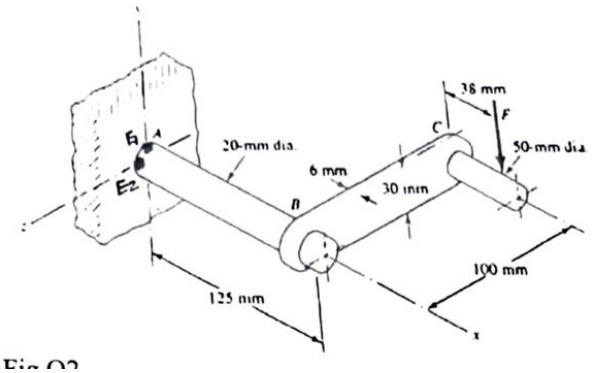


Fig.Q2

Original question: PYQ 2023-24 Compre B Q2

Problem. A crank loaded by F = 2 kN twists and bends a 20 mm shaft fixed at the origin A. The shaft AB runs 125 mm along x from the wall; a crank arm BC (6 × 30 mm) is 100 mm long; at C a pin parallel to x carries F (downward, −y) at 38 mm along the pin. (a) Draw the FBD of AB and find the reactions at A. (b) Find the stresses on elements E_1 (on the y-axis) and E_2 (on the z-axis) at A. (c) Show them on 2-D elements. (d) Find the maximum and minimum normal stresses and the maximum shear at E_1.

Approach. The load point is at x = 125 + 38 = 163 mm from A and offset 100 mm in z (reading of the figure).

Solution.

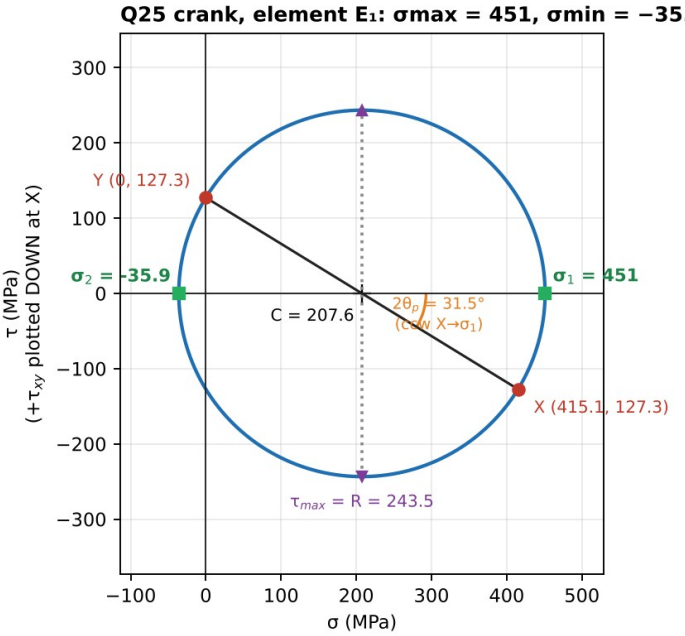
1. *(a) Reactions at A:* $R_y = +2$ kN. Twisting moment T = F(100) = 200 N·m. Bending moment $M_z = F(163) = 326$ N·m (hogging: top in tension).
2. *(b) Stresses at A:* $\sigma_b = \frac{32(326\,000)}{\pi(20)^3} = 415$ MPa; $\tau_T = \frac{16(200\,000)}{\pi(20)^3} = 127.3$ MPa; $\tau_V = \frac{4(2000)}{3\pi(100)} = 8.5$ MPa.

Element	σx	τ
E_1 (top, on y-axis)	+415 MPa (tension)	127.3 MPa (torsion only; transverse shear is 0 at the top)
E_2 (side, on z-axis = neutral axis)	0	127.3 + 8.5 = 135.8 MPa (they add on this side; 118.8 MPa on the opposite side)

3. *(c)* E_1 is viewed from +y, with σ along x and τ on the x-faces in the z direction. E_2 is viewed from +z, with τ on the x-faces in the y direction.
4. *(d) At E1:* centre 207.5, $R = \sqrt{207.5^2 + 127.3^2} = 243.5$, so σmax = 451 MPa, σmin = −35.9 MPa and τmax = 243.5 MPa.

$$E_1: \sigma_{max} = 451\text{ MPa}, \sigma_{min} = -35.9\text{ MPa}, \tau_{max} = 243.5\text{ MPa}; E_2: \tau = 135.8\text{ MPa}$$

Exam takeaway. The examiner tests *where* each stress peaks: bending at E_1, transverse shear at E_2, torsion equal at both.

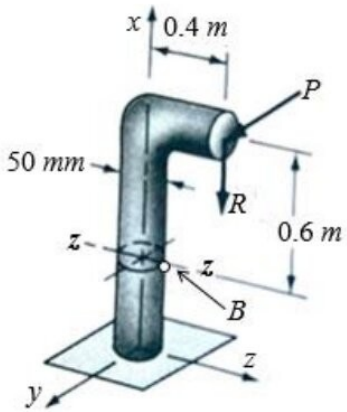


Mohr's circle at element E_1

Q27 (PYQ 2024-25 Compre B Q2) — L-shaped post: F, V, M and T at z–z; stresses at B; principal stresses; von Mises FS (hard, 18 marks)

Q2. Loads $P = 400\pi$ N and $R = 800\pi$ N applied at the free end of the 50 mm diameter post as shown in **FigQ2**. Determine
[18M]

(a) the axial force (F), shear force (V), bending moment (M_b), and twisting moment (T) at a plane z-z which is at 0.6 m from free end of post.
(b) the stresses acting on a stress element at point B. The point B is along the diameteral line z-z and on the outer surface of the post. **Show these stresses on a 2D element**.
(c) the prinicipal stresses and maximum shear stress at point B.
(d) the factor of saftey using *von-misses* yield criterion at point B, if the yield strength of the material is 140 MPa



FigQ2

Original question: PYQ 2024-25 Compre B Q2

Problem. Loads P = 400π N and R = 800π N act at the free end of a 50 mm diameter post, which has a vertical stem (axis x) and a 0.4 m horizontal arm at the top. Section z–z is 0.6 m below the free end. Point B is on the outer surface, on the diametral line z–z. σY = 140 MPa. Find (a) F, V, Mb and T at z–z, (b) the stresses at B on a 2-D element, (c) the principal stresses and τmax at B and (d) the von Mises factor of safety.

Approach. Reading of the figure: the arm runs along z, P acts along −y and R downward (−x). If your figure differs, rerun the same steps.

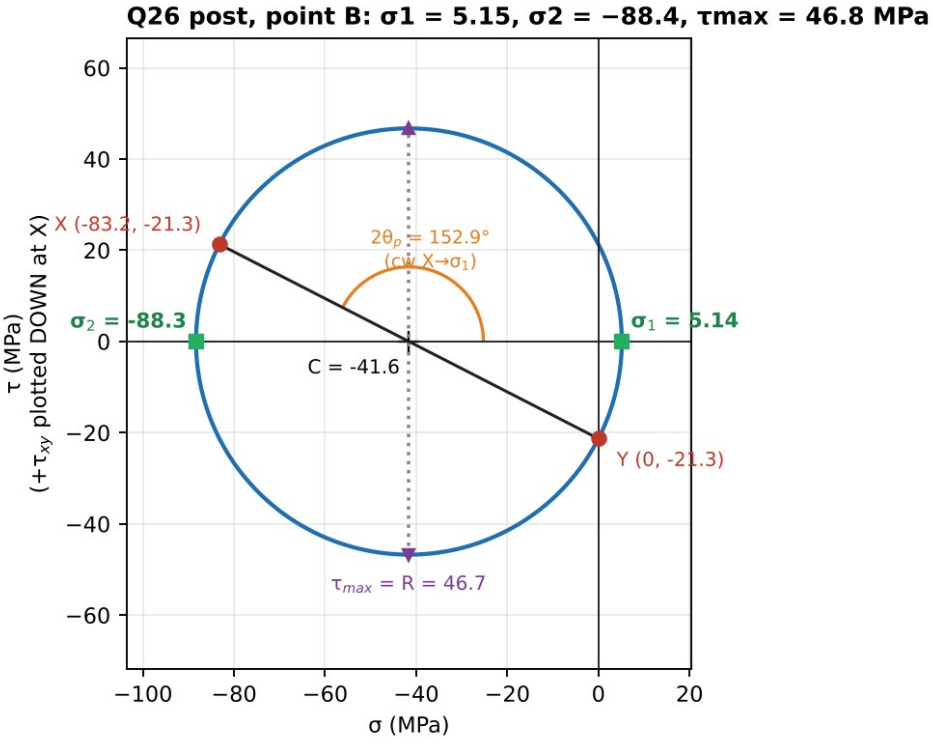
Solution.

1. *(a) Section forces* ($M = r \times F$, with $r = (0.6, 0, 0.4)$ m and $F = (-800\pi, -400\pi, 0)$ N): axial F = −800π = −2513 N (compression); shear V = 400π = 1257 N (y-direction); $M = (160\pi, -320\pi, -240\pi)$ N·m, so T = 160π = 502.7 N·m, $M_y = 320\pi = 1005$ N·m and $M_z = 240\pi = 754$ N·m (resultant $M_b = 400\pi = 1257$ N·m).
2. *(b) Stresses at B* (on the z-axis, so only My bends B, and B is on the neutral axis of Mz, where the y-shear peaks). With A = 625π mm² and πd³/32 = 3906π mm³ (the π's cancel, which is why they are in the data):
   - $\sigma_{axial} = -800\pi/625\pi = -1.28$ MPa
   - $\sigma_{bend} = -320\pi \times 10^3/3906\pi = -81.92$ MPa (the downward R on the +z arm compresses the +z side)
   - $\tau_T = 160\pi \times 10^3/7812\pi = 20.48$ MPa
   - $\tau_V = 4V/3A = 0.853$ MPa (adds to torsion at B)

   So σx = −83.2 MPa and τ = 21.3 MPa.
3. *(c)* Centre −41.6, R = 46.75, so σ₁ = 5.15 MPa, σ₂ = −88.35 MPa and τmax = 46.75 MPa.
4. *(d)* $\sigma_{vm} = \sqrt{83.2^2 + 3(21.3)^2} = 91.0$ MPa, so FS = 140/91.0 = 1.54.

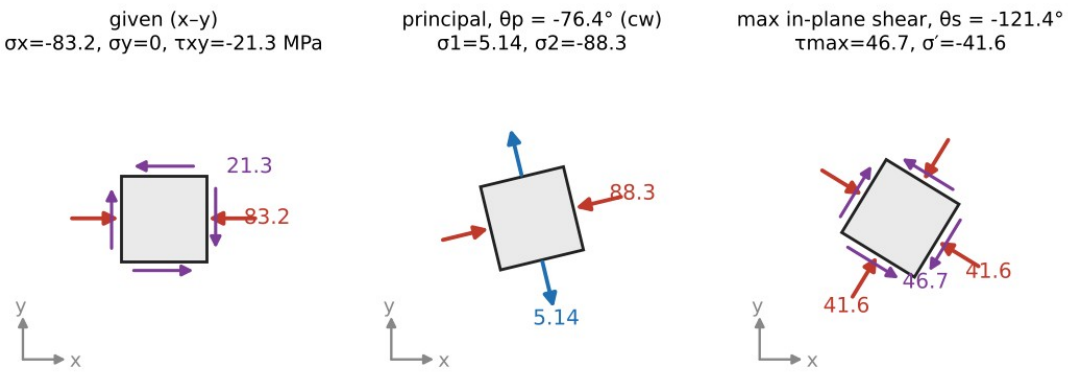
$$\sigma_x = -83.2\,\text{MPa}, \tau = 21.3\,\text{MPa}; \sigma_{1,2} = 5.15, -88.35\,\text{MPa}; \tau_{max} = 46.8\,\text{MPa}; FS = 1.54$$

Exam takeaway. π-multiples in the data are a hint that the areas and moduli will cancel nicely; it is also a check that you used the right formula.



Mohr's circle at B

Q26 stress element at B (x = post axis)



Stress elements at B

Chapter 6 — Formula and convention sheet

Item	Formula
Kinematics	sections stay plane and undistorted; $\gamma = r\,d\phi/dz$
Torsion formula	$T/J = \tau/r = G\phi/L$
Solid	$J = \pi d^4/32$; $\tau_{max} = 16T/\pi d^3$
Hollow	$J = \pi\left(d_o^4 - d_i^4\right)/32$; $\tau_{min} = \tau_{max} d_i/d_o$
Thin tube	$J \approx 2\pi r_m^3 t$, $\tau \approx T/\left(2\pi r_m^2 t\right)$
Twist	$\phi = \sum T_i L_i/\left(G_i J_i\right)$ with **internal**, signed torques
Stiffness	$k_T = GJ/L$
Power	$P = T\omega = 2\pi nT/60$ (n in rpm)
Gears	$T_C/T_B = r_C/r_B$; $r_B\phi_B = r_C\phi_C$; far shaft flexibility × (gear ratio)^2
Fixed–fixed, one torque	$T_A = T_0 b/L$, $T_D = T_0 a/L$
Composite shaft	$T_i = T\left(GJ\right)_i/\sum\left(GJ\right)$
Energy	$U = \sum T^2 L/\left(2GJ\right)$
Pure-torsion element	principal ±τ at 45°; gauge at 45° reads γ/2
Combined σ + τ	$\sigma_{1,2} = \sigma/2 \pm \sqrt{\left(\sigma/2\right)^2 + \tau^2}$; vM $\sqrt{\sigma^2 + 3\tau^2}$; Tresca $\sqrt{\sigma^2 + 4\tau^2}$
Bending + torsion (solid)	$\tau_{max} = \frac{16}{\pi d^3}\sqrt{M^2 + T^2}$, $\sigma_1 = \frac{16}{\pi d^3}\left(M + \sqrt{M^2 + T^2}\right)$
Transverse shear (solid circle)	$\tau_{V,max} = 4V/3A$ at the neutral axis

Course summary

Part 1 — One-page formula and convention sheet (Ch 1–6)

Conventions you must not mix up

Item	Convention used in the slides/Crandall
Stress/force sign	+ on a **positive face in a positive direction** (or −face, −direction). Tension +.
Shear force V (Ch 3)	Crandall: $V = -\Sigma(\text{up-forces left of cut})$, $dV/dx = -q$, $dM/dx = -V$. **Opposite sign to Gere/Hibbeler.** M is the same (sagging +).
Angle θ (Ch 4)	counter-clockwise from x to the **normal** of the face
Mohr's circle	σ to the right; **+τ_xy plotted downward at X**, upward at Y; rotating 2θ on the circle = rotating θ on the element, **same sense**
Strain circle	vertical axis = **γ/2**
Torque	right-hand rule; internal torque + if its vector points out of the cut face

Ch 1 — Statics

$\sum F_x = \sum F_y = \sum M = 0$ · two-force member: forces collinear · three-force: concurrent · $f \le \mu N$ (= at impending) · incline: $T = W(\sin\alpha \pm \mu\cos\alpha)$ · belt: $T_2/T_1 = e^{\mu\beta}$ · truss: $m = 2j - 3$; sections: cut ≤ 3 members, moments about the intersection of two.

Ch 2 — Axial members and energy

$\delta = \sum \frac{P_i L_i}{A_i E_i}$ (internal P) · $k = AE/L$ · parallel: k adds; series: $1/k$ adds · indeterminate: $\delta_{load} + \delta_{redundant} = 0$ · displacement diagram: $\delta_{bar} = u \cdot e \cdot U = P^2 L/2AE$ · Castigliano $\delta_i = \partial U/\partial F_i$; truss $\delta = \sum F \frac{\partial F}{\partial P}\frac{L}{AE}$; beam $\int \frac{M}{EI}\frac{\partial M}{\partial P} ds$ · thin ring $\sigma = pr/t$, $\delta_R = pr^2/Et$.

Ch 3 — Internal forces

Resultant of distributed load = area at the centroid (**reactions only**) · $V' = -q$, $M' = -V$ · M_{max} where V = 0 · contraflexure where M changes sign · clockwise couple: M jumps up · singularity: $\langle x-a\rangle^n$, point load $\langle\rangle_{-1}$, couple $\langle\rangle_{-2}$, no constants · standard: SS + W centre $WL/4$; SS + UDL $wL^2/8$; SS + W at a Wab/L; cantilever tip PL; cantilever UDL $wL^2/2$ · curved cantilever (tip P): $N = P\cos\theta$, $V = P\sin\theta$, $M = PR\cos\theta$.

Ch 4 — Stress/strain transformation

$\sigma_{x'} = \frac{\sigma_x+\sigma_y}{2} + \frac{\sigma_x-\sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$ · $\tau_{x'y'} = -\frac{\sigma_x-\sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$ · $\sigma_{x'} + \sigma_{y'}$ = const · $\sigma_{1,2} = \acute{\sigma} \pm R$, $R = \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$, $\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x-\sigma_y}$ (substitute back to identify σ1) · $\tau_{max} = R$ at $\theta_p \pm 45°$, with $\acute{\sigma}$ on those faces · **absolute** $\tau_{max} = (\sigma_{max} - \sigma_{min})/2$ including $\sigma_3 = 0$ · strain: replace σ→ε, τ→γ/2 ·

$\epsilon_x = \partial u/\partial x,\ \gamma_{xy} = \partial u/\partial y + \partial v/\partial x \cdot 45°$ rosette $\gamma = 2\epsilon_b - \epsilon_a - \epsilon_c \cdot 60°$ rosette $\epsilon_y = \frac{1}{3}\left(2\epsilon_b + 2\epsilon_c - \epsilon_a\right),$

$\gamma = \frac{2}{\sqrt{3}}\left(\epsilon_b - \epsilon_c\right).$

Ch 5 — Constitutive law and yield

$\epsilon_x = \frac{1}{E}\left[\sigma_x - \nu\left(\sigma_y + \sigma_z\right)\right] + \alpha\,\Delta T \cdot \gamma = \tau/G \cdot$ plane stress: $\sigma_x = \frac{E}{1-\nu^2}\left(\epsilon_x + \nu\,\epsilon_y\right),\ \epsilon_z = -\frac{\nu}{E}\left(\sigma_x + \sigma_y\right) \cdot G = \frac{E}{2(1+\nu)},$

$K = \frac{E}{3(1-2\nu)} \cdot \Delta V/V = \frac{1-2\nu}{E}\sum\sigma \cdot$ fixed bar heated: $\sigma = -E\,\alpha\,\Delta T \cdot u = \frac{1}{2E}\left[\sum\sigma_i^2 - 2\nu\sum\sigma_i\sigma_j\right] \cdot$ **Tresca** $\sigma_{max} - \sigma_{min} = \sigma_Y$ (σ–τ: $\sqrt{\sigma^2 + 4\tau^2}$) · **von Mises** $\sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2} = \sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2}$ (σ–τ: $\sqrt{\sigma^2 + 3\tau^2}$) · FS $= \sigma_Y/\sigma_{eq}$; Tresca is always more conservative.

Ch 6 — Torsion

$\gamma = r\,d\phi/dz \cdot \frac{T}{J} = \frac{\tau}{r} = \frac{G\phi}{L} \cdot J = \frac{\pi d^4}{32},\ J_h = \frac{\pi\left(d_o^4 - d_i^4\right)}{32} \cdot \tau_{max} = \frac{16T}{\pi d^3} \cdot \phi = \sum\frac{TL}{GJ} \cdot P = \frac{2\pi nT}{60} \cdot$ gears: $T \propto r,$

$r_B\phi_B = r_C\phi_C \cdot$ fixed–fixed: $T_A = T_0\,b/L \cdot U = \sum\frac{T^2L}{2GJ} \cdot$ pure shear: principal ±τ at 45° · combined:

$\sigma = \frac{F}{A} \pm \frac{32M}{\pi d^3},\ \tau = \frac{16T}{\pi d^3} \pm \frac{4V}{3A}$ (at the neutral axis).

Units cheat

N, mm, MPa (= N/mm²) are consistent; GPa = 10^3 MPa; kN·m = 10^6 N·mm; angles in **radians** inside formulas; P = Tω needs ω in rad/s.

Part 2 — PYQ question types ranked by frequency

Counted across the 7 papers supplied (2014-15 Test-1, Test-2, Compre; 2019-20, 2021-22, 2023-24, 2024-25 Compre). "Appearances" counts distinct questions, including sub-parts embedded in bigger questions.

Rank	Question type	Appearances	Where	Solved in
1	**Euler buckling of columns** (often combined with a beam that loads the column)	8	19 B1, 21 A3, 21 B3, 23 B1, 24 A3, 24 B3, 14 C-A20, 14 C-B4	*not in Ch 1–6 slides*
2	**Torsion of circular shafts** (allowable T, twist, hollow vs solid, gears, fixed–fixed)	7	14 T2-Q4, 14 C-A5, 14 C-B2, 19 B4, 21 A1, 23 A1, 24 B1	Ch 6 Q11–Q13, Q17, Q18, Q20; Ch 4 Q30
3	**Beam deflection / statically indeterminate beams** (integration, singularity functions, settlement, beam + rod)	7	14 C-A9/10/15/16, 19 B3, 21 B2, 21 B3, 23 B3, 24 B3	*not in Ch 1–6 slides* (SFD/BMD groundwork in Ch 3)
4	**Combined loading**	6	14 T2-Q1, 19 B2, 21	Ch 6 Q25–Q27; Ch 4

Rank	Question type	Appearances	Where	Solved in
	at a point (find F, V, M, T at a section → σ, τ on a 2-D element → principal stresses → FS)		A2, 21 B4, 23 B2, 24 B2	Q15 (19 B2 and 21 B4 need bending I of box/T sections, so they are not solved)
5	**Principal stresses / Mohr's circle / transformation** (stand-alone)	5	14 C-A12, 14 C-A13, 14 C-B1, 19 A2, 14 T2-Q2	Ch 4 Q13, Q14, Q16, Q31, Q33
5	**Yield criteria and factor of safety** (Tresca, von Mises)	5	14 C-A3, 14 C-A4, 19 A4, 21 A2c, 24 B2d	Ch 5 Q23–Q25; Ch 6 Q25, Q27
5	**Bending stress / curved beams / eccentric loads** (σ = My/I, T-sections, chain link)	5	14 C-A19, 14 C-B3, 19 A1b, 23 A2, 24 A1	partly (Ch 3 Q28 uses $\sigma = 32M/\pi d^3$); rest *not in slides*
8	**Generalized Hooke's law: Δ-dimensions, Δt, ΔV, circle → ellipse**	4	14 C-A2, 21 B1, 23 A3, 24 A2	Ch 5 Q5, Q9, Q10; Ch 4 Q32
8	**SFD/BMD, contraflexure, singularity functions**	4 papers (6 Qs)	14 T1-Q4, 14 C-A6/7/14/17/18, 19 A1	Ch 3 Q7, Q15, Q25, Q27, Q28
10	**Strain energy / Castigliano**	3	14 T1-Q2, 19 A3b, 24 B1c	Ch 2 Q25, Q26; Ch 6 Q20
11	**Thermal stresses**	2	14 T2-Q3, 14 C-A1	Ch 5 Q12, Q21
12	Trusses (sections) · friction + axial (rope on drum) · thin ring · strain gauge/rosette · axial stress ratio	1 each	14 T1-Q1, 14 T1-Q3, 14 C-A11, 21 A1, 14 C-A8	Ch 1 Q26, Q16; Ch 2 Q28, Q11; Ch 4 Q30

What this tells you for the compre:

1. **Combined loading + Mohr + yield** shows up in every recent Part B (2019, 2021, 2023, 2024), worth 18–20 marks each time. Practise the 6-step procedure in Ch 6 Concept 6.6 until it's automatic.
2. **Torsion** appears in nearly every paper, usually as one clean 10–12 mark question (fixed–fixed shaft, gear pair, or allowable T + twist).
3. **Hooke's-law plate questions** (Δt, ΔV, etched circle) are a near-certain 10 marks in Part A (2021, 2023, 2024).
4. **Buckling and beam deflection** together are the biggest block, and they are **outside** these slides. Study them next.
5. Test-1 style (2014) puts trusses, friction, Castigliano and singularity SFD/BMD together, so Ch 1–3 matter most for the mid-sem.

Part 3 — Errors found in the slides (double-check with your instructor)

Location	Slide says	Correct value	Why
Ch 1, Warren truss (2000	BE = 500 N (C)	**577.4 N (C)**	500 N is the *vertical*

Location	Slide says	Correct value	Why
N/4000 N) table			*component*; member force = 500/sin 60°
Ch 1, A-frame truss table	member "AB"	**AC**	there is no member AB
Ch 2, brass bar 50/80/40/10 kN	δ = 0.2857 mm	**0.114 mm**	0.2857 is the AB segment alone; BC (−30 kN) and CD (+10 kN) were dropped
Ch 2, member ABCD	0.4903 **m**	0.491 **mm**	units typo
Ch 4 Part 2, rosette example	ε_2 = **246** − 119.2	**153** − 119.2 = 33.8 μ	153 is the circle centre; final value is right
2023-24 Compre A-Q3	E = 200 **MPa**	E = 200 **GPa**	steel

Part 4 — PYQ questions not covered here (topics after Ch 6)

Paper	Question	Topic needed
2014-15 Compre A	Q9, Q10 (cantilever tip deflection/slope), Q15–16 (propped cantilever), Q19 (bending/shear ratio), Q20 (buckling)	Ch 7 (bending), Ch 8 (deflection), Ch 9 (buckling)
2014-15 Compre B	Q3 (inverted-T beam stresses), Q4 (buckling from deflection data)	Ch 7, Ch 9
2019-20	A-Q1(b) (partly done in Ch 3 Q28), B-Q1 (two columns), B-Q2 (box-beam combined loading needs I and Q of a box section), B-Q3 (beam + rod, 4th-order ODE)	Ch 7–9
2021-22	A-Q3, B-Q2, B-Q3, B-Q4	Ch 7–9
2023-24	A-Q2 (eccentric pull on a post), B-Q1 (spreader buckling), B-Q3 (singularity deflection with settlement)	Ch 7–9
2024-25	A-Q1 (curved chain link), A-Q3, B-Q3	Ch 7–9

Say the word and the next session can cover Ch 7–9 (bending stresses, beam deflection, buckling) in the same format, starting with these questions.